PROJECT UDBHAVA

Team 316 Project Technical Report for the 2026 IREC



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Project *Udbhava* is a single-stage sounding rocket developed by Team Sammard for the IREC 2026 10,000 ft SRAD solid propulsion category. The vehicle has a length of 3.06 m, a diameter of 153 mm, and a liftoff mass of 36.25 kg. It is powered by an N2622-class SRAD solid rocket motor using KNSB propellant, delivering a total impulse of 10,384 Ns. The design targets a 10,000 ft apogee, a maximum velocity of 278 m/s, and a peak acceleration of 7.74 g. The rocket employs a composite structure comprising filament-wound fiberglass airframes, a fibreglass nose cone, and hybrid composite fins designed to ensure aeroelastic stability. A dual-deployment recovery system using black powder enables controlled descent and safe recovery. Avionics include COTS and redundant SRAD systems with onboard telemetry and GPS tracking. The payload utilizes a bioprinter system designed to operate under high dynamic flight conditions, capable of printing during flight using a specialized bio-ink formulation.

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TABLE OF CONTENTS

I. Nomenclature

A. Chemicals....3
B. Acronyms....3
C. Variables....4

II. Introduction

A. Academic Program....5
B. Stakeholders....5
C. Team Management Strategies....5
D. Team Structure....6
E. Report Preparation....6

III. System Architecture Overview

A. Propulsion Subsystem....8
B. Aerostructures Subsystem....11
C. Avionics Subsystem....21
D. GroundStation Subsystems....24
E. Recovery Subsystems....29
F. Payload Subsystems....32
G. Airbrakes Subsystem....34

IV. Mission Concept of Operations ……..38

V. Conclusions and Lessons Learned....39

VI. Acknowledgements....40

VII. Appendix

A. Systems Weights, Measures and Performance Data Appendix....41
B. Project Test Reports....185
C. Hazard Analysis....237
D. Risk Assessment....241
E. Assembly, Preflight, Launch, Recovery and Off-Nominal Checklists....255
F. Engineering Drawings....266

VIII. References

I.Nomenclature

I.A Chemicals

ABS	Acrylonitrile Butadiene Styrene
Al	Aluminum
BP	Black Powder
PLA	Polylactic Acid
SS	Stainless Steel
KNSB	Potassium Nitrate–Sorbitol (Propellant)
EPDM	Ethylene Propylene Diene Monomer (Rubber Insulation)

I.B Acronyms

AGL	Above Ground Level
AoA	Angle of Attack
AoS	Angle of Sideslip
CONOPS	Concept of Operations
CAD	Computer-Aided Design
CFD	Computational Fluid Dynamics
CFRP	Carbon Fiber Reinforced Polymer
CNC	Computer Numerical Control
COTS	Commercial Off-The-Shelf
Cp	Centre of Pressure
Cg	Centre of Gravity
CSV	Comma-Separated Values
DoF	Degrees of Freedom
DTEG	Design, Test & Evaluation Guide
dt	Time Step
ESRA	Experimental Sounding Rocket Association
FOS	Factor of Safety
FPS	Frames Per Second
FSI	Fluid-Structure Interaction
GFRP	Glass Fiber Reinforced Polymer
GPS	Global Positioning System
GPU	Graphics Processing Unit
GSM	Grams per Square Meter
GUI	Graphical User Interface
IMU	Inertial Measurement Unit
IREC	International Rocket Engineering Competition
LED	Light Emitting Diode
Li-ion	Lithium-ion Battery
LiPo	Lithium Polymer Battery
MP	Mega Pixel
PCB	Printed Circuit Board
PDB	Power Distribution Board

SD	Secure Digital
SDL	Space Dynamics Laboratory
SRAD	Student Researched and Developed

I.B Variables

A	Cross-sectional area (m^2)
AR	Aspect Ratio (-)
c	Chord length (m)
Cd	Coefficient of Drag (-)
D_0	Outer diameter (mm)
D_i	Inner diameter (mm)
f, g	Generic functions (-)
F	Force (N)
F_load	Load force (N)
F_max	Maximum force (N)
F_x	Force in X-direction (N)
F_γ	Force in Y-direction (N)
F_z	Force in Z-direction (N)
h	Height (m)
p	Pressure (Pa)
r	Radius (mm)
S	Surface area (m^2)
t	Wall thickness (pressure vessel) (mm)
V	Velocity (m/s)
V_f	Fin flutter velocity (m/s)
V_d	Desired descent velocity (m/s)
V_i	Velocity at main deployment (m/s)
W	Weight (kg)

II. INTRODUCTION

Team Sammard is a student-led aerospace team based at the Vellore Institute of Technology (VIT), Vellore, India, focused on the design, analysis, manufacturing, and testing of high-performance sounding rockets. For IREC 2026, Team Sammard is competing in the 10,000 ft Solid Rocket Motor (SRAD) category with project *Udbhava*, marking the team's first transition to a fully student-developed propulsion system. In addition to achieving the target apogee, a key objective of this mission is the design, validation, and flight testing of the SRAD motor, with emphasis on predictable performance and safe operation.

The launch vehicle is powered by a Class N SRAD solid rocket motor, utilizing a BATES grain configuration with a KNSB propellant formulation. The propulsion system is designed for stable combustion, structural integrity, and repeatable manufacturing.

A major advancement in this iteration is the implementation of a dual communication architecture, comprising a 4-stack Yagi antenna system (915 MHz) for long-range telemetry and a high-gain horn antenna system (5.8 GHz) for live video transmission. These systems are integrated with an Antenna Tracking System (ATS), enabling reliable signal acquisition throughout flight.

The payload for this mission is an in-flight bioprinter system, designed to operate under high dynamic conditions. The system utilizes a specialized bio-ink formulation and performs controlled printing during flight, enabling the study of bioprinting behavior under varying acceleration and environmental conditions.

The primary mission objectives include achieving the target apogee within tolerance, maintaining stable flight, successfully deploying the recovery system, validating SRAD propulsion performance, operating the payload during flight, and ensuring safe recovery of the rocket and onboard systems. Beyond mission success, the team aims to strengthen its capabilities in SRAD propulsion development, subsystem integration, and experimental payload validation.

A. Academic Program

Team Sammard operates as an extracurricular, interdisciplinary engineering initiative, comprising students from mechanical, electronics, electrical, computer science, and instrumentation backgrounds. In the absence of a dedicated aerospace program, members engage in self-driven learning and hands-on development, bridging the gap between theoretical knowledge and real-world application.

The team utilizes university facilities, including manufacturing workshops and project laboratories, to design, fabricate, and test its systems. This environment promotes collaborative problem-solving and provides members with exposure to **practical engineering workflows and system-level design challenges**.

B. Stakeholders

The development of Team Sammard's launch vehicle is supported through a combination of **institutional resources, industry partnerships, and alumni contributions**. The Vellore Institute of Technology (VIT) provides access to manufacturing workshops, laboratory spaces, and testing facilities for design, fabrication, and validation activities.

The team is supported by industry partner **Convergent Science**, which enhances technical capabilities through access to advanced engineering tools for simulation and analysis.

Additionally, **team alumni** contribute through mentorship and technical guidance, supporting design refinement and knowledge transfer. Funding is secured through institutional support, sponsorships, and team-led initiatives.

C. Team Management Strategies

Team Sammard follows a **structured management approach** centered on communication, accountability, and technical rigor. The Team Captain and Vice Captain serve as the primary interface between the team and the Vellore Institute of Technology's Students' Welfare Office, managing essential tasks such as fund procurement, financial oversight, and obtaining university permissions for testing. For IREC 2026 and Project *UDBHAVA*, the team is led by a hierarchy of Project Lead, Department Heads, and Sub-leads. While Department Heads oversee the broader deadlines and strategy for their divisions, Sub-leads provide direct supervision by monitoring daily progress and assisting members with specific technical tasks. This structure ensures a high level of accountability and provides a consistent support system for all members.

The team's operations are rooted in the principles of punctuality and technical competence, supported by a robust communication network. Open dialogue is maintained through daily meetings held at 9:30 PM in the university's Project Labs, which allow for real-time collaboration across all departments. These physical meetings are

supplemented by dedicated digital groups in WhatsApp Messenger for quick inquiries and weekly management reviews to ensure the project remains aligned with the competition timeline. To sustain the team's legacy, leads with Sub-leads conduct regular training sessions and technical presentations to mentor junior members. At the end of each tenure, the leadership conducts a thorough review of achievements and challenges, using these insights to refine the team structure and transition responsibilities to the next generation of student leaders.

D. Team Structure

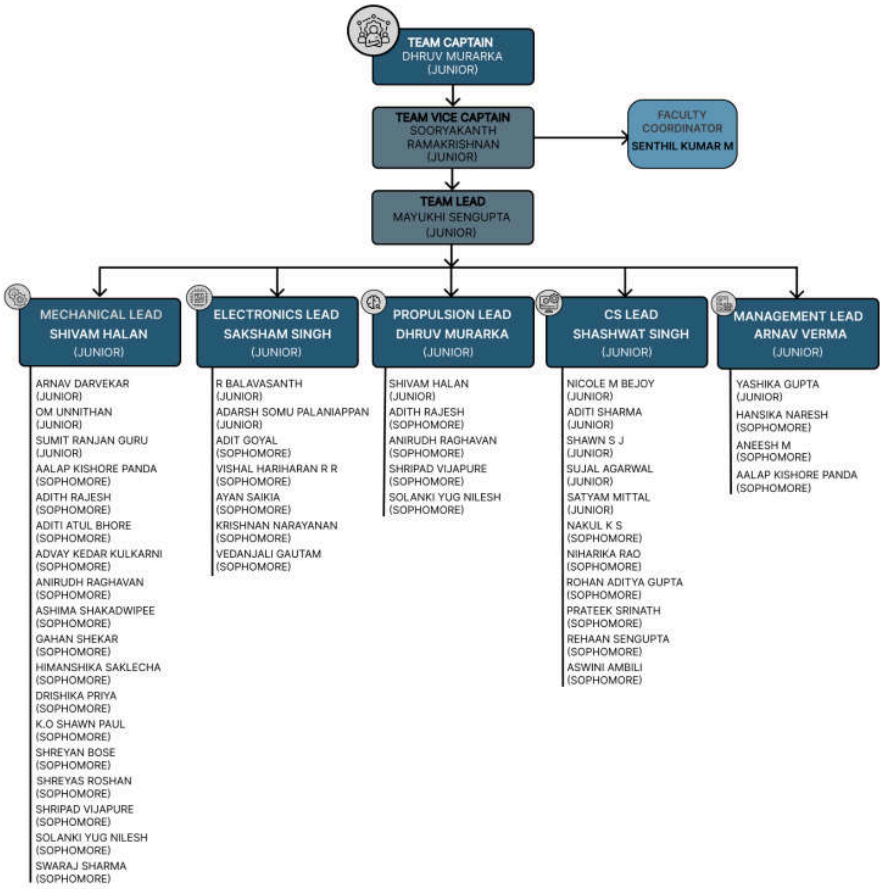


Figure 1: Team Structure

E. Report Preparation

The Project Technical Report was prepared as a **collaborative effort** under Project UDBHAVA, with each subsystem team responsible for drafting their respective sections based on design, analysis, and testing activities. Subsystem leads ensured technical accuracy within their domains, while a centralized documentation team maintained consistency in formatting and presentation. Final review and editing were conducted by the project leadership to ensure clarity, coherence, and alignment with competition requirements.

III. SYSTEM ARCHITECTURE OVERVIEW

Project UDBHAVA is designed as an integrated multi-subsystem launch vehicle, where each subsystem is developed to operate both independently and in coordination with the overall system architecture. The primary subsystems include propulsion, aero-structures, avionics, recovery, payload, and ground systems, each fulfilling a specific functional role within the mission profile. The architecture emphasizes modularity, reliability, and ease of integration, enabling efficient development, testing, and iterative refinement.

Table 1: Udbhava high - level Specifications

Overall Rocket Parameters	**Measurements**
Airframe Length with Nosecone	3056 mm
Airframe Length without Nosecone	2058 mm
Airframe Diameter	153 mm
Number of Fins	4
Fin Length	Root Chord 240 mm, Tip Chord 75 mm
Vehicle Weight	36250g
Motor Type	SRAD Solid
Motor Letter Characterization	N
Propellant Weight	9542 g
Number of Stages	1

The propulsion subsystem provides the thrust required to achieve the target apogee, while the aero-structures subsystem ensures structural integrity and aerodynamic stability throughout flight. The avionics system serves as the central control unit, handling sensor data acquisition, state estimation, telemetry transmission, and recovery actuation. The recovery subsystem enables controlled descent and safe landing through a dual-deployment parachute system. The payload subsystem, consisting of an in-flight bioprinter, operates under dynamic flight conditions to achieve mission-specific objectives. The ground systems, including telemetry reception, antenna tracking, and data visualization platforms, ensure continuous monitoring and post-flight analysis.

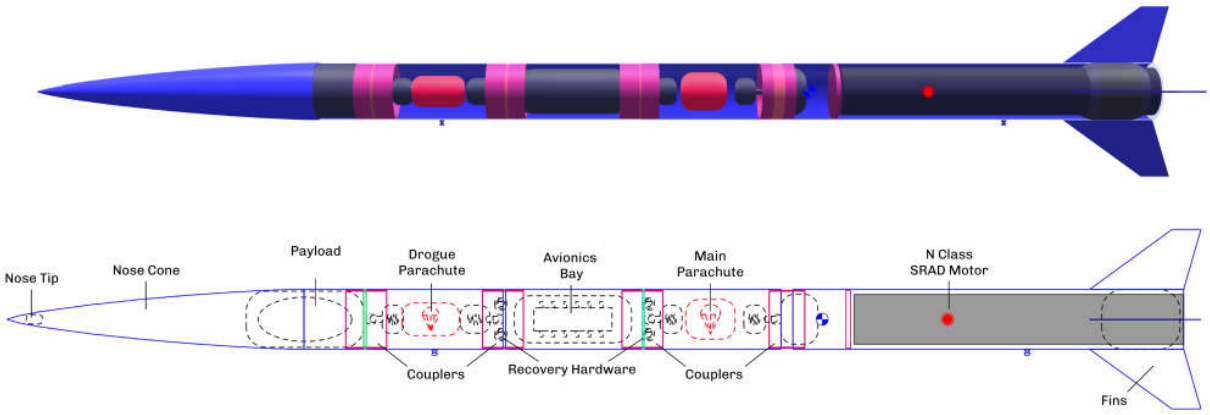


Figure 2: Annotated Rocket Configuration

The system architecture is designed with a focus on robust communication, redundancy, and fault tolerance, ensuring reliable operation across all mission phases. Interfaces between subsystems are carefully defined to minimize integration complexity while maintaining flexibility for future design iterations. This structured approach enables coordinated subsystem performance, contributing to overall mission success.

A. Propulsion Subsystem

The propulsion system is designed to meet the performance requirements of a 10,000 ft apogee mission. The *Udbhava* rocket is powered by an in-house developed Class N solid rocket motor employing a BATES grain configuration. The motor utilises six KNSB propellant grains with a 65-35 oxidiser-to-fuel ratio, delivering the total impulse necessary for the mission profile. This design builds upon the team's prior experience in 10,000 ft class rockets demonstrated at the IREC.

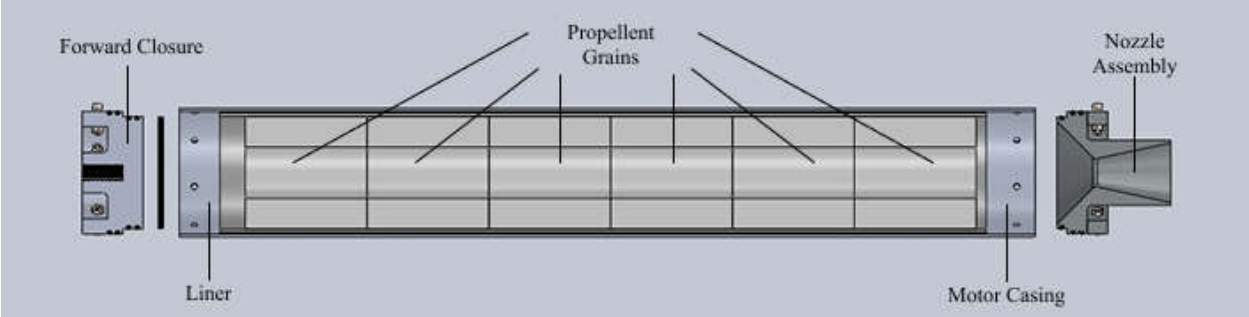


Figure 3 : Motor CAD section view

The propellant grains were cast with institutional permission from the Vellore Institute of Technology. The propulsion system underwent three static fire tests to validate performance characteristics, in addition to a dedicated hydrostatic pressure test of the motor casing to verify structural integrity and safety margins under pressurised conditions.

Propellant Chemistry

The propulsion system uses a KNSB (Potassium Nitrate–Sorbitol) propellant with a 65:35 oxidizer-to-fuel ratio. Potassium nitrate (KNO_3) provides oxygen for combustion and pressure generation, while sorbitol ($C_6H_{14}O_6$) acts as both fuel and binder, ensuring structural integrity and stable regression. A small amount of SLES is added as a surfactant to improve particle wetting and mixture uniformity.

```
  Code                                  WEIGHT     D-H  DENS     COMPOSITION
      0 POTASSIUM NITRATE                65.000   -1167  0.07670   1 N    3 O    1 K
      0 SORBITOL                         35.000   -1776  0.05310   6 C   14 H    6 O

THE PROPELLANT DENSITY IS  0.06638 LB/CU-IN OR    1.8373 GM/CC
THE TOTAL PROPELLANT WEIGHT IS  100.0000 GRAMS

NUMBER OF GRAM ATOMS OF EACH ELEMENT PRESENT IN INGREDIENTS

    2.689677 H
    1.182719 C
    0.642877 N
    3.081345 O
    0.642877 K

****************************CHAMBER RESULTS FOLLOW   ****************************
  T(K)  T(F)  P(ATM)   P(PSI) ENTHALPY  ENTROPY   CP/CV    GAS   RT/V
  1586  2396   27.41   403.00  -138.02   173.21  1.1378  2.523  10.866

SPECIFIC HEAT (MOLAR) OF GAS AND TOTAL =    10.303    14.610
NUMBER MOLS GAS AND CONDENSED =     2.523     0.309

 9.219702e-001 H2O        4.788265e-001 CO         4.105365e-001 H2         3.651240e-001 CO2
 3.213719e-001 N2         3.090022e-001 K2CO3*     2.327988e-002 KHO        1.910166e-003 K
 1.263047e-004 KEH2O2     7.966791e-005 NH3        9.136407e-006 H          6.702189e-006 KH
 4.680709e-006 CH4        5.181767e-006 KCN        1.671407e-006 CH2O       1.499353e-006 CHN

THE MOLECULAR WEIGHT OF THE MIXTURE IS    35.313

****************************EXHAUST RESULTS FOLLOW   ****************************
  T(K)  T(F)  P(ATM)   P(PSI) ENTHALPY  ENTROPY   CP/CV    GAS   RT/V
  1150  1611     1.00    14.70  -160.11   173.21  1.1630  2.458   0.400
```

Figure 4: Equilibrium Solver results from ProPEP 3

Thermochemical performance was evaluated in ProPEP at a chamber pressure of 3.5 MPa under sea-level conditions. Results predict an adiabatic flame temperature of 1586 K, characteristic velocity of 980 m/s, and specific impulse of 114.3 s, with a specific heat ratio of ~1.138 and exhaust molecular weight of 35.3 g/mol.
The propellant density is 1.837 g/cc, consistent with KNSB, indicating stable combustion with manageable thermal loads while accounting for two-phase flow losses.

The propellant uses a BATES grain geometry for near-neutral thrust curve. Each grain has an outer diameter of 110 mm, core diameter of 50 mm, and length of 120 mm, giving a web thickness of 30 mm. This balances radial burn area growth with length reduction, maintaining nearly constant chamber pressure and thrust. Detailed manufacturing procedures for the propellant grains are provided under *Grain Casting Process* in *Appendix A*.



Figure 5 : Top view of the grain



Figure 6: Cast KNSB BATES propellant

Internal Ballistics

The OpenMotor internal ballistics simulation shows that the 6-segment BATES grain operates within safe limits, with an average chamber pressure of 3.34 MPa and a peak of 3.52 MPa, indicating stable operation. The thrust profile is near-neutral, reaching 3000 N peak thrust and delivering a total impulse of 10,995 Ns over a 3.93 s burn.

The Kn varies from 283.66 to 296.95, indicating controlled burn area evolution and supporting near-steady thrust. The port-to-throat ratio of 3.861 is near the upper limit but acceptable, limiting gas velocity and reducing erosive burning risk. This is supported by a peak mass flux of 1193.23 kg/(m²·s), which remains within safe limits for the operating pressure.

Overall, the simulation results indicate that the motor design maintains a good balance between performance and structural/combustion constraints, with no significant anomalies observed in pressure, thrust, or mass flow behaviour. The results can be seen in Fig 4.

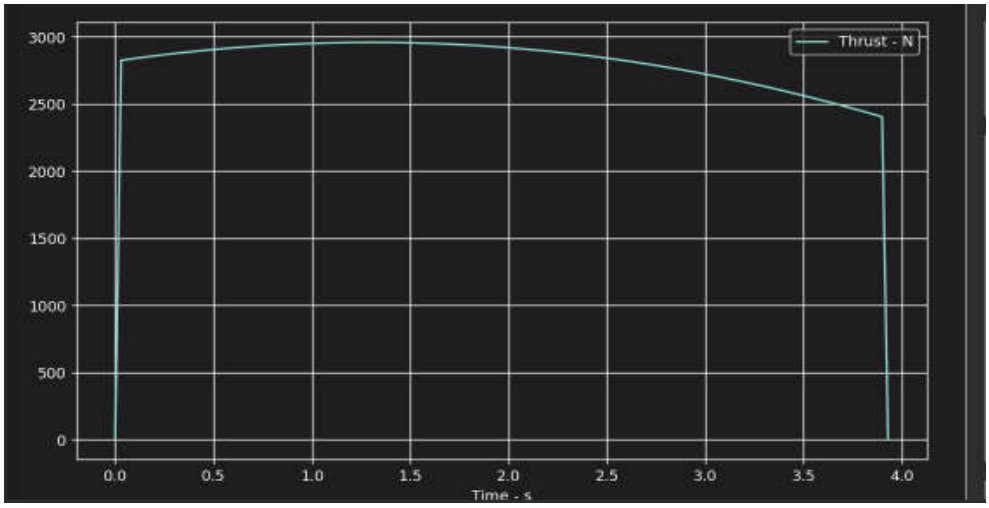


Figure 7: Thrust Curve obtained from OpenMotor

Motor Casing

The combustion chamber is a cylindrical thin-walled pressure vessel made from Aluminum 6061-T6, chosen for its high strength-to-weight ratio, machinability, corrosion resistance, and ductility. With an outer diameter of 127 mm, an inner diameter of 121 mm, and a wall thickness of 3 mm, it meets thin-walled assumptions, allowing simplified stress analysis.

Hoop stress governs the design, and using a factor of safety of 1.5 on the yield strength (276 MPa), the allowable stress is limited to 184 MPa. An Ansys Explicit Dynamics simulation [refer *Propulsion Simulation II* in *Appendix A*] confirmed a burst pressure of *9.34 MPa*, driven by plastic instability and midsection bulging consistent with hoop-stress-dominated failure, yielding a burst factor greater than 2 and confirming adequate structural margin.



Figure 8: Assembled motor casing and nozzle.

Thermal Liner

The liner consists of an internal configuration of a fibreglass (FG) liner with an outer diameter of 118 mm and an inner diameter of 111 mm, which interfaces concentrically with the casing. The annular gap between the casing inner diameter (121 mm) and the FG liner outer diameter (119 mm) is filled with a 1 mm-thick EPDM layer, which acts as a compliant thermal and mechanical interface.



Figure 9: Thermal Liner



Figure 10:EPDM application on filament-wound liner

The EPDM layer accommodates manufacturing tolerances, provides thermal insulation, dampens vibrations, and reduces stress from differential thermal expansion between the aluminium casing and FG liner. The FG liner serves as the primary structural interface with the propellant grain while also contributing to thermal resistance.

Forward Closure

The forward closure is machined from Aluminum 6063-T6, selected for its favourable strength-to-weight ratio, machinability, and corrosion resistance. It interfaces with the motor casing via 8 M6 bolted joints and incorporates 2 O-ring grooves for sealing against the motor casing, alongside a further 2 O-ring grooves on its shoulder to ensure a reliable pressure seal against the thermal liner. A layered EPDM disc, bonded using high-temperature epoxy, is seated at the closure's aft face and interfaces directly with the motor liner to provide thermal protection. A central M16 thread is machined into the closure to mate with the thrust plate, which serves as the primary load path for thrust transmission to the rocket structure.



Figure 11: Application of Red RTV to Forward closure

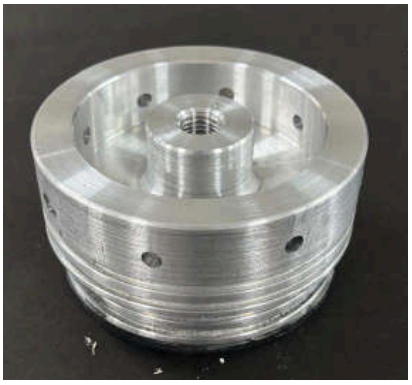


Figure 12: Machined Forward closure

Nozzle

The nozzle is a converging–diverging design for efficient gas expansion and thrust generation. It is made from SS304, chosen for high-temperature strength, oxidation resistance, and durability under high-pressure, erosive flow. Though non-ablative, it provides sufficient thermal resistance for the burn duration, validated through testing, while maintaining dimensional stability and reusability. The nozzle has a throat diameter of 28.5 mm, optimized for chamber pressure and expansion. It includes an O-ring groove on a shoulder extrusion that interfaces with the thermal liner, and is secured at the aft end by a bolted retention ring for reliable load transfer and easy assembly.

Appropriate sealing elements, i.e., high-temperature RTV silicon and Nitrile O rings, are incorporated to prevent gas leakage, and the design accounts for differential thermal expansion between the steel nozzle and aluminium casing/closure, minimising stress concentrations during operation.

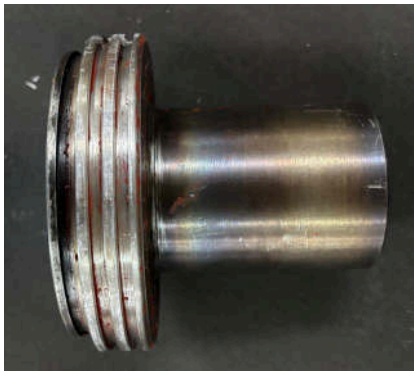


Figure 13: Machined SS304 Nozzle

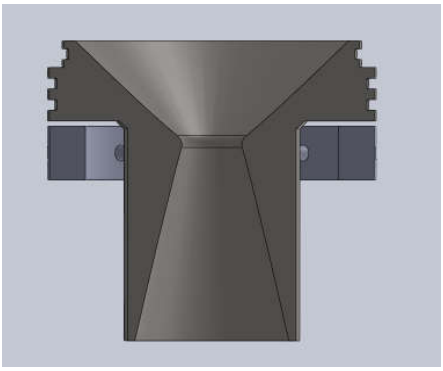


Figure 14: Nozzle and Retaining Ring CAD Section View

Both the forward closure and nozzle are retained by 8 M6 screws each, selected through numerical validation [Refer *Propulsion Analysis IV* of *Appendix A*] against ASME Boiler and Pressure Vessel Code formulae , confirming a minimum factor of safety of 1.5 across all relevant parameters.
2D CFD and 3D FSI simulations were performed in ANSYS Fluent and Ansys Mechanical to assess compressible flow behaviour through the nozzle and the resulting structural deformation under pressure loading, with detailed results provided in section *Propulsion Simulation I* and *Propulsion Simulation II* of Appendix A.

B. Aero-structures Subsystems

For the *Udbhava* project, the Aerodynamics and Structures team employed an integrated design framework. This approach simultaneously optimized vehicle performance to achieve the 10,000-foot apogee target and ensured robust structural integrity against dynamic flight loads. The core design principles focused on maintaining aerodynamic stability and structural integrity across all flight phases, which was essential for meeting flight safety requirements.

1.Airframes

The *Udbhava* launch vehicle utilizes a lightweight composite airframe designed to withstand both aerodynamic loads during ascent and mechanical loads during recovery, including ejection forces and parachute deployment. The primary structure consists of two cylindrical body tubes that form the load-bearing backbone of the rocket. These tubes are responsible for resisting axial compressive loads from motor thrust, tensile loads during recovery, and aerodynamic forces throughout flight.



Figure 15: Manufactured Body Tubes



Figure 16: Top View of Manufactured Tubes

The body tubes are manufactured using a filament winding process, selected for its ability to achieve a high strength-to-weight ratio and uniform material distribution. This method enables precise fiber alignment along the axial direction, improving load-bearing efficiency and structural consistency for high-performance applications.

Table 2:Fin Geometric Parameters and Material Properties

Fin Parameters	Sizes
Root Chord	240mm
Tip Chord	75mm
Semi-Span	152mm
Sweep Angle	54.1
Thickness	4.5mm
Aspect Ratio(AR)	0.965
Taper Ratio**(λ)**	0.3125
Shear Modulus	884730.2 PSI

Structural validation was conducted through compression testing, which yielded a maximum load capacity of 21,539 N (18.4 MPa) prior to buckling failure. This behavior is consistent with expected failure modes for thin-walled composite cylinders. When compared to the maximum expected motor thrust (~3,700 N), the structure exhibits a factor of safety of approximately 5.8, confirming adequate structural margin and reliability under flight conditions

2. **Nosecone**

The nose cone is a critical aerodynamic component of *Udbhava,* designed to minimise drag and enhance overall flight performance, particularly during the transonic ascent phase (Mach 0.8–1.2). Within this regime, the vehicle experiences significant wave drag due to the formation of shock waves at the forward section of the nose cone. This leads to a sharp pressure rise in the stagnation region near the tip, followed by a gradual pressure decrease along the aft section, thereby increasing aerodynamic resistance.

To mitigate these effects, the nose cone geometry is carefully optimised to reduce wave drag and delay the onset of strong shock formation. Smooth geometric transitions between the nose cone and adjoining body tubes help minimize pressure drag, skin friction drag, and boundary layer separation phenomena that are particularly pronounced in the transonic regime. In this context, the LD-Haack (von Kármán) profile, with a fineness ratio of 5:1, offers near-optimal aerodynamic performance by ensuring a gradual pressure distribution and reduced drag characteristics.

Design and Profile Comparison

Five nose cone profile families were evaluated across a fineness ratio range of 4:1 to 6:1 and a Mach number range of 0.7 to 1.3 at zero degrees angle of attack: Von Kármán (LD-Haack), LV-Haack, Tangent Ogive, Power Series (0.5 and 0.75 exponents), and Conical. The objective was to identify the geometry yielding the lowest drag coefficient and maximum altitude. CFD simulations were executed using ANSYS Student 2025 R2 with an axisymmetric density-based solver.

Table 3: CFD Boundary Conditions

Category	Details
Boundary Type	Pressure far-field
Freestream Mach Range	0.8 – 1.2 Mach
Turbulence Viscosity Ratio	10
Ambient Pressure	101325 Pa
Static Temperature	300 K
Wall Condition	Adiabatic, No-Slip
Axis of Symmetry	X-Axis

Table 4: Numerical Settings

Category	Details
Spatial Discretization	Second-Order
Convergence Criteria	1e-6

The isentropic compressible flow relations used in the CFD setup are:

$$\frac{p}{p_t} = \left(1 + \frac{\gamma-1}{2}M^2\right)^{-\frac{\gamma}{\gamma-1}}$$

$$\frac{T}{T_t} = \left(\frac{p}{p_t}\right)^{\frac{\gamma-1}{\gamma}}$$

Contours For Selected Nose Cone at MaxQ Conditions:

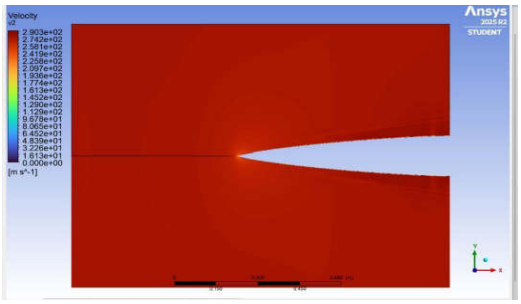


Figure 17:Velocity Contours At MaxQ

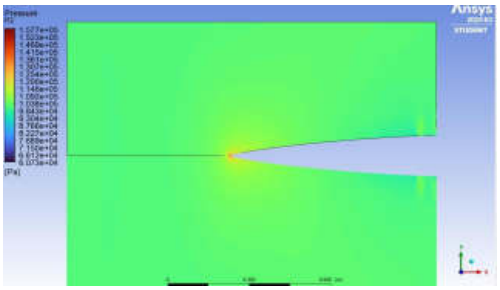


Figure 18:Pressure Contours At MaxQ

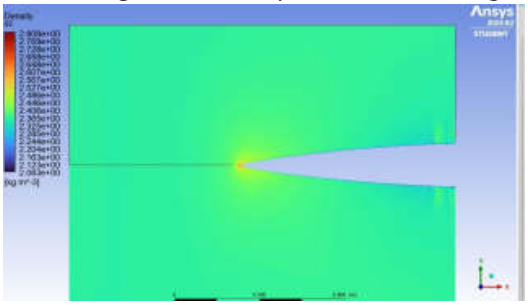


Figure 19: Density Contours At MaxQ

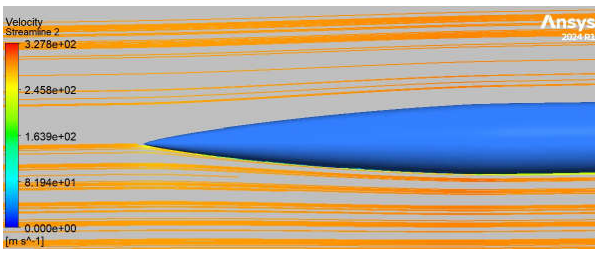


Figure 20: Streamlines at MaxQ

Manufacturing

The nosecone was manufactured via a hand layup process utilizing 240 GSM fiberglass. To form the composite shell, a multi-part, 3D-printed mold was assembled using M3 and M4 fasteners (See Figure 1). For optimal aerodynamic performance, the nosecone tip was precision-machined from Aluminum 6061 on a CNC Lathe and bonded to the cured composite shell using J-B Weld Metal Epoxy Adhesive. Comprehensive post-processing was conducted to ensure strict profile accuracy and to eliminate any surface discontinuities introduced during the layup phase.



Figure 21: Multi-Part 3D-Printed PLA Mold Assembly

To manufacture the nosecone, ten 3D-printed PLA mold components were mechanically fastened with M4 screws and sanded to a smooth finish before applying an industry-standard release system consisting of three buffed layers of mold release wax and a liquid Polyvinyl Alcohol (PVA) film barrier. The composite layup utilized five layers of 240 GSM fiberglass wetted out with BhorBond HIFC Epoxy mixed at a strict 100:45 weight ratio, a system deliberately chosen for its low-exotherm 16-hour cure to prevent the PLA from thermal warping.

Specialized fiberglass rollers were actively used to consolidate the laminate and eliminate air voids before allowing the epoxy to semi-cure to a "green state" for 3-4 hours, at which point protruding fibers were cleanly flush-trimmed to the mold flanges. The mold halves were then mated using M3 screws and left to fully cure at ambient temperature, after which the demolded composite was sanded, and any resin-deficient regions were structurally reinforced using a localized chopped carbon fiber slurry.Finally, to complete the assembly, the mating surfaces of the composite shell and a precision-machined Aluminum 6061 tip were heavily abraded and degreased, ensuring maximum mechanical interlock when securely bonded with J-B Weld to establish a precise aerodynamic profile.



Figure 22: Pre-Processed Composite Shell



Figure 23: Wetted-Out Fiberglass Layup



Figure 24: Fully Integrated Nosecone

2. Fins

Fin Attachment

The *Udbhava* rocket employs a fin-can based attachment system to ensure structural rigidity, precise alignment, and repeatability. The fin-can consists of four CNC-machined 6061-T6 aluminium plates mounted onto the coupler using M5 fasteners. Each fin is inserted into slots between adjacent plates and secured using M4 fasteners, forming a rigid and flush-mounted assembly that maintains aerodynamic smoothness and minimizes external protrusions .
This integrated design distributes aerodynamic and structural loads efficiently while ensuring consistent fin alignment, which is critical for flight stability.



Figure 25: Fin Can Assembly

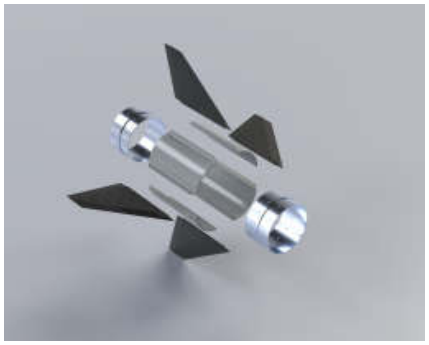


Figure 26: Exploded View of Fin Assembly



Figure 27: Manufactured Fin with Integrated Fin Plate

Fin Geometry and Stability

Fin geometry was optimized to achieve a stable flight profile while minimizing aerodynamic penalties. The primary stability metric used is the stability margin (SM):

$$SM = (X_{CP} - X_{CM}) / \text{Diameter}$$

The design targets a stability margin of 1.8–2.4 calibres at launch and 3–4.5 calibres at apogee, ensuring adequate stability without excessive weathercocking. Parametric iterations were conducted to balance aerodynamic efficiency and stability requirements.

Material Selection and Testing

The fins are manufactured using a hybrid composite of forged carbon fiber and fiberglass, selected for its quasi-isotropic behavior and ability to withstand multi-directional aerodynamic loads. The compression moulding process enables consistent manufacturing with good surface finish and structural repeatability.
Material characterization was performed through tensile testing, yielding:

- Young's Modulus: 14.9 GPa
- Ultimate Tensile Strength: 204 MPa
- Shear Modulus: 6.1 GPa

These properties were used for structural sizing and aeroelastic analysis.

Table 5:Fin Geometric Parameters and Material Properties

Fin Parameters	Sizes
Root Chord	240mm
Tip Chord	75mm
Semi-Span	152mm
Sweep Angle	54.1
Thickness	4.5mm
Aspect Ratio(AR)	0.965
Taper Ratio(**λ**)	0.3125
Shear Modulus	884730.2 PSI

Fin Flutter and CFD validation

A fin flutter analysis was performed using experimentally obtained material properties and geometric parameters. The predicted flutter velocity is 625.88 m/s (Mach 1.89), providing a safety margin of ~2.15 relative to maximum flight velocity, ensuring safe operation within the flight envelope.

$$V_f = a\sqrt{\frac{2G(AR+2)}{1.337\,AR^3\,P\,(\lambda+1)\left(\frac{t}{c}\right)^3}}$$

CFD simulations at Mach 0.875 (Max-Q) confirm attached flow over the fin surface with no significant separation. Pressure distribution and aerodynamic loading remain within acceptable limits, validating the aerodynamic design. Modal analysis further indicates that structural natural frequencies are well separated from expected excitation frequencies, eliminating the risk of resonance during flight.

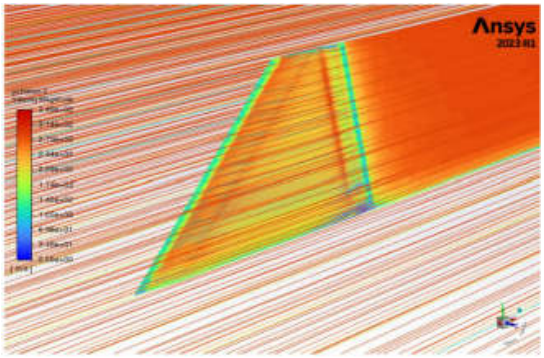


Figure 28: Velocity Pathlines

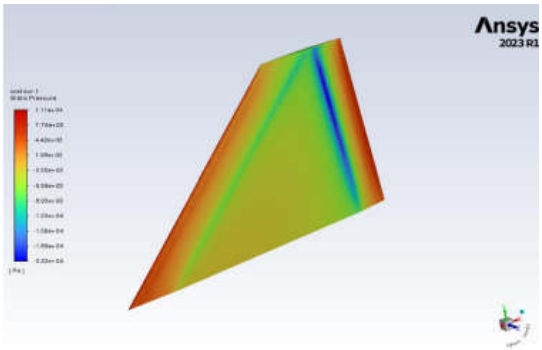


Figure 29: Static Pressure Contours

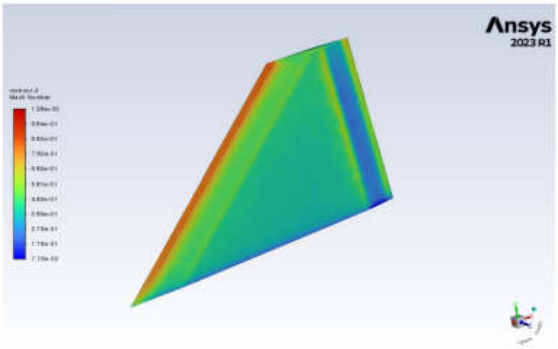


Figure 30: Mach Contours

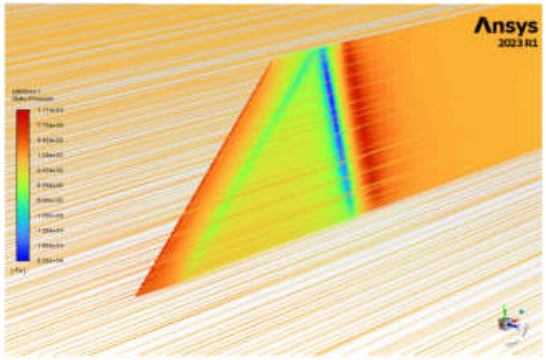


Figure 31: Static Pressure Pathlines

The results indicate smooth flow attachment over the fin surface with no significant flow separation. The pressure distribution remains stable, and the aerodynamic loading is within acceptable limits, confirming the suitability of the fin geometry.

A modal analysis was performed to evaluate the dynamic response of the fin under aerodynamic excitation

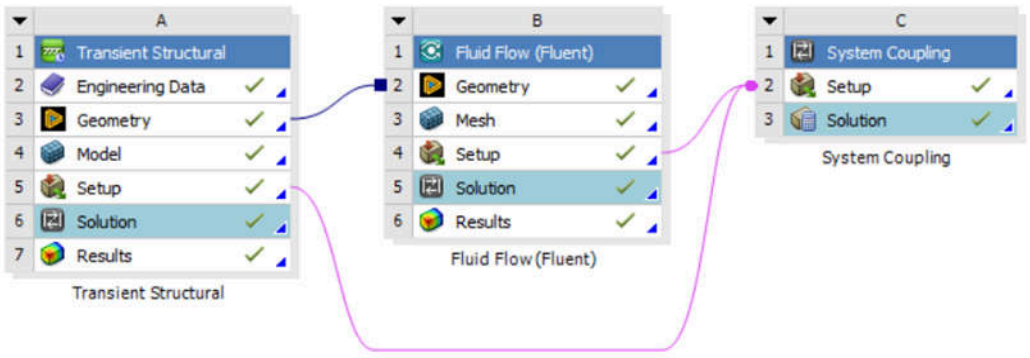


Figure 32: FSI Workflow

The modal frequencies are sufficiently separated from expected aerodynamic excitation frequencies, indicating that resonance conditions are unlikely during flight. This confirms the structural stability of the fin design under dynamic loading.

Table 6: Modal Analysis Results of Fin

Mode No. 01	Frequency(Hz)	Description
01	78.94	1st Mode of Bending
02	305.66	2nd Mode of Bending
03	385	3rd Mode of Bending
04	493.39	4th Mode of Bending
05	548	Torsion + Bending
06	666	Torsion

Fin Fabrication

The fin layup process began with 3D printing a four-part mould, consisting of two negative and two positive components, which were sanded to achieve a smooth surface.Three layers of PVA mould release wax were applied, followed by PVA release agent and silicon spray to ensure easy demoulding. Two trimmed fibreglass layers were placed in the negative mould and coated with BhorBond HIFC epoxy mixed in a 100:45 ratio.Chopped carbon fibre was evenly distributed and wetted with epoxy, followed by an additional fibreglass layer and a thin epoxy coat.
The positive mould was then placed, and the assembly was clamped using M3 screws to maintain uniform pressure during curing for 16 hours. After demoulding, the fins were inspected for defects such as dry spots and voids, followed by necessary post-processing.

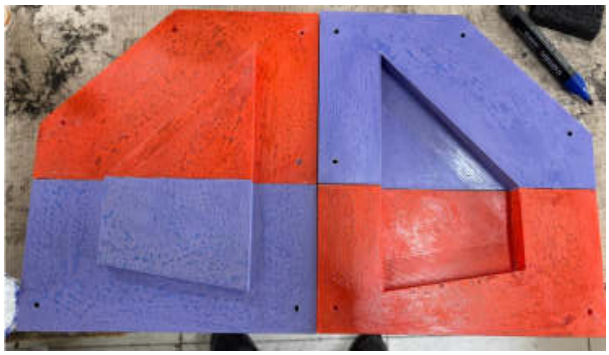


Figure 33: Fibre Layup in Mould



Figure 34 : Fibre Layup in Mould



Figure 35: Manufactured Fin



Figure 36: Cross Sectional Profile of Fins

3. Thrust Plate

The thrust plate is a precision-machined aluminium 6063-T6 bulkhead that closes the forward closure of the solid rocket motor, transferring axial thrust loads into the rocket structure via six M4 fasteners. The domed outer face and concave inner pocket form a thin-shell profile that efficiently resists chamber pressure while minimising mass. The forward closure attaches to the thrust plate via a central M16 threaded screw.

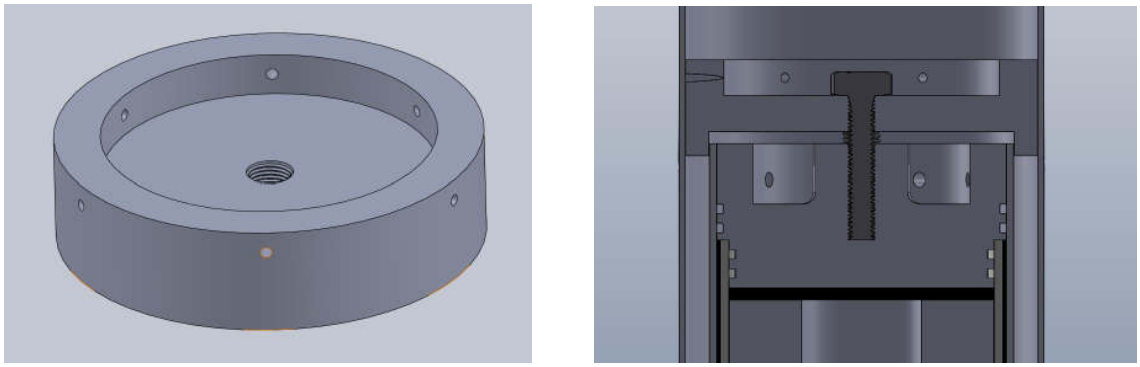


*Figure 37 **:** Thrust Plate CAD Model -Isometric View (Left) and Cross-Sectional Assembly View (Right)*

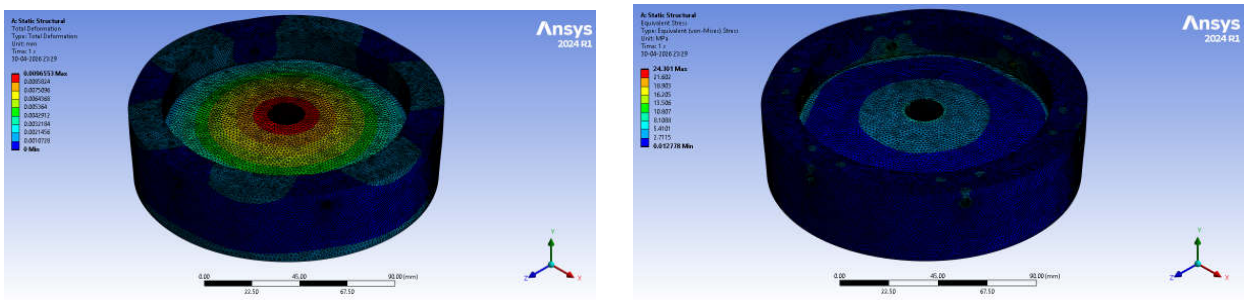


Figure 38: Thrust Plate static structural Results - Total Deformation (Left) and Von-Mises Stress Distribution (Right)

ANSYS Static Structural analysis was performed to verify the structural integrity of the thrust plate. Under an axial load of 5400 N (including a 1.5 safety factor), the maximum von Mises stress of 24.301 MPa occurs at the central bore and remains well below the 170 MPa yield strength of 6063-T6, confirming fully elastic behavior. The maximum equivalent elastic strain (3.7337×10^{-4} mm/mm) is also localized at the bore and decreases toward the fastener interfaces. A peak deformation of 0.0097 mm is observed at the bore face, while the M4 fastener regions remain effectively rigid, validating the structural adequacy under operational loads.

4. Motor Retaining System

The motor retaining mechanism is a single component designed made of 6061 T6 Aluminium to secure the propulsion motor within the rocket airframe.It consists of a circular flanged ring with a stepped, dual-diameter cross-section, featuring an evenly spaced bolt pattern around its outer flange for attachment to the coupler which interfaces with the fin can, and small raised locking tabs on its inner collar that interfaces with the retaining ring of the motor casing . Overall, this single component serves the purpose of firmly retaining the motor casing and centering it in place during flight while also allowing straightforward motor removal and replacement between flights. The retaining ring has slots made to account for the passive drag reduction systems i.e ,base bleed outlet ducts.

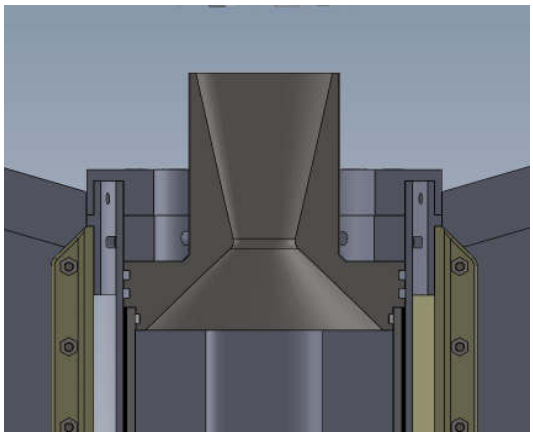


Figure 39 : Retaining System Assembly Interface View

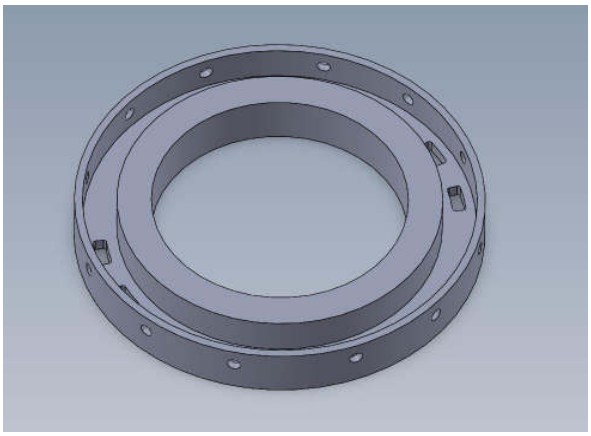


Figure 40: Retaining System - CAD Model

5. Rail Buttons

The rail buttons used for this rocket are airfoiled rail buttons designed for a standard 1515 launch rail sourced from Apogee Rocketry. The airfoil profile helps reduce aerodynamic drag during ascent while maintaining smooth guidance along the rail. These buttons are mounted securely to ensure structural integrity and alignment during launch.



Figure 41: Airfoiled Rail Buttons (Top View)



Figure 42: Airfoiled Rail Buttons with Stud

6. Passive Drag Reduction System

Sounding rockets operating in the transonic regime (Mach 0.8 -- 1.2) experience a significant increase in aerodynamic drag, with **base drag contributing up to 40% of the total drag** due to the formation of a low-pressure recirculating wake behind the rocket aft section. To mitigate this effect, a **passive base bleed system** is implemented in the UDBHAVA launch vehicle to improve aerodynamic efficiency and enhance apogee performance.



Figure 43: Base Bleed Channel Design

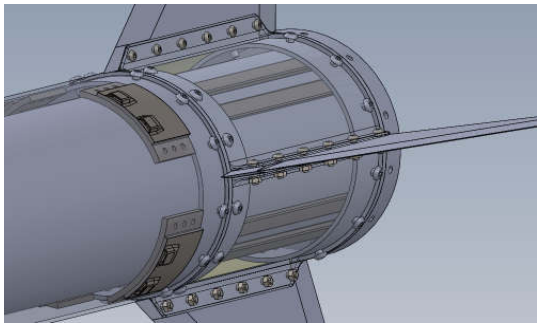


Figure 44:Base Bleed integrated with Rocket Assembly

The system operates by **redirecting high-momentum external airflow into the wake region** through an internal channel integrated within the aft coupler. This injected flow increases the base pressure and weakens the recirculation zone, thereby reducing the pressure differential responsible for base drag. The design is fully passive, requiring no additional propellant or active control, and is optimized to maintain structural integrity while avoiding interference with motor exhaust flow.

CFD simulations were conducted across the transonic regime (Mach 0.5–0.875), with refined meshing in the wake and outlet regions to capture flow behavior accurately. Results indicate a clear modification of the wake structure, with increased base pressure and reduced recirculation intensity. Comparative analysis shows a **maximum drag reduction of approximately 8.1% at peak velocity**, with effectiveness decreasing at lower Mach numbers.

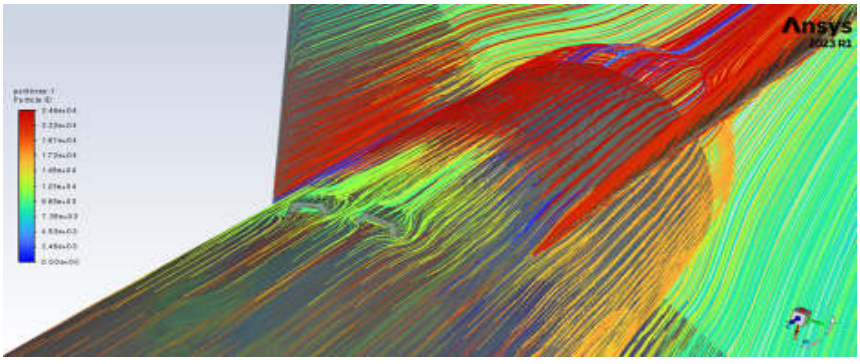


Figure 45: Pathlines around the Bleed Inlet

The system is integrated into the aft section with a **carbon fiber reinforced inlet, internal bleed channel, and outlet near the base**, ensuring minimal weight addition and compatibility with the existing airframe. Overall, the PDRS provides a **lightweight and effective solution for reducing transonic drag**, contributing to improved flight efficiency and enhanced mission performance.

C. Avionics Subsystem

The avionics subsystem is designed to ensure reliable recovery actuation, robust telemetry, and high-fidelity data acquisition under the dynamic conditions of rocket flight. The avionics bay houses all the critical electronic hardware required for the rocket's operation. It is designed to be modular while maintaining high structural integrity, ensuring both flexibility and durability during flight. Each of these sections can be individually removed from the main assembly, and the components within each section are accessible for local maintenance and replacement without needing to disassemble the entire avionics bay.



Figure 46: Avionics Bay Isometric View

COTS System

This section is centered around a COTS battery pack housing five 21700 cells, supported on both sides by COTS structural chassis with integrated grooves for secure positioning and a mount for a limit switch. The outer faces of the chassis provide mounting points for components such as a 9V battery, RRC3, Blue Raven, and Featherweight GPS. Battery connection plates are fixed to the side faces using M3 × 15 mm socket head screws, where the screw ends contact the cells and the opposite ends hold Cu-Ni bus bars for series-parallel connections. M3 nuts are used throughout to secure the bus bars and wiring, ensuring reliable electrical and structural integrity.

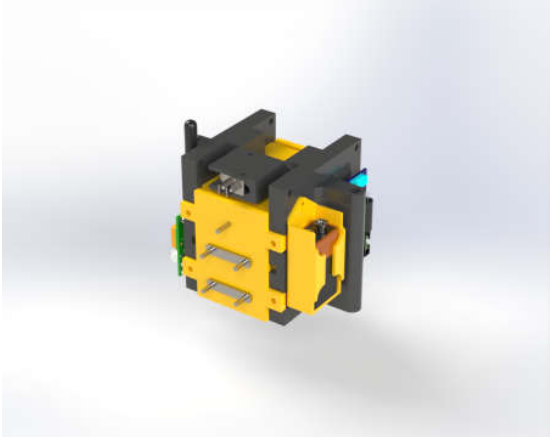


Figure 47: COTS Battery Pack

SRAD System

Complementing the COTS systems, the SRAD avionics architecture consists of two flight computers-Arceus and Sirius-along with a centralized power distribution system.

Power Distribution Board

The Power Distribution Board (PDB) serves as the centralized power management unit for the avionics bay, distributing regulated power to all SRAD subsystems. The primary source is a 10-cell 21700 lithium-ion battery pack, with regulated 3.3 V and 5 V rails provided for avionics electronics. Dedicated battery domains are employed for critical subsystems, including a 2S2P configuration for the flight computers and separate 3S supplies for the video transmitter (VTx) and FPV camera, reducing electrical interference between subsystems. Additionally, each SRAD subsystem incorporates local voltage regulation, allowing direct battery operation in the event of a centralized PDB failure.

To ensure clean power delivery to sensitive analog components, the VTx and FPV camera are powered through a separate dedicated PDB, which utilizes low-dropout (LDO) regulation to provide stable and low-ripple 5 V and 10 V outputs. This isolation minimizes noise coupling from digital and high-current switching loads, improving video transmission quality and overall system reliability.

Arceus (PRIMARY Flight Computer (FC))

Arceus is an SRAD FC designed to operate reliably in the harsh conditions associated with high-performance rocket launches. These include prolonged high-frequency vibration, peak axial and transverse accelerations, and rapid thermal transients. It is a 4-layer PCB designed in such a way as to minimize electromagnetic interference (EMI). The system architecture puts deterministic real-time performance with strong fault tolerance first.

The STM32F4 microcontroller is the jewel of Arceus. It is a high-performance ARM Cortex-M4 working at a maximum clock frequency of 168 MHz, a processing unit chosen for its ability to speed up floating-point calculations, support a wide range of peripherals, and its proven reliability in embedded aerospace applications. This is the main place where all the onboard sensing, telemetry processing, and actuation logic come together to make decisions and do calculations.

The BMI088, LIS2MDL, and ISM330DHCX are three 9-axis IMU sensors that make up the inertial measurement subsystem. This setup with multiple sensors checks the angular rate and linear acceleration data against each other, lowering the chance of a single-point sensor failing during important flight phases. The BMP280 barometric sensor measures atmospheric pressure, while the MS5611 serves as a high-resolution backup. There are two GNSS receivers, the L89 and the backup NEO-M8N, that help with positioning and navigation. They make sure that geolocation is always accurate, even if one receiver drops out or the signal gets weaker.

The FC houses two independent pyro channels, each designed to reliably trigger deployment charges at the designated flight events. Each channel employs a MAX9075 window comparator to monitor the EM line voltage, providing a safeguard against unintended igniter activation. The switching element for each channel is an AO3400A N-channel MOSFET, gated through a resistor divider network consisting of two 10 kΩ resistors, ensuring controlled and noise-immune switching. A dedicated PYRO EN signal governs each MOSFET, preventing inadvertent firing. Each channel is also equipped with an LED indicator with current-limiting resistors, providing a visual continuity check to verify igniter connectivity before launch. A 100 nF decoupling capacitor is placed on the comparator supply line to filter any supply noise that could cause spurious triggering. The pyro battery input is monitored via a resistor divider on the BAT_IN line, enabling onboard voltage sensing to confirm sufficient charge is available before firing. Both channels are accessible through a 4-pin connector, with independent EM+ and EM− lines for each channel.

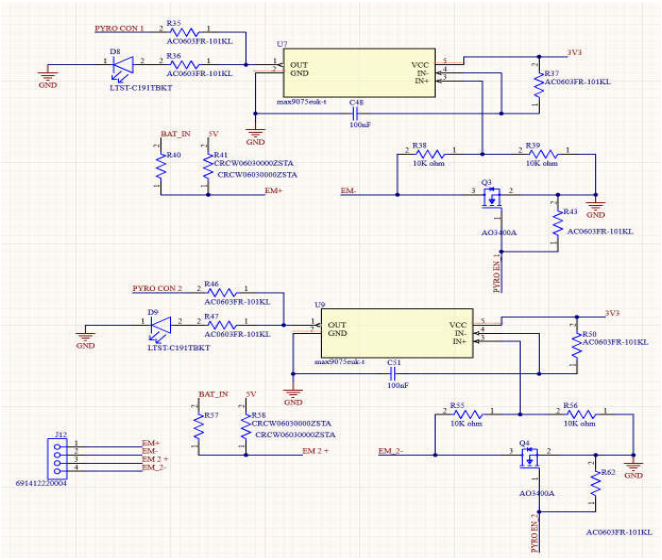


Figure 48 - Pyrochannel Schematics

Sirius - (REDUNDANT Flight Computer (FC))

The Sirius Flight Computer serves as the primary data acquisition system throughout the flight, integrating a comprehensive suite of sensors to deliver high-fidelity flight data. The board incorporates multiple IMUs, magnetometers, and barometric pressure sensors, with sensor fusion algorithms employed across these redundant sensors to maximize measurement accuracy and reliability. As Sirius operates without any pyro channels, it is entirely isolated from the voltage transients and electrical noise associated with pyrotechnic events, ensuring a clean and stable power supply to all onboard sensors. This makes Sirius the ideal candidate for telemetry transmission to the ground station, as the integrity of sensor readings is maintained throughout the entirety of the flight.

Sirius uses a dual-storage architecture for persistent data logging. It has a micro SD card using SDIO protocol for recording high-capacity sequential flight data and an external W25Q128 NOR flash memory module for non-volatile, high-speed buffered storage. This makes it resistant to power outages and mechanical shocks that could make the SD card hard to reach during flight.

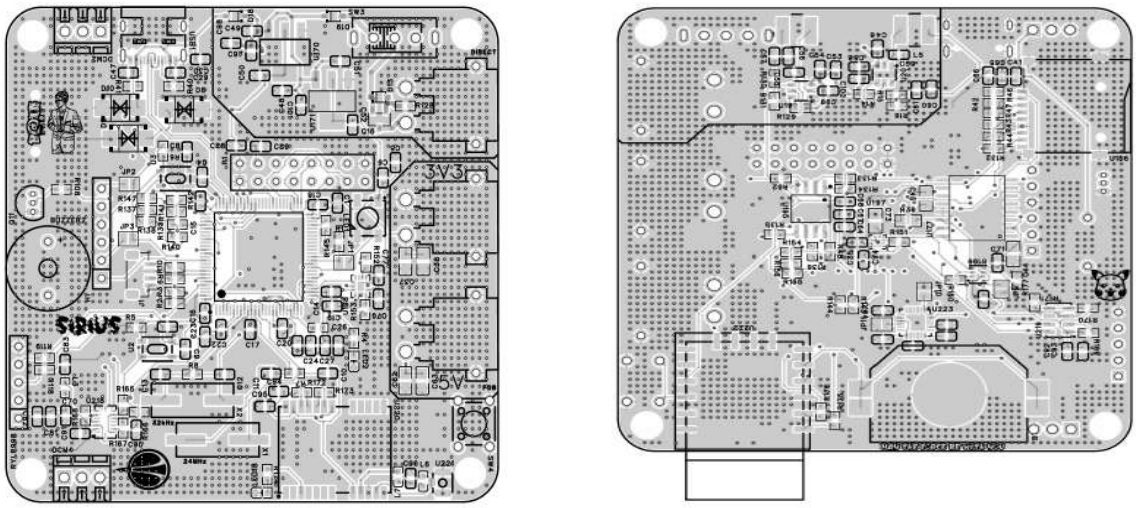


Figure 49 - Sirius PCB Layout

4.Avionics Bay Wiring Strategy and Arming Mechanisms

The avionics bay consists of two independent arming mechanisms to ensure safe and reliable system activation. The primary mechanism utilizes a mechanical lever-type limit switch actuated by a pull-pin. Before launch, the pin holds the lever switch in a depressed state, keeping the system disarmed. Upon removal of the pin at the launch pad, the lever is released, closing the circuit and arming the avionics bay. This provides a simple, reliable, and physically intuitive means of arming the system immediately before flight. Housing five independent mechanical lever-type limit switches, each controlled by a dedicated pull-pin mechanism. One switch governs power delivery from the PDB to the Flight Computers, one controls the voltage supply to the RRC3+ pyro channels, and the remaining three individually arm each of the three COTS components - the RRC3+, Blue Raven, and Featherweight GPS.

As a backup, a magnetically triggered switch is implemented using the Honeywell SS360PT Hall-effect latch. The switch is controlled via a power MOSFET-based inverter circuit. This magnetic arming mechanism requires no physical contact or mechanical actuation, offering a non-invasive and highly reliable backup means of arming and disarming the avionics bay without requiring direct access to the system internals.



Figure 50: Pull pins and SRAD

SRAD PCBs and Pull Pin Switch Housing

This section is built around a chassis designed to house the avionics circuitry and switching system. Side grooves accommodate both flight computer PCBs, while a rear slot holds the PDB. These components are mounted on plates, slid into the grooves, and secured with M3 screws for proper alignment and stability. The center of the chassis includes mounting points for four pull-pin switches arranged in a stacked configuration, enabling compact and organized integration of the switching system.

SRAD batteries and VTX housing

This section is centered around a high-capacity battery system comprising a central pack of ten 21700 cells, supported by two chassis structures with integrated grooves for secure positioning. The outer faces of the chassis provide mounting points for the VTX module, antenna, and VTX PDB, ensuring compact integration of the transmission system. Battery connection plates are mounted on the side faces using M3 × 15 mm socket head screws for cell contact, with Cu–Ni bus bars on the opposite side to establish electrical connections, all secured with M3 nuts for reliable performance and structural stability.

D. GroundStation Subsystem

Flight Software

The Flight Software is an autonomous, high-reliability system designed for real-time telemetry processing, state-aware decision-making, and mission-critical event triggering. Built on the STM32F4 platform, it manages everything from sensor fusion to parachute deployment with a focus on recovery and data integrity.

Key Objectives

- **Autonomous State Management:** Identify six distinct flight phases and trigger recovery hardware (drogue/main chutes) based on redundant sensor confirmation.

- **Precision Telemetry:** Real-time tracking of altitude, orientation, and velocity using custom-written drivers for high-frequency sampling.
- **Fault Tolerance:** Implement a "State Recovery" system using non-volatile flash memory to resume operations instantly after power interruptions.
- **Redundant Logging:** Simultaneous data capture to SDIO SD cards and SPI Flash to prevent data loss during high-dynamic events.

Workflow & Architecture

The software operates as a deterministic State Machine governed by sensor thresholds and "trust counters" to filter noise:

1. **Launch Pad:** Initializes sensors and recovers the last state; detects launch at >5m altitude gain.
2. **Ascent:** Monitors for peak altitude (Apogee); triggers when altitude fails to increase for 3 cycles, deploying the drogue parachute.
3. **Descent:** Tracks descent rate; triggers the main parachute at 457m (1500ft) after 2 cycles of confirmation.
4. **Main Descent:** Monitors for touchdown; transitions when altitude remains <10m for 3 cycles.
5. **Landing:** Verifies static status (accelerometer approx ≈ 1g and Δalt < 1m) for 5 continuous seconds.
6. **Landed:** Stops telemetry to conserve power and activates audio/visual recovery beacons.

Tech Stack & Algorithms

The system is developed in C++ using the STM32 HAL library, allowing for deterministic control of the STM32F4 microcontroller via STM32CubeMX. To maintain high-fidelity attitude estimation, the software implements a Kalman/Complementary filter that fuses raw inputs from the BMI088 IMU to smooth sensor noise during high-dynamic flight. The core logic is governed by a finite state machine utilizing "trust counters" to validate events before triggering deployment. To ensure continuity, a state-recovery algorithm writes the mission phase to W25Q128 SPI Flash after every transition, while simultaneously logging CSV telemetry to an SD card via SDIO.

Data Formats

Transmission & Storage: The system utilizes a structured 23-field packet for both SD logging and LoRa telemetry:

```
TEAM_ID, PACKET_CNT, TIME, NUM_SATS, GPS_SPEED, LAT, LONG, ALT,
RTC_TIME, ACC_X, ACC_Y, ACC_Z, PRESSURE, BARO_ALT, GYRO_X, GYRO_Y,
GYRO_Z, TRIGGER, TEMP, VOLT, STATE, EVENT_FLAG
```

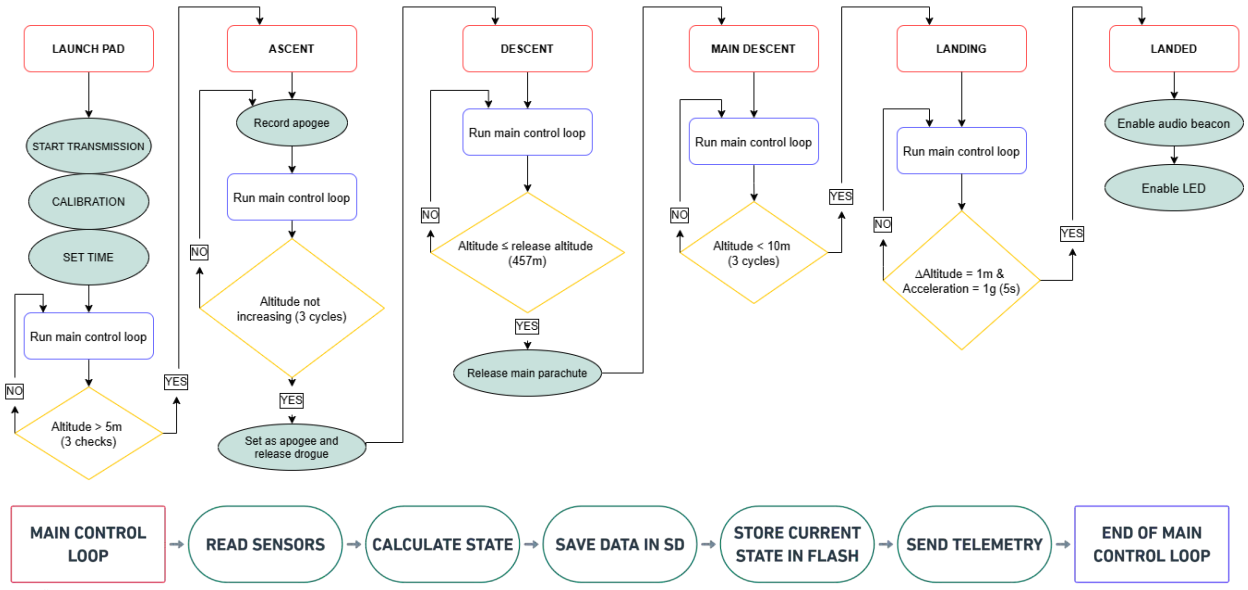


Figure 51: Flowchart & State Diagram

Cyberdeck

The Cyberdeck is a custom-built ground support system designed to host and operate the Ground Control Station (GCS) along with live video monitoring during rocket missions. It serves as a compact, portable, and integrated hardware platform that brings together all essential components required for real-time telemetry reception, visualization, and situational awareness in field conditions.

The system is built around a Raspberry Pi, which runs the GCS software and interfaces with multiple peripherals through USB connections. These include a telemetry receiver PCB (SHROTA) for acquiring flight data via LoRa communication and an ESP32 for transmitting GPS data to a mobile application for recovery purposes. The Cyberdeck also incorporates a display unit, keyboard, USB hub, and an uninterrupted power supply (UPS), ensuring continuous and reliable operation during outdoor missions.

In addition to telemetry handling, the system integrates a separate video reception setup using another Raspberry Pi connected to an EasyCAP video receiver. This enables real-time video feed monitoring alongside telemetry data, enhancing overall mission awareness.

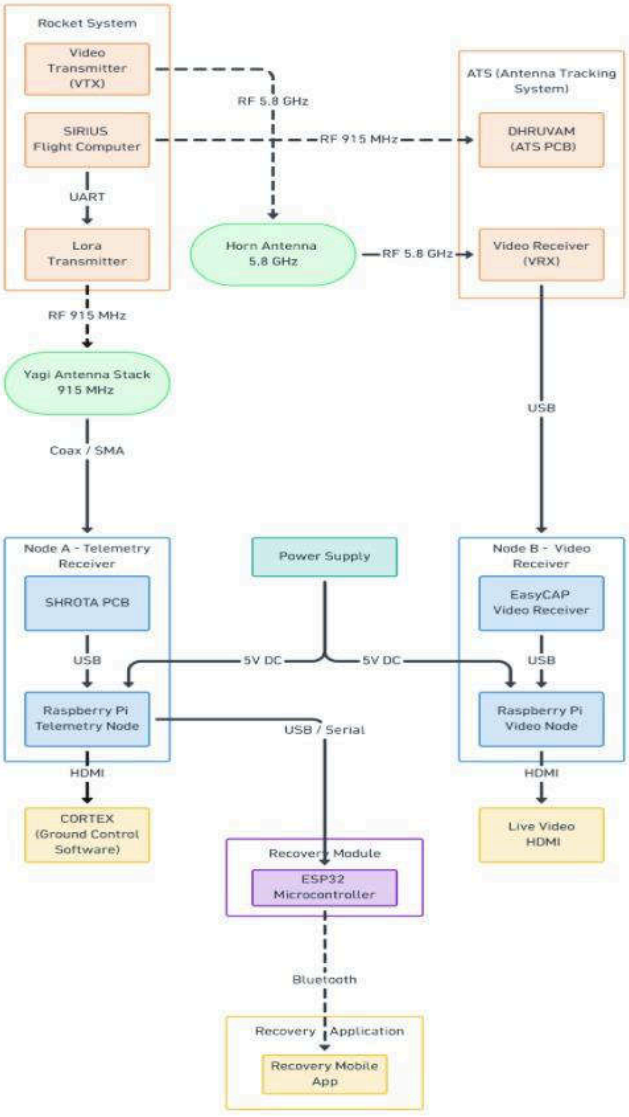


Figure 52 - Workflow Diagram

CORTEX-Ground Control Software

CORTEX is a custom-engineered Ground Control Station (GCS) designed for real-time rocket flight monitoring and recovery. It integrates high-speed data processing with a modular web-based interface to provide mission-critical situational awareness.

Workflow & Data Flow
The mission sequence begins when the SIRIUS flight computer transmits telemetry via 915 MHz LoRa. These signals are captured at the ground segment by a Yagi antenna stack and a SHROTA receiver, which converts the RF signal into a serial data stream for a Raspberry Pi. A Node.js backend serves as the central engine, utilizing an event-driven, non-blocking architecture to parse and validate incoming packets while filtering corrupted data to ensure integrity. Once processed, the telemetry is broadcast simultaneously to two endpoints: a React frontend via WebSockets for instantaneous dashboard updates, and an ESP32 via Bluetooth, which relays GPS coordinates to a mobile application for field recovery operations.

Technologies & Concepts
The system leverages a modern full-stack architecture optimized for high-frequency data. The frontend is built with React and Vite for performance, using Zustand for lightweight state management and Tailwind CSS for a responsive, field-ready UI. Data visualization is handled through a suite of specialized libraries, including ECharts for dynamic graphing, Three.js (via React Three Fiber) for 3D trajectory rendering, and Leaflet for geospatial mapping. To ensure precision, the system implements Kalman filtering for signal smoothing and noise reduction, while the backend utilizes the SerialPort API for hardware communication and PapaParse for structured CSV data handling. The hardware integration of Raspberry Pi and ESP32 ensures a robust bridge between long-range LoRa communication and mobile-ready recovery tools.

System Functionalities
- **Home Page:** The central command hub. Features a categorized telemetry panel, mission timers, 3D orientation view, live GPS map, and real-time sparklines for altitude and pressure.
- **Graphs Page:** Dedicated analysis suite for dynamic plotting. Supports multi-parameter comparisons (e.g., Velocity vs. Acceleration) with interactive scaling and numerical overlays.
- **GPS Page:** Focused recovery and tracking interface. Combines a high-resolution map with 3D flight path visualization and specific coordinate data (Lat/Long, Satellite Count).
- **Table Data Page:** A structured data grid featuring sorting and filtering. Used for granular inspection of live packets and post-flight CSV analysis.
- **Settings Page:** System configuration portal. Allows toggling between Metric and Imperial units and customizing display thresholds for voltage, temperature, and speed.

Data Specifications
Transmission & Storage Format: The system processes a 113-byte packet string containing the following parameters:

```
TEAM_ID, PACKET_COUNT, GPS_TIME, GPS_SATS, GPS_SPEED, GPS_LAT, GPS_LONG,
GPS_ALT, RTC_TIME, ACCEL_X, ACCEL_Y, ACCEL_Z, PRESSURE, ALTITUDE, GYRO_X,
GYRO_Y, GYRO_Z, TRIGGER, TEMPERATURE, VOLTAGE, STATE_CURRENT, PYRO_E1_E2
```

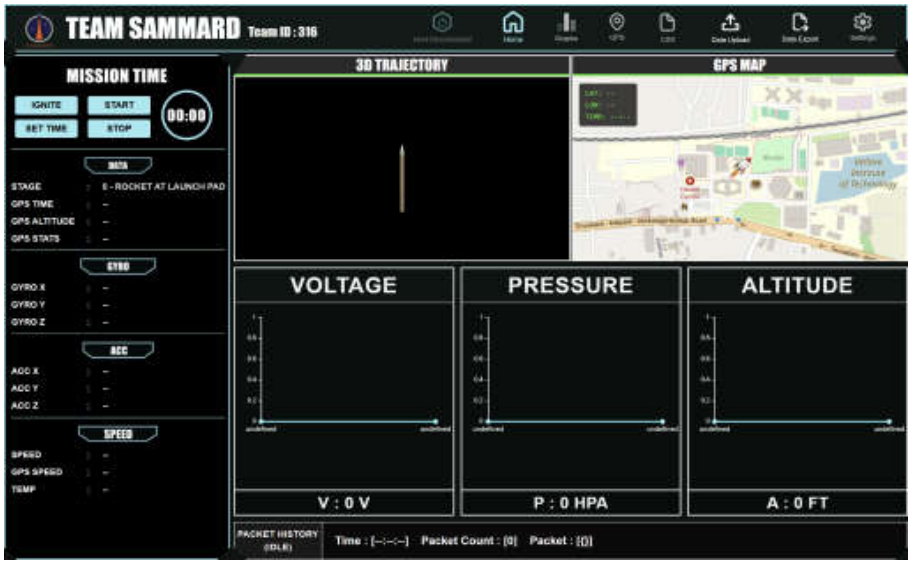


Figure 53: GCS Dashboard Page

Antenna Tracking System (ATS)

The Antenna Tracking System (ATS) is developed to maintain continuous alignment of the ground-based receiving antenna with the rocket throughout flight, ensuring reliable reception of high-frequency video and telemetry signals. The system estimates the rocket's real-time position using incoming GPS and IMU data transmitted via LoRa, which are processed through an Extended Kalman Filter (EKF) to obtain a smooth and accurate trajectory estimate. Based on this estimation, the required azimuth and elevation angles are computed and translated into precise motor commands. The tracking mechanism employs two NEMA-23 stepper motors, driven via dedicated motor drivers, to control horizontal (azimuth) and vertical (elevation) motion of the antenna. The system operates on a Teensy 4.1 microcontroller, enabling high-speed computation and real-time response. Continuous updates to position and orientation allow dynamic tracking, while fallback logic such as search patterns is implemented to recover signal in case of temporary telemetry loss. This closed-loop tracking approach significantly improves signal strength and reduces packet loss during flight.



Figure 54: Horn antenna used for Live Video Reception

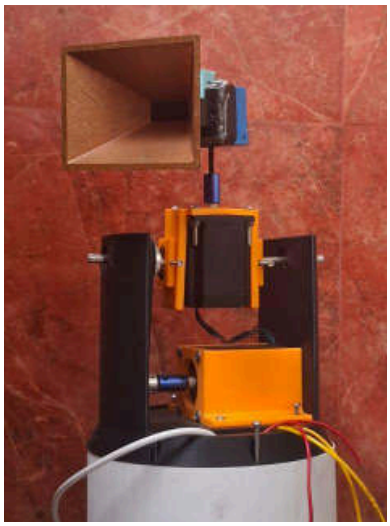


Figure 55: Full Hardware assembly of the ATS

RecovY- The Rocket Recovery App

The RecovY mobile application is developed as a dedicated recovery support system to assist in locating and retrieving the rocket post-landing. Built using Flutter, the application receives real-time telemetry data relayed from the ground station via an ESP32 module over Bluetooth Low Energy (BLE), enabling low-power and reliable communication in field conditions. The app processes incoming GPS coordinates and integrates them with the device's onboard location and orientation sensors to provide intuitive navigation toward the rocket's landing site. It features map-based tracking with offline capability, directional guidance with distance estimation, and a structured interface for monitoring key recovery parameters. By combining real-time data streaming with user-friendly visualization, RecovY enhances recovery efficiency and reduces search time, particularly in low-connectivity environments.

E. Recovery Subsystems

The recovery subsystem uses a dual-deployment parachute system with a pyrotechnic separation mechanism. A drogue parachute deploys at apogee for stabilization and initial deceleration, followed by a main parachute at ~1500 ft for safe landing. Avionics trigger staged black powder charges – first separating the drogue tube, then the main parachute tube.

This approach reduces loads, limits drift, and ensures controlled recovery. Both parachutes are in-house Nylon 6,6; the drogue uses 16 gores and the main 6 for stable performance.

Parachute Calculations

The terminal descent velocity of a parachute is calculated using the drag equilibrium condition:

$$V_t = \sqrt{\frac{2mg}{\rho A C_d}}$$

Table 7: Parachute Sizing and Performance Data

Parachute parameters	Drogue	Main
Surface Area (m^2)	1.05	8
Dimensions (m)	1.15 side	3 diameter
Descent Speed (m/s)	28.6	10
Type	Square	Hemispherical
Weight (g)	110	194
Volume (m^3)	636	874
Material	Nylon 6,6	Nylon 6,6
GSM	28	28
Gores	16	16
Shroud line material	Dyneema	Dyneema
Breaking strength of shroud lines (kg)	90	90



Figure 56: Manufactured Main Parachute



Figure 57: Main Parachute During Testing



Figure 58: Deployed Drogue Parachute



Figure 59: Manufactured Drogue Parachute

Shock Cords

The launch vehicle uses a dual-deployment recovery system with two primary harness assemblies. Both employ 5/8-inch Kevlar shock cords sourced from Apogee Rockets with a 4,500 lb test strength.



Figure 60: Sourced Kevlar Shock cords

The recovery system uses a dual-deployment configuration with defined line lengths to ensure controlled separation and minimize entanglement. A **5 m** line connects the drogue tube to the drogue parachute, followed by **1.5 m** to the avionics bay. The main section uses **5.5 m** from the avionics bay to the main parachute, and **3 m** to the main body tube.These lengths balance compact packing with adequate separation to prevent recontact and ensure reliable deployment, and were determined using an in house shock cord sizing calculator.

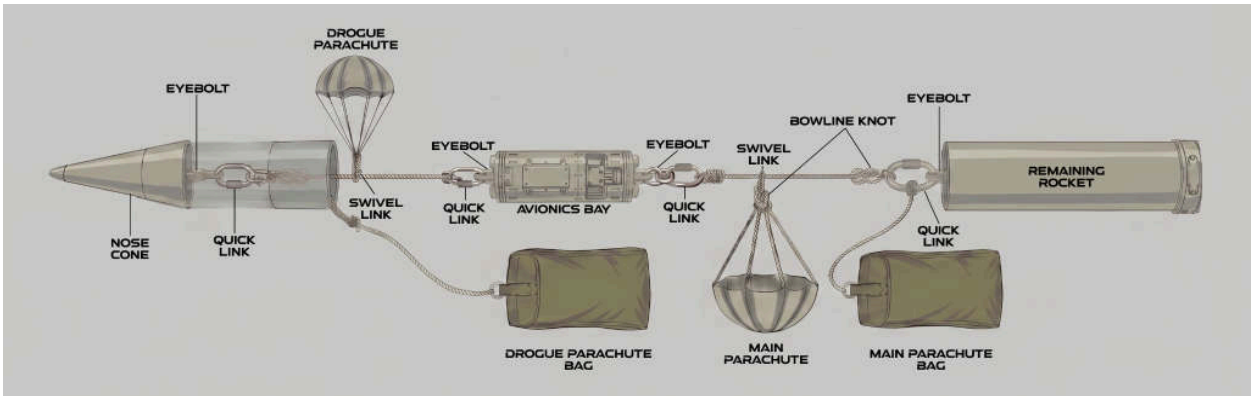


Figure 61: Rocket Recovery Line Diagram

Hardware and Black Powder Calculations

The recovery system uses a dual mid-body deployment mechanism, creating two separation events on either side of the avionics bay [Figure 7]. Both parachutes are enclosed in two-layer stitched fibreglass bags, which protect the canopy from black powder flames and embers while aiding reliable deployment and also contributing structurally. Each separation interface includes a primary and redundant charge well, each loaded with **5 g** of black powder. Recovery wadding is used to provide additional thermal protection. After installing e-matches and wadding, the charge wells are sealed with hot glue and masking tape to form a pressurized chamber, improving ejection efficiency and reliability.



Figure 62: Manufactured Recovery charge wells Integrated with coupler



Figure 63: Manufactured Parachute Bag

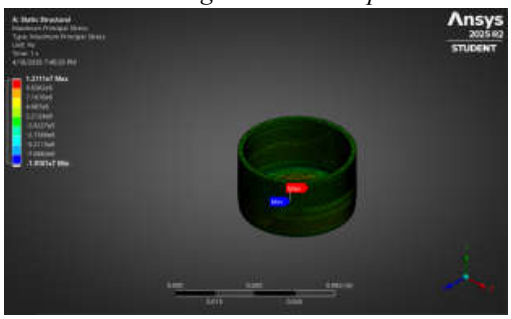


Figure 64: Charge well analysis

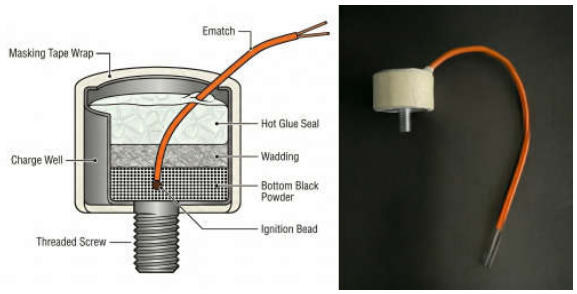


Figure 65: Charge well Configuration

The mass of black powder was calculated taking into account the force required to shear 6-Nylon M4 shear screws to be used in the coupler, each requiring a force of **300N** to shear. During ejection, the chargewells are able to generate an internal pressure of about **167KPa** within the parachute body tubes which produces a total force of **2800N** to shear all 6 shear pins. Analysis shows that at 167KPa of pressure the maximum stress subjected onto the chargewell is **12MPa** which is well within the ultimate tensile strength of Aluminium 6061-T6 alloy (310MPa)
The pressure generated by the black powder charge is estimated using the ideal gas relation, given by:

$$PV = mRT$$

Thus, the generated pressure ($1.668 \times 10^5\, Pa$) exceeds the required pressure ($1.056 \times 10^5\, Pa$), indicating that the selected black powder mass is sufficient to shear the screws and ensure reliable deployment.

Recovery Avionics

Apogee Deployment

The primary recovery sequence is initiated by the Missileworks RRC3+ altimeter, deploying the drogue at apogee. Redundancy is provided by a Blue Raven altimeter, which triggers deployment 2 seconds after a primary failure, followed by an SRAD flight computer that activates 2 seconds after a Blue Raven failure.

Main Deployment

The main parachute deployment charges are initiated by the RRC3+ at 1,500 ft AGL. To guarantee a highly reliable and redundant deployment sequence, the system includes backup mechanisms: the Blue Raven activates after a 2-second delay if the primary RRC3+ fails, followed by the SRAD Flight Computer if necessary.

Power & Wiring

The recovery avionics use independent power systems for redundancy and reliability. The RRC3+ is powered by a 9V logic battery and a separate 3.7V 18650 (2S) pyro battery for deployment charges. The Blue Raven uses a 3.7V 21700 (2S) battery, while the SRAD Flight Computer uses a 3.7V 21700 (2S2P) battery pack, both of which also power their respective charge firing systems. This ensures fault tolerance during critical recovery events

F. Payload Subsystems

The payload for Project UDBHAVA is a compact 3D micro bioprinter system designed to operate under high dynamic and reduced-gravity flight conditions encountered during rocket ascent. The system aims to investigate the behavior of bioinks and the feasibility of controlled extrusion-based printing under transient acceleration environments, providing insights for future aerospace and biomedical manufacturing applications.
The payload incorporates a miniaturized, autonomous architecture consisting of precision motion control, a controlled extrusion mechanism, and onboard process logic. The system is designed to maintain print fidelity despite external disturbances such as vibration and varying acceleration profiles during flight.

Objectives

The primary objectives of the payload are:

- Evaluate the effect of flight-induced acceleration on bioink flow and deposition behavior
- Assess printing precision and structural integrity under dynamic conditions
- Validate the feasibility of automated bioprinting during flight
- Generate experimental data for future biomanufacturing system development



Figure 66: Fully Assembled Payload

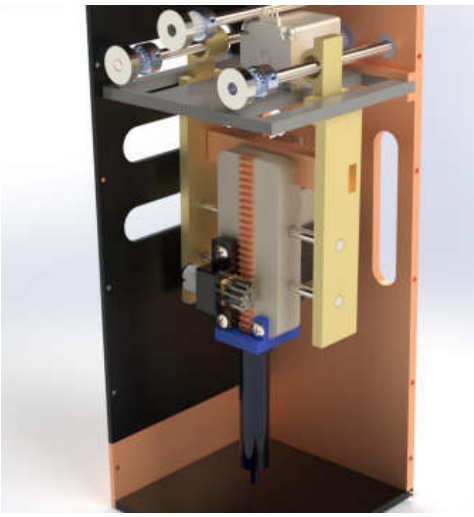


Figure 67: Internal Assembly of Payload

System Design

1. Motion System

The motion system provides controlled planar positioning of the extrusion assembly using two orthogonal axes:

- X-axis: Lead screw mechanism driven by a stepper motor, enabling precise and repeatable linear motion
- Y-axis: Rack and pinion mechanism directly coupled to a stepper motor, reducing mechanical complexity and moving mass

This configuration ensures accurate positioning while maintaining structural simplicity and reliability under flight conditions.

Extrusion Mechanism

Bio-ink extrusion is achieved using a syringe-based dispensing system, where a controlled actuation mechanism drives the plunger to produce a continuous and proportional flow of material.

The closed-system architecture minimizes the risk of leakage and ensures consistent extrusion behavior under varying acceleration. The design enables predictable flow characteristics, making it suitable for operation in reduced-gravity conditions.

Materials and Bioinks

The system is designed to support hydrogel-based bioinks, including:

- PEGDA (photopolymerizable bioink)
- Alginate-based formulations
- Collagen–gelatin composites

These materials are selected for their stability, controllable rheological properties, and suitability for extrusion-based printing, enabling characterization of flow behavior and structural formation during flight.

Hardware Architecture

The hardware architecture is centered around an onboard microcontroller interfaced with stepper drivers, sensors, and storage modules. Motion control is achieved using DRV8825 drivers configured for high-resolution microstepping, enabling smooth and precise positioning. A barometric sensor provides altitude estimation for flight-phase detection, while onboard storage ensures reliable data logging for post-flight analysis. The control system operates in a deterministic loop, enabling synchronized execution of motion commands and sensor acquisition.

Concept of Operations (CONOPS)

The payload operates through the following sequence:

1. Initialization: The system establishes holding torque and performs baseline calibration prior to launch
2. Ascent Detection: Motion remains locked until ascent conditions are confirmed

3. Homing: The system performs a predefined homing routine to establish a reference position
4. Printing Phase: The bioprinting operation is executed during the ascent phase to minimize disturbances
5. Failsafe: Upon detection of apogee or descent, the system halts all motion to prevent structural damage

This approach ensures that printing occurs within the most stable portion of the flight while protecting the system during recovery events.

Safety and Reliability

Safety and reliability are ensured through a combination of hardware and software safeguards. The bioink system is fully enclosed to prevent leakage, while the control logic incorporates operational limits and a failsafe shutdown triggered by flight conditions. Continuous monitoring of system parameters ensures that the payload operates within safe limits throughout the mission.These features ensure that the payload operates safely within the constraints of the flight environment.

The 3D micro bioprinter payload serves as a proof-of-concept platform for studying extrusion-based manufacturing under dynamic flight conditions. By combining compact mechanical design, controlled material deposition, and autonomous operation, the system enables experimental evaluation of bioprinting processes beyond static environments.

The data obtained from this mission will contribute to the development of future advanced manufacturing systems for aerospace and biomedical applications.

G. Airbrakes Subsystems

The Airbrake mechanism consists of four main components:

*1. Cam profile-based linear actuation***:** The cam profile provides near-linear radial displacement with servo rotation, ensuring predictable aerodynamic response. A servo-driven cam plate with four symmetric spiral slots deploys the airbrake leaves. Each leaf travels on a precision-guided linear rail on the coupler for smooth motion and effective load transfer, with ball bearings minimizing friction in the rail blocks and actuator connections.

2. Airbrake leaf geometry: Airbrake leaves are plates that sit flush when retracted and extend radially outward when deployed to increase drag. Their dimensions are determined by the airframe diameter, internal mechanism constraints, and maximum actuation displacement.

3. Integrated coupler design: The airbrake mechanism is integrated into a structural coupler connecting rocket body tubes. It serves as both a structural component and a housing, supporting the linear rails, cam slot, servo mount, and control electronics, using threaded rods for low mass and high integrity.

4. Control electronic mounting: The servo motor is mounted on an actuation plate, connected to the cam plate via a ball-bearing supported central shaft. Control electronics (PCB and batteries) are on a separate electronic plate. Both plates are connected to the coupler by four threaded rods.



Figure 68: Airbrake Assembly (External View)

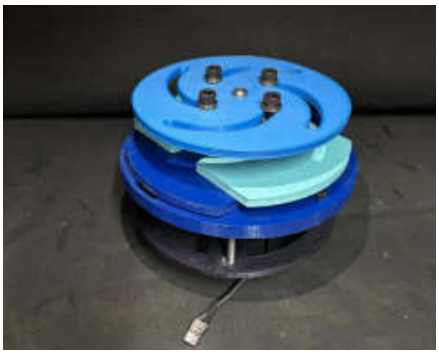


Figure 69: Airbrake Assembly (Internal Mechanism View)

Aerodynamic simulations and Validation

Extensive CFD analysis was performed on airbrakes at different Mach regimes from Mach 0.875 to Mach 0.3 during coasting phase of the rocket, this analysis was performed at every deployment level from 40 percent deployment to 100 percent deployment. To perform transonic analysis of whole rocket geometry integrated with airbrakes, poly

-hex core mesh was performed on the geometry using Fluent meshing software, to use density based method of solution and SST K-omega turbulence modelling **y+ of +1** was maintained across all the rocket walls.

Initially geometry of the rocket was prepared in Space Claim software with BOI provided at specific locations like wake region, cylinder around the rocket body .

Table 8: CFD Setup and Boundary Conditions

Parameter	Value
Solver	Density-based
Flow Model	Ideal Gas
Inlet	PressureFar Field(M= 0.875, P= 90,912, T= 276.28K)
Outlet	Pressure Outlet
Wall	Adiabatic, No Slip
Geometry	ANSYS SpaceClaim
Mesh	BOI(wake + body)

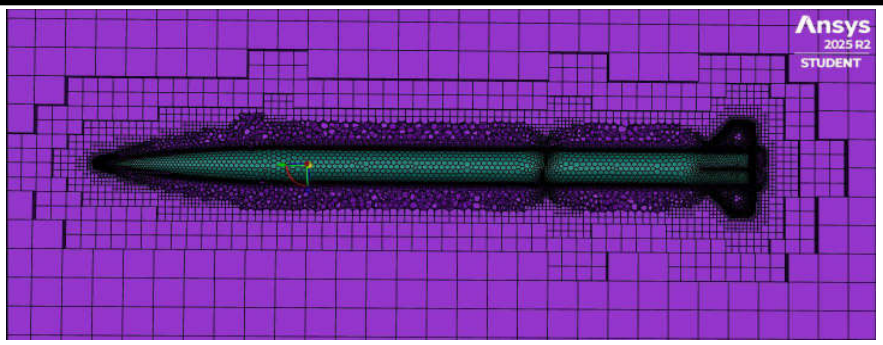


Figure 70: Poly-hexcore Mesh of Rocket with Airbrakes Deployed

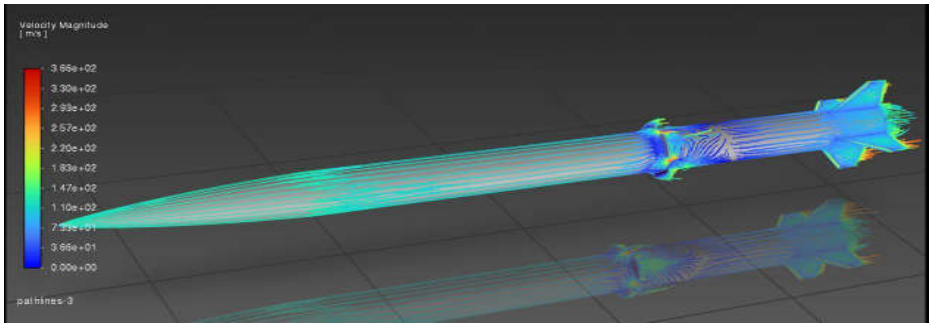


Figure 71: Velocity Magnitude Contours Around Airbrake Region

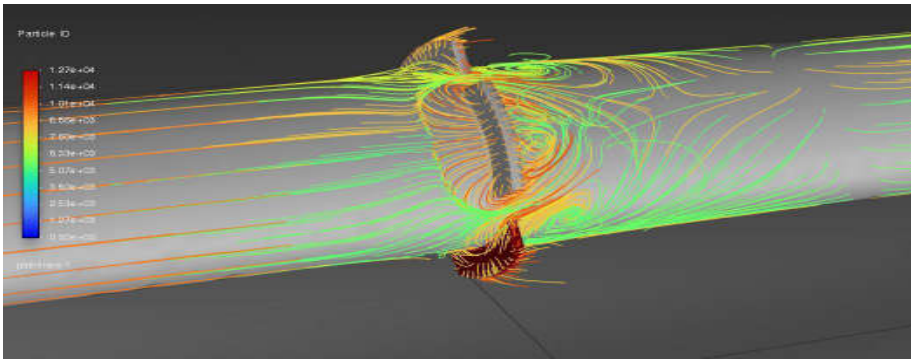


Figure 72: Flow Separation and Recirculation Near Deployed Airbrakes

The results show strong flow separation and recirculation behind the deployed airbrakes, leading to increased pressure drag. Streamlines indicate vortex formation and turbulent wake development, confirming effective drag generation while maintaining aerodynamic symmetry.
The drag coefficient data presented in **Table 9** was obtained from CFD simulations across varying Mach numbers and airbrake deployment levels. This dataset was used to generate a lookup table within the flight control software, enabling real-time mapping between airbrake deployment percentage and the corresponding drag coefficient. During flight, the control algorithm references this precomputed CFD dataset to estimate aerodynamic drag for a given deployment state, allowing accurate apogee prediction and dynamic adjustment of the airbrake position. This approach reduces onboard computational load while maintaining high prediction accuracy.

Table 9: Cd vs Drag based on percentage deployment

CD	100% deployment	90% deployment	80% deployment	70% deployment	60% deployment	50% deployment	40% deployment
0.875	**1.08**	**0.966**	**0.871**	**0.85**	**0.7190**	**0.648**	**0.63**
0.7	**0.996**	**0.947**	**0.85**	**0.8341**	**0.6830**	**0.652**	**0.60**
0.6	**0.993**	**0.932**	**0.8277**	**0.802**	**0.6856**	**0.638**	**0.586**
0.5	**0.96**	**0.9276**	**0.8839**	**0.799**	**0.6796**	**0.60**	**0.574**
0.4	**0.9769**	**0.924**	**0.8032**	**0.821**	**0.668**	**0.6123**	**0.589**
0.3	**0.945**	**0.906**	**0.823**	**0.789**	**0.656**	**0.6156**	**0.576**

Airbrakes Avionics

BHAGAT-C (Borne Hardware for Apogee Guidance, Airbrakes Tuning & Control) is a custom SRAD electronic system designed for precise apogee control during the coast phase after motor burnout. The vehicle is initially boosted to a velocity exceeding that required to reach the target apogee of 10,000 ft AGL.

After burnout, BHAGAT-C functions as a closed-loop drag control system. It uses redundant sensors to measure altitude, orientation, rates, and acceleration, which are processed by an onboard microcontroller running a custom control algorithm. Based on this, a rugged servo-via dual redundant channels-actuates the airbrake mechanism to regulate aerodynamic drag. Key design principles for BHAGAT-C were field accessibility and redundancy, ensuring reliable and consistent operation across multiple flight campaigns.

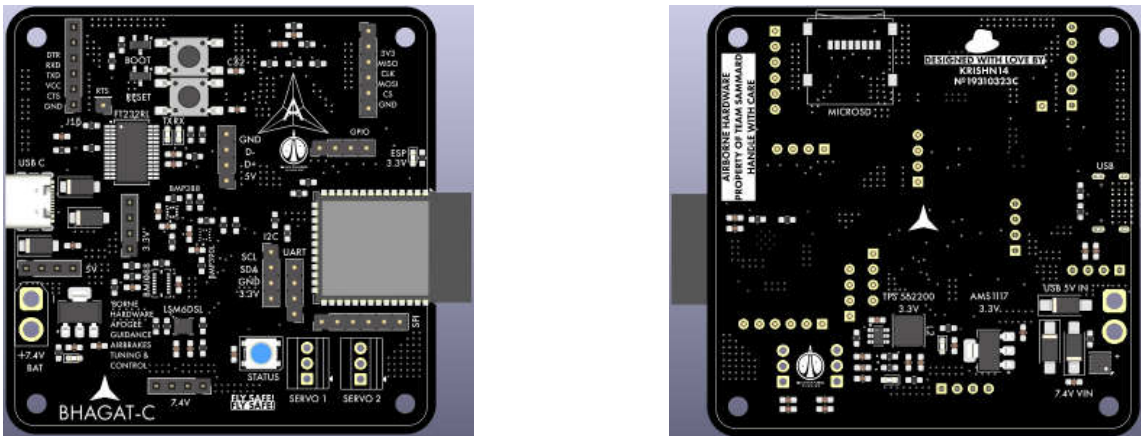


Figure 73: Bhagat C Printed Circuit Board Design

Table 10: Major Components overboard

ESP-32 S3-WROOM-1	BMP390L	BMP388	BMI088	LSM6DSL
AMS117 (3.3V, 5V Linear Dropout Regulator)	TPS562200 (3.3V Buck Converter)	microSD Interface	USB-C Interface	FTDI 232RL USB-UART Interface

Airbrake software

The airbrake system actively modulates vehicle drag during the unpowered ascent phase to bring the rocket to apogee within the competition target of 10,000 ft. Flight simulation alone cannot account for all sources of trajectory uncertainty - variations in motor burn profile, atmospheric density, and aerodynamic drag coefficients collectively produce apogee dispersions that passive ballistic flight cannot correct. By continuously computing a real-time predicted apogee from onboard sensor data and adjusting the deployed area of four drag panels accordingly, the system closes a feedback loop around these uncertainties.The airbrake operates exclusively during the coasting ascent phase. Deployment prior to motor burnout would destabilise the vehicle; deployment during descent has no corrective effect on apogee. The flap mechanism therefore remains fully retracted in all other flight phases.

The flight computer is implemented on an ESP32-S3-WROOM-1 microcontroller operating at 240 MHz with a hardware floating-point unit. The firmware employs a dual-rate loop architecture: a 500 Hz inner loop for inertial measurement and attitude estimation, and a 100 Hz outer loop for barometric state correction, apogee prediction, and servo actuation. The overall system is partitioned into four layers, as illustrated in Figure 1.

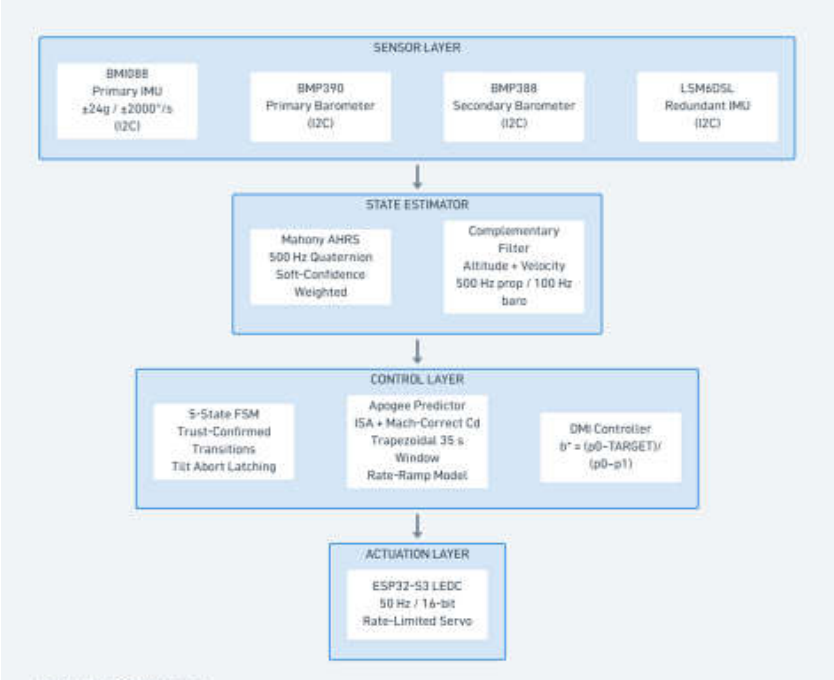


Figure 74: System Architecture

2. State Estimator

A) Barometric Altimeter -

The primary barometer (BMP390, I2C) provides a low-bandwidth but drift-free measurement of altitude above ground level (AGL). It is used for two purposes: (1) periodically correcting the complementary filter's altitude and velocity estimates, which would otherwise accumulate velocity drift from IMU bias over the coast phase, and (2) supplying the absolute altitude input to the apogee predictor.The secondary barometer (BMP388, I2C) operates in parallel as a crosscheck. A divergence of more than 15 m between the two sensors generates a serial warning; the secondary reading does not feed the estimator directly.

B) Mahony Quaternion AHRS - Role in the Estimator

The Mahony filter maintains a quaternion representation of vehicle attitude at 500 Hz. Its output serves two critical functions: extracting the true vertical component of acceleration for the complementary filter, and computing the vehicle tilt angle for the safety abort logic.

C) Complementary Filter

The complementary filter fuses inertial propagation at 500 Hz with barometric correction at 100 Hz to produce altitude and velocity estimates for the controller. The filter is chosen over an Extended Kalman Filter (EKF) because it achieves equivalent accuracy at approximately 5% of the computational cost, with no covariance matrix tuning risk

3. Apogee Predictor

The predictor integrates the equations of motion forward in time from the current state, returning the altitude at which velocity reaches zero. It is invoked twice per control tick: once with brakes fully retracted (yielding p0, the natural-trajectory apogee) and once fully deployed (yielding p1, the maximum-drag apogee). These two values bracket the reachable apogee space and are the sole inputs to the controller inversion

IV. MISSION CONCEPT OF OPERATIONS

The mission profile of the UDBHAVA launch vehicle follows a sequence of controlled flight phases, as illustrated in Figure X, from launch through recovery. The system is designed for fully autonomous operation with continuous telemetry transmission and ground-based monitoring.

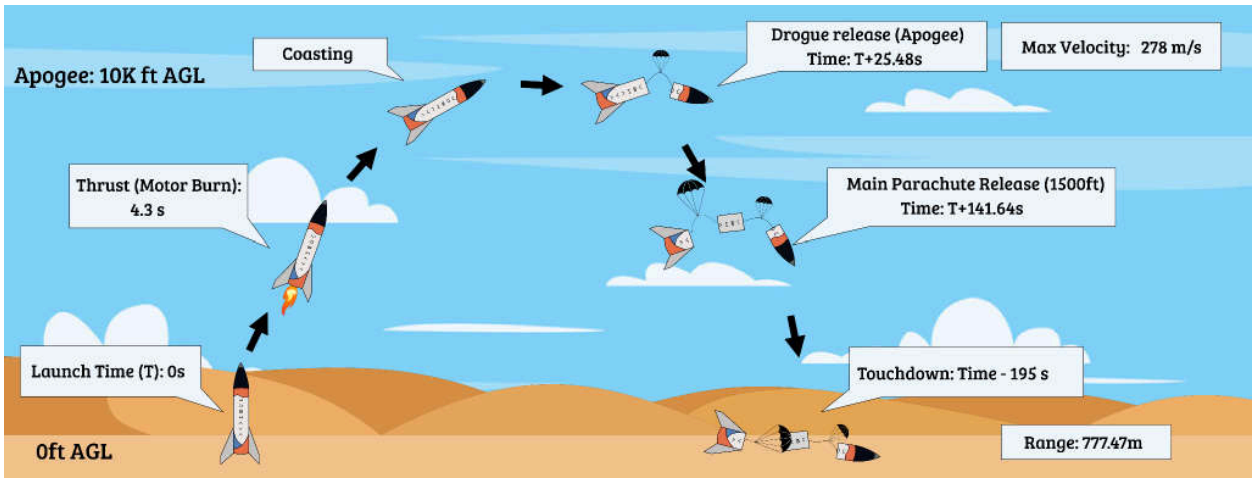


Figure 75 - Mission CONOPS

1. Launch and Boost Phase (T = 0 – 4.3 s)

At T = 0 s, the rocket is ignited and accelerates along the launch rail under thrust from the SRAD solid rocket motor. The motor burns for approximately 4.3 seconds, generating the required thrust to achieve the target trajectory. During this phase, the avionics system records high-frequency sensor data and transmits real-time telemetry to the ground station.

2. Coasting Phase

Following motor burnout, the rocket enters the coasting phase, where it continues to ascend under inertia. Aerodynamic forces dominate during this period, and the vehicle reaches a maximum velocity of approximately 278 m/s before gradually decelerating as it approaches apogee.

3. Apogee and Drogue Deployment (T ≈ 25.48 s)

The rocket reaches an apogee of approximately 10,000 ft AGL at T ≈ 25.48 s. Apogee is detected using onboard altitude measurements and validated through multi-cycle logic. Upon confirmation, the flight computer triggers the drogue parachute deployment, stabilizing the vehicle and initiating controlled descent while minimizing drift.

4. Descent Phase and Main Deployment (T ≈ 141.64 s)

During descent, the vehicle continues to transmit telemetry while descending under the drogue parachute. At a predefined altitude of approximately 1500 ft AGL, the system deploys the main parachute at T ≈ 141.64 s, significantly reducing descent velocity to ensure safe landing conditions.

5. Landing and Recovery (T ≈ 195 s)

The rocket lands at approximately T ≈ 195 s, with a predicted ground range of ~777 m from the launch site. Landing is confirmed through altitude stabilization and low acceleration readings. Upon touchdown, the GPS data is used to guide the recovery team to the landing location.

V. CONCLUSION AND LESSONS LEARNED

Conclusion

Project UDBHAVA represents a major advancement for Team Sammard, marking the team's transition to a fully student-researched and developed (SRAD) propulsion system while integrating a complex multi-subsystem architecture. The project successfully combines propulsion, aero-structures, avionics, recovery, payload, and ground systems into a cohesive launch vehicle designed for reliability and performance. Emphasis on modularity, subsystem validation, and system-level integration enabled the team to develop a robust design capable of meeting mission objectives.

The inclusion of an experimental bioprinter payload and enhanced communication systems further demonstrates the team's progression toward handling high-complexity aerospace systems. Overall, the project establishes a strong foundation in SRAD development, subsystem coordination, and mission-driven engineering, positioning the team for more advanced future missions.

Lessons Learned

The development of UDBHAVA provided significant insights across both technical and team management domains. The transition to a fully SRAD propulsion system underscored the critical importance of repeatable manufacturing processes, disciplined grain casting procedures, and rigorous static fire validation in achieving predictable and consistent motor performance. A key lesson drawn from this subsystem was the necessity of an incremental development approach, where each component and process is individually validated before integration, ensuring that risks are identified and mitigated progressively rather than compounded at the system level.

In aero-structures, precise manufacturing and alignment were found to be critical, particularly for load-bearing components and fin integration. Avionics development highlighted the need for robust state-machine logic, sensor redundancy, and noise-resistant algorithms to ensure reliable event detection and recovery system actuation.

The payload subsystem introduced challenges associated with operating precision systems under dynamic conditions, reinforcing the importance of mechanical simplicity, vibration tolerance, and deterministic control strategies. Ground system development demonstrated the value of modular software architecture, real-time data validation, and efficient telemetry pipelines in maintaining situational awareness during flight.

From a team management perspective, the project underscored the effectiveness of structured workflows, early integration testing, and cross-functional collaboration. Iterative design reviews and parallel subsystem development helped reduce integration risks and improve overall efficiency.

Failures encountered during the project, including manufacturing inconsistencies, integration mismatches, and communication limitations, were systematically analyzed and used to refine both design and testing approaches. These experiences reinforced the importance of designing for manufacturability, validating assumptions early, and maintaining flexibility for iteration.

To ensure continuity, Team Sammard implemented structured knowledge transfer practices, including comprehensive documentation, version-controlled repositories, and mentorship of junior members by senior team leads. This approach ensures that critical design knowledge, testing experience, and lessons learned are preserved and effectively passed on to future team members.

VI. ACKNOWLEDGEMENTS

Team Sammard expresses its heartfelt gratitude to Dr. Senthil Kumar M, Faculty Coordinator, for his constant mentorship throughout our journey to the International Rocket Engineering Competition (IREC) 2026. His belief in our potential has been pivotal in driving the team toward technical excellence. We also extend our sincere gratitude to the Vellore Institute of Technology (VIT) for providing the infrastructure and resources necessary for this ambitious endeavor. Specifically, we thank Dr. Naiju C D, Director of Students' Welfare, and Dr. Rajasekhara Reddy Mutra, Assistant Director of Students' Welfare, for providing a dedicated workspace and facilitating the permissions required to advance our project. The university's commitment to student-driven research continues to inspire our team to bring complex engineering ideas to life.

We are deeply grateful to Dr. Rahul Manohar O, Assistant Professor (Senior Grade) from the School of Electronics Engineering (SENSE), for his invaluable technical guidance. His contributions were essential in the fabrication of our horn antenna and the development of the 2×2 Yagi stack antenna for long-range telemetry. We specifically thank him for providing critical materials, granting access to the anechoic chamber for testing, and validating our power divider designs for low-cost fabrication. His support has been instrumental in our growth within the technical community.

We would also like to thank Dr. Anjaneyulu U, Assistant Professor Sr. Grade 1 from Centre of Biomaterials, Cellular and Molecular Theranostics, Vellore Institute of Technology, Vellore, India for his support in the development of our 3D bioprinter payload specifically for the procurement of the bio-ink.

Our sincere thanks go to Convergent Science India LLP, our Platinum Sponsor, for providing access to their advanced simulation software. This tool allowed us to perform critical Computational Fluid Dynamics (CFD) analyses, essential for understanding the aerodynamic and thermodynamic behaviors of our rocket.

For the complex logistics of international competition, we thank Mr. Edward Samuel, Senior Manager of Purchase at VIT, alongside Mr. Salaudin from GEODIS India Pvt. Ltd. and Mr. Avinash from Flomic Global Logistics. Their prompt support in handling the shipment of our rocket from India to the USA was vital to ensuring the project's timely arrival. We are also thankful to Mr. Robert Dickinson, our Flyer of Record, whose insights during the certification process ensured our compliance with all safety protocols. Finally, we thank the Experimental Sounding Rocket Association (ESRA) for fostering a global platform that allows us to engage deeply with the international aerospace community.

Appendix A.
Systems Weights, Measures and Performance Data

1. Rocket Information

The Rocket Information can be seen in Table A.1:

Table A.1: Rocket Specifications

Parameters	Value	Units
Number of Stages	1	-
Length	3060	mm
Airframe Diameter	153	mm
Number of Fins	4	-
Fin Semi-Span	152	mm
Fin Tip Chord	75	mm
Fin Thickness	4.5	mm
Weight On Launch Rail	36.25	Kg
Propellant Weight	9.5	Kg
Empty Motor Casing/Structure Weight	9.4	Kg
Payload Weight	4	Kg
Liftoff Weight	36.25	Kg
Centre of Pressure	2407	mm
Centre of Gravity	2086	mm
Launch Rail Length	17	ft

2. Motor Parameters

The design parameters associated with the motor are listed in Table A.2.

Table A.2: Specifications of the Motor

Parameters	Data	Units
Propellant	KNSB	-
Dry Mass	9.4	Kg
Propellant Mass	9.5	Kg
Loaded Mass	18.9	Kg
Theoretical Burn Time	4.3	s
Theoretical Total Impulse	10384	N-s
Theoretical Specific Impulse	111.46	s
Theoretical Maximum Thrust	2794.8	N
Theoretical Average Thrust	2622	N
Theoretical Maximum Pressure	3.52	MPa
Expansion Ratio	3.861	-

2. Flight Dynamics Simulation Results

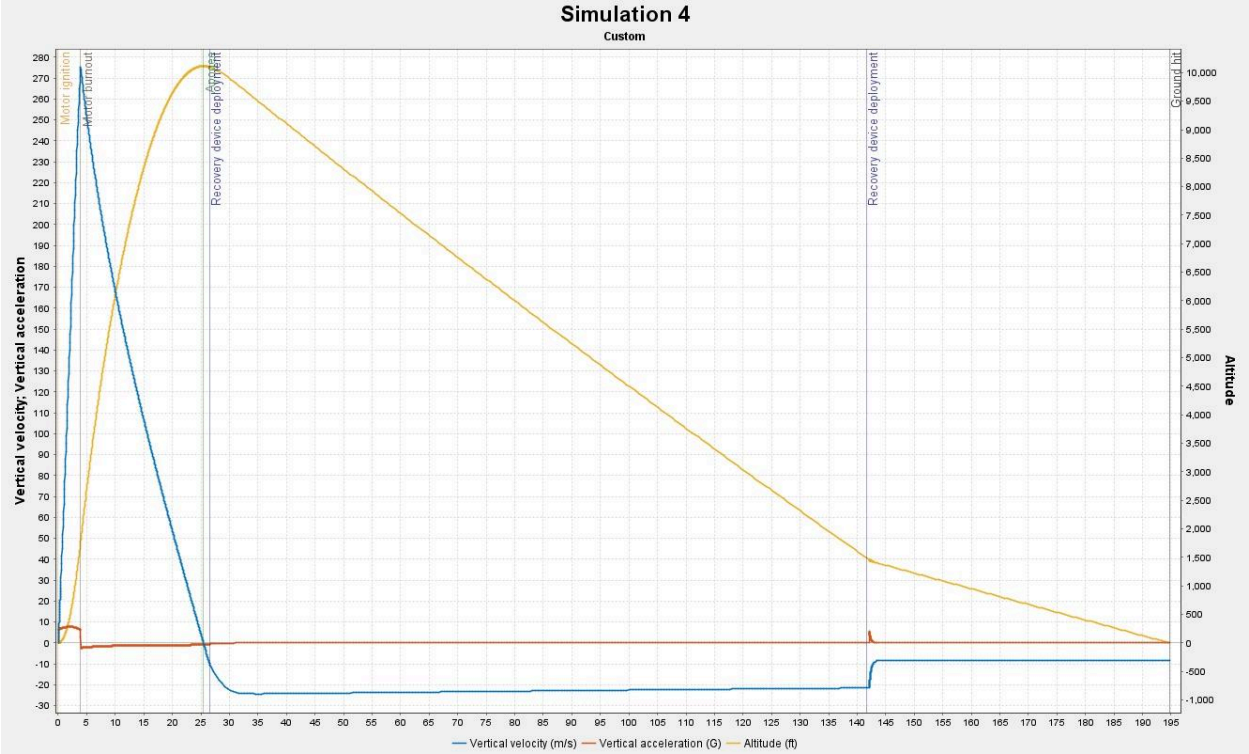


Figure A.1: Altitude,Vertical Velocity, Vertical Acceleration VS Time

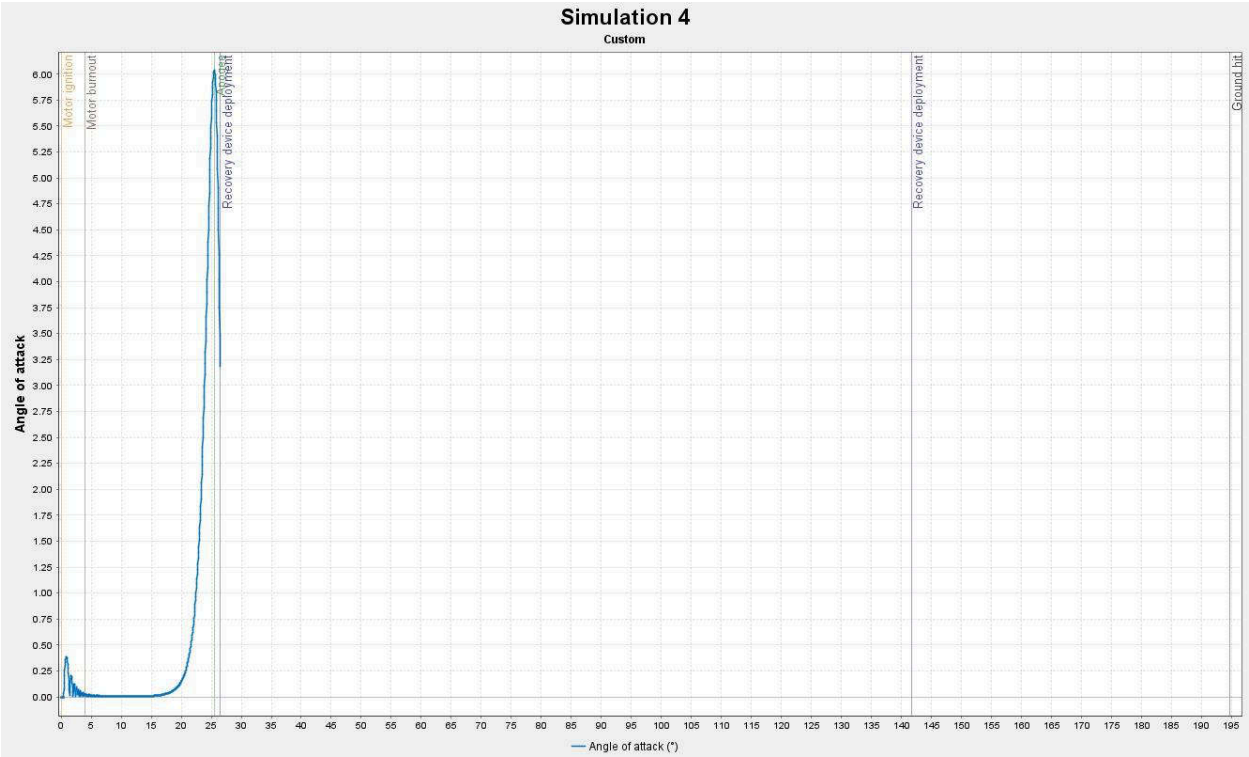


Figure A.2: Angle of Attack of VS Time

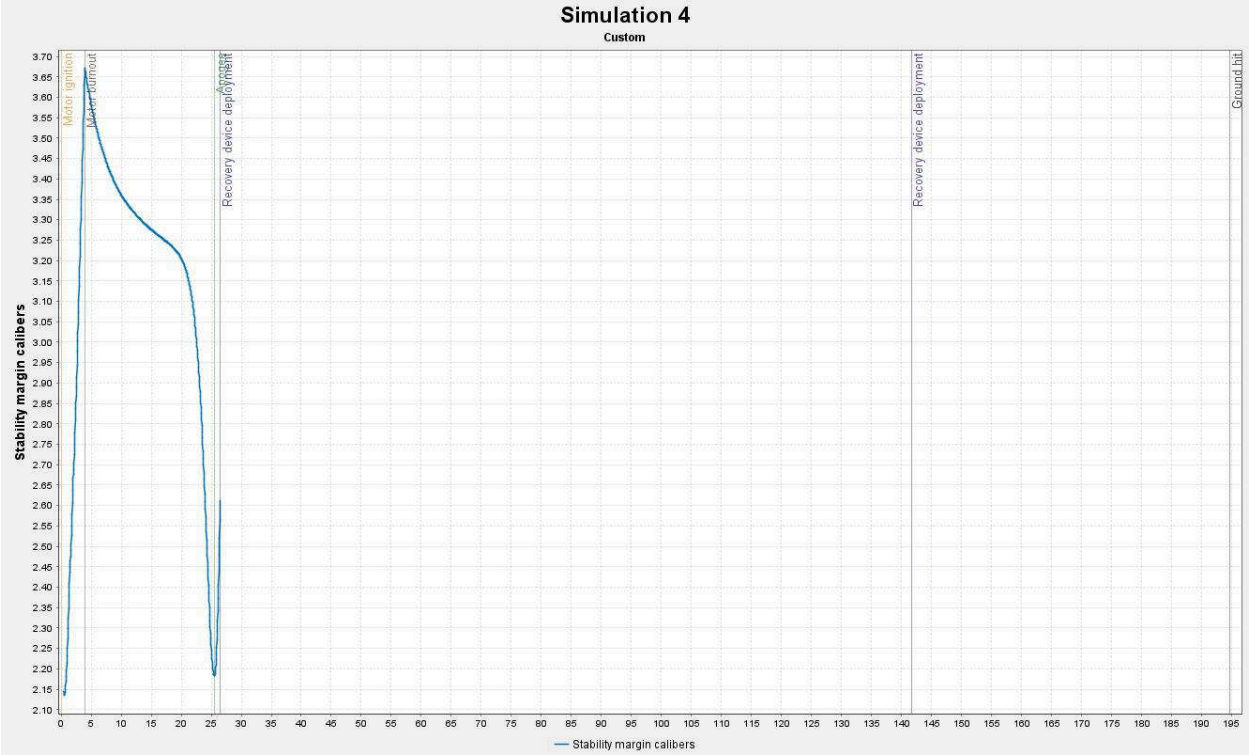


Figure A.3: Variation of Static Margin VS Time

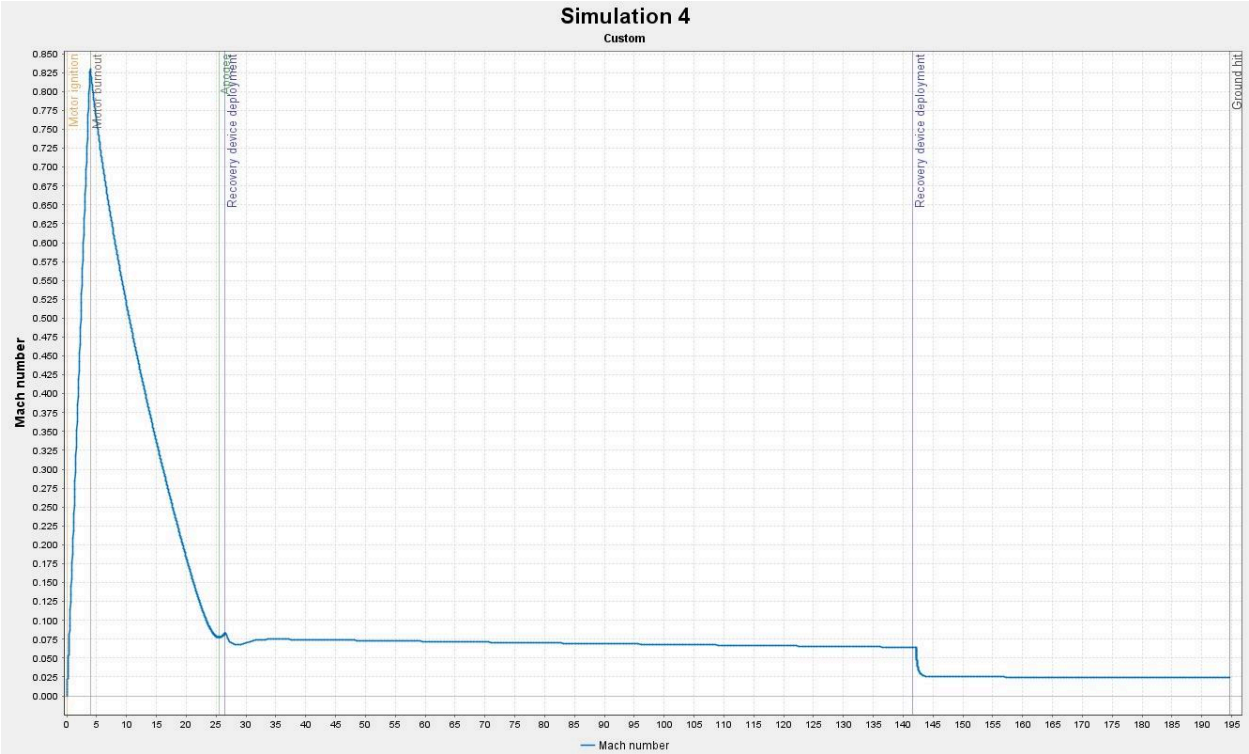


Figure A.4: Mach Number VS Time

The predicted flight performance parameters are summarized in Table A.3.

Table A.3: Predicted Flight Performance Parameters

Parameters	Value
Liftoff Thrust-Weight Ratio	7.86
Rail Departure	26.5 m/s
Minimum Static Margin	2.14 cal
Maximum Acceleration	75.9 m/s^2 or 7.74G
Maximum Velocity	278m/s or Mach 0.829
Fin Flutter Velocity	627m/s
Target Apogee	10,000ft
Predicted Apogee	10,115ft

3. Recovery Information

The Recovery specifications can be seen in Table A.4:

Table A.4: Recovery Information

Parameters	Value
Primary Altimeter	RRC3+
Redundant Altimeter	Blue Raven & Tertiary Flight Computer
Shear Pins	M4 Nylon 6.6 Screws
Shock Cord	5/8-inch Flat Kevlar Webbing 4500 lbs
Drogue Parachute	SRAD Rectangular Chute
Drogue Primary Deployment Charge	5g
Drogue Backup Deployment Charge	8g
Drogue Deployment Altitude	Apogee
Drogue Descent Rate	30m/s
Drogue Parachute Protection	Glass Fibre
Main Parachute	SRAD Elliptical Chute
Main Primary Deployment Charge	5g
Main Backup Deployment Charge	8g
Main Deployment Altitude	1500 ft
Main Parachute Protection	Glass Fibre
Connection Point Type	Eyebolt
Linkages	Oval 8mm Carabiner - Screw Lock

4. Propulsion Specifications

Propulsion Simulation I

2D Nozzle Analysis

Table A.5: Ansys Fluent setup Parameters

Category	Parameter	Specification / Value
Objective	Analysis Type	Compressible flow
	Purpose	Evaluate pressure, temperature, velocity, and Mach number variations under strong pressure gradients
Solver Settings	Solver Type	Density-based, implicit
	Time Formulation	Steady-state
	Spatial Discretization	Second-order upwind
Turbulence Model	Model Used	SST k-ω
	Justification	Accurate prediction of boundary layer behaviour and adverse pressure gradients
Geometry	Domain Type	Axisymmetric
	Flow Type	Internal compressible nozzle flow
Mesh Details	Cell Type	Structured quadrilateral
	Number of Cells	56,784
	Number of Faces	114,215
	Number of Nodes	57,432
Mesh Quality	Minimum Orthogonal Quality	0.707
	Maximum Aspect Ratio	5.68
Material Properties	Fluid	Air (modelled as KNSB exhaust equivalent)
	Phase	Single-phase
Boundary Conditions	Inlet Condition	Total Pressure = 3.5 MPa
	Inlet Temperature	1600 K
	Outlet Condition	Atmospheric pressure
	Wall Condition	No-slip, adiabatic
Solution Controls	Iterations Run	1000
	Convergence Status	Not fully converged
Post-Processing	Key Parameters Evaluated	Pressure, Temperature, Velocity, Mach Number

The following were the results for the residuals after 1000 iterations

Table A.6:CFD Solver Residual Values

Parameter	continuity	x-velocity	y-velocity	energy	k	omega
Value	0.0003885197	0.0001196266	0.716558	0.0002031105	2.94E-05	1.47E-05

The results of the simulations are represented in the form of report plots and contours.

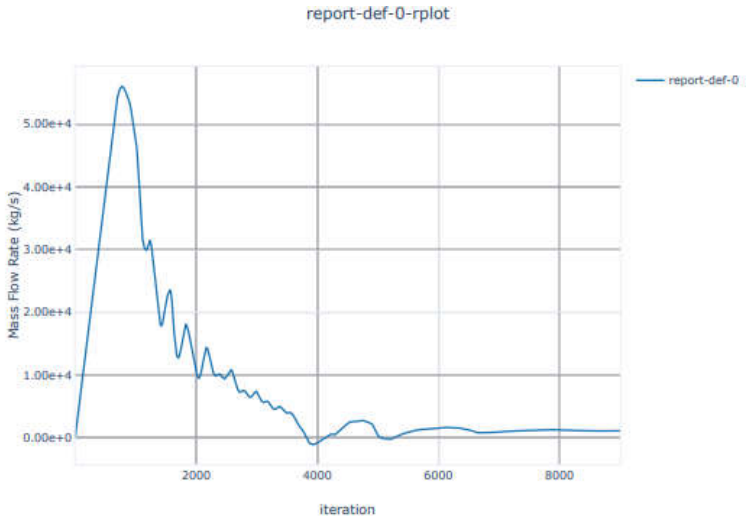


Figure A.5 : Mass Flow Rate Plot

The steady-state mass flow rate for this simulation is recorded at **1074.171 kg/s**, achieving physical stability. The mass flow rate plot reveals a highly volatile initialisation phase where mass flow rates peaked = before undergoing decaying oscillations and a brief reversal near iteration 4,000. Stabilisation occurred after approximately **6,000 iterations**, aligning with the formation of distinct shock structures.

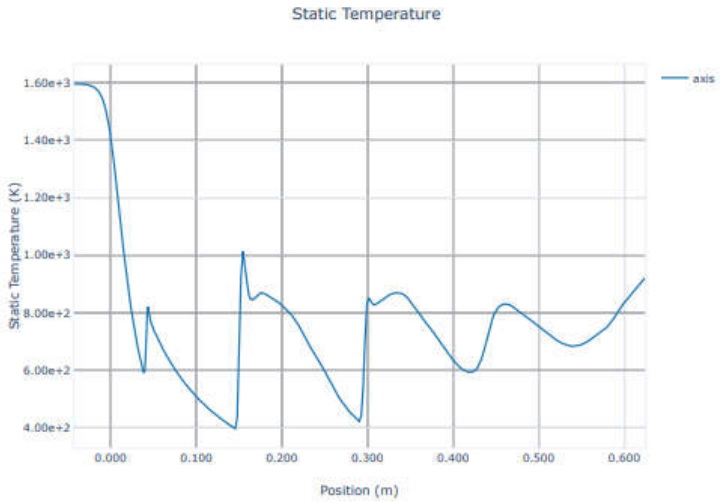


Figure A.6: Static Temperature Plot

The static temperature profile illustrates the conversion of thermal energy into kinetic energy during expansion. From a stagnant total temperature of **1600 K**, the temperature drops sharply to a minimum of approximately **400 K** at the end of the primary expansion zone. Subsequent spikes in the plot, such as the rise back toward **1000 K**, indicate localised heating caused by shock reflections within the plume or nozzle extension.

The Mach number profile confirms a highly supersonic flow field, peaking at approximately **Mach 3.9**. The "sawtooth" pattern observed after the initial expansion represents a series of shock diamonds where the flow periodically decelerates and accelerates. Each sharp drop in Mach number perfectly correlates with the pressure and temperature spikes, suggesting that the **SST k-omega** turbulence model is effectively capturing the compressible flow structures

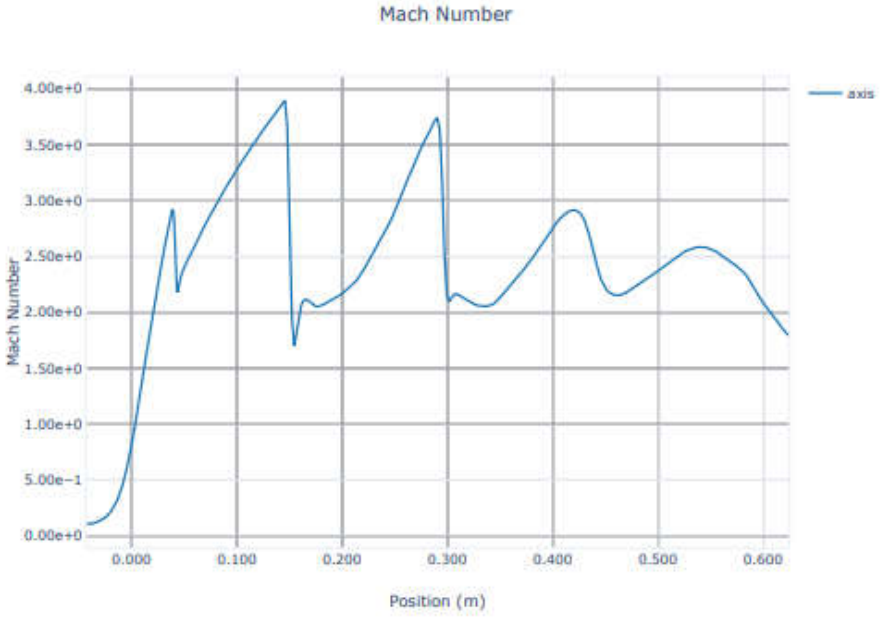


Figure A.7 : Mach Number Plot

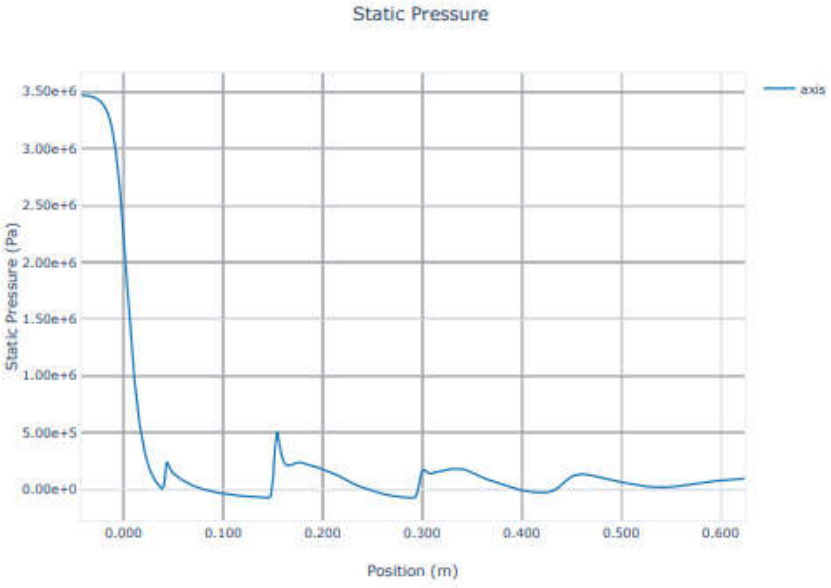


Figure A.8 : Static Pressure Plot

Starting from an inlet gauge pressure of **3.5 MPa**, the static pressure rapidly decays as the fluid expands into the ambient environment. The profile reaches a near-vacuum or ambient state quickly, followed by small-amplitude pressure recovery peaks. These localised pressure increases confirm the presence of oblique shock waves, validating that the density-based implicit solver has reached a physically consistent solution for this high-pressure ratio scenario.

The following are the contour plots obtained . These plots verify the result inferences that have been described and inferred from the above plots

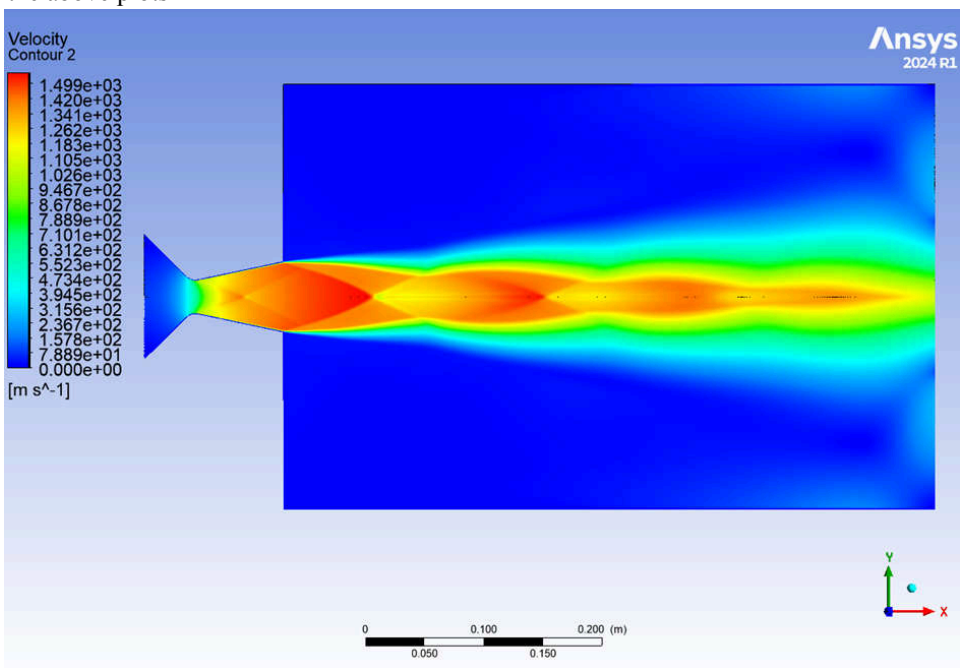


Figure A.9- Velocity Contours

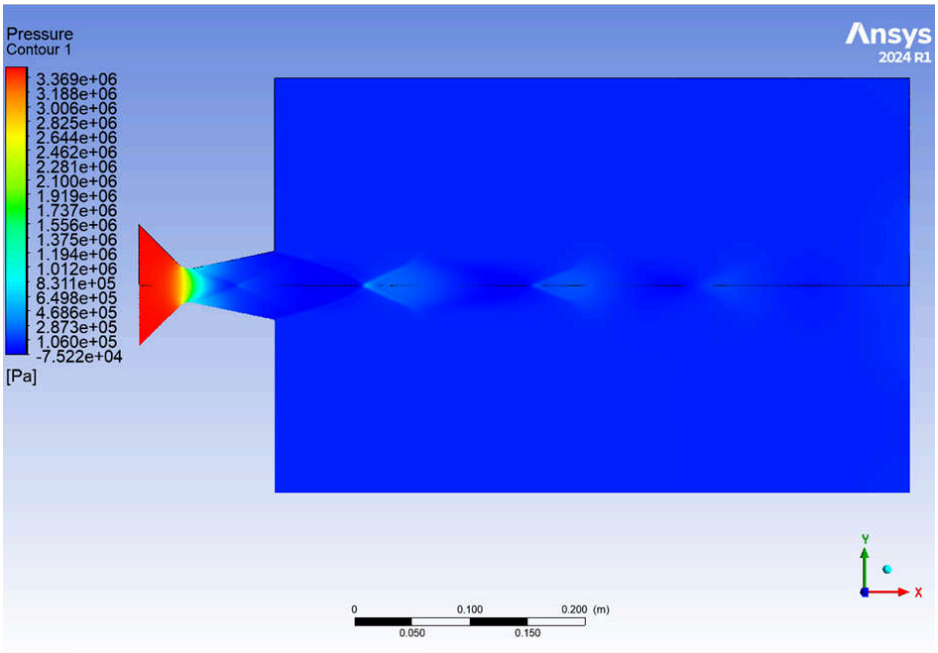


Figure A.10: Static Pressure Contour

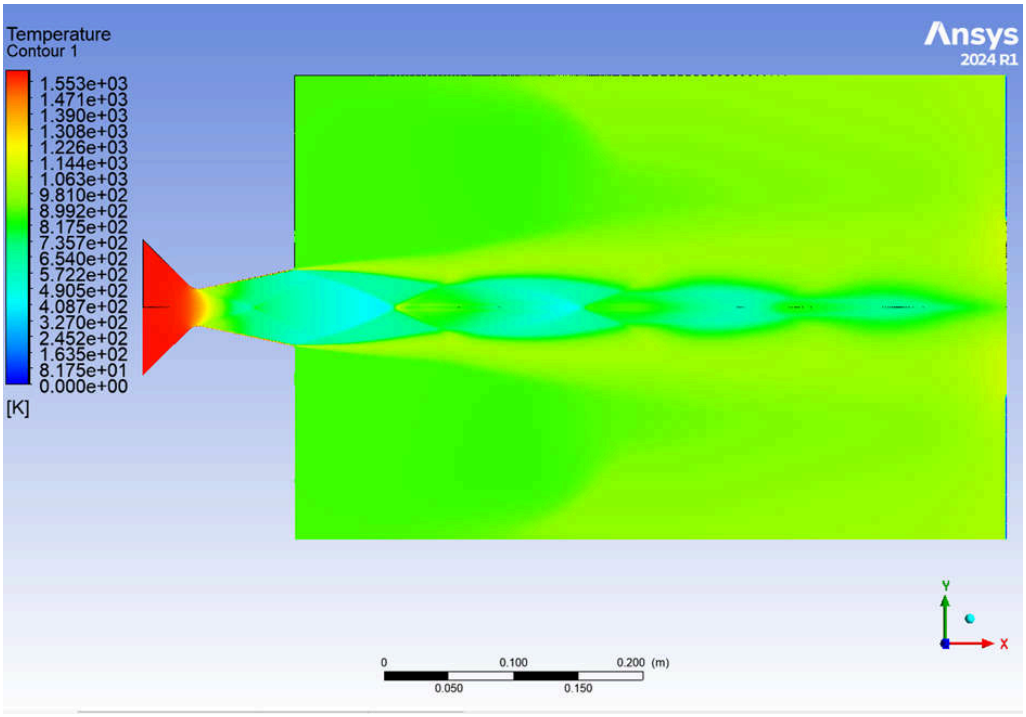


Figure A.11: Static Temperature Contour

Propulsion Simulation II

Motor Casing Burst Pressure Analysis:

Ansys Explicit Dynamics to examine the transient structural behaviour of the motor casing with the accommodation made for the radial holes where the forward closure and nozzle are attached.
The critical feature of this domain is a series of radial fastener holes located at the assembly interface. Unlike static analysis, this explicit approach accounts for the propagation of stress waves and high-strain-rate material responses that occur during the initial ignition transient. The simulation focuses on a representative section of the casing to analyse the localised effects of high-pressure transients on these orifices.

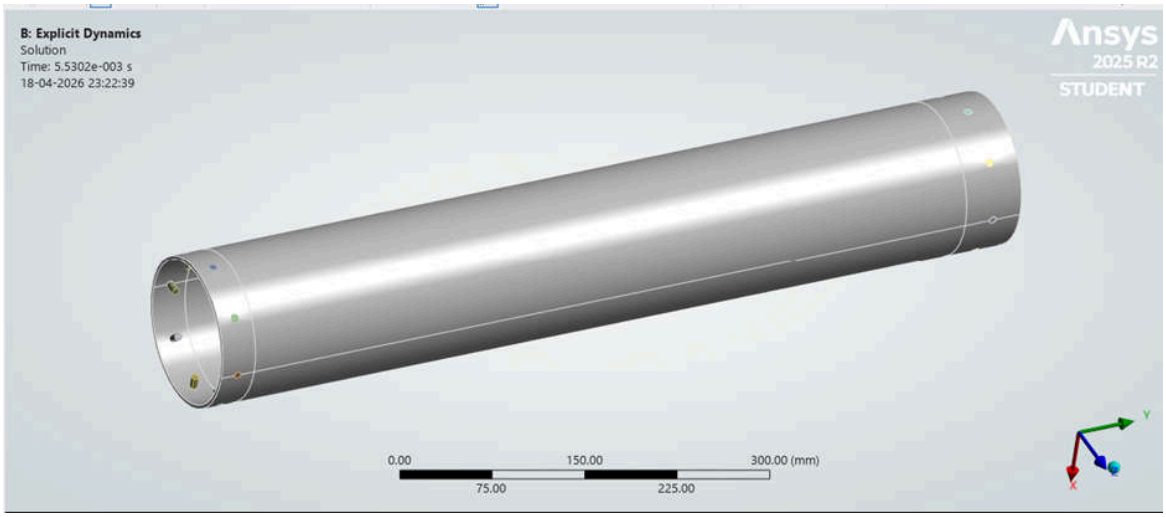


Figure A.12: Geometry Description of Motor Casing

Mesh Characteristics
To ensure numerical accuracy where stress gradients are steepest, a specialised meshing strategy was employed. A High-fidelity hexahedral/quad-dominant mesh was used. The model utilises approximately 56,784 cells, optimised for a balance between computational cost and spatial resolution. A localised face sizing was applied to the fastener holes, ensuring that the curvature of the hole boundary is captured by at least 12–16 elements to prevent artificial stiffness.
The mesh maintains a minimum orthogonal quality of **0.7075**, ensuring that the explicit solver remains stable during high-deformation time steps.

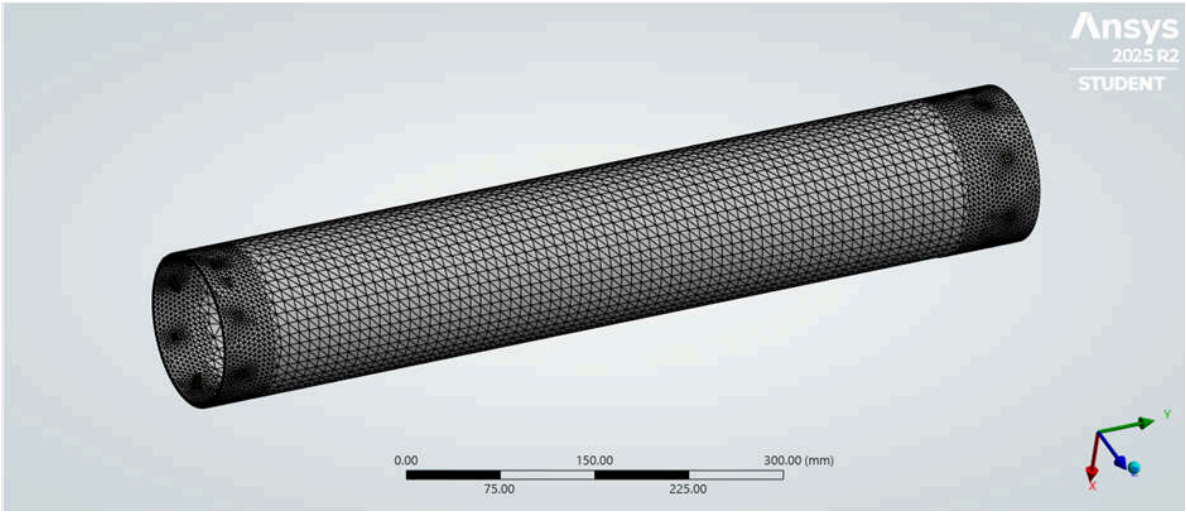


Figure A.13: Mesh Model

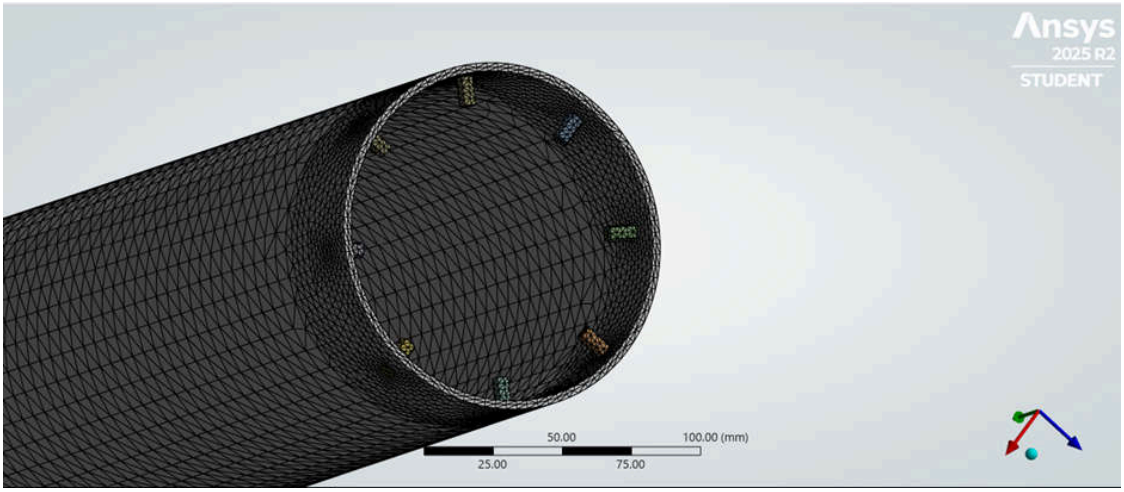


Figure A.14: Localised Mesh Refinement at the Forward Closure Bolt Attachments

The setup utilises an Explicit Dynamics solver, which is ideal for capturing shock wave propagation and transient structural responses.
High-strength Aluminum/Steel alloy modelled with bilinear isotropic hardening. This allows the simulation to track permanent plastic deformation once the material exceeds its yield point.
An internal pressure ramp was applied to simulate the ignition spike. The pressure ramp function was based on the calculation made for pressure analysis .

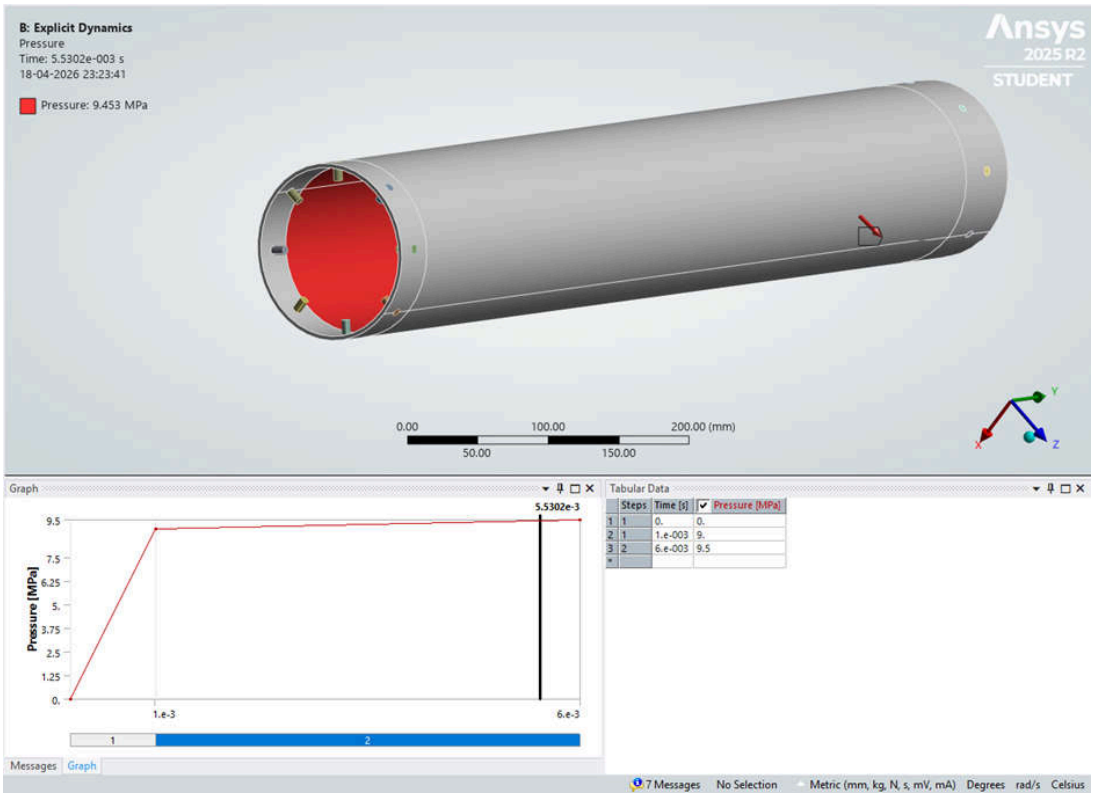


Figure A.15: Internal Pressure Loading Profile Applied to the Motor Casing

The forward edge is treated as a fixed support, while the radial boundaries are permitted to expand.

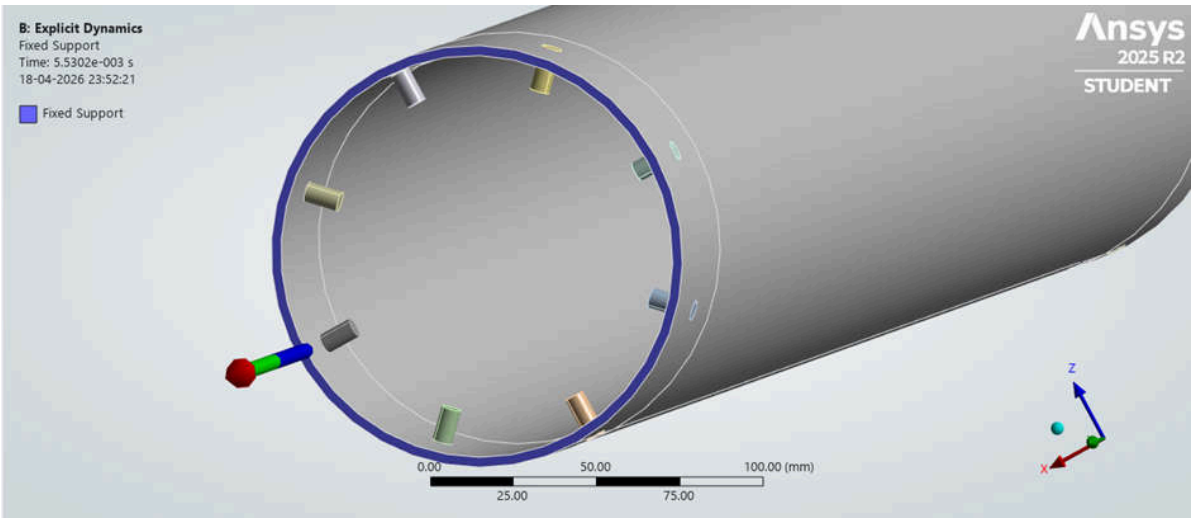


Figure A.16: Fixed Support Boundary Condition Applied to the Forward Edge of the Motor Casing

Solution

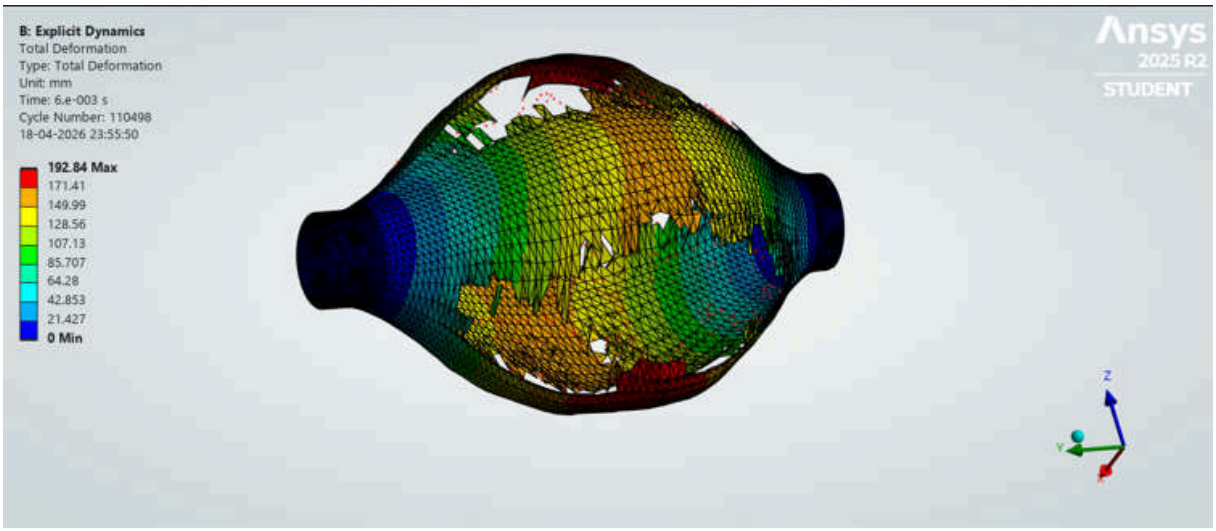


Figure A.17: Total Deformation Contour of the Motor Casing at Burst

The contour plot represents the explicit dynamic response of the casing in terms of total deformation under increasing internal pressure, culminating in a burst at approximately 9.34 MPa.
The deformation pattern shows a pronounced bulging at the central region, where the maximum deformation reaches about 192 mm, indicated by the red zones, while the ends remain relatively undeformed (blue), suggesting they are constrained or structurally stiffer. This behaviour is characteristic of a thin-walled pressure vessel where hoop stresses dominate and lead to maximum expansion at the midsection.

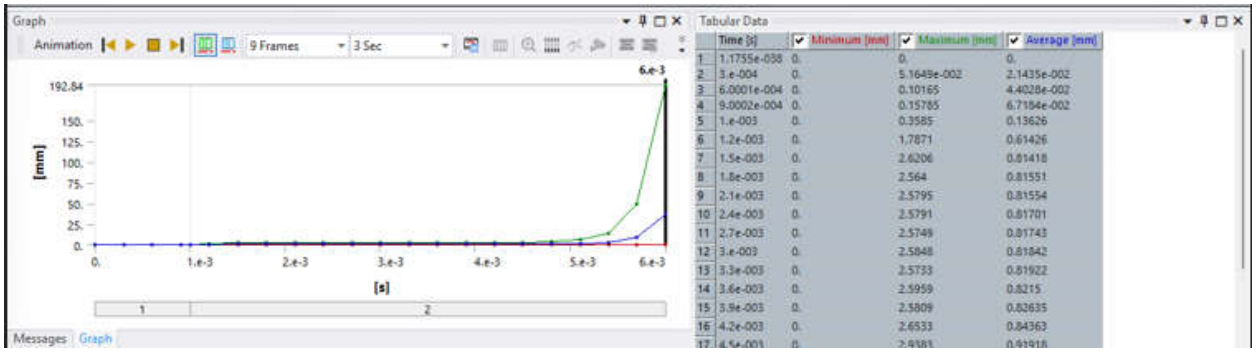


Figure A.18: Total Deformation Time-History Graph Showing Onset of Plastic Instability and Rapid Failure

The time-history graph further supports this, showing minimal deformation initially followed by a rapid, nonlinear increase near 6 ms, indicating the onset of plastic instability and sudden failure. The irregular deformation and localised distortions in the contour suggest ductile bursting, possibly accompanied by element distortion or deletion in the simulation.

Overall, the casing fails due to excessive plastic deformation and instability rather than brittle fracture, and the identified burst pressure of 9.34 MPa represents the structural limit under the given loading conditions

Propulsion Simulation III

3D FSI Nozzle Analysis

A one-way Fluid-Structure Interaction (FSI) simulation was conducted on the nozzle. The simulation was executed using ANSYS 2024 R1, employing ANSYS Fluent for the compressible computational fluid dynamics (CFD) analysis and ANSYS Mechanical in its Static Structural mode for the subsequent structural integrity assessment.

The one-way FSI methodology was selected as the coupling approach. In this method, fluid pressure loads computed by the CFD solver are mapped and transferred to the structural mesh as boundary conditions, but no feedback from the structural deformation is returned to the fluid domain.
This approach is physically justified when the structural displacements are sufficiently small relative to the nozzle dimensions so as not to meaningfully alter the flow field (the computation is simplified since the material for the nozzle is SS304 , which can be expected to undergo extremely low deformation on application of large load).

The workflow followed a strict sequential order:
First the fluid domain was meshed and solved to convergence, then the converged pressure field on the nozzle walls was exported and imported into the Mechanical module as an applied load, after which the static structural analysis was run to obtain deformation, stress, and strain fields throughout the nozzle body.

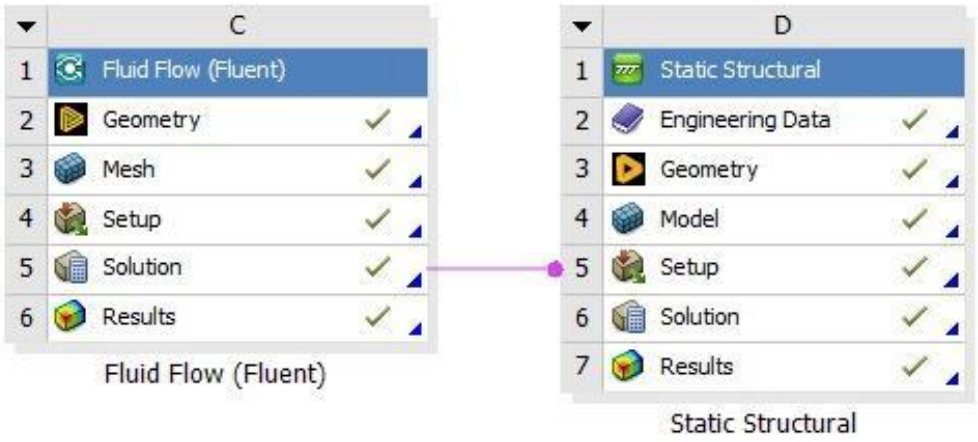


Figure A.19 - ANSYS Workbench FSI Project Schematic Showing One-Way Coupling Between Fluid Flow (Fluent) and Static Structural Systems

Geometry Description

The nozzle geometry is of axisymmetric convergent-divergent (CD) form. The CAD model is divided into two distinct bodies assembled together: the fluid domain (the internal gas passage through which combustion products flow) and the structural nozzle body (the physical hardware surrounding the flow passage).

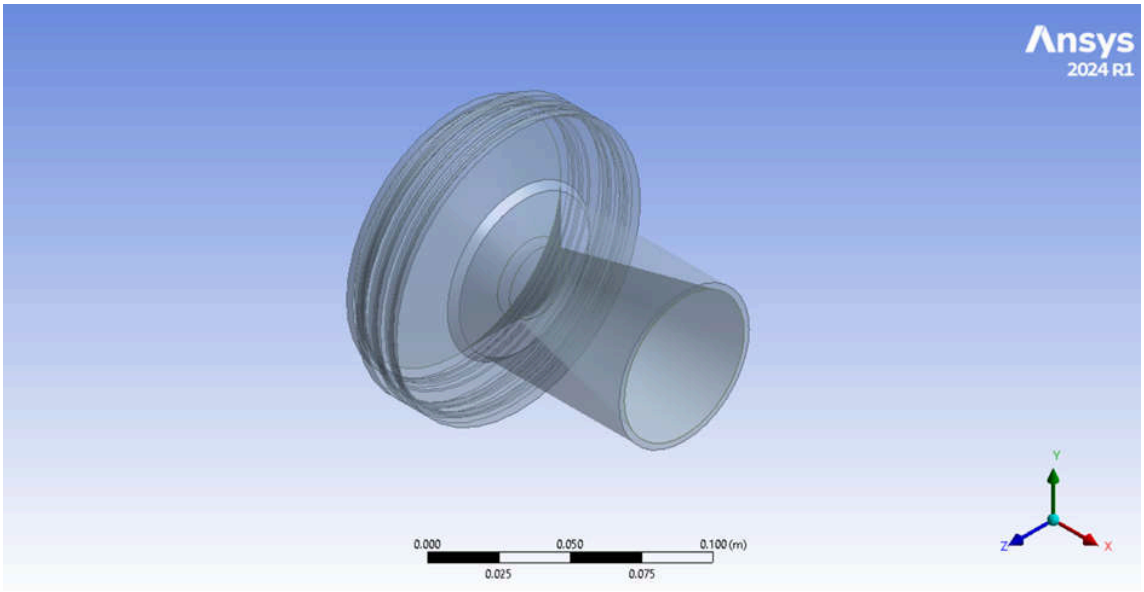


Figure A.20- 3D Nozzle Geometry Model in ANSYS Workbench

Meshing Strategy

Fluid Domain Mesh

The fluid domain mesh was generated using a structured hexahedral topology. This approach was intentionally selected over unstructured tetrahedral meshing because structured hexahedral meshes offer superior accuracy for compressible flow simulations, particularly in capturing boundary layers, shock waves, and steep gradients in supersonic regions. The mesh exhibits high orthogonality with cells aligned to the predominant flow direction, reducing numerical diffusion.

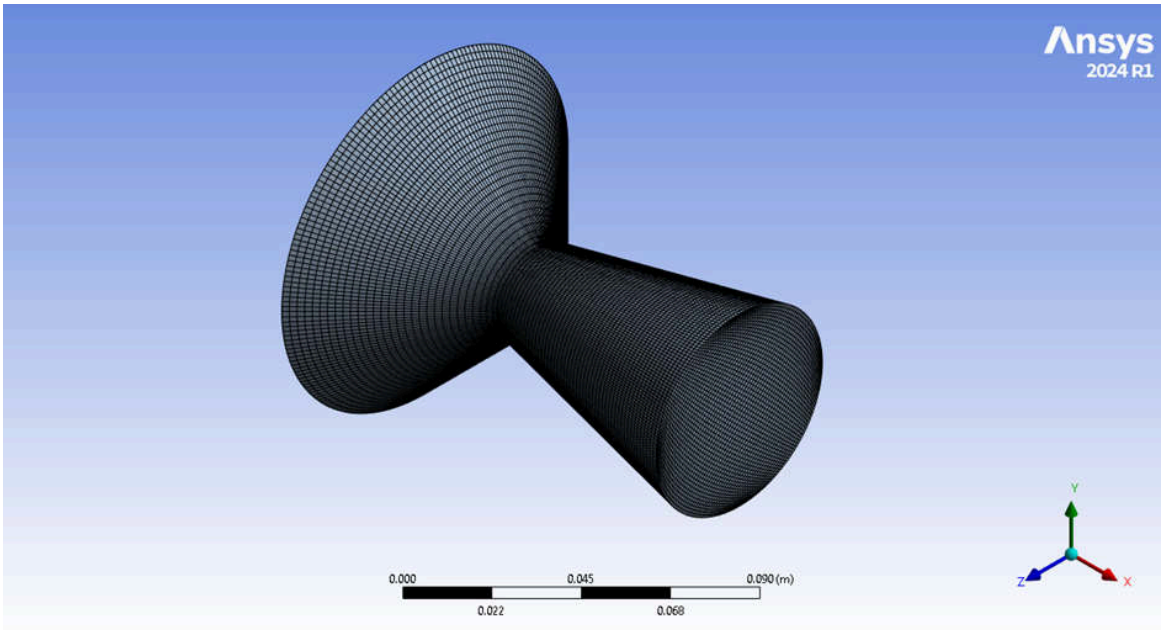


Figure A.21- Structured Hexahedral Fluid Domain Mesh of the Nozzle Assembly

Inspecting the cross-sectional cut views of the fluid mesh , the internal mesh shows a well-resolved O-grid topology in the circular cross-sections, transitioning smoothly through the convergent region into the throat and divergent section.

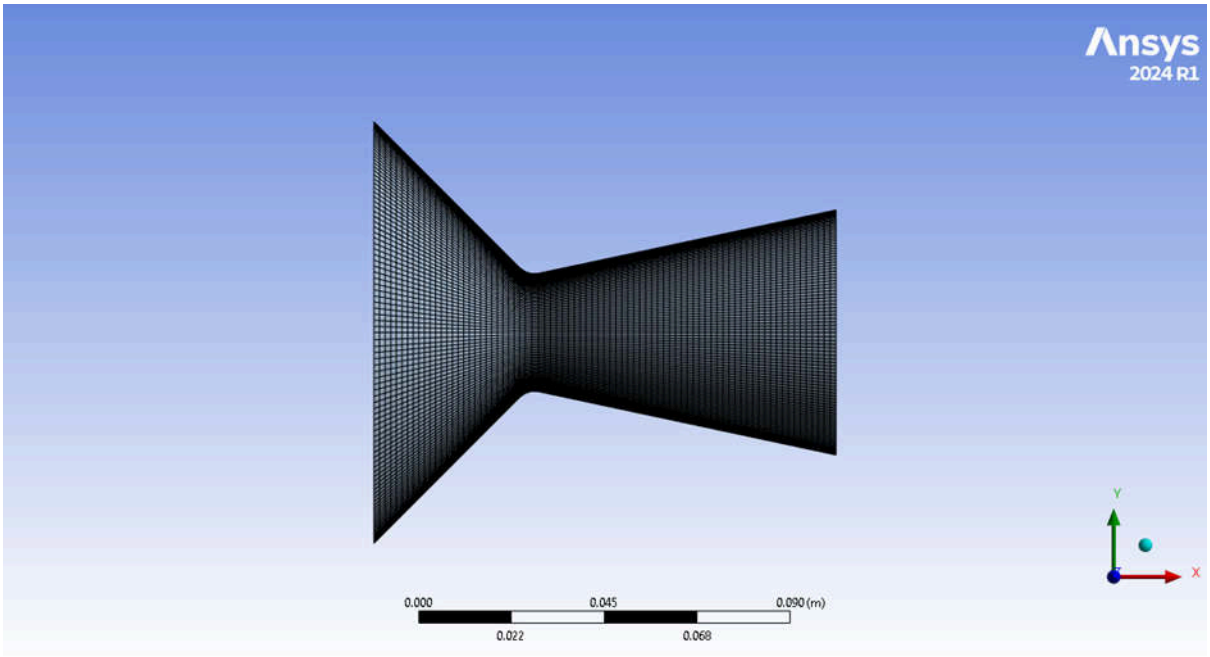


Figure A.22 - Sectional View of the Nozzle Fluid Domain Mesh Showing O-Grid Topology

The mesh density is visually higher near the throat region, which is appropriate since this is where the strongest velocity gradients, pressure gradients, and the sonic transition occur. The front face view of the fluid inlet confirms a concentric ring pattern of structured quadrilateral cells, consistent with an O-grid blocking strategy. The outlet face similarly shows a full-face structured quadrilateral mesh.

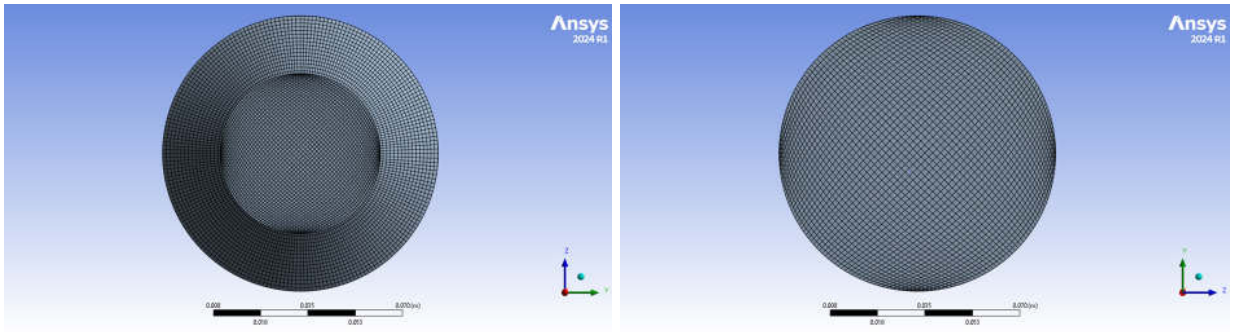


Figure A.23 - Inlet and Outlet Face Views of the Fluid Domain Mesh

The mesh quality in the fluid domain is judged to be high based on visual inspection. No obvious cell distortion or skewness is apparent. The consistent sizing through the axial direction and the smooth gradation from the convergent to divergent regions indicate that cell aspect ratios are well controlled. The structured hexahedral approach also ensures that the mesh is well suited for the k-omega SST turbulence model, which requires adequate near-wall resolution to resolve the turbulent boundary layer.

Structural Domain Mesh

The structural nozzle body mesh is also predominantly structured and hexahedral, consistent with best practices for structural FEA . The isometric view and the multiple face views confirm a highly regular, fine hexahedral mesh over the entire external and internal surfaces of the nozzle body.

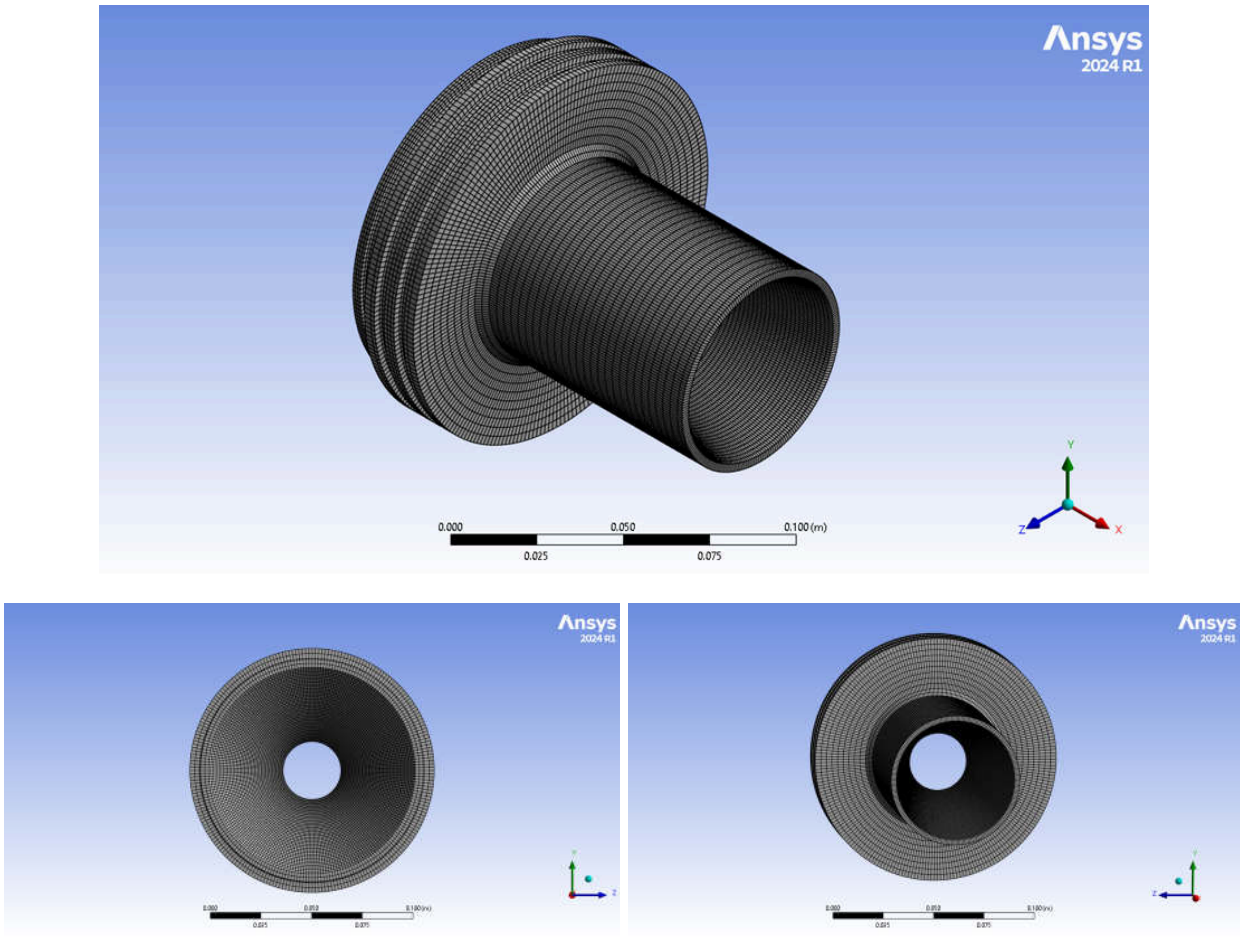


Figure A.24 - Structured Hexahedral Mesh of the Nozzle Structural Domain -Isometric, Inlet, and Outlet Face Views

The structural mesh extends over the full solid domain, including the inlet flange, the convergent outer wall, the throat region, and the divergent outer wall.

The structural mesh density is fine to ensure accurate resolution of the imported pressure loading from the fluid analysis, as the mapping of pressure from the fluid mesh to the structural mesh requires sufficient structural mesh resolution at all loaded surfaces. The overall quality of the structural mesh is high, with minimal visible distortion in any of the displayed views.

CFD Solver Configuration and Settings

Turbulence Model

The SST k-omega model is a two-equation eddy-viscosity model that blends the k-omega formulation near the walls (where it performs well in adverse pressure gradients) with the k-epsilon formulation in the free stream, making it one of the most reliable and widely used turbulence models for aerodynamic and nozzle flow simulations involving boundary layer separation and shock-boundary layer interactions.

The simulation produced Mach numbers reaching 3.10 at the nozzle exit and inlet pressures on the order of 3.5 MPa with inlet temperatures around 1600 K, the flow solver was configured for compressible, high-speed gas dynamics. The energy equation was activated ,enabling the solver to compute temperature variations due to compressible heating and cooling effects throughout the nozzle. The working fluid is KNSB exhaust gas from a solid propellant, modeled as an ideal gas.

Boundary Conditions

Based on the resulting flow field, the inlet total pressure was 3.5 MPa and the total temperature was 1600 K. At the outlet boundary, the simulation allowed the flow to exit supersonically into ambient conditions, consistent with a pressure outlet set to ambient back pressure.
The nozzle walls were treated as no-slip viscous walls. The mass flow rate at the inlet was computed to be 2.1699772 kg/s, and at the outlet -2.1699767 kg/s, yielding a net mass imbalance of only 4.77e-7 kg/s, which confirms near-perfect mass conservation across the fluid domain

Result Assessment

The residual convergence plot covers iterations from 5,000 to 10,000 and displays all seven solved transport equations. The continuity residual stabilised at approximately 1e-2, the x-velocity and z-velocity residuals converged to around 1e-3, and the y-velocity residual, the highest among the momentum equations , oscillated around 1e-2. The energy equation residual achieved the best convergence among all equations, stabilising at approximately 1e-6.
Both the turbulent kinetic energy (k) and specific dissipation rate (omega) residuals also converged to approximately 1e-6, indicating excellent turbulence model convergence.

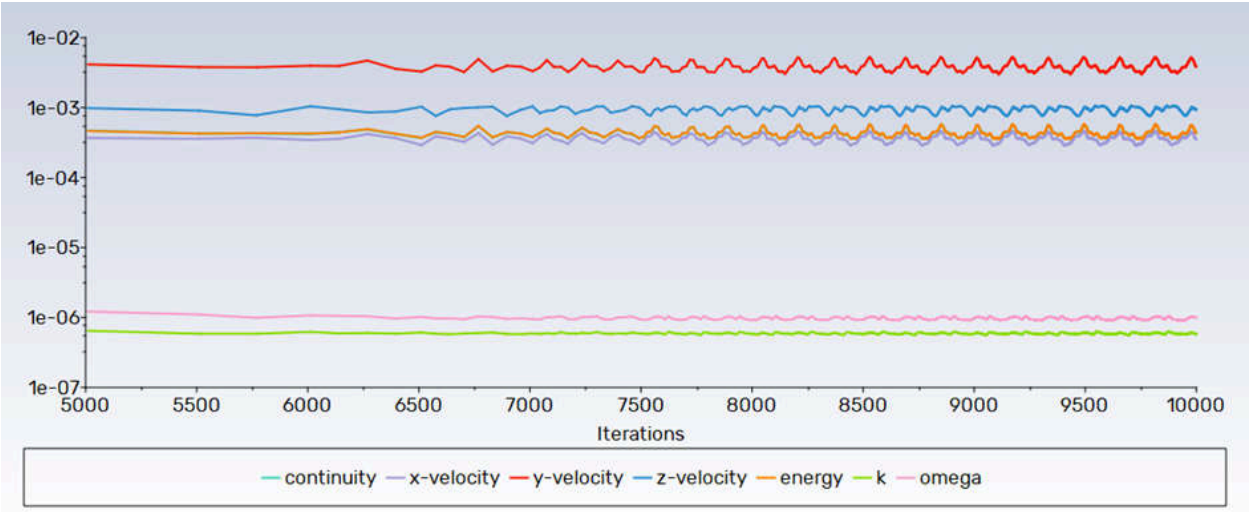


Figure A.25 - The residual convergence plot

The residual levels for the continuity and velocity equations, while not reaching the classical 1e-4 or 1e-6 thresholds, are characteristic of supersonic nozzle simulations with internal shock structures.
In such flows, the presence of shock waves, shock-boundary layer interactions, and supersonic expansion fans naturally produce oscillatory residuals that plateau at higher values than purely subsonic flow cases. The oscillatory behaviour visible in all residual curves, particularly between iterations 7,000 and 10,000, is consistent with periodic shock position oscillation within the divergent section

Mass Flow Rate Assessment

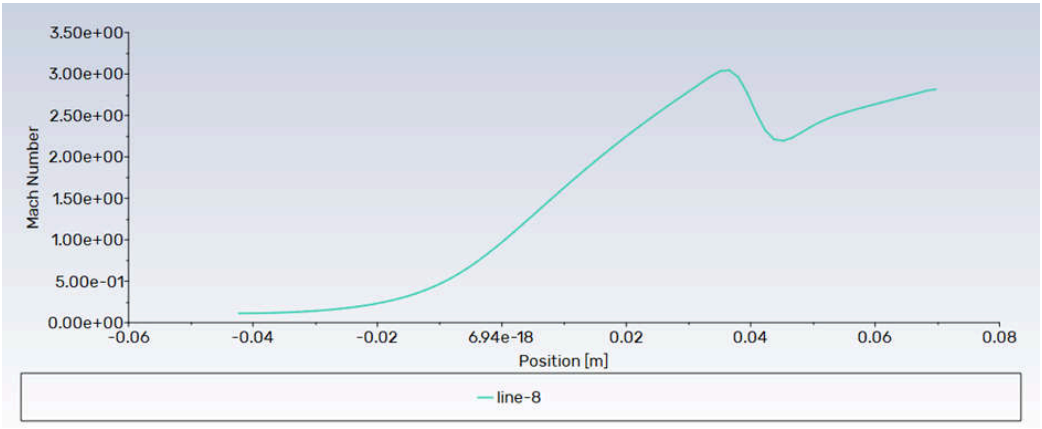


Figure A.26 - Mach Number Distribution Along the Nozzle Centreline

The mass flow rate convergence monitor tracks the net mass imbalance The monitor shows an initial transient peak around iterations 5,500 to 6,000, followed by a damped oscillation that eventually settles to a very small amplitude oscillation around zero. The final steady-state net mass flow imbalance was reported numerically as 4.7661119e-07 kg/s ,less than one millionth of the inlet mass flow rate of 2.17 kg/s ,confirming mass conservation and a physically meaningful converged solution. This mass balance error of approximately 0.00002% is well within acceptable engineering tolerance for compressible CFD simulations.

Mach Number Distribution

The Mach number contour plot provides a clear and physically consistent picture of the flow acceleration through the nozzle.In the convergent section, the flow approaches the throat at subsonic Mach numbers, remaining predominantly in the low-Mach regime indicated by blue and green colors. The flow transitions to exactly Mach 1 at the throat cross-section. In the divergent section, the flow expands supersonically, rapidly accelerating to the observed peak Mach number of approximately 3.10.

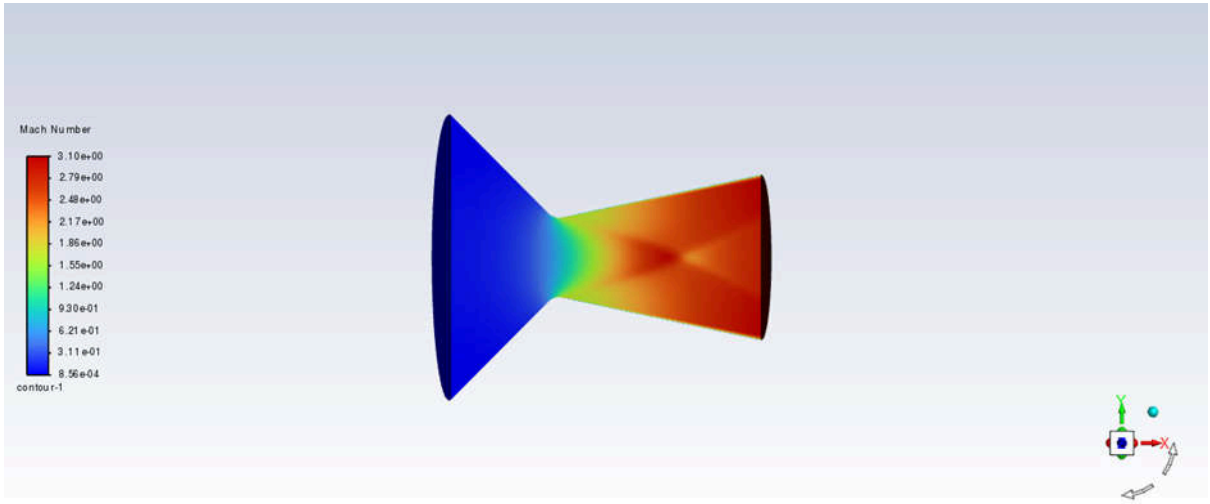


Figure A.27 - The Mach number contour plot

The Mach number centerline profile graph confirms this behavior quantitatively.. The curve begins near zero at the far left, rises gradually through the convergent section, transitions through unity at the throat and then accelerates steeply in the divergent section reaching a peak of approximately 3.0 to 3.1. A characteristic dip and recovery in the Mach profile is consistent with the presence of a normal shock or an oblique shock system within the divergent section. This shock structure causes a momentary deceleration of the supersonic flow followed by re-acceleration downstream.

Static Pressure Distribution

The static pressure contour shows the pressure field throughout the nozzle cross-section. The pressure scale ranges from 8.09e4 Pa at the nozzle exit to 3.50e6 Pa at the inlet. The convergent section carries the full chamber pressure, which drops sharply through the throat as the flow accelerates and converts pressure energy into kinetic energy. The divergent section shows continued pressure reduction as the gas expands supersonically, with very low pressures near the exit.

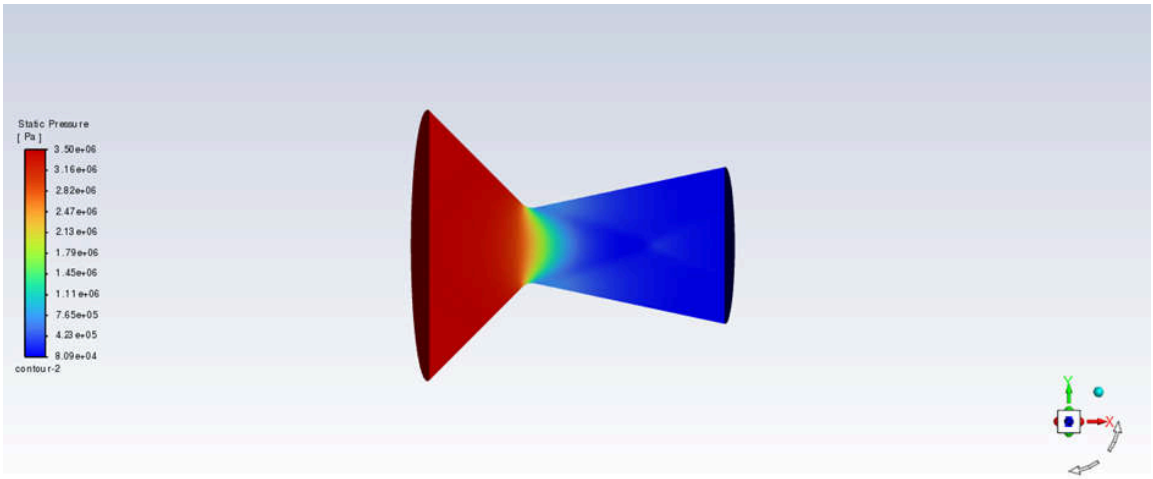


Figure A.28 - The static pressure contour

The static pressure centerline profile quantifies this pressure drop. The pressure starts at approximately 3.35 MPa at the inlet face and remains relatively flat through the convergent section before dropping sharply at and downstream of the throat. The pressure falls to near-ambient levels within a short distance of the throat exit plane.

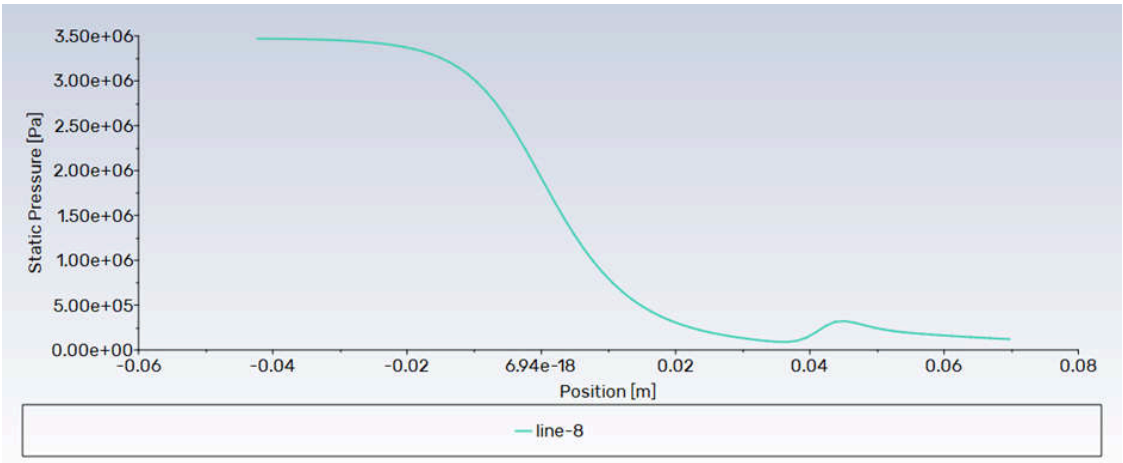


Figure A.29 - The static pressure centerline profile

Static Temperature Distribution

The static temperature contour displays the temperature field on the nozzle cross-section. The color scale ranges from 5.47e2 K (547 K) at the exit to 1.84e3 K (1840 K, approximately 1567 degrees C, deep red) at the inlet.The high inlet temperature of approximately 1840 K is consistent with the combustion temperatures achieved by solid composite propellants.

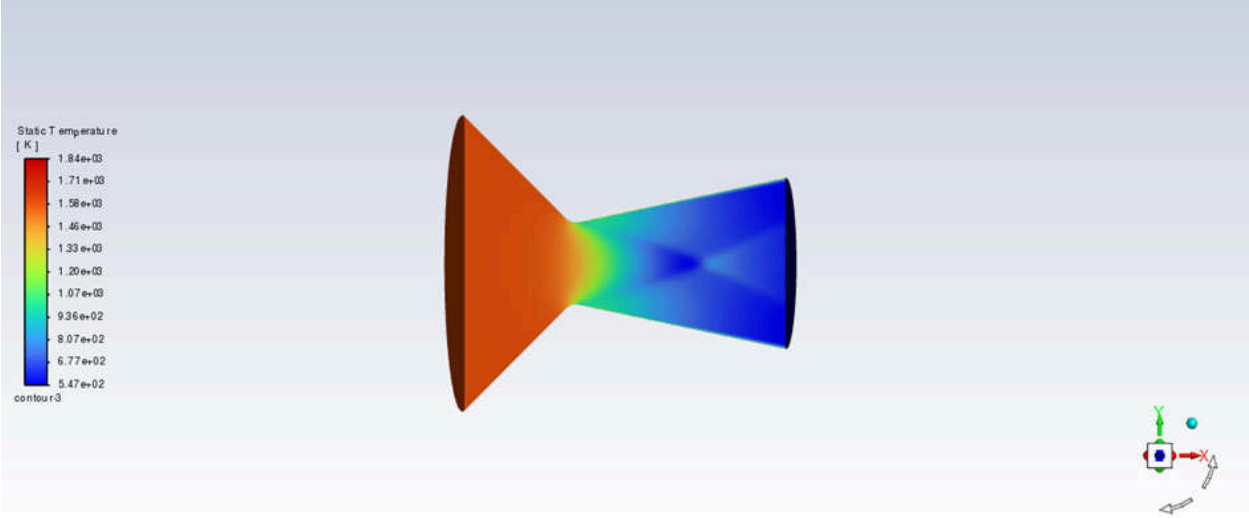


Figure A.30 - The static temperature contour

The temperature centerline profile shows the temperature starting at approximately 1600 K at the inlet, decreasing through the convergent section, reaching a minimum of approximately 580 K just downstream of the throat, then recovering slightly to approximately 800 K around the shock position before decreasing again to a final exit temperature of approximately 600 K. The temperature rise at the shock location is physically correct a normal shock converts kinetic energy back into thermal energy, causing a local temperature increase. These temperature levels confirm that the structural analysis must account for severe thermal loading in the material selection and that the imported CFD pressures are derived from a thermodynamically realistic compressible flow solution.

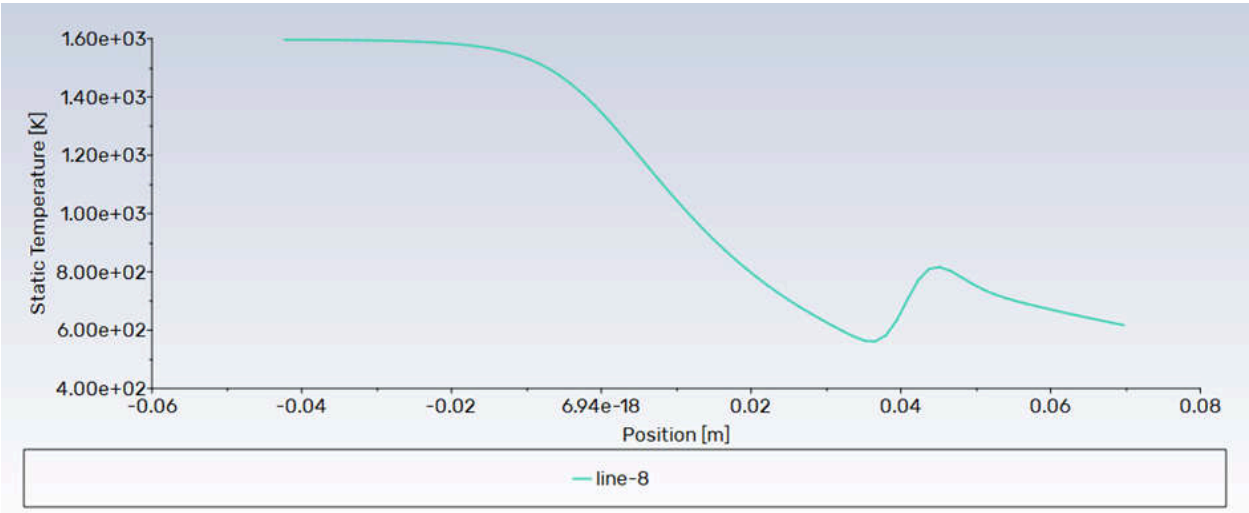


Figure A.31 - The temperature centerline profile graph

Structural Analysis Setup and Boundary Conditions

Pressure Import and Mapping

The one-way coupling between the CFD and structural analyses was accomplished by exporting the converged pressure field from Fluent and importing it into the ANSYS Mechanical Static Structural module. The pressure mapping was executed across three separate surface zones of the nozzle interior, corresponding to the convergent inner wall, the throat inner wall, and the divergent inner wall. This can be seen by the three distinct Imported Pressure labels visible in the images

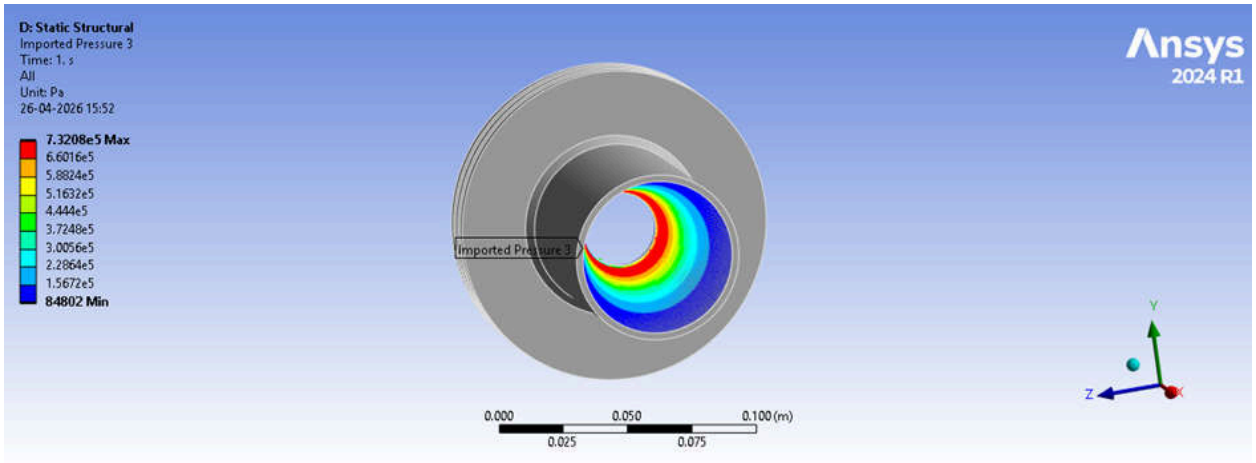


Figure A.32 - Imported Pressure Load Mapped onto the Divergent Section

The first imported pressure load on Figure A.32 corresponds to the nozzle throat to divergent region. The pressure range mapped onto this surface spans from a minimum of 8.48e2 Pa to a maximum of 7.32e5 Pa (732 kPa). The rainbow color distribution on this surface area shows significant spatial variation, as expected in the throat region where the flow transitions from subsonic to supersonic and pressure gradients are steepest.

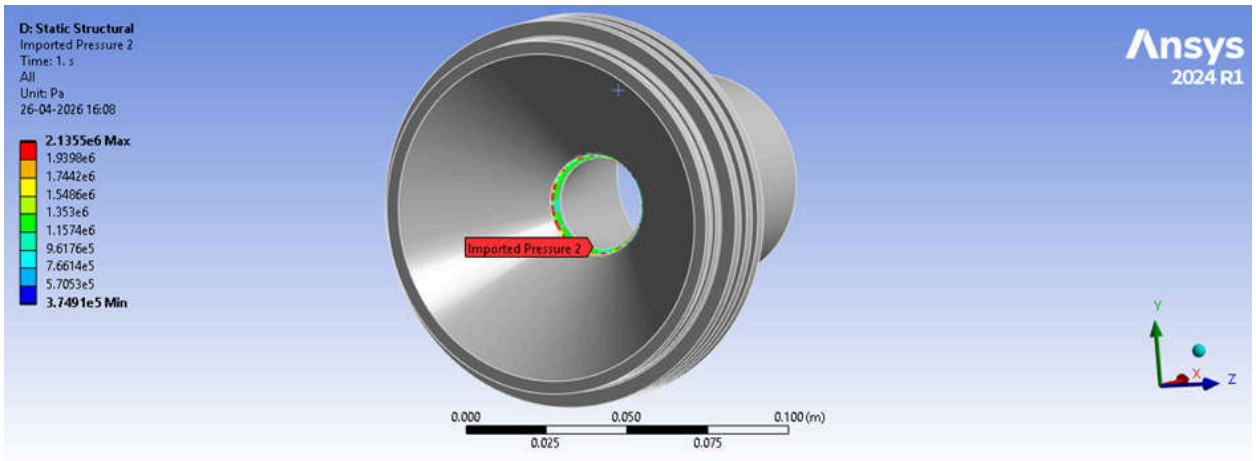


Figure A.33 - Imported Pressure Load Mapped on the throat

The Figure A.33 covers the throat region . This load ranges from a minimum of 3.749e5 Pa (374.9 kPa) to a maximum of 2.1355e6 Pa .The broad range reflects the significant pressure drop from the across the throat.

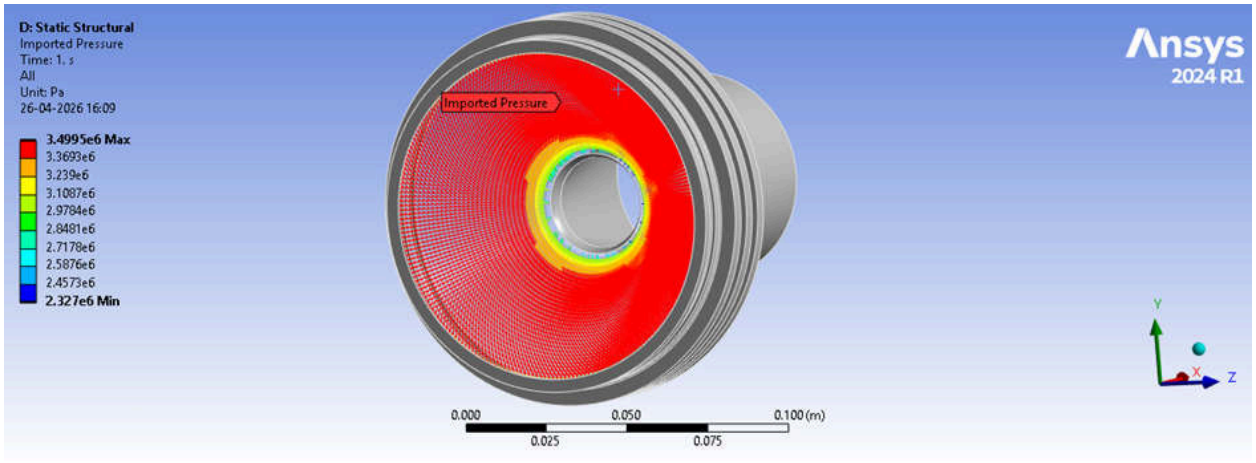


Figure A.34 - Imported Pressure Load Mapped onto the convergent region

The Figure A.34 covers the convergent face of the nozzle. This shows a pressure range from 2.327e6 Pa to 3.4995e6 Pa, predominantly in red, indicating near-uniform high pressure across this surface. This is consistent with the nearly flat pressure profile observed in the CFD results within the subsonic convergent region near the inlet, where the pressure has not yet begun its steep drop. The structural model therefore experienced realistic spatially-varying pressure loads faithful to the CFD solution on all three internal surface zones.

Structural Boundary Conditions

The Static Structural analysis required mechanical boundary conditions in addition to the imported pressure loads. The inlet flange face was constrained with a fixed support .This prevents rigid body motion and provides the reaction ground for the pressure loads to generate internal stresses. The nozzle outer surfaces not in contact with the flange were left as free surfaces.

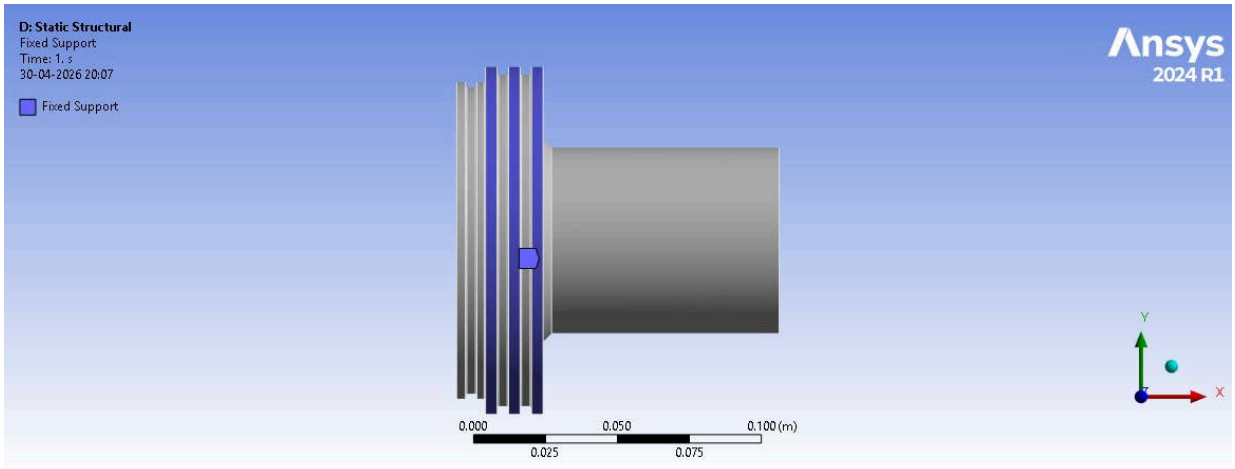


Figure A.35 - Applied static boundary retention

Structural Analysis Results

Total Deformation

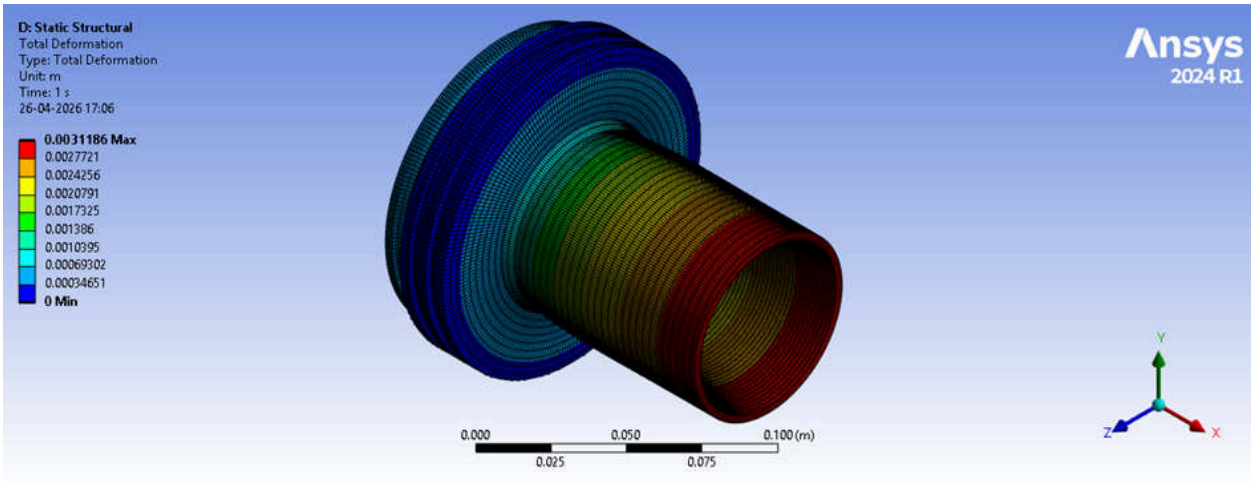


Figure A.36 - Total Deformation Contour of the Nozzle Structure Under Imported Pressure Loading

The total deformation results are displayed from two viewing angles. The maximum total deformation in the nozzle structure was computed to be 3.1186 mm (0.0031186 m), while the minimum deformation is 0 mm at the constrained flange face. The color map transitions from deep blue (zero deformation at the fixed flange face) through green, yellow, and orange to deep red (maximum deformation).

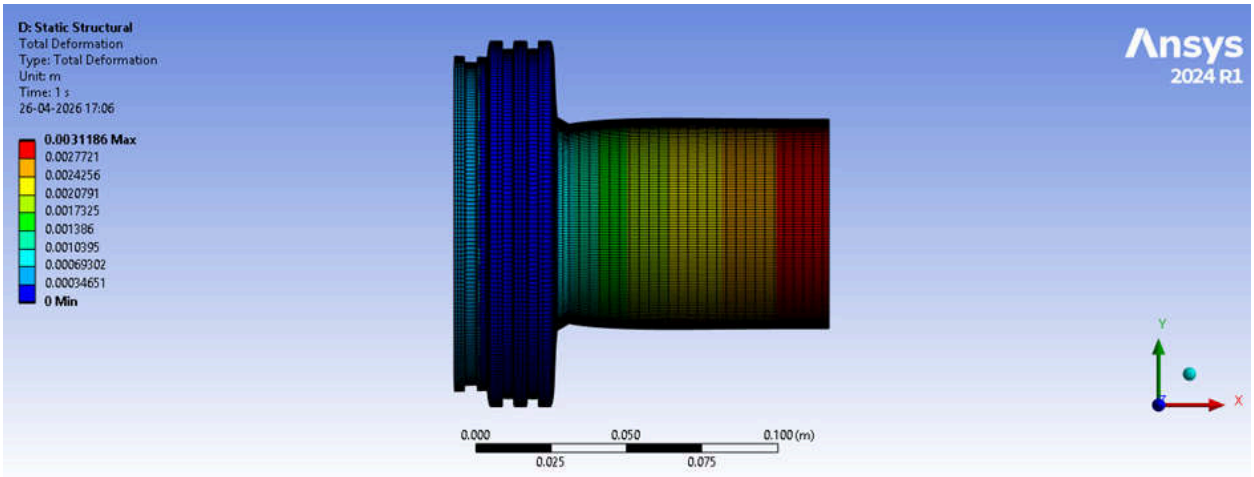


Figure A.37 - Side view of the Total Deformation Contour of the Nozzle Structure

The deformation field reveals a physically consistent pattern: the regions with maximum displacement are the divergent exit end of the nozzle tube, which is the farthest point from the constrained flange and experiences significant pressure-induced bending moments, and portions of the outer nozzle barrel.
The inlet flange, being the constrained boundary, shows effectively zero deformation, while the free end of the divergent section exhibits the largest radial and axial displacements.

Equivalent (von-Mises) Stress

The equivalent von-Mises stress distribution is presented from two viewing angles. The maximum von-Mises stress was 2.0543e10 Pa (20.54 GPa) and the minimum was 6.1076e6 Pa (6.1 MPa). The stress field is dominated by a concentrated high-stress band located at the transition zone between the large-diameter inlet flange and the smaller-diameter nozzle barrel. This is the geometric stress concentration region where the abrupt change in cross-sectional area creates a bending stress riser under the combined pressure and structural loads.

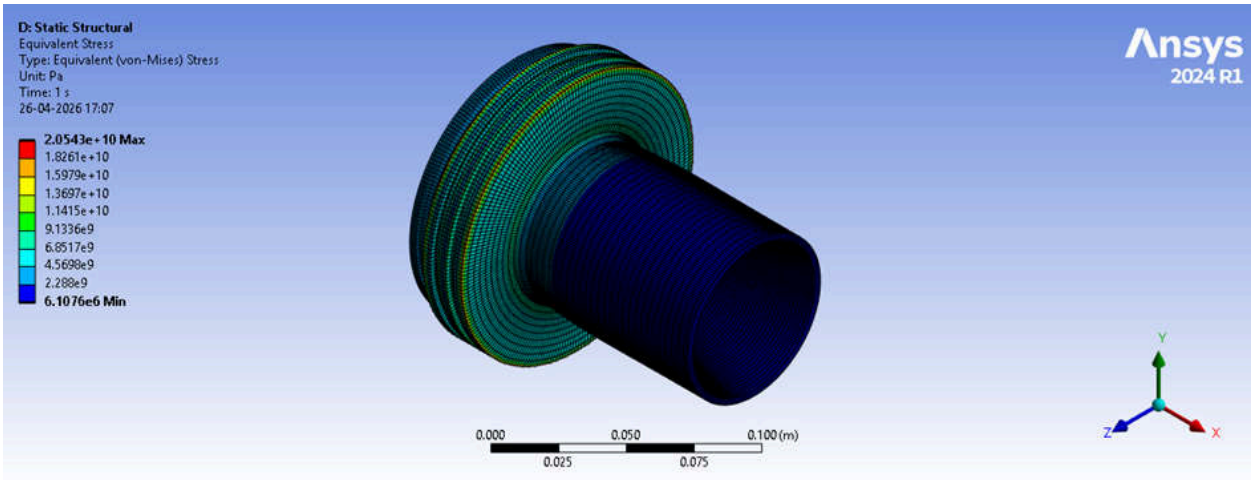


Figure A.38 - Equivalent Von-Mises Stress Distribution

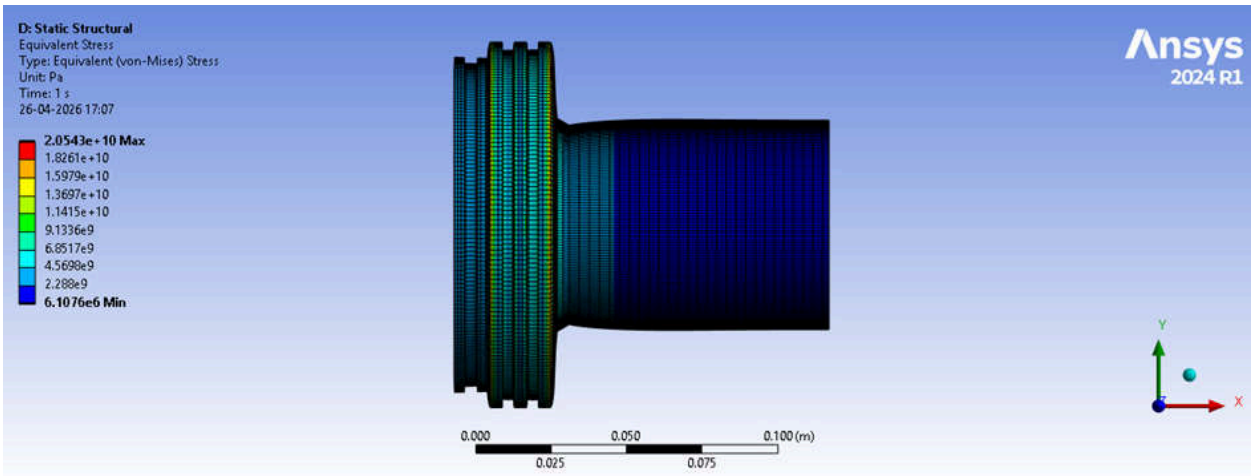


Figure A.39 - Side View of Equivalent Von-Mises Stress Distribution

The remainder of the nozzle structure ,the barrel of the convergent section, the throat, and the divergent section , is predominantly in the low-to-moderate stress regime. The flange outer disk also shows low stress relative to the transition zone.The peak stress is highly localized and may be a consequence of mesh singularity at a sharp geometric corner. The region however is also supported by a retaining ring which if the scenario is apparent will off load the stresses developed.

Interpretation

CFD solution convergence, pressure data transfer, and structural assessment. The CFD results are physically self-consistent: the nozzle chokes at the throat (Mach 1), expands supersonically to Mach 3.1 in the divergent section, pressure drops from 3.5 MPa at inlet to near-ambient at exit, and temperature decreases from 1840 K to approximately 580 K at the throat minimum. The presence of a shock structure in the divergent section, evidenced by the local dip in the Mach profile and the corresponding temperature and pressure recovery around the same axial position, indicates that the nozzle may be operating in a slightly off-design condition.

The structural results indicate that the bulk of the nozzle body experiences moderate deformation (maximum 3.12 mm at the free end) and low-to-moderate stress in the primary structural regions. The critical design concern identified is the stress concentration at the flange-to-barrel junction. This region is inherently a high-stress zone due to the bending moment created by the pressure differential across the nozzle and the geometric discontinuity. Design mitigation strategies included material reinforcement in this area through the retaining ring, a slight taper in wall thickness from the flange to the barrel to reduce the bending stress rise

Propulsion Analysis IV

Bolt Analysis

The axial ejection for the nozzle load path as per design is going to be in the following sequence:-

P(t) acts on the nozzle aft face

- Nozzle flat face(below) pushes axially on the SS304 retainer ring top face
- The retainer ring tries to eject axially forward
- Bolts resist this in direct tension (nut is present on the inside face)
- Bolt shank bears on the Al tube wall (3 mm bearing length)
- Load transferred to motor tube and airframe

Table A.7: Associated stress value of the motor components.

Component	Material	Property	Symbol	Value (MPa)
Motor Tube	Al 6061-T6	Yield Strength	σ_y	276
		Ultimate Tensile Strength	σ_u	310
		Bearing Allowable	$\sigma_{bear,\ allow}$	414
Retainer Ring	SS 304	Yield Strength (min)	σ_y	170
		Ultimate Tensile Strength	σ_u	485
		Localised Bearing Allowable	$\sigma_{bear,\ allow}$	255
		Bending Allowable	S_{bend}	113
Bolts	A4-80 SS	Proof / Yield Strength	σ_y	600
		Ultimate Tensile Strength	σ_u	800
		Bolt Tensile Allowable	$S_{bolt,\ allow}$	420

Load Calculations

Peak ejection force

The ejection force is the net axial force on the nozzle aft face at peak chamber pressure, taken over the full bore area.

$$\begin{aligned} F_{\text{eject}} &= P \cdot \frac{\pi}{4} \cdot D_{\text{bore}}^2 \\ &= 3.5 \times 10^6 \cdot \frac{\pi}{4} \cdot (0.121)^2 \\ &= 3.5 \times 10^6 \cdot 0.01150 \\ &= 40{,}247 \text{ N} = 40.25 \text{ kN} \end{aligned}$$

Force per bolt at peak static load

The eight bolts share the ejection force equally.

$$\begin{aligned} F_{\text{bolt}} &= \frac{F_{\text{eject}}}{n} \\ &= \frac{40{,}247}{8} \\ &= 5{,}031 \text{ N} = 5.03 \text{ kN} \end{aligned}$$

Bolt tensile stress

Each bolt carries its share of the peak static ejection force in direct tension.

$$\sigma_{\text{bear,tube}} = \frac{F_{\text{bolt}}}{d_{\text{bolt}} \cdot t_{\text{wall}}} = 279.5 \text{ MPa}$$

$$\sigma_{\text{bear,allow}} = 1.5 \times 276 = 414 \text{ MPa}$$

$$\text{FoS} = \frac{414}{279.5} = 1.48 \quad \Rightarrow \text{PASS}$$

Bearing stress on the Al tube wall

The bolt shank bears on the 3 mm Al 6061-T6 tube wall at each hole. Bearing length equals the tube wall thickness.

$$\sigma_{\text{bear,tube}} = \frac{F_{\text{bolt}}}{d_{\text{bolt}} \cdot t_{\text{wall}}} = 279.5 \text{ MPa}$$

$$\sigma_{\text{bear,allow}} = 1.5 \times 276 = 414 \text{ MPa}$$

$$\text{FoS} = \frac{414}{279.5} = 1.48 \quad \Rightarrow \text{PASS}$$

Bearing stress on the SS316 retainer ring

The bolt shank bears on the SS316 retainer ring over the full 20 mm ring thickness.

$$\sigma_{\text{bear,ring}} = \frac{F_{\text{bolt}}}{d_{\text{bolt}} \cdot t_{\text{ring}}} 41.9 \text{ MPa}$$

$$\sigma_{\text{bear,allow}} = 1.5 \times 170 = 255 \text{ MPa}$$

$$\text{FoS} = \frac{255}{41.9} = 6.08 \quad \Rightarrow \text{PASS}$$

Tube wall tearout

Edge distance e = 18 mm axially from bolt centreline to tube aft end.

$$\sigma_{\text{tear-out}} = \frac{F_{\text{bolt}}}{2 \cdot e \cdot t_{\text{wall}}} = 46.58 \text{ MPa}$$

$$\sigma_y = 276 \text{ MPa}$$

$$\text{FoS} = \frac{276}{46.85} = 5.89 \quad \Rightarrow \text{PASS}$$

Propulsion V

Grain Production Process

The casting mould for the KNSB BATES grain was developed as a two-part 3D printed assembly to enable ease of fabrication, alignment, and demolding, while maintaining dimensional accuracy. The two halves are joined to form a cylindrical cavity, within which a kraft paper tube of 1 mm thickness is inserted to act as the primary casting interface for the KNSB slurry. This paper liner not only provides a smooth internal surface finish but also serves as a sacrificial barrier that prevents direct adhesion of the propellant to the mould, thereby simplifying post-casting

extraction.

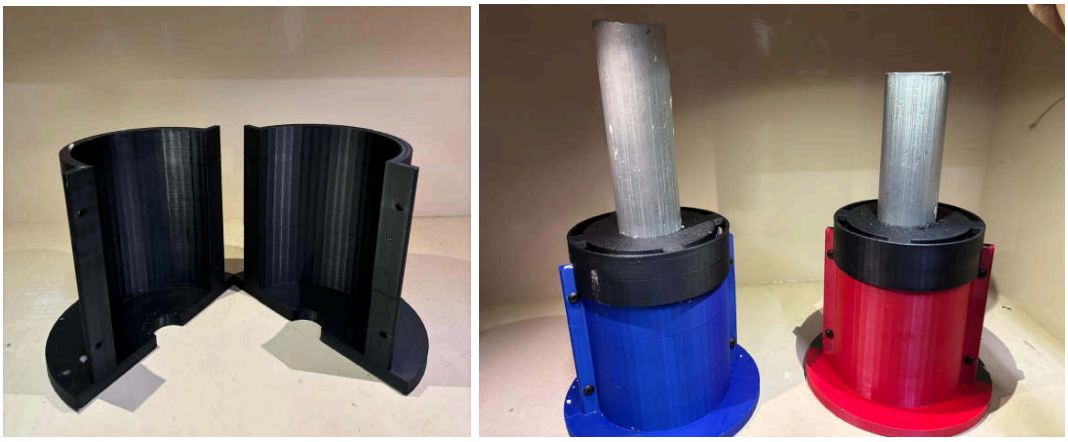


Figure A.40 - Split Mould Halves (Left) and Assembled Mould with Central core (Right)

The grain diameter is 110 mm; accordingly, each half of the mould is designed with a radius of 55.5 mm, resulting in an assembled internal diameter of 111 mm. This slight oversizing accounts for the Kraft paper thickness and ensures proper interfacing with the fibreglass liner during subsequent integration.

The grain length is maintained at 120 mm, consistent with the BATES geometry requirements for achieving a near-neutral burn profile.

Each grain has an approximate mass of 1.6 kg, consisting of about 1.1 kg of potassium nitrate and 0.55 kg of sorbitol. The potassium nitrate is first finely ground to achieve a uniform particle size distribution, which is essential for consistent combustion and efficient packing within the binder matrix. It is then preheated to approximately 40°C in a pan to remove residual moisture and reduce thermal gradients during subsequent mixing.



Figure A.41: Propellant Mixing Operation

Sorbitol is added to the preheated oxidiser and mixed thoroughly to initiate partial melting and coating of particles. The mixture is gradually brought to around 90°C, at which point the Sodium Lauryl Sulfate is introduced. Continuous heating and stirring are maintained to ensure homogeneous mixing, prevent localised overheating, and

avoid caramelisation or decomposition of sorbitol. The temperature is gradually increased to approximately 120°C, at which stage the mixture transitions into a fully molten, viscous slurry suitable for casting.

The mould assembly is prepared in parallel to ensure efficient handling of the molten propellant. A 5 mm thick aluminium base plate is friction-fitted at the bottom of the mould, providing structural rigidity and a flat reference surface. The kraft paper tube is then inserted concentrically within the mould cavity.
An aluminium core of 50 mm diameter is positioned centrally using a precisely machined hole in the base plate, ensuring accurate alignment and uniform port geometry throughout the grain length. The use of aluminium for both the base plate and core is intentional, as it provides good thermal conductivity, dimensional stability, and ease of removal after casting due to its low adhesion with the propellant.

Once the propellant reaches the desired temperature and consistency, it is carefully poured into the mould in a controlled manner to avoid air entrapment.



Figure A.42: Propellant Casting Operation

Following casting, an upper cap consisting of a 3D printed ABS component integrated with an aluminium interface plate is pressed into the mould. This step serves multiple purposes: it applies slight compression to the molten propellant, helps expel trapped air, ensures proper filling of the mould extremities, and promotes a more uniform density distribution. The assembly is then allowed to cool and solidify under controlled conditions to minimise thermal stresses, shrinkage-induced defects, and internal cracking.

After complete solidification, the mould is disassembled, and the grain is extracted. Since the propellant interfaces only with aluminium and the kraft paper liner, demolding is facilitated through light mechanical tapping without causing structural damage to the grain. The kraft paper liner remains adhered to the propellant surface, providing additional mechanical protection and aiding in handling and integration. This entire casting methodology ensures repeatability, dimensional accuracy, and structural reliability of the BATES grain, which are essential for predictable combustion behaviour and safe motor operation.

Propulsion VI

Propulsion Testing Setup

The setup for SRAD motor testing was developed by the team as a vertical thrust stand. The primary frame is constructed from 20 × 20 mm T-slot aluminium extrusion, forming a lightweight but stiff rectangular tower. This extrusion system allows precise alignment of components and easy adjustment of mounting points, which is particularly useful when accommodating different motor geometries or test configurations.

At the motor interface, the propulsion unit is secured using U-ring clamps, which provide circumferential constraint while allowing quick installation and removal. These rings ensure that the motor remains aligned along the thrust axis, minimising off-axis loads that could otherwise corrupt measurement accuracy or induce bending in the frame.



Figure A.43: Static Fire Test Setup

The thrust is transmitted downward through the structure to a 1 Ton cylindrical Load cell, which acts as the primary data acquisition system.

Data Acquisition System

Sarabhai is an in-house developed data acquisition PCB built around a custom ESP32-S3 design, integrating all sensor interfacing and signal conditioning onto a single board purpose-built for SRAD N-class motor static fire testing. The PCB directly accommodates an HX711 24-bit ADC for high-resolution load cell measurements, dedicated analog input headers for pressure transducer connectivity, and thermocouple interface circuitry enabling

simultaneous acquisition of thrust force, chamber and pressure.
The ESP32-S3 sits at the heart of the design, its dual-core Xtensa LX7 processor managing concurrent sensor polling across all three measurement channels while the onboard Wi-Fi stack streams timestamped telemetry to a ground station in real time, with local flash logging running in parallel as a redundancy measure.
Routing a 24-bit ADC, analog pressure inputs, and thermocouple lines on the same PCB demands careful layout discipline analog and digital ground planes must be separated, decoupling capacitors placed close to the HX711 supply pins, and thermocouple traces kept away from high-frequency switching signals to avoid injecting noise into the microvolt-level temperature signals all of which make Sarabhai a non-trivial custom hardware design rather than a simple breakout-board assembly. transmission and logging during test operations.

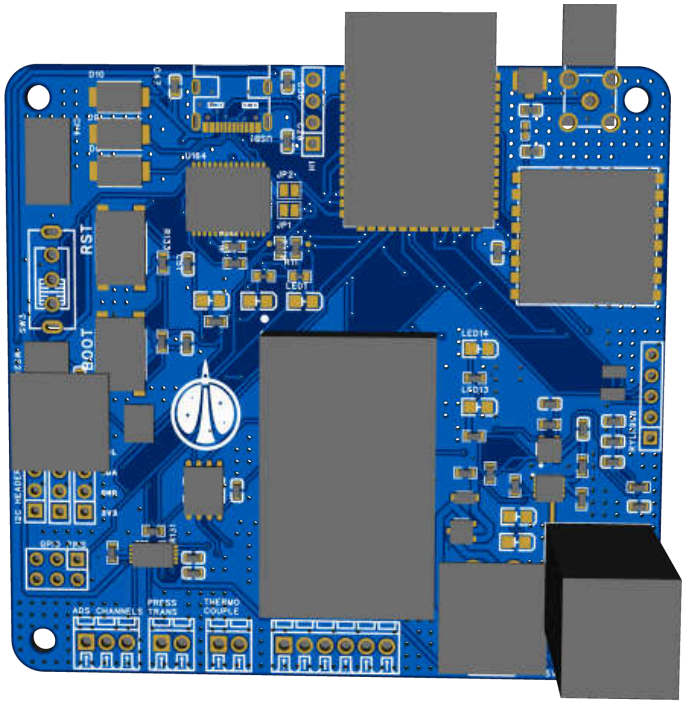


Figure A.44 - PCB Rendered Image

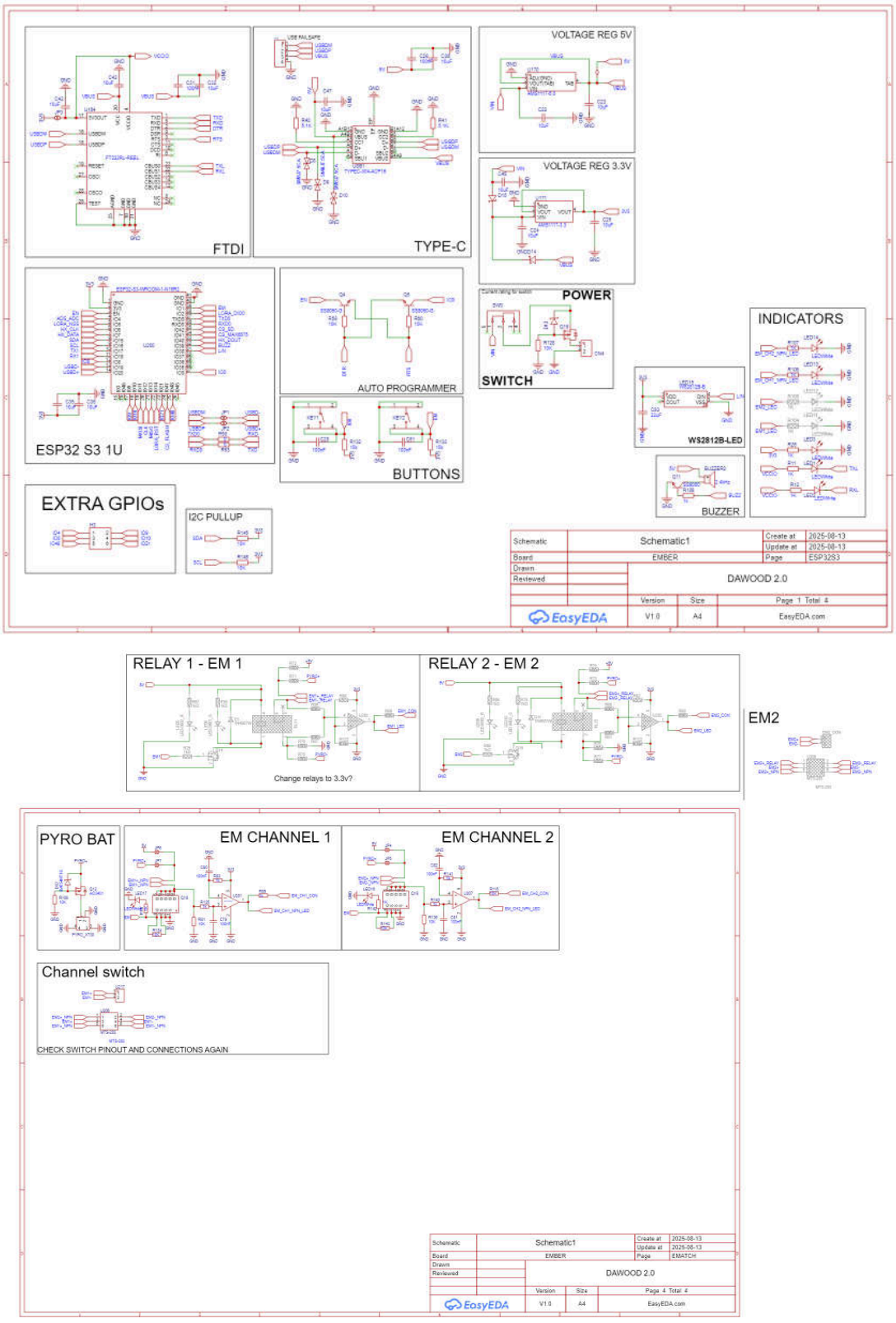
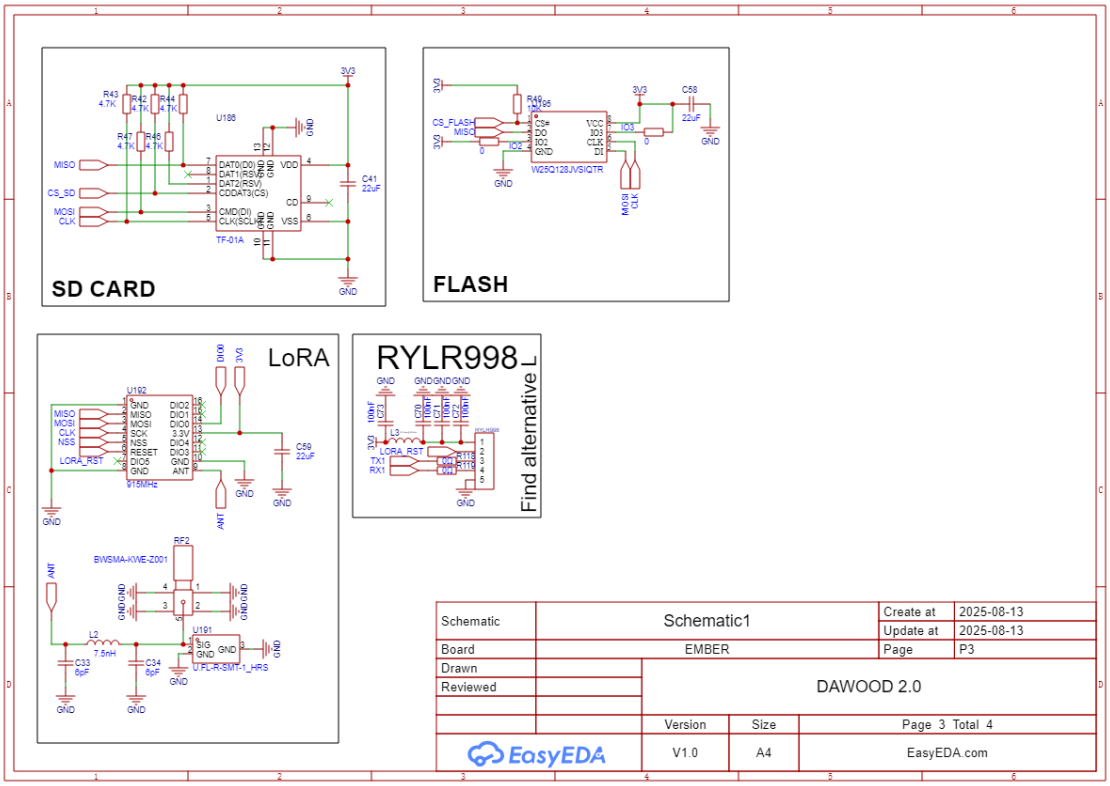


Figure A.45 - PCB Schematics



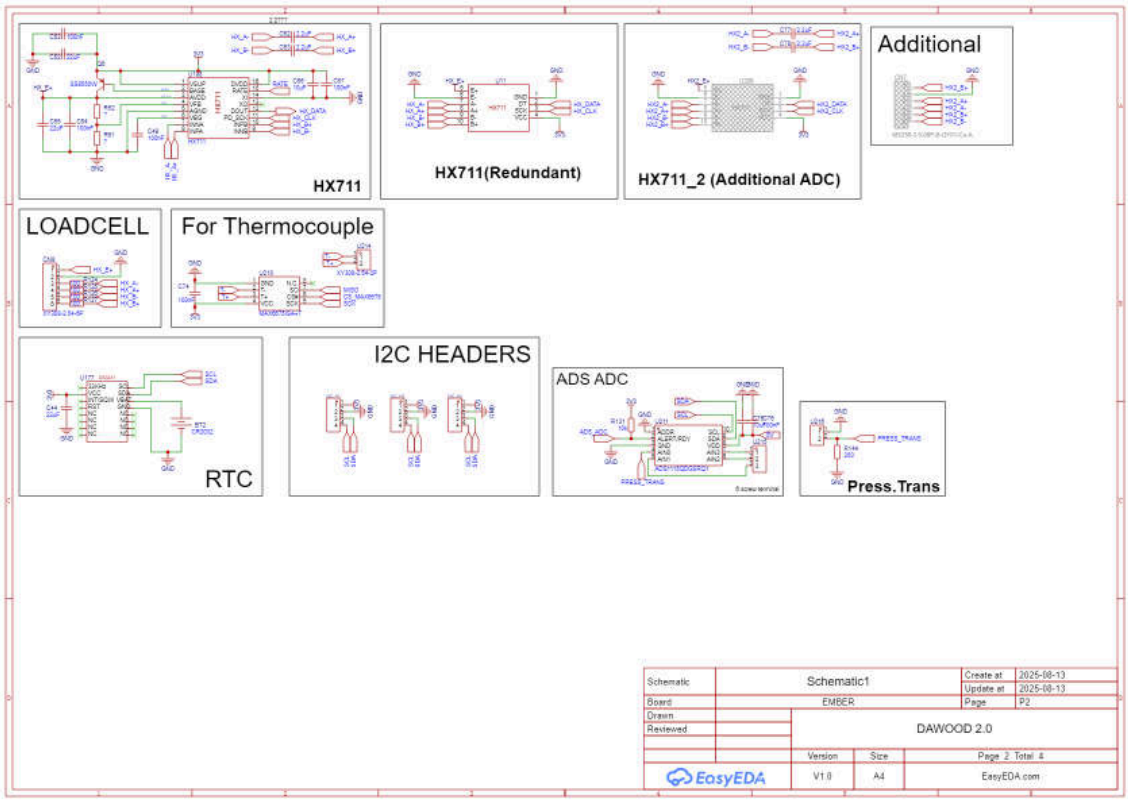


Figure A.46 - PCB Schematics

5. Nose Cone Simulation

CFD studies were performed for multiple nose cone profiles to evaluate aerodynamic performance across transonic and supersonic regimes.

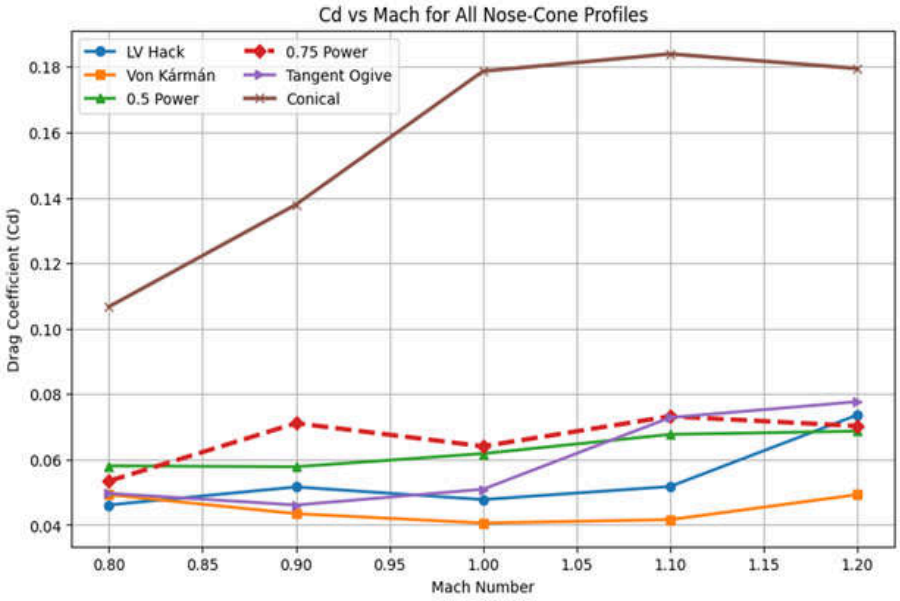


Figure A.47: Drag Coefficient vs Mach for All Nose Cone Profiles

From the above plot, the Von Kármán profile demonstrated the lowest drag coefficients across all simulated Mach numbers, with no evidence of flow separation or detached shock wave formation, unlike other profiles, such as conical and power series. The Von Kármán profile was selected for further fineness ratio optimization.

The Von Karman profile demonstrated the lowest drag coefficients across all simulated Mach Numbers, and hence a varying range of fineness ratio values from 4:1 to 6:1 was tested across a Mach Number range of 0.7-1.3 Mach. There were two major metrics used for finalizing a fineness ratio for the nose cones- CD/Weight v/s Mach Number and CD v/s Mach Number.

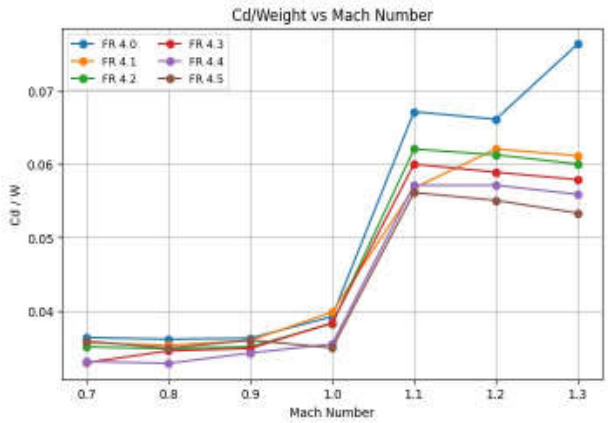


Figure A.48: Drag Coefficient/Weight vs Mach for FR 4:1 to 4.5:1

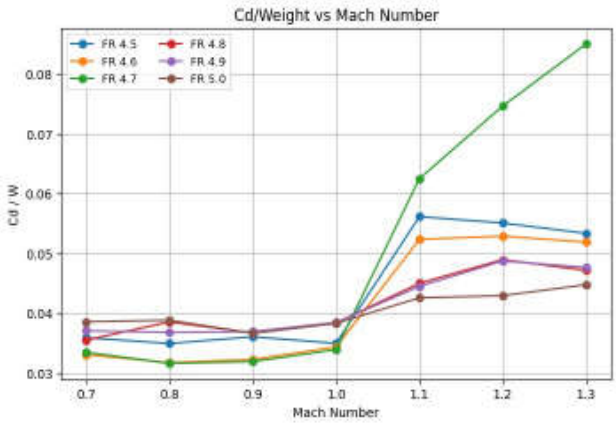


Figure A.49: Drag Coefficient/Weight vs Mach for FR 4.5:1 to 5:1

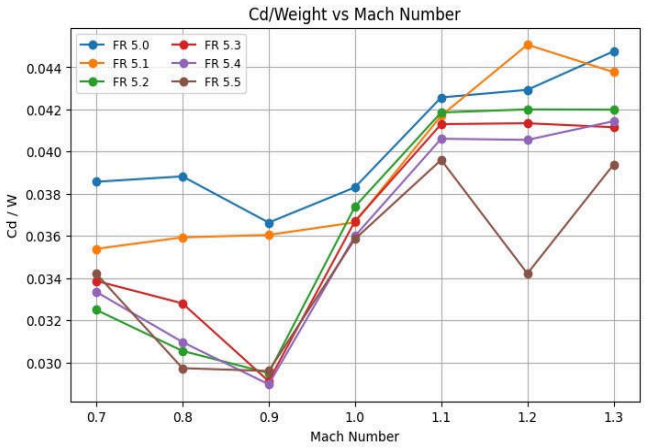


Figure A.50: Drag Coefficient/Weight vs Mach For FR 5:1 to 5.5:1

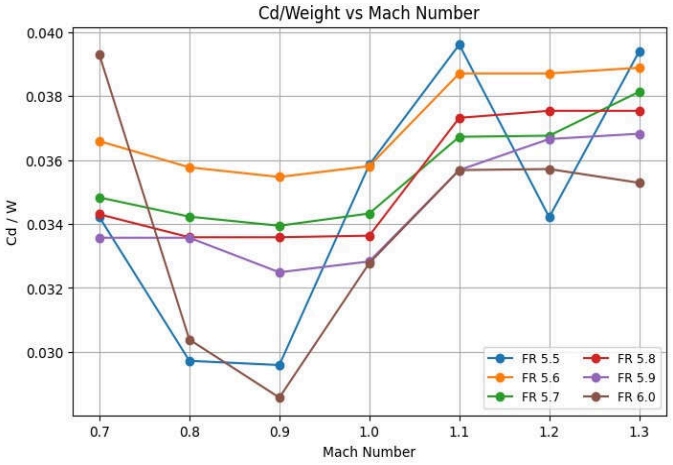


Figure A.51: Drag Coefficient/Weight vs Mach for 5.5:1 to 6:1

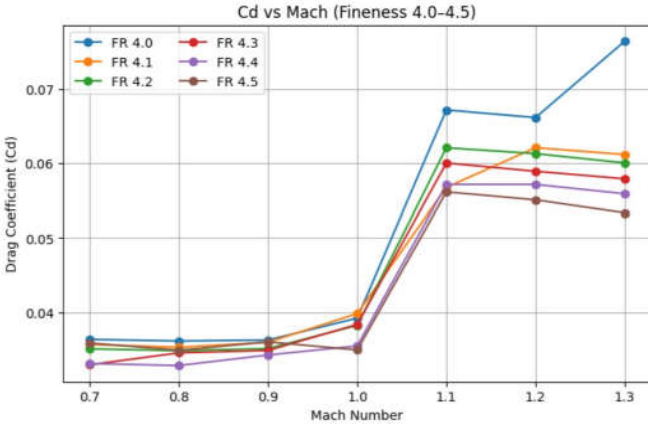


Figure A.52 : Drag Coefficient vs Mach for FR 4:1 to 4.5:1

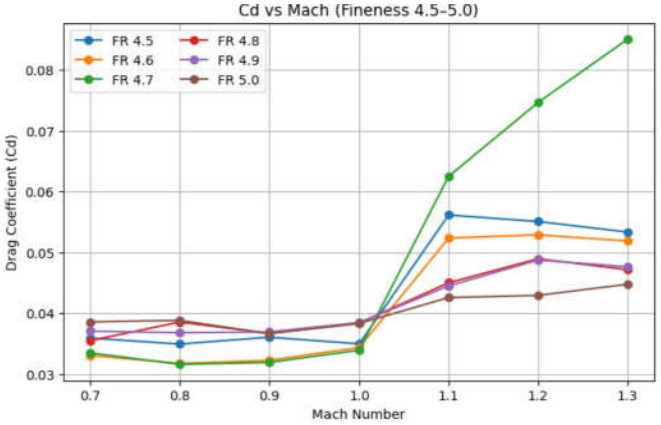


Figure A.53: Drag Coefficient vs Mach for 4.5:1 to 5:1

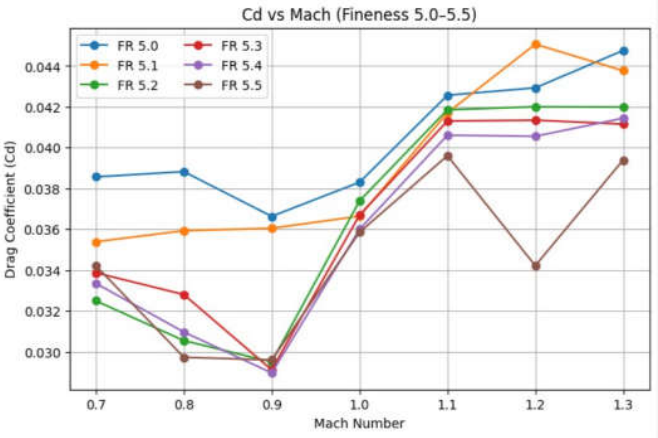


Figure A.54 : Drag Coefficient vs Mach for FR 5:1 to 5.5:1

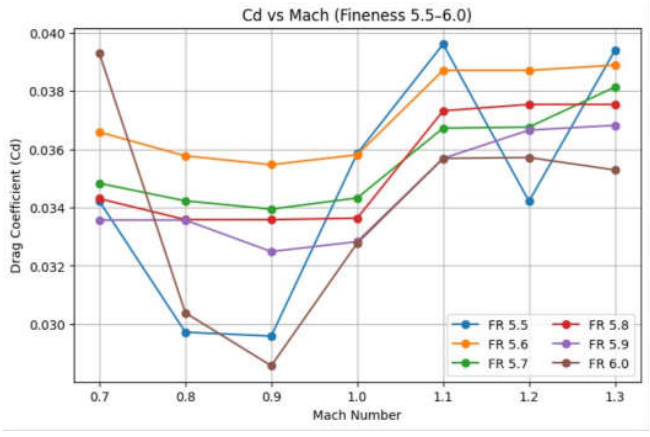


Figure A.55: Drag Coefficient vs Mach for 5.5:1 to 6:1

Simulations were conducted to analyze **velocity, pressure, and density distributions** for the selected Von Kármán nose cone profile at five freestream Mach numbers within the transonic regime. The resulting contour plots demonstrated smooth flow behavior with favorable pressure gradients and minimal shock-induced disturbances. These results support the selection of the **Von Kármán profile with a fineness ratio of 5:1** for the *Udbhava* nose cone configuration.

The simulations were performed using ANSYS Fluent. The operating conditions and solver settings used for the analysis are summarized below.

Table A.8: Simulation Operating Conditions

Parameters	**Setting**
Software	ANSYS Fluent- ANSYS Student 2025 R2
Geometry & Space	2D Axisymmetric
Mach Number Range	0.8 - 1.2 Mach
Solver Type	Density Based
Static Temperature	300K
Convergence Criteria	E-06
Turbulence Model	Spallart allamaras
Discretisation	Second Order Upwind
Density	Ideal Gas

Contours for Selected Nose Cone Profile:-

At Freestream Velocity of Mach 0.8

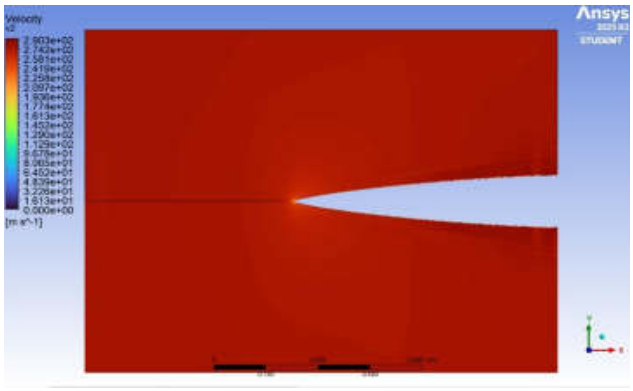


Figure A.56: Velocity Contour

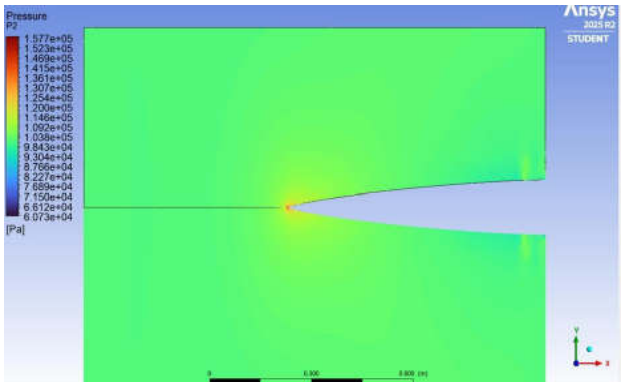


Figure A.57 : Pressure Contour

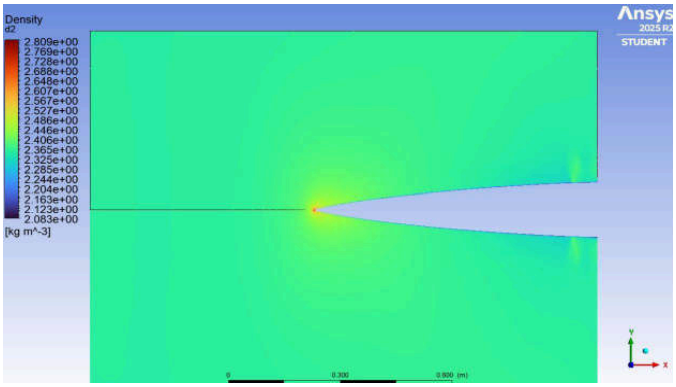


Figure A.58 : Density Contour At Freestream Velocity of Mach 0.9

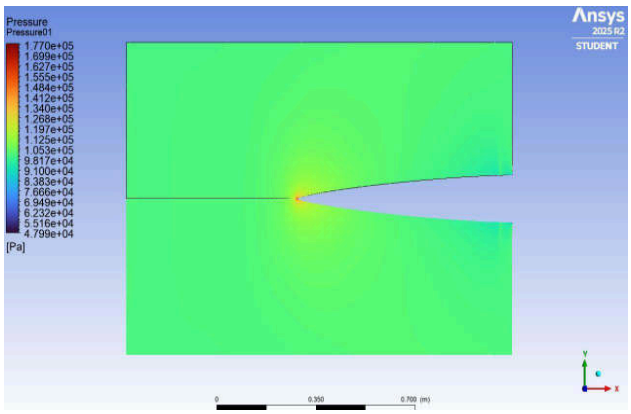


Figure A.59 : Velocity Contour

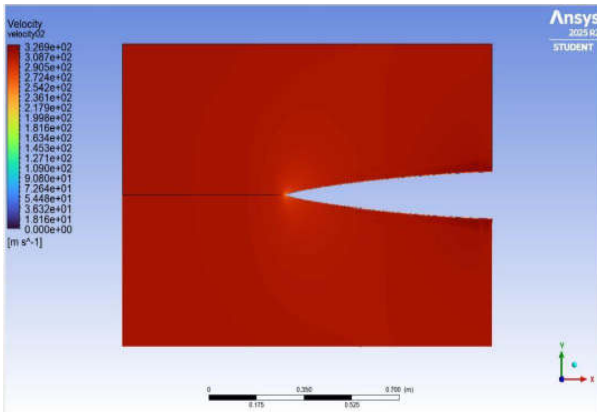


Figure A.60 : Pressure Contour

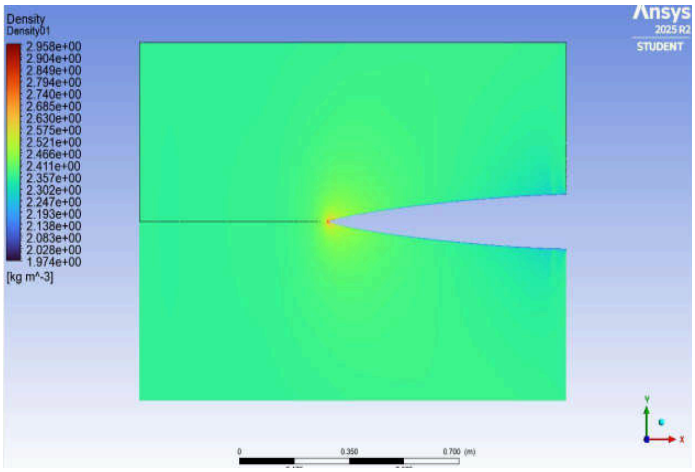


Figure A.61 : Density Contour

At Freestream Velocity of Mach 1.0

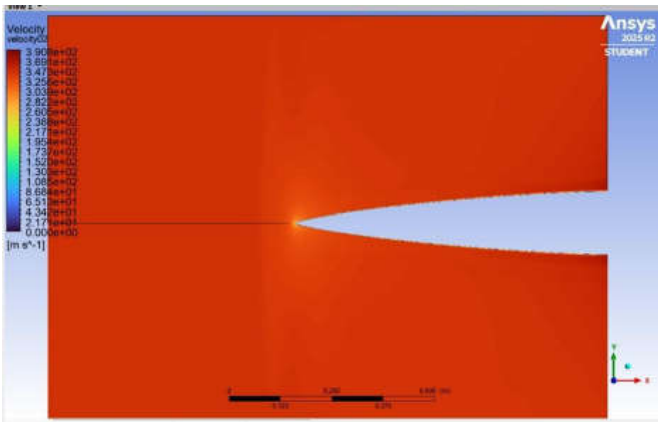


Figure A.62 : Velocity Contour

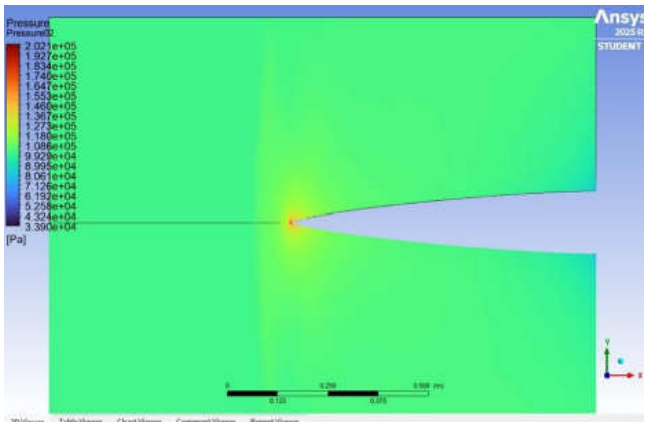


Figure A.63 : Pressure Contour

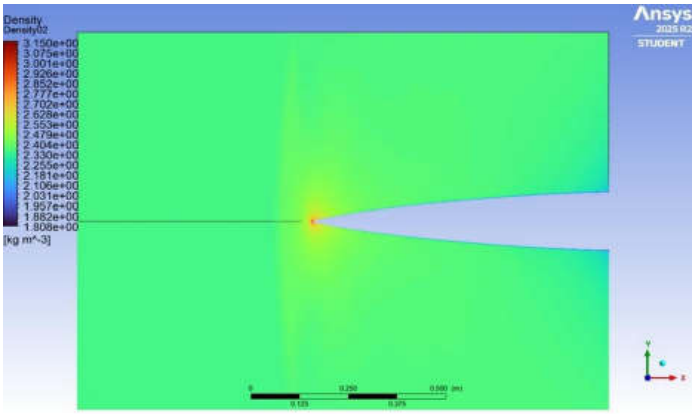


Figure A.64 : Density Contour

At Freestream Velocity of Mach 1.1

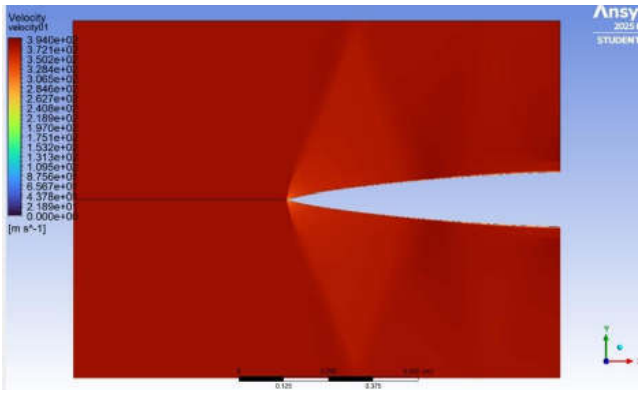


Figure A.65 : Velocity Contour

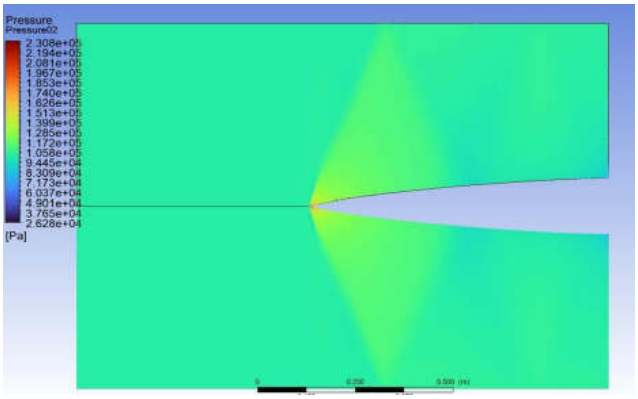


Figure A.66 : Pressure Contour

At Freestream Velocity of Mach 1.2

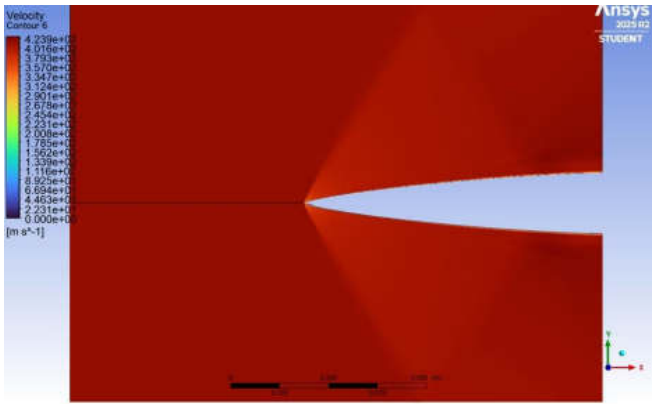


Figure A.67 : Velocity Contour

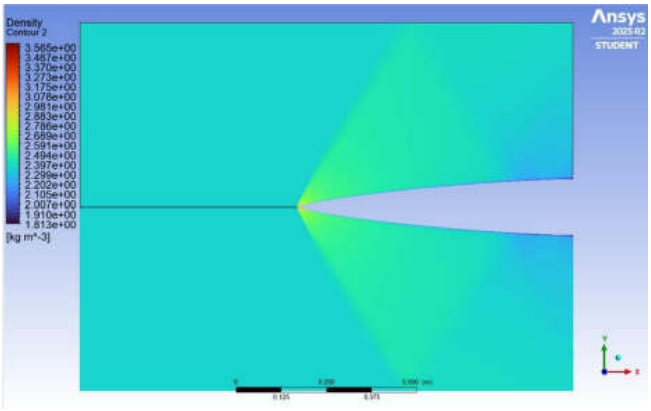


Figure A.68 : Density Contour

The contour results validate the selection of the Von Karman profile at a fineness ratio of 5:1 for the nosecone

6. Fin Specifications

Fin Elimination Matrix
The fin elimination matrix serves as the initial filtering stage, where a wide range of fin geometries is evaluated and unsuitable configurations are eliminated based on stability and fundamental flight performance requirements.

Obtain Basic Rocket Data
Key rocket parameters, including the center of mass, moment of inertia, component positions, and motor thrust curve, were obtained by modeling the vehicle in OpenMotor. Additionally, power-on and power-off drag characteristics were extracted using RASAero.

Pre-Processing of Data
Initial fin design parameters were defined using mean values derived from OpenRocket simulations. Variations around these values were introduced using standard deviations based on conventional design heuristics. A probabilistic sampling approach was employed to generate and evaluate a wide range of geometries, eliminating infeasible configurations.

The elimination process was carried out in three sequential stages:

- Broad exploration
- Focused search
- Fine tuning

Simulation Framework
A physics-based simulation framework was developed incorporating four major modules:

- Environment
- Motor
- Rocket
- Flight

These modules were integrated into a rocket generation function, where fin parameters were systematically varied. For each configuration, key performance metrics were evaluated, including:

- Maximum velocity
- Apogee
- Off-rail conditions
- Drag
- Pitch behavior
- Stability (calibers)

Ranking Algorithm Based on Monte Carlo Optimization
A multi-parameter ranking algorithm based on Monte Carlo sampling was implemented to compare fin geometries. Each configuration was evaluated against multiple performance criteria, including moment of inertia, initial mass, vehicle length, and thrust characteristics.

The algorithm assigned a composite score to each configuration, enabling the selection of optimal fin designs for further analysis.

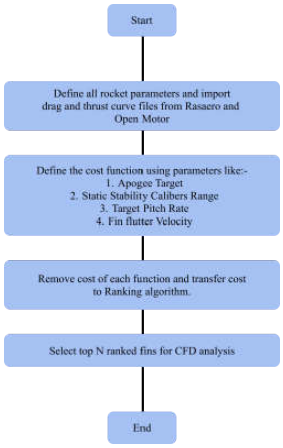


Figure A.69 : Fin Elimination Matrix

Fin Decision Matrix:

A combined 2D and 3D CFD analysis was conducted to evaluate the aerodynamic performance of the fin profiles (5°–20° bevel, double wedge, biconvex, and NACA 0012). The 3D simulations assessed overall flow behavior using pressure, temperature, and velocity contours along with pathlines to evaluate wake development. Complementary 2D simulations captured detailed transonic flow features, including flow separation and trailing-edge pressure recovery. The combined results provided a consistent dataset for the fin elimination matrix.

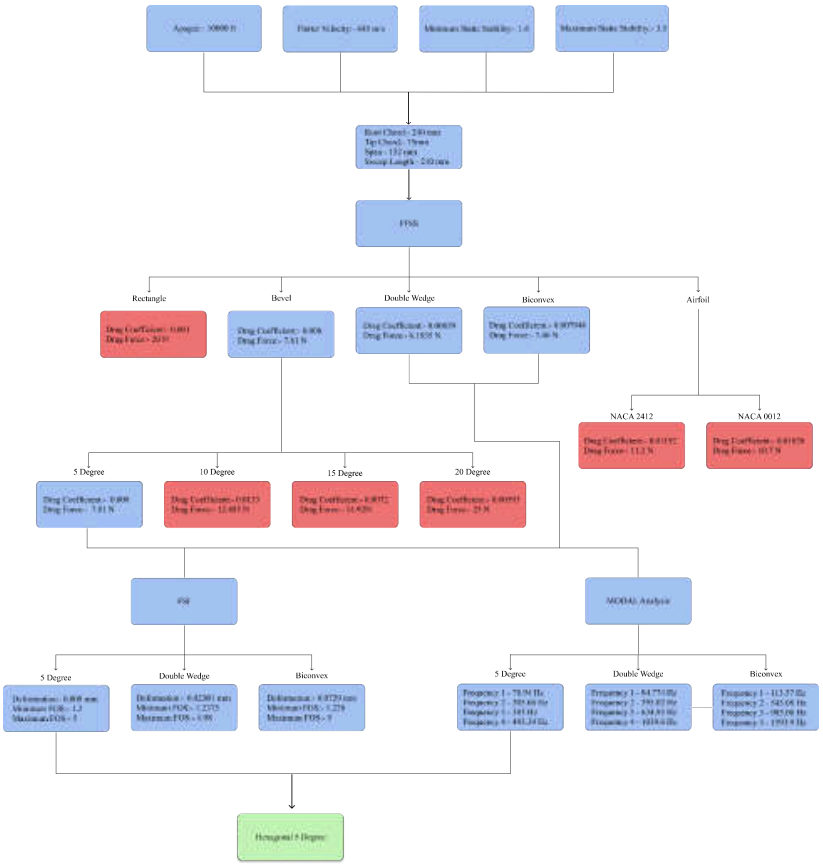


Figure A.70 : Fin Decision Matrix

Based on this combined evaluation, the 5° bevel, Double Wedge, Biconvex profile fins were chosen as the optimal configuration due to lower drag, smooth pressure recovery and favourable flow characteristics.

2D & 3D Simulations

CFD study conducted a visualized analysis to decide the most optimum fin profiling. 3D pressure gradients and pathlines clearly define the wake structures.

The following solver settings were used as a standard for conducting all of the simulations required on the fins. The solver used was ANSYS Fluent. The list below presents the operating conditions used:

Table A.9: Simulation Operating Conditions

Parameters	**Setting**
Geometry & Space	2D Axisymmetric
Mach Number Range	0.8 - 1.2 Mach
Solver Type	Density Based
Static Temperature	281K - 279.48K
Turbulence Model	SST K-omega
Density	Ideal Gas

3D Simulations of 10º Bevel

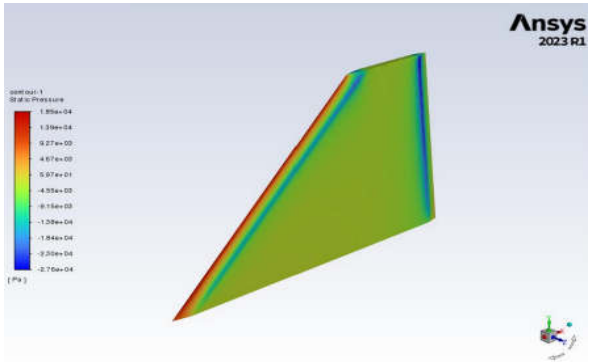


Figure A.71 : Pressure Contours for 10º Bevel

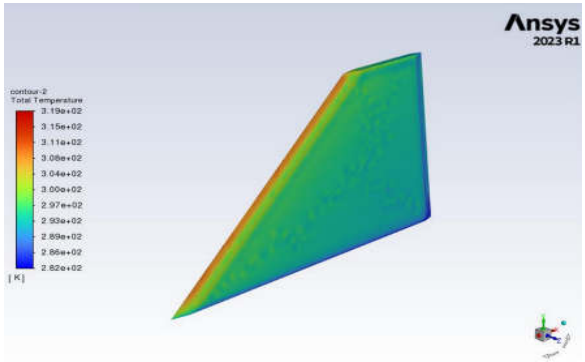


Figure A.72 : Temperature Contours for 10º Bevel

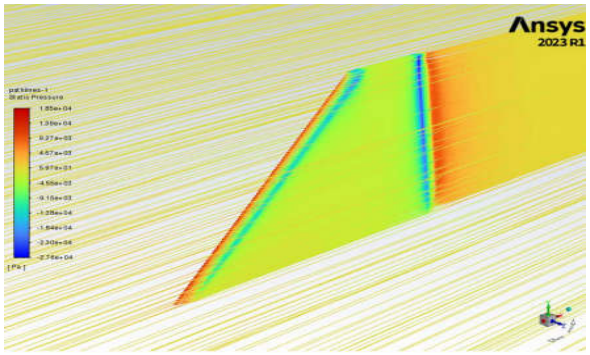


Figure A.73 : Pressure Pathlines for 10º Bevel

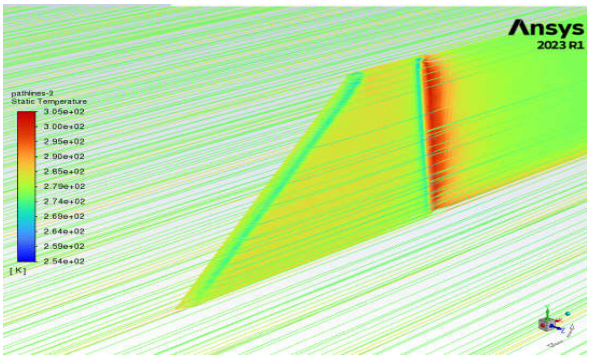


Figure A.74 : Temperature Pathlines for 10º Bevel

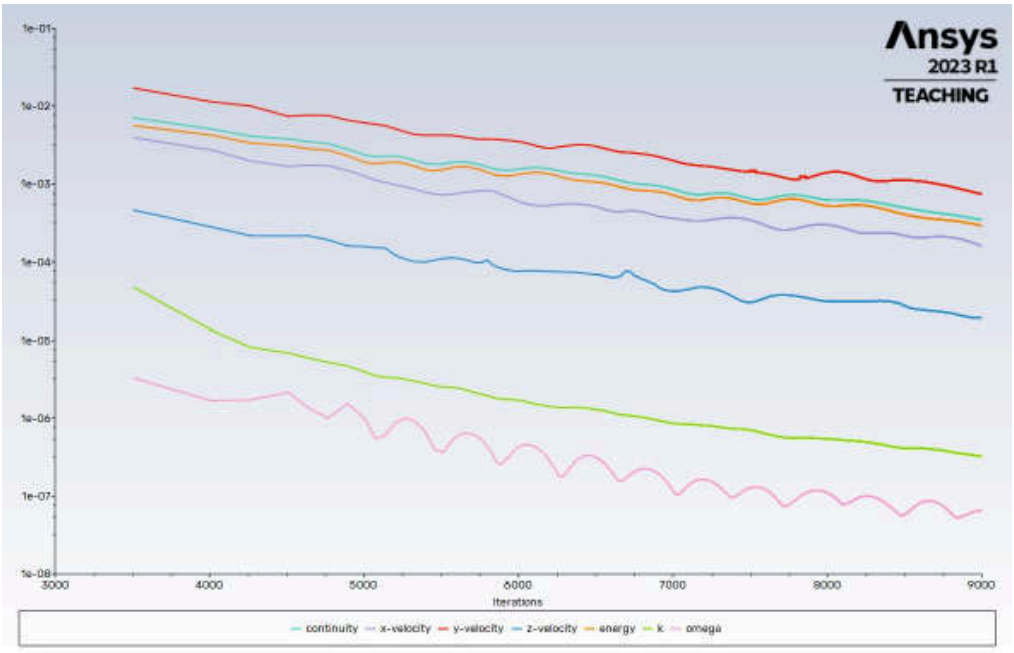


Fig A.75 : Scaled Residuals for 10º Bevel

3D Simulations of 20º Bevel

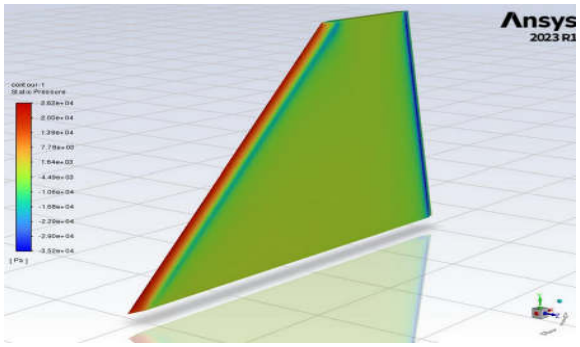


Figure A.76 : Pressure Contours for 20º Bevel

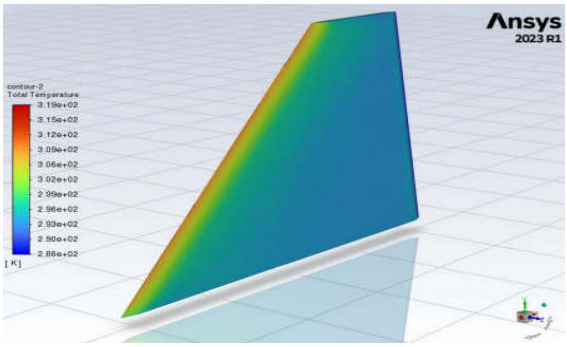


Figure A.77 : Temperature Contours for 20º Bevel

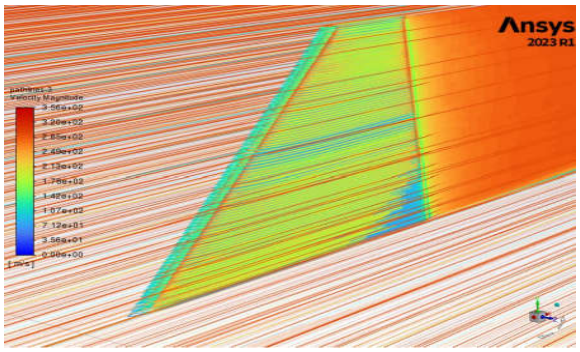


Figure A.78 : Pressure Pathlines for 20º Bevel

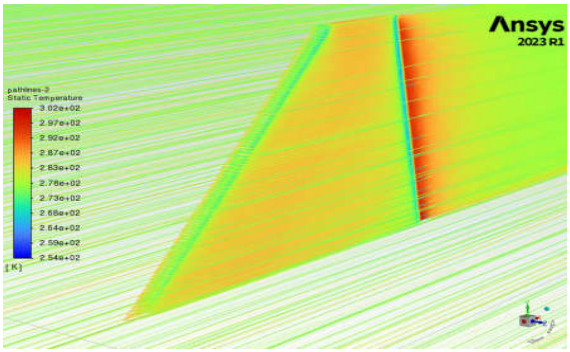


Figure A.79 : Temperature Pathlines for 20º Bevel

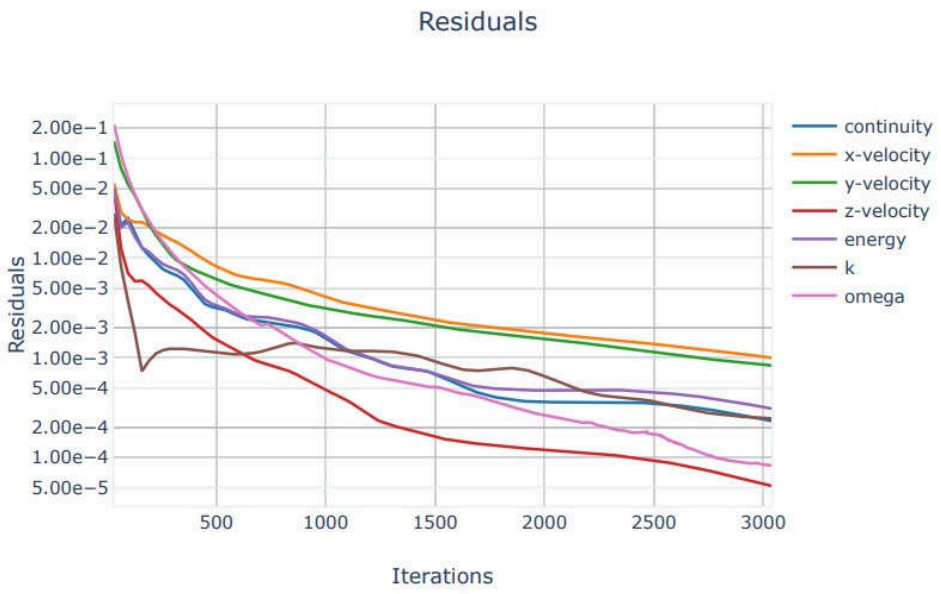


Figure A.80 : Scaled Residuals for 20º Bevel

3D Simulations of Biconvex

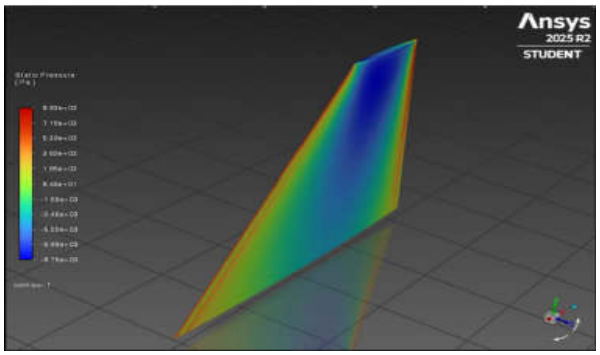


Figure A.81 : Pressure Contours for Biconvex

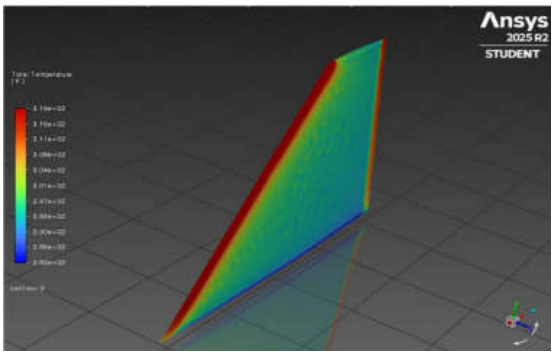


Figure A.82 : Temperature Contours for Biconvex

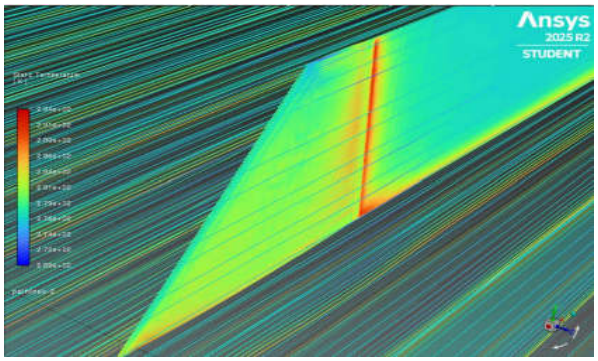


Figure A.83 : Temperature Pathlines for Biconvex

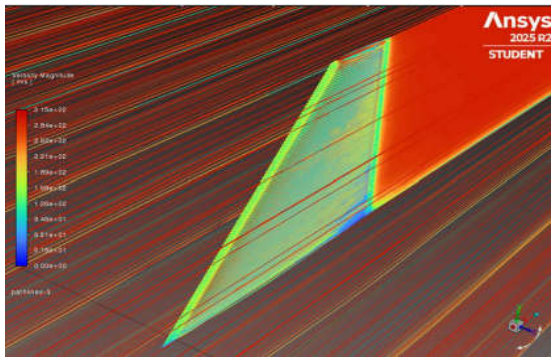


Figure A.84 : Velocity Pathlines for Biconvex

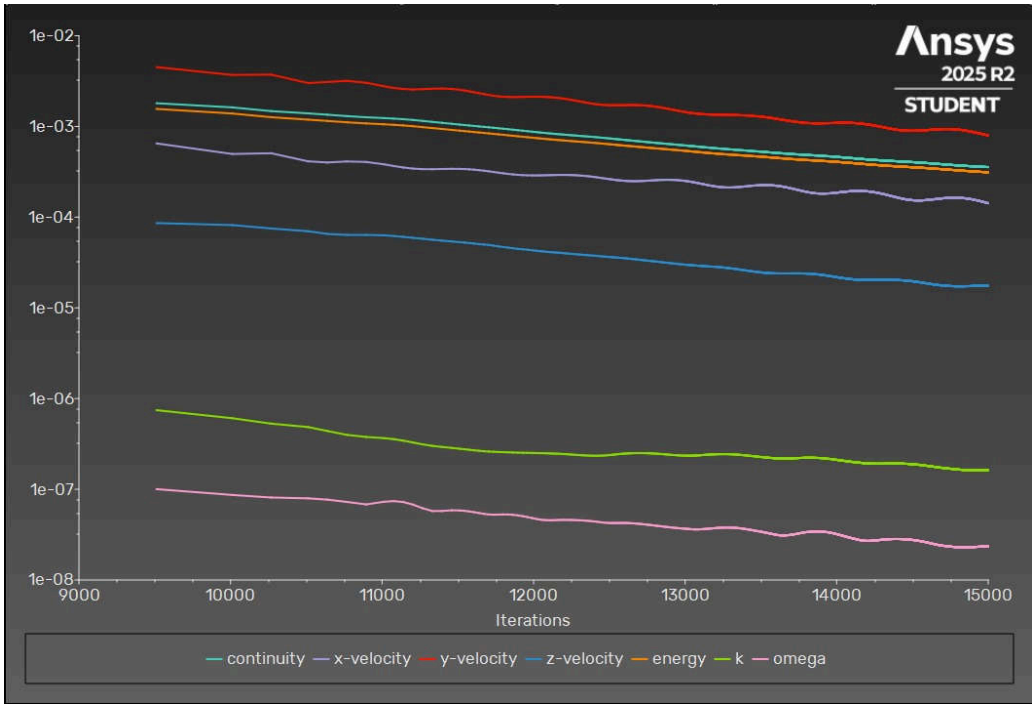


Figure A.85 : Scaled Residuals for Biconvex

3D Simulations of Double Wedge

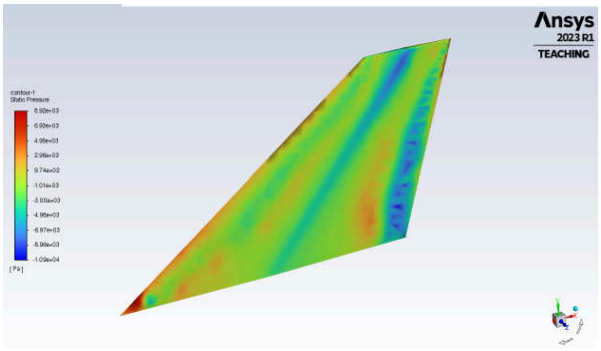


Figure A.86: Pressure Contours for Double Wedge

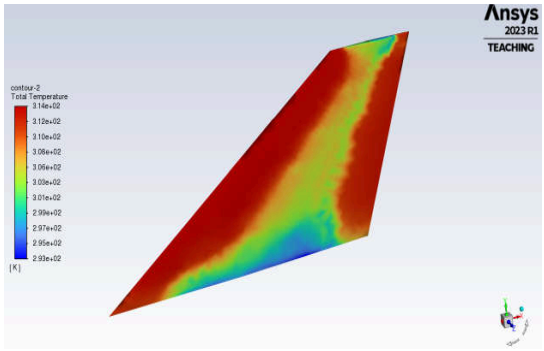


Figure A.87: Temperature Contours for Double Wedge

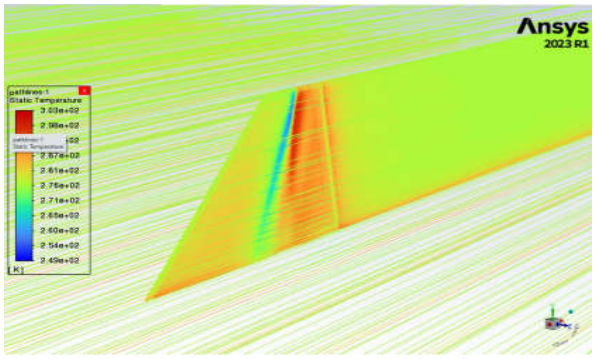


Figure A.88: Temperature Pathlines for Double Wedge

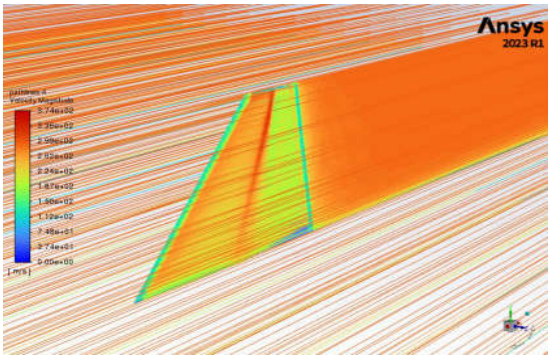


Figure A.89: Velocity Pathlines for Double Wedge

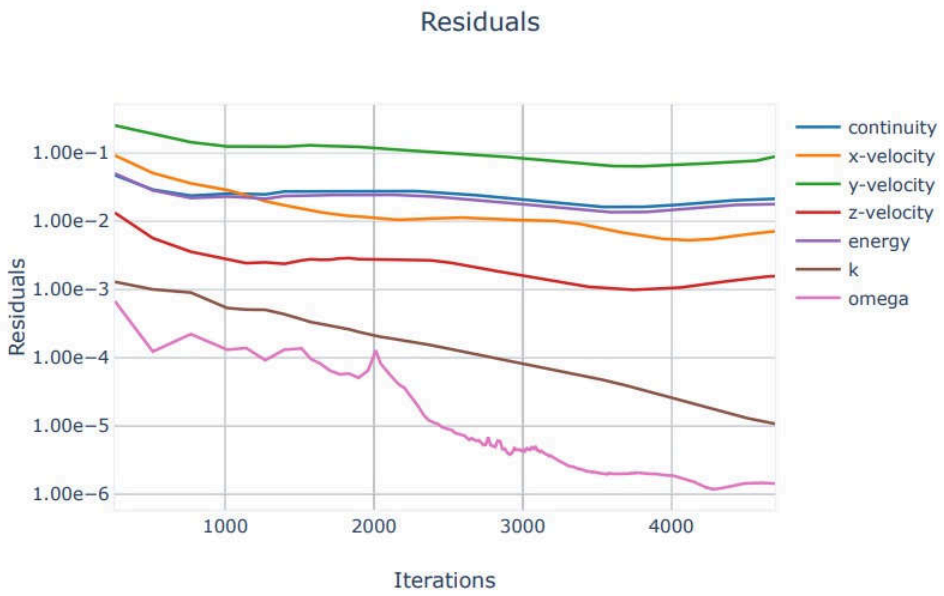


Figure A.90 : Scaled Residuals for Double Wedge

3D Simulations of NACA 0012

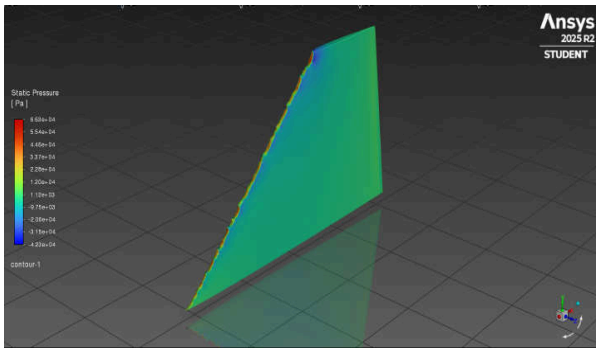


Figure A.91 : Pressure Contours for NACA 0012

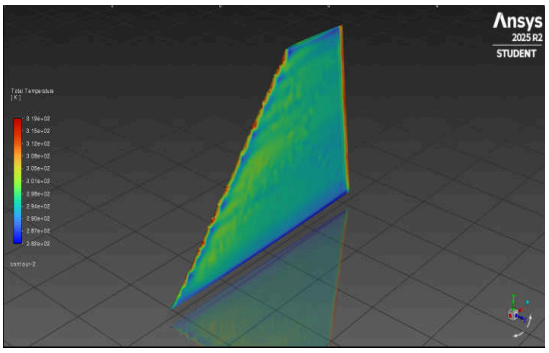


Figure A.92 : Temperature Contours for NACA 0012

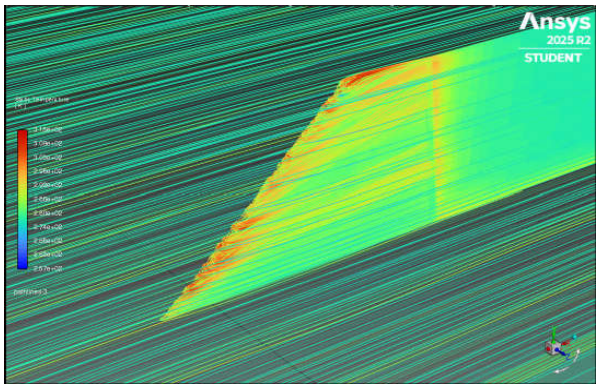


Figure A.93 : Temperature Pathlines for NACA 0012

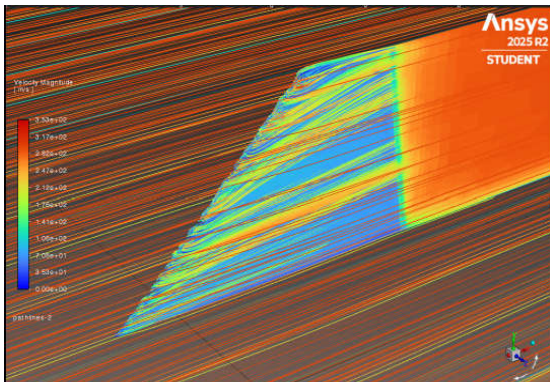


Figure A.94 : Velocity Pathlines for NACA 0012

3D Simulations of NACA 2412

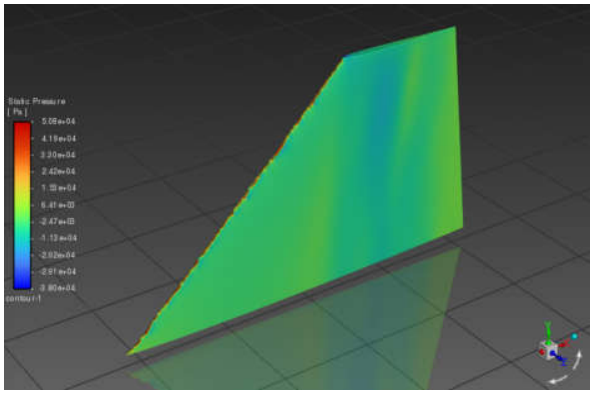


Figure A.95: Pressure Contours for NACA 2412

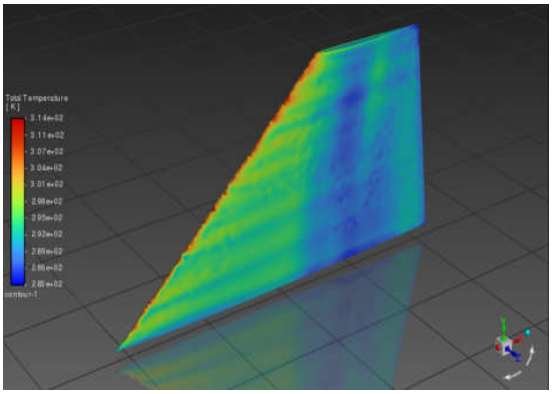


Figure A.96: Temperature Contours for NACA 2412

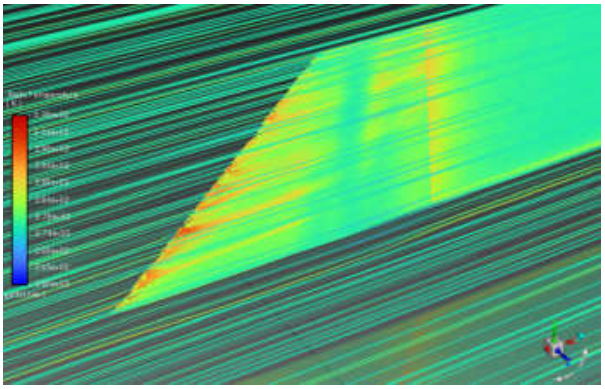


Figure A.97: Temperature Pathlines for NACA 2412

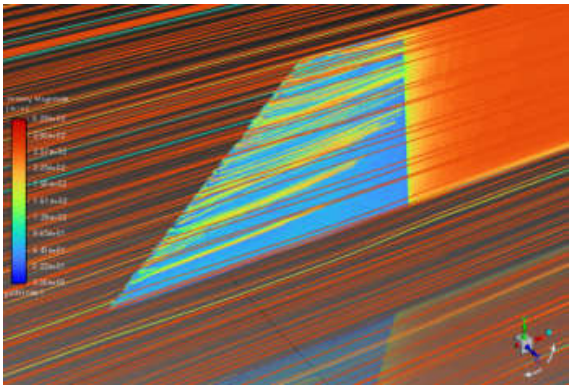


Figure A.98: Velocity Pathlines for NACA 2412

2D simulations were performed on selected fin profiles to study key aspects such as pressure contours, shock behaviour and onset of flow separation. Pressure recovery and other key aerodynamic factors were identified from these simulations.

2D Simulations of 5º Bevel Fins

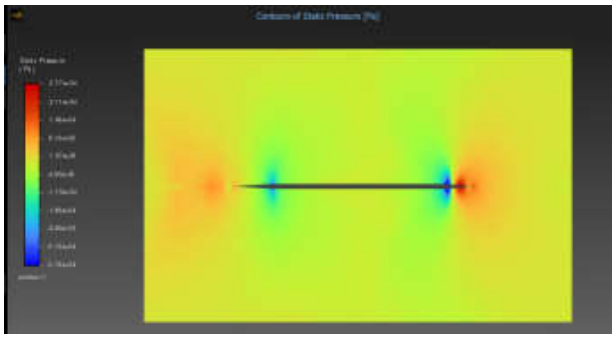


Figure A.99: Static Pressure Contours for 5 º Bevel

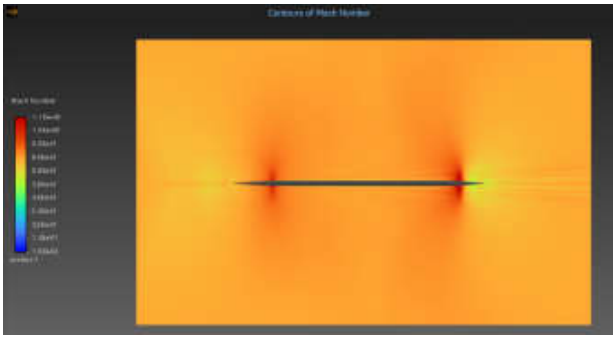


Figure A.100: Mach Number Contours for 5 º Bevel

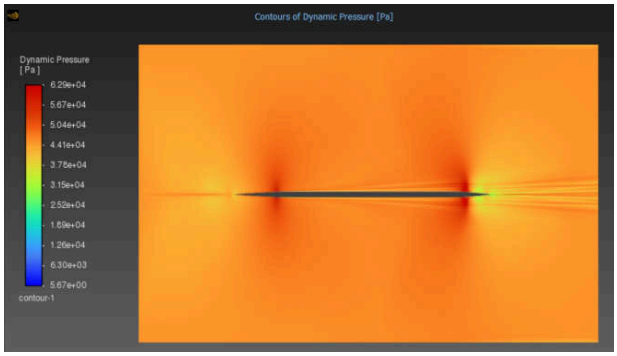


Figure A.101: Dynamic Pressure Contours for 5 º Bevel

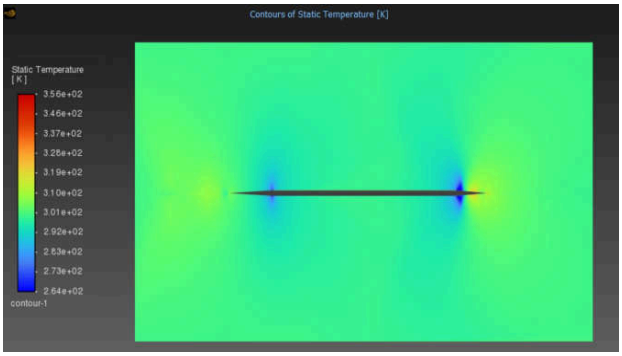


Figure A.102: Static Temperature Contours for 5 º Bevel

2D Simulations of 10º Bevel Fins

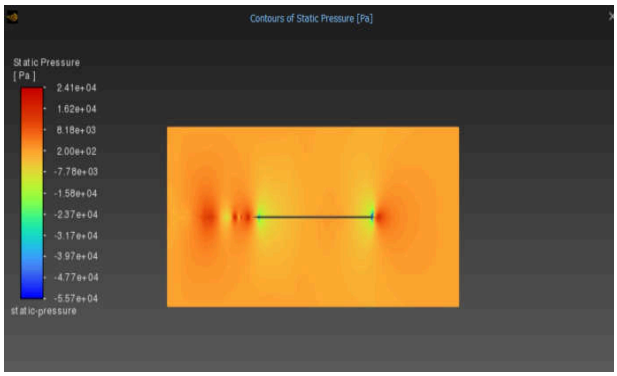


Figure A.103: Static Pressure Contours for 10 º Bevel

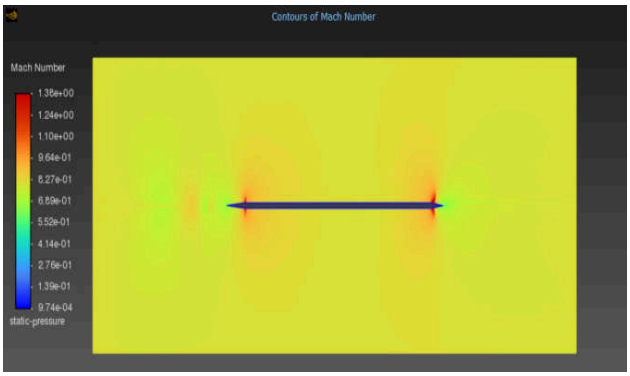


Figure A.104: Mach Number Contours for 10 º Bevel

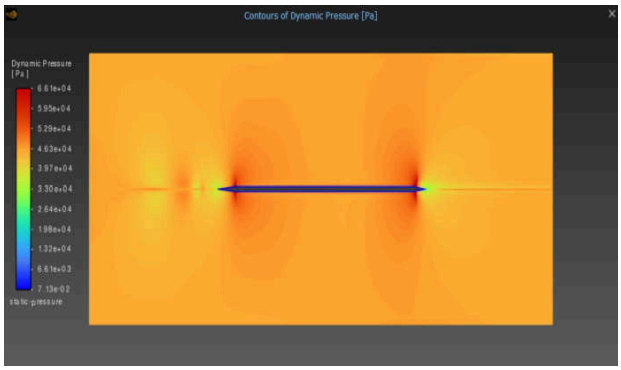


Figure A.105: Dynamic Pressure Contours for 10 º Bevel

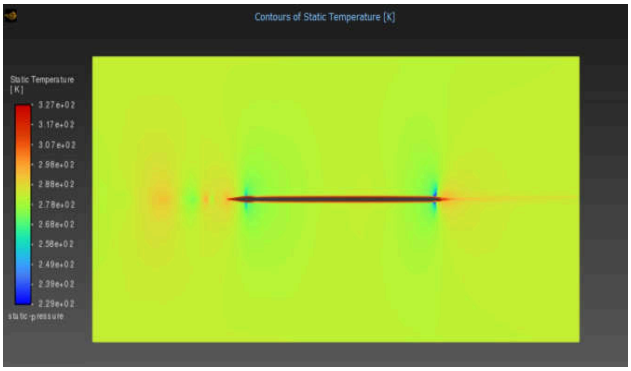


Figure A.106: Static Temperature Contours for 10 º Bevel

2D Simulations of 15º Bevel Fins

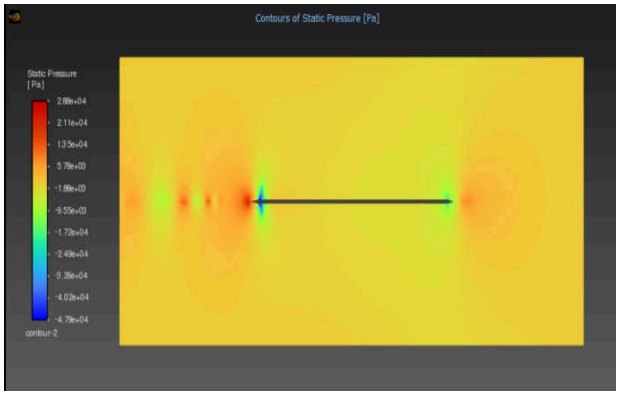
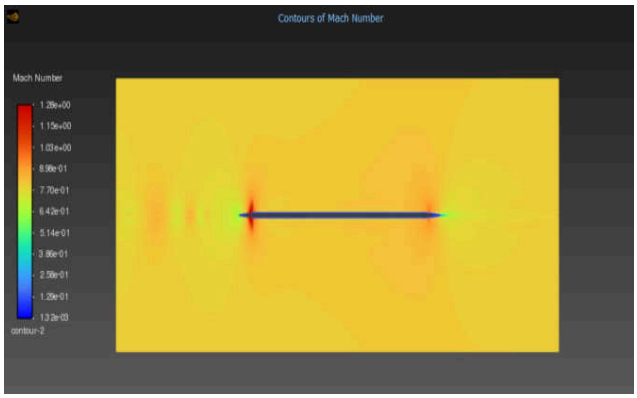


Figure A.107: Static Pressure Contours for 15 º Bevel



FigureA.108: Mach Number Contours for 15 º Bevel

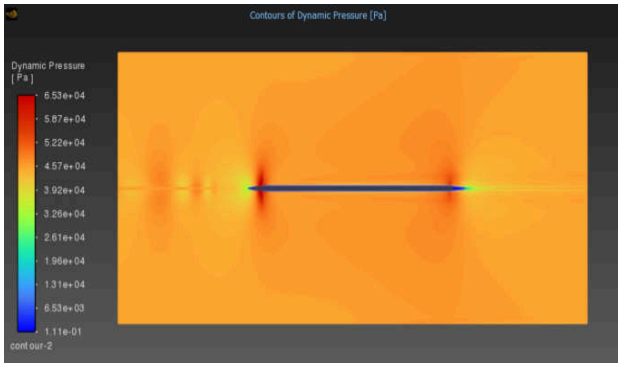


Figure A.109: Dynamic Pressure Contours for 15 º Bevel

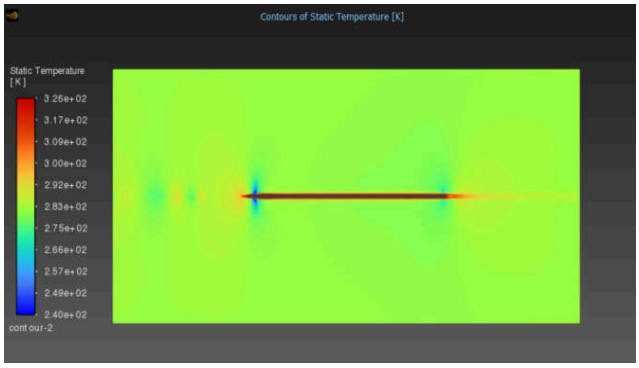


Figure A.110: Static Temperature Contours for 15 º Bevel

2D Simulations of Bi Convex Fins

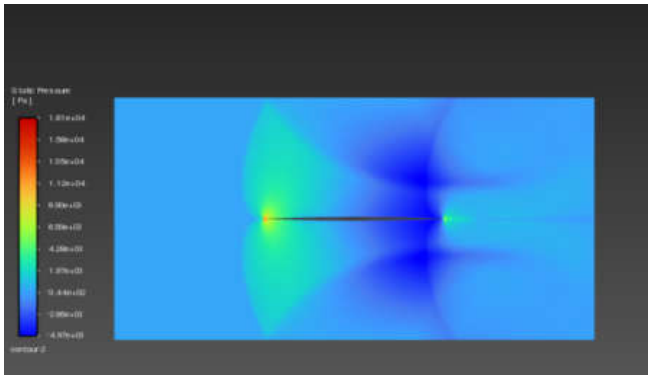


Figure A.111 : Static Pressure Contours for Biconvex



Figure A.112 : Mach Number Contours for Biconvex



Figure A.113 : Dynamic Pressure Contours for Biconvex

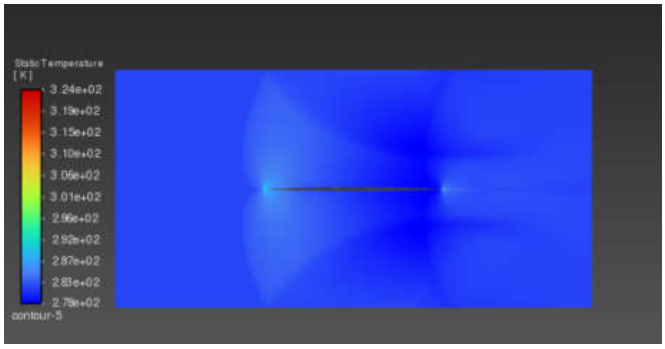


Figure A.114 : Static Temperature Contours for Biconvex

Fluid Structure Interaction

Based on the combined evaluations of 2D & 3D flow analysis, the 5° bevel, Double Wedge, Biconvex profile fins were selected amongst the set for choosing the finalized fin profile based on their drag, smooth pressure recovery and favourable flow characteristics.

A 2-way fluid structure Interface (FSI) was conducted on the selected profile to identify the profile undergoing the least deformation due to Aeroelastic instability suffered by the fins due to Aerodynamic, Inertia and Elastic forces acting on the body due to high Mach regimes experienced during flight.

Out of the 3 selected profiles, the 5 degree Bevel triumphed with a Maximum Deformation of 0.01 mm over 0.02301 mm (Double Wedge) and 0.07209 mm (Biconvex) for a time step of 0.0002sec. Hence, **5 degree Bevel Fin** was chosen as the finalized fin profile for our *Udbhava*.

2-Way FSI for Double Wedge fins

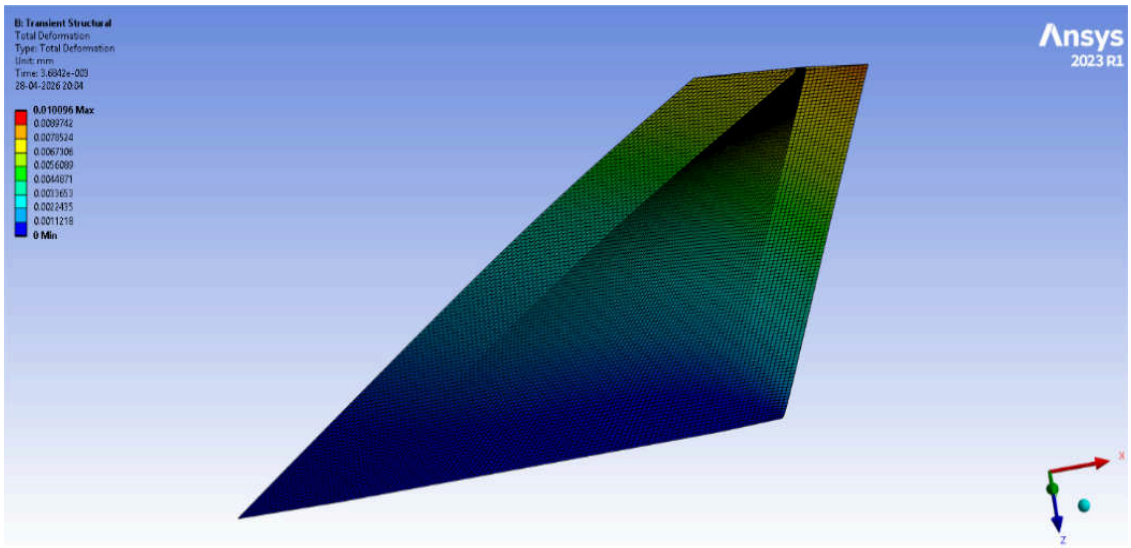


Figure A.115 : Total Deformation of 5 degree bevel Fins under aerodynamic loads

The maximum deformation observed was **0.01009** mm, indicating negligible structural displacement under the applied loads. The minimum factor of safety was **1.2512**, confirming that the structure remains within safe operational limits.

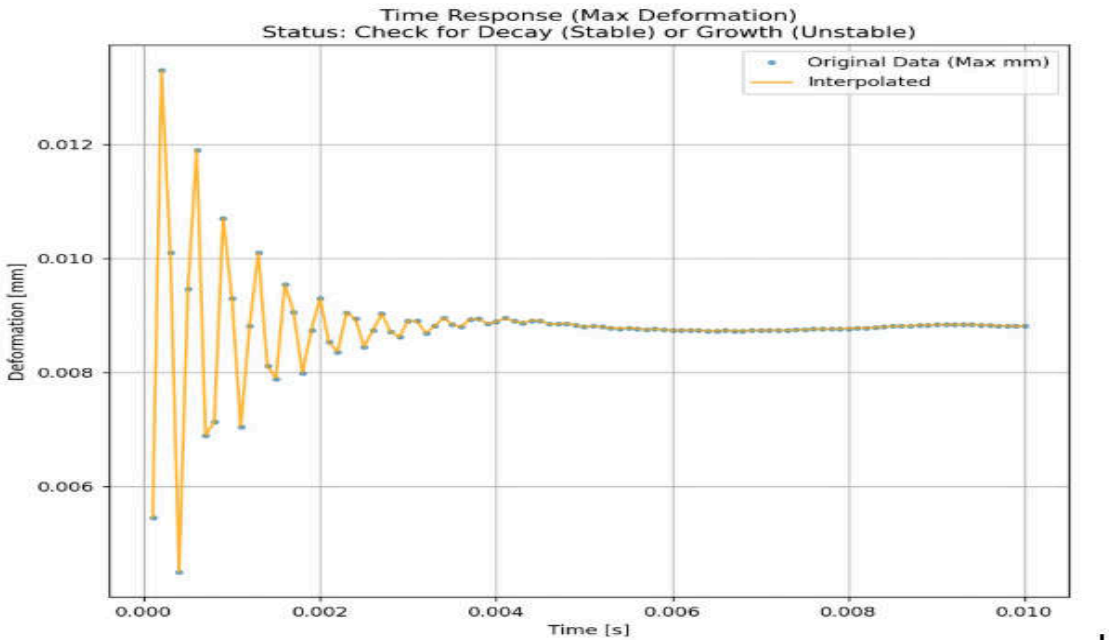


Figure A.116 : Deformation v/s time plot of 5 degree bevel fins under aerodynamic loads

FFT (Fast Fourier Transform) was conducted via Python script to evaluate the deformation v/s time graph obtained from the transient results and to obtain the nature of the plot and the peak flutter frequency.

The 5 degree bevel fins showed a Decay (stable) curve which indicated that the fin did not hit flutter during Mach 0.8 (MaxQ) hence evaluating our fins and inhibiting favourable conditions for the fin profile selection.

2 -way FSI for Double Wedge fins

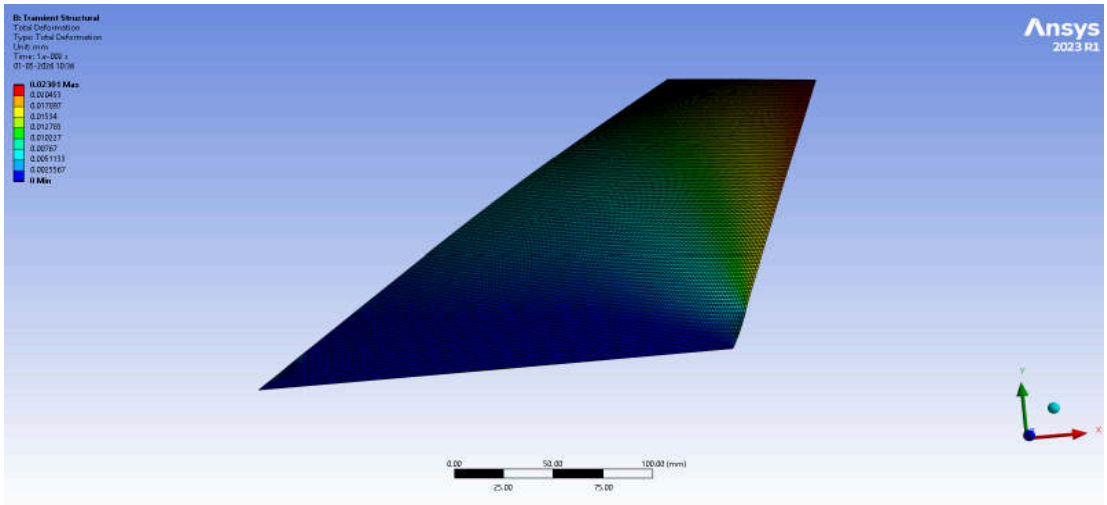


Figure A.117 : Total Deformation of Double Wedge Fins under aerodynamic loads

The maximum deformation was **0.02301 mm**, indicating minimal structural deflection under the applied loads. The minimum factor of safety was **1.24**, confirming that the design remains within acceptable safety limits.

FFT (Fast Fourier Transform) was conducted via Python script to evaluate the deformation v/s time graph obtained from the transient results and to obtain the nature of the plot and the peak flutter frequency.

While the 5 degree bevel fins showed a Decay (stable) curve which indicated that the fin did not hit flutter during Mach 0.8 (MaxQ), the **double wedge fins** showed a deformation growth (Unstable) of flutter frequency 100 Hz at Mach 0.8, making it one more reason not to be selected for the fin profile of UDBHAVA.

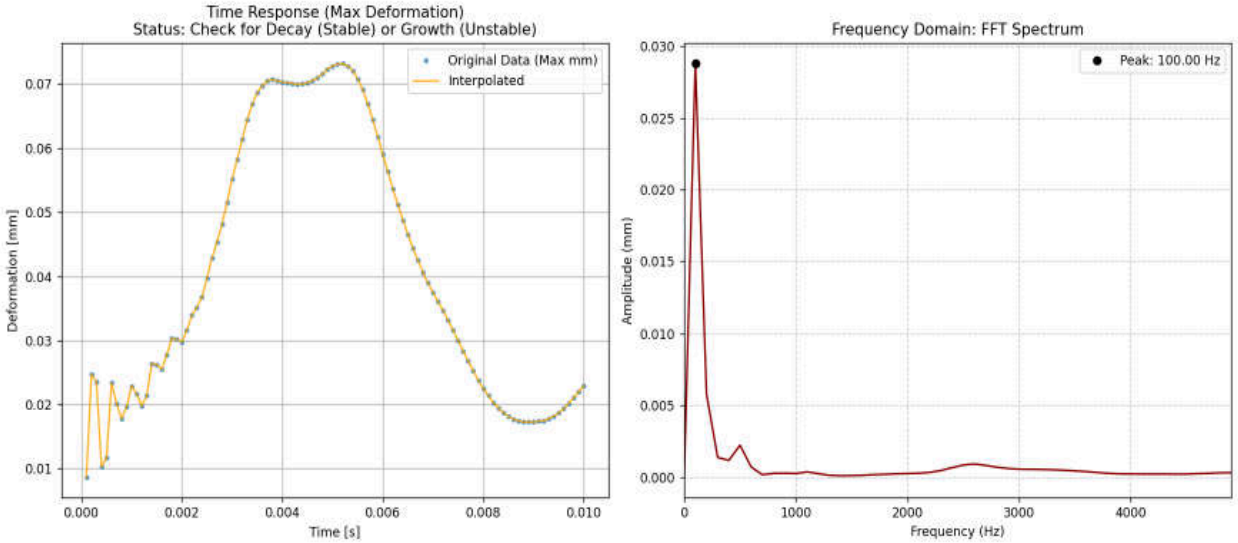


Figure A.118: Deformation v/s time plot of double wedge fins under aerodynamic loads With Maximum flutter frequency

2 -way FSI for Bi-Convex fins

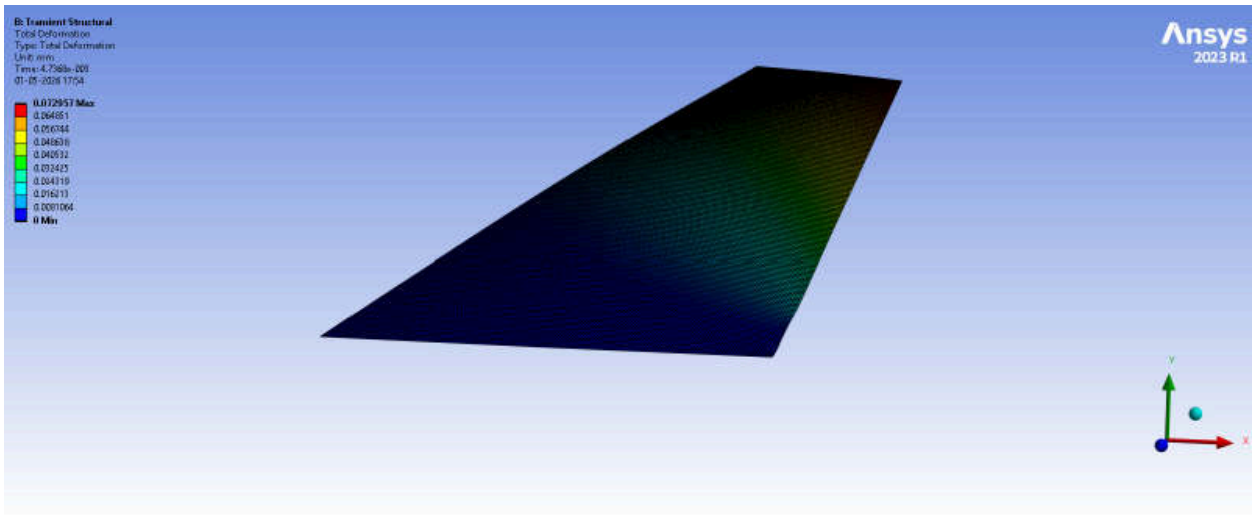


Figure A.119: Total Deformation of Biconvex Fins under aerodynamic loads

The maximum deformation was 0.07209 mm, indicating low structural deflection under the applied loads. The minimum factor of safety was 1.23, suggesting that the design remains within acceptable limits, though with a relatively low safety margin.

Fin Flutter Calculations

A preliminary fin flutter analysis was conducted using experimentally determined material properties. A minimum flutter factor of safety of 1.5 was targeted to ensure adequate stability under all conditions. The calculation was carried out as follows:

Aspect Ratio:

$$AR = \frac{b^2}{S}$$

$$S = \frac{1}{2}\, b\left(c_r + c_t\right) = \frac{1}{2} \times 0.152 \times (0.24 + 0.075)$$

$$AR = \frac{0.152^2}{\frac{1}{2} \times 0.152 \times (0.315)} = 0.965$$

Taper Ratio:

$$\lambda = \frac{c_t}{c_r} = \frac{0.075}{0.24} = 0.3125$$

Temperature at Altitude:

$$T = 59 - (0.00356 \times 300)$$

$$T = 48.32^{\circ} F$$

Pressure:

$$P = 2116\left(\frac{T+459.7}{518.8}\right)^{5.256}$$

$$P = 2116\left(\frac{48.32+459.7}{518.8}\right)^{5.256} = 1898.74\, lb/ft^2$$

Speed of Sound:

$$a = \sqrt{1.4 \times 1716.59 \times (T + 460)}$$

$$a = \sqrt{1.4 \times 1716.59 \times (48.32 + 460)} = 1105.26\, ft/s$$

Shear Modulus:

$$G = 884730.2\, psi$$

Flutter Velocity:

$$V_f = a\sqrt{\frac{G}{1.337\, AR^3\, P\, (\lambda+1)} \cdot \frac{2(AR+2)}{\left(\frac{t}{c}\right)^3}}$$

$$V_f = 1400.29\, mph = 625.98\, m/s \approx 626\, m/s\ (Mach \sim 1.8)$$

The calculated fin flutter velocity sums up to 626 m/s (corresponding to Mach 1.89). This results in an effective safety margin of approximately **2.15** when compared to the maximum flight velocity. The fin configuration remains stable and operates well below the onset of fin flutter.

Modal Analysis of 5º Bevel Fins

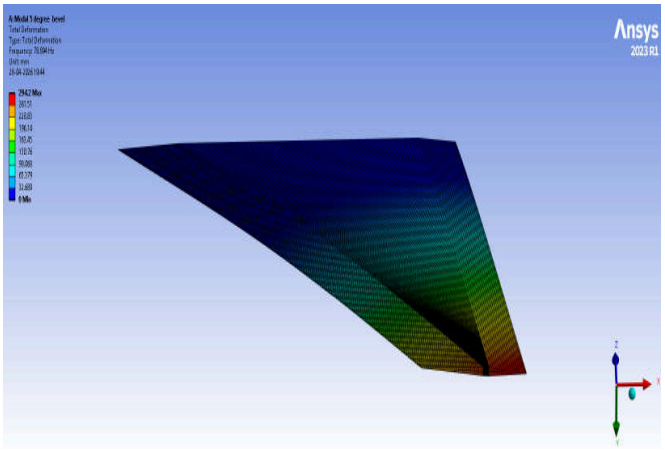


Figure A.120: Bending Mode 01

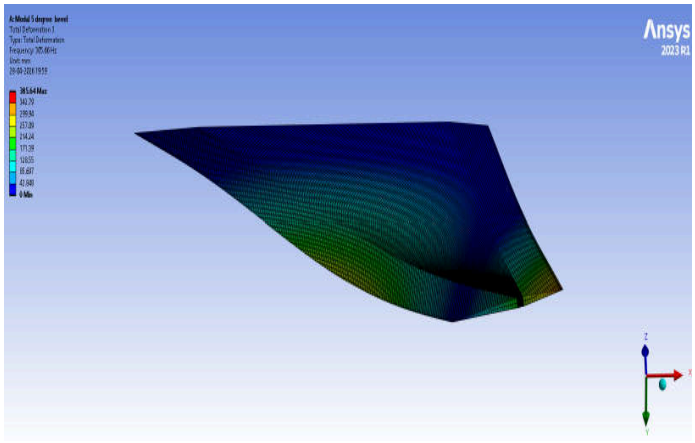


Figure A.121: Bending Mode 02

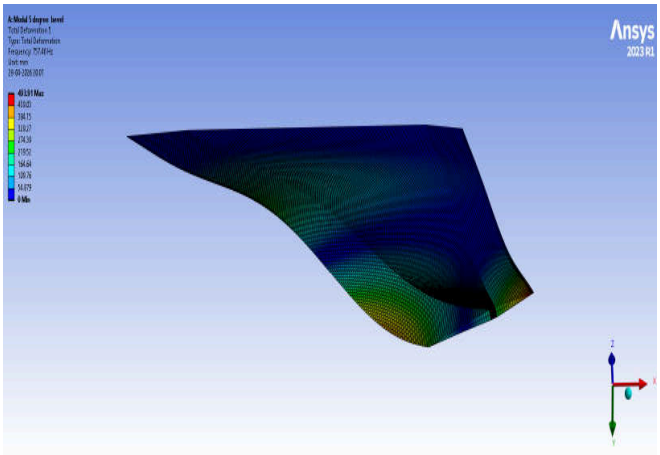


Figure A.122: Bending Mode 03

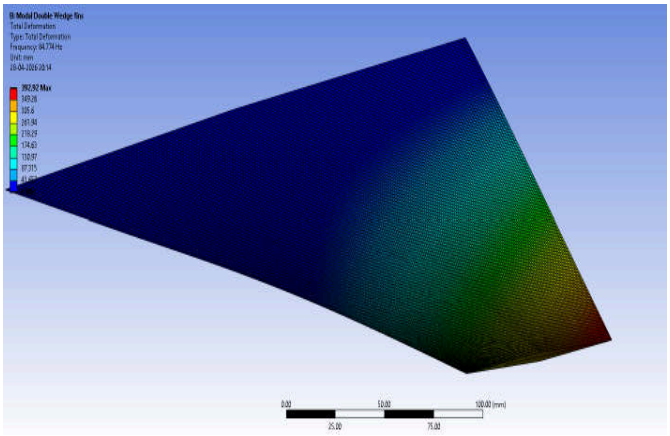


Figure A.123: Bending Mode 04

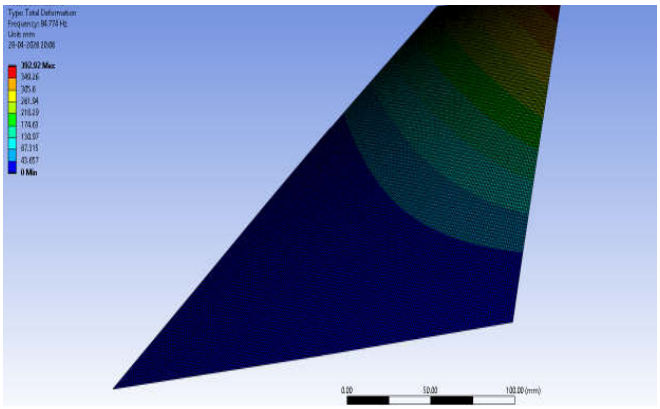


Figure A.124: Bending Mode 04

7. Passive Drag Reduction System (PDRS)

CFD analysis was conducted to evaluate the aerodynamic performance of the rocket with an integrated base bleed system during the coasting phase. Simulations were performed across the transonic Mach range of 0.5 to 0.875 (Max Q), where base drag effects are most significant. The complete rocket geometry, including the base bleed system, was modeled in ANSYS SpaceClaim, and a poly-hexcore mesh was generated using ANSYS Fluent Meshing. Critical regions such as the wake and base bleed zone were refined using body of influence (BOI) and localized face sizing to accurately capture wake dynamics.

A density-based solver with the SST k–ω turbulence model was employed to resolve separated flow structures and wake behavior under transonic conditions. A near-wall resolution of $y^+ \approx 1$ was maintained to ensure accurate boundary layer modeling. Flow field contours of pressure, dynamic pressure, density, and velocity were analyzed. Additionally, pathline visualizations were used to examine wake structure and jet–wake interactions downstream of the rocket base.

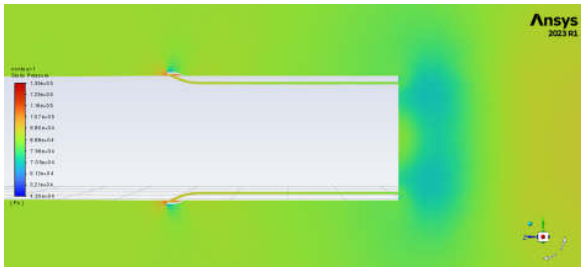


Figure A.125: Wake pressure contours at Aft region



Figure A.126: Wake Density Contours

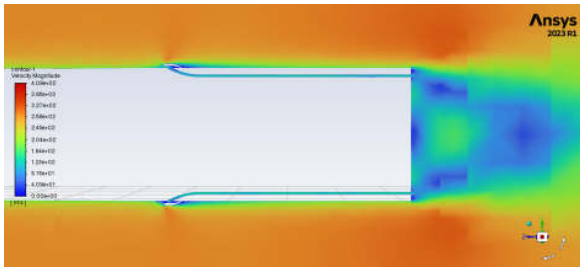


Figure A.127: Wake velocity Contours

Pathlines

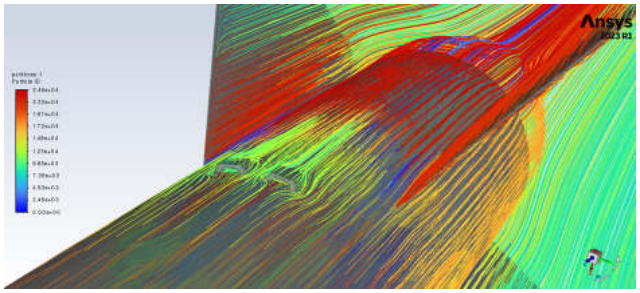


Figure A.128: Pathlines around the Bleed Inlet

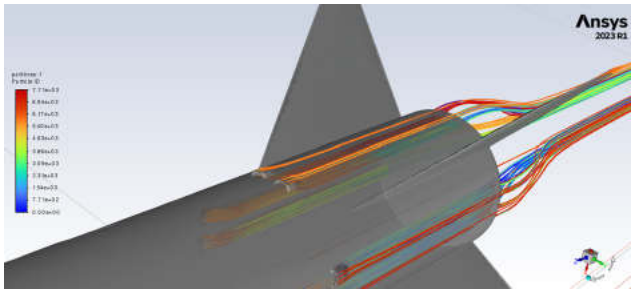


Figure A.129: Base Bleed Jet-Wake Interaction Pathlines

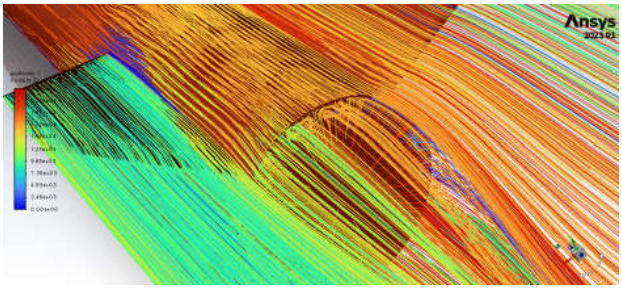


Figure A.130: Wake Circulation Pathlines

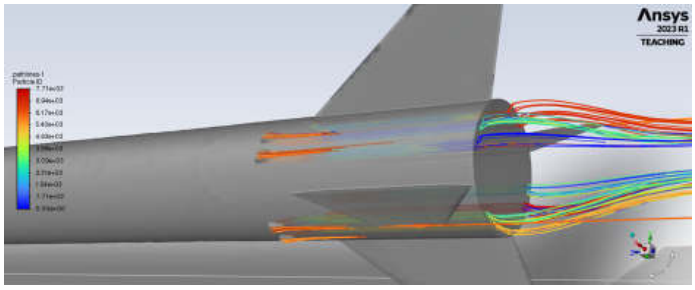


Figure A.131: Base Bleed Jet-Wake Interaction Along Rocket Aft

The simulation results show a clear modification of the wake region behind the rocket. Velocity contours indicate that high-speed transonic airflow is channelled through the bleed inlet and directed into the wake region. This additional flow increases the pressure in the base wake, which helps reduce the pressure deficit responsible for base drag.

This effect is also visible in the pressure contours, where the region near the bleed outlets shows an increase in static pressure compared to the surrounding wake region. The increased base pressure results in a reduction of the overall aerodynamic drag acting on the rocket.

Flow visualisation using pathlines further illustrates the behaviour of the system. The pathlines show airflow entering through the bleed inlet, travelling through the internal bleed channel, and exiting at the base outlet into the wake region. This confirms that the bleed geometry successfully directs airflow into the separated wake flow to modify the base pressure distribution.

Table A.10 : Comparison of Cd at different Mach Number

Mach Number	Cd with DRS	Cd without DRS
0.829	0.464	0.505
0.7	0.455	0.465
0.6	0.450	0.446
0.5	0.427	0.43

Final Geometry and Design

The final base bleed system, designed from CFD analysis, effectively channels external airflow into the rocket's wake to increase base pressure and decrease the recirculation zone. Integrated into the aft section, it has a bleed inlet above the fin can coupler, an internal bleed channel within the coupler structure, and a bleed outlet at the base next to the nozzle. The inlet is manufactured with carbon fiber/epoxy reinforcement, and the internal channel is made of carbon fiber manufactured by the lay-up process.

Table A.11: Base Bleed Inlet and Outlet Geometric Specifications

Dimension	Value
Inlet height	2.5 mm
Inlet width(bottom)	5.5 mm
Inlet width(top)	4.25 mm
Outlet arc angle	10 deg
Outlet height	3 mm
Outlet Duct length	270 mm

Validation and conclusion

Drag values obtained from the CFD simulations were compared with the drag predictions obtained from OpenRocket (ORK) trajectory simulations. The comparison showed that the base bleed configuration resulted in an approximately 8.11881% reduction in drag compared to the baseline configuration at maximum velocity, while the effectiveness decreases as velocity decreases. The CFD results indicate a measurable reduction in pressure drag due to the modification of the wake region behind the rocket base. These results confirm that the implemented base bleed geometry contributes to reducing base drag during the transonic coasting phase, which can lead to improved aerodynamic efficiency and potentially higher apogee performance.

Overall, the base bleed system demonstrates a measurable improvement in aerodynamic performance during the transonic coasting phase. By reducing base drag without requiring additional propellant or active flow control systems, the design provides a lightweight and passive solution that can contribute to improved flight efficiency and potentially higher apogee performance.

Simulations at 0.5 Mach Regime

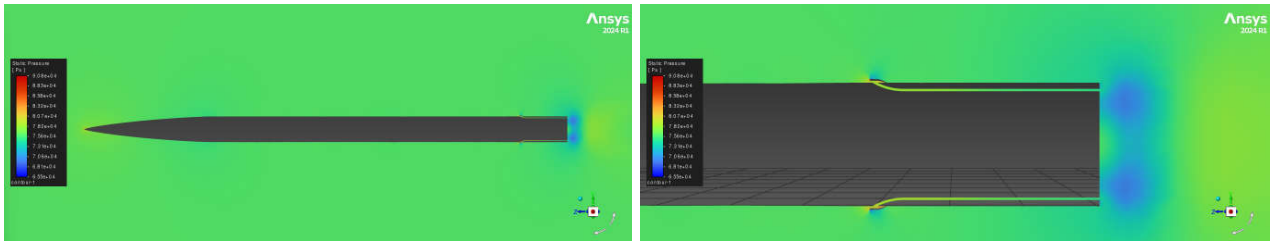


Figure A.132: Static Pressure Contours for 0.5 Mach Regime

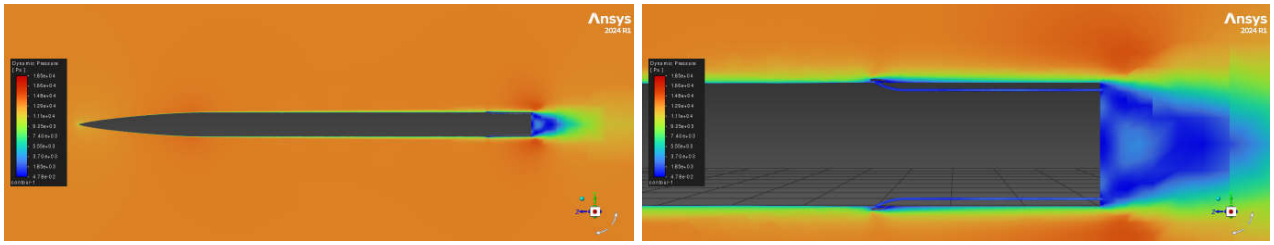


Figure A.133 : Dynamic Pressure Contours for 0.5 Mach Regime

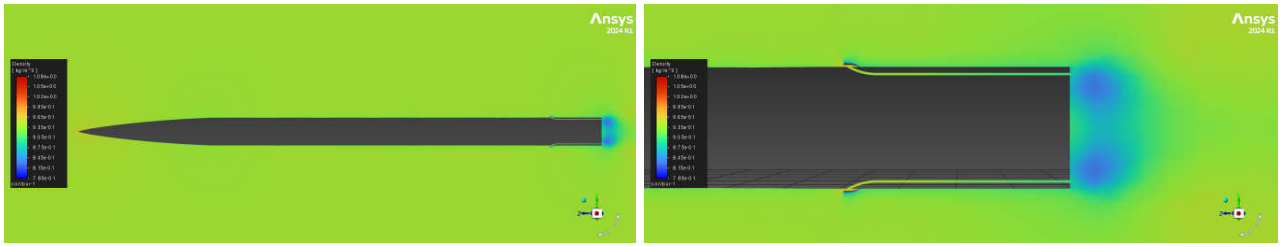


Figure A.134 : Density Contours for 0.5 Mach Regime

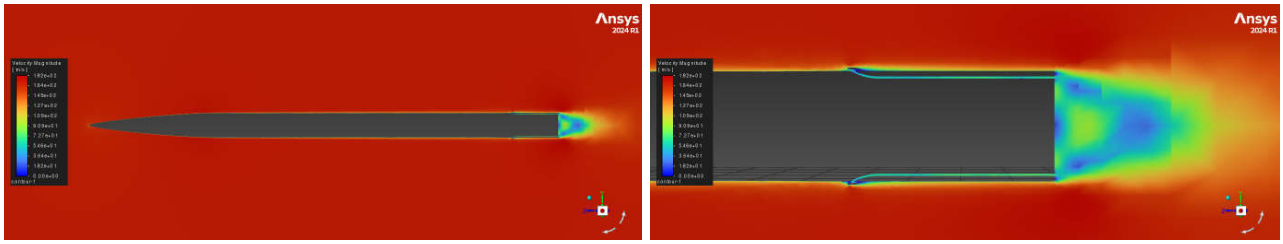


Figure A.135 : Velocity Magnitude Contours for 0.5 Mach Regime

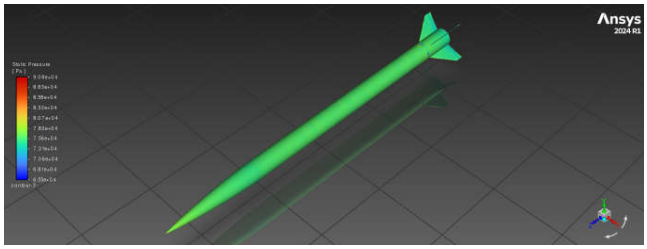


Figure A.136 : Static Pressure Contours for 0.5 Mach

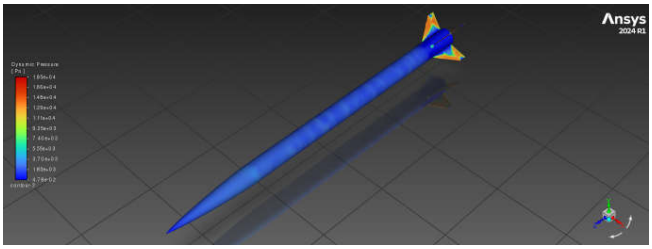


Figure A.137 : Dynamic Pressure Contours for 0.5 Mach

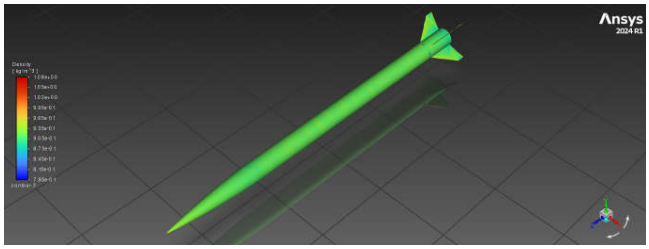


Figure A.138 : Density Contours for 0.5 Mach

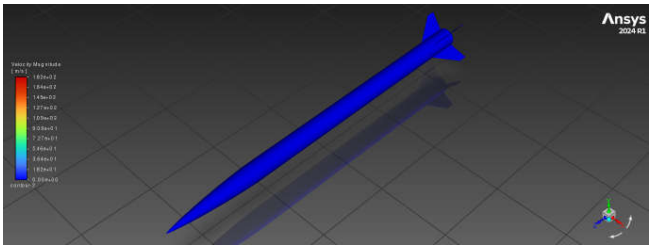


Figure A.139 : Velocity Magnitude Contours for 0.5 Mach

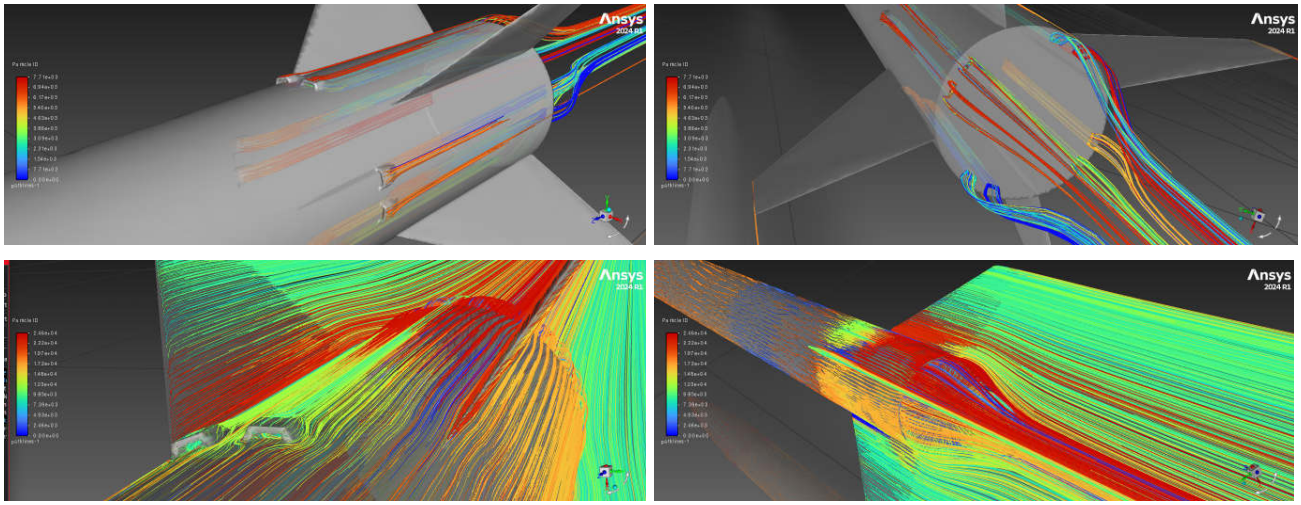


Figure A.140: Pathline Contours for 0.5 Mach Regime

Simulations at 0.6 Mach Regime

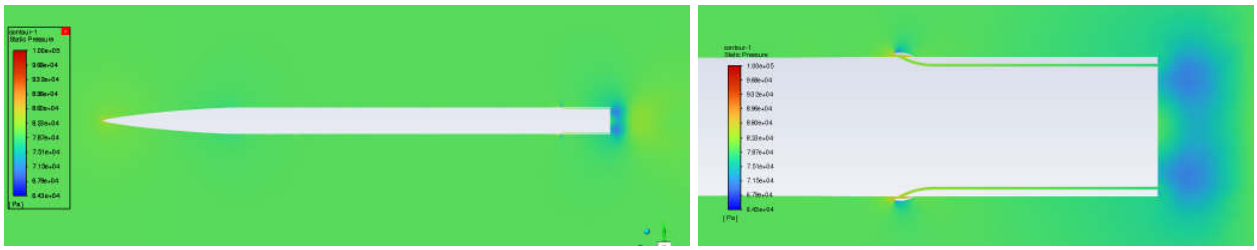


Figure A.141: Static Pressure Contours for 0.6 Mach Regime

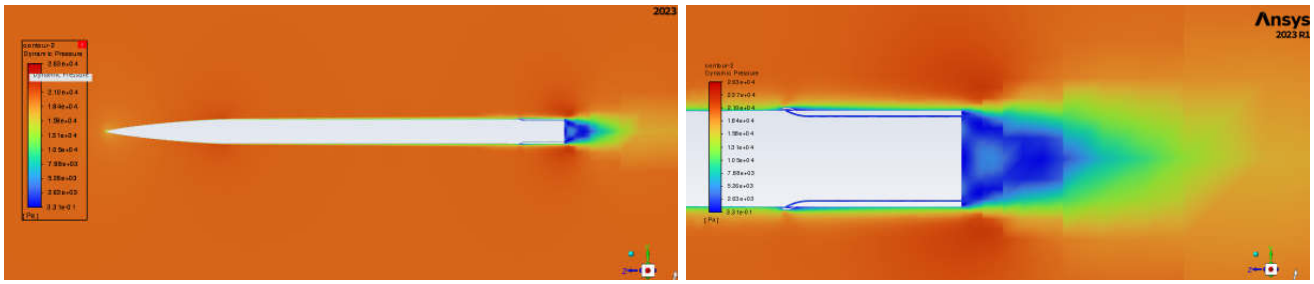


Figure A.142: Dynamic Pressure Contours for 0.6 Mach Regime

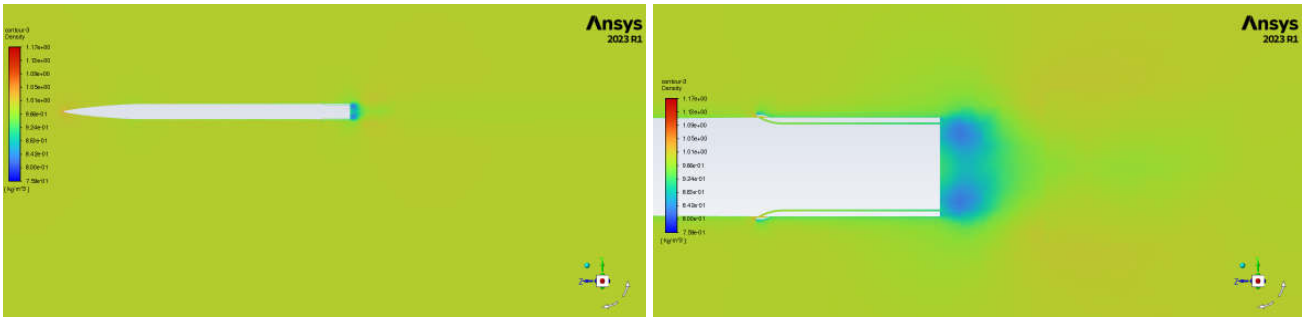


Figure A.143: Density Contours for 0.6 Mach Regime

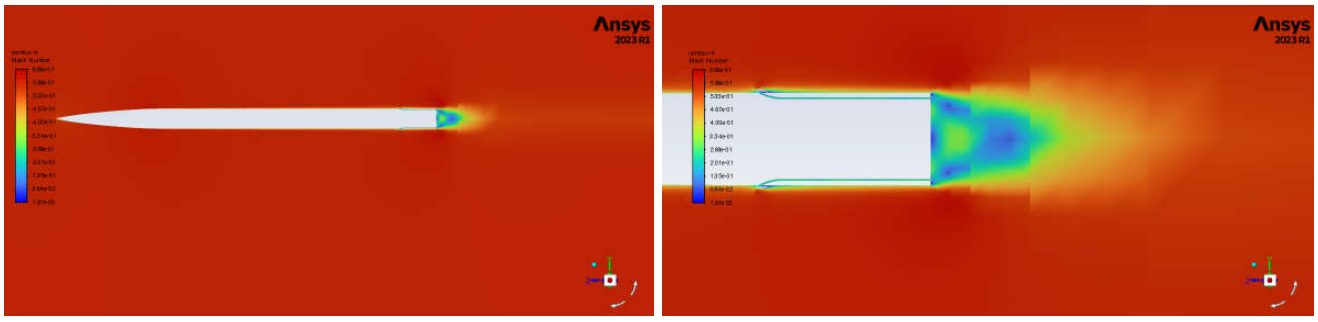


Figure A.144: Mach Contours for 0.6 Mach Regime

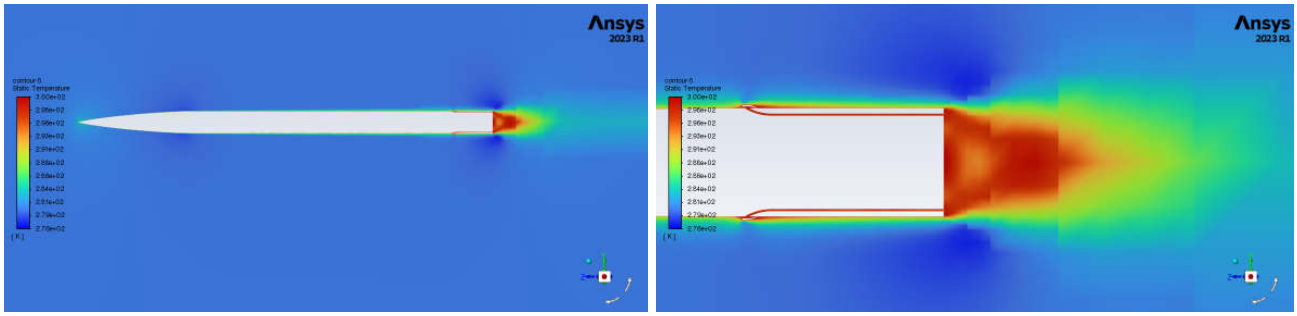


Figure A.145: Static Temperature for 0.6 Mach Regime

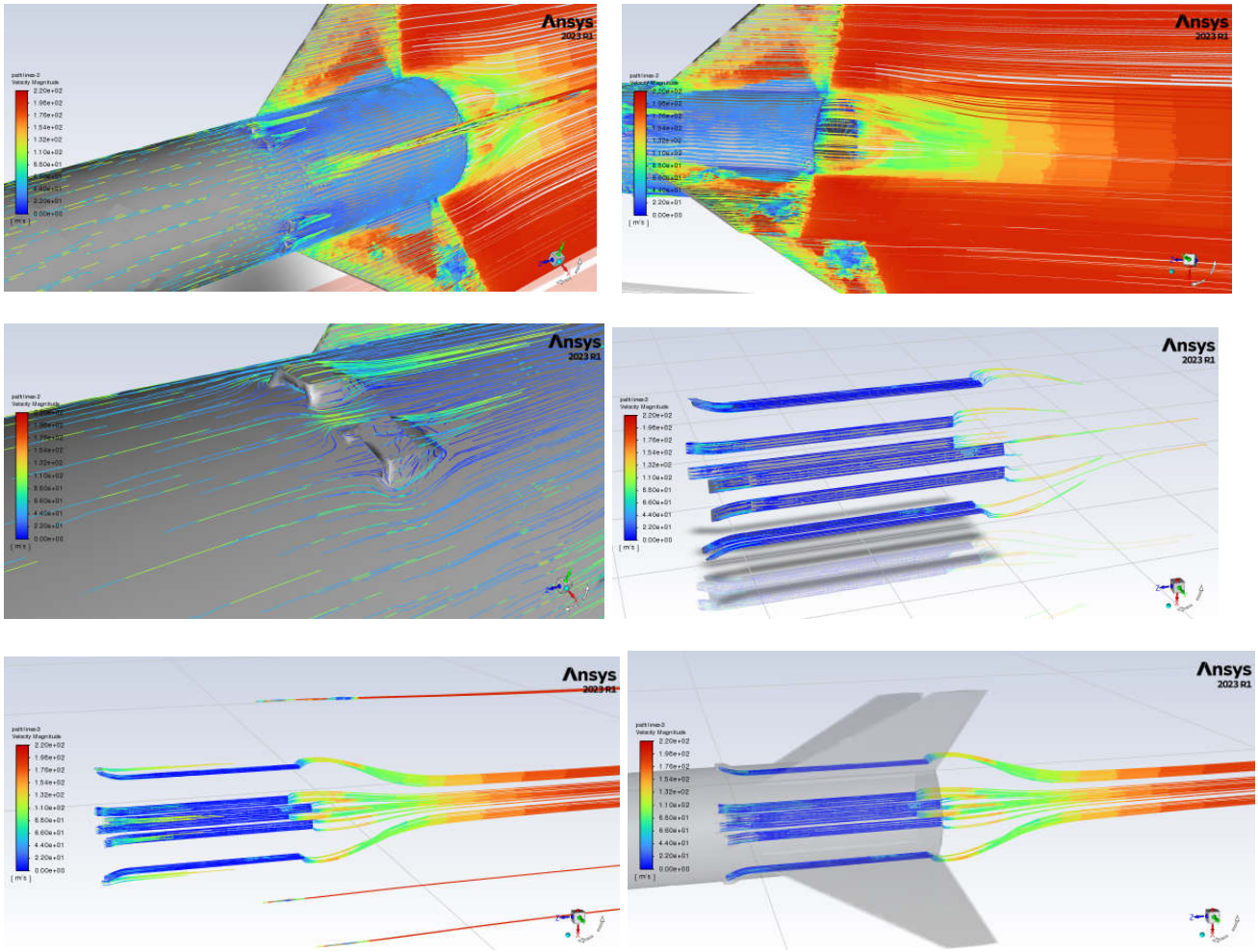


Figure A.146: Velocity Magnitude pathlines for 0.6 Mach Regime

Simulations at 0.7 Mach Regime

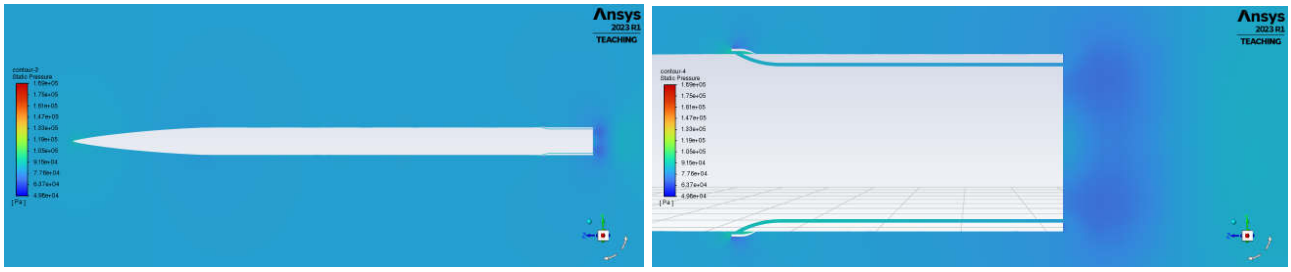


Figure A.147: Static Pressure Contours for 0.7 Mach Regime

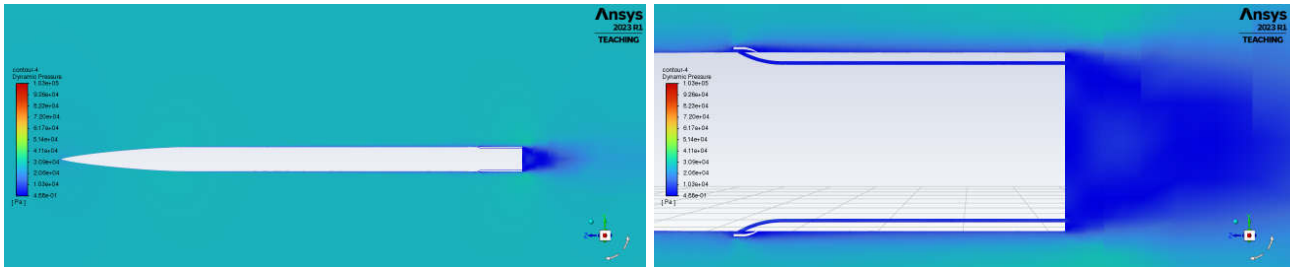


Figure A.148: Dynamic Pressure Contours for 0.7 Mach Regime

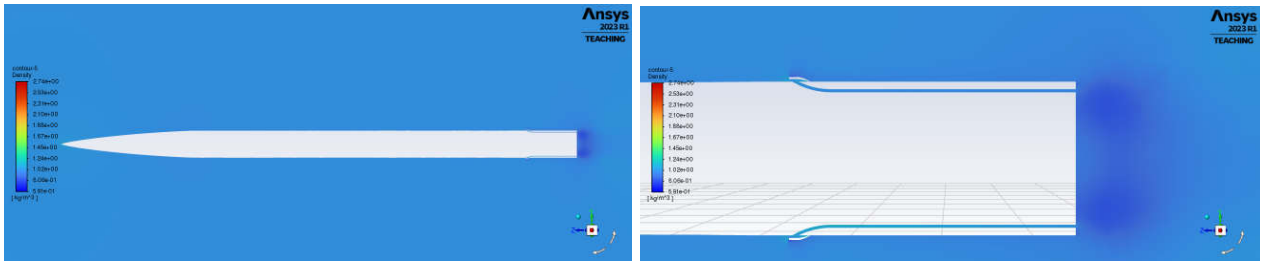


Figure A.149: Density Contours for 0.7 Mach Regime

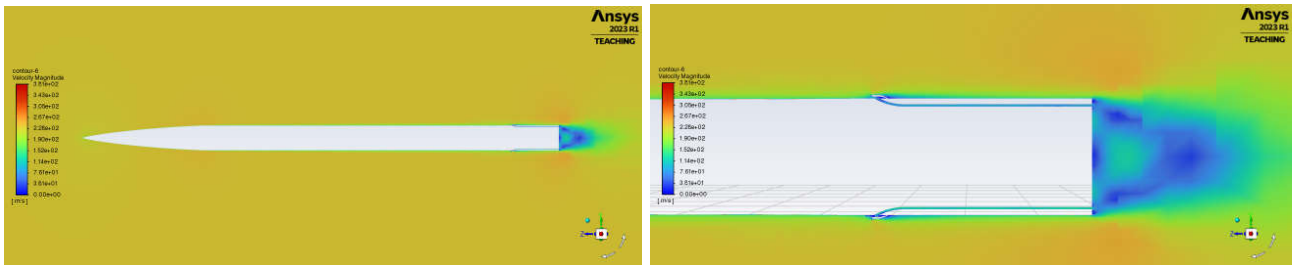


Figure A.150: Velocity Magnitude Contours for 0.7 Mach Regime

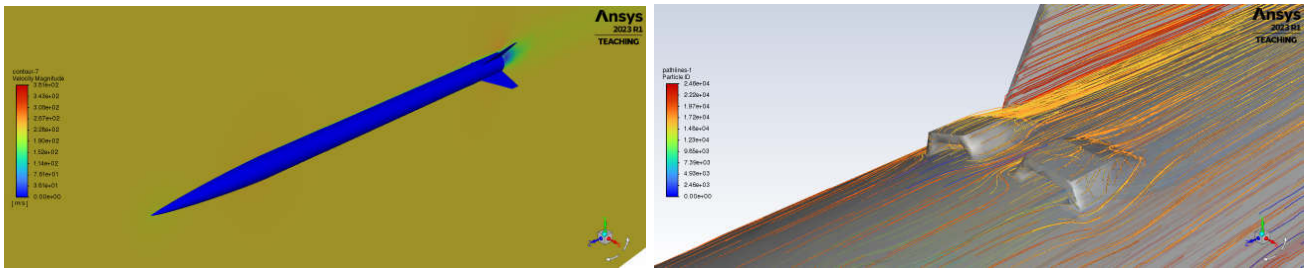


Figure A.151: Velocity Magnitude Contours for 0.7 Mach Regime

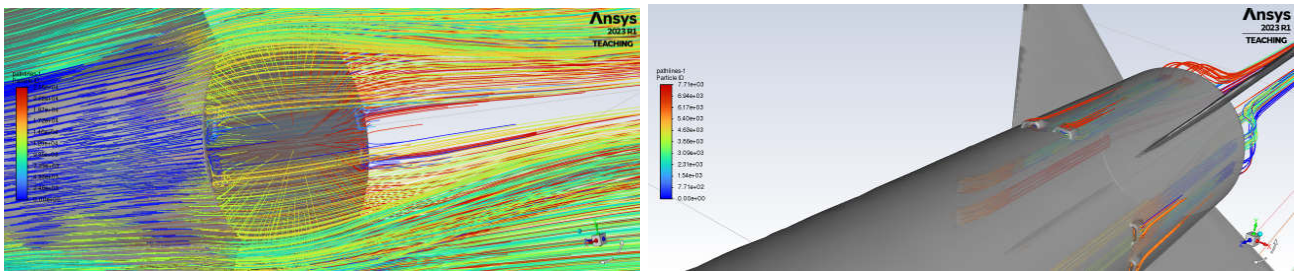


Figure A.152: Pathlines for 0.7 Mach Regime

Simulations at 0.875 (Max Q) Mach Regime



Figure A.153: Static Pressure Contours for 0.875 Mach Regime

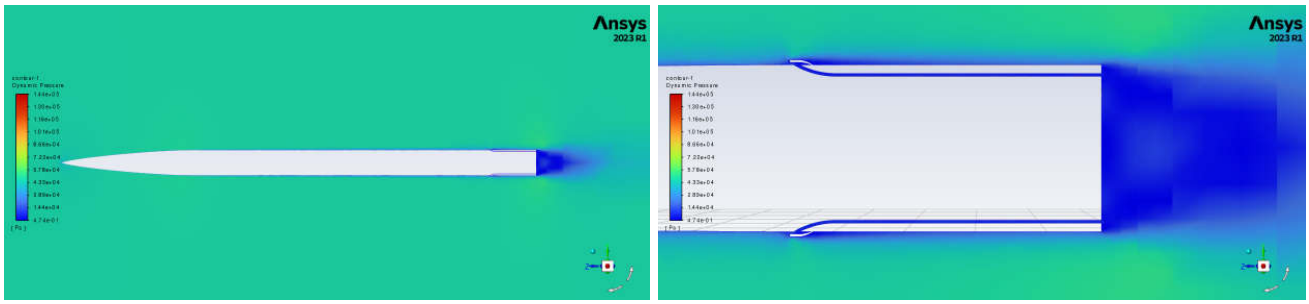


Figure A.154 : Dynamic Pressure Contours for 0.875 Mach Regime

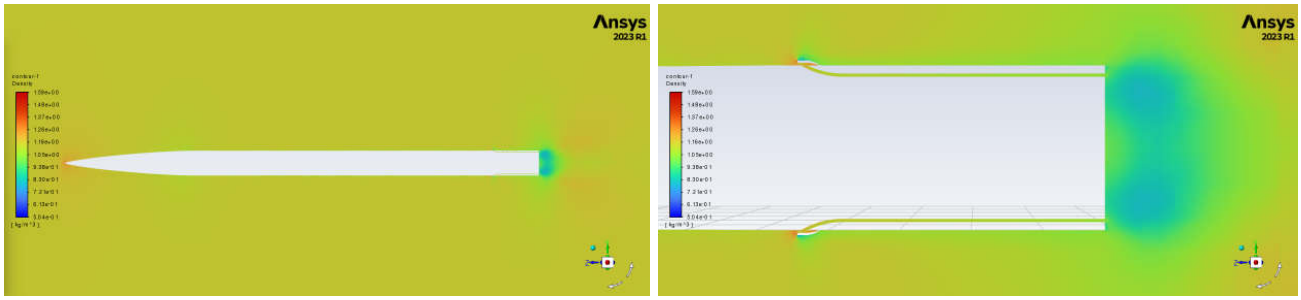


Figure A.155 : Density Contours for 0.875 Mach Regime

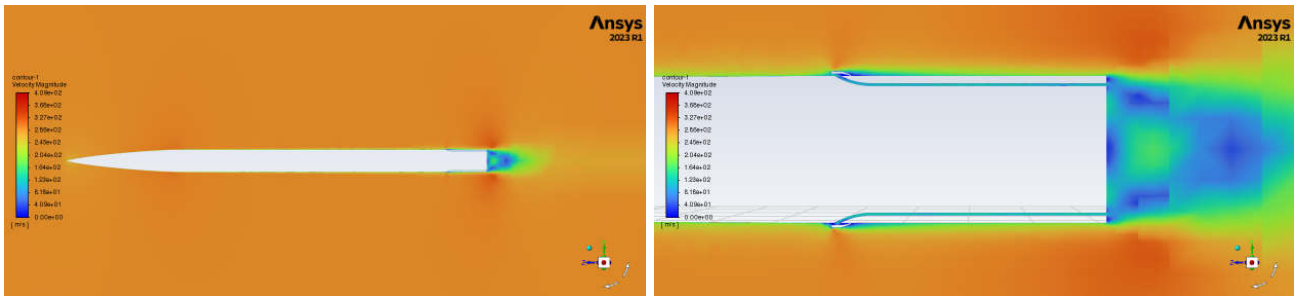


Figure A.156 :Velocity Magnitude Contours for 0.875 Mach Regime

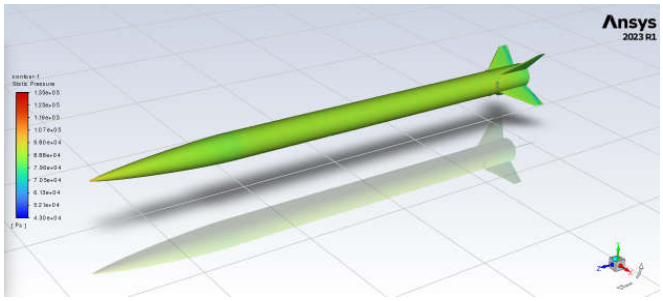


Figure A.157: Static Pressure Contours for 0.875 Mach

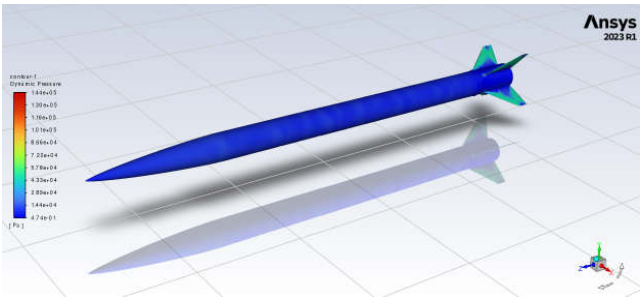


Figure A.158: Dynamic Pressure Contours for 0.875 Mach

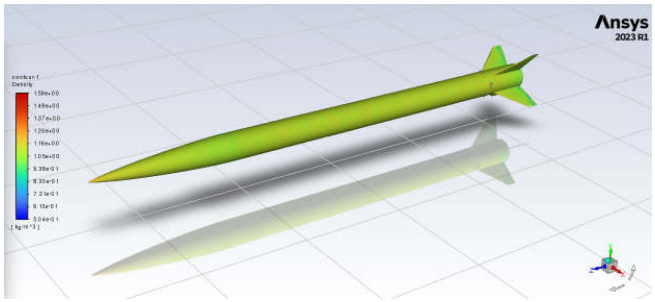


Figure A.159: Density Pressure Contours for 0.875 Mach

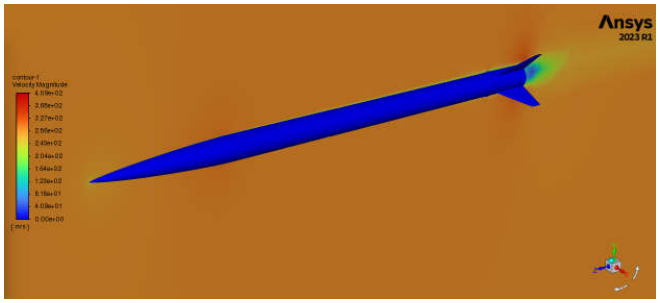


Figure A.160: Velocity Magnitude Contours for 0.875 Mach

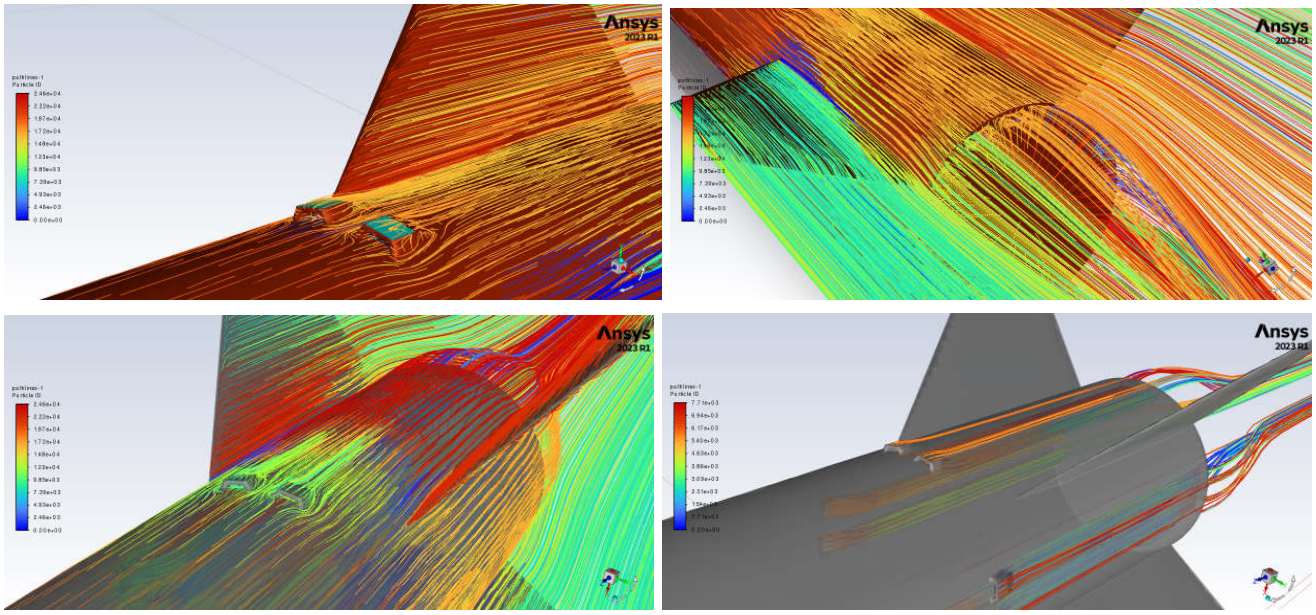


Figure A.161 : Pathlines for 0.875 Mach Regime

8 .Recovery Hardware and Couplers

Couplers

Finite Element Analysis (FEA) was performed on the coupler, which also functions as a load-bearing component of the recovery hardware, to evaluate its structural performance under representative operating conditions. The simulation incorporated maximum expected loads and material properties from the ANSYS materials library to ensure realistic and reliable results. The conditions considered were:

- 300 Nm bending moment applied on either side, transferred from the body tube
- 200 kPa internal pressure generated due to the black powder ejection charge
- Body tube material: glass fiber (ANSYS materials library)
- Coupler material: aluminum alloy (ANSYS materials library)
- Screws material: steel (ANSYS materials library)

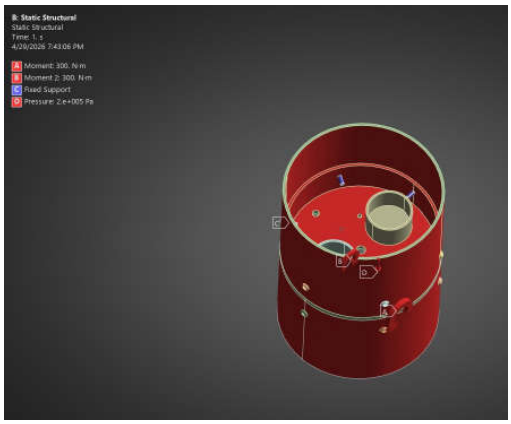


Figure A.162: Coupler Static Structural Simulation Setup

The results indicate that the coupler operates within acceptable structural limits under the specified loading conditions. The maximum deformation of 0.12 mm is minimal, ensuring that geometric integrity is maintained without significant distortion during operation. The maximum stress developed (45.3 MPa) is well below the tensile strength of aluminum (310 MPa), resulting in a safety margin of approximately 6.8. This confirms that the coupler is structurally robust and capable of reliably withstanding the applied bending moments and internal pressure loads.

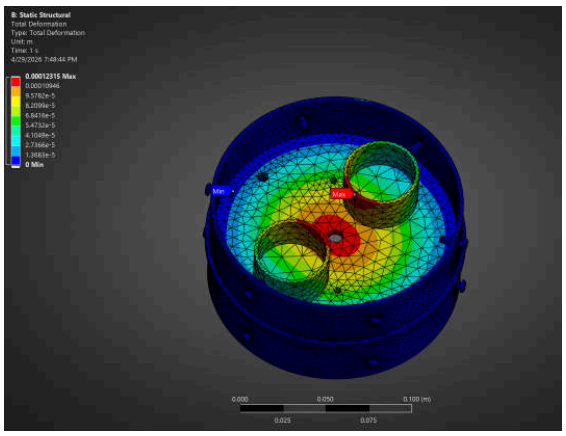


Figure A.163: Total deformation Contours

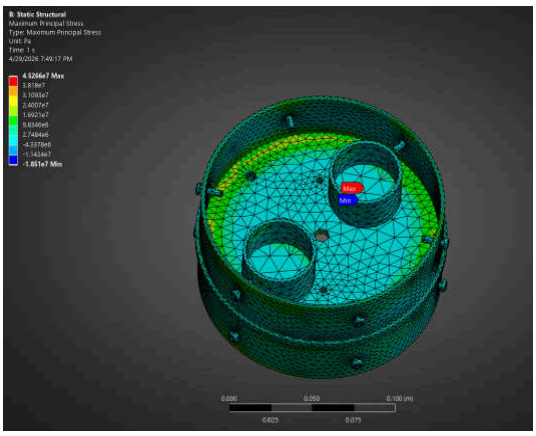


Figure A.164: Maximum Principal Stress Contours

9. Chargewells

FEA analysis was carried out to represent the loads generated during black powder ejection in the charge wells. The applied pressure value accounts for the peak pressure developed during combustion, ensuring a conservative and safe assessment. The boundary conditions were defined to reflect how the component is constrained within the assembly.

- **167 kPa** internal pressure (representing black powder ejection in charge wells)
- Fixed support at the bottom to represent structural constraint

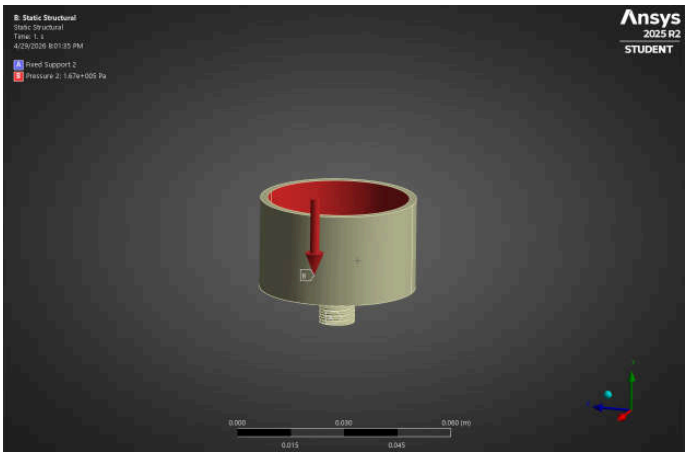


Figure A.165: Chargewell Static Structural Simulation Setup

The results indicate that the charge well performs well within acceptable structural limits under the applied loading conditions. The maximum deformation of **8 µm (0.008 mm)** is negligible, ensuring that the component maintains its geometry during operation. The maximum stress developed (**14.7 MPa**) is significantly lower than the tensile

strength of aluminum (**310 MPa**), resulting in a safety margin of approximately **21**. These results confirm that the charge well is structurally robust and capable of safely withstanding the pressures generated during black powder ejection.

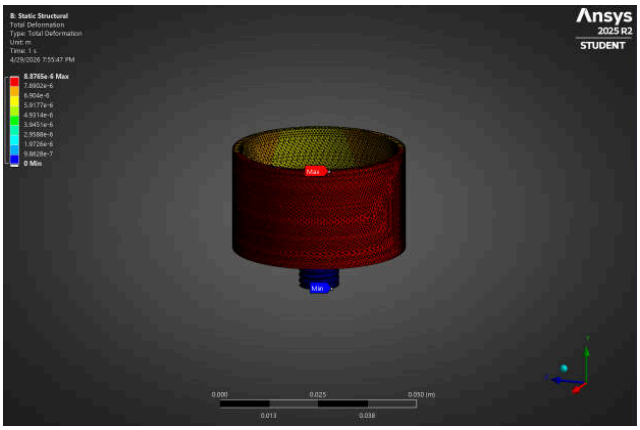


Figure A.166: Total Deformation Contours

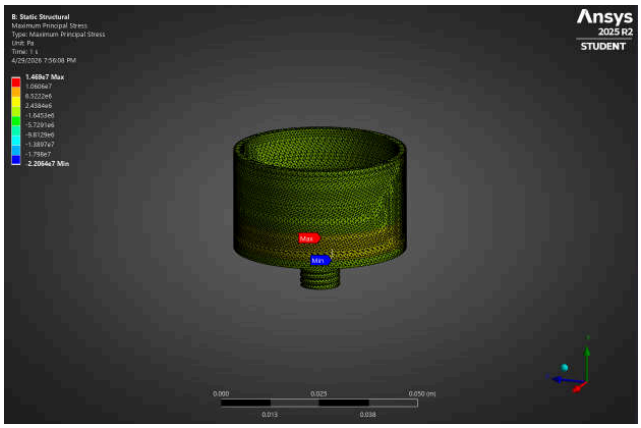


Figure A.167: Maximum Principal Stress Contours

10. Shock loading calculations

Total Velocity at Apogee

$\mathbf{V_{total}}$ = Total velocity at apogee (m/s)
$\mathbf{V_{horizontal}}$ = Horizontal velocity component at apogee = 28.16 m/s
$\mathbf{V_{vertical}}$ = Vertical velocity component at apogee = 19.62 m/s

$$V_{total} = \sqrt{V_{horizontal}^{2} = V_{vertical}^{2}}$$

$$V_{total} = \sqrt{28.16^{2} + 19.62^{2}}$$

$$V_{total} = 34.3 m/s$$

Shock Loading during Drogue Parachute Deployment

$\mathbf{F_D}$ = Shock load during drogue deployment (N)
m = Mass of the rocket (kg) = 35 kg
g = Acceleration due to gravity (m/s²) = 9.81 m/s²
$\mathbf{V_i}$ = Initial velocity – total velocity at apogee (m/s) = 34.3 m/s
$\mathbf{V_t}$ = Terminal descent velocity of drogue parachute (m/s) = 30 m/s

$$F_D = mg(\frac{V_i}{V_t})^2$$

$$F_D = (35kg)(9.81m/s^2)(\frac{34.3m/s}{30m/s})^2$$

$$F_D = 450N$$

Shock Loading during Main Parachute Deployment

F_M = Shock load during main parachute deployment (N)
m = Mass of the rocket (kg) = 35 kg
g = Acceleration due to gravity (m/s²) = 9.81 m/s²
V_i = Initial velocity – drogue terminal descent velocity (m/s) = 30 m/s
V_2 = Terminal descent velocity of main parachute (m/s) = 10 m/s

$$F_M = mg(\frac{Vi}{V2})^2$$

$$F_M = (35kg)(9.81m/s^2)(\frac{30m/s}{10m/s})^2$$

$$F_M = 3090.15N$$

Load rating on shock cords = 4500 lbs = **20,000 N**

$$3090.15N << 20,000N$$

11. Black Powder Pressure calculations

The pressure generated by the black powder charge is estimated using the ideal gas relation, given by:

$$PV = mRT$$

The volume of the chamber is calculated assuming a cylindrical geometry:

$$V = \frac{\pi}{4}D^2h$$

The force developed due to pressure is related by:

$$F = PA$$

For FFFF black powder, the gas constant is taken as $R = 118.9 J/kg{\cdot}K$. A mass of $m = 5\,g$ (i.e., $0.005\,kg$) is considered. The chamber dimensions are a diameter $D = 0.1473\,m$ and height $h = 0.35\,m$, with an assumed combustion temperature of $T = 1673\,K$.

Rearranging the ideal gas equation gives:

$$P = \frac{mRT}{V}$$

The chamber volume is calculated as:

$$V = \frac{\pi}{4}(0.1473)^2(0.35) = 5.964 \times 10^{-3}\,m^3$$

Substituting into the pressure equation:

$$P = \frac{0.005 \times 118.9 \times 1673}{5.964 \times 10^{-3}} = 1.668 \times 10^{5}\, Pa$$

To determine whether this pressure is sufficient for deployment, the force required to shear six nylon M4 screws is calculated. Assuming each screw has a shear strength of 300 N:

$$F_R = 6 \times 300 = 1800\, N$$

The corresponding pressure required to generate this force over the cross-sectional area is:

$$P_R = \frac{F_R}{A} = \frac{1800}{\frac{\pi}{4}(0.1473)^2} = 1.056 \times 10^{5}\, Pa$$

Thus, the generated pressure ($1.668 \times 10^{5}\, Pa$) exceeds the required pressure ($1.056 \times 10^{5}\, Pa$), indicating that the selected black powder mass is sufficient to shear the screws and ensure reliable deployment.

12. Fin Attachment

Simulation

The fin assembly structure was analyzed using a one-way Fluid–Structure Interaction (FSI) workflow to evaluate the aerodynamic loads and the resulting shear and principal stresses acting on the system. The pressure distribution obtained from the fluid simulation was transferred to the structural domain for stress analysis.

Geometry Setup

The fin-can model was initially assembled in SolidWorks and exported as an ***.igs*** file. A simplified, symmetric version of the geometry was then created in DesignModeler to prepare the model for analysis in ANSYS Fluent.

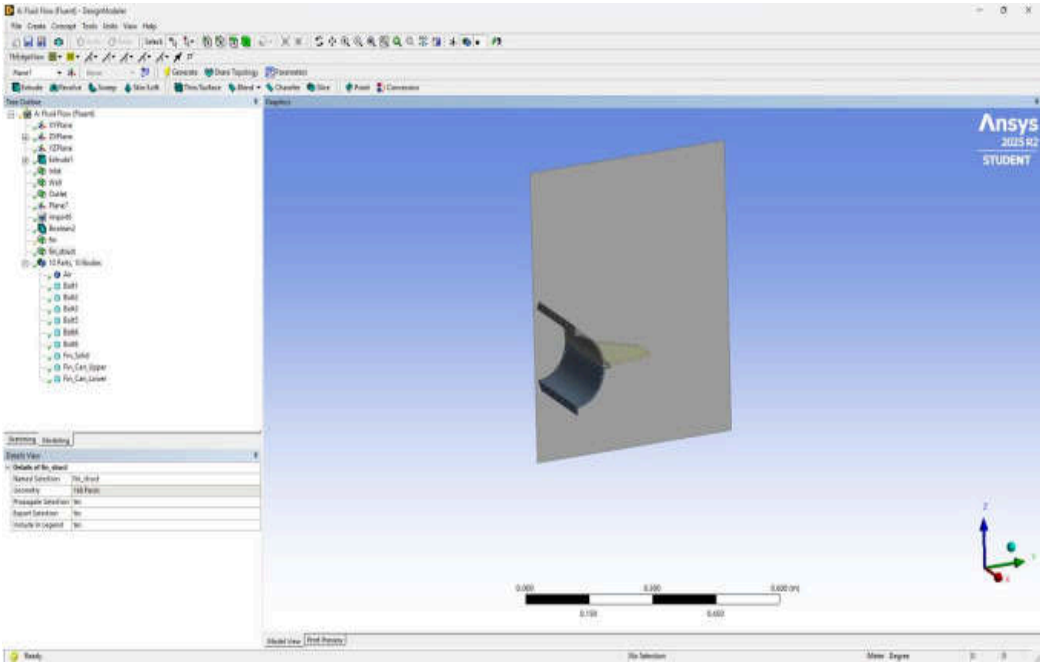


Figure A.168: Geometry Setup

FSI Analysis – Setup

The fluid domain was meshed, and appropriate boundary conditions were applied at the predefined inlet and outlet regions.

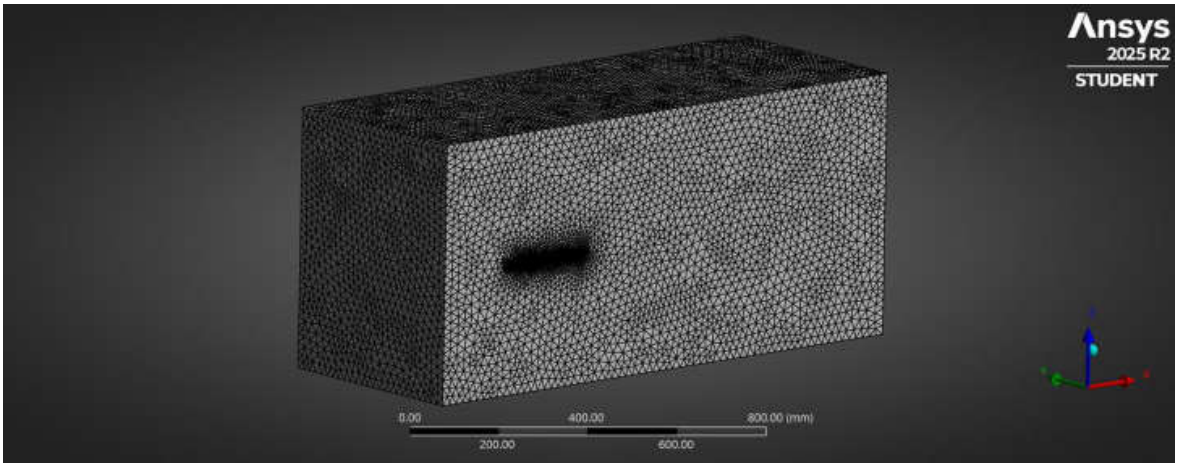


Figure A.169: Fluent Setup

Table A.12: Simulation Parameters

Parameter	Value
Operating Pressure	0 Pa (gauge reference)
Fluid Model	Ideal Gas (Air at ~2,300 ft conditions)
Cross-sectional Area	0.0304 m²
Inlet Gauge Pressure	93,000 Pa
Inlet Mach Number	1.5

The obtained drag coefficient is slightly elevated due to the inclusion of the fin-can attachment structure within the fluid domain. Overall, the solution shows good numerical stability, indicating that the results are reliable for further structural (FSI) analysis.

Table A.13: Results and Convergence

Parameter	Value
Drag Coefficient (Cd)	0.282
Iterations to Convergence	2000
Convergence Validation	Constant area-weighted average observed

The simulation demonstrated stable convergence and produced consistent aerodynamic results for the fin–can configuration. The slightly elevated drag coefficient is attributed to the inclusion of the fin assembly attachment structure within the computational domain. Overall, the solution exhibits good numerical stability and is considered reliable for subsequent structural (FSI) analysis.

FSI Analysis - Static Structural

The structural model was analyzed using a one-way FSI workflow by importing the fluid loads into the static structural module. Contact conditions were modified to ensure numerical stability and realistic load transfer.

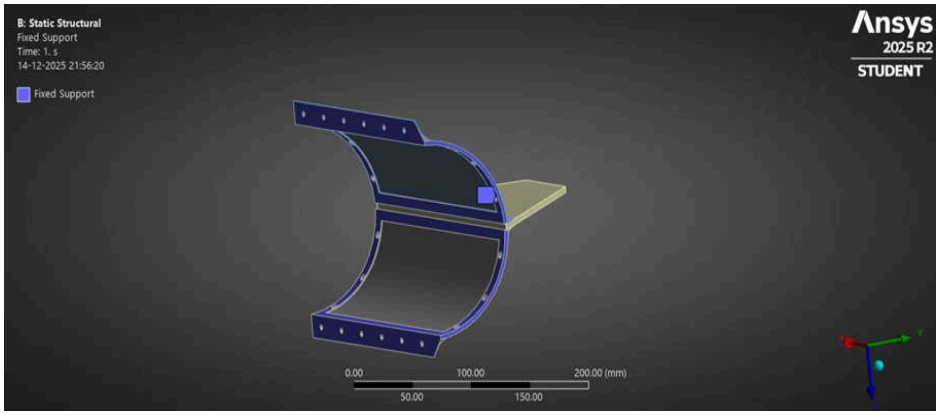


Figure A.170: Static Setup

The structural analysis was performed for a configuration utilizing six M4 bolts, and the resulting stress distribution indicates that the design performs adequately under the applied loading conditions. The maximum shear stress was found to be **62.3 MPa**, while the maximum principal stress reached **69.9 MPa**. These values are well within allowable limits, confirming that the fin assembly attachment possesses sufficient strength to withstand the projected loads. Based on a mild steel shear strength of **300 MPa**, the system achieves a **Factor Of Safety of approximately 4.8** in shear. It is important to note that the simulations were conducted at load conditions corresponding to a multiple of the expected Max-Q velocity, thereby providing a conservative assessment of the design's structural integrity.

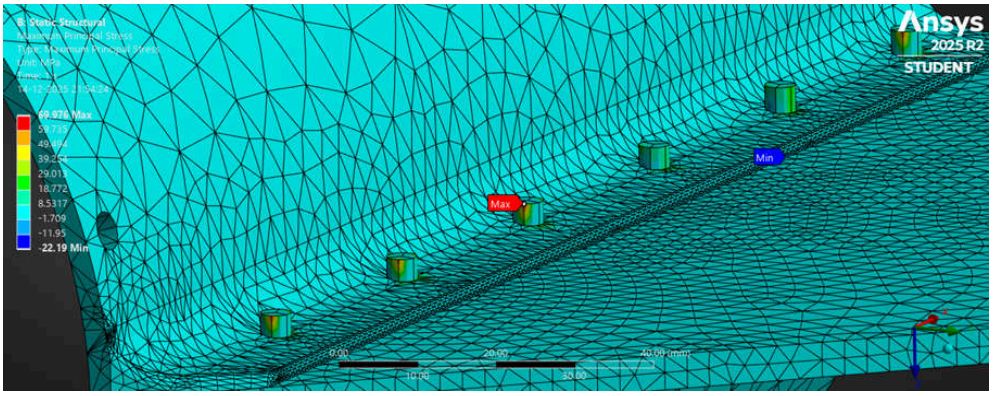


Figure A.171: Stress Plot

The same analysis methodology was applied to the complete structure to evaluate bolt loads in the coupler section and non-symmetric bolt–nut regions, ensuring comprehensive and reliable structural validation of the entire assembly.

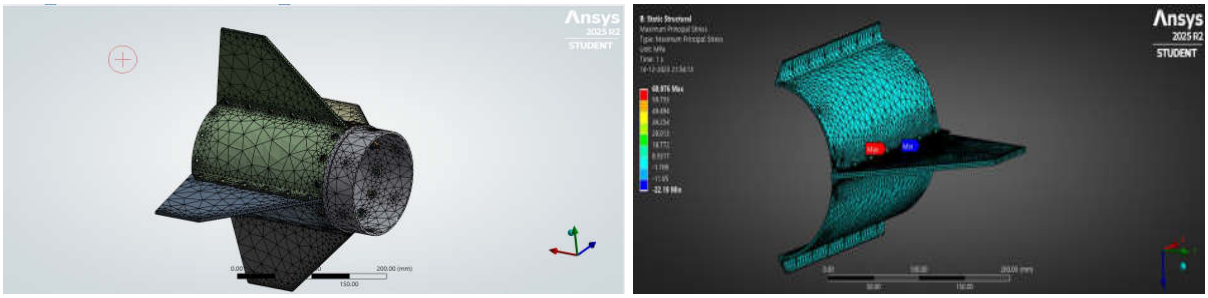


Figure A.172: Mesh and imported loads

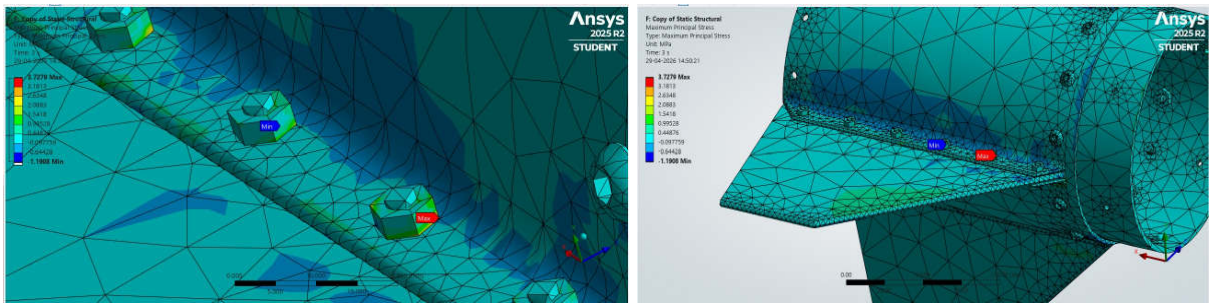


Figure A.173: Resultant Maximum Stress Plots

The extended structural analysis of the complete assembly confirms that load transfer through the fin assembly and coupler regions is well distributed, with no critical stress concentrations observed in the bolt nut interfaces or adjacent structural components.

The stress levels remain within allowable limits, indicating that the joints and fasteners are adequately sized for the applied loads. The non-symmetric regions were also evaluated to account for realistic loading scenarios, and the results demonstrate consistent structural performance across the assembly. Overall, the analysis validates the robustness of the design, confirming its ability to safely withstand operational loads under both symmetric and asymmetric conditions.

13. MATLAB-Based Flight Dynamics and Bending Moment Analysis

Four MATLAB/Simulink models were developed to evaluate bending moment loads at each of the six major coupler joints along the Udhbava airframe during powered ascent. The analysis progressed through three phases of increasing fidelity, starting from a simplified 1 DOF proof of concept model and advancing to a full 3 DOF six joint solver, and concluded with a continuous Bending Moment Diagram (BMD) across the full airframe span.

Structural Layout and Coupler Joint Stations

The rocket airframe is divided into six major structural segments, each separated by a coupler joint. Bending moments are evaluated at each joint location, measured from the nose tip. Segment masses reflect the full assembly of components within each zone, as the mass distribution directly governs inertial loading.

Table A.14: Data of Structural Layout

S. No.	Position (m)	Joint Name	Seg. Mass (kg)	Components
1	0.864	Nose-Payload Interface	5.647	Nose cone, Payload Bay
2	1.214	Drogue Coupler	2.693	Coupler + Drogue Tube
3	1.570	Avionics Bay	5.799	Coupler + Avionics Tube
4	1.810	Airbrake Interface	1.451	Coupler + Airbrake
5	1.897	Recovery Joint	2.300	Recovery Tube
6	1.996	Motor Mount	2.885	Motor Casing + Mount

Phase 1-Simplified Single-Axis (1-DOF) Model

The baseline model uses a single vertical axis with a fixed angle of attack (α = 7°) and a static center of gravity (CG). A seven point thrust approximation (peak 3,500 N, burnout time t = 4.5 s) drives the flight dynamics. Dynamic pressure q is extracted from Simulink at each time step and fed into:

$$M = q(S_{ref})(Cn_{\alpha})(\alpha)(|CP - CG|)$$

$$[Cn_{\alpha} = 2.0,\ \alpha = 7^{\circ}]$$

The Simulink model is invoked programmatically via Simulink.SimulationInput. Output: Bending Moment vs. Altitude and vs. Time, with peak annotated. Peak ~16.9 N·m occurs at t ≈ 2.5 s / 220 m.

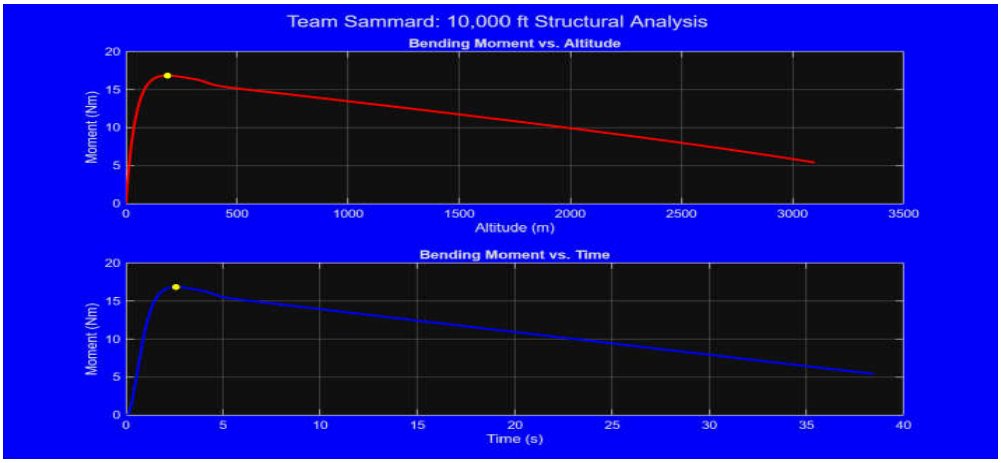


Figure A.174: BM vs. Altitude & Time

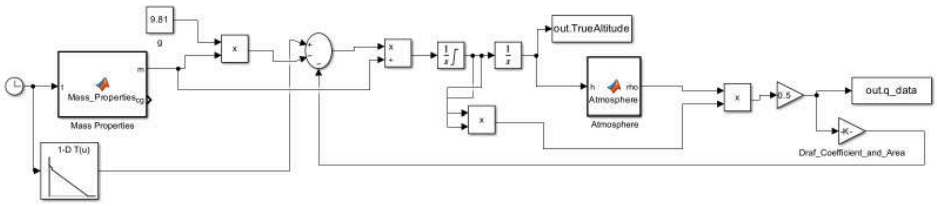


Figure A.175:Phase 1 block diagram

Phase 2 -Two-Axis (2-DOF) Model with Dynamic Parameters

Phase 2 introduces two key improvements: (1) a dynamic CG that shifts aft as propellant is consumed, and (2) a live angle-of-attack computed from velocity components rather than a fixed value. The launch angle is set to 84°, enabling downrange tracking. The updated bending moment formula is:

$$M = q(S_{ref})(Cn_{\alpha})(\alpha_{dynamic})(|CP - CG_{dynamic}|)$$

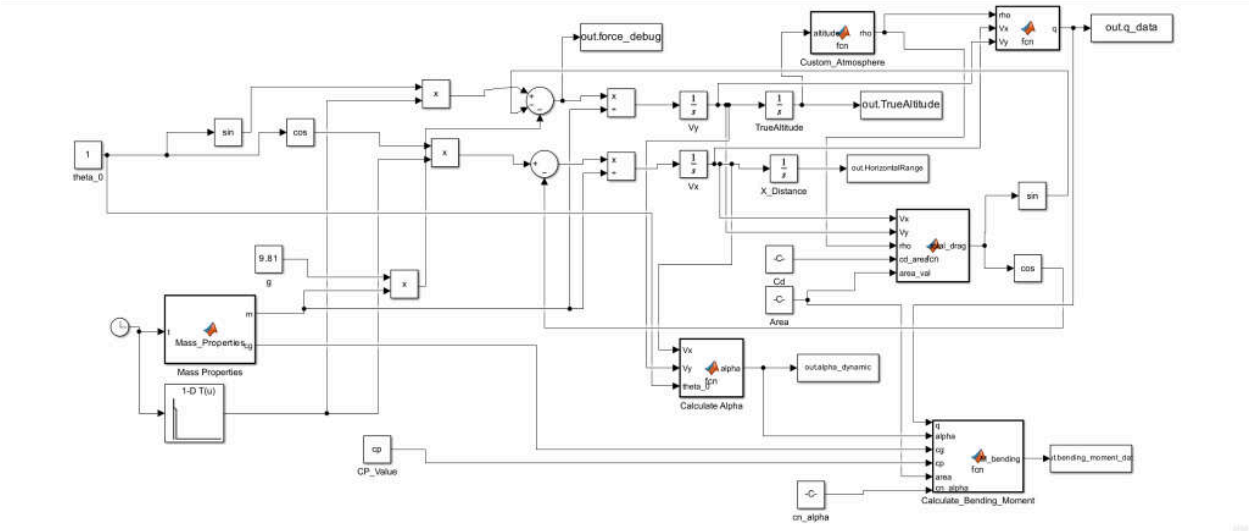


Figure. A.176: Phase 2 simulation block diagram two-axis flight dynamics with dynamic CG tracking, live angle-of-attack computation (Calculate Alpha subsystem), custom atmosphere model, and bending moment calculation subsystem.

The Phase 2 model produces bending moment as a function of altitude and, where available, a two-dimensional flight path plot showing downrange distance vs. altitude. The improved lever arm and angle-of-attack computation yield a more physically accurate estimate of structural loading throughout the burn.

Phase 3 -Full Three-Axis (3-DOF) High-Fidelity Model

The Phase 3 model is the highest-fidelity simulation. It computes bending moments simultaneously at all six coupler joints using the complete 168-point N2221 motor thrust curve, providing one data point every 30 ms from ignition (t = 0.03 s) to burnout (t = 5.04 s). Peak thrust is approximately 2,547 N at t ≈ 1.14 s. This resolution accurately captures the thrust rise, plateau, and tail-off phases.

Aerodynamic and Atmosphere Model

The COESA standard atmosphere is used for density, temperature, and pressure as a function of altitude. A time-varying normal force coefficient Cn is interpolated from four data points mapped to the flight simulation -increasing from 0.30 at ignition to 0.80 near apogee (t = 28.5 s) -to capture the aerodynamic evolution through transonic deceleration. Drag is evaluated through a dedicated Drag Logic subsystem using the optimised Cd = 0.33. The simulation halts automatically at apogee.

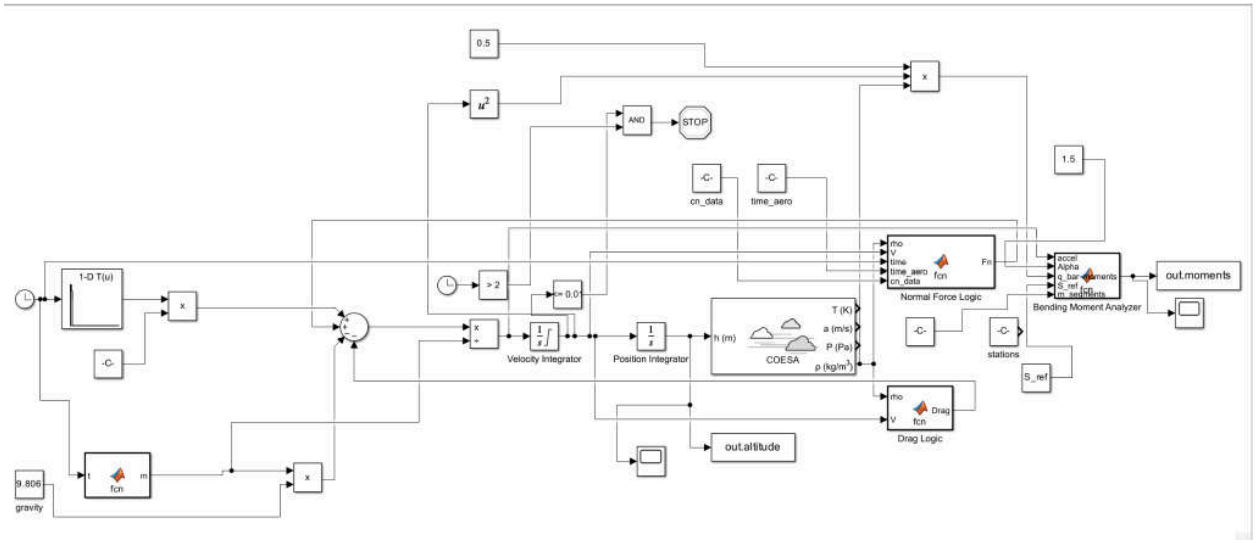


Figure A.177: Phase 3 simulation block diagram -full 3-DOF flight dynamics with COESA atmosphere, Normal Force Logic, Drag Logic, and Bending Moment Analyzer evaluating all six coupler joints simultaneously.

```
Station: Nose-Payload (0.864 m)
  >> Peak Moment: 72.59 N*m
  >> Occurred at: 4.20 s
  >> Altitude at Peak: 596.06 m (1955.56 ft)

Station: Drogue Coupler (1.214 m)
  >> Peak Moment: 119.16 N*m
  >> Occurred at: 4.57 s
  >> Altitude at Peak: 705.12 m (2313.40 ft)

Station: Avionics Bay (1.570 m)
  >> Peak Moment: 192.76 N*m
  >> Occurred at: 4.42 s
  >> Altitude at Peak: 659.94 m (2165.16 ft)

Station: Airbrake Interface (1.810 m)
  >> Peak Moment: 229.20 N*m
  >> Occurred at: 4.65 s
  >> Altitude at Peak: 729.74 m (2394.17 ft)

Station: Recovery Joint (1.897 m)
  >> Peak Moment: 273.42 N*m
  >> Occurred at: 4.76 s
  >> Altitude at Peak: 764.49 m (2508.16 ft)

Station: Motor Mount (1.997 m)
  >> Peak Moment: 322.55 N*m
  >> Occurred at: 4.80 s
  >> Altitude at Peak: 776.85 m (2548.73 ft)

----------------------------------------
```

Figure A.178: Console Output

Table A.15 : Phase 3 Results

Station Name and Coordinate (m)	Peak Moment
Nose-Payload (0.864m)	72.59 Nm
Drogue Coupler (1.214m)	119.16 Nm
Avionics Bay (1.570m)	192.76 Nm
Airbrake Interface (1.810m)	229.20 Nm
Recovery Joint (1.897m)	273.42 Nm
Motor Mount (1.997m)	322.55 Nm

Full-Length Continuous Bending Moment Diagram

A separate analysis generates a publication-quality BMD spanning the 2.712 m length of the rocket. Rather than evaluating only at the six discrete coupler joints, this model distributes the total aerodynamic side force across all structural segments proportionally to their mass, then accumulates moment contributions from upstream segments at each station:

$$M = \Sigma[F_{total}(\frac{m_j}{m_{total}})(x_i - x_j)] \ \ for\ any\ j\ upstream\ from\ station\ positioned\ at\ i$$

The angle of attack is derived from crosswind geometry at each timestep: α = atan(v_wind / v) for velocities above 10 m/s, with a conservative 5° floor at low speeds. Atmospheric density is modelled using the COESA standard atmosphere. The peak loading condition is identified within the powered flight window (t = 0.5 to 6.0 s), and a smooth 1,000-point interpolated curve is generated using piecewise cubic Hermite interpolation (pchip).

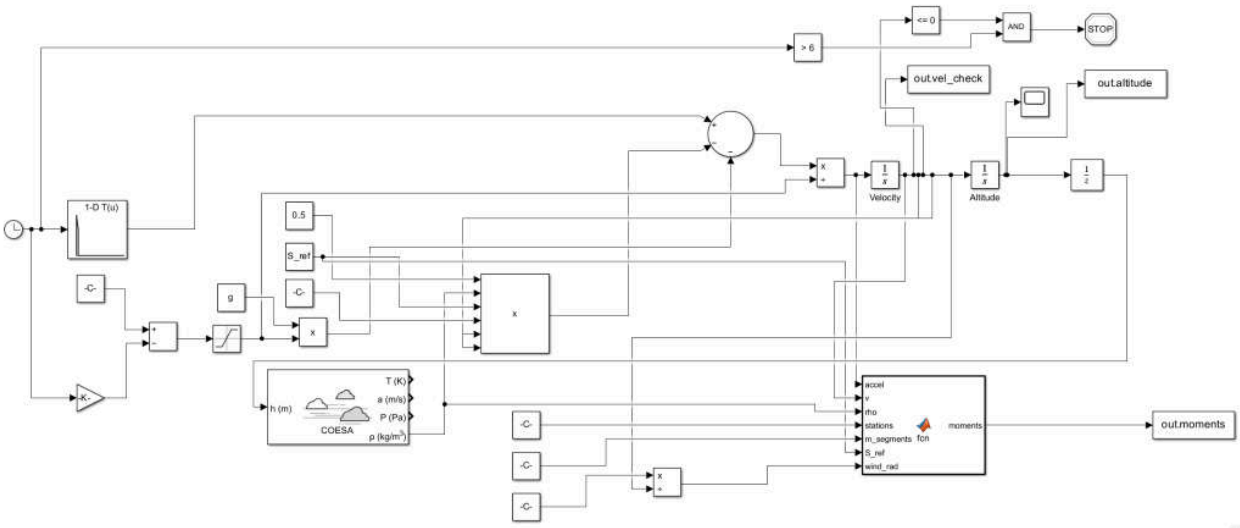


Figure. A.179: Continuous BMD simulation block diagram -full flight dynamics with COESA atmosphere, N2221 thrust lookup, velocity and altitude integrators, and a dedicated bending moment computation subsystem receiving wind speed, segment masses, and station positions.

Continuous BMD Output

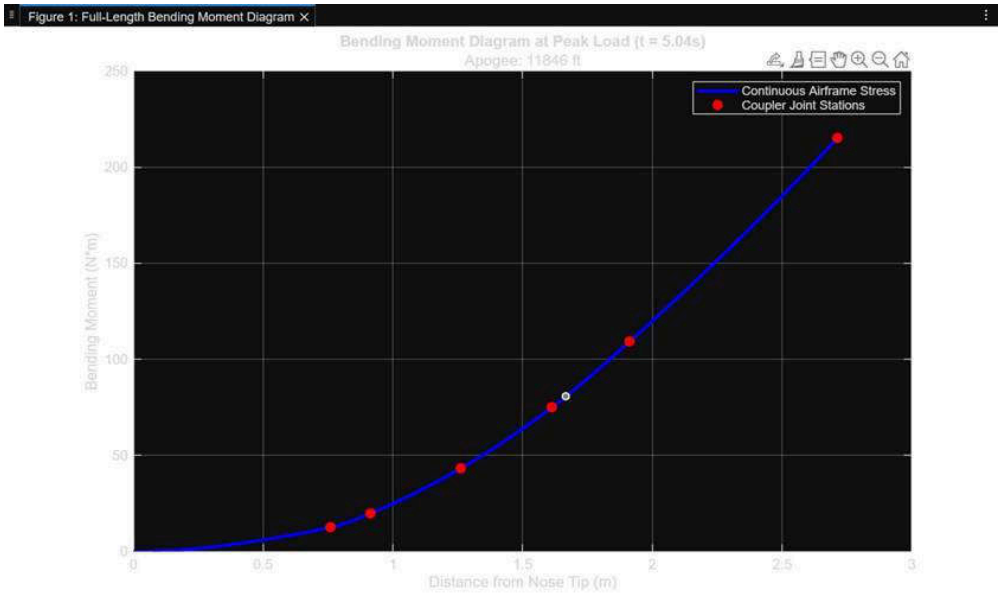


Figure A.180: Full-length Bending Moment Diagram at peak load (t = 5.04 s). Predicted apogee: 11,846 ft. Maximum bending moment of approximately 215 N·m occurs at the motor mount station (2.712 m). Red dots indicate the six coupler joint stations.

Key Parameters and Constants

The following table summarises the physical constants and aerodynamic parameters used across all simulation phases.

Table A.16 : Key Parameters of Rocket

Parameters	Symbol	Value	Applied In
Gravitational Acceleration	g	9.80665 m/s²	All Phases
Reference Area	S_{ref}	0.018146 m²	All Phases
Drag Coefficient (optimised)	Cd	0.33	Phase 3, Continuous BMD
Normal Force Coeff. (full model)	Cn_a	9.58	Continuous BMD
Normal force Coeff. (simplified)	Cn_α	2.0	Phase 1, Phase 2
Sea-level air density	ρ_o	1.225 kg/m³	Continuous BMD
Constant crosswind speed	v_{wind}	5.0 m/s	Continuous BMD
Launch angle	θ_o	84° from horizontal	Phase 2, Phase 3
Motor peak thrust	T_{max}	~2547 N at t ≈ 1.14 s	Phase 3
Motor burnout time	t_{burn}	5.04 s	Phase 3
CP location from nose	x_{CP}	2.117 m	Phase 1, Phase 2
CG location from nose (static)	x_{CG}	1.405 m	Phase 1

Assumptions and Limitations

Assumptions

- Atmospheric density decreases with altitude following the COESA standard atmosphere (Phase 3 and Continuous BMD); an exponential approximation is used in Phase 1/2.
- Aerodynamic forces are applied quasi-statically at each timestep; no unsteady or transient aerodynamic effects are modelled.
- Crosswind is modelled as a constant 5 m/s throughout the entire flight -a conservative simplification.
- Each segment's mass is assumed to be concentrated at its geometric centroid rather than distributed continuously.
- Bending moment contributions are accumulated from structural segments upstream of each joint, consistent with a cantilever beam loading model from the nose.

Limitations

- Phase 1 uses a simplified 7-point thrust approximation which may slightly under- or over-estimate peak dynamic pressure compared to the full 168-point dataset.
- No fatigue analysis, vibration analysis, or fin bending analysis is performed -only static peak bending moments at the main airframe joints are assessed.
- The angle of attack in Phase 1 does not account for wind gusts or aerodynamic damping effects from the fins.
- The simulation is limited to the ascent phase (up to apogee); descent structural loads are not evaluated.

14. Material Analysis

Characterization of Composite Fin Material

Udbhava's fins are manufactured using 3D-printed compression moulds and feature a hybrid composite of forged carbon fibre and fibreglass. This combination was selected due to its quasi-isotropic behaviour, arising from the random fibre orientation within the epoxy matrix, making it well-suited for handling multi-directional aerodynamic and structural loads.The compression moulding process enables near net-shape manufacturing of complex fin geometries using low-cost 3D-printed tooling, while ensuring good surface finish and repeatability.

To characterize the material, composite specimens were subjected to tensile testing using a Universal Testing Machine (UTM). A uniaxial load was applied at a constant strain rate of 2 mm/min until failure. The resulting engineering stress–strain data was used to determine the Young's Modulus, which was subsequently used to estimate the Shear Modulus for fin flutter analysis.



Figure A.181: Tensile Testing of Composite Specimen in UTM



Figure A.182: Fractured Specimens

The tensile composite sandwich specimen exhibited a Young's modulus of 14.9 GPa, an ultimate tensile strength of 204 MPa, a total elongation of 3.14%, and a shear modulus of 6.1 GPa. These properties were used for structural sizing and fin flutter margin evaluation to ensure reliable performance under flight conditions.The fractured specimens showed a break displacement of 4.71 mm, with visible fibre breakage and delamination, indicating typical brittle failure behaviour of composite materials.

Compression Testing of Body Tubes

Compression testing of the body tube indicated a maximum load-bearing capacity of **21,539 N**, corresponding to a compressive stress of **18.4 MPa**, prior to buckling failure.



Figure A.183: Failure During Compression Testing



Figure A.184: Load Indicator Showing Peak Compression Value

The compression test results confirm that the filament-wound body tube possesses sufficient axial load-bearing capacity, with failure governed by buckling as expected for thin-walled composite cylinders. The measured strength aligns with typical imperfection-sensitive behaviour, validating the structural adequacy of the design for flight conditions.Comparing the experimental failure load (**21,539 N**) to the maximum expected motor load (**~3,700 N**) yields a factor of safety of approximately **5.8**, indicating a robust structural margin against axial buckling

15. Shrota Pro - SRAD Recovery module

Shrota Pro is our custom-designed telemetry receiver and tracker system engineered for mission-critical tracking applications. Driven by the STM32H7 ARM Cortex-M7 microcontroller running at up to 480 MHz and powered by a 7.4V 2S 18650 lithium-ion battery bus with independent regulated power rails, it delivers stable, high performance operation across all subsystems. For positioning, Shrota Pro employs a dual GNSS architecture featuring the u-blox NEO-M8N and the L89 GNSS module for redundant satellite coverage, further complemented by an onboard BMM150 geomagnetic sensor for accurate heading estimation independent of GNSS availability.

The radio subsystem features a dual radio architecture integrating the LoRa RYLR998 for low-power wide-area communication, and the LoRa LLCC68 module resonant at 915 MHz band. A full-colour SPI-driven touchscreen provides real-time monitoring of coordinates, GNSS lock status, radio link quality, and system health without requiring any external device, while an onboard SD card interface enables local data logging for post-mission analysis.

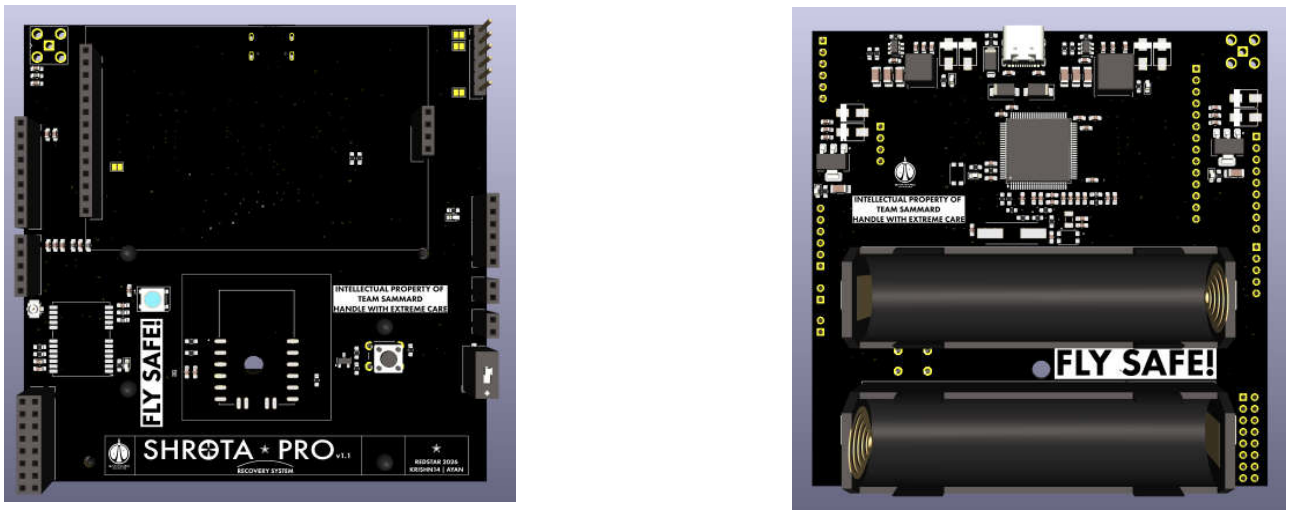


Figure A.185: Shrota Pro PCB

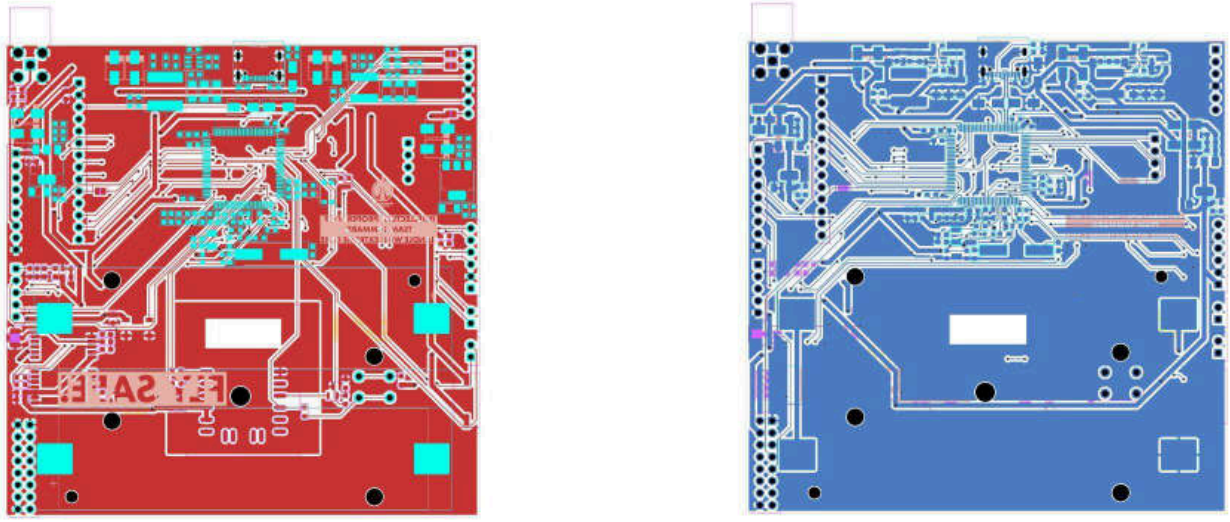


Fig A.186: Shrota Pro PCB Layout & Routing

16. Shrota Dongle - SRAD Telemetry Receiver

The Shrota dongle is the miniature counterpart of the Shrota Pro, designed to be easily installed onto the CyberDeck. It uses the STM32F4 ARM Cortex-M4 microcontroller at 168 MHz with an RYLR998 LoRa module interfaced with the four-stack Yagi antenna to receive data from the FC with minimal packet loss. The Shrota dongle uses a USB Type-C cable to receive power from the CyberDeck and communicate serially to the Raspberry Pi on the CyberDeck.

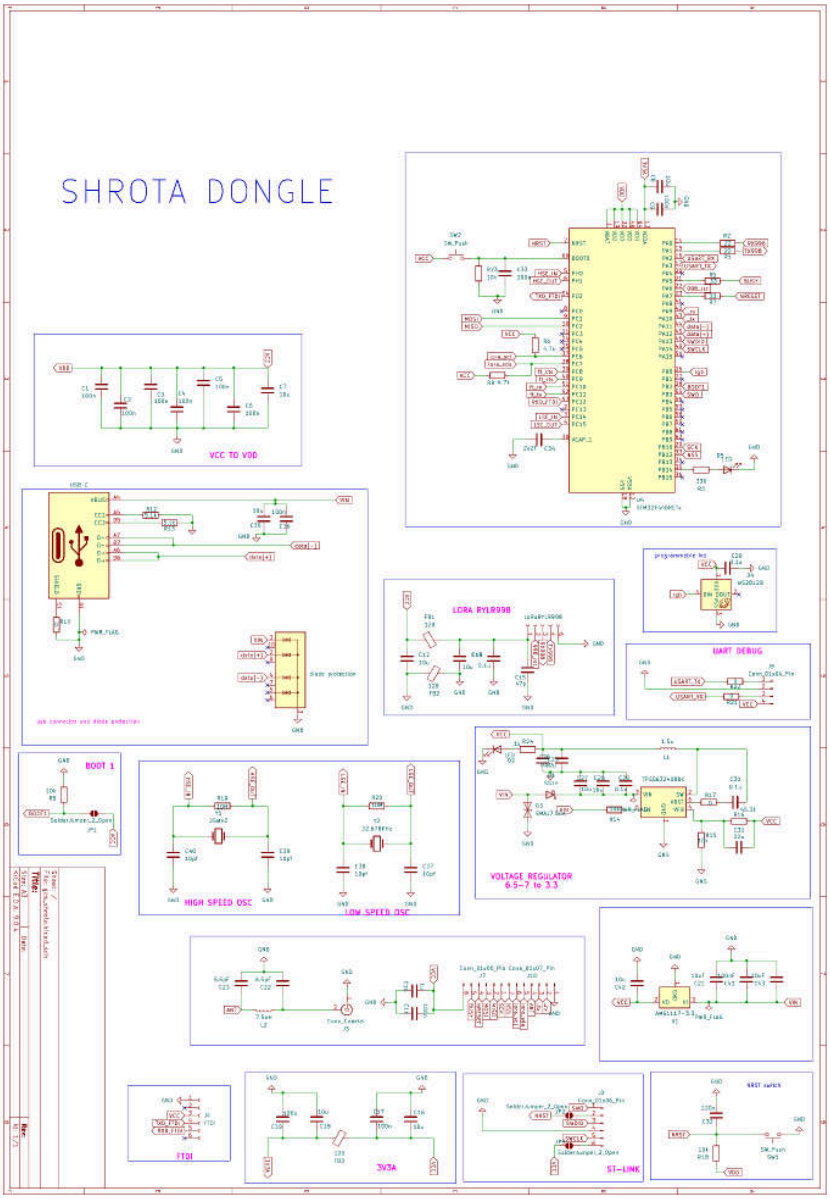


Figure A.187: Shrota Dongle Schematics

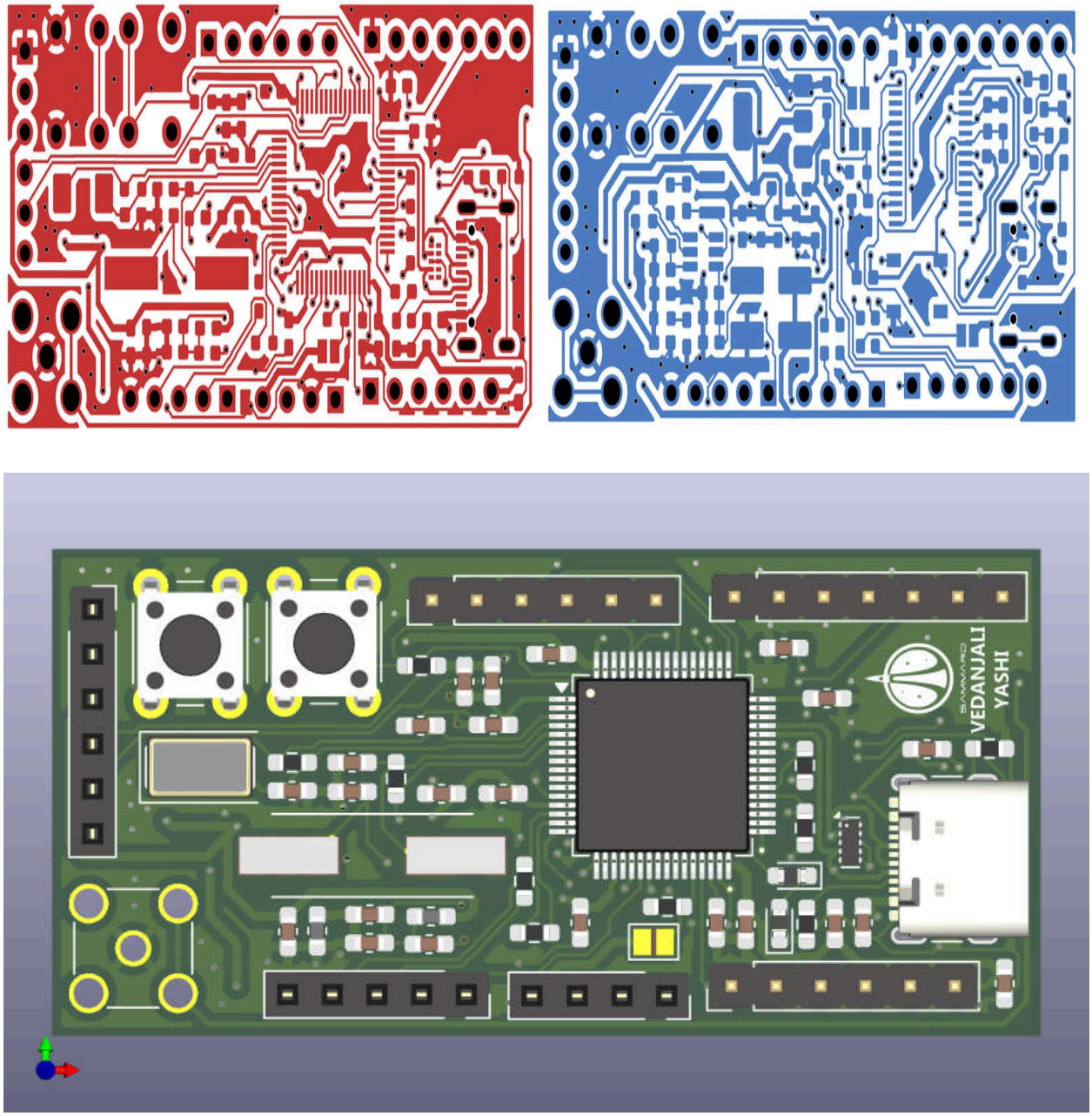


Figure A.188: Shrota Dongle PCB Layout and Routing

17. Live Video Transmission

To attempt the live rocket video challenge, our rocket houses an Analog Video transmission sub-system made out of COTS components which transmits video from an Analog FPV Camera (FPV Mini Camera COLOR 1200TVL) using an Analog Video Transmitter (TBS Unify Pro 5G8 V3) both of which receive their power from a miniaturized PDB solely for the purpose of providing 5V to the VTx (Video Transmitter) and 11.1V to the Camera. The Unify Pro is attached to a 5.8GHz omni-directional antenna that transmits the video feed at 5.8GHz 800mW to the pyramidal horn antenna mounted on the ATS to ensure lossless and high-speed video transmission at 480p.

This video feed is then displayed on the CyberDeck which hosts our GCS software which displays and records the live video feed for the live video challenge and future analysis.

Long Range Communication Systems : We have two antenna based subsystems for long range communication with the rocket that have separate functionalities but work together hand in hand to form an integrated system. These subsystems are: (a)SRAD Yagi 4 Stack Telemetry System and (b) SRAD Horn Antenna For Live Video System

18. SRAD Yagi 4 Stack Telemetry System

To achieve reliable live telemetry over extended ranges, our team developed a Yagi stack antenna system comprising four Yagi antennas arranged in a 2×2 configuration, with a vertical separation of 75 cm and a horizontal separation of 86 cm between elements.
The 4-stack architecture was selected for its increased effective aperture, improved signal-to-noise ratio, and enhanced directivity and gain. Given that the onboard transmitter employs an omnidirectional helical antenna at 915 MHz, the high-gain ground station receiver ensures robust signal acquisition across the full operational range.
A SRAD RF combiner PCB was developed in-house to coherently combine signals from all four antenna elements into a single output, alongside a COTS RF combiner for comparative evaluation. The combined signal is processed by our Shrota Ground Receiver PCB, which renders real-time telemetry and forwards flight parameters to the ATS system.
The array was simulated in Ansys HFSS, yielding a gain of approximately **16.5 dBi at 915 MHz** and S(1,1) parameter was found to be -15dB, indicating a matching efficiency of 97%.

SRAD Yagi 4 Stack Antenna Simulation Results:

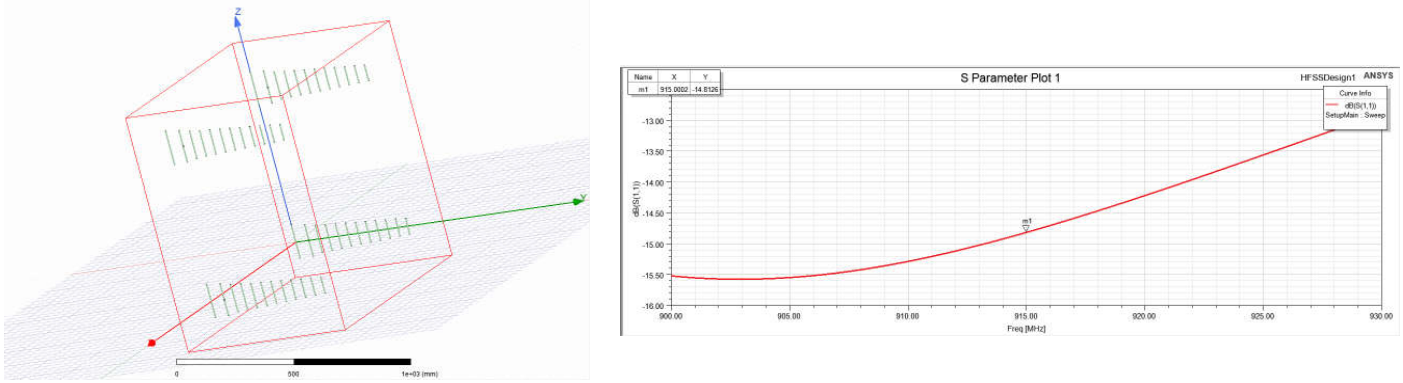


Figure A.189: Yagi 4 - Stack Simulation with S(1,1) return loss plot in dB.

The S(1,1) parameter is almost -15dB indicating a matching efficiency of almost 97%.

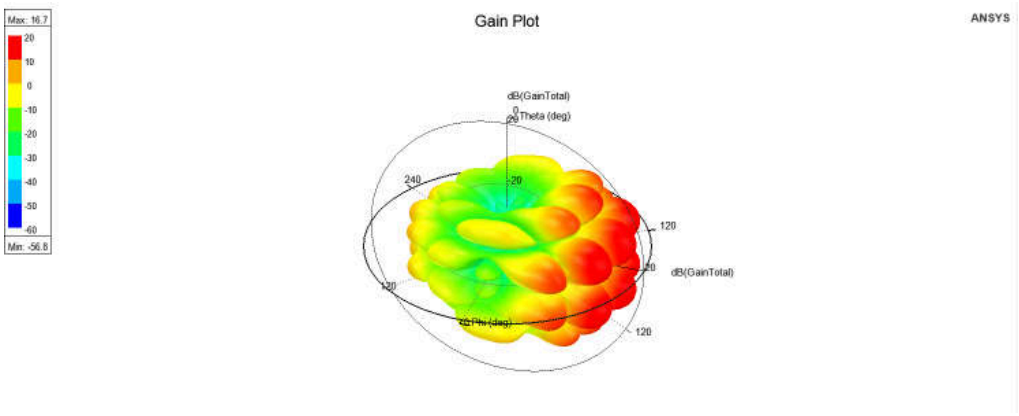


Figure A.190: Gain and Directivity of 4 - Stack Yagi Antenna

Photos of the SRAD RF Combiner are attached below:

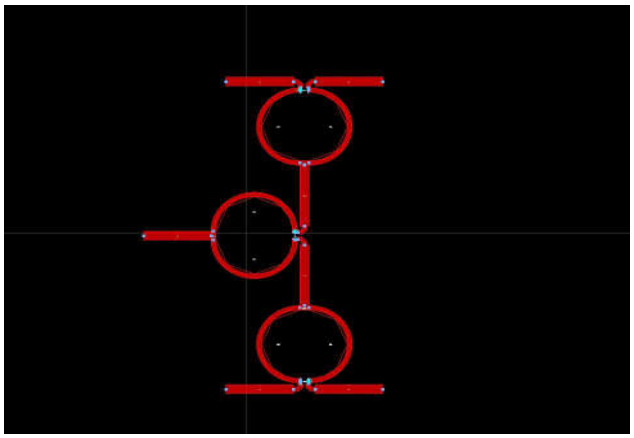


Figure A.191: RF Combiner PCB Routing

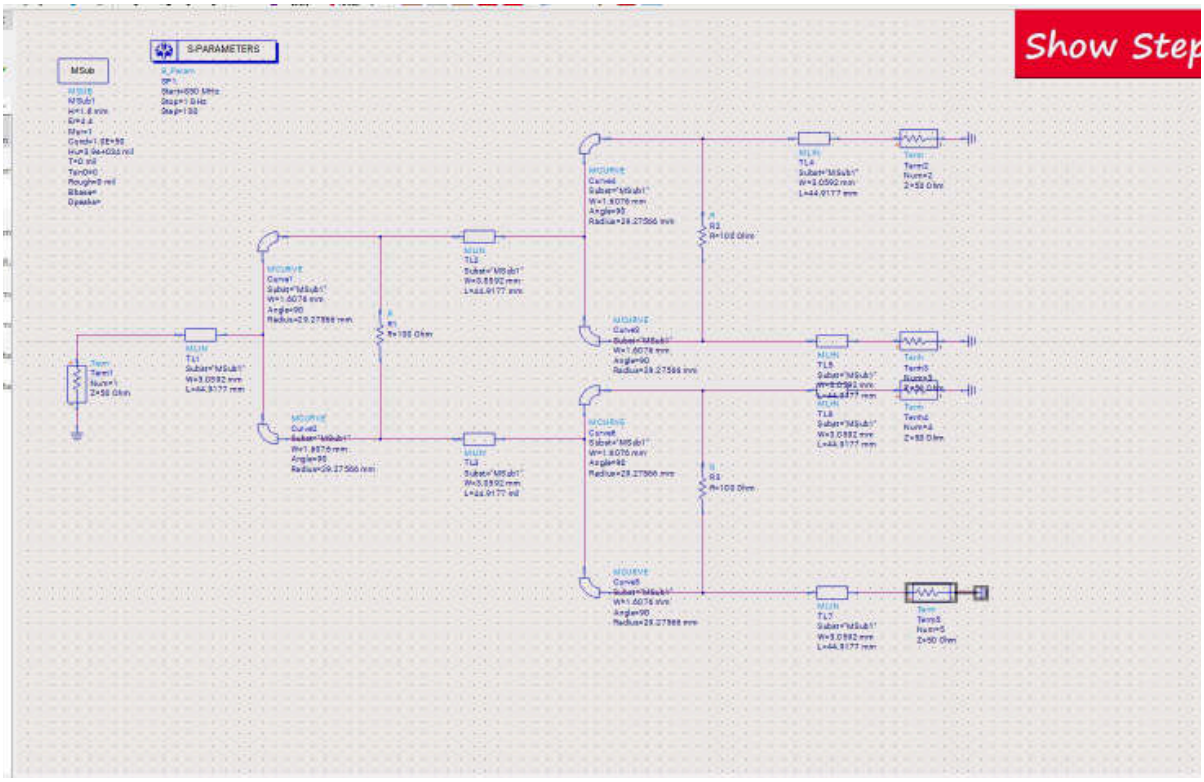


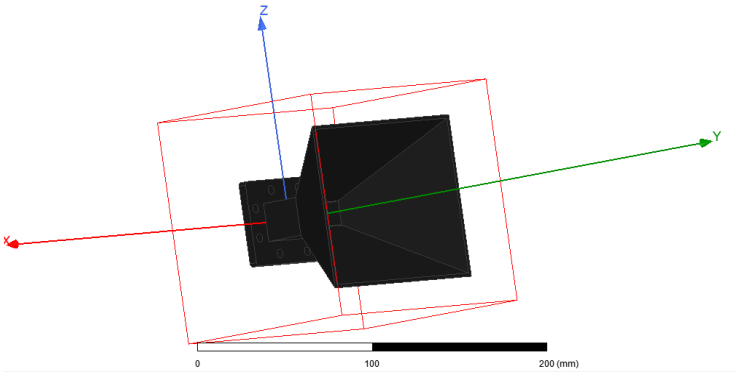
Figure A.193 - RF Combiner PCB Schematics

19. SRAD Horn Antenna for Live Video

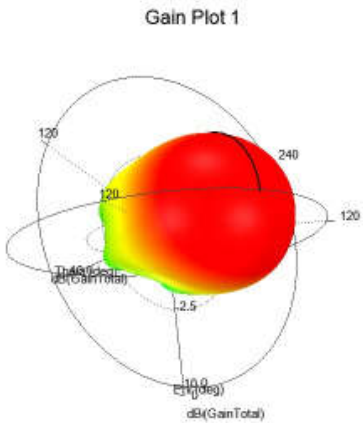
For IREC 2026, we came up with another objective: To capture live footage from our rocket in addition to storing video onboard. The SRAD Horn Antenna was developed for this purpose - to receive live video from onboard VTx with FPV Camera at 5.8GHz.

The horn antenna was specifically chosen for its directionality, stable radiation pattern, and high efficiency - extremely low reflection losses. The antenna is small and compact ensuring not only good gain values but also easy transportation. It is resonant at 5.8Ghz. The antenna was designed in such a way that it works for a range of frequencies from 4.5GHz to 7GHz. The antenna, since highly directional, does require proper positioning for signal acquisition and hence the antenna tracking system (ATS) was developed. This horn antenna would be mounted onto the ATS system.

The SRAD Horn antenna was simulated in Ansys HFSS and manufactured by 3D printing the horn shape out of PLA. Thin Copper Sheets were stuck onto the 3D printed surface, and a wr-137 waveguide adapter was attached to the bottom of the antenna to perform signal transition to a coaxial cable. The gain of this antenna was found to be 13.5dBi.



Fig*ure* A.194: Horn Antenna Simulation



Fig*ure* A.195: Gain and Directivity Plot of Horn Antenna

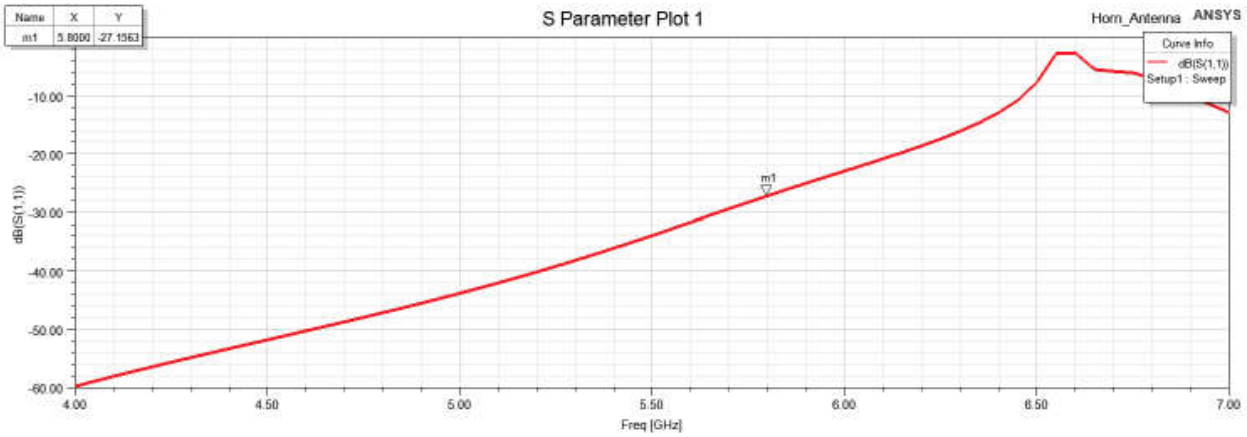


Figure A.196: S(1,1) Return Loss Parameter plot of the Horn Antenna

20. Sensor Suite for *BHAGAT* - Airbrakes PCB

The sensor suite consists of two precision barometric sensors and two inertial measurement units operating on a shared I2C bus interfaced with the ESP32-S3 WROOM 1 Microcontroller, along with auxiliary SPI, I2C, and UART General-Purpose Input/Output Pins (GPIOs) for interfacing any additional sensors.

Table A.16: Components List

Component	Manufacturer	Function	Sensitivity/Range	Key Features
Primary IMU: BMI088	 Bosch Sensortec	Primary inertial sensing for precise motion tracking.	Accel: up to ±24 g Gyro: up to ±2000°/s	Superior vibration resistance, stability under high dynamic loads, robust onboard filtering.
Secondary IMU: LSM6DSL	 STMicroelectronics	Secondary inertial sensing.	Accelerometer ±16 g Gyroscope ±2000 dps	Low-power operation, high-speed FIFO buffer, ensures uninterrupted data capture.
Primary Barometer: BMP390	 Bosch Sensortec	Primary barometric pressure sensing for altitude determination .	Pressure: 300–1250 hPa Relative Accuracy: ±0.03 hPa (±25 cm alt.)	High-precision, low noise floor, integrated IIR filter, critical for apogee detection and airbrake timing.
Secondary Barometer: BMP388	 Bosch Sensortec	Secondary barometric pressure sensing.	Pressure: Similar to BMP390 Relative Accuracy: ±0.08 hPa (±0.66 m alt.)	Independent measurement channel for continuous cross-validation, ensures uninterrupted sensing.

Flight Computer ESP32-S3-WROOM-1



The raw data from all four sensors is processed using sensor fusion techniques on the ESP32-S3-WROOM-1, a highly integrated wireless microcontroller module developed by Espressif Systems. Designed for low-power operation with deep-sleep modes and high processing throughput, the module ensures energy-efficient data acquisition and real-time decision-making throughout flight. Native support for Bluetooth Low Energy, Wi-Fi, and ESP-NOW wireless protocols further extends the module's utility for ground communication and pre-flight diagnostics. The module's rich I/O set and hardware encryption support provide the flexibility and security required for a flight-critical embedded system.

Data Logging SD Card Interface



An SD card interface provides removable, high-capacity storage for flight data. Operating over SPI, the interface supports high-frequency logging operations, enabling comprehensive post-flight analysis and data retrieval with 32MB of space.

Servo Actuation

BHAGAT integrates two independent servo channels, each designed to provide the necessary voltage supply, relay PWM control signals, and sustain the high currents demanded during both nominal operation and servo stall conditions. The resulting PWM commands drive a 7.4V digital high-torque servo that actuates the airbrake mechanism at or above 100 Hz, enabling real-time closed-loop drag control to precisely target the vehicle's apogee altitude.

Power Management and Reverse Polarity Protection

BHAGAT is powered by a 7.4V Battery during its functioning inside the vehicle through a reverse-polarity protection circuit, though it can be powered through 5V from a USB-C Cable on the ground for flashing code at the testbench. A TPS562200 Switching Regulator steps down these voltages to a stable 3.3V power supply that can be used to power the electronics on the board. An AMS1117-3.3V Linear Dropout Regulator is also available as a backup 3.3V supply. Another AMS1117-5V Linear Dropout Regulator powers the 5V bus. Auxiliary power pins are included for interfacing peripherals with any of the three voltages, respectively.

Layout

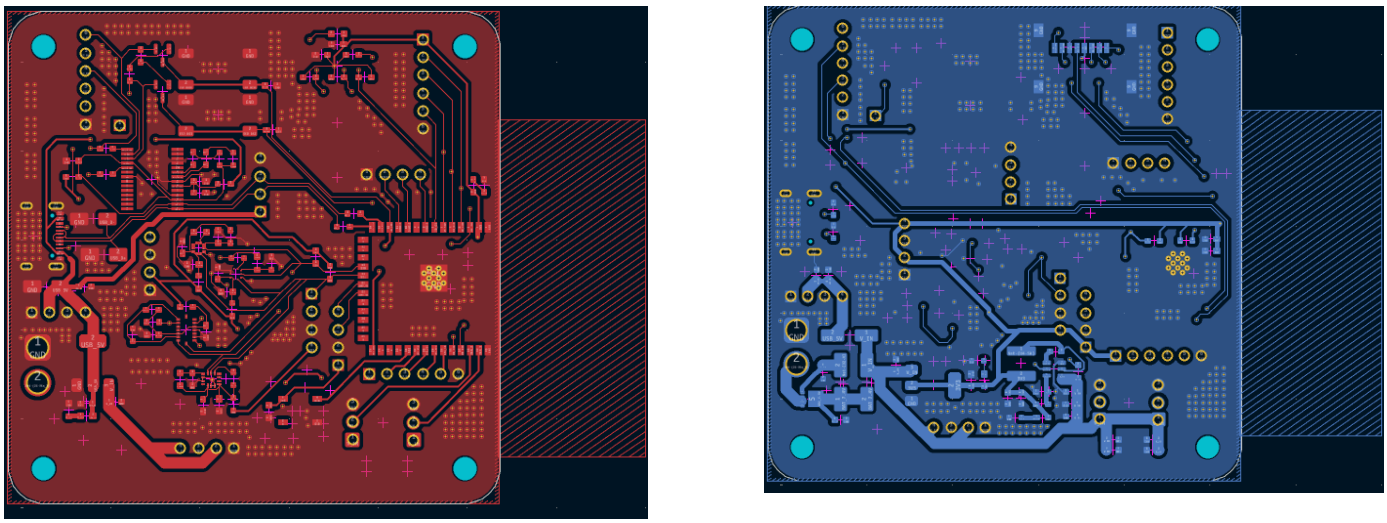


Figure A.197 - Bhagat Layout and Routing

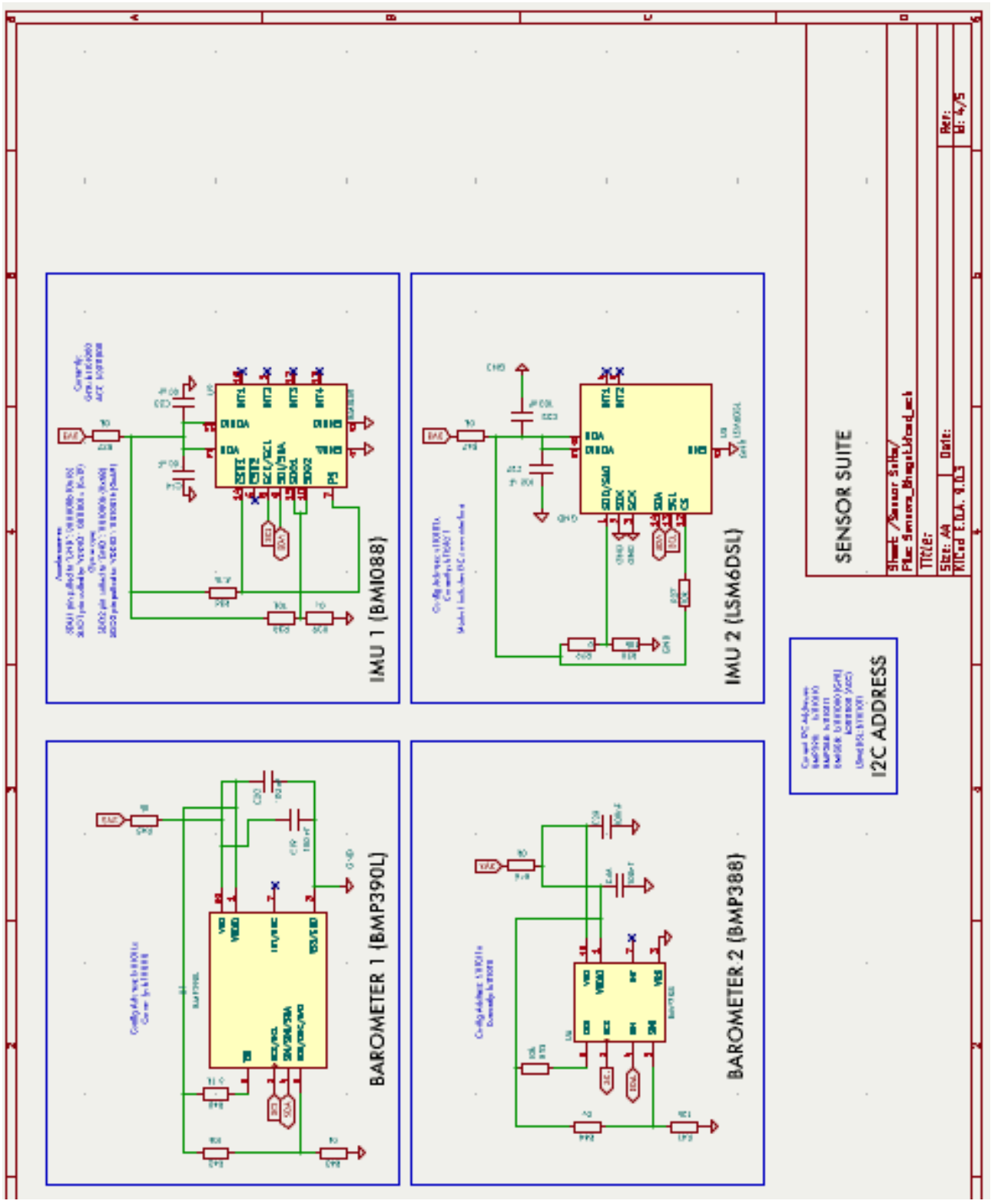


Figure A.198 - Bhagat PCB Schematics

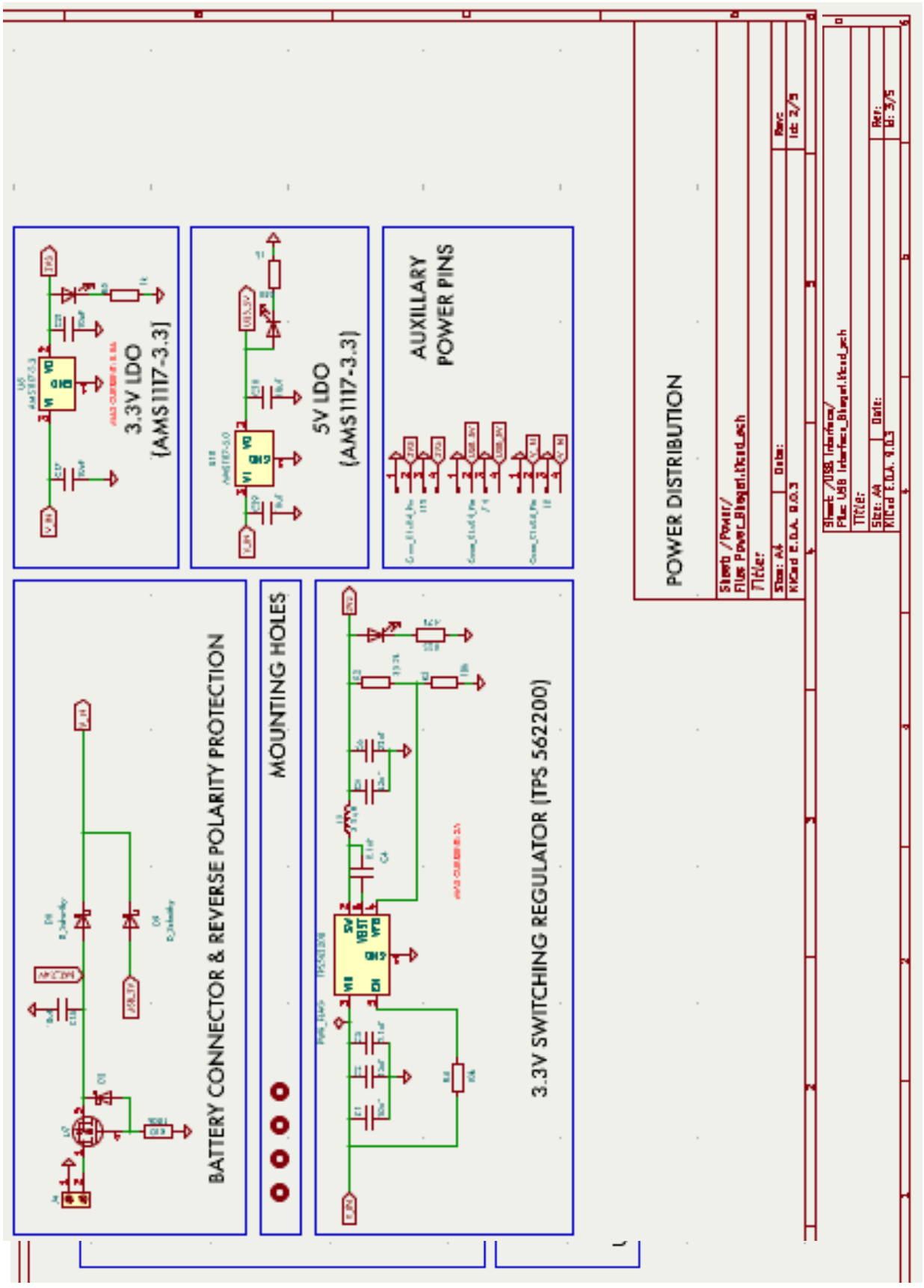


Figure A.199 - Bhagat PCB Schematics

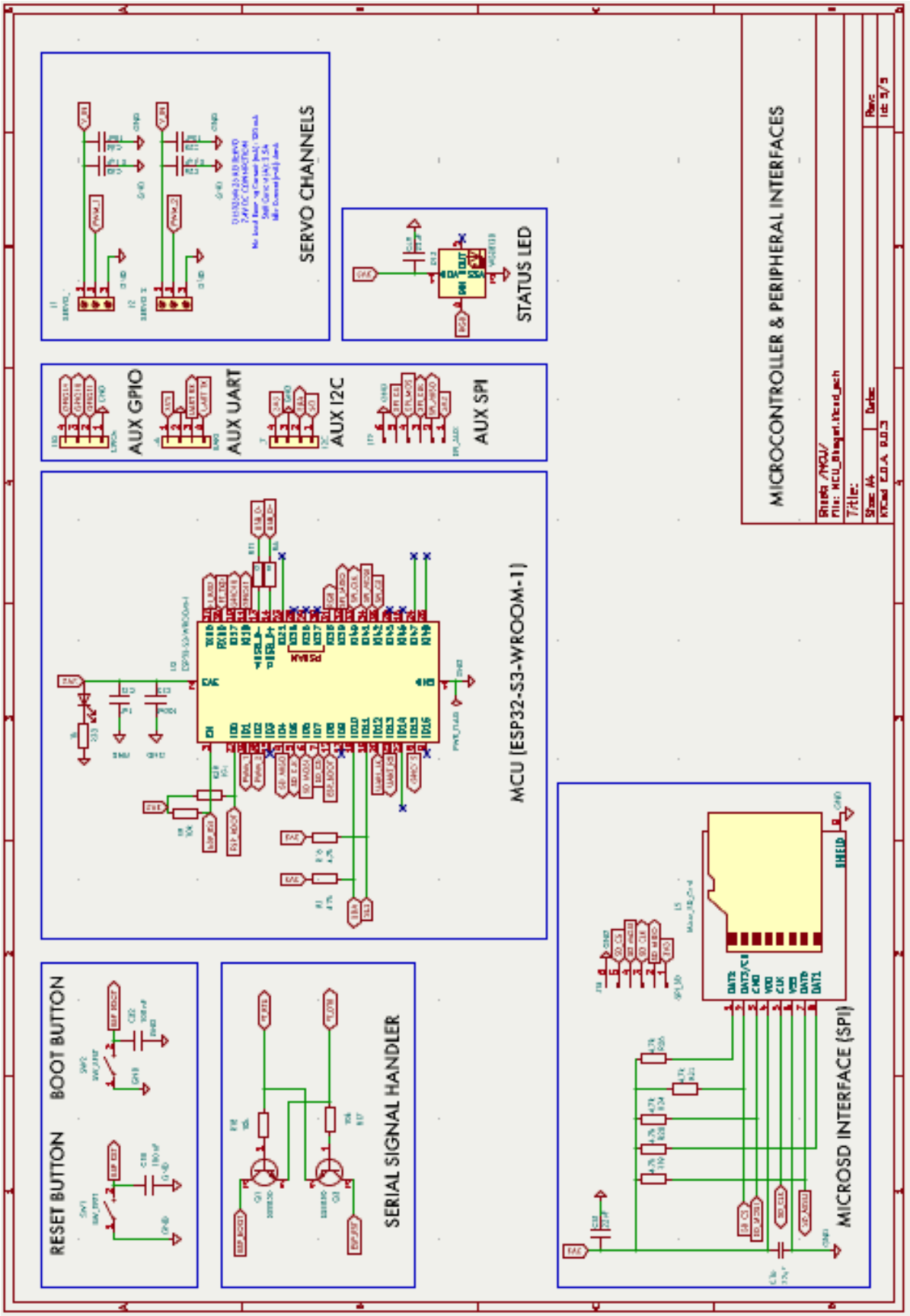


Figure 4.200 - Bhagat PCB Schematics

21. Arceus FC

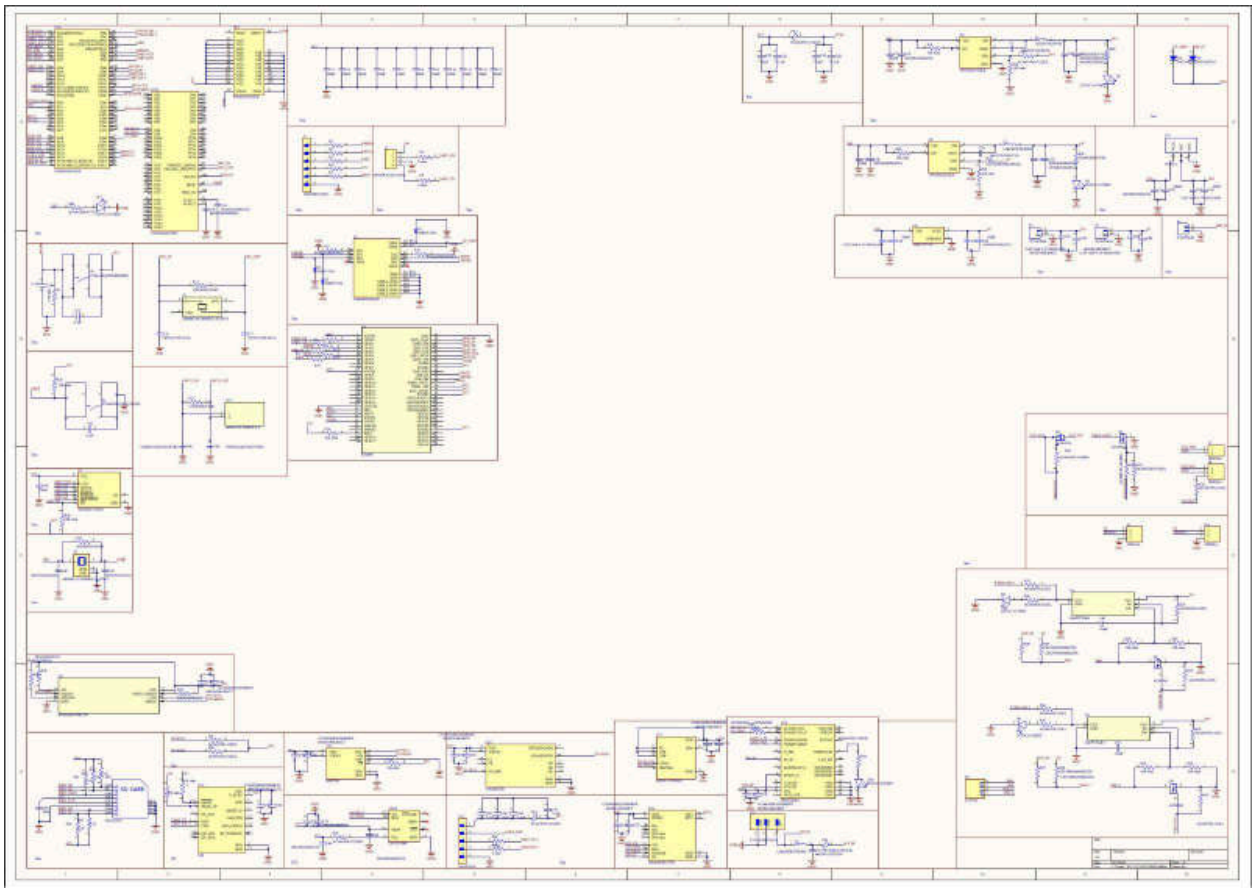


Figure A.201 - Arceus PCB Schematics

22. Sirius FC

The Sirius Flight Computer serves as the primary data acquisition system throughout the flight, integrating a comprehensive suite of redundant sensors to deliver high-fidelity flight data. Because Sirius operates without any pyro channels, it is entirely isolated from the voltage transients and electrical noise associated with pyrotechnic events, ensuring a clean and stable power supply to all onboard sensors. This makes Sirius the ideal candidate for telemetry transmission to the ground station, as the integrity of sensor readings is maintained throughout the entirety of the flight.

The inertial measurement subsystem consists of four IMUs operating in parallel. The BMI088 is a vibration-resistant 6-axis IMU optimized for high-vibration applications, offering a ±24g accelerometer range and ±2000°/s gyroscope -characteristics well-suited to the high-g loads and acoustic interference of motor burn. The ISM330DHCX is a high-performance 6-axis IMU featuring an on-chip machine learning core and anti-aliasing filters, providing excellent noise immunity and long-term stability under thermal stress. The LIS2MDL is a 3-axis magnetometer with a ±50 gauss full-scale range and built-in hard-iron compensation, pairing with the other IMUs to complete the 9-DOF orientation solution. The MPU-6050 provides an additional redundant 6-axis inertial reference, rounding out the subsystem with a widely characterized sensor whose behavior is well understood across temperature ranges. Sensor fusion algorithms run continuously across all four devices, cross-checking angular rate and linear acceleration data to suppress single-sensor anomalies and reduce the probability of a measurement fault during critical flight phases such as max-Q, apogee, and descent.

Barometric altitude is derived from two pressure sensors running simultaneously. The BMP280 serves as the primary device, while the MS5611 acts as a high-resolution backup -with 24-bit ADC resolution and a pressure RMS noise of approximately 0.012 mbar, it delivers superior altitude resolution in thermally dynamic conditions where temperature gradients could otherwise degrade barometric accuracy. Their altitude estimates are continuously compared to detect drift or sensor failure. Two GNSS receivers further support positioning and navigation throughout the flight: the L89 serves as the primary receiver with integrated multi-constellation tracking, while the NEO-M8N backup concurrently receives GPS, GLONASS, and Galileo signals. Running both in parallel ensures geolocation remains accurate even if one receiver loses lock due to high dynamic rotation, antenna shadowing, or signal attenuation during motor burn.

For ground station communication, Sirius primarily relies on the RYLR998 LoRa module, which operates in the 915 MHz band and offers a low-power, long-range link well-suited to the power budget and communication requirements of a flight computer. Its spread spectrum modulation provides resilience against narrowband interference common in launch environments. Where longer range or higher throughput is required, the RFD900x is available as an optional telemetry upgrade. All sensor streams and telemetry frames are logged to a dual-storage architecture: a micro SD card over SDIO provides high-capacity sequential recording, while an external W25Q128 NOR flash module provides non-volatile, high-speed buffered storage with no mechanical parts -offering resilience against power interruptions and mechanical shock events that might otherwise make SD card data inaccessible after flight.

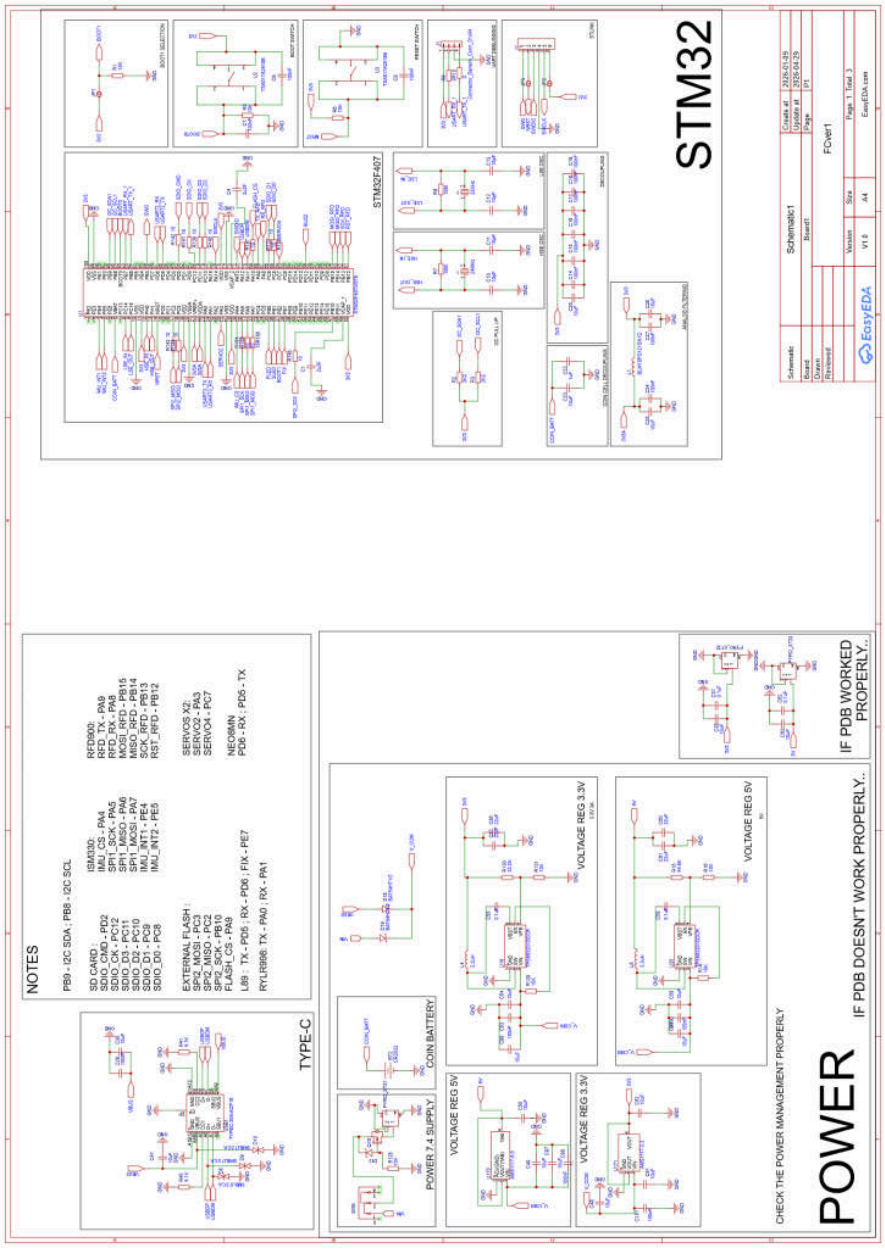


Figure A.202 - Sirius PCB Schematics

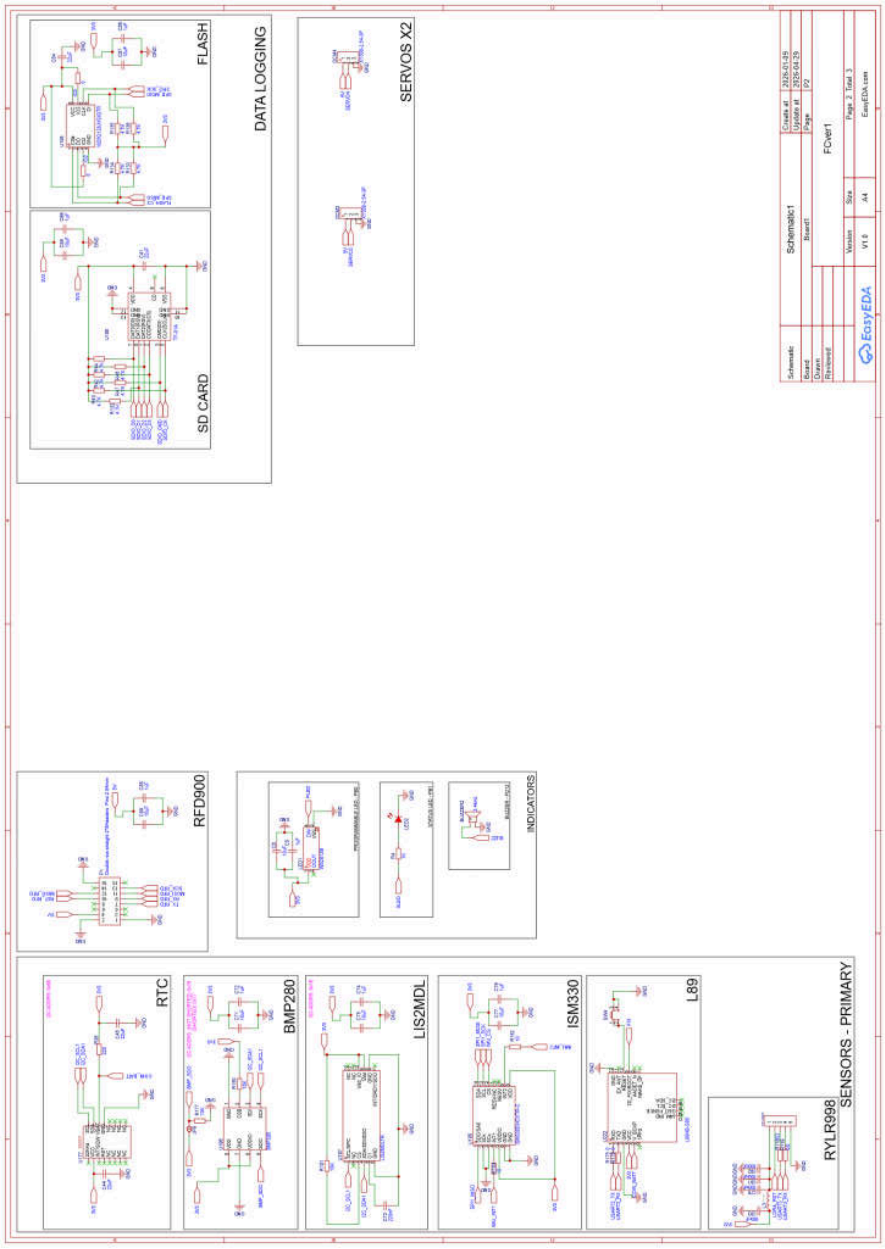


Figure A.203 - Sirius PCB Schematics

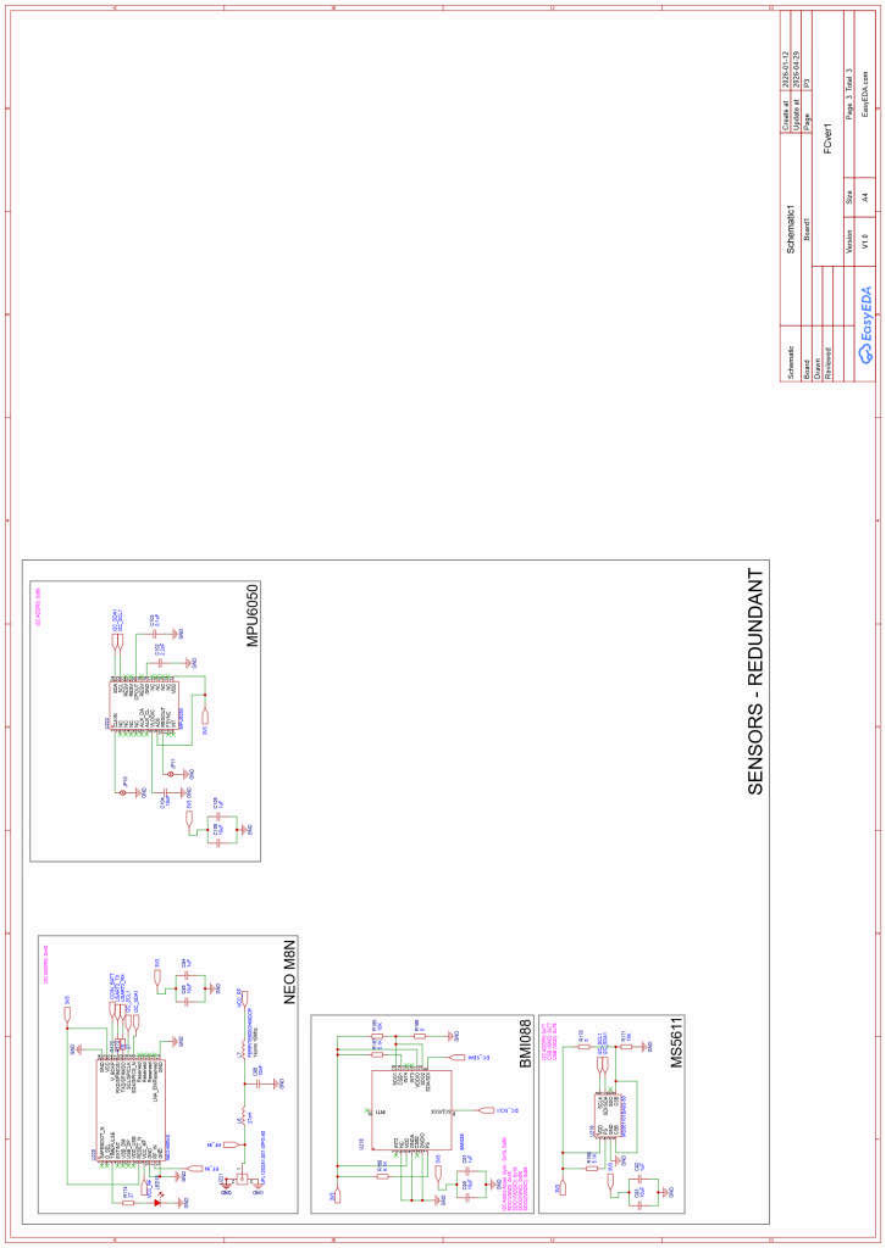


Figure A.204 - Sirius PCB schematics

23. PDB:

The power distribution board manages all onboard voltage regulation and power routing from a single LiPo input. The input stage uses an XT30PW-M30 connector with bulk decoupling capacitors to accept raw LiPo voltage, providing a stable input rail before any regulation occurs. From this input, the board derives two regulated output rails -5V and 7V -each generated through multiple parallel regulation paths to ensure adequate current capacity and redundancy. The 5V rail is produced by two independent regulators: an LM1117-5.0 linear regulator handling lower-current loads, and a TPS563251DDCR synchronous buck converter for higher-efficiency, higher-current 5V distribution, the latter featuring a 3.3µH inductor and a feedback network with 54.9kΩ and 10kΩ resistors to set the output voltage. The 7V rail is similarly dual-sourced, with two LM317CM3ASM5Y adjustable linear regulators each configured with a 1.5kΩ and 390Ω resistor divider network to set the output, providing regulated 7V to downstream loads. Each output rail is further buffered through XT30PW-M30 output connectors with local bypass capacitors -typically 22µF, 100nF, and 1µF in parallel -placed close to each output to suppress high-frequency noise and maintain rail stability under transient load conditions. The separation of switching and linear regulation across the two rails reflects a deliberate design choice: the buck converter handles the efficiency-critical 5V loads while the linear regulators on the 7V rail prioritize low-noise output, important for sensitive sensor and RF circuitry downstream.

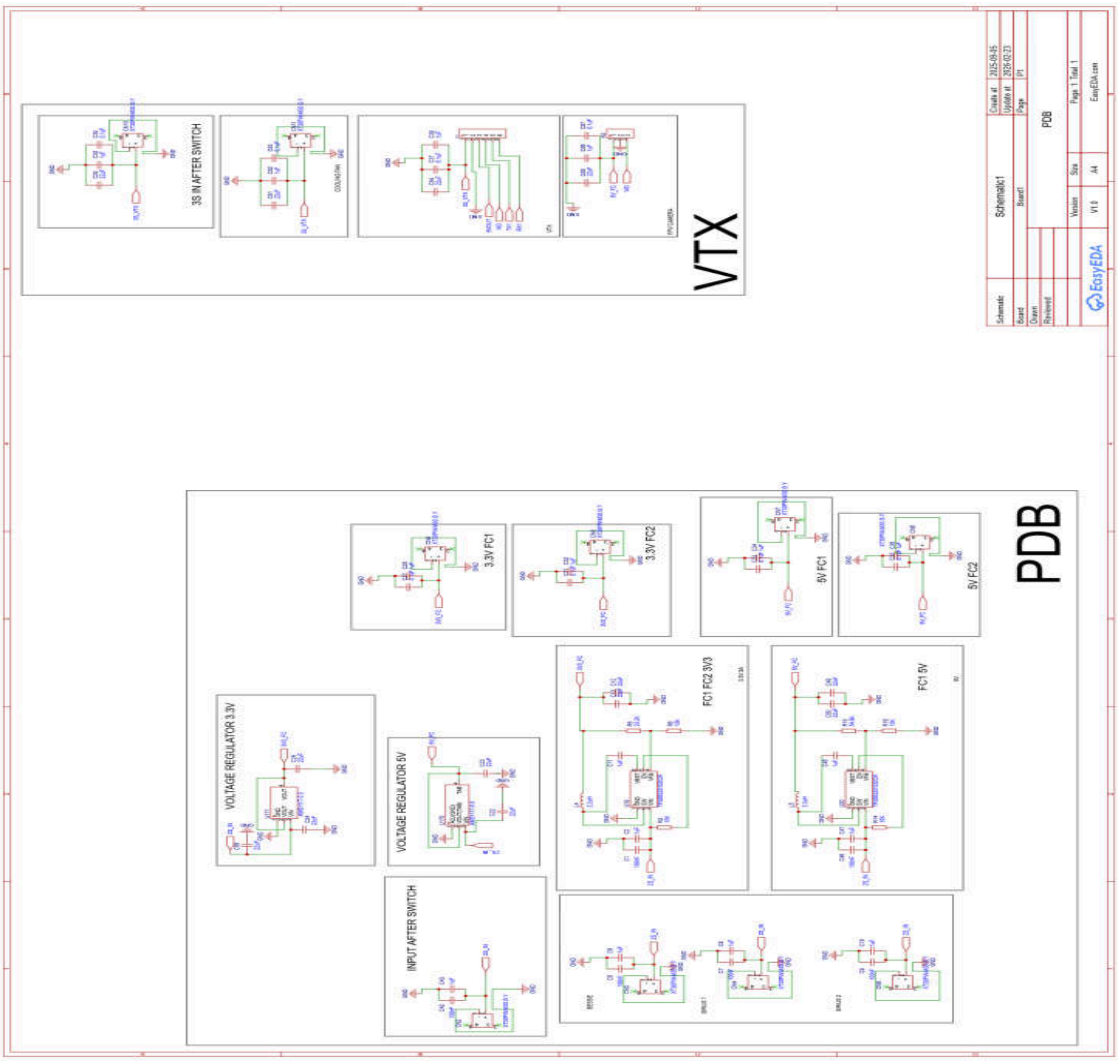


Figure A.205: PDB Schematics

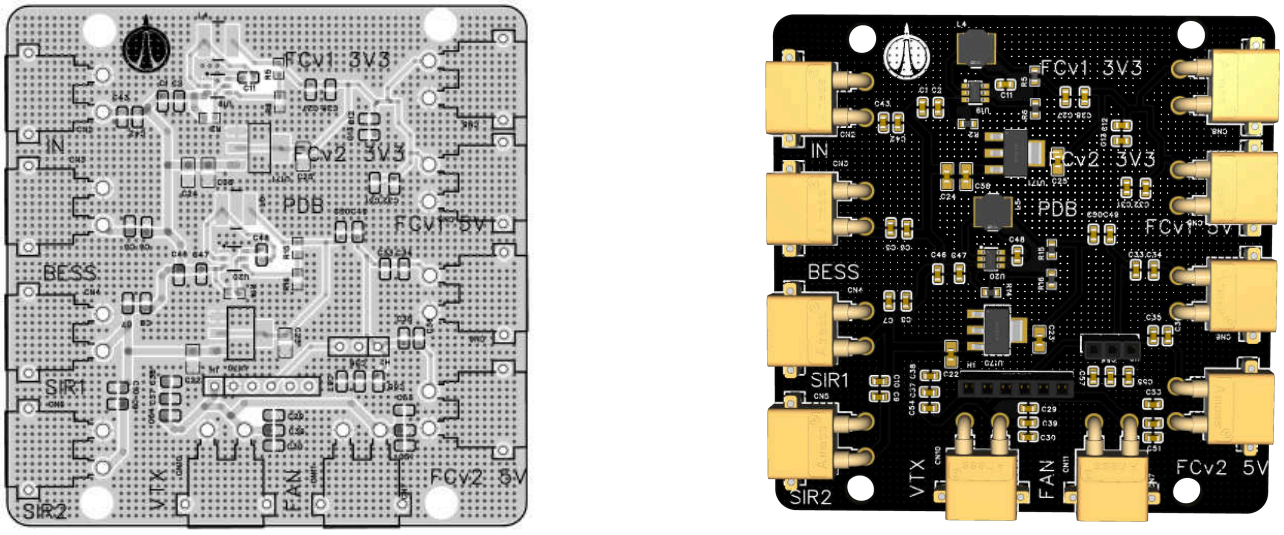


Figure A.206: PDB PCB

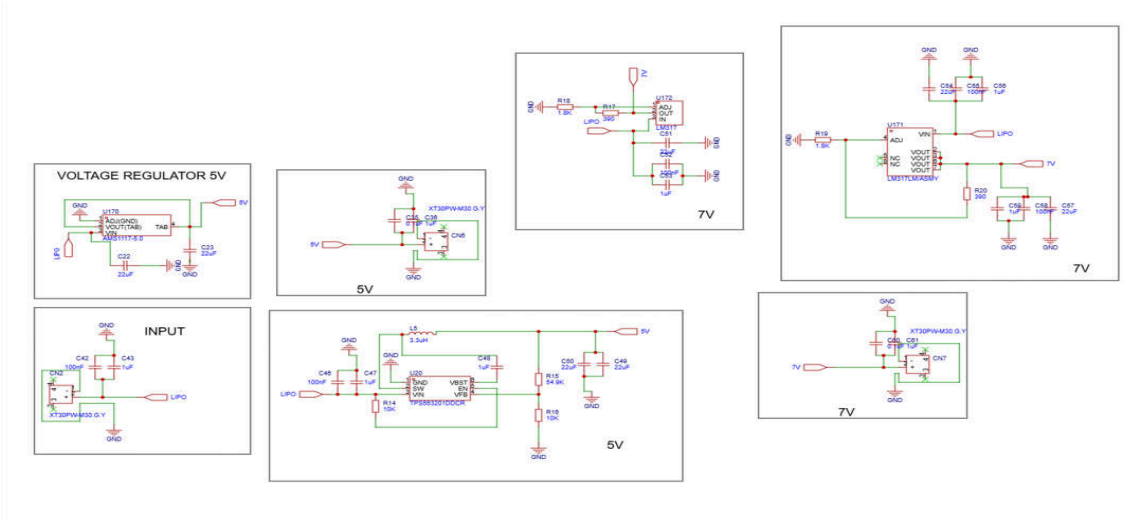


Figure A.207: PDB PCB Schematics

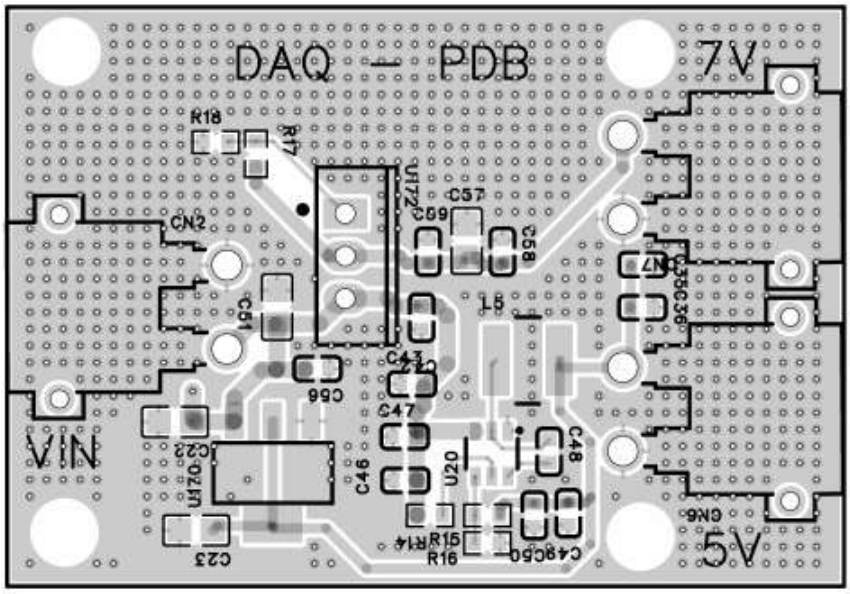


Figure A.208: PDB Pcb Layout and Routing

24. Flight Software:

The Flight Software System is developed to autonomously log, process, and transmit telemetry during all phases of flight. Running on the STM32F4 microcontroller, the software integrates real-time sensor data collection and storage, onboard state-machine decision-making, and LoRa-based telemetry for reliable communication with the ground station. All drivers are developed in-house in C++ using the STM32 HAL library, configured via STM32CubeMX.

Key Objectives:

• Track altitude, orientation, position, and velocity in real-time using onboard sensors, with consistent sampling and processing for reliable state estimation.
• Identify and manage key flight phases: Launch Pad, Ascent, Descent, Main Descent, Landing, and Landed, using a defined state machine based on sensor thresholds and timing conditions.
• Log and transmit mission-critical data using redundant storage (SDIO SD Card and W25Q128 SPI Flash), with buffering and timestamping to improve data reliability.
• Serve as the primary flight computer to autonomously fire drogue and main parachute deployment charges based on altitude thresholds and sensor confirmation, with safety checks to avoid false triggering.
• Recover last known state from flash memory on power-on reset to handle power interruptions mid-flight, allowing the system to resume operation without losing continuity.

Workflow and Architecture:

State 1: Launch Pad
On power-on, the system initializes all sensors and reads the last saved state from the W25Q128 SPI Flash. If the last state is IDLE, the system waits for a calibration command from the ground station over the RYLR998 LoRa link. The main control loop then runs continuously, handling sensor updates and communication. A launch is detected when altitude exceeds 5 m for 3 consecutive cycles, at which point the system transitions to the Ascent state.

State 2: Ascent
During Ascent, the system continuously records the apogee altitude by tracking the highest altitude reading. The main control loop executes each cycle, ensuring consistent data logging and transmission. Apogee is detected when altitude does not increase for 3 consecutive cycles, indicating peak altitude has been reached. The system then sets the apogee reference, fires the drogue parachute deployment charge, and transitions to the Descent state.

State 3: Descent
In the Descent state, the system monitors altitude against the main parachute release threshold of 457 m (1500 ft). The main control loop continues to execute, maintaining telemetry transmission and data logging. When altitude is at or below 457 m for 2 consecutive cycles, the system triggers the main parachute deployment charge and transitions to the Main Descent state.

State 4: Main Descent
During Main Descent, the system continues running the main control loop while monitoring altitude trends for touchdown detection. A landing condition is triggered when altitude drops below 10 m for 3 consecutive cycles, at which point the system transitions to the Landing state.

State 5: Landing
The Landing state verifies final touchdown by checking that the absolute altitude change is less than 1 m and the accelerometer reads approximately 1g (static) for a continuous period of 5 seconds. These conditions help ensure the system is no longer in motion. Once confirmed, the system transitions to the Landed state.

State 6: Landed
Upon reaching the Landed state, the system activates an audio beacon to assist in recovery and enables the LED

indicator for visual identification. Telemetry transmission is stopped to conserve power, marking the successful completion of the mission.

State Diagram:

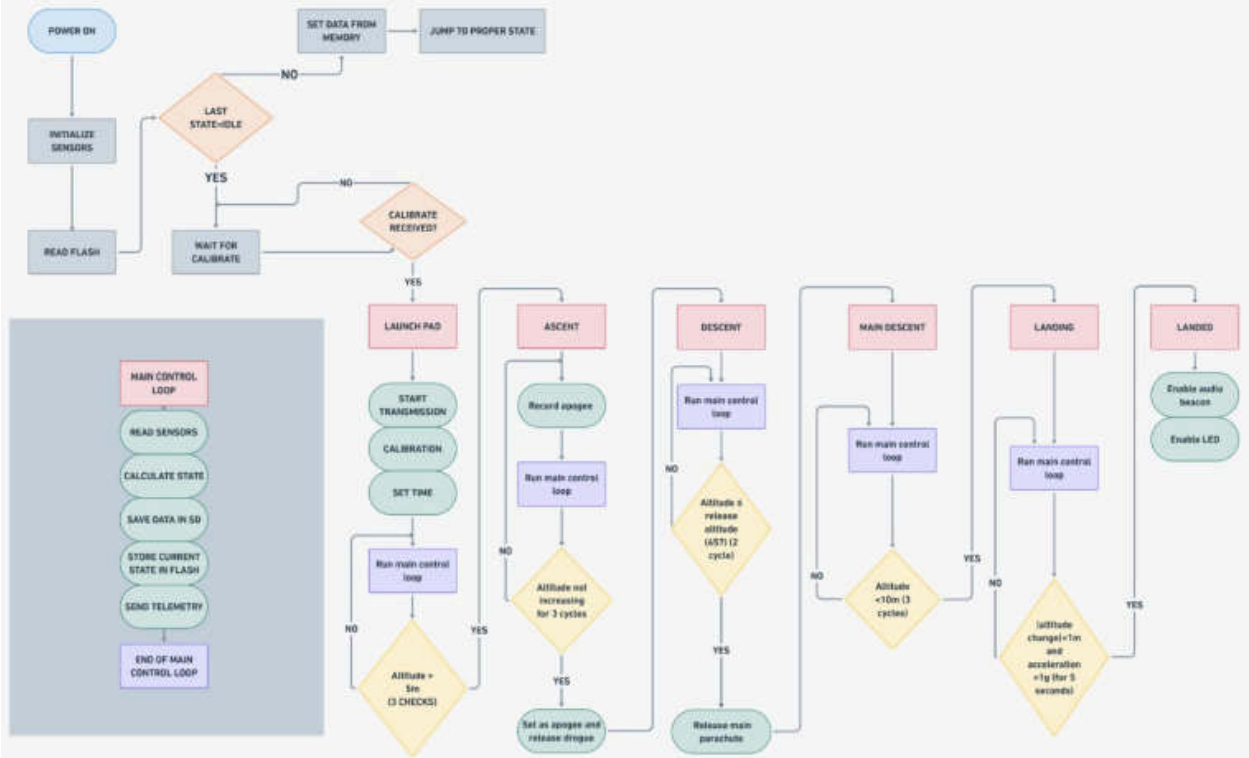


Figure A.209: Flowchart & State Diagram

Technologies and Concepts:

Tech Stack:

The flight software is written in C++ using the STM32 HAL library, developed and configured using the STM32CubeMX IDE, and flashed via the ST-Link Debugger. The flight computer is based on the STM32F4 microcontroller, which handles all real-time operations and peripheral coordination. All peripheral drivers are custom-written in-house to ensure greater control, flexibility, and reliability compared to generic implementations.

Table A.18: Hardware interfaces and peripherals used

Peripheral	Interface	Driver
NEO-M8N GPS Module	UART	CustomNMEA parser
L89 GPS (backup)	UART	CustomNMEA parser
SD Card	SDIO (1-bit)	HAL FatFS
W25Q128 Flash Memory	SPI	Custom Driver
BMI088 IMU(Accel+Gyro)	I2C	Custom Driver
DS3231 RTC	I2C	Custom Driver
BMP280 Barometric Sensor	I2C	Custom Driver
MS5611 Barometric Sensor (Backup)	I2C	Custom Driver
RYLR998 LoRa Module	UART	Custom AT command Driver

Data Storage Format:

All telemetry data is logged to the SD card in CSV format for post-flight analysis. The stored data packet contains the following fields:

TEAM_ID, PACKET_CNT, TIME (HH:MM:SS), NUM_SATS, GPS_SPEED, LATITUDE, LONGITUDE, ALTITUDE, RTC_TIME (HH:MM:SS), ACCEL_X, ACCEL_Y, ACCEL_Z, PRESSURE, BARO_ALTITUDE, GYRO_X, GYRO_Y, GYRO_Z, TRIGGER, TEMPERATURE, VOLTAGE, CURRENT_STATE, EVENT_FLAG

Data Transmission Format:

Telemetry is transmitted over LoRa (RYLR998) using a comma-delimited packet formatted as follows:

"%d,%d,%d:%d:%d,%d,%3.1f,%f,%f,%.2f,%d:%d:%d,%06.2f,%06.2f,%06.2f,%.2f,%.2f,%06.2f,%06.2f,%06.2f, %d,%.2f,%.2f,%d,%d"

Table A.19: Telemetry description

Field	Format	Description
team_id	%d	Team Identification number
packet_cnt	%d	Sequential packet counter
hour:minute:second	%d:%d:%d	GPS time (HH:MM:SS)
num_sats	%d	Number of GPS satellites locked
gps_speed	%3.1f	GPS ground speed (m/s)
latitude	%f	GPS latitude (decimal degrees)
longitude	%f	GPS longitude (decimal degrees)
altitude	%.2f	GPS altitude (m)
RTC hour:min:sec	%d:%d:%d	RTC timestamp (HH:MM:SS)
accel.x	%06.2f	Accelerometer X-axis (m/s²)
accel.y	%06.2f	Accelerometer Y-axis (m/s²)
accel.z	%06.2f	Accelerometer Z-axis (m/s²)
pressure	%.2f	Barometric pressure (hPa)
altitude (baro)	%.2f	Barometric altitude (m)
gyro.x	%06.2f	Gyroscope X-axis (deg/s)
gyro.y	%06.2f	Gyroscope Y-axis (deg/s)
gyro.z	%06.2f	Gyroscope Z-axis (deg/s)
trigger	%d	Deployment trigger status flag
temperature	%.2f	Temperature(°C)
voltage	%.2f	Battery voltage(V)
current_state	%d	Current flight state (0-5)
event_flag (e)	%d	Miscellaneous event flag

Algorithms:

The system employs a robust state machine to identify six flight phases-Launch Pad, Ascent, Descent, Main Descent, Landing, and Landed-using altitude thresholds combined with multi-cycle confirmation checks (trust counters) to eliminate transient noise and ensure reliable event triggering such as apogee detection and parachute deployment. For orientation estimation, sensor fusion is implemented using accelerometer, gyroscope, and magnetometer data, where a Kalman/Complementary filter processes raw inputs from the BMI088 to reduce noise and maintain stable attitude estimation under dynamic flight conditions. The telemetry encoding scheme transmits 23 structured parameters as a lightweight, comma-delimited ASCII packet optimized for LoRa communication via the RYLR998 module, with sequential packet counting and embedded state information ensuring consistency and integrity. To enhance reliability, redundant storage is implemented by logging full telemetry data to an SD card (CSV format) for post-flight analysis, while critical flight states are simultaneously stored in W25Q128 SPI Flash after each transition, enabling recovery and seamless continuation of the state machine in case of mid-flight power interruptions.

Kalman / Complementary Filter :

Sensor fusion is applied to combine accelerometer, gyroscope, and magnetometer data to compute accurate orientation values. The Kalman filter plays a key role in smoothing raw IMU inputs from the BMI088, reducing noise and improving stability. This allows the system to maintain reliable attitude estimates throughout different phases of flight, even under rapid motion and changing dynamics.

Telemetry Encoding :

The telemetry packet consists of 23 fields transmitted as a structured, comma-delimited ASCII string, optimized for LoRa communication over the RYLR998 module. The format is designed to be lightweight and consistent for easy parsing on the ground station. Packet integrity is supported through sequential packet counting, along with the inclusion of state information in each transmission.

Redundant Storage :

Mission-critical data is protected through dual logging mechanisms. All telemetry is written to an onboard SD card (via SDIO) in CSV format for detailed post-flight analysis. At the same time, the current flight state is stored in the W25Q128 SPI Flash after every state transition. On power-on, the system reads the last saved state from flash and resumes from the corresponding state in the state machine, allowing the system to recover smoothly from mid-flight power interruptions.

25. Cyberdeck

The Cyberdeck is a custom-built ground support system designed to host and operate the Ground Control Station (GCS) along with live video monitoring during rocket missions. It serves as a compact, portable, and integrated hardware platform that brings together all essential components required for real-time telemetry reception, visualization, and situational awareness in field conditions.

The system is built around a Raspberry Pi, which runs the GCS software and interfaces with multiple peripherals through USB connections. These include a telemetry receiver PCB (SHROTA) for acquiring flight data via LoRa communication and an ESP32 for transmitting GPS data to a mobile application for recovery purposes. The Cyberdeck also incorporates a display unit, keyboard, USB hub, and an uninterrupted power supply (UPS), ensuring continuous and reliable operation during outdoor missions.

In addition to telemetry handling, the system integrates a separate video reception setup using another Raspberry Pi connected to an EasyCAP video receiver. This enables real-time video feed monitoring alongside telemetry data, enhancing overall mission awareness. All components are securely housed within a custom 3D-printed structure, providing durability, organization, and ease of deployment in field environments.

By combining telemetry reception, live video streaming, power management, and portability into a single unified system, the Cyberdeck acts as a robust and efficient control hub, supporting both monitoring and recovery operations throughout the mission lifecycle.

CORTEX - Centralised Operations & Real-time Telemetry Exchange (Ground Control Station Software)

The CORTEX (Centralized Operations & Real-time Telemetry Exchange) is a custom-developed Ground Control Software designed to monitor rocket flights in real time and support critical decision-making during mission operations. The system is implemented as a lightweight and modular platform that integrates embedded hardware with a web-based user interface.
The GCS interfaces with the flight computer via LoRa communication, receiving continuous telemetry data throughout the mission. It provides real-time visualization of key flight parameters through an intuitive interface, enabling operators to monitor system performance, detect anomalies, and respond effectively during flight.
In addition to real-time monitoring, the system supports structured telemetry parsing and optional data logging for post-flight analysis. A dedicated recovery pipeline is also integrated, allowing GPS data to be transmitted to a mobile device for efficient rocket retrieval. By combining reliable communication, real-time visualization, and recovery support, the GCS plays a vital role in mission success.

Workflow and Architecture

The CORTEX (GCS) is designed to efficiently manage real-time telemetry reception, processing, and visualization using a distributed architecture.

Telemetry data is transmitted from the rocket using LoRa modules and received at the ground station through SHROTA (receiver) connected to a Raspberry Pi. The received data is accessed through a serial interface (e.g., /dev/ttyUSB*), enabling continuous telemetry streaming.

The telemetry stream is handled by a backend built using Node.js. The backend is responsible for serial communication, packet parsing, and validation. Incoming telemetry packets are processed to ensure correct formatting, while corrupted or incomplete packets are filtered out to maintain data integrity.The backend follows an event-driven, non-blocking architecture, allowing it to handle continuous data streams efficiently without performance degradation. Processed telemetry data is transmitted to the frontend using WebSockets, enabling real-time updates.The frontend is developed using React and provides a responsive and user-friendly interface. It displays telemetry data through graphical and tabular views, allowing operators to quickly interpret system behavior.For recovery operations, GPS data is extracted and transmitted to an ESP32, which relays the information to a mobile application via Bluetooth. This enables efficient tracking and retrieval of the rocket after landing.

Various Functionalities of Cortex (GCS)

Home Page

The CORTEX homepage serves as the central dashboard for monitoring real-time flight telemetry, providing a structured and intuitive interface for effective decision-making during all mission phases.
The left panel displays key telemetry parameters such as gyroscope, accelerometer, GPS data, and speed metrics in a categorized format for quick access. The mission control section includes controls for ignite, start, stop, and time-setting, along with a mission timer for tracking flight duration.
The central display is divided into two panels: a 3D trajectory view on the left for visualizing the rocket's orientation and flight path, and a GPS map on the right for real-time location tracking.
The lower section contains real-time graphs for parameters such as voltage, pressure, and altitude, each accompanied by numerical values for precise monitoring. A packet history section logs incoming telemetry data, enabling tracking of packet count and data continuity. Overall, the homepage integrates multiple data representations into a unified interface for efficient monitoring and analysis of flight performance.

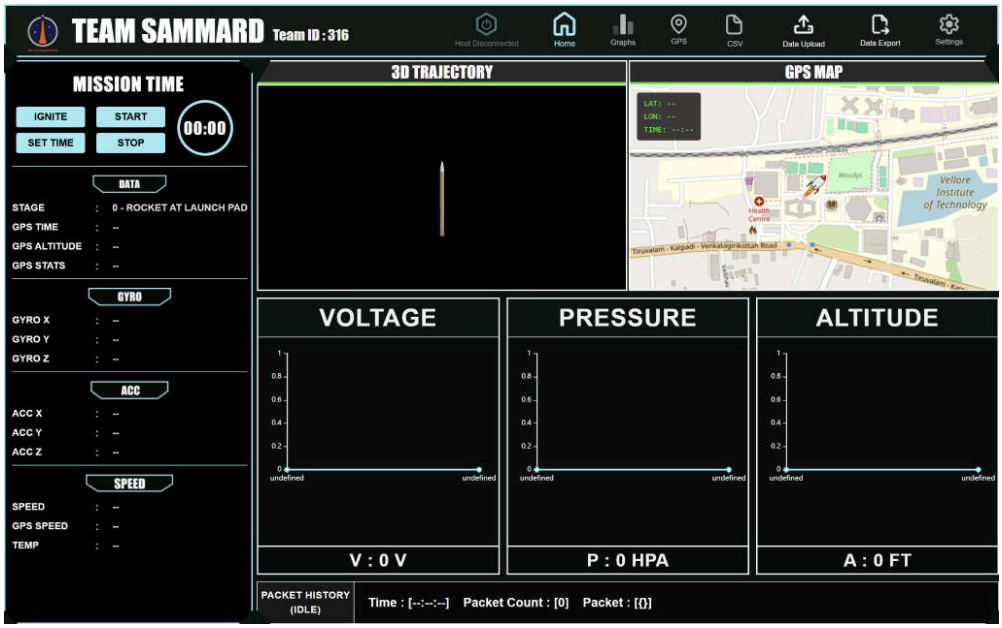


Figure A.210 : Home Page

Graphs Page

The Graphs Page provides a dedicated interface for real-time visualization and analysis of telemetry data. It displays dynamic plots of key parameters such as altitude and pressure, enabling operators to monitor trends and variations throughout the flight.

Each graph is continuously updated and accompanied by numerical values for precise interpretation. Interactive controls allow users to switch between different parameters, including speed, acceleration, gyroscope data, displacement, voltage, and temperature.

The layout supports simultaneous display of multiple graphs, allowing comparative analysis of flight behavior. Overall, the page enables detailed monitoring and efficient analysis of the rocket's performance during and after the mission.



Figure A.211 : Graphs Page

GPS Page

The GPS Page provides real-time tracking of the rocket's geographical position, supporting both in-flight monitoring and post-flight recovery. It features an interactive map that displays the rocket's live location using latitude and longitude coordinates, updating continuously with incoming telemetry data.
A 3D trajectory view is also included to visualize the rocket's orientation and motion, offering additional insight into flight behavior. A telemetry panel displays key GPS-related parameters such as latitude, longitude, altitude, satellite count, and speed for precise monitoring.
Overall, the GPS Page combines map-based tracking, 3D visualization, and telemetry data to enable accurate tracking and efficient recovery operations.

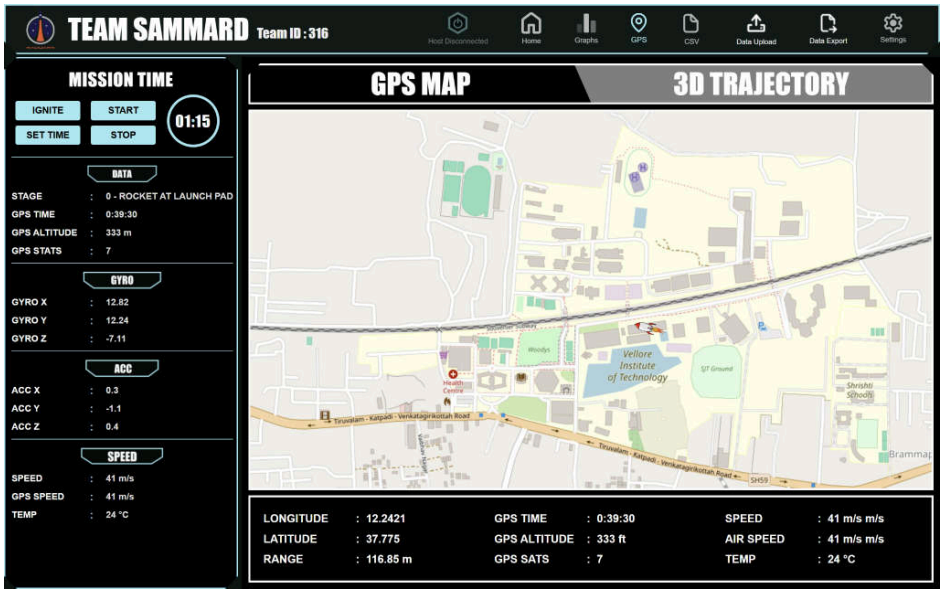


Figure A.212 : GPS Page(1)

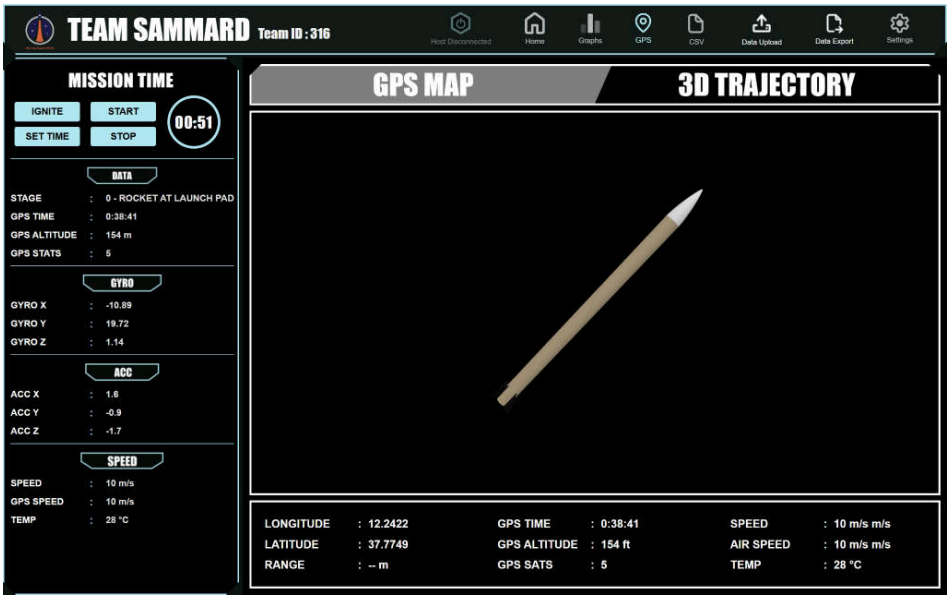


Figure A.213 : GPS Page (2)

Table Data Page

The CSV/Data Page provides a structured interface for viewing and analyzing telemetry data in a tabular format. It displays key parameters such as team ID, packet count, GPS time, satellite count, and speed, allowing detailed inspection of flight data.The table includes features like sorting, filtering, and search, enabling efficient navigation and analysis of large datasets. It updates in real time with incoming telemetry, ensuring synchronization with the live mission feed.
Additionally, the page supports viewing stored data for post-flight analysis, making it useful for debugging and performance evaluation. Overall, it offers a clear and interactive way to access and analyze telemetry data throughout the mission lifecycle.

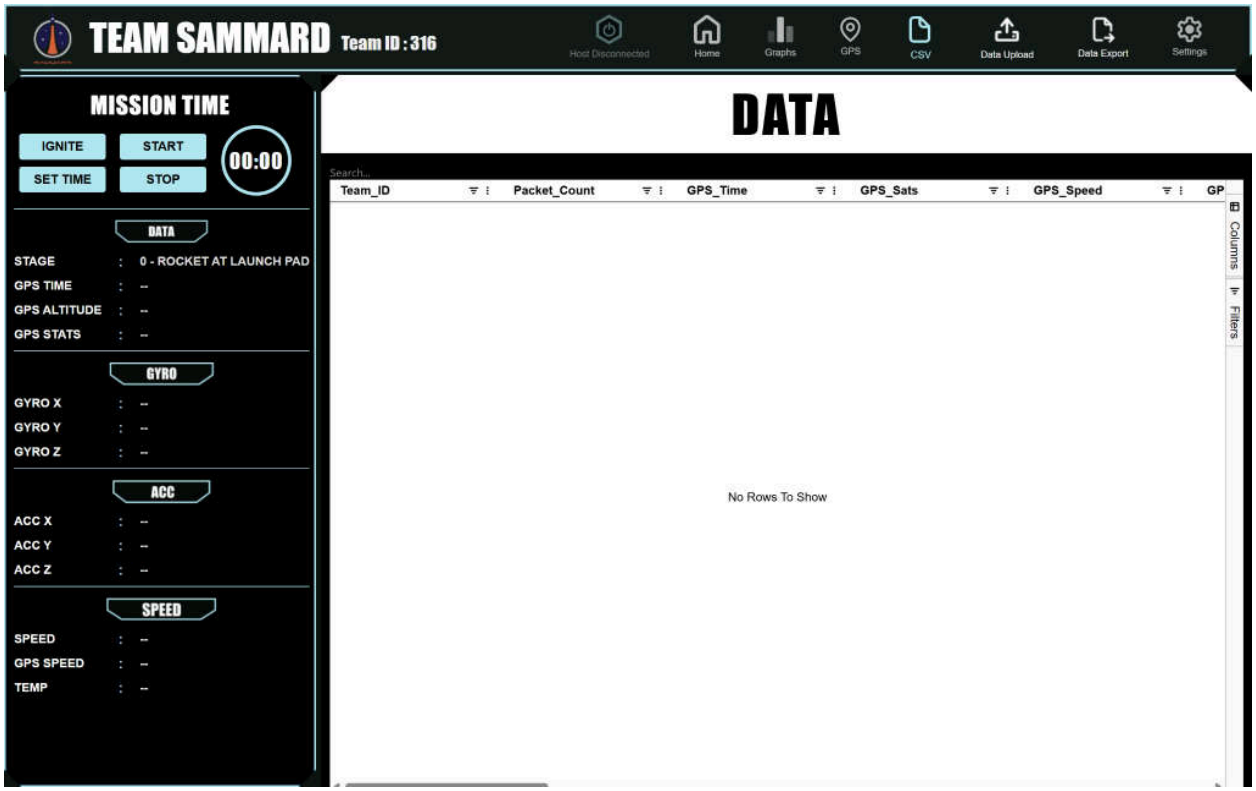


Figure A.214 : Table Data Page

Settings Page

The Settings Page provides a centralized interface for configuring system parameters and customizing telemetry data representation. It allows users to select unit systems such as metric, imperial, or custom options for parameters including speed, acceleration, altitude, temperature, and voltage.The interface is organized for easy selection, enabling precise control over how data is displayed. An “Apply All Changes” option ensures that selected configurations are consistently reflected across all pages of the CORTEX (GCS).

Overall, the Settings Page improves usability and ensures accurate interpretation of telemetry data during both real-time monitoring and post-flight analysis.

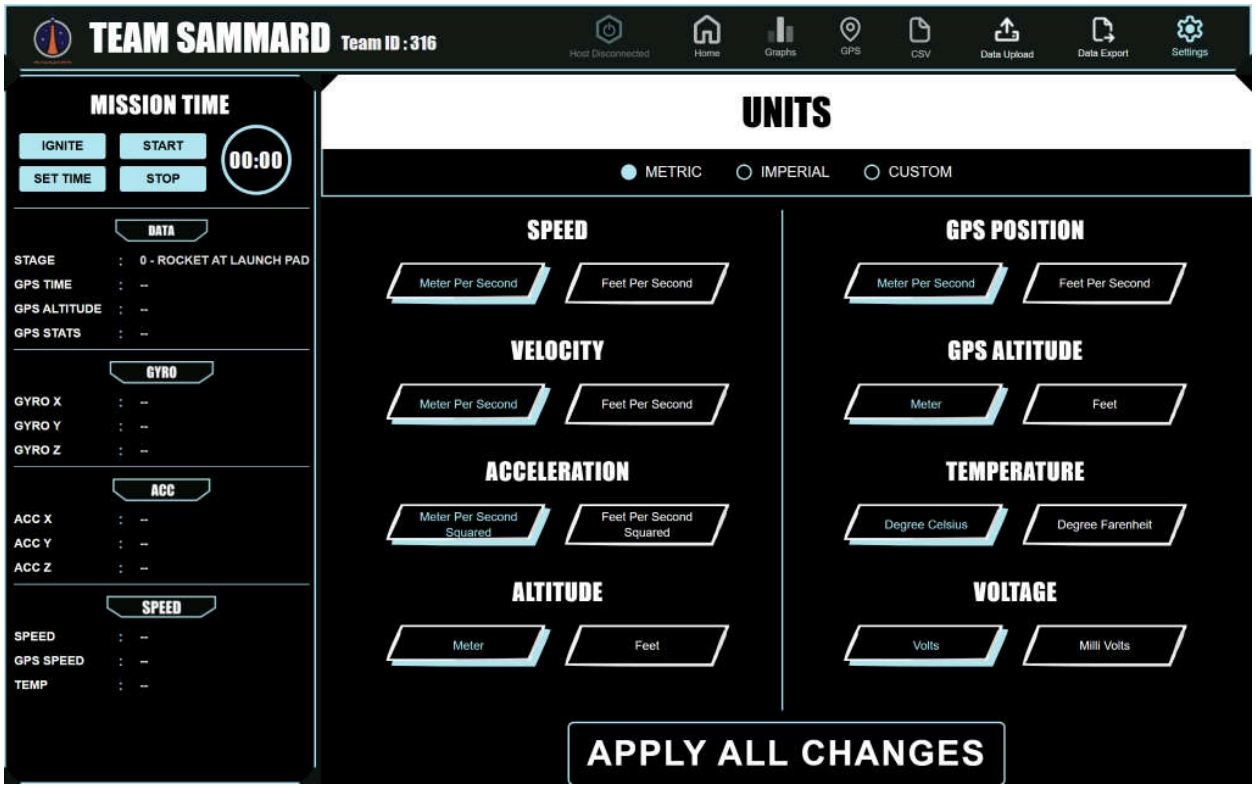


Figure A.215 : Settings Page

Technologies and Concepts used

The CORTEX system integrates modern web technologies with embedded systems to enable efficient real-time telemetry monitoring and visualization. The frontend is developed using React with Tailwind CSS for a responsive UI, while Vite ensures fast builds and development. Data visualization is achieved using ECharts and AG Grid, with GPS tracking via Leaflet and React Leaflet, and 3D trajectory rendering using Three.js, React Three Fiber, and Drei. State management is handled using Zustand for efficient data flow, while telemetry parsing and CSV handling are implemented using PapaParse, alongside Kalman filtering for signal smoothing and noise reduction. The backend is built on Node.js, leveraging its event-driven, non-blocking architecture, with WebSockets enabling real-time communication and serial interfaces handling telemetry input from the LoRa receiver. On the hardware side, a Raspberry Pi manages telemetry reception and processing, while an ESP32 handles recovery communication by transmitting GPS data via Bluetooth. LoRa modules provide long-range, low-power telemetry transmission. Together, this integrated stack ensures a scalable, reliable, and high-performance system for real-time monitoring and post-flight analysis.

Hardware Requirements:

The CORTEX requires a LoRa module for receiving telemetry data from the rocket. This module is connected to a Raspberry Pi via USB or UART, enabling direct serial communication. The Raspberry Pi acts as the primary interface for telemetry reception and forwarding. An ESP32 is used for recovery communication, transmitting GPS data to a mobile device via Bluetooth.
A laptop or workstation is used to run the frontend dashboard, providing real-time visualization and control. The system is optimized to run efficiently under field conditions.

Data Flow

Telemetry data is transmitted from the flight computer via LoRa at 915 MHz and received at the ground station using a Yagi antenna stack for enhanced range and directionality. The SHROTA receiver module converts the RF signal into serial data, which is forwarded to a Raspberry Pi via USB. The Raspberry Pi reads and passes this data to the backend, where it is parsed, validated, and processed for reliability. Processed telemetry is streamed in real time to the frontend dashboard using WebSockets for continuous monitoring and visualization. Simultaneously, GPS data is sent to an ESP32, which transmits it via Bluetooth to a mobile application for accurate tracking and efficient rocket recovery.

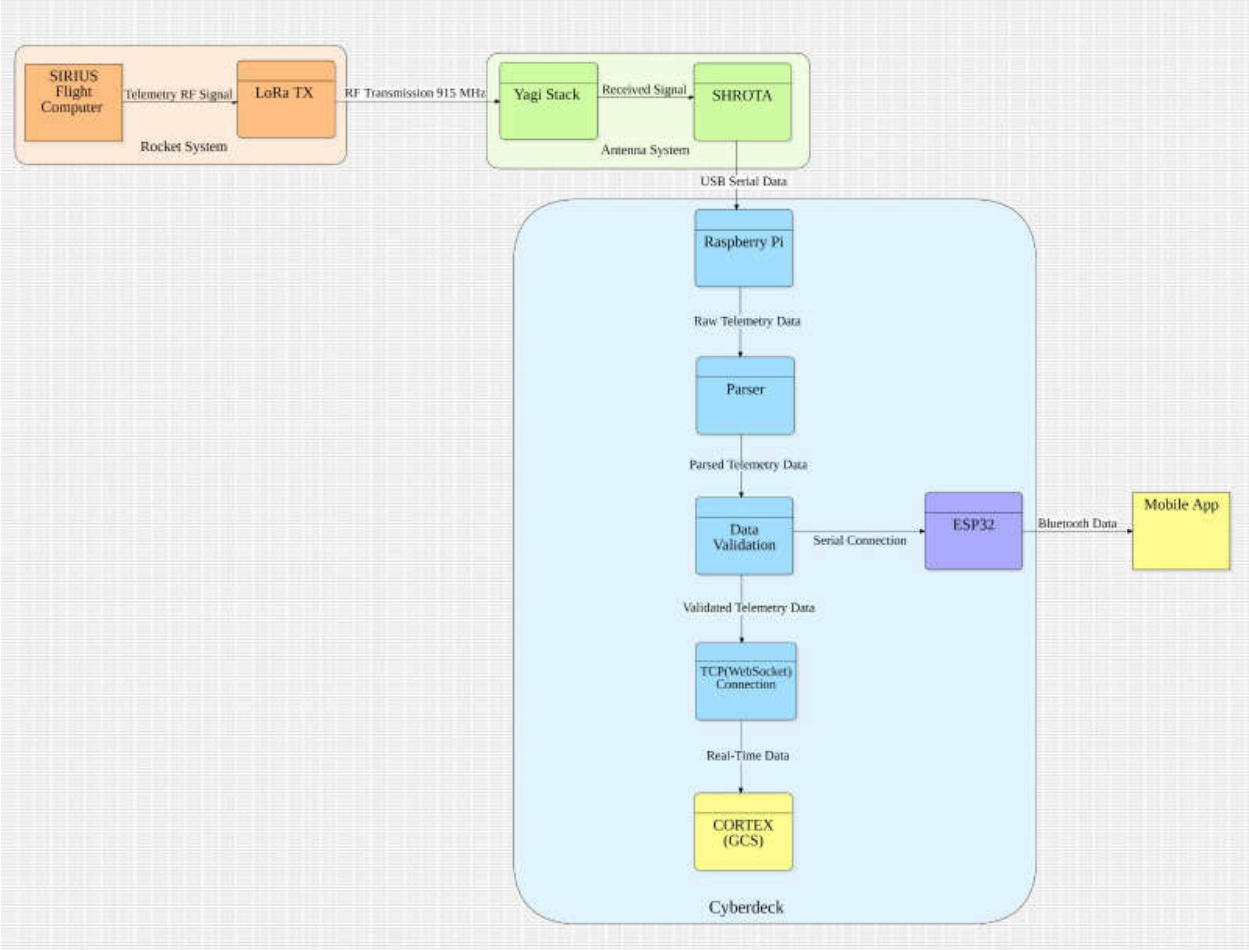


Figure A.216 : Data Flow Diagram

Data Storage Format :

TEAM_ID, PACKET_COUNT, GPS_TIME, GPS_SATS, GPS_SPEED, GPS_LAT, GPS_LONG, GPS_ALT, RTC_TIME, ACCEL_X, ACCEL_Y, ACCEL_Z, PRESSURE, ALTITUDE, GYRO_X, GYRO_Y, GYRO_Z, TRIGGER, TEMPERATURE, VOLTAGE, STATE_CURRENT, PYRO_E1_E2

Data Transmission Format : (113 bytes)

TEAM_ID, PACKET_COUNT, GPS_TIME, GPS_SATS, GPS_SPEED, GPS_LAT, GPS_LONG, GPS_ALT, RTC_TIME, ACCEL_X, ACCEL_Y, ACCEL_Z, PRESSURE, ALTITUDE, GYRO_X, GYRO_Y, GYRO_Z, TRIGGER, TEMPERATURE, VOLTAGE, STATE_CURRENT, PYRO_E1_E2

26. Antenna Tracking System (ATS)

The Antenna Tracking System (ATS) is used to continuously track and point the Video Receiving horn antenna towards the rocket, to maintain consistent live video reception from the rocket throughout its flight. It processes GPS and IMU data from the rocket using the Extended Kalman Filter to estimate its position and motion and calculates the required elevation and azimuth angles. Two NEMA-23 stepper motors are then moved by those angles to align the antenna accordingly.

Workflow and Architecture

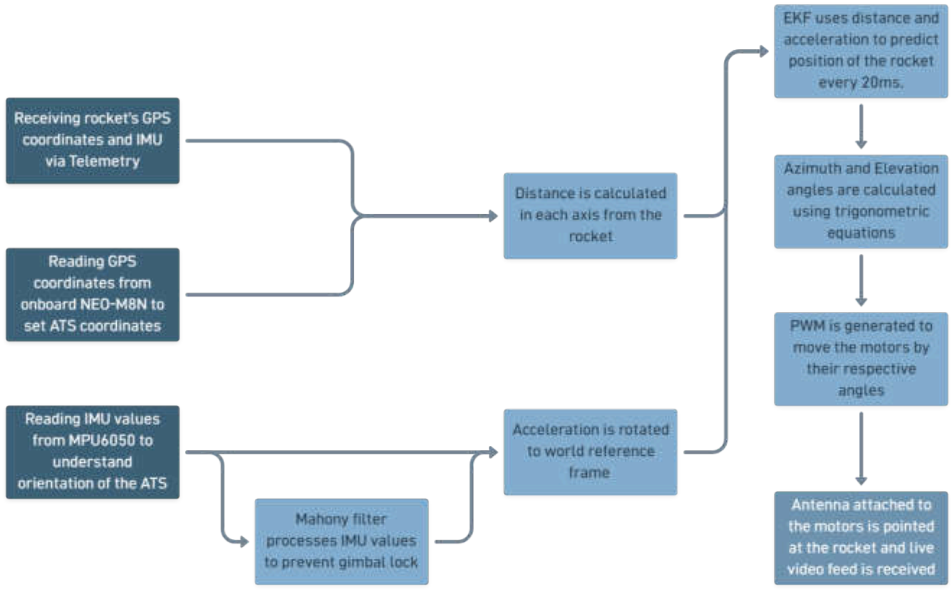


Figure A.217 : Software Workflow of ATS

Hardware:

The ATS is a 3D-printed assembly with all electronics housed inside the base enclosure. The antenna is mounted on the azimuth motor, allowing horizontal rotation. This motor-antenna subassembly is then rotated in elevation by a separate elevation motor using a belt drive. The brain of the ATS works on Teensy 4.1, which is the primary microcontroller that controls all operations. The setup consists of two NEMA-23 stepper motors that move the horn antenna based on the calculated azimuth (left to right) and elevation (up and down) angles. Each motor is driven by a TB6600 stepper motor driver. The rocket's position, acceleration and orientation are received through a RYLR998 LoRa module. A NEO-M8N GPS module is used to set the coordinates of the ATS on initialisation and an MPU6050 IMU that provides the ATS's orientation to compensate for reorientation of the ATS base.



Figure A.218 : Horn antenna used for Live Video Reception

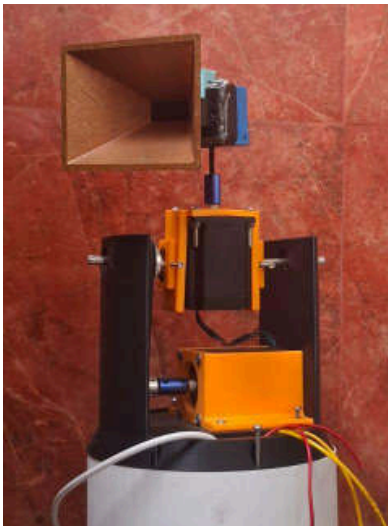


Figure A.219 : Full Hardware assembly of the ATS

Software Workflow:

Initialization:

The system configures communication modules, initializes and calibrates the GPS and IMUs and sets the base reference coordinates (lat_0, lon_0, alt_0) of the ATS. The Extended Kalman Filter (EKF) is initialized and allowed to converge using initial sensor readings. Once stable, azimuth and elevation offsets are calculated to align the tracking system with the target's initial position.

Telemetry:

The system continuously receives telemetry data which includes GPS (latitude, longitude, altitude) and IMU data (acceleration and angular velocity) from the rocket. The data is parsed and validated, and sensor readings are converted into usable units. This ensures real-time awareness of the target's motion and orientation.

State Estimation:

The system performs sensor fusion using the EKF. Acceleration is rotated into the world frame using orientation (quaternion update via Mahony filter), and gravity is removed. The EKF predicts the target's position and velocity, and corrects it using incoming GPS measurements, providing a smooth and accurate estimate of motion.

Angle Computation:

Using the estimated position, the system computes the required azimuth and elevation angles. Offset correction and angle unwrapping are applied to ensure smooth tracking without discontinuities. This step converts spatial coordinates into pointing directions for the antenna.

Motor Control:

The computed angles are converted into stepper motor movements. The system drives azimuth and elevation motors synchronously to align the antenna with the target. Continuous updates ensure real-time tracking. If telemetry is lost, the system can switch to the fallback search mode (spiral scan).



Figure A.220 : Motor Assembly of the ATS

Simulation:

The plot below shows the azimuth and elevation angles calculated in a comparison between two different algorithms for the EKF. One where acceleration is calculated using only GPS coordinates and the time in between packets; and another where acceleration is received via telemetry from the flight computer.

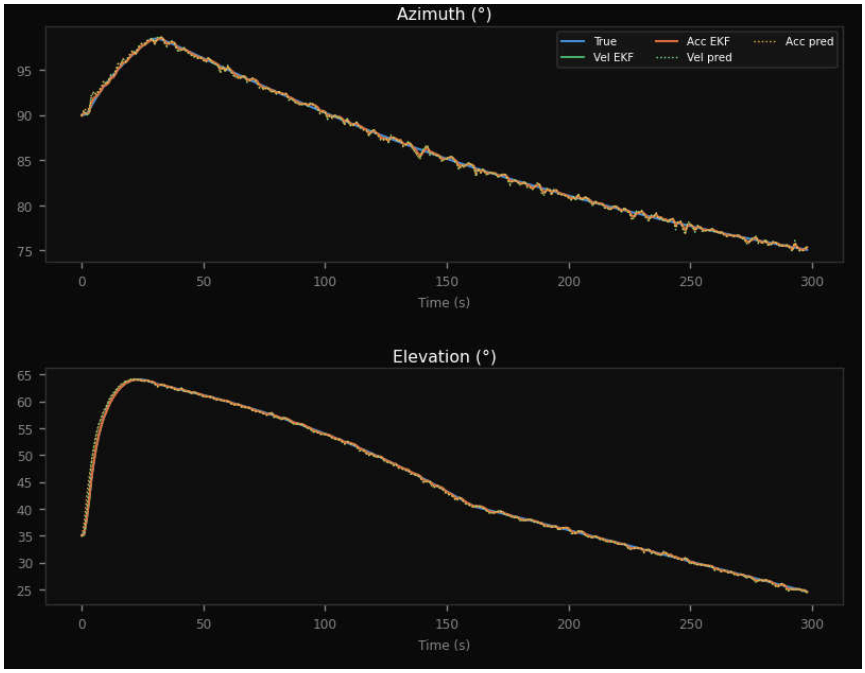


Figure A.221 : Simulation results of the different algorithms

The True trajectory of the rocket had a simulated gaussian error of ±30m. The simulations' root mean square errors are listed below.

Position RMSE: Calculated Acceleration EKF: 15.43 m
Telemetry Acceleration EKF: 14.46 m

Azimuth angle RMSE: Calculated Acceleration EKF: 12.34°
Telemetry Acceleration EKF: 4.11°

Since the telemetry acceleration algorithm had performed better in the simulations, it was the one that was finally used in the ATS.

Technologies & Concepts

Tech Stack:

The flight software is written in C++ using the Arduino framework. The Teensy 4.1 microcontroller interfaced with GPS, IMU, and LoRa communication modules. Real-time telemetry is received via LoRa and processed onboard. Sensor fusion is implemented using an Extended Kalman Filter (EKF) to predict the position of the rocket real-time, while orientation estimation uses a Mahony filter on quaternions to avoid gimbal lock. Stepper motors are controlled by generating a PWM for precise azimuth and elevation tracking. The GPS and LoRa modules work on the UART communication protocol and the coordinates GPS are parsed using NMEA parsing. The onboard MPU6050 is controlled using the Adafruit MPU6050 library.

Input Telemetry Format:

GPS_latitude, GPS_longitude, altitude, acceleration_x, acceleration_y, acceleration_z, angular_velocity_x, angular_velocity_y, angular_velocity_z

Concepts:

The Extended Kalman Filter (EKF) plays a central role in ensuring accurate and stable tracking of the target by sensor fusing the noisy sensor inputs - GPS position, acceleration, and angular velocity. The prediction step uses acceleration data (transformed into the world frame) to estimate position and velocity, while the update step corrects this estimate using GPS measurements. This fusion significantly reduces the impact of noise, latency, and GPS dropouts, providing a smooth and continuous estimate of the target's trajectory.

To support accurate state estimation, a Mahony filter is implemented for real-time orientation tracking. This filter combines gyroscope readings with accelerometer-based gravity correction to maintain a stable quaternion representation of orientation. Using this orientation, body-frame accelerations are rotated into the world frame, allowing gravity to be removed and enabling correct motion modeling for the EKF. This ensures that dynamic motion is accurately captured even during rapid changes in direction or attitude.

A structured tracking logic governs the computation of azimuth and elevation angles required to point the antenna toward the target. The estimated position is converted into angular coordinates using trigonometric relationships, and offset calibration is applied based on initial alignment. To ensure smooth motion, angle unwrapping techniques are used to eliminate discontinuities caused by angular wraparound (±180° transitions), enabling continuous rotation of the azimuth axis without sudden jumps.

The motor control algorithm converts computed azimuth and elevation angles into precise stepper motor commands. Direction and step control signals are generated using PWM, with synchronized stepping ensuring coordinated motion of both axes. The system calculates the required step difference from the current position and executes movement with controlled timing to maintain smooth and accurate tracking. Deadband thresholds are applied to prevent unnecessary micro-adjustments and reduce mechanical jitter.

A fallback recovery mechanism is implemented to handle telemetry loss or signal degradation. In such scenarios, the system switches to a deterministic search strategy based on an Archimedean spiral scan in azimuth and elevation space. The scan expands gradually within predefined angular limits, ensuring efficient coverage of the search region until the signal is reacquired. This prevents complete loss of tracking and enables autonomous recovery without manual intervention.

27. RecovY- The Rocket Recovery App

RecovY - The Rocket Recovery App is a mobile application designed to assist in locating and recovering the rocket after landing. It establishes a seamless data flow from onboard rocket telemetry to a user-friendly mobile interface, enabling real-time tracking and navigation, even in offline environments. Built using Flutter, RecovY follows a modular architecture that ensures efficient data handling, smooth performance, and easy maintenance, making it a reliable tool for recovery operations.

Workflow and Architecture

RecovY follows a structured data flow from the rocket to the user interface. Telemetry data generated onboard the rocket is first received by the CORTEX (GCS) running on a Raspberry Pi. This data is then transmitted to an ESP module via serial communication, which further relays it to the mobile application using Bluetooth Low Energy (BLE).
On the mobile side, the application continuously receives and processes this data, extracting key parameters such as latitude, longitude, and altitude. These values are then utilized across different modules, including navigation and GPS tracking. The navigation system combines real-time location data with device orientation to guide the user toward the rocket's landing point.

The overall architecture is modular, with separate components handling communication (Bluetooth), data processing, navigation, mapping, and logging. This modular design ensures smooth data flow, easier maintenance, and efficient performance during recovery operations.

Technologies and Concepts

RecovY is developed using the Flutter framework and Dart programming language, enabling cross-platform mobile development with a single codebase. Flutter provides a responsive user interface and smooth performance, which are essential for real-time tracking applications.
Bluetooth Low Energy (BLE) is used for communication between the ESP module and the mobile device, ensuring low power consumption and reliable data transfer. GPS technology is used to determine the user's location and track the rocket's position, while digital compass sensors help in determining direction.
The application also supports offline map tiles, enabling navigation in low-connectivity environments. Concepts such as real-time data streaming, modular architecture, and local data storage are applied to ensure efficient data handling, continuous operation, and structured documentation of recovery details.

Functionalities of RecovY

Home Page

The Home Page of RecovY provides a basic overview of the rocket and its key specifications. It serves as the initial screen of the application and presents essential information in a simple and clear layout.

The page displays the rocket name at the top and features a 3D visual representation of *Udbhava*. Alongside this, key rocket parameters are presented, including:

- Rocket Length
- Airframe Diameter
- Vehicle Weight
- Predicted Apogee

Overall, the Home Page provides a straightforward and organized view of basic rocket details without complex interactions.



Figure A.222: RecovY Home Page

Navigator Page

The Navigator Page provides directional guidance to help the user locate the rocket after landing. It combines GPS location data and device orientation to guide the user toward the target position.

The page allows users to set target coordinates manually or receive them automatically from incoming telemetry. Once the target is set, the system continuously updates the user's current location and calculates the direction and distance to the rocket.

A central circular indicator displays the navigation direction using a rotating arrow, which adjusts in real time based on the user's orientation. Along with this, simple directional instructions such as *Go Straight*, *Turn Left*, *Turn Right*, or *Turn Back* are provided for ease of use.

The page also displays the distance to the target, helping the user estimate how far they are from the rocket.

Overall, the Navigator Page provides a simple and effective directional system for guiding the recovery team to the rocket.

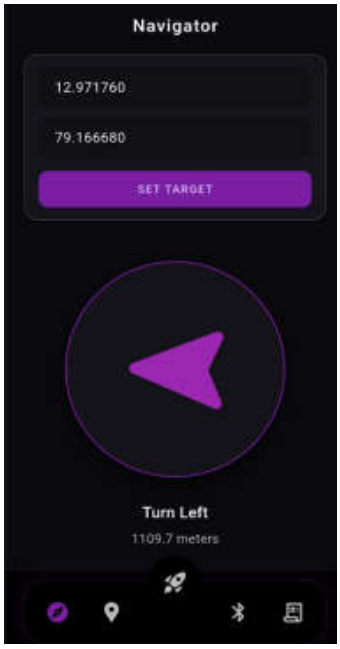


Figure A.223: RecovY Navigator Page

GPS Page

The GPS Page provides a map-based view for tracking the rocket's location and guiding the user during recovery. The page displays the user's current position and the rocket's location on a map. A line is drawn between these two points to indicate the path to the rocket. The user's position is represented with a directional arrow, while the rocket location is marked separately.The system continuously updates the user's position using GPS and calculates the distance and direction to the rocket in real time. An estimated time to reach the rocket is also displayed based on a constant movement speed.

The map uses preloaded offline tiles, allowing it to function without internet connectivity. Basic controls such as centering and zooming are handled automatically to keep both the user and rocket in view.
Additionally, a small information panel displays:

- Rocket coordinates
- Distance to the rocket
- Estimated time to reach (assuming speed = 2m/s)

Overall, the GPS Page provides a clear and practical map interface for tracking and reaching the rocket during recovery operations.

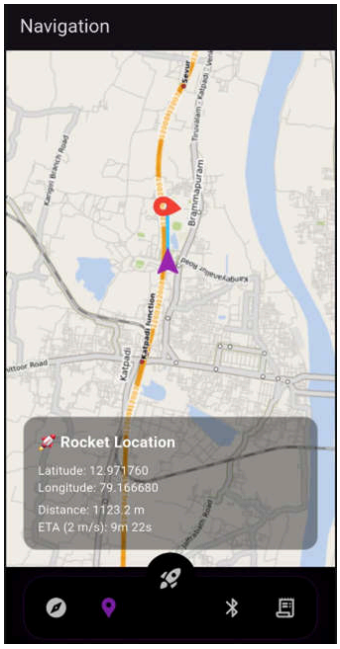


Figure A.224: RecovY GPS Page

Bluetooth Page

The Bluetooth Page handles communication between the mobile application and the ESP device using Bluetooth Low Energy (BLE). Telemetry data from the rocket is received in the CORTEX (GCS) running on the Raspberry Pi. This data is then sent to an ESP module via serial communication, which transmits it to the mobile application over BLE. The page allows users to scan for nearby Bluetooth devices and connect to the required ESP device. Once connected, the application begins receiving telemetry data in real time.

The incoming data is parsed and key parameters such as:
-Latitude
-Longitude
- Altitude

are extracted and shared with other modules like navigation and tracking. The page also provides controls to start, stop, and reset data reception, allowing the user to manage the data stream. Additionally, the interface displays the current rocket coordinates and altitude for quick reference.

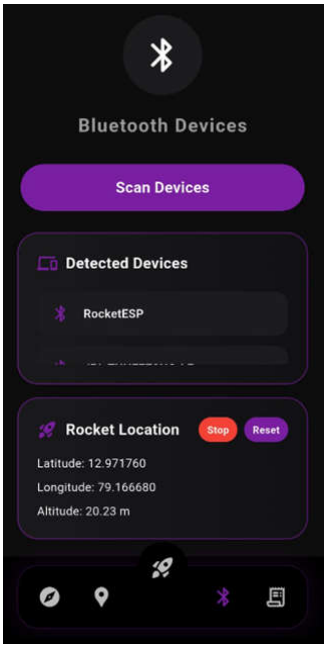


Figure A.225 : RecovY Bluetooth Page

Details Log Page

The Details Log Page of RecovY is designed to document the condition of rocket components after recovery. It provides a structured interface where users can inspect and record the status of different parts such as the nose cone, fins, parachutes, payload, and other critical elements.

Each component is presented in an expandable section, allowing users to input three key pieces of information:

- Condition (selected from predefined options)
- Description (manual text entry)
- Images (captured using camera or selected from gallery)

The page ensures that all necessary data is collected before saving, helping maintain consistency in reporting. All entries are stored locally, enabling data persistence even after the app is closed. Additionally, the page supports generating a detailed PDF report. This report includes all recorded conditions, descriptions, images, along with the date, time, and recorder's name. This feature is useful for post-recovery analysis, documentation, and sharing within the team.

Overall, the Details Log Page provides a simple and organized way to capture and preserve recovery-related information without unnecessary complexity**.**

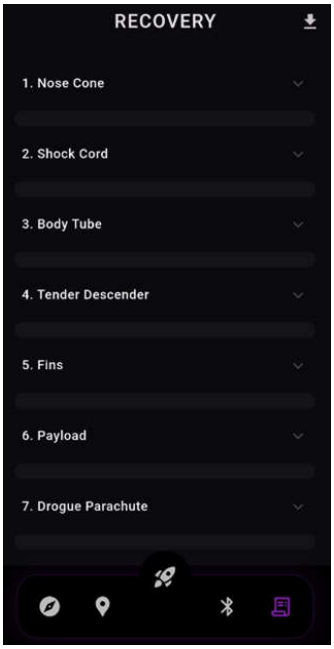


Figure A.226 : RecovY Details Log Page

28. Payload Subsystem

System Design

The motion system employs a two-axis planar configuration to provide controlled positioning of the extrusion assembly. Linear motion along the X-axis is achieved using a lead screw mechanism driven by a stepper motor through a belt-pulley arrangement, allowing precise displacement with inherent self-locking characteristics. The Y-axis utilizes a rack and pinion mechanism directly coupled to a stepper motor, reducing moving mass and simplifying mechanical complexity. Guide rods constrain the motion assembly and prevent rotational misalignment, ensuring consistent toolpath accuracy throughout operation. Under reduced-gravity conditions, the absence of gravitational loading reduces drivetrain resistance, improving motion smoothness and repeatability.

X-Axis Motion:

X-axis translation of the extrusion assembly is achieved through a lead screw drive system actuated by a stepper motor. The motor transmits torque to a pair of lead screws via a timing belt and pulley arrangement, with pulleys mounted directly onto the lead screws. Each lead screw interfaces with a nut carrying one arm of the extrusion assembly, which is housed within the Y-axis motion subassembly. Rotational input from the motor is thereby converted into precise linear displacement of the nuts along the lead screw axes, providing controlled and repeatable X-axis positioning of the extrusion assembly. The inherent self-locking characteristics of the lead screw mechanism further ensure positional stability during the microgravity phase of flight, where the absence of gravitational loading removes any restoring bias that would otherwise act on the drivetrain.

Y-Axis Motion:

Y-axis motion is implemented using a rack and pinion drive, with the pinion coupled directly to a stepper motor mounted on the extrusion assembly itself. This arrangement converts motor rotation directly into Y-axis translation without intermediate linkages, reducing mechanical complexity and minimising the moving mass of the assembly in accordance with DFMA principles. Critically, the absence of gravitational loading in the microgravity environment eliminates sag and gravity-induced moment loads on the extrusion assembly, allowing the rack and pinion to operate as the sole load-bearing drive element without the frictional and wear penalties that such a configuration might incur under terrestrial conditions. Two guide rods oriented along the Y-axis provide lateral constraint and prevent rotation of the assembly about the drive axis, ensuring print path accuracy is maintained throughout the microgravity window.

Extrusion Mechanism:

Bio-ink extrusion is achieved through a syringe-based dispensing system. A 5 mL syringe serves as both the ink reservoir and the extrusion element, selected to simplify in-flight refill logistics and improve handling safety during pre-launch integration. Actuation is provided by a rack and pinion mechanism in which a translating rack depresses the syringe plunger at a controlled rate; the syringe body is mounted on the rack such that linear displacement produces a proportional and continuous volumetric output to the print head.

A key design consideration in the microgravity context is the elimination of hydrostatic pressure within the bio-ink volume. Under terrestrial conditions, the weight of the ink column within the syringe contributes a variable back-pressure that must be compensated for in extrusion rate calibration. In microgravity, this hydrostatic component vanishes, yielding a pressure environment governed solely by the mechanical actuation of the rack. This simplifies flow modelling and enables more predictable extrusion behaviour, as the output becomes a direct and near-linear function of rack displacement rather than a compound of mechanical actuation and fluid-column pressure. The syringe-based architecture is further well-suited to the microgravity environment in that it forms a closed fluid system, eliminating free surface effects and the risk of ink migration under microgravity conditions that would be associated with open reservoir designs.

Control of the motion system is implemented through a lightweight G-code interpretation framework, allowing predefined toolpaths to be executed without modifying firmware. The following implementation demonstrates the parsing of G-code commands and their conversion into stepper motor movements:

G-Code Parser Snippet:

```C/C++
// String parsing engine natively handles standard G-Code inputs
void executeGcodeLine(String line) {
  if (line.startsWith("G1")) {
    int xIndex = line.indexOf('X');
    int yIndex = line.indexOf('Y');

    // Extracts magnitude and polarity for the X-Axis
    if (xIndex != -1) {
      int spaceIndex = line.indexOf(' ', xIndex);
      if (spaceIndex == -1) spaceIndex = line.length();
      int xSteps = line.substring(xIndex + 1, spaceIndex).toInt();
      moveMotor(STEP_X, DIR_X, EN_X, (xSteps >= 0), abs(xSteps));
    }

    // Extracts magnitude and polarity for the Y-Axis
    if (yIndex != -1) {
      int spaceIndex = line.indexOf(' ', yIndex);
      if (spaceIndex == -1) spaceIndex = line.length();
      int ySteps = line.substring(yIndex + 1, spaceIndex).toInt();
      moveMotor(STEP_Y, DIR_Y, EN_Y, (ySteps >= 0), abs(ySteps));
    }
  }
}
```

Motor actuation is achieved through step generation using precise timing control, as shown below:

Kinematic Step Generation Snippet:

```C/C++
// General Kinematic Control Function
void moveMotor(int stepPin, int dirPin, int enPin, bool forward, int steps) {
  digitalWrite(enPin, LOW); // Engage axis holding torque
  digitalWrite(dirPin, forward ? HIGH : LOW); // Set vector direction

  // Generate PWM square wave to advance stepper rotor
  for (int i = 0; i < steps; i++) {
```

```
      digitalWrite(stepPin, HIGH);
      delayMicroseconds(STEP_DELAY);
      digitalWrite(stepPin, LOW);
      delayMicroseconds(STEP_DELAY);
    }
  }
```

This approach enables deterministic control of motion, ensuring accurate positioning and repeatable deposition under dynamic conditions. The use of parameterized step control allows scalability across different toolpaths and printing geometries.

29. Air Brakes Subsystem

Structural Analysis

The airbrake mechanism is integrated into the structural coupler that connects two sections of the body tube. The coupler functions both as a load bearing element, as well as the housing for the airbrakes actuation system. The internal structure of the coupler supports the linear rails, cam plate, servo mounting interface and the controlling electronics. This integration minimizes the additional structural mass while maintaining the mechanical integrity of the airframe.

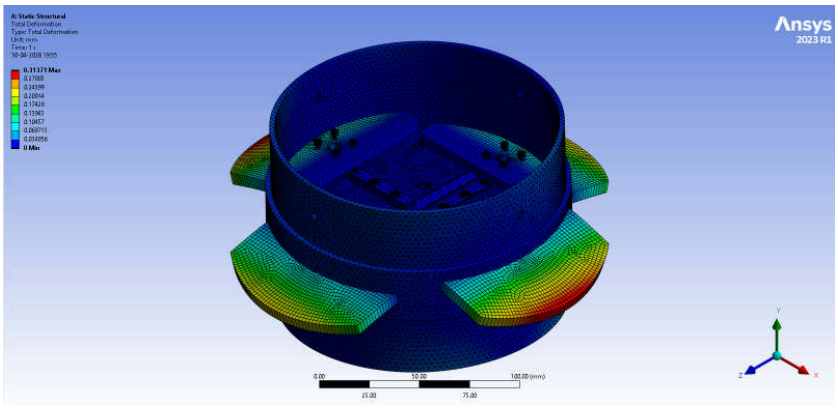


Figure A.227 : Total Deformation Analysis of Airbrakes Coupler

Structural analysis of the coupler was performed to evaluate its performance. The results indicate a maximum displacement of 0.31mm, demonstrating minimum deformation. The minimum factor of safety was found to be 1.39, confirming that the design remains stable under applied loads throughout flight.

CFD Validation

We performed a comprehensive computational fluid dynamics (CFD) analysis of the complete rocket configuration of Udbhava during the ascent (coasting/casting) phase, with particular emphasis on the aerodynamic effects of airbrake deployment. The simulations were conducted across multiple discrete deployment levels to capture the variation in flow behavior and aerodynamic coefficients as a function of airbrake extension

The study involved evaluating key aerodynamic parameters, including drag coefficient (Cd), pressure distribution, shock formation, and flow separation characteristics under transonic and supersonic regimes. Special attention was given to the interaction between the deployed airbrakes and the surrounding flow field, as these surfaces introduce significant perturbations, leading to changes in boundary layer behavior and wake structure.

For the transonic analysis of the complete rocket configuration with integrated airbrakes, a poly-hexcore mesh was generated using Fluent Meshing. The geometry was initially created in SpaceClaim, with a body of influence region defined at critical locations such as the wake region, and a cylindrical domain surrounding the rocket body to achieve refined meshing.

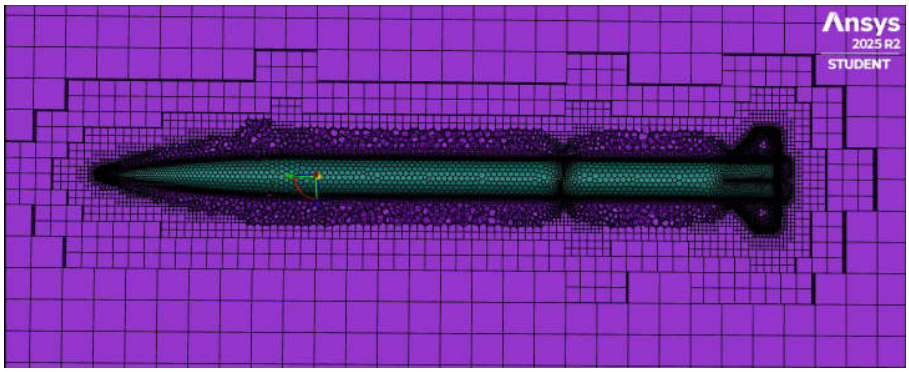


Figure A.228 : Poly Hex-Core Mesh formed Around Rocket

By systematically varying the airbrake deployment percentage, the analysis provides a detailed understanding of how incremental deployment influences overall vehicle drag and aerodynamic stability. The results are intended to support the development of an accurate drag model for trajectory control algorithms and to ensure reliable prediction of apogee modulation during flight.

At 40% Deployment



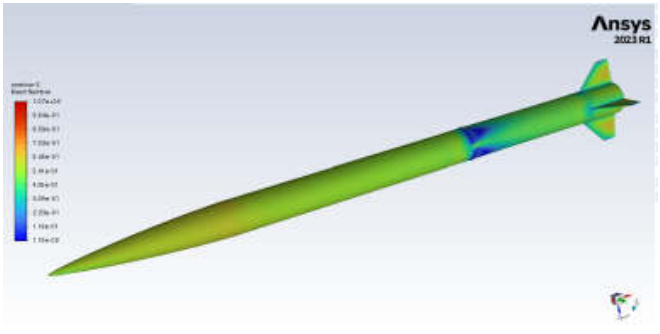


Figure A.229 : Static Pressure Contours for 40% Deployment *Figure A.230 : Mach Number Contours for 40% Deployment*

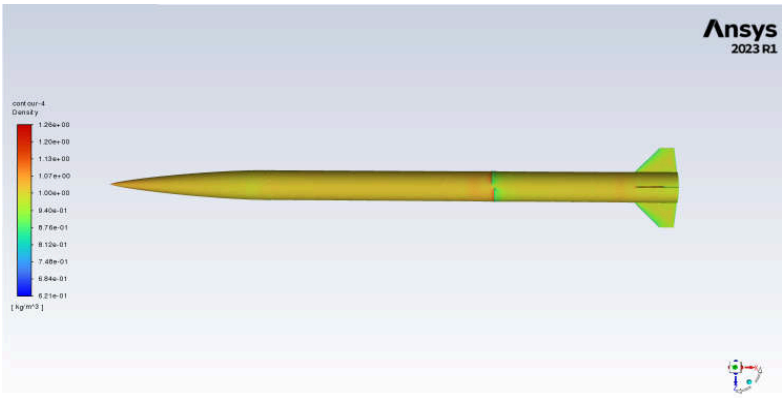


Figure A.231 : Density Contours for 40% Deployment

At 50% Deployment

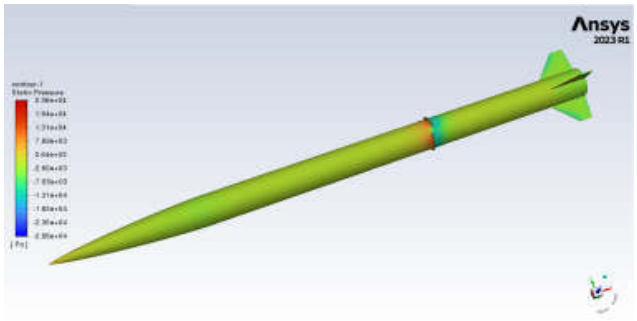


Figure A.232:Static Pressure Contours for 50% Deployment

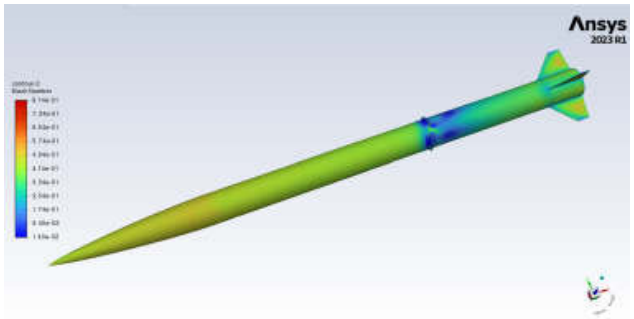


Figure A.233: Mach Number Contours for 50% Deployment

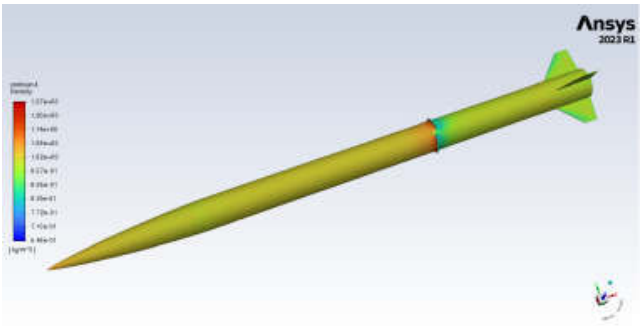
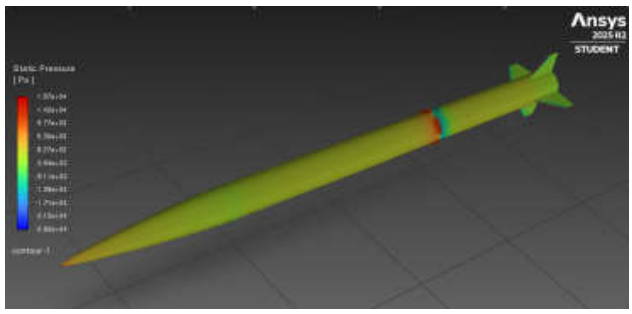
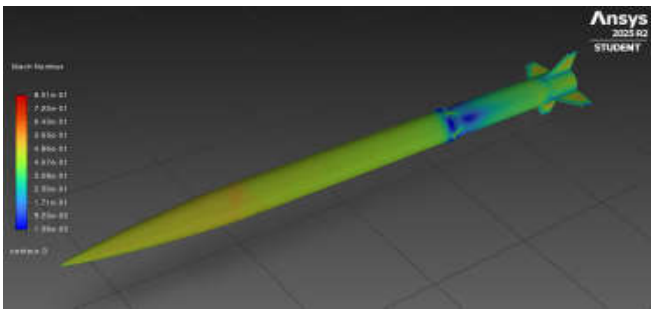


Figure A.234 : Density Contours for 50% Deployment

At 60% Deployment



FigureA.235: Static Pressure Contours for 60% Deployment



FigureA.236: Mach Number Contours for 60% Deployment

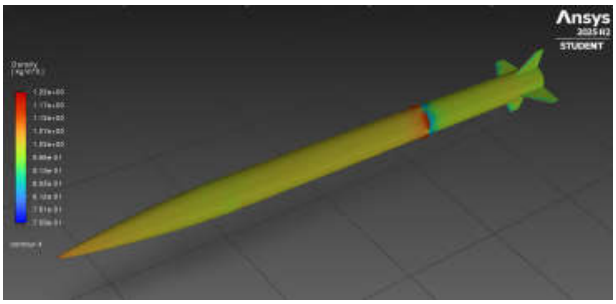


Figure A.237 : Density Contours for 60% Deployment

At 70% Deployment

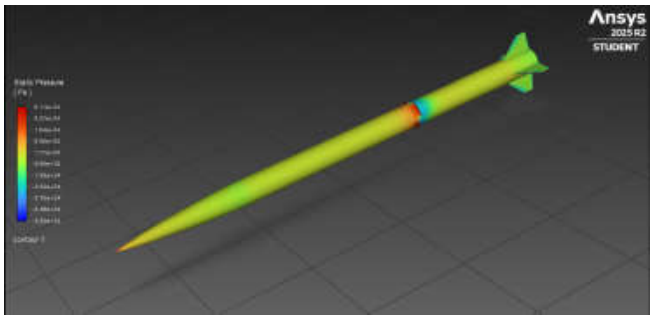


Figure A.238:Static Pressure Contours for 70% Deployment

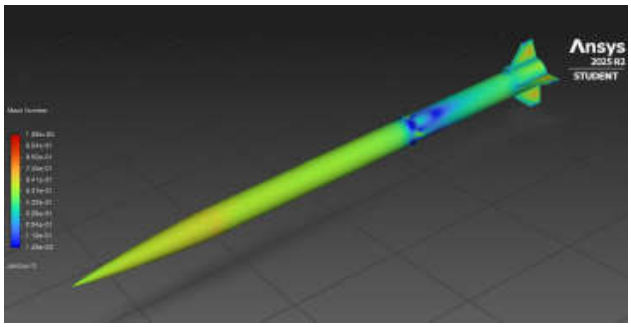


Figure A.239: Mach Number Contours for 70% Deployment

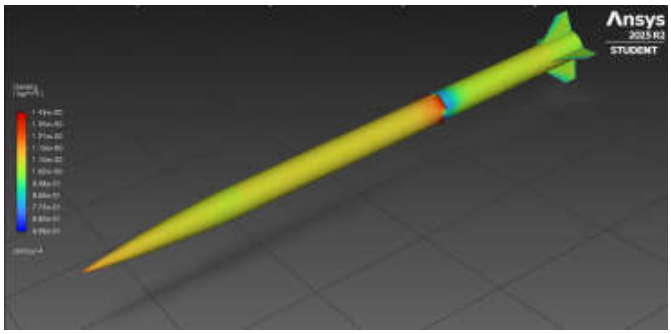
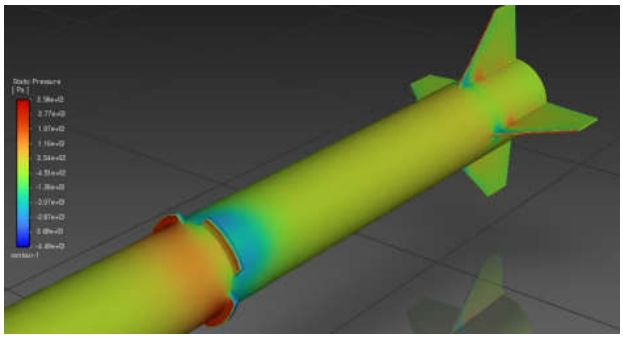


Figure A.240: Density Contours for 70% Deployment

At 80% Deployment



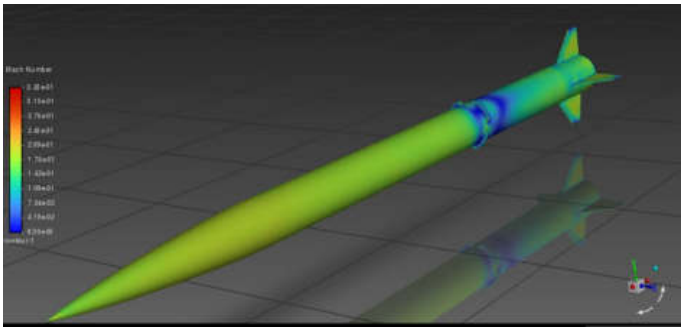


Figure A.241:Static Pressure Contours for 80% Deployment *Figure A.242:Mach Number Contours for 80% Deployment*

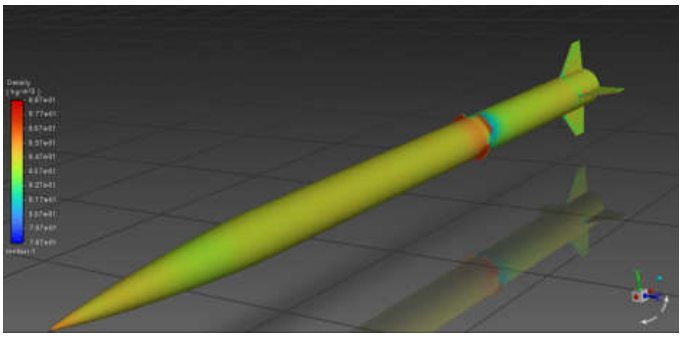
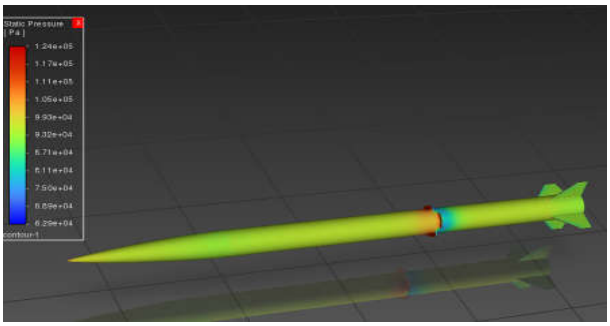


Figure A.243: Density Contours for 80% Deployment

At 90% Deployment



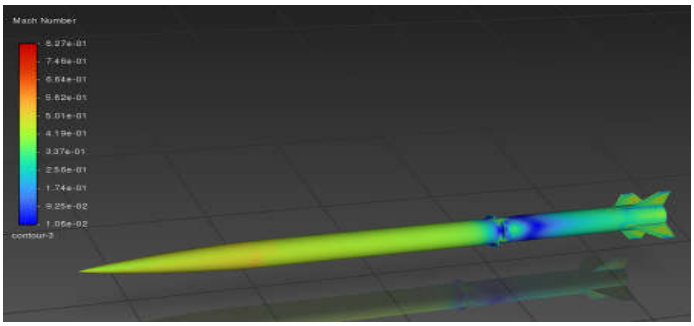


Figure A.244 :Static Pressure Contours for 90% Deployment *Figure A.245: Mach Number Contours for 90% Deployment*

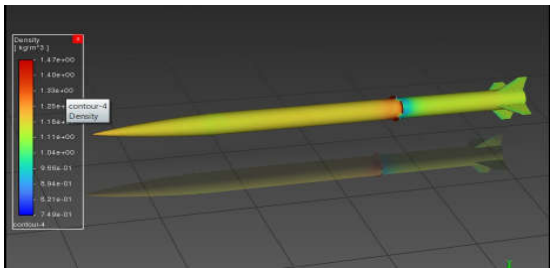


Figure A.246: Density Contours for 90% Deployment

At 100% Deployment

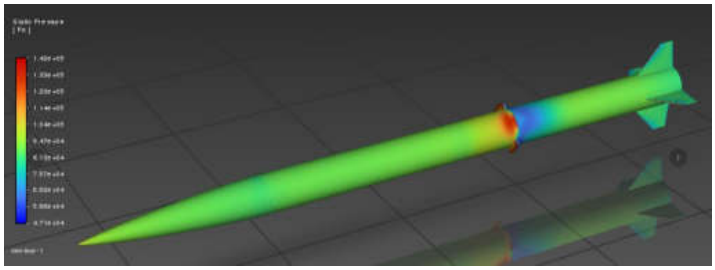


Figure A.247 : Static Pressure Contours for 100% Deployment

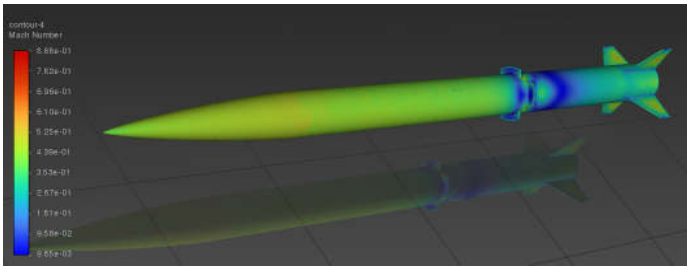


Figure A.248:Mach Number Contours for 100% Deployment

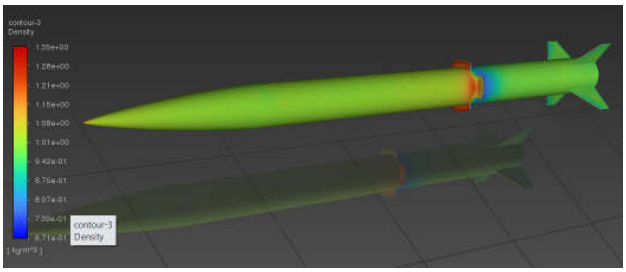


Figure A.249: Density Contours for 100% Deployment

Following the contour visualizations, a quantitative analysis was conducted by plotting the Coefficient of Drag with Mach Number for different airbrake deployment levels. The results are summarized in the plot below.

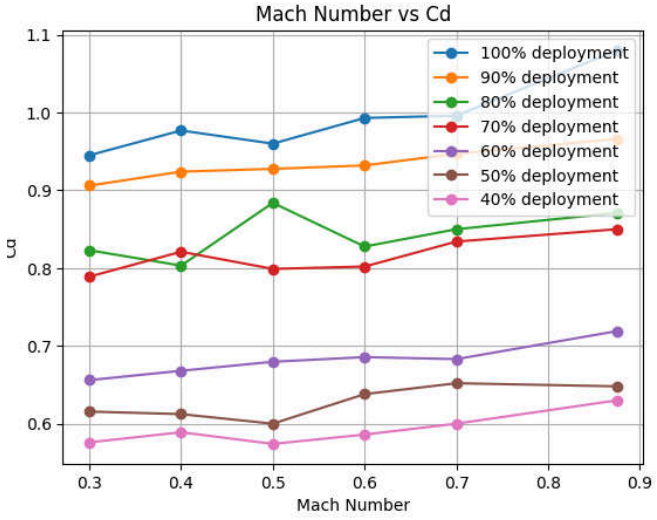


Figure A.250: Mach Number vs CD plot for various percentages of deployment

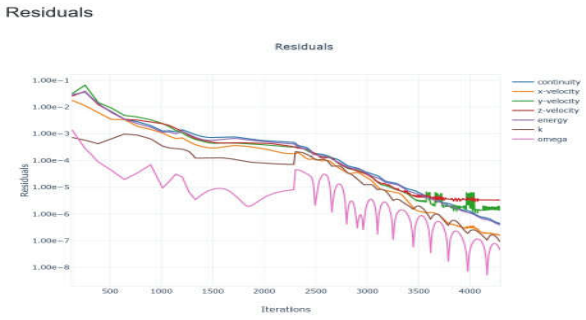


Figure A.251: Scaled Residuals

Iterations: 4294

	Value	Absolute Criteria	Convergence Status
continuity	4.305457e-07	1e-06	Converged
x-velocity	1.577059e-07	1e-06	Converged
y-velocity	1.404854e-06	1e-06	Not Converged
z-velocity	3.337174e-06	1e-06	Not Converged
energy	3.859607e-07	1e-06	Converged
k	8.962877e-08	1e-06	Converged
omega	4.374273e-08	1e-06	Converged

Figure A.252: Console Convergence Status

Deployable Airbrake System Overview

The airbrake operates exclusively during the coasting ascent phase. Deployment prior to motor burnout would destabilise the vehicle; deployment during descent has no corrective effect on apogee. The flap mechanism therefore remains fully retracted in all other flight phases.

The flight computer is implemented on an ESP32-S3-WROOM-1 microcontroller operating at 240 MHz with a hardware floating-point unit. The firmware employs a dual-rate loop architecture: a 500 Hz inner loop for inertial measurement and attitude estimation, and a 100 Hz outer loop for barometric state correction, apogee prediction, and servo actuation. The overall system is partitioned into four layers, as illustrated in Figure 1.

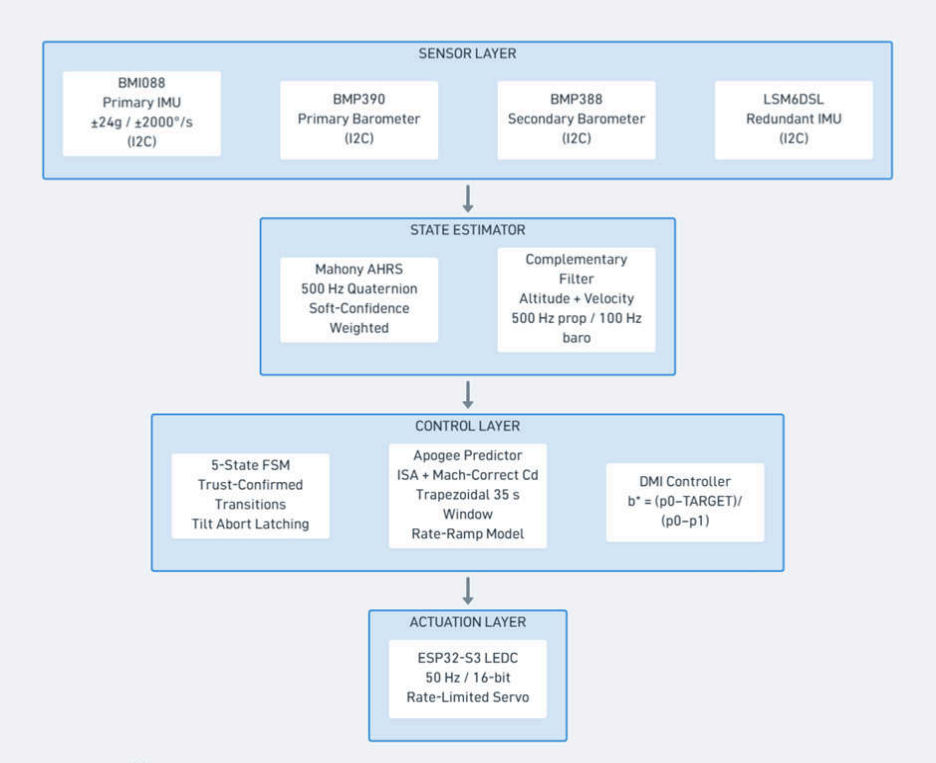


Figure A.253: System Architecture

2. Loop Architecture

The main loop is paced by a microsecond hardware timer that enforces a strict 2 ms IMU period. A divide-by-five counter triggers the slower 10 ms control window. Execution within each window is as follows:

500 Hz IMU tick (every 2 ms): imu_read() (~0.3 ms) reads both inertial sensors, performs per-sensor plausibility checks, fuses valid outputs, and updates the Mahony AHRS quaternion. cf_propagate() advances the complementary filter altitude and velocity estimates using the latest world-frame vertical acceleration. fsm_update() evaluates all phase-transition conditions against trust-confirmed counters.

100 Hz control tick (every 10 ms, on the fifth IMU tick): baro_read() acquires a fresh barometric altitude and passes it through a rate-plausibility gate. cf_baro_correct() applies the barometric correction to the complementary filter. controller_update() calls the apogee predictor twice and computes the Direct Model Inversion (DMI) brake command. print_telem() emits a CSV telemetry line.

The worst-case execution budget for a 10 ms control window is approximately 6.0 ms, leaving a 40% margin before the next boundary. Loop timing is logged in every telemetry packet; values of imu_loop_us > 2000 µs or ctrl_loop_us > 9000 µs indicate a timing overrun requiring investigation.

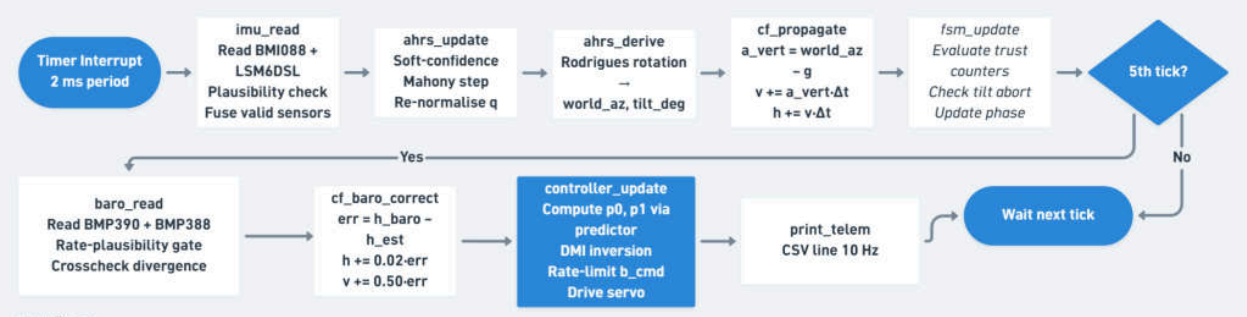


Figure A.254: Main Loop Flowchart

3. State Estimator

A Barometric Altimeter -Role in the Estimator

The primary barometer (BMP390, I2C) provides a low-bandwidth but drift-free measurement of altitude above ground level (AGL). It is used for two purposes: (1) periodically correcting the complementary filter's altitude and velocity estimates, which would otherwise accumulate velocity drift from IMU bias over the coast phase, and (2) supplying the absolute altitude input to the apogee predictor.

Ground calibration establishes the launch-site reference pressure P_0 and temperature T_0 from a 100-sample average of BMP390 readings with 2σ outlier rejection. All in-flight altitude calculations are relative to this baseline, ensuring that AGL altitude is correctly referenced regardless of launch-site elevation. Omitting this calibration and assuming sea-level ISA conditions would introduce an approximately 9% air density error, producing a prediction offset of approximately 20 m at the target altitude [ICAO, 1993].

Each barometric reading is passed through a rate-plausibility gate before it is accepted into the state estimate. The gate computes the implied altitude rate of change from the previous accepted sample; any reading implying a rate exceeding 350 m/s is discarded and flagged in telemetry. This threshold exceeds the rocket's actual maximum speed (~300 m/s) while reliably rejecting sensor glitches or pressure transients caused by airbrake panel deployment.

The secondary barometer (BMP388, I2C) operates in parallel as a crosscheck. A divergence of more than 15 m between the two sensors generates a serial warning; the secondary reading does not feed the estimator directly.

B Mahony Quaternion AHRS -Role in the Estimator

The Mahony filter maintains a quaternion representation of vehicle attitude at 500 Hz. Its output serves two critical functions: extracting the true vertical component of acceleration for the complementary filter, and computing the vehicle tilt angle for the safety abort logic.

Without attitude knowledge, only the body-frame axial accelerometer reading is available, which underestimates true vertical acceleration by cos(θ) at any tilt angle θ. For a rocket that has pitched 10° at burnout, this introduces a systematic velocity error of ~1.5%. The quaternion corrects this by rotating the full three-axis accelerometer vector into the world frame, yielding the true vertical specific force regardless of tilt.

Algorithm. Each 500 Hz tick the filter executes six steps:

1. **Soft accel confidence weighting:** The accelerometer magnitude |a| is compared to the expected gravitational norm G. A confidence scalar is computed as:

   $$clamp\left(1 - \frac{|\,\|\mathbf{a}\| - g\,|}{2g},\ 0,\ 1\right)$$

   At 1g the confidence is 1.0 (full correction active). At 2g it is 0.5 (half-gain correction). Above 3g it drops to zero (gyro-only propagation). A hard floor of zero is applied below 0.5g to reject free-fall and dead-sensor conditions. This soft weighting replaces a previously used hard gate of [0.5g, 2g], which caused the filter to propagate on gyroscope data alone for the entire motor burn -accumulating unchecked gyro bias that manifested as attitude error at coast entry.

2. Gravity reference. The expected gravity direction in the body frame is extracted from the third column of the rotation matrix encoded in quaternion

   $$q = [q_w, q_x, q_y, q_z]$$

   $$g_{body} = \begin{bmatrix} 2(q_x q_z - q_w q_y) \\ 2(q_y q_z + q_w q_x) \\ q_w{}^2 - q_x{}^2 - q_y{}^2 + q_z{}^2 \end{bmatrix}$$

3. Cross-product error

   $$\mathbf{e} = \hat{\mathbf{a}} \times \mathbf{g}_{\text{body}}, \qquad \hat{\mathbf{a}} = \frac{\mathbf{a}}{||\mathbf{a}||}$$

4. PI feedback applied to gyroscope reading, with gains scaled by confidence:

$$\omega_c = \omega + K_p \cdot \text{conf} \cdot \mathbf{e} + \int K_i \cdot \text{conf} \cdot \mathbf{e}\, dt$$

$$K_p = 2.0, \qquad K_i = 0.005$$

5. Quaternion integration.

$$\dot{\mathbf{q}} = \frac{1}{2}\mathbf{q} \otimes \begin{bmatrix} 0 \\ \omega_c \end{bmatrix}$$

$$\mathbf{q} \leftarrow \mathbf{q} + \dot{\mathbf{q}}\,\Delta t$$

```
C/C++

q[0] += (-q_x·ω_x - q_y·ω_y - q_z·ω_z) × 0.5·Δt

q[1] += (+q_w·ω_x + q_y·ω_z - q_z·ω_y) × 0.5·Δt

q[2] += (+q_w·ω_y - q_x·ω_z + q_z·ω_x) × 0.5·Δt

q[3] += (+q_w·ω_z + q_x·ω_y - q_y·ω_x) × 0.5·Δt
```

6. Re-normalisation to the unit sphere, preventing floating-point drift outside SO(3).

World-frame vertical acceleration. Only the Z component of the rotated acceleration vector is needed. The Rodrigues rotation formula yields this in 15 multiplications without constructing the full 3×3 rotation matrix:

```
C/C++
t  = 2 × (q.xyz × a_body)

a_world_z = a_body_z + q_w·t_z + (q_x·t_y - q_y·t_x)
```

Tilt angle. Derived directly from the quaternion, avoiding the atan2-based method which is unreliable during boost:

$$\cos\theta = q_w^2 - q_x^2 - q_y^2 + q_z^2, \qquad \theta = \arccos\big(\text{clamp}(\cos\theta,\, -1,\, 1)\big)$$

```
C/C++
cos(tilt) = q_w² - q_x² - q_y² + q_z²
```

```
tilt_deg  = arccos(clamp(cos_tilt, -1, 1)) × 180/π
```

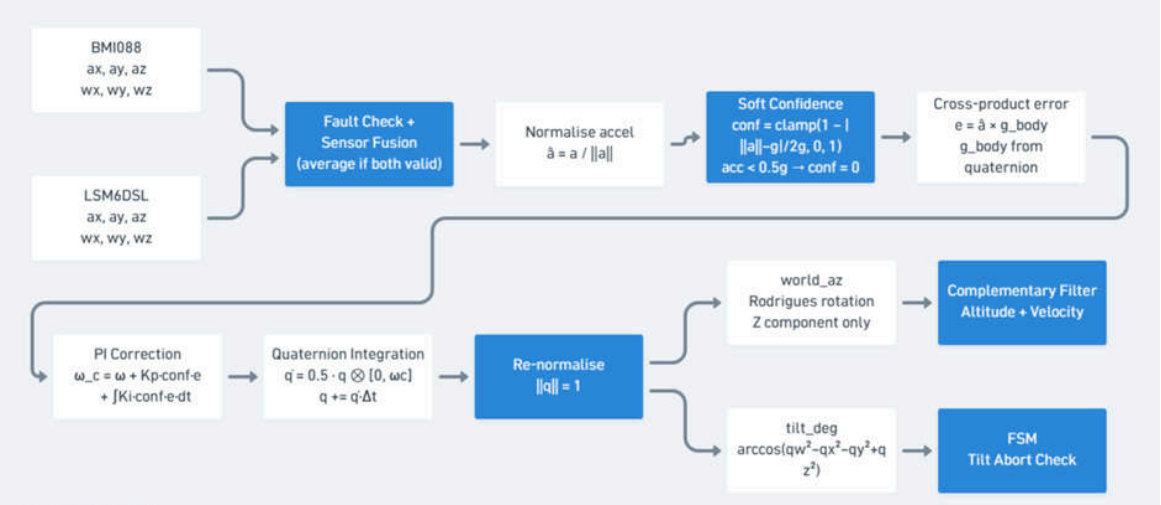


Figure A.255: Mahony AHRS Signal Flow

C Complementary Filter

The complementary filter fuses inertial propagation at 500 Hz with barometric correction at 100 Hz to produce altitude and velocity estimates for the controller. The filter is chosen over an Extended Kalman Filter (EKF) because it achieves equivalent accuracy at approximately 5% of the computational cost, with no covariance matrix tuning risk [Mahony et al., 2008].

Propagation step (500 Hz):

$$v \mathrel{+}= (a_{\text{world},z} - g)\,\Delta t, \qquad h \mathrel{+}= v\,\Delta t$$

```
C/C++
a_vert = a_world_z - g

v  +=  a_vert × Δt_imu

h  +=  v × Δt_imu
```

Barometric correction step (100 Hz):

$$\varepsilon = h_{\text{baro}} - h_{\text{est}}, \quad h \mathrel{+}= K_h\,\varepsilon, \quad v \mathrel{+}= K_v\,\varepsilon$$

```
C/C++
err = h_baro - h_est

h  += CF_BARO_GAIN × err      (CF_BARO_GAIN = 0.02)

v  += CF_VEL_GAIN  × err      (CF_VEL_GAIN  = 0.50)
```

The two-pole observer characteristic equation is $s^2 + 0.50 \cdot s + 0.02 = 0$, giving $\omega_n \approx 3.2$ rad/s and $\zeta \approx 0.78$ -well-damped and stable. The altitude time constant is approximately 0.1 s (≈10 Hz crossover). The velocity gain of 0.50 yields a critically-damped observer response.

4. Apogee Predictor

The predictor integrates the equations of motion forward in time from the current state, returning the altitude at which velocity reaches zero. It is invoked twice per control tick: once with brakes fully retracted (yielding p0, the natural-trajectory apogee) and once fully deployed (yielding p1, the maximum-drag apogee). These two values bracket the reachable apogee space and are the sole inputs to the controller inversion.

The predictor accepts both the current brake command and the target brake fraction, ramping between them at MAX_BRAKE_RATE (2.0 fraction/s) per simulation step. This models the physical servo deployment rate inside the prediction, making the Direct Model Inversion self-consistent with actual actuator behaviour.

The simulation window is 700 steps at 0.05 s each, covering a 35-second coast phase. If the window is exhausted with positive velocity (indicating an unusually slow coast), the function returns the current altitude as a conservative lower bound and raises a telemetry warning. In nominal flight this condition does not arise.

A Integration -Trapezoidal Rule

Trapezoidal integration is used in preference to the symplectic Euler method. Symplectic Euler evaluates each new position using the updated velocity, introducing a consistent upward bias of approximately 3 m over the coast phase [Hairer et al., 2006]. The trapezoidal rule eliminates this bias at the cost of one additional addition per step:

$$h \mathrel{+}= \frac{v_{\text{old}} + v_{\text{new}}}{2} \Delta t$$

B Atmosphere Model (ISA, Site-Calibrated)

A full International Standard Atmosphere model is used, seeded with the P_0 and T_0 measured during ground calibration. Site calibration is critical at altitude: a launch site at 1500 m elevation has approximately 9% lower air density than sea level, directly affecting the available drag authority.

Troposphere ($T > 216.65$ K):

$$T(h) = T_0 - L\,h, \qquad P(h) = P_0 \left(\frac{T}{T_0} \right)^{\frac{g}{RL}}$$

```
C/C++
T(h) = T₀ - L·h

P(h) = P₀·(T/T₀)^(g/R·L)
```

Stratosphere (T ≤ 216.65 K):

```
C/C++
P(h) = P_tp · exp(-g·(h - h_tp) / (R·T_strat))
```

Air density at each step: ρ = P/(R·T).

C Drag Model and Mach Correction

The local speed of sound is computed from ISA temperature at each simulation step, ensuring altitude-dependent Mach number accuracy:

$$M = \frac{v}{\sqrt{\gamma\, RT(h)}}$$

At 3000 m ISA, T ≈ 268 K, giving c_s ≈ 328 m/s versus the sea-level value of 343 m/s. For a rocket travelling at 300 m/s, this corrects the predicted Mach number from 0.87 to 0.91 -a shift of approximately one table interval in the Cd lookup, equivalent to a 3–5% drag coefficient correction.

The total drag coefficient interpolates linearly between the fully-retracted and fully-deployed tables based on the instantaneous brake fraction:

Cd(M, b) = lerp(CD_BODY(M), CD_BRAKE_FULL(M), b)

The lookup tables are presented in Table 20

Table A.20: Cd Lookup Tables (abridged -values from OpenRocket export)

Mach	CD body (b=0)	CD full brakes (b = 1)
0.0 - 0.2	0.390	0.670
0.3	0.397	0.677
0.5	0.421	0.701
0.7	0.464	0.744
0.9	0.535	0.815
1.0	0.560	0.840

5. Control Design

A Direct Model Inversion

The controller employs Direct Model Inversion (DMI) as its sole control strategy. Rather than closing a feedback loop on altitude error, DMI directly inverts the predictor to find the brake fraction b* such that predict(h, v, b*) = TARGET. Since p0 and p1 are already computed, the solution requires only linear interpolation in prediction space:

```
C/C++
if p0 ≤ TARGET  →  b_dmi = 0      (natural trajectory reaches target
without brakes)

if p1 ≥ TARGET  →  b_dmi = 1      (full brakes required)

otherwise       →  b_dmi = (p0 - TARGET) / (p0 - p1)
```

This inversion is physically exact -it uses the full Mach-dependent, ISA-correct drag model rather than a linearised approximation, and therefore requires no gain parameters. An integral correction term previously included in earlier firmware revisions was removed following Monte Carlo analysis. The integral accumulated the difference (p0 − TARGET) -a noisy predicted signal, not a true tracking error -and introduced a systematic undershoot bias of 30–50 m across all uncertainty levels tested. Pure DMI achieves a mean absolute error (MAE) ≤ 12 m with 100% of Monte Carlo runs within ±50 m across ±5–15% systematic drag and mass uncertainty. Table 2 summarises the comparison.

Table A.21: Monte Carlo v2 Controller Comparison (systematic Cd/area bias)

Controller	MAE (σ = 5%)	MAE (σ = 15%)	Within ±50 m	Tuning params
DMI only	1.5 m	12.0 m	100%	0
DMI + Integral	37.8 m	52.4 m	50-85%	3
Adaptive PID	304 m	316 m	0%	4

B Rate Limiter and Final Command

The brake command is rate-limited to MAX_BRAKE_RATE = 2.0 fraction/s, corresponding to full deployment in 0.5 s. This value was derived from a bench measurement of 0.33 s mean deployment time for the complete actuator assembly, with a 1.5× safety factor applied to account for aerodynamic loading on the panels in flight:

$$t_{\text{limit}} = t_{\text{meas}} \times 1.5 = 0.33\,\text{s} \times 1.5 = 0.50\,\text{s} \quad \Rightarrow \quad \dot{b}_{\max} = \frac{1}{t_{\text{limit}}} = 2.0\,\text{fraction/s}$$

```
C/C++
t_limit = 0.33 s × 1.5 = 0.50 s
```

```
MAX_BRAKE_RATE = 1 / 0.50 = 2.0 fraction/s
```

The final command update per 10 ms control tick:

$$b^* = \frac{p_0 - h_{\text{target}}}{p_0 - p_1}, \qquad b^* \in [0,\ 1]$$

```
C/C++
Δb = clamp(b_dmi - b_cmd,  -0.02,  +0.02)   per tick  (= 2.0 frac/s × 0.01
s)

b_cmd += Δb
```

The controller is active only when the vehicle is in the COAST phase, the post-burnout entry delay has elapsed, altitude exceeds 500 m AGL, and vertical velocity remains positive. Outside these conditions the brake command is held at zero.

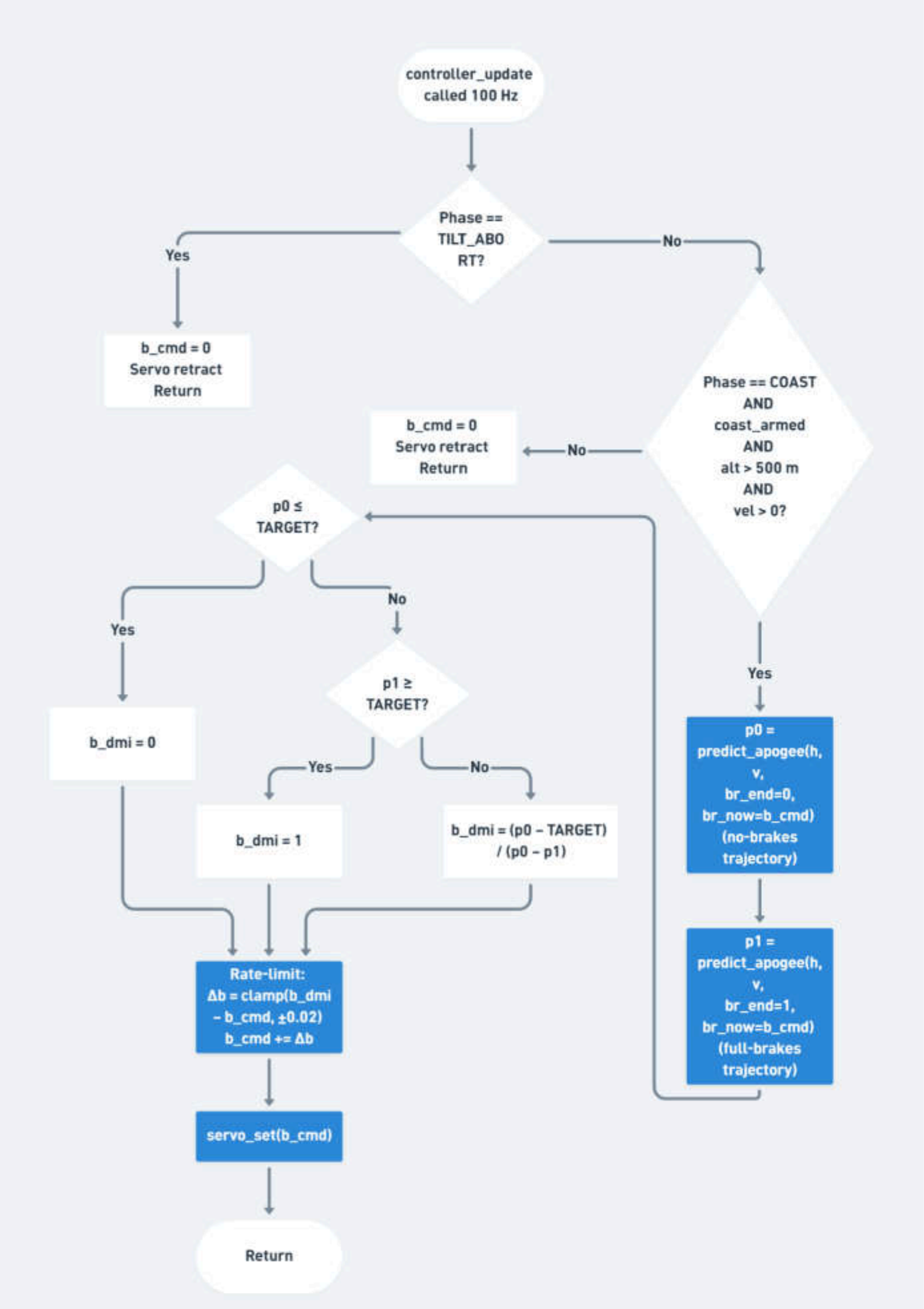


Figure A.256: controller_update() Logic Flowchart

Table A.22: Control Parameters

Parameter	Value	Basis
TARGET_APOGEE_M	3048 m	Competition target (10,000 ft)
MAX_BRAKE_RATE	2.0 fraction/s	Bench-measured 0.33 s deploy x 1.5 x safety factor
CONTROL_ALT_M	500 m	Minimum altitude for controller activation
PRED_STEPS	700	35 s prediction window at 0.05 s per step
PRED_DT	0.05 s	Simulation time step
COAST_ENTRY_DELAY	100 ms	Post-burnout stabilisation delay

6. Flight Phase State Machine

The firmware implements a five-state FSM evaluated at every 500 Hz IMU tick. A key feature of the implementation is that every state transition -including the tilt abort -requires its triggering condition to remain satisfied for three consecutive ticks (FSM_TRUST_COUNT = 3) before the transition fires. At 500 Hz, three ticks represent 6 ms of confirmation. This "trust counter" mechanism eliminates false transitions caused by single-sample sensor spikes that can otherwise produce premature coast entry, premature descent detection, or spurious abort events.

The trust counter for each transition increments when the condition holds and resets to zero when it does not. The transition only fires when the counter reaches three. Separate counters are maintained for each transition (g_boost_n, g_coast_n, g_descent_n, g_tilt_n), ensuring that a condition that intermittently satisfies the threshold does not accumulate credit across multiple events.

A second important feature is that the burnout detection condition uses the world-frame vertical specific force (g_smoothed_world_az) rather than the body-frame axial accelerometer reading. At motor burnout a rocket that has pitched by even 5° shows a cos(5°) ≈ 0.996 error in the body-frame reading, but at 10° this reaches approximately 1.5% and at 15° approximately 3.4%. The world-frame value, derived from the AHRS quaternion rotation, remains correct at all tilt angles.

The smoothed world-frame signal is obtained via a first-order low-pass filter applied at every 500 Hz tick:

$$\tilde{a}_{w,z}[k] = \alpha\, a_{w,z}[k] + (1-\alpha)\, \tilde{a}_{w,z}[k-1], \qquad \alpha = 0.05$$

This corresponds to a time constant of approximately 40 ms, sufficient to reject single-sample noise while responding promptly to the step-change in acceleration at burnout.

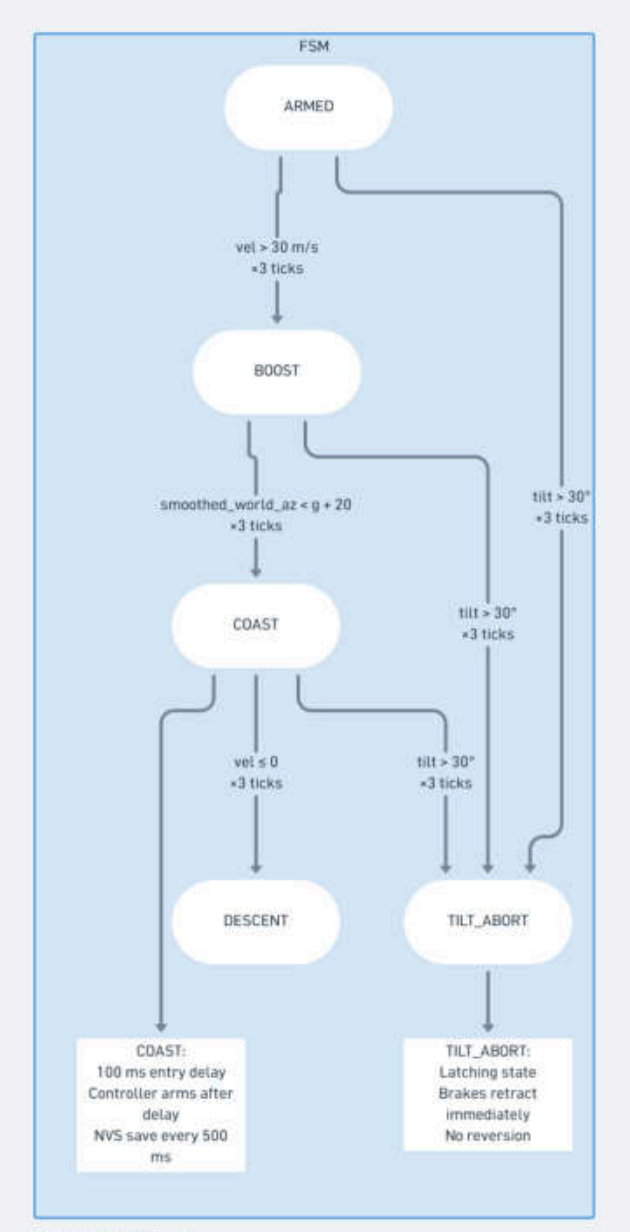


Figure A.257:Flight Phase FSM

Table A.23: FSM State Descriptions (all transitions require 3 consecutive confirmations)

State	Entry Condition (× 3 ticks)	Brake	Notes
ARMED	Power-on default	0	Waiting for liftoff; boost_n counter active
BOOST	Vel > 30 m/s	0	Low-pass SF filter($\alpha = 0.05$) smooths for burnout; coast_n active
COAST	Smoothed SF < g+20 m/s²	DMI active	100ms entry delay; tilt_n and descent_n counters active
DESCENT	Vel ≤ 0	0	Passive - recovery takes over; no further actuation
TILT_ABORT	Quat tilt > 30° (any phase)	0 (immediate)	Latching - does not revert; I_corr and b_cmd zeroed

The tilt abort transition is latching: once the tilt_n counter reaches three and the transition fires, the state does not revert regardless of subsequent tilt measurements. The brake command is immediately zeroed and the servo commanded to the retracted position.

The 100 ms coast entry delay and the trust counter are independent gate conditions. The trust counter gates entry into the COAST phase; the entry delay subsequently gates when the controller arms within COAST. This double-check prevents the controller from responding to the noisy, transient measurements that characterise the period immediately following motor burnout.

7. IMU Fault Detection

Each 500 Hz IMU tick includes a plausibility check on both inertial sensors before their outputs are fused. The accelerometer vector magnitude is tested against bounds [0.5g, 35g] and the gyroscope magnitude against [0, 35 rad/s (≈ 2000 dps)]. The lower accelerometer bound detects all-zero output indicative of sensor power loss; the upper bound rejects obvious hardware faults or saturation.

If one sensor fails the plausibility check, the other operates in solo mode with no change to the AHRS or control logic. If both sensors fail on the same tick, the AHRS update is skipped entirely for that tick and stale quaternion values are retained -a safe default that prevents corrupted attitude estimates from propagating into the controller. Fault flags are sticky until the sensor recovers, and a rate-limited warning is emitted to telemetry on first detection.

8. Power-Loss State Persistence

The flight computer writes critical state to the ESP32-S3's Non-Volatile Storage (NVS) flash partition, which is wear-levelled and survives hard power removal. This enables mid-flight recovery in the event of a power interruption.

Save schedule. State is saved on every FSM phase transition and periodically every 500 ms during the COAST phase. Saving is not performed during BOOST, as state changes too rapidly to be useful for recovery.

Saved state. The SavedState structure (approximately 120 bytes) contains: launch-site reference (P_0, T_0), complementary filter state (h_est, v_est), AHRS quaternion and Mahony bias terms, smoothed world-frame acceleration, last accepted barometric reading, current brake command, FSM phase, and metadata (save count, save timestamp).

Integrity. A 32-bit FNV-1a checksum over all structure bytes (excluding the checksum field itself) and a magic sentinel (0xAB3048AB) must both validate before any restoration is attempted. A struct size or field-order change between firmware versions automatically produces a checksum mismatch and falls back to a cold start.

Recovery logic. On boot, the firmware performs a full NVS dump to serial before proceeding. If the saved phase is COAST, the barometer is re-read ten times to obtain a fresh altitude, and velocity is estimated from the altitude difference divided by an assumed reboot duration (RECOVERY_BOOT_MS = 3 s). If the saved phase is DESCENT or TILT_ABORT, the brakes remain retracted and the estimator resumes from the saved state. Saved phases of ARMED or BOOST are discarded in favour of a fresh cold start.

9. Ground Calibration

Prior to arming, the firmware collects 100 barometric samples from the BMP390 and applies a two-pass 2σ outlier rejection procedure to establish P_0 and T_0:

Pass 1. Compute the mean ($\bar{P}$, $\bar{T}$) and standard deviation (σ_P, σ_T) of all collected samples.

Pass 2. Discard any sample for which $|P_i - \bar{P}| > 2\sigma_P$ or $|T_i - \bar{T}| > 2\sigma_T$. Recompute the mean from the surviving samples.

A minimum of 30 surviving samples is required; if this threshold is not met, the firmware halts with a fatal error. This procedure protects against wind gusts partially blocking the pressure port, power-on sensor transients, or any other mechanism that could bias the ground reference and consequently introduce a fixed offset into all AGL altitude calculations.

Following calibration, the IMUs are warmed up for 5 seconds to allow the Mahony filter to converge. A telemetry line reports the settled tilt angle; a value below 5° confirms correct axis mapping and sign convention.

10. Servo Actuation

The brake panels are driven by the ESP32-S3's LEDC peripheral at 50 Hz with 16-bit resolution. No external PWM library is required. The brake command fraction maps to a physical deployment angle, which maps to a PWM pulse width:

```
C/C++
angle    = RETRACTED + b_cmd × (DEPLOYED - RETRACTED)

pulse_µs = MIN_US + (angle / 180°) × (MAX_US - MIN_US)

count    = (pulse_µs / 20000) × 65535
```

The SERVO_ANGLE_RETRACTED_DEG and SERVO_ANGLE_DEPLOYED_DEG values must be measured on the physical mechanism before each flight. The pulse width limits (SERVO_MIN_US, SERVO_MAX_US) must be verified on a servo tester, not assumed from the datasheet.

11. Telemetry

The firmware emits a CSV line prefixed "CSV," at 10 Hz over UART at 115,200 baud. The 28 fields include: timestamp and phase, both barometer readings, per-sensor and fused IMU values, world-frame vertical acceleration, complementary filter altitude and velocity, brake command percentage, both predictor outputs (p0 and p1), tilt angle from both the quaternion method and the direct body-frame method, all four quaternion components, IMU fault status bits, and loop timing in microseconds.

12. Pre-Flight Parameter Checklist

The following parameters must be verified before every flight -several of them depend on physical measurements that change between builds.

Table A.24: Pre-Flight Checklist

Parameter	Required Action
ROCKET_MASS_KG	Set to actual burnout mass from motor data, not estimated
ROCKET_AREA_M2	$\pi \times \text{body_radius}^2$ -measure the actual airframe radius
TARGET_APOGEE_M	Match to competition waiver altitude
SERVO_MIN/MAX_US	Calibrate on servo tester - do not use datasheet defaults
SERVO_ANGLE_RETRACTED/DEPLOYED_DEG	Measure physically with brakes in both positions
CD_BODY[] / CD_BRAKE_FULL[]	Update from latest OpenRocket Cd vs Mach export / CFD run
BURNOUT_ACCEL_THR	Measure (peak boost SF - coasting SF) / 2 from a previous flight if available
TILT_ABORT_DEG	30° default - tighten only if pad is reliably vertical
MAHONY_KP / KI	After 3 s warmup on the pad, telemetry tilt should read < 5°
ALL I2C / SPI addresses and GPIOs	Cross-check against PCB schematic - do not rely on memory
MAX_BRAKE_RATE	Re-measure deploy time with actual servo + mechanism; apply 1.5 × safety factor

Appendix B.
Project Test Reports

Test I

Recovery Testing 1

Test Lead: Shivam Halan

Test Date: February 6, 2025			
Participant List:	**Material List:**	**Test Location:**	**Test Result:**
Shivam Halan Saksham Singh Aneesh Muralidharan Adith Rajesh	Rocket Tubes Recovery Hardware SRAD e-matches Batteries	Outdoor Stadium, VIT Vellore	Partial Success

Test Purposes:

The test was conducted to evaluate the performance of the charge well concept using an ABS 3D printed assembly integrated into one of our previously developed rockets. The objective was to validate the functionality and reliability of both the in house developed black powder and the charge well system under operational conditions.

Desired Result:

The recovery mechanism should open efficiently because of the black powder explosion, ensuring that the body tube separation is reliable enough to guarantee the recovery mechanism can deploy efficiently.

Test Procedure:

1. The SRAD e-matches are securely installed within the recovery mechanism.
2. Black powder is then placed inside recovery charge wells.
3. The base, through which the wiring passes, is sealed using M-seal to prevent any leakage.
4. The e-match is manually triggered by supplying voltage from the battery.

Test Observation:

The recovery mechanism failed to deploy fully due to incomplete combustion of the black powder charge, tolerance problems and gas leakage from the containment area. This resulted in insufficient pressure generation, causing only partial actuation of the mechanism and preventing the recovery system from functioning as intended.

Project Lead Comments:

Failure of this test was primarily due to improper machining of our Recovery Mechanism. A new recovery mechanism must be machined.



Conclusion:

The recovery mechanism should be machined to a higher standard to withstand explosive forces and it should be more robust while allowing for easy assembly and replacement when required. The black powder should be made finer and its composition should be adjusted to better mimic the performance of FFFF-grade black powder.

Test II

Recovery Testing 2

Test Lead: Shivam Halan

Test Date: February 9, 2025

Participant List:	Material List:	Test Location:	Test Result:
Shivam Halan Saksham Singh Aneesh Muralidharan Adith Rajesh	Rocket Tubes Recovery Hardware SRAD e-matches Batteries	Outdoor Stadium, VIT Vellore	Partial Success

Test Purposes:

The test was conducted to evaluate the performance of a new iteration of the charge well concept using a new ABS 3D printed assembly integrated into one of our previously developed rockets. The objective was to validate the functionality and reliability of both the in house developed black powder and the charge well system under operational conditions. The quality of the black powder was also improved.

Desired Result:

The recovery mechanism should open completely and reliably because of the black powder explosion, ensuring that all deployment components are fully released and that the recovery system activates without obstruction.

Test Procedure:

1. The SRAD e-matches are securely installed within the recovery mechanism.
2. Black powder is then placed inside recovery charge wells.
3. The base, through which the wiring passes, is sealed using M-seal to prevent any leakage.
4. The e-match is manually triggered by supplying voltage from the battery.

Test Observation:

The explosion of the black powder resulted in a powerful separation force. However, the coupler setup was not strong enough to withstand the explosive forces, leading to failure. As a result, the separation momentum was insufficient for effective parachute ejection.

Project Lead Comments:

The density of the coupler should be increased further to better withstand the applied forces.
The performance of the black powder has improved, but it should be enhanced further for optimal results.



Conclusion:

The pressure generated was sufficient to create adequate momentum between the two body tubes for deploying the parachutes, but it should be increased further. The design of the coupler should be improved and the quality of the black powder should be further refined.

Test III:
Recovery Testing 3

Test Lead: Shivam Halan

Test Date: February 14, 2026

Participant List:	**Material List:**	**Test Location:**	**Test Result:**
Shivam Halan Saksham Singh Aneesh Muralidharan Adith Rajesh	Rocket Tubes Recovery Hardware SRAD e-matches Batteries	Outdoor Stadium, VIT Vellore	Partial Success

Test Purposes:

The test was conducted to evaluate whether the pressure generated by the charge wells is sufficient to shear four M4 nylon shear pins and produce enough momentum between the body tubes despite the added restraint of the pins. Another iteration of an ABS 3D-printed assembly was designed to assess robustness and functionality. The quality of the black powder was also improved.

Desired Result:

The body tubes should be able to separate effectively, breaking all four shear pins. The black powder is expected to perform better with less residue.

Test Procedure:

1. The SRAD e-matches are securely installed within the recovery mechanism.
2. Black powder is then placed inside recovery charge wells.
3. The base, through which the wiring passes, is sealed using M-seal to prevent any leakage.
4. The e-match is manually triggered by supplying voltage from the battery.

Test Observation:

The improved performance of the black powder resulted in a powerful separation force against the four shear pins. However, the coupler setup was not strong enough to withstand the explosive forces, leading to failure and allowing pressure to escape through the bottom body tube along with the combustion flame. As a result, the separation momentum was insufficient for effective parachute ejection.

Project Lead Comments:

The material of the coupler needs to be changed, as ABS is not strong enough to withstand the pressure. The quality of the black powder should be improved to achieve a stronger and cleaner explosion.



Conclusion:

Improvements can be seen in the quality of the black powder, but it needs further refinement. ABS is not strong enough to withstand the pressure and heat and therefore should be replaced. Any minor leaks in the mechanism should also be sealed more effectively.

Test IV

Recovery Testing 4

Test Lead: Shivam Halan

Test Date: March 8, 2025

Participant List:	**Material List:**	**Test Location:**	**Test Result:**
Shivam Halan Saksham Singh Aneesh Muralidharan Adith Rajesh	Rocket Tubes Recovery Hardware SRAD e-matches Batteries	Outdoor Stadium, VIT Vellore	Partial Success

Test Purposes:

The test was conducted to evaluate the performance of the charge well concept using an machined aluminium assembly with one of our previously tested rockets. The objective was to validate the functionality and reliability of both the in house developed black powder and the charge well system under operational condition also accounting for the previous issues such as printing tolerances and gas leaks. The aluminium assembly improved reusability greatly which helped for later tests.

Desired Result:

The recovery mechanism should open completely and reliably because of the black powder explosion, ensuring that all deployment components are fully released and that the recovery system activates without obstruction.

Test Procedure:

1. The SRAD e-matches are securely installed within the recovery mechanism.
2. Black powder is then placed inside recovery charge wells.
3. The base, through which the wiring passes, is sealed using M-seal to prevent any leakage.
4. The e-match is manually triggered by supplying voltage from the battery.

Test Observation:

The recovery mechanism was more structurally sound, with most of the pressure effectively contained within the top body tube. The explosion impulse generated was sufficient to break all four M4 shear screws and would theoretically result in effective parachute ejection. Minor leaks were still observed during deployment and should be sealed. The robustness of the mechanism should also be improved, as there were difficulties in assembly after each test.

Project Lead Comments:

Failure of the mechanism was primarily due to minor leaks at various locations within the assembly. The assembly procedure and sealing of the recovery body tube must be improved to ensure consistent performance. The quality of the black powder should also be further enhanced.



Conclusion:

Greater focus should be placed on sealing the recovery body tube and improving the quality of the black powder.

Test V

Recovery Testing 5

Test Lead: Shivam Halan

Test Date: March 11, 2025

Participant List:	Material List:	Test Location:	Test Result:
Shivam Halan Saksham Singh Aneesh Muralidharan Adith Rajesh	Rocket Tubes Recovery Hardware SRAD e-matches Batteries Blueraven Altimeter	Outdoor Stadium, VIT Vellore	Partial Success

Test Purposes:

The test was conducted to evaluate the performance of the charge well concept using a machined aluminum assembly with a previously tested rocket. The objective was to validate the reliability of the in-house developed black powder and the charge well system under operational conditions, while addressing earlier issues such as printing tolerances and gas leaks. The test also assessed whether the charge well mechanism could successfully deploy the full recovery system. In addition, the Blue Raven Altimeter needs to be tested in a simulated environment to verify its performance. The aluminum assembly improves reusability, making it suitable for repeated testing.

Desired Result:

The recovery mechanism should open completely and reliably because of the black powder explosion, ensuring that all deployment components are fully released and that the recovery system activates without obstruction.

Test Procedure:

1. The SRAD e-matches are securely installed within the recovery mechanism.
2. Black powder is then placed inside recovery charge wells.
3. The parachute is packed within its in house parachute bag along with its shockcords based on the true setup.
4. The base, through which the wiring passes, is sealed using M-seal to prevent any leakage.
5. The e-match is triggered by supplying voltage from the battery based on a signal from the Blueraven altimeter.

Test Observation:

The full setup did not deploy and most of the pressure leaked through the shear screw holes. However, without the parachutes, the ejection mechanism caused a clean separation and generated enough impulse that could have deployed the parachutes. The black powder quality was very high due to milling in a custom-made ball mill.

Project Lead Comments:

The failure of this test was primarily due to the loss of energy and pressure as the gas passed through the fabric of the parachute bag and the parachute itself. The shear screws need to be tightened further to provide an effective seal.



Conclusion:

The amount of black powder used should be increased to compensate for the energy loss as the gases pass through the parachute bag and the parachute. The parachute body tube should also be sealed more effectively.

Test VI

Recovery Testing 6

Test Lead: Shivam Halan

Test Date: April 9, 2025

Participant List:	Material List:	Test Location:	Test Result:
Shivam Halan Saksham Singh Aneesh Muralidharan Adith Rajesh	Rocket Tubes Recovery Hardware SRAD e-matches Batteries Blueraven Altimeter	SJT Ground, VIT Vellore	Partial Success

Test Purposes:

The purpose of this test is to achieve a fully successful deployment with the recovery mechanism integrated in its actual configuration. This will demonstrate the effectiveness of the developed ejection mechanism and establish a scalable performance and design metric for the main rocket.

Desired Result:

The recovery mechanism should open completely and reliably due to the black powder explosion and the parachute should be able to exit the parachute bag effectively, demonstrating a clean deployment at the intended location along the shock cord.

Test Procedure:

1. The SRAD e-matches are securely installed within the recovery mechanism.
2. Black powder is then placed inside recovery charge wells.
3. The parachute is packed within its in house parachute bag along with its shockcords based on the true setup.
4. The base, through which the wiring passes, is sealed using M-seal to prevent any leakage.
5. The e-match is triggered by supplying voltage from the battery based on a signal from the Blueraven altimeter.

Test Observation:

The recovery mechanism was able to fully deploy as required, with the impulse creating a separation of over 20 ft. This test demonstrated the complete functionality of the recovery system. However, the result could not be consistently replicated due to failure of the recovery bag attachment and the recurring issue of energy absorption by the parachute and parachute bag fabrics.

Project Lead Comments:

Two layers of fiberglass proved to be very effective in protecting the parachute from the black powder flame. The tolerances of the coupler can be tightened to achieve a better friction fit. Additionally, the screw holes for the shear screws on the body tube should be made more accurate to reduce leakage.



Conclusion:

The tolerancing of the coupler needs to be revised to eliminate all potential leak points. The amount of black powder should also be increased to compensate for pressure losses in the fully packed configuration. In addition, the packing and placement of the parachutes should be improved to increase the effective volume within the parachute body tube, allowing for more efficient gas expansion and ensuring the required ejection impulse.

Test VII

Recovery Testing 7

Test Lead: Shivam Halan

Test Date: April 19, 2025

Participant List:	Material List:	Test Location:	Test Result:
Shivam Halan Saksham Singh Aneesh Muralidharan Adith Rajesh	Rocket Tubes Recovery Hardware SRAD e-matches Batteries Blueraven Altimeter	Outdoor Stadium, VIT Vellore	Partial Success

Test Purposes:

The purpose of this test was to evaluate the new 6-inch coupler that will be integrated into the main rocket, along with the exact packing volume of the parachute body tube. The opposite end of the parachute body tube was sealed using a similar coupler for the eyebolt attachment. Unused holes were sealed with M-seal to prevent any leakage. The test was conducted using six M4 shear pins and without parachutes to verify the performance of the coupler with the new dimensions and packing volumes.

Desired Result:

The recovery mechanism should open completely and reliably due to the black powder explosion and the parachute should be able to exit the parachute bag effectively, demonstrating a clean deployment at the intended location along the shock cord.

Test Procedure:

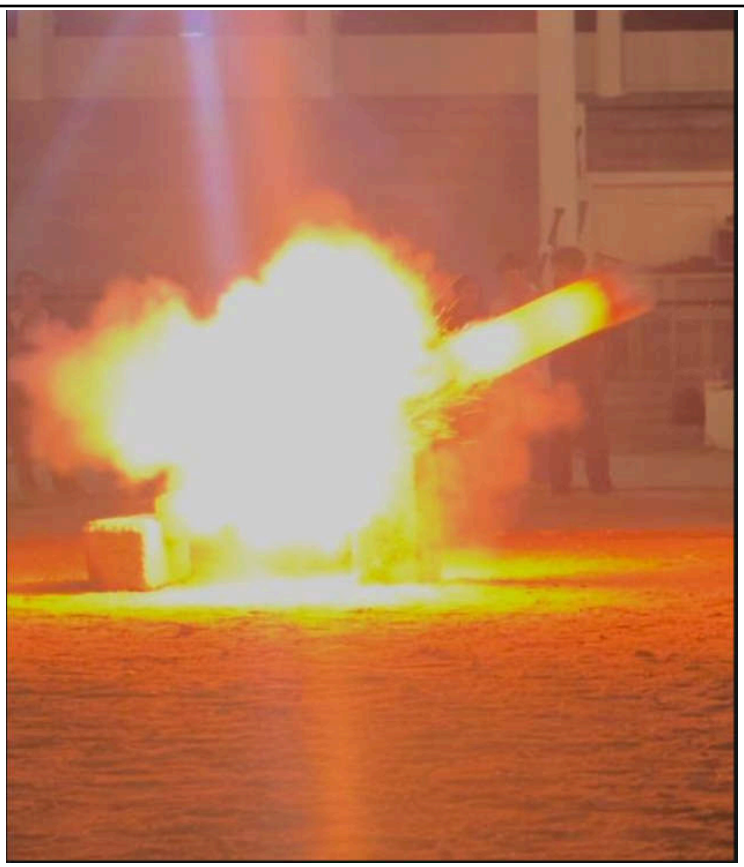
1. The SRAD e-matches are securely installed within the recovery mechanism.
2. Black powder is then placed inside recovery charge wells.
3. The parachute is packed within its in house parachute bag along with its shockcords based on the true setup.
4. The base, through which the wiring passes, is sealed using M-seal to prevent any leakage.
5. The e-match is triggered by supplying voltage from the battery based on a signal from the Blueraven altimeter.

Test Observation:

The mechanism successfully achieved ejection, although leakage was observed. The black powder performed effectively, generating an impulse with just 8g that was sufficient to propel the body tube roughly 20 feet and achieve full tension in the shock cord.

Project Lead Comments:

The new assembly had tolerance and alignment issues, which allowed gases to escape at the coupler–body interface as well as through other screw interfaces. Another major pressure leak point was identified at the e-match wire hole, which also needs to be sealed with M-seal.



Conclusion:

The amount of black powder should be reduced, considering that the final rocket will use FFFF-grade powder. The e-match wire port needs to be secured thoroughly and the body tube coupler interface should be properly sealed to retain pressure.

Test VIII

Recovery Testing 8

Test Lead: Shivam Halan

Test Date: April 26 , 2025

Participant List:	**Material List:**	**Test Location:**	**Test Result:**
Shivam Halan Saksham Singh Aneesh Muralidharan Adith Rajesh	Rocket Tubes Recovery Hardware SRAD e-matches Batteries Blue Raven	SJT Ground, VIT Vellore	Partial Success

Test Purposes:

The purpose of this test was to verify that the complete ejection system functions as intended in its final configuration using the specified amount of black powder. It also aimed to identify any major issues with the full assembly so they can be addressed before launch.

Desired Result:

The mechanism should be integrated quickly and efficiently, and it should produce an effective parachute deployment that replicates real operational conditions. No leaks should be observed.

Test Procedure:

1. The SRAD e-matches are securely installed within the recovery mechanism.
2. Black powder is then placed inside recovery charge wells.
3. The parachute is packed within its in house parachute bag along with its shockcords based on the true setup.
4. The base, through which the wiring passes, is sealed using M-seal to prevent any leakage.
5. The e-match is triggered by supplying voltage from the battery based on a signal from the Blueraven altimeter.

Test Observation:

The mechanism successfully deployed the parachute; however, minor leaks were observed along the sides upon reviewing the ejection footage. This was primarily due to in complete

Project Lead Comments:

Minor failure was due to tolerancing issues with the coupler and body tube.



Conclusion:

Minor tolerances need to be corrected, and further testing should be conducted to achieve consistent, repeatable results.

Test IX

Fibreglass Compression Testing

Test Lead: Shivam Halan

Test Date: April 24, 2026

Participant List:	**Material List:**	**Test Location:**	**Test Result:**
Shivam Halan Adith Rajesh Aditi Bhore	Rocket Body Tubes Universal Testing Machine	Strength Of Materials Lab, VIT Vellore	Partial Success

Test Purposes:

The test was conducted to find out the compression strength of the filament winded body tube using a Universal Testing Machine. The objective was to determine the maximum axial load that the body tubes can withstand before failure and to also validate that the following body tubes meet the required safety margins for the same.

Desired Result:

The body tubes should be able to withstand around 150 MPa of force without exhibiting any type of failure, again demonstrating safety margins with respect to flight loads and confirming that the manufacturing provides consistent results, i.c, that the filament winding angles and strengths are reliable.

Test Procedure:

1. Body Tube specimens are measured and inspected for any visual defects. Basic parameters such as inner and outer diameter and length are measured.
2. Specimen is placed vertically between the two plates of the UTM Machines, ensuring proper alignment.
3. Load is applied axially at a controlled displacement rate and load and data are logged.
4. Test is continued till failure load, where the specimen breaks or an initial crack is spotted.

Test Observation:

The fibreglass body tube demonstrated compressive strength of 18.4 MPa, falling short of the required load. This provided insights on critical data required to make changes in the next iterations of body tubes with respect to filament winding angles and manufacturing processes to further optimize axial loading capacity of the body tubes.

Project Lead Comments:

Failure of the test was primarily due to the fiberglass tube buckling at 18.4 MPa. Revised layup with n manufacturing techniques will be employed for new iterations.




Conclusion:

The test successfully identified the structural limit of the fiberglass tube, with buckling observed at 18.4 MPa. This result provided a clear validation benchmark for the design and informed improvements in the laminate layup and manufacturing process. Revised layup configurations and enhanced fabrication techniques will be implemented in subsequent iterations to achieve higher structural performance and reliability.

Test X
Composite Tensile Testing

Test Lead: Shivam Halan

Test Date: April 24, 2026

Participant List:	**Material List:**	**Test Location:**	**Test Result:**
Shivam Halan Adith Rajesh Aditi Bhore	Tensile Testing Specimens- Forged CF/FG Composite UTM Machine	Materials Engineering and Technology Lab, VIT Vellore	Success

Test Purposes:

The test was conducted to find out the tensile mechanical properties of forged carbon fibre and fibreglass composite sandwich specimens. Two specimen configurations were evaluated, one with 2 layers of fibreglass and chopped carbon fibre and the other with only a single layer of fibreglass with chopped carbon fibre. Results were used for the material selection of the fins based on the shear modulus and other properties that were calculated from results from this test.

Desired Result:

Both composite specimens should demonstrate good tensile strength and stiffness to meet structural load and fin flutter requirements, with fibre breakage as the primary failure criteria. Higher values of ultimate stress are expected.

Test Procedure:

1. Each tensile test specimen is measured and inspected for surface defects prior to testing. Geometry and other parameters are noted.
2. The specimen is clamped within the grippers of the Universal Testing Machine, ensuring that the specimen is centered within the load axis.
3. The UTM Machine is calibrated and a tensile load is applied at the strain rate of 2mm/min. Force, position, stress and strain data is continuously recorded by the data logger. These parameters are later used to form the stress-strain graph.
4. Test is continued till specimen failure, after which the failure load and fracture location is documented and stress-strain curve is reviewed.

Test Observation:

Both tensile specimens demonstrated fibre breakage as the primary failure mode. Specimen 1 1 (composite with 2 layers) achieved a higher ultimate stress of 204MPa compared to Specimen 2 (1 FG layer), at 125 MPa. Young’s Modulus, Total Elongation and other parameters were also evaluated with these tests. Results will be used to validate the material selections and manufacturing processes for the fins.

Specimen-1:

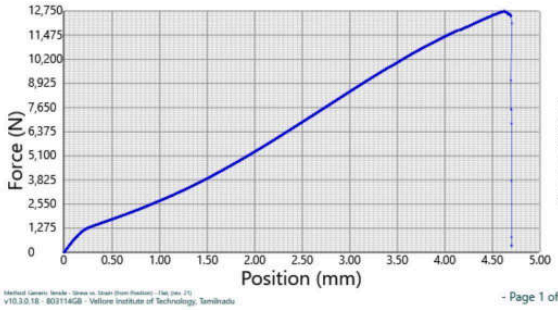
Department of Mechanical Engineering
Vellore Institute of Technolgy
Vellore, Tamilnadu-632014

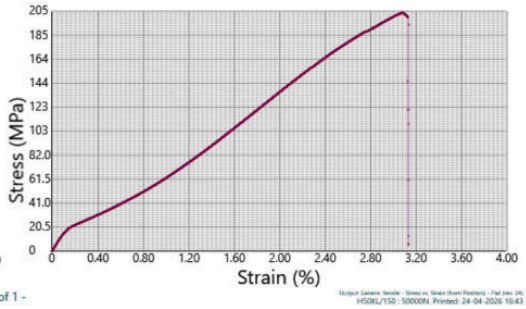
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Method Name: Generic Tensile - Stress vs. Strain (from Position) - Flat

Output Name: Generic Tensile - Stress vs. Strain (from Position) - Flat

	Width mm	Thickness mm	Area mm²	Ultimate Force N	Ultimate Stress MPa	Break Distance mm	Total Elongation %	Modulus MPa
	25.0	2.50	62.5	12800	204	4.71	3.14	14900
Average				12800	204	4.71	3.14	
SD				N/A	N/A	N/A	N/A	





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Specimen-2:

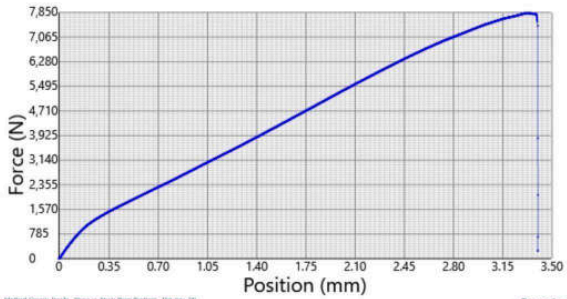
Department of Mechanical Engineering
Vellore Institute of Technolgy
Vellore, Tamilnadu-632014

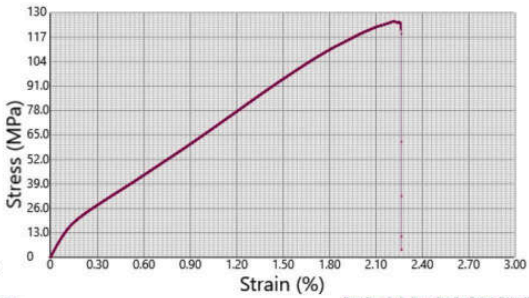
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(Deemed to be University under section 3 of UGC Act, 1956)

Method Name: Generic Tensile - Stress vs. Strain (from Position) - Flat

Output Name: Generic Tensile - Stress vs. Strain (from Position) - Flat

	Width mm	Thickness mm	Area mm²	Ultimate Force N	Ultimate Stress MPa	Break Distance mm	Total Elongation %	Modulus MPa
	25.0	2.50	62.5	7840	125	3.40	2.27	16000
Average				7840	125	3.40	2.27	
SD				N/A	N/A	N/A	N/A	





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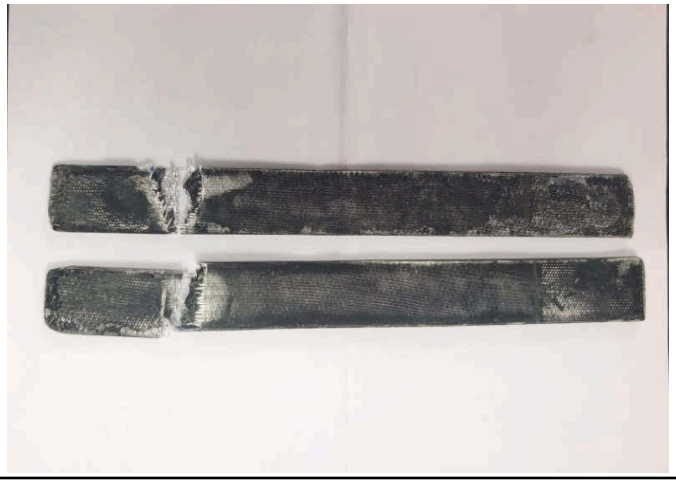
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Project Lead Comments:

Both specimens failed via fiber breakage at their ultimate loads. Specimen 1 (2 FG layers) outperforme Specimen 2 (1 FG layer) with 204 MPa vs 125 MPa. The two-layer FG configuration is recommended structural airframe components.





Conclusion:

Tensile test conducted on the composite sandwich specimens was successful. Both specimens exhibited predominantly linear stress strain graph till failure and showed brittle fracture behaviour expected of fibre reinforced composites. Specimen 1 incorporating 2 layers of fiberglass demonstrated superior tensile performance and was hence selected as the material configuration for Udbhava's fins.

TEST XI

RRC3 Testing

Test Lead: Saksham Singh

Test Date: April 2, 2026			
Participant List:	**Material List:**	**Test Location:**	**Test Result:**
Saksham Singh	RRC3 9V battery 8V pyrotechnic battery mDACS	VIT Vellore Campus	Successful

Test Purposes:

To verify the functionality of the RRC3 module in igniting SRAD e-matches using pyrotechnic batteries, triggered through simulated commands from the mDACS software.

Desired Result:

The RRC3 should successfully receive firing signals from the mDACS simulation and activate the pyrotechnic output to ignite the e-matches.

Test Procedure:

1. Set up the RRC3 with a 9V battery and connect it to the SRAD e-matches.
2. Integrate the system with the mDACS simulation environment.
3. Initiate the firing sequence through mDACS software.
4. Monitor the RRC3's response and e-match activation.

Test Observation:

The RRC3 responded correctly to the simulated firing command from the mDACS software. The pyrotechnic output successfully ignited the SRAD e-matches, confirming the system's readiness for deployment.

Project Lead Comments:

Proper functioning of system observed. Nothing of significance to note.



Conclusion:

The RRC3 successfully received firing signals from the mDACS simulation and activated the pyrotechnic output to ignite the e-matches.

Test XII

Blue Raven Testing

Test Lead: Saksham Singh

Test Date: April 5, 2026			
Participant List:	**Material List:**	**Test Location:**	**Test Result:**
Saksham Singh	Blue Raven 9 V batteries E- matches	VIT Vellore Campus	Successful

Test Purposes:

To validate the operation of the Blue Raven module in triggering SRAD e-matches using a 9V battery, controlled via simulated firing signals from the Featherweight software.

Desired Result:

The Blue Raven should correctly interpret the simulated command from the software and deliver sufficient output to ignite the connected e-matches.

Test Procedure:

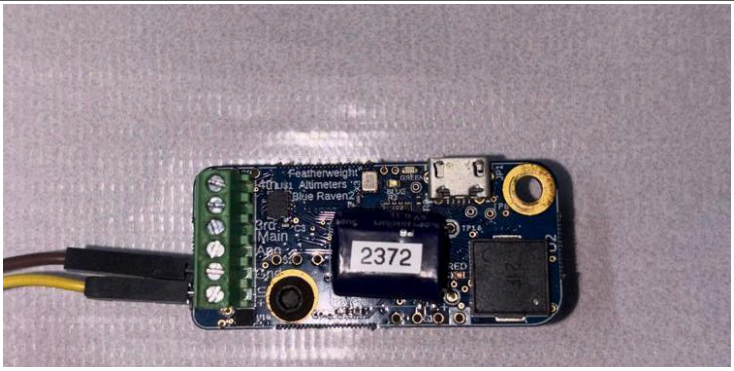
1. Connect the Blue Raven to a 9V battery and interface it with SRAD e-matches.
2. Configure and run the Featherweight simulation environment.
3. Issue the firing command via the software.
4. Observe the module's behaviour and check for ignition of the e-matches.

Test Observation:

The Blue Raven successfully received the firing command from the Featherweight wFIP 2.0 simulation and ignited the SRAD e-matches as expected, confirming full functionality

Project Lead Comments:

Proper functioning of system observed. Nothing of significance to note.



Conclusion:

The Blue Raven correctly interpreted the simulated command from the featherweight software and delivered sufficient output to ignite the connected e-matches.

Test XIII

Featherweight GPS Tracker Testing

Test Lead: Saksham Singh

Test Date: January 16, 2026

Participant List:	Material List:	Test Location:	Test Result:
Adit Goyal	Featherweight GPS Tracker Featherweight Ground Station 3.7V battery Featherweight User Interface Phone App	VIT Vellore Campus	Successful

Test Purposes:

To verify the functionality of the Featherweight GPS Tracker in transmitting real-time GPS location data to the Featherweight Ground Station and displaying it on the Featherweight User Interface Phone App.

Desired Result:

The GPS Tracker should successfully communicate its positional data to the ground station, and the data should be accessible on the mobile app interface within a reasonable operational range.

Test Procedure:

1. Power the Featherweight GPS Tracker using a 3.7V battery.
2. Set up the Featherweight Ground Station and pair it with the GPS Tracker.
3. Connect the ground station to the Featherweight User Interface Phone App.
4. Move the GPS Tracker to increasing distances and monitor GPS data transmission to the app.

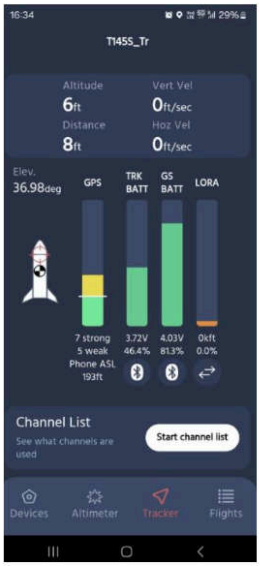
Test Observation:

The GPS data was successfully transmitted from the Featherweight GPS Tracker to the Ground Station and displayed on the mobile app. Data reception was confirmed up to a distance of 900 meters, validating effective communication and tracking range.

Project Lead Comments:

Range is still a glaring issue but GPS functionality has been verified. Range improvement techniques must be continued to looked into.





Conclusion:

The GPS Tracker successfully communicated its positional data to the ground station, and the data is accessible on the mobile app interface within a reasonable operational range.

Test XIV
SRAD Yagi 4 Stacked Antenna Range Test

Test Lead:Saksham Singh

Test Date: April 24 , 2025			
Participant List:	**Material List:**	**Test Location:**	**Test Result:**
Vedanjali Gautam Ayan Saikia Adit Goyal	SRAD 4 stack yagi antenna RF connectors RF combiner Lora Modules Shrota PCB for Rx Sirius flight computer for Tx	National Highway 40, Near Vellore	Success

Test Purposes:

To validate the working of 4 stacked yagi antenna systems at long range under real world conditions. This test would validate the working of the 4 stacked yagi and has more emphasis as the anechoic chamber test for the same stacked structure could not be conducted due to anechoic chamber constraints.

Desired Result:

The antenna should receive live telemetry without substantial packet losses from long range >5km without Line of Sight. This would ensure that the antenna would definitely pick up signals if it has line of sight at a greater distance.

Test Procedure:

1. Sirius, the Flight Computer was configured to send data packets at fixed intervals.
2. The SRAD Yagi 4-stack antenna system was mounted on a stable platform and aligned toward the transmitter.
3. Shrota, the receiving PCB, was used to monitor incoming telemetry data in real time.
4. The Flight Computer was moved progressively farther away from the ground station to evaluate range performance.
5. At each distance interval, parameters such as packet reception rate, signal strength (RSSI), and data integrity were recorded.
6. Tests were conducted under different orientations to observe the directional characteristics of the antenna system.
7. Environmental conditions such as line-of-sight, obstructions and interference were noted during testing.

Test Observation:

Yagi 4 stacked antenna system (4 individual yagi antennae + RF Combiner + Shrota) received data from 5.8km without any packet losses **without** line of sight.
Past 5.8km, substantial losses occurred and after 6.5km, no data was received.
A Low Noise Amplifier can be used to increase the range further.

Project Lead Comments:

Decent Range was acquired by the use of the Yagi stack setup. In order to increase the range an LNA can be used.



Conclusion:

The long-range field test conducted on April 24, 2025, successfully validated the performance of the SRAD 4-stack Yagi antenna system as a viable ground station solution for telemetry reception. By achieving a stable, zero-loss data link at a distance of 5.8 km under non-line-of-sight (NLOS) conditions, the system surpassed the primary success criterion of >5 km.

Test XV
SRAD Horn Antenna Test at Anechoic Chamber

Test Lead:Saksham Singh

Test Date: February 3, 2026			
Participant List:	**Material List:**	**Test Location:**	**Test Result:**
Adit Goyal Saksham Singh	SRAD Horn Antenna , N(m) to SMA antenna adapter wire	Anechoic Chamber, Wireless Test Facility - VIT Vellore	Success

Test Purposes:

The purpose of this test was to verify the working of the horn antenna at the required frequency (5.8Ghz) for live video reception. The test was conducted to verify whether parameters such as S(1,1) Return loss, Gain and directionality were as simulated.

Desired Result:

The antenna was expected to work well between frequencies of 5-7Ghz. Gain of antenna was expected to be around 13.5dBi, with good directionality and minimum side lobes to ensure that reception of live video occurs seamlessly. The S(1,1) or return loss was expected to be at least -13db, ensuring at least 95% efficiency.

Test Procedure:

S11 Return Loss Parameter Measurement

1. Switch ON VNA and allow warm-up (~10–15 min).
2. Set frequency sweep range covering expected antenna band.
3. Perform OSL (Open-Short-Load) calibration at cable end.
4. Connect calibrated RF cable to antenna under test (AUT).
5. Place AUT in anechoic chamber or reflection-free area.
6. Ensure the antenna is kept away from metallic objects.
7. Measure reflection coefficient S11 over frequency range.
8. Identify resonant frequency where S11 is minimum.
9. Determine bandwidth where S11 ≤ −10 dB

Directivity and gain Measurement

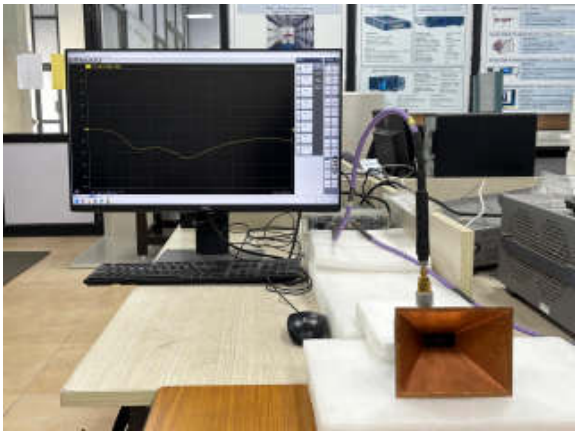
1. Connect the transmitting antenna to a signal generator or VNA.
2. Place AUT at a far-field distance.
3. Align both antennas with the same polarization.
4. Keep transmitted power constant.
5. Rotate AUT in azimuth plane and record received power.
6. Plot radiation pattern for first cut (usually E-plane).
7. Rotate AUT by 90°.
8. Measure radiation pattern again (H-plane).
9. Determine HPBW for E-plane (θE).
10. Determine HPBW for H-plane (θH).
11. Compute directivity from beamwidths.
12. Calculate gain using efficiency.

Test Observation:

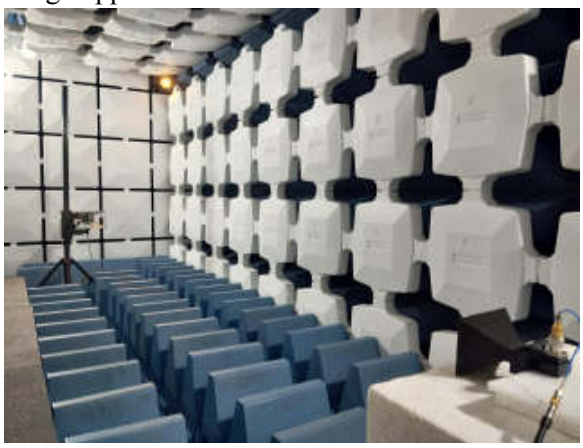
The antenna was resonating between 5-7Ghz frequency. The gain of the antenna was found to be 13.5dBi, with good directionality. The return loss was found to be -17dB, indicating that the efficiency of the antenna is 98%. Hence long range live video can be theoretically received if the VRX has decent transmission range.

Project Lead Comments:

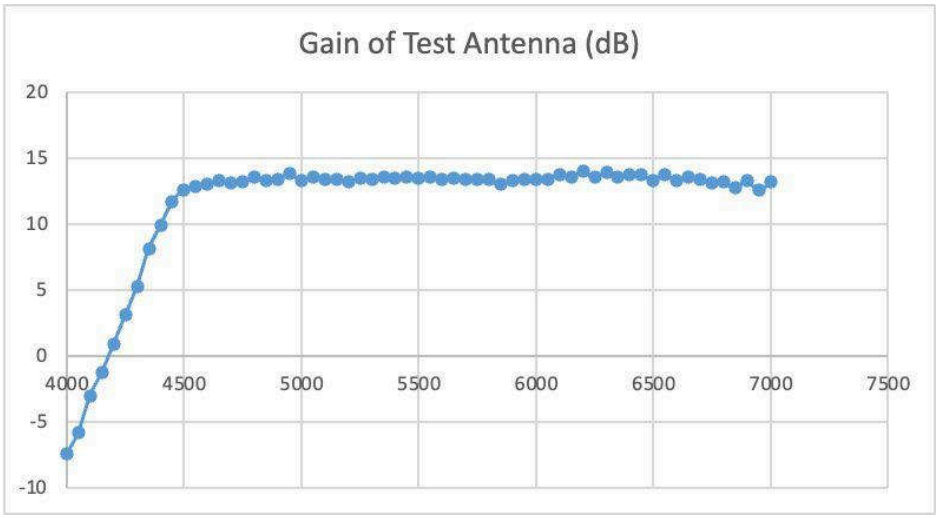
This Horn antenna faces a problem of copper oxidation over time. This can be fixed and gain of the antenna can be further increased by uniformly sticking copper sheets over the oxidized areas.



S11 Parameter Test

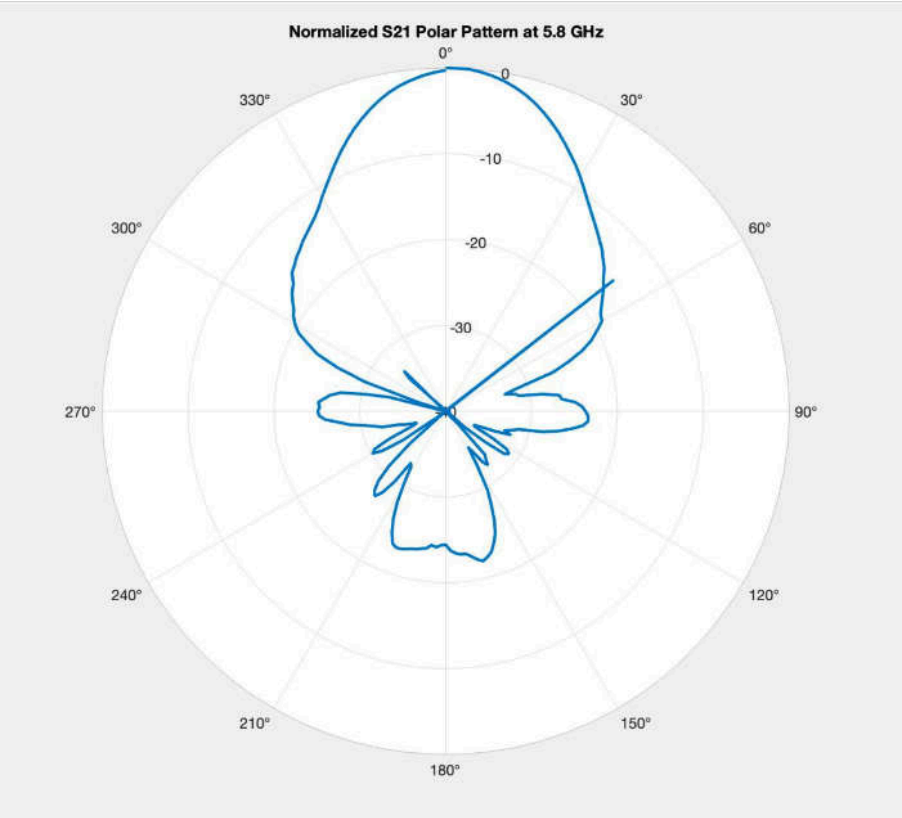


Antenna Parameters - Directionality and Gain test



Gain is consistent between 13.5 to 14dBi - for a wide range of frequencies from 4.5 to 7Ghz. Hence it is a broadband antenna.

Test



The antenna is directional and side lobes are minimal.

We thank VIT Vellore EMC/Wireless Lab for providing the test facility.

Conclusion:

The anechoic chamber testing conducted on February 3, 2026, at the VIT Vellore Wireless Test Facility has successfully validated the SRAD Horn Antenna for high-frequency video transmission. The experimental results not only met but exceeded the initial design specifications, confirming the antenna's suitability for 5.8GHz live video telemetry.

Test XVI
SRAD Single Yagi Antenna Test at Anechoic Chamber

Test Lead:Saksham Singh

Test Date: April 17th, 2025			
Participant List:	**Material List:**	**Test Location:**	**Test Result:**
Adit Goyal	SRAD Single Yagi Uda Antenna , N(m) to SMA antenna adapter wire	Anechoic Chamber, Wireless Test Facility - VIT Vellore	Success

Test Purposes:

The purpose of this test was to verify the working of one of the yagi antennas of the yagi 4 stack system at the required frequency (915MHz) for long range telemetry reception. The test was conducted to verify whether parameters such as S(1,1) Return loss, Gain and directionality were as simulated earlier in Ansys HFSS.

Desired Result:

The antenna was expected to work well at precisely 915Mhz. Gain of single antenna was expected to be around 11 to 12dBi, with good directionality and minimum side lobes to ensure that reception of live video occurs seamlessly. The S(1,1) or return loss was expected to be at least -10db, ensuring at least 90% efficiency.

Test Procedure:

S11 Return Loss Parameter Measurement

1. Switch ON VNA and allow warm-up (~10–15 min).
2. Set frequency sweep range covering expected antenna band.
3. Perform OSL (Open-Short-Load) calibration at cable end.
4. Connect calibrated RF cable to antenna under test (AUT).
5. Place AUT in anechoic chamber or reflection-free area.
6. Ensure the antenna is kept away from metallic objects.
7. Measure reflection coefficient S11 over frequency range.
8. Identify resonant frequency where S11 is minimum.
9. Determine bandwidth where S11 ≤ -10 dB

Directivity and gain Measurement

1. Connect the transmitting antenna to a signal generator or VNA.
2. Place AUT at a far-field distance.
3. Align both antennas with the same polarization.
4. Keep transmitted power constant.
5. Rotate AUT in azimuth plane and record received power.
6. Plot radiation pattern for first cut (usually E-plane).
7. Rotate AUT by 90°.
8. Measure radiation pattern again (H-plane).

9. Determine HPBW for E-plane (θE).
10. Determine HPBW for H-plane (θH).
11. Compute directivity from beamwidths.
12. Calculate gain using efficiency.

Test Observation:

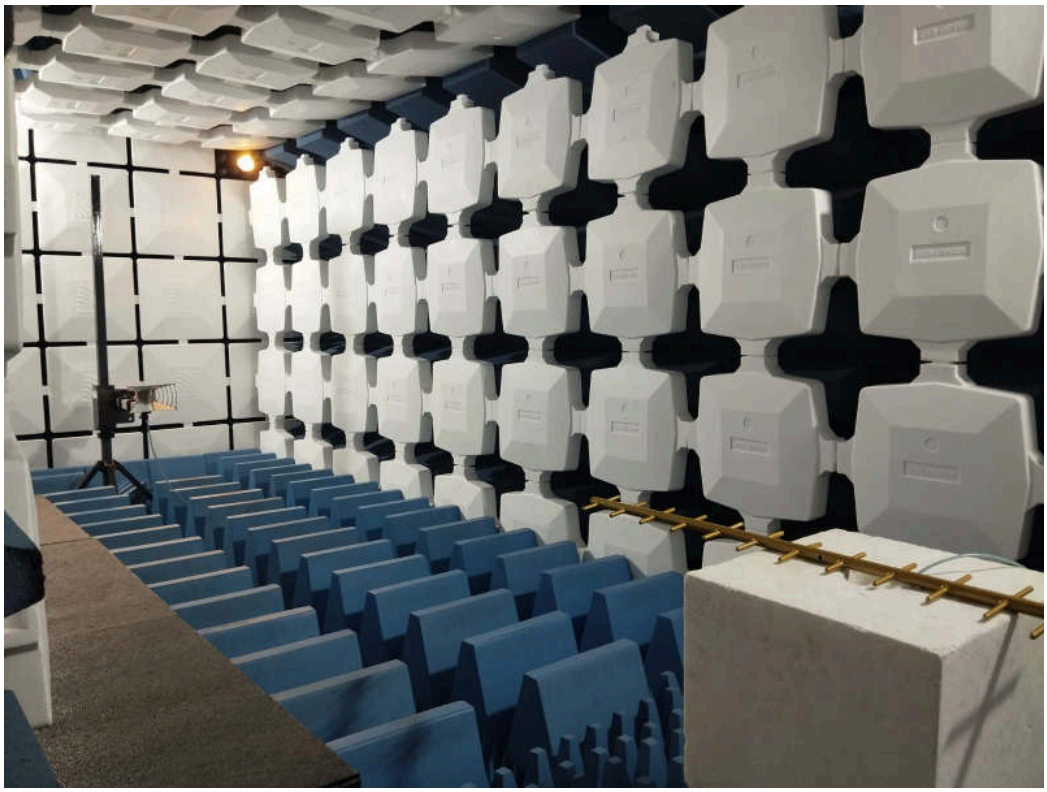
The antenna was resonating between 855Mhz to 970MHz frequency. The gain of the single yagi uda antenna was found to be between 8-9dBi, with good directionality. The directionality was slightly off centered as seen in the images below. The return loss was found to be -12.6dB at 915Mhz, indicating that the efficiency of the antenna is 94.5%. Hence long range telemetry can be theoretically received and received with better results, since we are using a yagi 4 stack with constructive interference, resulting in better gain (~16.7dBi) as simulated in Ansys HFSS.
The 4 stack yagi uda antenna that is developed could not be tested at the anechoic chamber due to size and space constraints of the chamber. Hence elaborate and detailed range tests were conducted instead to validate its working in different conditions.

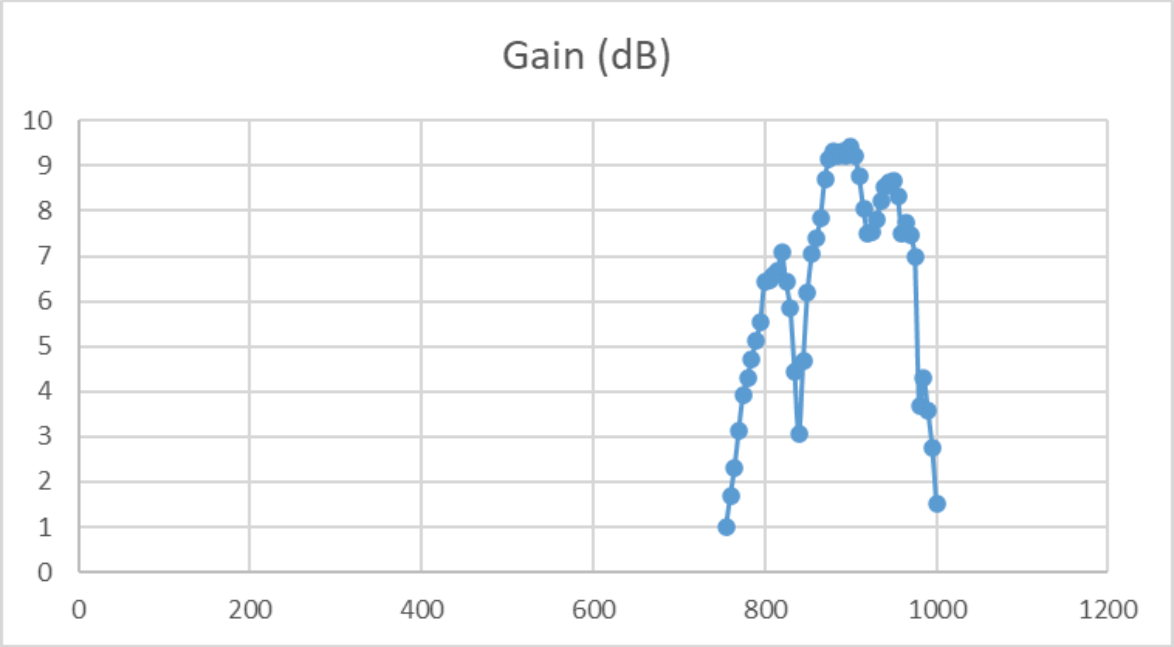
Project Lead Comments:

The antenna performed slightly lower than expected, giving between 8-9dBi of gain instead of the simulated 11dBi. This may be due to slight manufacturing defects, such as slightly bent directors and less precise drilling. This could probably be ignored, as the 4 stack yagi would compensate for the slight imperfections. Regarding the directionality of the individual yagi, that too would be compensated by constructive interference of the 4 yagi stack.



S11 Parameter Test





Gain is between 8 to 9 dBi - for a range of frequencies between 870 to 950Mhz . Hence it is a narrow, frequency specific resonating antenna.

Test XVII

GROUND CONTROL STATION TEST

Test Lead: Shashwat Singh

Test Date:April 28, 2026			
Participant List	**Material List:**	**Test Location:**	**Test Result:**
Niharika Rao Nakul K S	SHROTA (receiver module) Flight Computer (Telemetry Source) Laptop (Running GCS Backend + Frontend - SHROTA Setup)	VIT Vellore	Success

Test Purpose:

The purpose of this test was to verify that the Ground Control Station (CORTEX) , running on a local laptop setup, could reliably receive telemetry data from the flight computer and accurately measure communication latency in real time.

Desired Result:

- Telemetry packets should be received without loss or corruption.
- The GCS should correctly parse and display incoming telemetry data.
- Real-time updates should be reflected across the dashboard (graphs, values, and map).
- Communication latency should be computed accurately using onboard timestamps.
- Packet loss, if any, should be detected and logged.

Test Procedure:

- The STM32 flight computer was configured to transmit telemetry packets over LoRa at fixed intervals.
- Each packet included a timestamp generated using the onboard timer (HAL_GetTick()).
- The GCS backend and frontend were run locally on a laptop (SHROTA setup).
- Telemetry data was received via serial communication on the laptop.
- Incoming data was parsed to extract sensor values and timestamps.
- Latency was calculated using relative time differences between the flight computer timestamp and the GCS reception time.
- Simulated telemetry data (GPS, IMU, altitude, etc.) was varied to validate real-time updates.
- Packet count sequencing was monitored to detect packet loss.

Test Observation:

- Telemetry packets were successfully received and parsed in real time without corruption.
- CORTEX displayed live altitude, packet count, and sensor data accurately.
- Latency computation stabilized with an average of ~2.5 ms and a maximum of ~19 ms.
- Packet loss detection functioned correctly, with observed loss ranging from ~9% to 16%.

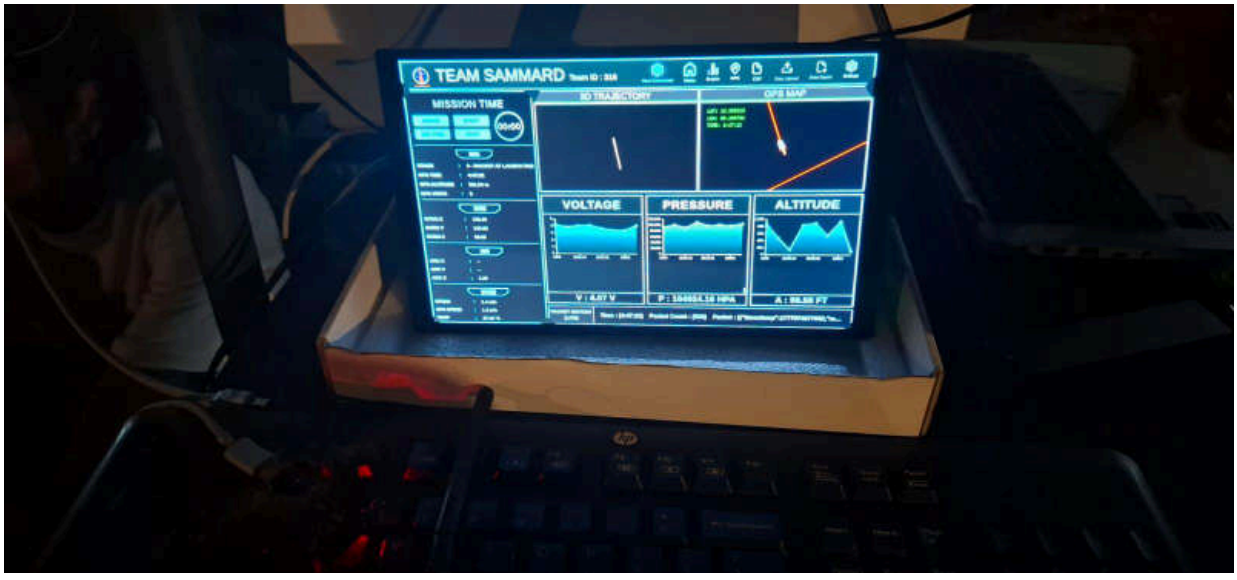
- The system remained stable under continuous telemetry flow with no crashes or performance issues.

Project Lead Comments:

The system demonstrated reliable end-to-end telemetry acquisition and processing under test conditions. Latency estimation using relative timestamps proved effective without requiring clock synchronization. Packet loss detection and logging were accurate and consistent during continuous operation. The modular architecture of CORTEX ensures scalability, robustness, and readiness for further field testing.

```
── GCS Status ──────────────────────
Uptime        : 264s
Serial        : Connected (COM3)
Packets Recv  : 243
Packets Bcast : 243
WS Clients    : 0
USB Outputs   : 0
ATS UART      : Connected
Avg Latency   : 2.51ms
Max Latency   : 19ms
────────────────────────────────────

[ATS] write success
[Telemetry] #2278 | Alt: 783.06m | Lat: 0ms | Loss: 9.43%
[ATS] write success
[Telemetry] #2279 | Alt: 42.5m | Lat: 0ms | Loss: 9.39%
[ATS] write success
[Telemetry] #2280 | Alt: 642.85m | Lat: 17ms | Loss: 9.35%
[ATS] write success
[Telemetry] #2281 | Alt: 318m | Lat: 0ms | Loss: 9.31%
[ATS] write success
[Telemetry] #2282 | Alt: 307.04m | Lat: 0ms | Loss: 9.27%
[ATS] write success
[Telemetry] #2283 | Alt: 260.56m | Lat: 0ms | Loss: 9.24%
[ATS] write success
```



Test XVIII
ATS Test I

Test Lead: Rehaan Sengupta

Test Date: March 15, 2026			
Participant List	**Material List:**	**Test Location:**	**Test Result:**
Rehaan Sengupta Rohan Aditya Gupta	ATS GPS and IMU Transmitter	VIT Vellore	Failure

Test Purpose:

The purpose of this test was to evaluate whether the ATS could accurately orient the horn antenna toward the transmitter and maintain stable and continuous alignment under condition of rapid positional changes.

Desired Result:

The motors were expected to move at the correct expected angles and the antenna would continuously point towards the transmitter being carried around a roundabout.

Test Procedure:

- A transmitter setup with gps coordinates and imu values was used to test the ATS
- The transmitter was carried around a big arc centered at Hathway roundabout, VIT Vellore, around the ATS.

Test Observation:

Initially for a few seconds the ATS would track the transmitter correctly but over time it would lose track and the motor movement angle would be very negligible compared to the actual angle the transmitter was at from the ATS base.

Project Lead Comments:

The anomaly was expected to be due to incorrect assumption of the transmitting GPS's uncertainty values. To improve on the algorithm further, all the received data had been stored to be used later in our lab for testing by simulating it using the transmitter and the ATS.





Test XIX
ATS Test II
Test Lead: Rehaan Sengupta

Test Date: March 15th, 2026			
Participant List	**Material List:**	**Test Location:**	**Test Result:**
Rehaan Sengupta Prateek Srinath	ATS GPS and IMU Transmitter	Hathway Roundabout, VIT Vellore	Success

Test Purpose:

The purpose of this test was to determine whether the updated ATS algorithm would calculate the movement angles correctly and point the antenna towards the transmitter constantly as it is being carried around.

Desired Result:

The movement angles were expected to be correct and the antenna was expected to point towards the GPS transmitter that was being carried around a roundabout.

Test Procedure:

- Significant changes were made to how the ATS calculated angles and trust values of sensors were updated.
- The transmitter would send gps coordinates and imu values specifically to test the ATS
- A person imitating the rocket would carry the transmitter following a big arc around the ATS which was centered at Hathway Roundabout.

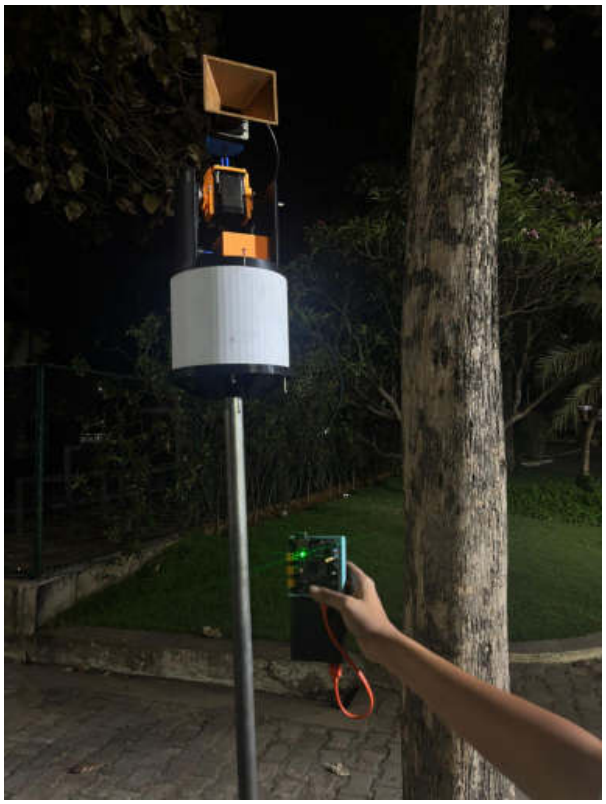
Test Observation:

The ATS algorithm with the updated uncertainty values had performed successfully. The motors moved at correct angles and the antenna was pointed correctly at the transmitter.

Project Lead Comments:

The last test's inferences had been correct. The transmitter was made to imitate the flight computer, Sirius, which had a L89 GPS onboard. According to its datasheet, the L89's has a default error of ±5m, whereas on testing it was found out to be ±0.5m. Correcting these values had significantly improved the ATS.





Test XX
Live Video Test

Test Lead: Adarsh Somu Palaniappan

Test Date: April 30, 2026

Participant List	Material List:	Test Location:	Test Result:
Nakul K S Rehaan Sengupta Ayan Saikia Vedanjali Gautam Niharika Rao Aswini Ambili	FPV Mini Camera COLOR 1200TVL, TBS Unify Pro 5G8 V3(Vtx), RD945(Vrx),SRAD Horn antenna	VIT Vellore	Success

Test Purpose:

The purpose of this test was to evaluate the end-to-end live analog video transmission capability of the onboard FPV sub-system, specifically to verify that the onboard FPV Mini Camera COLOR 1200TVL paired with the TBS Unify Pro 5G8 V3 Video Transmitter (VTx) operating at 5.8 GHz and 800 mW could establish a stable, continuous video link to the ground station over a meaningful distance.

Also the ground station equipped with the RD945 Video Receiver (VRx) interfaced with the SRAD Horn Antenna (designed and manufactured in-house) was tested, which is resonant at 5.8 GHz and provides a gain of 13.5 dBi across a usable bandwidth of 4.5 GHz to 7 GHz.

Desired Result:

The analog FPV video transmission system was expected to achieve and maintain a stable, continuous video feed from the onboard camera to the ground station receiver throughout the test duration. Specifically, the system was expected to demonstrate the following:

- The RD945 VRx, connected to the SRAD Horn Antenna, should successfully detect and lock onto the 5.8 GHz RF signal transmitted by the TBS Unify Pro 5G8 V3 VTx onboard within a short warm-up window after power-on.
- The received video feed, when rendered on the connected laptop, should be visually coherent and stable, with an analog video signal of sufficient quality to clearly discern real-world imagery without persistent noise or complete signal blackout.
- End-to-end video latency (camera capture to display rendering) should fall within the expected analog FPV range of approximately 40 ms to 100 ms, consistent with analog transmission characteristics at 5.8 GHz.
- Frame drop incidence should remain low throughout the observation window, with any momentary disruptions being transient and self-correcting, rather than indicative of systematic RF link failure.
- The test should confirm the RF link budget is sufficient for the expected operational range during flight, and that the SRAD Horn Antenna performs adequately as the primary receive element on the ATS.

Test Procedure:

- The onboard FPV analog video sub-system (FPV Mini Camera COLOR 1200TVL + TBS Unify Pro 5G8 V3 VTx + miniaturized PDB), was powered on and allowed to initialize. The VTx was confirmed to be transmitting at 5.8 GHz at 800 mW output power via the omni-directional antenna mounted on the rocket airframe.
- At the ground station, the RD945 VRx was connected to the SRAD Horn Antenna via a coaxial cable. The VRx output was connected to the laptop via a USB video capture interface to enable real-time display and screen recording of the incoming video feed.
- The camera and VTx assembly was physically separated from the ground station and moved to a distance representative of a near-field operational test. The camera unit was carried ahead of the ground station in an open-sky environment at VIT Vellore to simulate a moving transmitter scenario.
- The SRAD Horn Antenna was manually oriented towards the transmitter to ensure boresight alignment for maximum received signal strength, simulating the directional lock that the ATS would provide during flight operations.
- Once the RF link was established and a stable video feed was confirmed on the laptop, the recording was initiated. The live feed was observed and recorded continuously over the test observation window.
- The recorded video was subsequently reviewed and evaluated for frame drops, signal blackouts, visual noise levels, and perceived end-to-end latency.

Test Observation:

The analog FPV video transmission system performed adequately during the test, demonstrating a functional end-to-end RF video link from the onboard camera to the ground station. The following observations were recorded:

Signal Acquisition

Upon powering on the VTx, the RD945 VRx connected to the SRAD Horn Antenna acquired the 5.8 GHz signal within approximately 3 to 5 seconds. The video feed appeared on the laptop display shortly thereafter with no manual channel scanning required, as the channel had been pre-configured on both the VTx and VRx prior to the test.

Video Quality & Continuity

The live video feed was generally coherent and intelligible throughout the observation window. The image from the FPV Mini Camera COLOR 1200TVL was rendered at 480p resolution with reasonable color fidelity and adequate contrast for identifying ground features. Occasional mild analog noise (visible as brief snow-like interference) was observed, particularly when the transmitter was moved further from the horn antenna's boresight axis, consistent with expected behavior for an analog link approaching its margin limits. No complete signal blackouts were recorded during the primary observation period.

Frame Drops

Review of the recorded video revealed a low but non-trivial number of frame drops during the test session. Approximately 2 to 4 frame drop events were observed per minute on average, with individual drops lasting between 1 and 3 frames (approximately 33 ms to 100 ms at standard analog rates). These drops were transient and the feed self-recovered immediately in each instance, with no sustained loss of signal. The drop rate was considered acceptable for a first-pass bench-level validation, though further optimization of antenna alignment and link budget headroom is recommended for flight conditions.

End-to-End Latency

Perceived end-to-end latency, assessed by visually comparing the camera feed to real-time motion in the scene, was estimated to be in the range of approximately 80 ms to 120 ms. This is slightly above the lower bound of the expected analog FPV latency window (40 ms to 100 ms), likely attributable to the USB video capture interface introducing a small additional processing delay on the laptop side. The latency was not perceptually disruptive and is within an acceptable range for telemetry and flight awareness purposes.

SRAD Horn Antenna Performance

The SRAD Horn Antenna demonstrated effective signal reception when aligned to the transmitter. Signal quality was noticeably better when the antenna's boresight was maintained toward the VTx, confirming the antenna's directional characteristics. Signal degradation was observed when the transmitter moved off-axis by more than approximately 15 to 20 degrees without compensating antenna reorientation, underscoring the necessity of the ATS for maintaining link quality during dynamic flight operations.

Project Lead Comments:

The observed frame drop rate of 2 to 4 drops per minute is within tolerable bounds for this stage of testing, and is likely to improve with tighter ATS boresight control during flight. The latency figure of approximately 80 to 120 ms is acceptable for live flight awareness purposes and consistent with the analog transmission architecture chosen for this sub-system.

The primary action item from this test is to ensure the ATS system maintains horn antenna alignment within the confirmed ±15° effective boresight tolerance during dynamic flight to prevent signal degradation at altitude. Overall, the live video sub-system is confirmed operational.



Test XXI
Propulsion Test 1

Test Lead: Dhruv Murarka

Test Date: April 17th, 2025

Participant List:	**Test Location:**	**Test Result:**
Shivam Halan Sooryakanth Ramakrishnan Saksham Singh Om Unnithan Yug Solanki Swaraj Sharma	VIT Outdoor Stadium	Success

Test Purposes:

The purpose of the test was to perform a vertical static fire of the motor to be used to power our rocket “Udbhava”.

Desired Result:

The motor was designed with the following thrust characteristics

Parameter	Value	Unit
Motor Designation	N2622	
Volume Loading	79.34	%
Impulse	10384.30	Ns
Delivered ISP	111.46	s
Burn Time	3.93	s
Ideal Thrust Coefficient	1.5	—
Average Pressure	3.34	MPa
Peak Pressure	3.52	MPa
Initial Kn	283.66	—
Propellant Mass	9.5	kg
Propellant Length	720	mm
Port/Throat Ratio	3.08	—

Peak Mass Flux	1,193.23	kg/(m²·s) G: 6

Introduction and Process

In order to validate the motor to comply with the IREC competition regulations, as outlined in section 5.16.5 of the IREC Design, Test, & Evaluation Guide document, a static hot-fire was performed .
The test aimed to collect data regarding the thrust and pressure of the motor through sensors on a static bench. The results obtained from the test were compared with data from simulations.
The N-class motor has a length of 850 mm and has an outer diameter of 127.0 mm. Six grain propellant segments are used, totaling 9.8 kg. The propellant adopts the formulation outlined by the Ballistic Test and Evaluation System (BATES), with a 65/35 KNSB composition ratio.
The KNSB grains were covered with epoxy adhesive and then slid into the filament wounded fibre glass liner. The necessary sealants were used and the assembly was done for the same.

The following was the setup used for the motor test



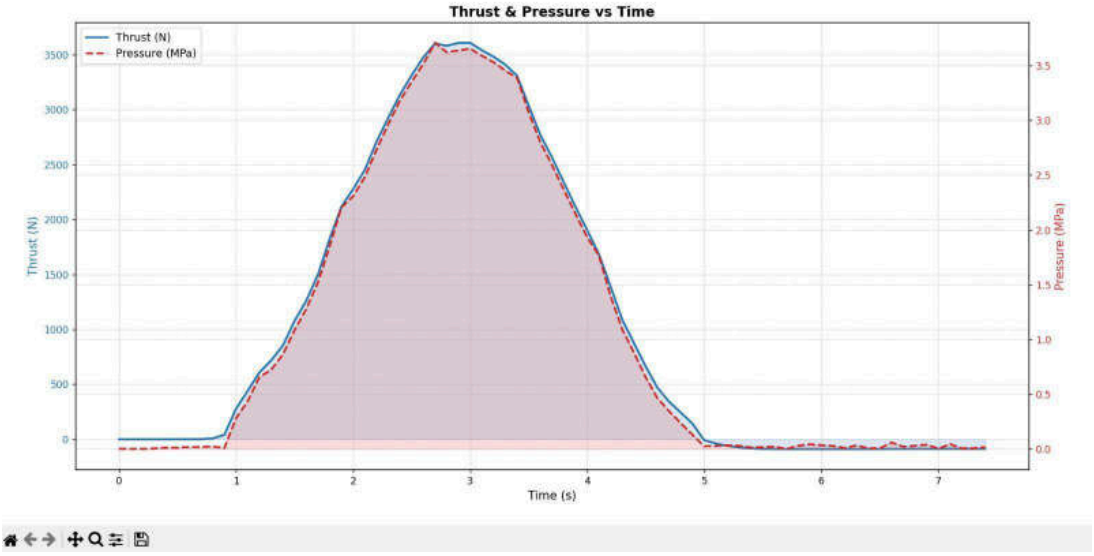


Test Data Analysis:

The motor test shows an initial ignition phase where the igniter activates and there is a short delay before full combustion, followed by a rapid pressurization visible as a sudden expansion and brightening of the exhaust plume.The ignition is clean, the plume transitions quickly from intermittent discharge to a steady, high-velocity jet without chuffing or sputtering.
As the burn progresses, a stable and symmetric plume indicates uniform regression of the grain and good internal flow alignment, while the brightness and color reflect combustion characteristics. During the peak thrust phase, the plume appears most coherent and intense, with minimal flickering suggesting

stable chamber pressure and efficient nozzle performance. As the motor approaches burnout, the plume shortens and becomes less intense, corresponding to the reduction in burning surface area and chamber pressure.

The following was the thrust curve obtained from the test.



Parameter	Value
Burn Time [s]	4.3 s
Maximum Thrust [N]	3609.82
Maximum Pressure [MPa]	3.65
Impulse [Ns]	10847

Project Lead Comments:

The test demonstrated the team's capability to develop N-class motors capable of meeting the performance requirements necessary to propel the rocket to the target apogee of 10,000 feet.



Test XXII
Propulsion Test 2

Test Lead: Dhruv Murarka

Test Date: April 18th, 2025		
Participant List:	**Test Location:**	**Test Result:**
Shivam Halan Sooryakanth Ramakrishnan Saksham Singh Om Unnithan Yug Solanki Swaraj Sharma Gahan Shekar Anirudh Raghavan Shripad Vijapure Advay Kulkarni	VIT Outdoor Stadium	Success

Test Purposes:

The purpose of the test was to do a repeatability analysis of the motor performance on the vertical static fire of the motor to be used to power our rocket “Udbhava” .

Desired Result:

The motor was expected to perform along the lines similar to that of the what was achieved in the previous motor test.

Introduction and Process

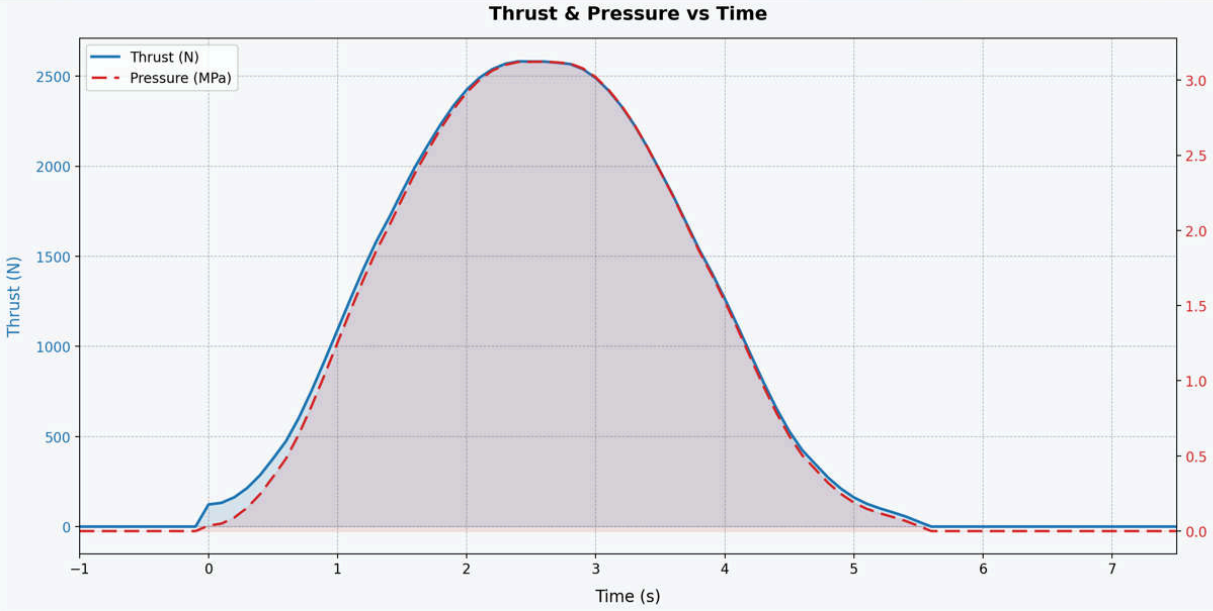
In order to revalidate the motor to comply with the IREC competition regulations, as outlined in section 5.16.5 of the IREC Design, Test, & Evaluation Guide document, a static hot-fire .
The test aimed to collect data regarding the thrust and pressure of the motor through sensors on a static bench and to revalidate the results that were obtained in the previous motor test .

The motor has the same hardware and components that were used in the previous motor test and the assembly for the motor was also carried out in a similar fashion.

Test Data Analysis:

The total burn duration lasted roughly 5.5 seconds, characterized by a smooth, progressive ramp-up and a symmetrical regressive tail-off, indicating good grain geometry and combustion stability. Visually, the test produces a stable exhaust plume with minimal oscillation, followed by a characteristic "black puff" at burnout, signaling the consumption of the final propellant segments and the insulation. This test re confirms the motor's structural integrity

The below is the characteristic curve obtained:-



Parameter	Value
Burn Time [s]	5.5 s
Maximum Thrust [N]	2671
Maximum Pressure [MPa]	3.1
Impulse [Ns]	9647

Project Lead Comments:

There was a slight discrepancy between these results and the design which is primarily driven by a lower-than-predicted burn rate, where the reduced peak pressure of 3.1 MPa compared to the ideal 3.7 MPa ,suggests the propellant burned more slowly and over a larger duration than planned.
This shift probably stems from actual propellant chemistry differing between the tests, such as moisture in the sorbitol or batch density variations (A component that is subjected variation based on cast of each time) . Because solid motor burn rates are pressure-dependent this lower operating pressure directly throttled the thrust output to 2,600 N while stretching the burn time to 5.5 seconds to consume the total propellant mass.The output impulse was within the comparable limits for a successful launch.

Test XXIII
GPS Test

Test Lead: Nakul K S

Test Date: April 30, 2026			
Participant List	**Material List:**	**Test Location:**	**Test Result:**
Nakul K S	Sirius Flight Computer	Hathway Roundabout, VIT Vellore	Success

Test Purpose:

The purpose of this test was to evaluate the onboard GPS functionality of the Sirius Flight Computer. Specifically, the test aimed to verify that the NEO-M8N GPS module with a patch antenna could acquire a satellite lock reliably, measure the Time to First Fix (TTFF) from power-on, and confirm that the module provides stable and consistent coordinate updates at its rated output frequency.

Desired Result:

The GPS module was expected to achieve a satellite lock within a reasonable cold-start TTFF window and thereafter output fresh GPS coordinates at a rate of 1 Hz (one update per second). The coordinates were expected to remain stable and consistent across successive readings, demonstrating the module's suitability for use in the Sirius flight computer's navigation and telemetry pipeline.

Test Procedure:

- The Sirius Flight Computer was powered on in an open-sky environment at VIT Vellore to ensure unobstructed satellite visibility.
- A timer was started simultaneously with power-on to measure the Time to First Fix (TTFF).
- The system was monitored until the NEO-M8N GPS module acquired a satellite lock and began transmitting valid coordinate data.
- Once a fix was established, live GPS output (Latitude, Longitude, Timestamp) was logged and observed to verify the 1 Hz update rate and coordinate stability.
- The raw serial output from the GPS module was captured and recorded as part of the test evidence.

Test Observation:

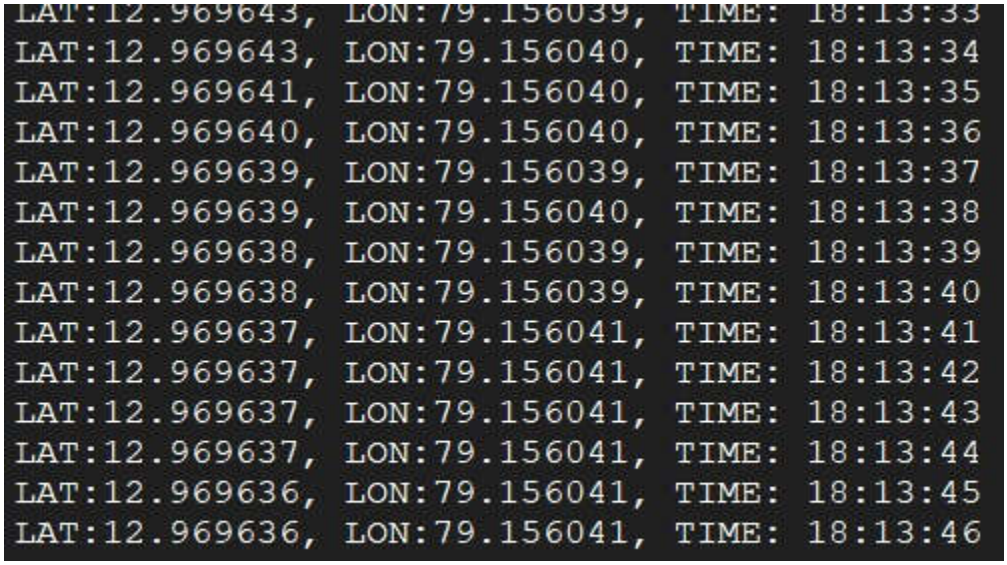
The NEO-M8N GPS module on the Sirius Flight Computer performed successfully. The module achieved a satellite lock after approximately 40 seconds of burn-in/power-on time, which is within the expected cold-start TTFF range for this module. Following the initial fix, the GPS module consistently delivered updated coordinate readings at exactly 1 Hz (one update per second). The latitude and longitude values remained stable across successive readings with minimal drift, indicating reliable satellite tracking. The serial output was clean and well-formed, confirming that the module's communication interface with the Sirius flight computer is functioning

correctly.

Project Lead Comments:

The NEO-M8N GPS module with the patch antenna performed well within expectations for this bench-level validation. The 40-second cold-start TTFF is consistent with the module datasheet and is acceptable for the Sirius flight profile. The 1 Hz update rate was confirmed to be stable and reliable throughout the observation window. Coordinate values showed minimal scatter. Overall, the GPS subsystem of Sirius is confirmed operational and ready for integration.

Images:



Live GPS serial output from NEO-M8N on Sirius showing stable 1 Hz coordinate updates

Appendix C.
Hazard Analysis

This appendix outlines the identification and preliminary assessment of hazards associated with the design, manufacturing, transportation, and launch phases of *Udbhava*. A structured risk analysis approach is employed, beginning with the formulation of a preliminary Hazard Scale (HS) to quantify risks in terms of individual safety, operational integrity, and potential damage to property. This scale provides a standardized framework to evaluate the severity and likelihood of each hazard. Utilizing this framework, a comprehensive Hazard Matrix is developed, detailing the nature of each hazard, its potential causes, the assigned risk level based on the HS, and the corresponding mitigation strategies. This process ensures early recognition and management of risks, enabling proactive safety planning and reinforcing the robustness of mission-critical systems throughout the lifecycle of **Udbhava**.

Table C.1 Hazard Scale

Risk	**Severity**
Minimal	Possess minimal hazard to personnel and property.
Low	Possess low hazard and a possibility for minor injury.
Intermediate	Possess intermediate hazard to personnel and property.
High	Possess high hazard to personnel and a slight possibility to sustain a critical injury or death.
Severe	Possess severe hazard to personnel and a significant possibility to sustain a critical injury or death.

Table C.2 Hazard Matrix

Hazard	Possible Cause	Hazard Level	Mitigation Approach (Process + Design)	Risk of Injury after Mitigation
Exposure to propellant chemicals during casting	Direct contact, poor ventilation	Medium	Process: PPE (gloves, masks), ventilated workspace Design: Enclosed mixing setup	Low
Improper storage of propellant	Exposure to heat/moisture	High	Process: Climate-controlled storage, labeling Design: Sealed containers resistant to environmental effects	Low
Unsafe transportation of propellant	Improper packaging or handling	High	Process: Approved transport containers, secure packaging Design: Shock-resistant casing	Low
Accidental ignition of ejection charges (black powder)	Static electricity, mishandling	High	Process: ESD precautions, separate storage, handling SOPs Design: Isolated charge compartments	Low
Premature activation of recovery system	Wiring fault, accidental arming	High	Process: Arm only at launch pad, checklist enforcement Design: Safe/arm switch, dual-deploy logic	Low
Electrical short circuit in avionics	Poor insulation, loose wiring	Medium	Process: Wiring inspection, continuity checks Design: Insulated wiring, fuse protection	Low
Accidental firing of pyrotechnics due to electronics failure	Software or hardware fault	High	Process: Preflight verification procedures Design: Redundant flight computers, independent power systems	Low
Structural failure during handling	Weak joints, improper assembly	Medium	Process: Assembly SOPs, inspection Design: Structural safety factor ≥ 2	Low
High-tension shock cords during deployment	Sudden load release	Medium	Process: Controlled packing procedures Design: Load-rated cords with FOS > 3	Low

Hazard	Possible Cause	Hazard Level	Mitigation Approach (Process + Design)	Risk of Injury after Mitigation
Parachute deployment during ground handling	Improper arming sequence	Medium	Process: Arm only at pad, strict checklist Design: Mechanical separation locks	Low
Launch proximity hazard	Personnel too close to rocket	High	Process: Safety perimeter enforcement, RSO compliance Design: Remote ignition system	Low
Misfire and delayed ignition hazard	Faulty ignition system	High	Process: Standard misfire procedures, wait-time enforcement Design: Reliable ignition circuitry	Low
Unfired ejection charges post-flight	Deployment failure	High	Process: Safe disarming procedures Design: Accessible charge compartments for safe removal	Low
Hot motor casing after landing	Residual thermal energy	Medium	Process: Cooling period before handling Design: Thermal-resistant materials	Low
Rough terrain during recovery	Environmental conditions	Low	Process: Team coordination, PPE Design: Robust structural design	Low

Appendix D.
Risk Assessment

This appendix provides a detailed assessment of potential risks associated with the pre-launch, launch, recovery, and landing stages of *Udbhava*. Each risk is systematically analysed based on its likelihood of occurrence and the extent of its impact across key parameters -namely, health and safety, system performance, timeline adherence, regulatory compliance, and financial burden. To facilitate a consistent evaluation methodology, a structured Probability Scale and a comprehensive Implication Scale have been defined. These tools support the development of a robust Risk Assessment Matrix, which enumerates each risk, its root causes, estimated probability before and after mitigation, and its multi-parameter implications. This risk assessment approach enables the identification of operational weaknesses and supports the implementation of effective mitigation strategies, thereby promoting mission success and ensuring a high level of safety and reliability across all phases of *Udbhava*'s deployment.

Table C.1 Probability Scale

Probability	**Implication**
Very Low	Aforementioned issue has a very low probability to occur (unlikely to occur)
Low	Aforementioned issue has a low probability to occur (slight chance to occur).
Medium	Aforementioned issue has a medium probability to occur (might occur).
High	Aforementioned issue has a high probability to occur (very likely will occur).

Table C.2 Implication Scale

	Concerned Parameter	**Level 1**	**Level 2**	**Level 3**	**Level 4**	**Level 5**
A	Health and Safety	No or Minor injuries fixable by first aid.	Moderate injuries or illness.	Significant injuries and/or incapacitation and/or impairment	Injuries which can subtend to permanent injuries.	Loss of life
B	System Operation	Issue of components unimportant to flight	Loss of components unimportant to flight	Issue/Damage to components important to flight	Loss of components important to flight	Loss of complete system operation
C	Schedule	Minor operational delays	Delays < 3hrs	3hrs < Delays < 10hrs	Delays of over a day	Significant Delay pertaining to launch
D	Cost	< 10$	10$ - 30$	30$ - 100$	100$ - 500$	>1000$

Phase	Risk	Probable Cause	Probability Before Mitigation	Mitigation Method	Probability After Mitigation	A	B	C	D
	Grain voids/cracks	Improper casting, trapped air	High	Controlled casting, inspection (visual/NDT), static fire validation	Low	L2	L5	L4	L5
	Grain debonding from casting tube	Poor surface prep, weak adhesion	Medium	Surface preparation, bonding validation, process control	Low	L2	L4	L3	L4
	Propellant density inconsistency	Improper mixing ratios	Medium	Standardized mixing procedure, batch control	Low	L2	L4	L3	L3
	Thermal liner delamination	Improper installation	Medium	Adhesive validation, installation checks	Low	L2	L4	L3	L4
	Incorrect CG/CP margin	Payload misplacement	Medium	CG measurement, simulation validation (>1.5 calibers)	Low	L1	L5	L3	L4
	Fastener loosening	Improper torque application	Medium	Torque specs, locking mechanisms (Loctite, lock nuts)	Low	L1	L3	L2	L2
	Structural joint failure	Poor bonding or machining defects	Medium	Structural inspection, load testing	Low	L2	L4	L3	L4
Pre-Launch	Avionics wiring fault	Loose or incorrect connections	Medium	Continuity checks, wiring	Low	L1	L3	L2	L2

Phase	Risk	Probable Cause	Probability Before Mitigation	Mitigation Method	Probability After Mitigation	A	B	C	D
				verification protocol					
	Battery failure	Improper charging, faulty cells	Medium	Battery health check, redundancy	Low	L1	L3	L2	L2
	Sensor miscalibration	Improper calibration procedure	Medium	Preflight calibration routine	Low	L1	L3	L2	L2
	Payload mounting failure	Weak interface or loose mounting	Medium	Mechanical locking, vibration testing	Low	L2	L4	L3	L3
	Wiring damage during assembly	Mechanical stress, routing issues	Medium	Proper routing, strain relief, inspection	Low	L1	L3	L2	L2
	Incorrect recovery packing	Human error during packing	High	Standard packing SOP, checklist validation	Low	L2	L5	L4	L5
	Incorrect ejection/deployment setup	Wrong configuration or connection	Medium	Preflight checklist, ground testing	Low	L2	L5	L4	L4
	Environmental exposure (humidity/temperature)	Poor storage conditions	Medium	Controlled storage, environmental monitoring	Low	L1	L3	L2	L2
	Launch rail misalignment	Improper setup	Medium	Rail alignment verification	Low	L1	L3	L2	L2
	Human error in assembly sequence	Lack of procedural compliance	Medium	Detailed checklist, supervised assembly	Low	L2	L4	L3	L3
Ignition	Igniter failure (no ignition)	Open circuit, faulty igniter	Medium	Continuity checks, verified	Low	L1	L3	L2	L2

Phase	Risk	Probable Cause	Probability Before Mitigation	Mitigation Method	Probability After Mitigation	A	B	C	D
				igniter, pre-launch testing					
	Ignition delay	Low igniter energy, poor contact	Medium	Igniter validation, proper seating, test firing	Low	L1	L2	L2	L1
	Ignition asymmetry	Off-center igniter placement	Medium	Centralized igniter positioning, mounting fixture	Low	L2	L3	L2	L2
	Partial ignition of grain	Incomplete flame spread	Medium	Multi-point ignition design / igniter optimization	Low	L2	L4	L3	L3
	Pressure spike at ignition	Rapid combustion onset	Low	Grain geometry optimization, ignition timing control	Low	L3	L4	L3	L4
	Wiring failure at launch	Loose connection, vibration	Medium	Secured connectors, strain relief, continuity verification	Low	L1	L3	L2	L2
	Electrical misfire (no current)	Power supply issue	Medium	Redundant power check, battery validation	Low	L1	L3	L2	L2
	Premature ignition (unsafe condition)	Accidental current, handling error	Low	Safe/arm system, arming only at pad, procedural control	Low	L4	L4	L3	L4
	Launch command failure	Signal transmission error	Low	Redundant triggering system, communication verification	Low	L1	L2	L2	L1

Phase	Risk	Probable Cause	Probability Before Mitigation	Mitigation Method	Probability After Mitigation	A	B	C	D
	Igniter ejection due to pressure	Improper retention	Low	Mechanical retention design, secure mounting	Low	L2	L3	L2	L2
Boost	Motor overpressure (MEOP exceedance)	Burn rate variation, grain geometry error	High	Static fire testing, burn rate validation, FOS ≥ 2.5	Low	L4	L5	L4	L5
	Casing rupture	Structural weakness, pressure spike	High	Structural analysis, hydrostatic proof testing	Low	L5	L5	L5	L5
	Grain burn-through	Thin web thickness	Medium	Grain geometry validation, simulation	Low	L4	L5	L4	L5
	Combustion instability	Grain irregularities, pressure oscillations	Medium	Grain design optimization, static testing	Low	L3	L4	L3	L4
	Nozzle erosion	High thermal loads	Medium	High-temp material selection, thermal analysis	Low	L3	L4	L3	L4
	Nozzle failure (crack or deformation)	Thermal stress, manufacturing defect	Medium	Material testing, machining precision	Low	L4	L4	L3	L4
	Thrust misalignment	Nozzle misalignment, structural error	Medium	Alignment fixtures, precision assembly	Low	L3	L4	L3	L3
	Excessive vibration	Combustion instability	Medium	Structural damping, testing	Low	L2	L3	L2	L3

Phase	Risk	Probable Cause	Probability Before Mitigation	Mitigation Method	Probability After Mitigation	A	B	C	D
	(thrust oscillation)								
	Fin structural failure	High aerodynamic loads	Medium	FEA validation, safety factor > 2	Low	L3	L5	L4	L4
	Fin detachment	Poor bonding or mounting	Low	Bonding validation, mechanical reinforcement	Low	L3	L5	L4	L4
	Airframe buckling	Axial compressive loads	Low	Structural design validation, testing	Low	L4	L5	L4	L5
	Avionics malfunction due to vibration	High dynamic loads	Medium	Vibration isolation, component testing	Low	L2	L4	L3	L3
	Sensor noise (IMU/altimeter)	High acceleration and vibration	Medium	Signal filtering, sensor fusion	Low	L1	L3	L2	L2
	Wiring disconnection	Mechanical stress	Medium	Strain relief, secure routing	Low	L2	L4	L3	L3
	CG shift during burn	Propellant mass reduction	Low	Stability margin > 1.5 calibers	Low	L2	L5	L4	L4
	Thermal damage to avionics	Heat transfer from motor	Low	Thermal insulation, placement optimization	Low	L2	L3	L2	L3

Phase	Risk	Probable Cause	Probability Before Mitigation	Mitigation Method	Probability After Mitigation	A	B	C	D
	Loss of static stability	CG/CP miscalculation, mass shift	Medium	Stability margin > 1.5 calibers, simulation validation	Low	L3	L5	L4	L4
	Fin flutter	High velocity, insufficient stiffness	Medium	Flutter analysis, safety margin > 1.5× max velocity	Low	L3	L4	L3	L3
	Fin misalignment effects	Manufacturing/assembly error	Medium	Precision machining, alignment checks	Low	L2	L4	L3	L3
	Weathercocking	High wind, low initial velocity	Medium	Launch angle optimization, wind assessment	Low	L1	L3	L2	L2
	Excess aerodynamic drag	Surface roughness, design inefficiency	Medium	CFD validation, smooth finishing	Low	L1	L3	L2	L2
	Off-nominal trajectory	Thrust misalignment or wind	Medium	Flight simulation validation	Low	L2	L4	L3	L3
	Roll instability	Fin asymmetry	Medium	Symmetry checks, balancing	Low	L2	L3	L2	L2
	Payload vibration damage	High dynamic loads	Medium	Vibration isolation, structural damping	Low	L2	L3	L2	L3
	Avionics malfunction (mid-flight)	Vibration, thermal stress	Medium	Redundant avionics, insulation	Low	L2	L4	L3	L3
	Altimeter drift / pressure error	Atmospheric variation, sensor noise	Medium	Sensor fusion, calibration	Low	L1	L3	L2	L2
	IMU noise / inaccurate data	High acceleration, vibration	Medium	Filtering algorithms	Low	L1	L3	L2	L2

Phase	Risk	Probable Cause	Probability Before Mitigation	Mitigation Method	Probability After Mitigation	A	B	C	D
	Telemetry signal loss	RF interference, antenna orientation	Medium	Antenna optimization, redundancy	Low	L1	L3	L2	L2
	Wiring disconnection mid-flight	Structural vibration	Medium	Secure routing, strain relief	Low	L2	L4	L3	L3
	Thermal effects on electronics	External heating	Low	Thermal insulation, proper placement	Low	L1	L2	L2	L2
	Premature recovery trigger	Sensor noise, logic error	Low	Filtering logic, redundancy	Low	L2	L5	L4	L4
	Failure to detect apogee	Sensor/algorithm error	Medium	Dual altimeters, validation testing	Low	L2	L5	L4	L4
	Failure to detect apogee	Altimeter error, sensor drift	Medium	Dual redundant altimeters, calibration	Low	L2	L5	L4	L4
	Incorrect apogee detection (early/late)	Noise, filtering error	Medium	Signal filtering, sensor fusion	Low	L2	L4	L3	L3
	Premature deployment	False trigger, vibration noise	Low	Threshold filtering, shielding	Low	L2	L5	L4	L4
	Delayed deployment	Sensor lag or logic delay	Medium	Algorithm tuning, testing	Low	L2	L5	L4	L4
	Separation failure	Shear pin mis-sizing, mechanical jam	Medium	Shear pin testing, tolerance validation	Low	L2	L5	L4	L4

Apogee

Phase	Risk	Probable Cause	Probability Before Mitigation	Mitigation Method	Probability After Mitigation	A	B	C	D
	Partial separation	Friction, misalignment	Medium	Mechanical clearance validation	Low	L2	L4	L3	L3
	Failure of deployment mechanism	Mechanical/electrical fault	Medium	Redundant deployment system, testing	Low	L3	L5	L4	L5
	Inadequate deployment force	Underestimated force requirement	Medium	Ground ejection testing, sizing validation	Low	L2	L4	L3	L3
	Structural damage during deployment	Excessive shock loads	Medium	Shock load analysis, reinforcement	Low	L3	L4	L3	L4
	Avionics reset during deployment	Power fluctuation	Low	Stable power system, redundancy	Low	L2	L4	L3	L3
	Wiring failure during separation	Mechanical stress	Medium	Strain relief, secure routing	Low	L2	L4	L3	L3
	Payload damage during separation	Sudden mechanical shock	Medium	Shock isolation design	Low	L2	L3	L2	L3
	Excessive rotation during separation	Aerodynamic imbalance	Medium	Mass balancing, design validation	Low	L2	L3	L2	L2
	Failure to deploy drogue parachute	Deployment mechanism failure	Medium	Redundant deployment logic, testing	Low	L3	L5	L4	L5
	Tangling during initial deployment	Poor packing, rotation	Medium	Packing SOP, deployment testing	Low	L2	L4	L3	L3

Phase	Risk	Probable Cause	Probability Before Mitigation	Mitigation Method	Probability After Mitigation	A	B	C	D
	Drogue parachute failure	Deployment mechanism failure	Medium	Redundant deployment logic, ground testing	Low	L3	L5	L4	L5
	Main parachute failure	Improper packing, deployment issue	High	Packing SOP, repeated ground deployment tests	Low	L3	L5	L4	L5
	Partial parachute inflation	Line entanglement, improper folding	Medium	Controlled packing procedures, testing	Low	L2	L4	L3	L3
	Parachute line snapping	Excessive load during opening	Medium	Load testing, safety factor > 3	Low	L3	L4	L3	L4
	Shock cord failure	High opening shock load	Medium	High-strength materials, load validation	Low	L3	L5	L4	L4
	Excessive descent velocity	Undersized parachute	Medium	Descent rate calculations, validation (~5–7 m/s)	Low	L3	L4	L3	L4
	Canopy damage (tear/burn)	High deployment shock or heat exposure	Medium	Material selection, deployment timing optimization	Low	L2	L4	L3	L3
	Delayed main deployment	Sensor or logic delay	Medium	Redundant altimeters, timing validation	Low	L2	L5	L4	L4
	Premature main deployment	Incorrect trigger	Low	Filtering logic, deployment conditions check	Low	L2	L4	L3	L3
	Rocket rotation during descent	Asymmetric deployment	Medium	Symmetric packing, design validation	Low	L2	L3	L2	L2

Phase	Risk	Probable Cause	Probability Before Mitigation	Mitigation Method	Probability After Mitigation	A	B	C	D
	Drift outside recovery zone	High wind conditions	Medium	Flight simulation, launch condition checks	Medium	L1	L3	L3	L2
	Entanglement between components	Improper separation or packing	Medium	Separation distance validation, packing SOP	Low	L2	L4	L3	L3
	Avionics damage during descent	Continuous oscillations	Medium	Shock mounting, damping design	Low	L2	L3	L2	L3
	Payload damage	High landing velocity or oscillation	Medium	Shock absorption design	Low	L2	L3	L2	L3
	Loss of visual tracking	High altitude drift	Medium	Bright color parachute, tracking aids	Low	L1	L2	L2	L1
	Hard Landing	High descent velocity, parachute inefficiency	High	Descent rate optimization, parachute sizing validation	Medium	L3	L4	L3	L4
	Structural damage on impact	High impact loads	Medium	Reinforced airframe, impact design validation	Low	L2	L3	L2	L3
	Payload damage	Insufficient shock absorption	Medium	Shock isolation, padding design	Low	L2	L3	L2	L3
	GPS tracking failure	Signal loss, antenna damage	Medium	Dual GPS + RF tracking redundancy	Low	L1	L3	L2	L2
	Complete loss of rocket (unrecoverable)	Combined GPS + visual failure	Medium	Redundant tracking + visual aids	Low	L1	L4	L3	L4

Recovery

Phase	Risk	Probable Cause	Probability Before Mitigation	Mitigation Method	Probability After Mitigation	A	B	C	D
	Data loss	Power failure, storage corruption	Medium	Data redundancy, onboard logging backup	Low	L1	L2	L2	L2
	Delayed recovery	Inaccurate location estimate	Medium	Improved tracking accuracy, ground coordination	Low	L1	L2	L3	L2
	Landing in hazardous terrain	Wind drift, trajectory deviation	Medium	Flight simulation, launch condition control	Medium	L2	L3	L3	L3
	Damage due to environmental exposure	Rain, heat, moisture	Medium	Protective casing, sealing	Low	L1	L2	L2	L2
	Unfired deployment mechanism hazard	Deployment failure	Medium	Safe disarming procedures, trained handling	Low	L3	L3	L2	L3
	Hot motor casing	Residual heat after landing	Medium	Cooling time before handling, PPE	Low	L2	L2	L1	L1
	Component separation on impact	Weak joints	Medium	Structural reinforcement, testing	Low	L2	L3	L2	L3
	Injury during recovery operations	Rough terrain, manual handling	Low	PPE, team coordination, safety briefing	Low	L2	L1	L1	L1
	Loss of visual identification	Small rocket visibility	Medium	Bright paint, flags, tracking beacons	Low	L1	L2	L2	L1

Appendix E.
Assembly, Preflight, Launch, Recovery and Off-Nominal Checklists

TEAM SAMMARD GROUND OPERATIONS

PROJECT Udbhava CHECKLIST FOR
ASSEMBLY, PRE-FLIGHT, RECOVERY AND
OFF-NOMINAL FLIGHT

Version 1.1

Updated 18/04/2026

AUTHOR: DHRUV MURARKA (CAPTAIN)
CO AUTHORS: SOORYAKANTH RAMAKRISHNAN (VICE CAPTAIN)

IREC 2026

Midland, Texas, USA

Material List

The following list is a checklist for all the required components, items, parts, tools, and materials required for IREC 2026. The team members are mandated to follow the checklist before leaving for the USA from India and in all the days of the competition.

Item Name	Quantity	Category		Item Name	Quantity	Category	
Soldering Iron	1	Avionics		21700 Batteries	20	Avionics	
Antenna	5	Avionics		9V Batteries	2	Avionics	
Flight Computer	2	Avionics		Eyebolt M12	4	Part	
Bessie	1	Avionics		Swivel	4	Part	
Shrota	1	Avionics		Quicklink	6	Part	
Power Distribution board	1	Avionics		M4X10mm screw	12	Part	
RRC3+	1	Avionics		M4X20mm screw	42	Part	
Raven 4	1	Avionics		M4X16mm screw	12	Part	
SRAD Recovery	1	Avionics		M3X16mm screw	84	Part	
Jumpers	-	Avionics		M3X12mm screw	8	Part	
Limit Switches	5	Avionics		Shear-pins M4X15mm	50(pack)	Part	
Magnetic Switches	7	Avionics		L bracket	16	Part	
Camera module	1	Avionics		Epoxy and hardener	1	Material	
Screws M3x10mm		Avionics		Drogue parachute	1	Component	
Master Key	1	Avionics		Main parachute	1	Component	
Shrota	2	Avionics		Shock Cord	1	Component	
Wire Cutter	1	Avionics		Drill	1	Tool	
Avionics Bay Printed	1	Avionics		Allen Keys Kit	2	Tool	
Extra Connectors	-	Avionics		Angle grinder	1	Tool	
Micro USB	2	Avionics		Vernier calliper	2	Tool	
cables							
Posters		Miscellaneous		Measuring tape	3	Tool	
Brochures		Miscellaneous		Tender Descender	5	Component	

Fiberglass Cone	2	Material		Markers	10	Miscellaneous	
Fiberglass tubes	3	Material		Black Powder	1 lb	Component	
1515 large rail buttons	2	Component		Fin Can	1	Component	
Pliers	2	Tool		Battery pack	1		
Mallet	1	Tool		Binoculars	2	Miscellaneous	
Clamp	2	Tool		Rubber Bands		Miscellaneous	
Masking tape	5	Miscellaneous		Zip Ties		Miscellaneous	
Drill bits		Tool		Extension boxes	2	Miscellaneous	
Sandpaper	5	Material		Orbital Sander	1	Tool	
Adapter	5	Miscellaneous		Camera	1	Miscellaneous	
Fevi-quick	10	Miscellaneous		Gloves		Miscellaneous	
Hot glue gun	1	Tool		Paper plates/bowls		Miscellaneous	
Weighing Scale	2	Tool		Electrical tape	3	Tool?	
Screw Driver	3	Tool		Masks		Miscellaneous	
Brush	5	Miscellaneous		M seal	3 boxes	Material	
X stands	2 sets	Miscellaneous		Araldite	1 box	Material	
Boric powder	1 box	Miscellaneous		Wire cutter	3	Tool	
Spray Paint		Miscellaneous					
Motor Aluminium Tube	2	Mechanical		Yagi Antenna	4	Electronics	
Forward Closure	2	Mechanical		Aluminium Boat tail	2	Mechanical	
Stainless Steel Nozzle	2	Mechanical		3D Printed Parts		Mechanical	
Fibreglass Liner		Mechanical		ATS PCB	1	Electronics	
Dremel	2	Mechanical		CyberDec	1	Computer Science	
Carbon Fibre Fins	2 sets	Mechanical		Nema Motor Drivers	2	Electronics	
Aluminum Coupler	6	Mechanical		Shroud Lines		Mechanical	
Aluminium Charge wells	8	Mechanical		PVC pipes	2 sets	Management	
EPDM Sheets	50m	Mechanical		Posters	4	Management	
Packing tape	5	Mechanical		Airbrakes 3D printed structures		Mechanical	

Nema 23 Servo	2	Electronics		Servo Motor	1	Electronics	
3D printed parts		Mechanical		Linear guid rails	4	Mechanical	
Horn antenna	1	Electronics		Leaf (Airbrakes)	4	Mechanical	
Sandpaper	2 sets	Mechanical		CAM slot plate	1	Mechanical	
Filer	3	Mechanical		Shaft		Mechanical	
Sanding Disks	1 set	Mechanical		Rail Buttons	8	Mechanical	
Cutting Disk	1 set	Mechanical		Hacksaw with blades	2 sets	Mechanical	
Tap set	1 set	Mechanical		Decal	2	Management	
Spirit Leveller	1	Mechanical		Yagi Stack Stand	1	Electronics	
ATS PCB	1	Electronics		Airbrakes PCB	1	Electronics	
Payload PCB	1	Electronics					

1. Assembly Checklist

1. Avionics Assembly
1.1. Limit Switches
1.1.1. Visually inspect for any physical abnormalities or damage on the Limit Switches
1.1.2. Install Pull-pin mechanism to hold the switches in OFF position
1.1.3. Connect the switches to the respective components
1.2. POWER DISTRIBUTION BOARD (PDB)
1.2.1. Visually inspect for any physical abnormalities or damage on the flight computer
1.2.2. Check the Potential of the 2S + 2P 21700 Battery pack and connect to the PDB
1.2.3. Check the Potential of the output from the PDB and connect it to the respective SRAD components
1.3. Flight Computer
1.3.1. Visually inspect for any physical abnormalities or damage on the flight computer
1.3.2 Check the Potential of the PDB source and connect it to the flight computer
1.3.3. Wait 2-5 seconds for two buzzer beeps confirming normal operation.
1.3.4. Repeat the same procedure for the secondary flight computer
1.4. Bessie
1.4.1. Visually inspect for any physical abnormalities or damage on the Bessie
1.4.2. Connect the RUBBER DUCK antenna to the SMA connector
1.4.3. Check the Potential of the PDB source and connect it to the Bessie
1.4.4. Wait 2-5 seconds for two buzzer beeps confirming normal operation.
1.4.5. Repeat the same procedure for the secondary Bessie

1.5. RRC3 +
1.5.1. Visually inspect for any physical abnormalities or damage on the RRC3 +
1.5.2. Check the Potential of the 9V alkaline battery or 2S 21700 battery pack and connect it to the RRC3 +
1.5.3. After 3-10 secs wait for a long beep for 5 secs indicating there is no post-fault error
1.6. RAVEN4
1.6.1. Visually inspect for any physical abnormalities or damage on the RAVEN4
1.6.2 Check the Potential of the 2S 21700 battery pack and connect it to the RAVEN4
1.6.3 Check for 9 fast beeps followed by slow beeps every 2 seconds
1.7. FEATHERWEIGHT GPS
1.7.1 Visually inspect for any physical abnormalities or damage on the FEATHERWEIGHT GPS
1.7.2 Check the Potential of the 1 21700 battery and connect it to the FEATHERWEIGHT GPS
1.7.3 After 3-5 secs connect the FEATHERWEIGHT GPS to the FeatherWeightUI App
1.8. VTx TBS Unify Pro 5G8 V3
1.8.1 Visually inspect for any physical abnormalities or damage on the VTx TBS Unify Pro 5G8 V3
1.8.2 Check the Potential of the VTx PDB and connect it to the VTx TBS Unify Pro 5G8 V3
1.8.3 Connect the VTx TBS Unify Pro 5G8 V3 to the camera installed in the Body Tube
1.8.4 Check for the constant RED and BLUE Light and Video feed on the GCS after connecting to the camera
2. Camera Assembly
2.1. Mark the location of the holes on the body tube, using the camera mount and Ensure the mount is fitting snug.
2.2. Drill the holes, and tap them, mount the camera print using screws.
2.3. Pass the wiring through the mount and tape the connection wire inside the tube to the walls.
2.4. Ensure the length of the wire in enough on the inside, to connect to the avionics
2.5. Connect the cameras to the wires and the mount using glue, and epoxy
2.6. Test the cameras by connecting to the avionics.
3. Wiring and Connectors
3.1. Battery to PDB
3.1.1 Connect the battery terminal in the required series and parallel configuration
3.1.2 Connect the required batteries to the input side of the PDB through limit switches
3.1.3 Make sure the batteries are connected in the correct designated connectors
3.2. Battery to COTS
3.2.1 Connect the battery terminal in the required series and parallel configuration
3.2.2 Connect the required batteries to the limit switches
3.2.3 Make sure the batteries are connected in the correct designated connectors
3.2. Battery to TBS Unify Pro 5G8 V3 and Camera

3.2.1 Connect the batteries terminal in the required series and parallel configuration
3.2.2 Connect the required batteries to the VTx PDB through the limit switches
3.2.3 Make sure the batteries are connected in the correct designated connectors
4. Payload Assembly
4.1. Payload electronic components are checked for any physical abnormalities or damages.
4.2. Payload structure is assembled ground up with its structural parts and components. (Example: PCBs, 3D-Printed Parts)
4.3 The 3D printed holder for the Bioprinter has to be assembled and connect the Batteries to power it
5. Motor Assembly
5.1. Inspect all the grains for any physical abnormalities or damages
5.2. Cut the liner according to the length of the grain
5.3 Drill holes on the motor casing for the forward closure and nozzle retaining ring with the help of a 3d printed jig
5.4 Now insert the liner in the motor casing
5.5. Attach the O-Rings to the forward closure and then assemble the forward closure to the motor casing with the help of M6 screws
5.6 Attach the O-Rings to the nozzle and assemble the nozzle to the motor casing with the help of M6 screws after inserting all the grains
6. Rocket Integration
6.1. Fin Can
6.1.1 Attach the individual plates of the Fincan to the Couplers
6.1.2 Fix the 12 M5 screws to the plates of the Fincan
6.1.3 Check alignment and complete the assembly of all 4 fins with the help of the M4 screws.
6.1.4 Attach the assembled Fincan with the boat-tail coupler
6.2. Thrust Plate Integration
6.2.1 Take the lower body tube and mark out the location of the thrust plate
6.2.2 Drill 6 equally spaced holes on the body tube at marked locations with the help of the jig
6.2.3 Place the thrust plate inside and align the holes with the ones on the body tube.
6.2.4 Screw the thrust plate with the help of M4 Screws.
6.3. Nose Cone Assembly
6.3.1 Attach the Aluminium Nose Cone tip to the Fiberglass Nose cone body using metal epoxy and let it cure.
6.3.2 The threaded rod from the Nose Cone tip is attached with the Payload to provide rigidity
6.3.3 The Payload is attached to the upper side of the nose cone coupler and an M12 eyebolt is screwed on to the other side of the coupler.
6.3.4 The Nose Cone is attached to the nose cone coupler using 6 M4 screws.
6.4 Parachute Preparation
6.4.1 Lay parachutes out flat to form a semi-circle.

6.4.2 Fold both parachute's ends to the center and repeat until a satisfied size is reached.
6.4.3 Fold the shroud lines inside with the end remaining out.
6.4.4 Fold the parachutes along their vertical axis in a zig.
6.4.5 Fold the length of the parachute in a zig-zag manner until the packing length is reached.
6.4.6 Insert the parachute inside the parachute bag and only the end outside the parachute
6.4.7 Using a quick link, connect the shroud lines to the shock cord.
6.5. Recovery Mechanism Preparation
6.5.1. Main Recovery Mechanism
6.5.1.1. Fill the charge well with black powder and insert the e-match
6.5.1.2 Cover the charge well with recovery wadding and seal it to avoid leaks
6.5.1.3 Ensure the e-match is connected properly to Wago connectors and the black powder is neatly sealed in the charge well
6.6. Recovery Mechanism Integration.
6.6.1. Attach the flat shock cord to the eyebolt of the nose cone bulkhead with an eyebolt using a bowline knot.
6.6.2. Attach the shroud lines of the drogue parachute to the flat shock cord using a quick link and a swivel.
6.6.3. Attach the other end of the flat shock cord to the Avionics bay coupler using an eyebolt.
6.6.4. Attach the parachute bag to the Avionics bay coupler using an eyebolt.
6.6.5. Attach another flat shock cord to the eyebolt of the Avionics bay coupler at the bottom using an eyebolt.
6.6.6. Attach the shroud lines of the main parachute to the flat shock cord using a quick link and a swivel.
6.6.7. Attach the other end of the shock cord to the bulkhead using an eyebolt
6.6.8. Bolt the eyebolt to the bulkhead using a nut and use a thread locker to fix it.
6.6.9. Tape the electrical wires along the body tube walls, with masking tape and grease the walls with talcum.
6.6.10. Pack the shock cords with the connections neatly to avoid tangling
6.7. Rail Buttons, Fincan & Motor Integration.
6.7.1. Mark the location of the Rail Buttons on the lower body tube.
6.7.2. Drill the holes at marked locations and attach the rail buttons.
6.7.3. Attach the Fincan to the lower body tube.
6.7.4. Take the motor, pass it through the fin can, and screw it to the thrust plate with a bolt.
6.7.5. Attach the Fins to the Fincan.
6.7.6. Verify if all parts are fixed.
6.8. Avionics Bay Integration.
6.8.1. Attach the COTS components RRC3+, RAVEN4, FEATHERWEIGHT GPS in the intended location using brass standoff.
6.8.2. Attach The SRAD components 2 x Flight Computers, BESSIE, Limit Switches in the intended location using brass standoff

6.8.3. Insert the batteries in the battery bay and make wire connections.
6.8.4. Connect the batteries in the necessary predetermined configuration for powering components.
6.8.5. Connect the components from the battery via the limit switches.
6.8.6. Ensure that there are no loose connections during the assembly process.
6.8.7. Verify that all the systems are powering on and operating nominally.
6.8.8. Connect the components from the battery via the limit switches.
6.8.9. Shake the fully assembled avionics bay and confirm the nominal operation of components.
6.9. Payload Integration.
6.9.1. Take the payload and bolt it to the nose cone tip using the threded rod.
6.9.2. The other end of the payload is attached to the coupler connecting the nose cone and the upper body tube.
6.9.3. Check if the payload is operational by turning it ON.
6.10. Body Tube Integration.
6.10.1. Inspect all body tubes, couplers, and bulkhead for structural integrity, dimensional accuracy, and absence of debris.
6.10.2. Identify and label the three couplers based on their positions:
6.10.3. For each coupler, mark and drill 6 equidistant holes approximately 2 inches from both ends to ensure uniform load distribution and improved structural strength.
6.10.4. Insert each coupler into its respective body tube sections and transfer the hole markings accurately onto the airframe
6.10.5. Drill and tap the marked locations on all corresponding body tube sections, ensuring proper alignment across modules
6.10.6. Integrate the avionics bay within the designated section and attach bulkheads on both ends, aligning them with the coupler interfaces.
6.10.7. Secure the middle coupler between the avionics and recovery section, ensuring proper sealing and structural continuity.
6.10.8. Integrate recovery system components (drogue/main sections) and ensure proper alignment within the body tube.
6.10.9. Attach and secure the aft coupler connecting the recovery section to the motor section, ensuring proper load transfer during thrust.
6.10.10. Drill necessary access holes for avionics switching and recovery system arming without compromising structural integrity.
6.10.11. Perform alignment checks across all sections to ensure a straight thrust axis and proper mating between all modules.
6.10.12. Verify avionics functionality by switching ON the system and checking telemetry; once verified, switch it back to the OFF position.
6.11. Final Integration Check
6.11.1. Verify the flexibility of fins.
6.11.2. Verify all external screws are flush.
6.11.3. Verify the Telemetry is being received.
6.11.4. Verify the Shear pins are properly attached.
6.11.5. Verify the Payload, Camera and Avionics is turned ON.

2. Pre-Flight Checklist

1. Ensure the completion of the following checklists.
1.1. Assembly.
2. Inspect rail buttons.
2.1. Verify that rail buttons are securely attached.
2.2. Verify that the rail buttons are positioned correctly.
2.3. Inspect for any indications of wear or damage.
3. Transport the rocket safely to the launch pad.
4. Prepare the launch rail.
4.1. Loosen all screws on the launch rail and position it horizontally.
4.2. Examine the launch rail for any damage or obstructions.
5. Apply dry graphite spray to lubricate the rail.
6. Position the rocket on the launch rail.
6.1. Carefully slide the rail button into the launch rail until the rocket reaches the end stop.
6.2. Ensure no twisting or lifting during insertion.
7. Elevate and adjust the launch rail.
7.1. Lift the launch rail with care.
7.2. Modify the launch angle as per conditions
8. Ground Control Station: System Setup
8.1 Power up the Cyberdeck
8.2 Launch the GCS Software, check for any errors.
9. Telemetry Link.
9.1 Confirm telemetry packets are being received.
9.2 Verify the stability of the packet rate.
10. Data Visualization & UI
10.1 Ensure the map view displays the current rocket location
10.2 Confirm 3D Rocket model is updating with telemetry
10.3 Confirm whether all graphs are updating dynamically with the telemetry data.

3. Arming Checklist

1. Flight Software.
1.1 Confirm system time is synced with GCS.
1.2 Start telemetry broadcast.
1.3 Confirm all sensors are reporting correctly.
2. Run a dry test for state transitions.
3. Final Check
3.1 Verify all systems are nominal.
3.2 Confirm readiness with Ground Control Station.
4. Proceed to Launch Checklist

4. Launch Checklist

1. Launch
1.1 Verify that all pre-flight and the Arming checklists have been completed.
2. Activate the Avionics.
2.1. Monitor the sound occurring every 5 seconds.
3. Activate the Camera.
4. Activate the Payload.
5. Verify the establishment of a connection between Avionics and Telemetry.
5.1. If the connection has not been established, examine the channel and baud rate values on the ground station interface and rectify them accordingly.
6. Verify that Avionics is connected to a minimum of 4 satellites.
7. Ensure the altimeter is operational.
8. Final Checks: GCS
8.1 Confirm telemetry link is active
8.2 Reconfirm GPS Fix is valid
8.3 Final Status on UI: all sensors are sending values.

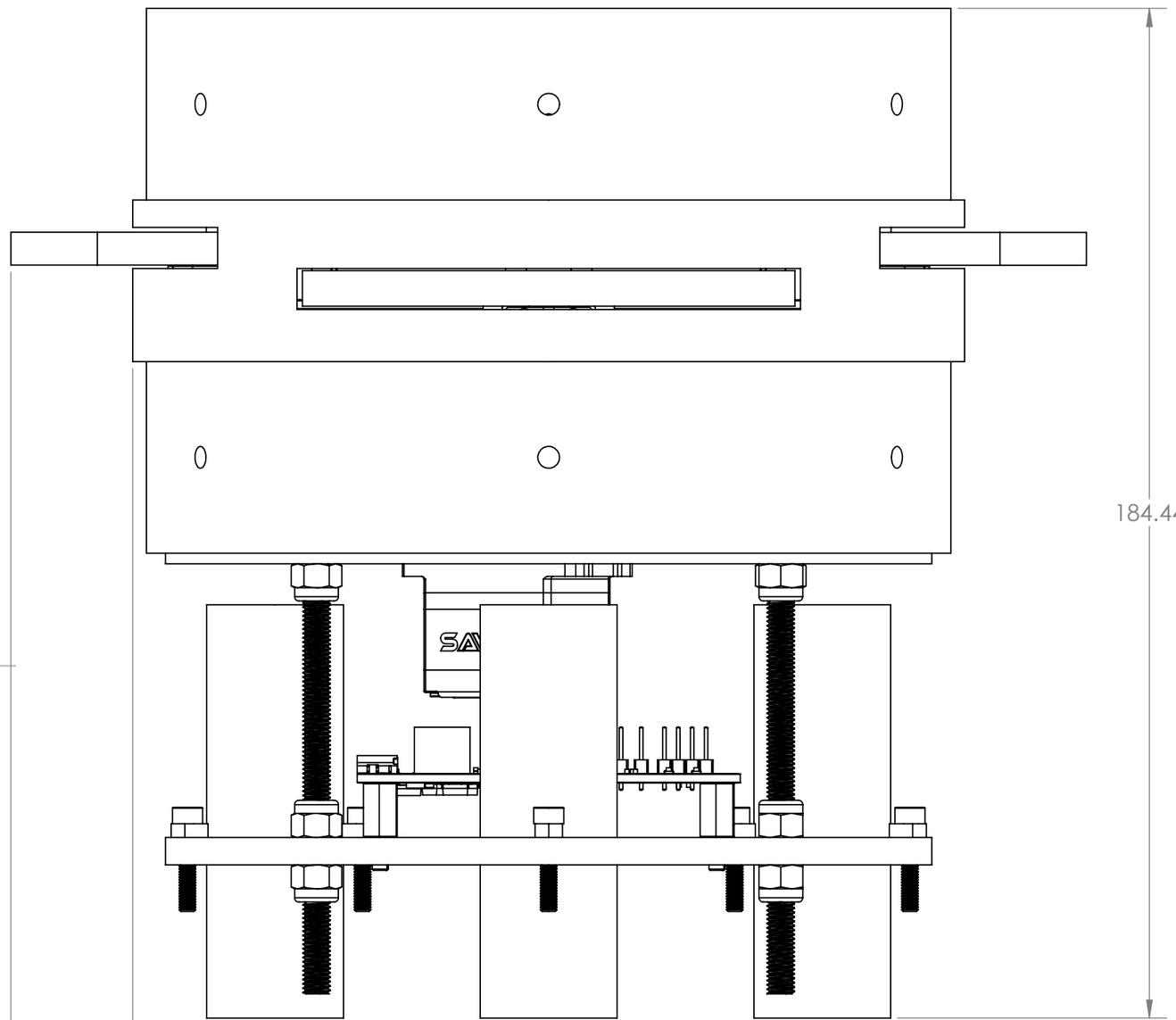
5. Recovery Checklist

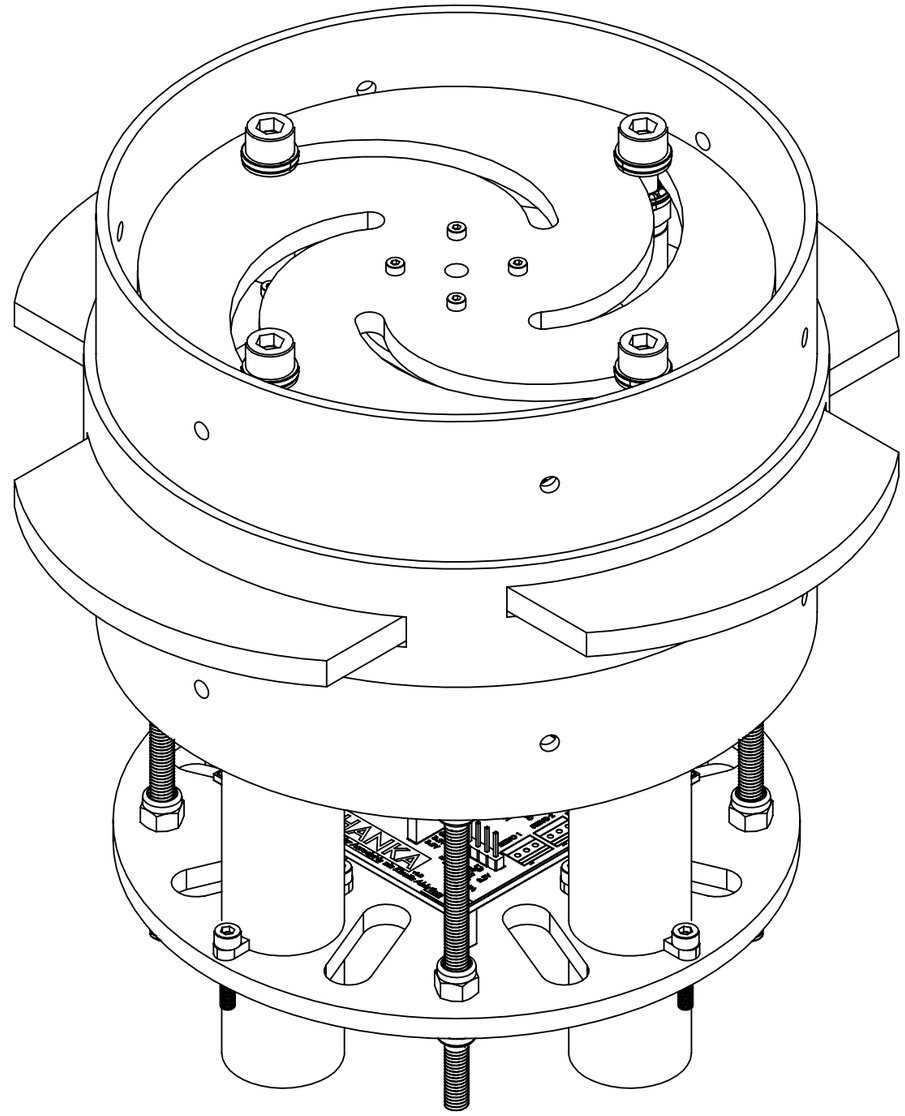
1. Approach the rocket cautiously and ensure it is safe to handle.
2. Examine the rocket splash zone for damage sustained or missing parts during flight or landing.
3. Cut any exposed wires, to disconnect the recovery mechanisms.
4. Document the splash zone as it is and observe any damages, existing and new.
5. Gather all rocket components, including any detached parts or debris.
6. Provide competition officials with pertinent flight data, including achieved altitude, flight duration and recovery system performance.
7. Dispose of used motors and igniters safely and clean blackened surfaces to check for damage.
8. Safely store the rocket and other equipment for future use.
9. Open the recovery application on team members' mobile phones
10. Input or receive the last known rocket coordinates
11. The arrow points towards the coordinates of the rocket
12. Once recovered, take screenshots of recovered components and submit the final recovery log for the final report

6. Off Nominal Checklist

1. Launch Scrub
1.1 Scrub the launch in case of failure to pass Final Flight Safety Review.
1.2 Scrub the launch in case if COTS Altimeter (main and redundant) are non-responsive.
1.3 Scrub the launch in case of improper/non-ignition of rocket motor.
1.4 Scrub the launch in case of any structural damage to the body of the rocket impacting its aerodynamics.
2. Safety Violation
2.1 In case of any safety violations pointed out in the Flight Review, address all issues promptly following required safety protocols and re-attempt FSR.
2.2 In case of any critical system malfunction which cannot be handled within necessary time frame, disengage all active mechanisms and prepare for launch scrub.
3. Meteorological Issues
3.1 Consult with meteorological experts and local weather app to predict impact of weather on the rocket trajectory.
3.2 Make necessary predictions and precautions for abnormal meteorological conditions.
3.3 In case of conditions being too drastic for launch, proceed to scrubbing launch.

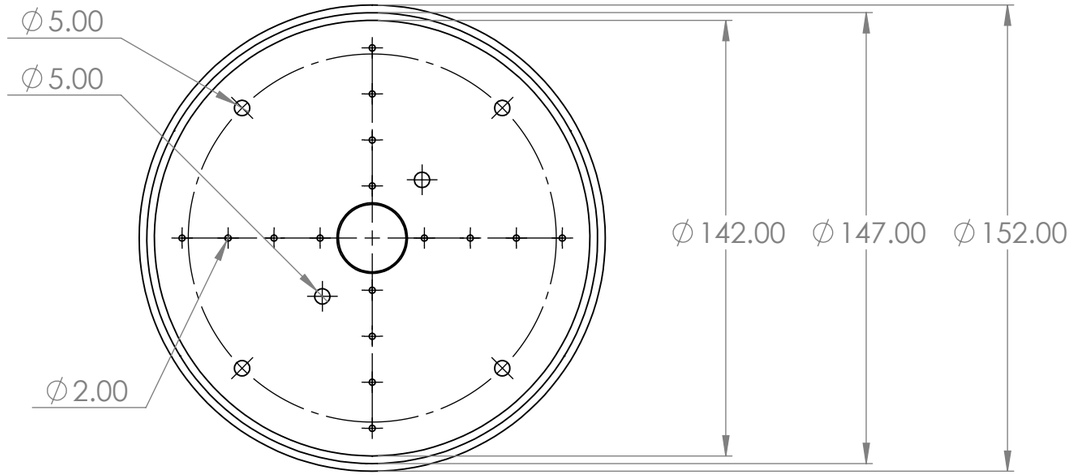
Appendix.F
Engineering Diagrams

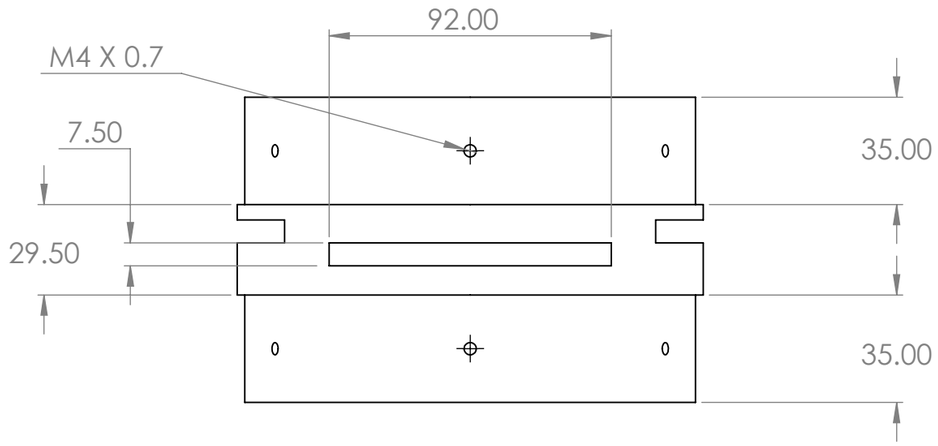


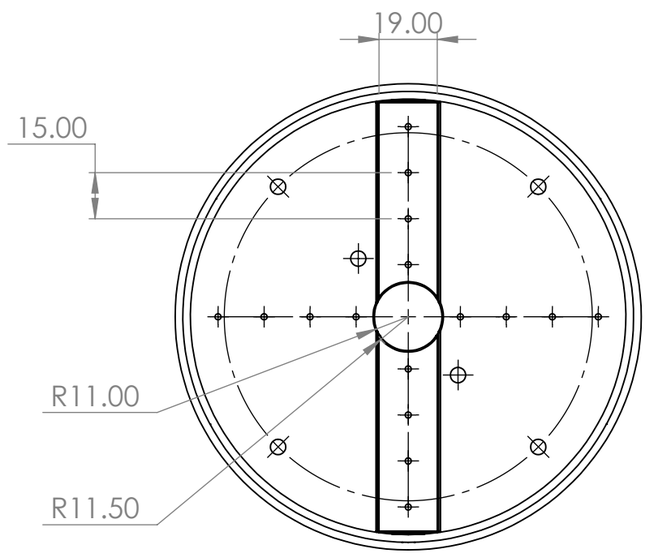


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Proprietary and confidential The information contained in this drawing is the sole property of Team Sammard. Any reproduction in part or as a whole without consent/permission of Team Sammard is prohibited.	TITLE: AIRBRAKE ASSEMBLY	
MATERIAL:	DWG NO.	A3
WEIGHT:	SCALE:1:4	SHEET 1 OF 1







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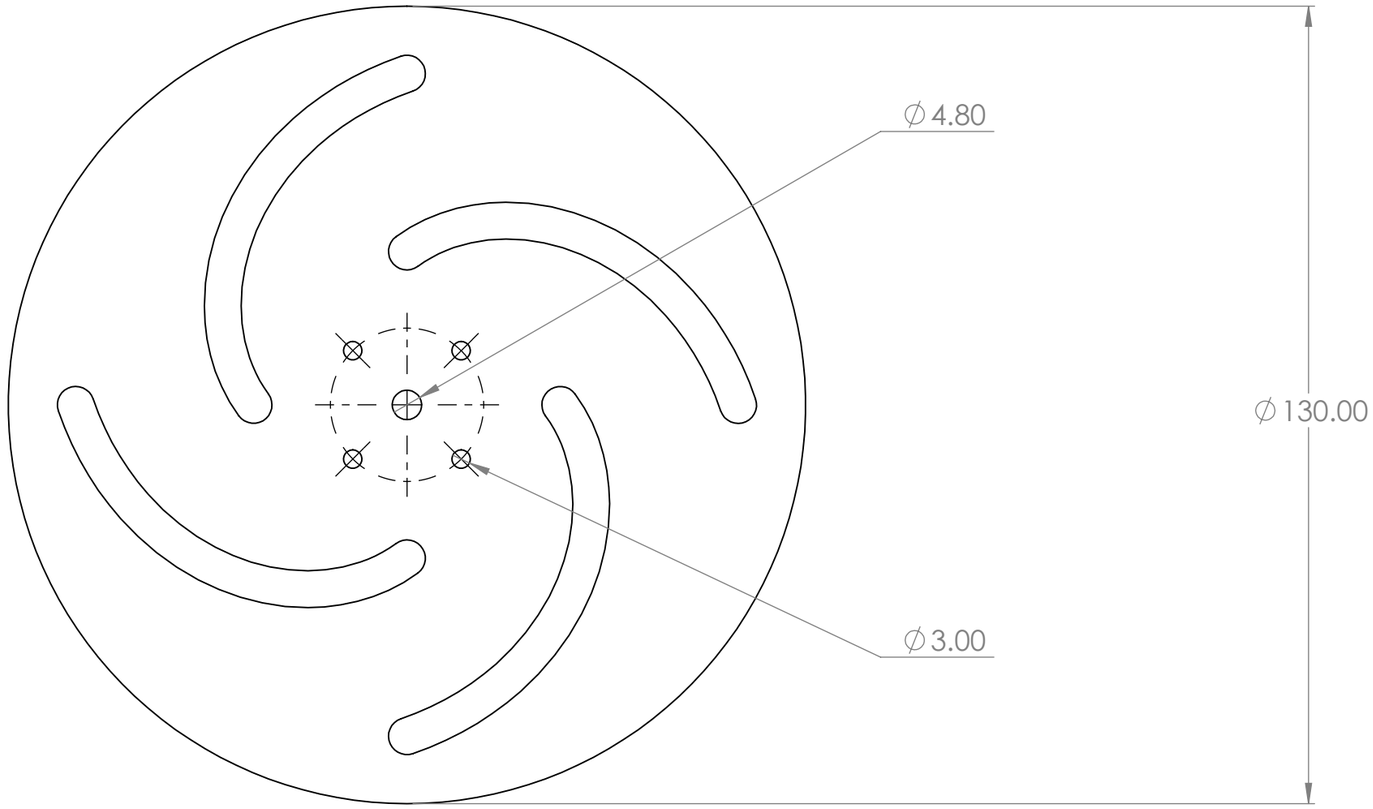
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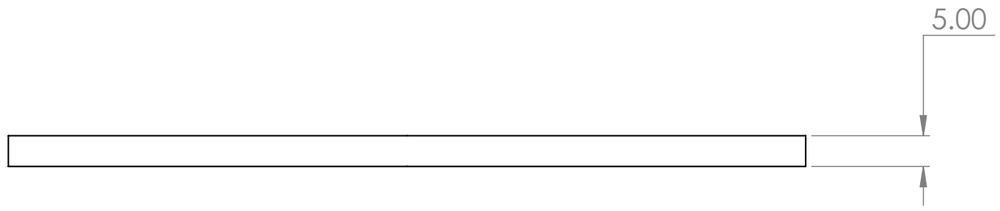
TITLE: Airbrakes Coupler

MATERIAL: Aluminium 6061-T6

DWG NO. | A3

WEIGHT: | SCALE:1:2 | SHEET 1 OF 6





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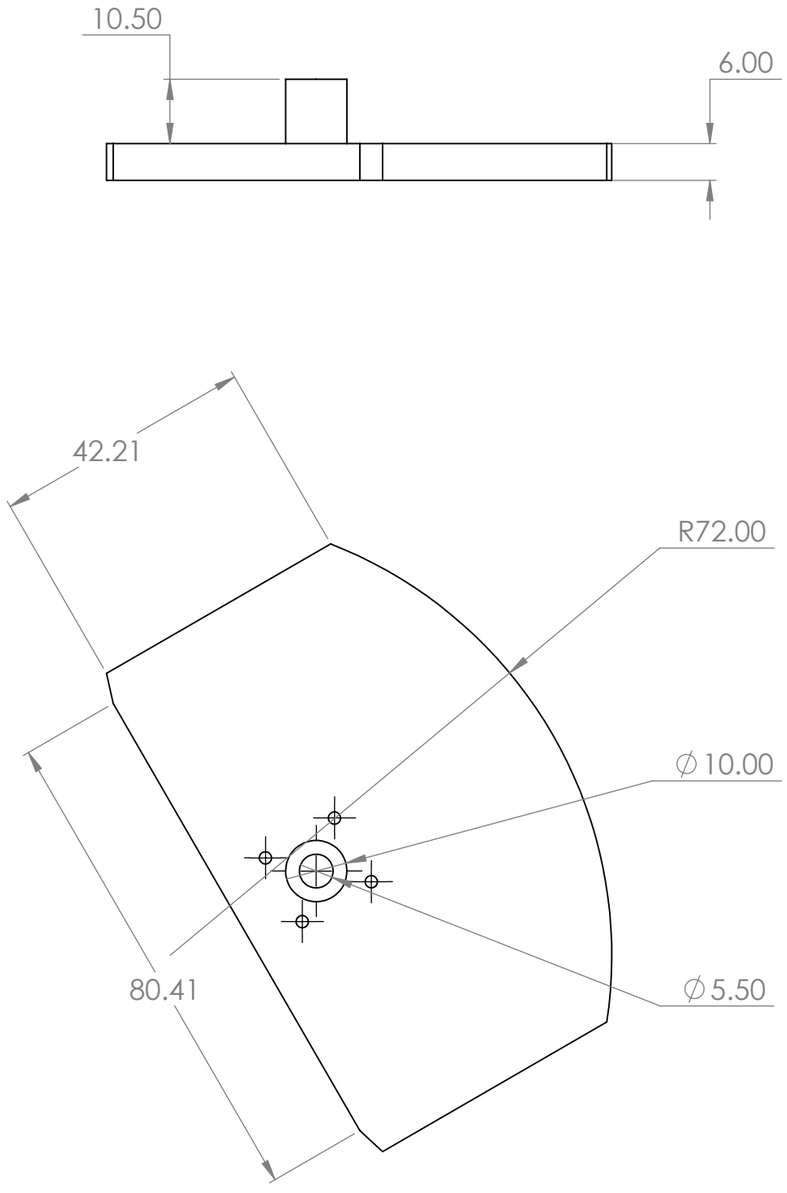
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TITLE: Actuating Plate

MATERIAL: Aluminium 6061-T6

DWG NO. | A3

WEIGHT: | SCALE:1:2 | SHEET 2 OF 6



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TITLE: Airbrake leaf

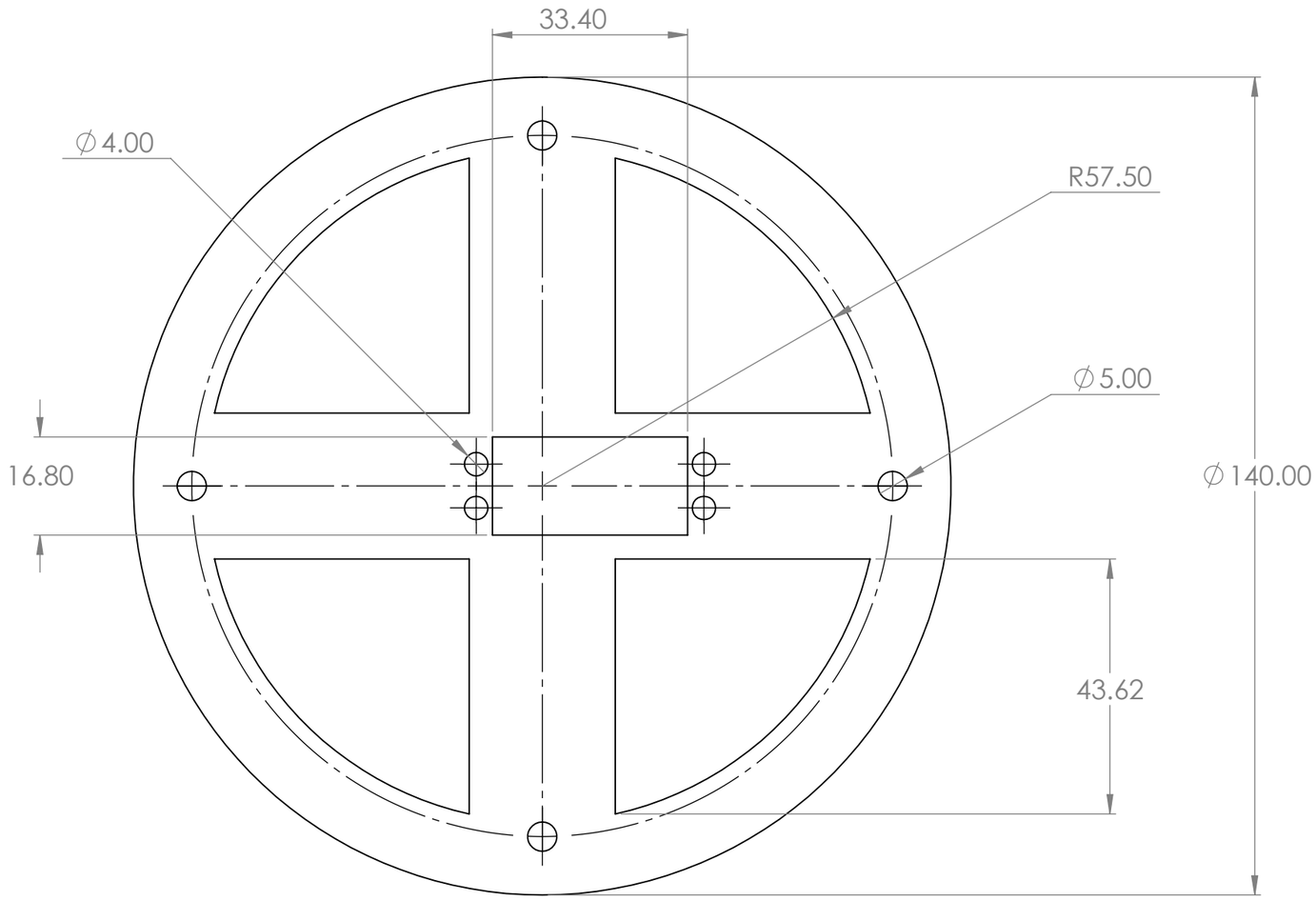
MATERIAL: Aluminium 6061-T6

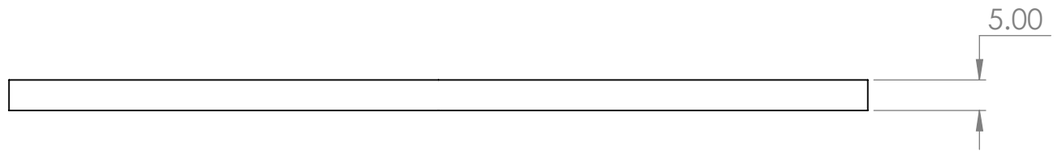
DWG NO. A3

WEIGHT:

SCALE:1:1

SHEET 3 OF 6





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TITLE: Servo Plate

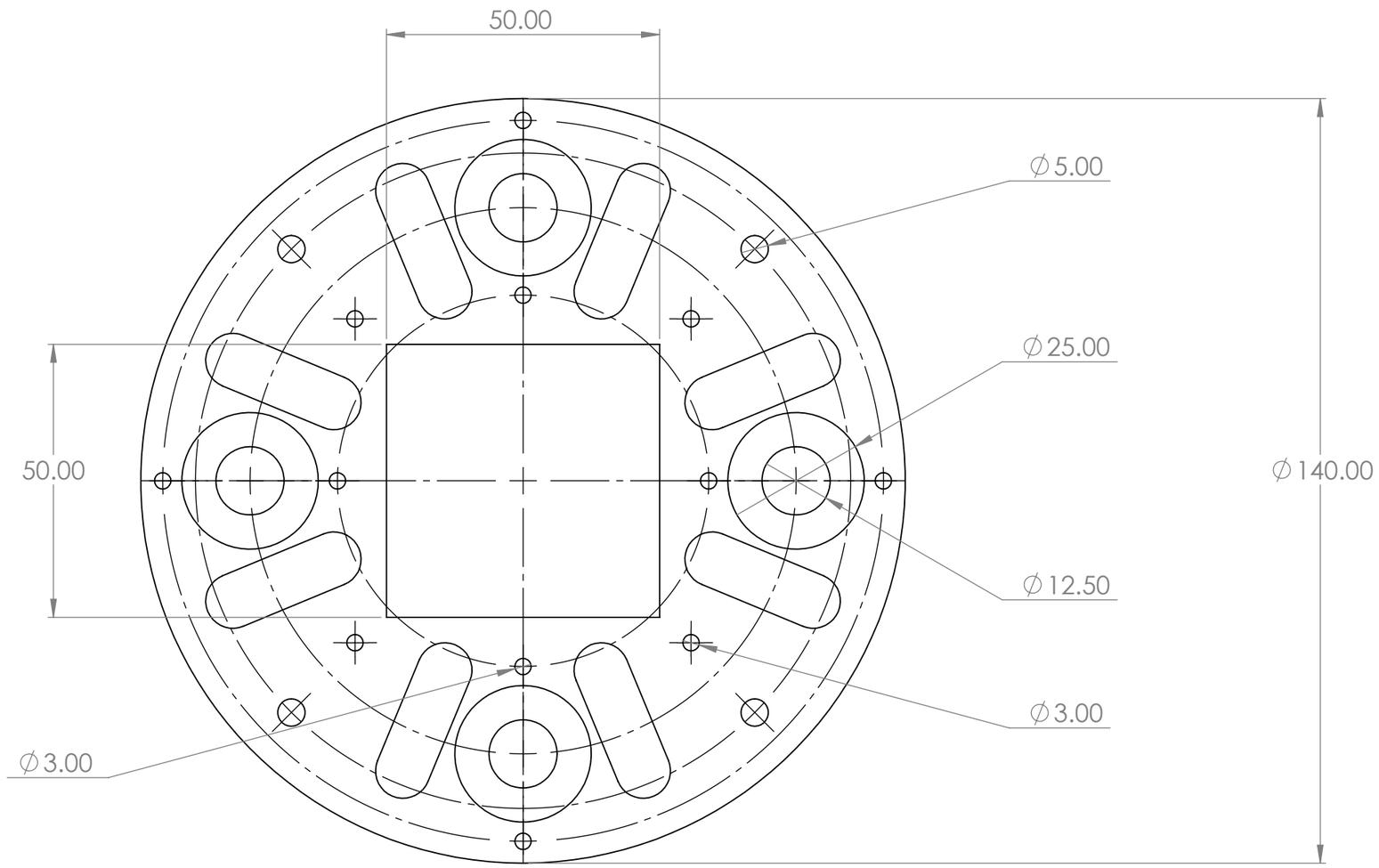
MATERIAL: Aluminium 6061-T6

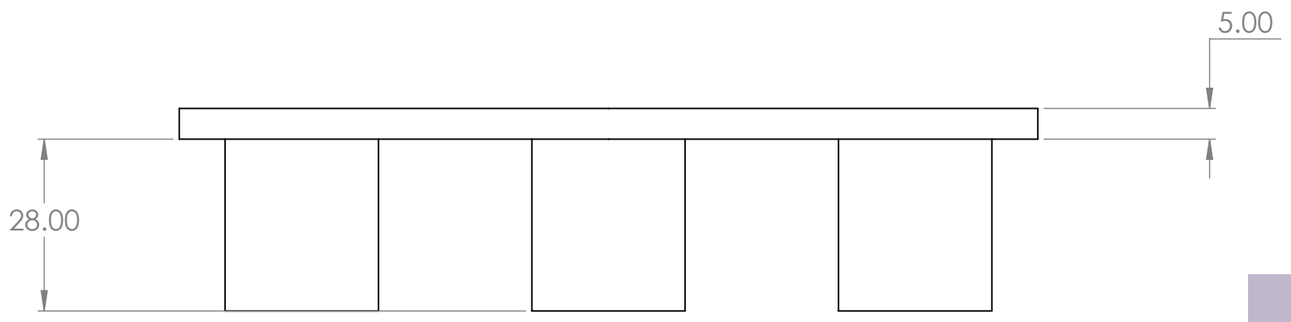
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SHEET 4 OF 6





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TITLE: Electronics Plate

MATERIAL: Aluminium 6061-T6

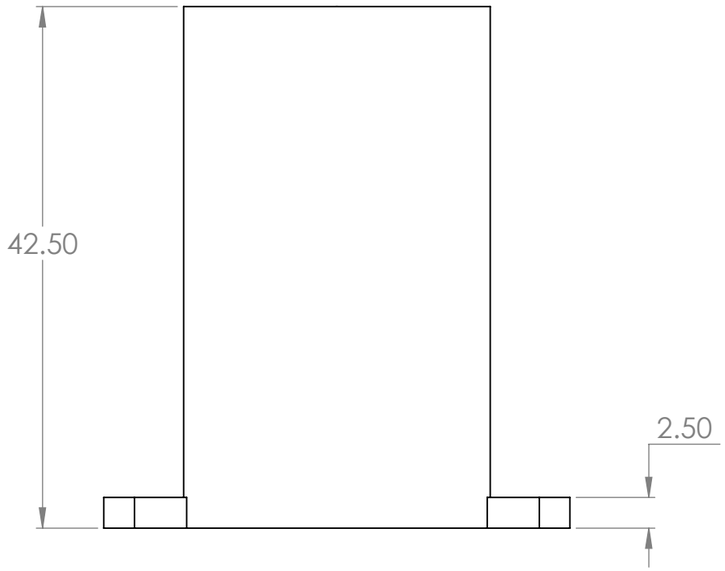
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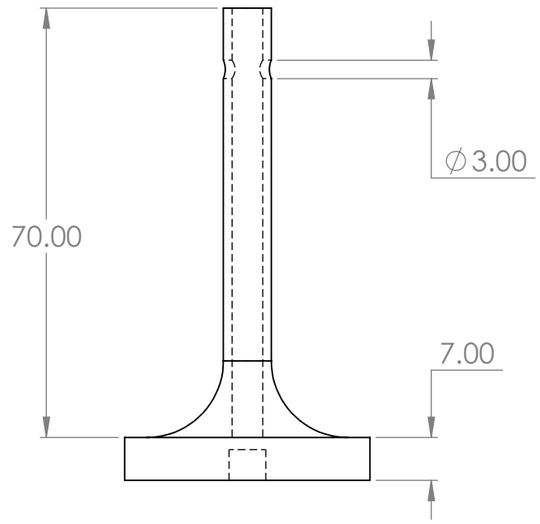
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SHEET 5 OF 6

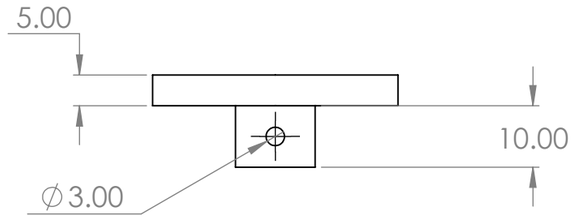
Battery Enclosure

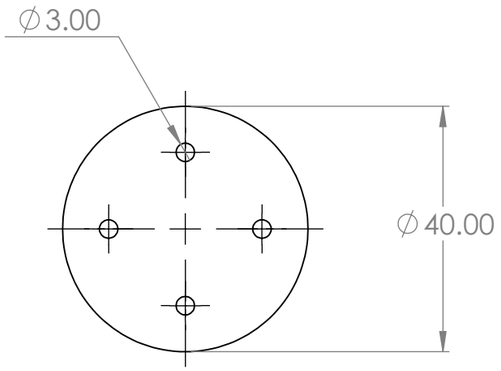


Servo Shaft



Shaft bearing





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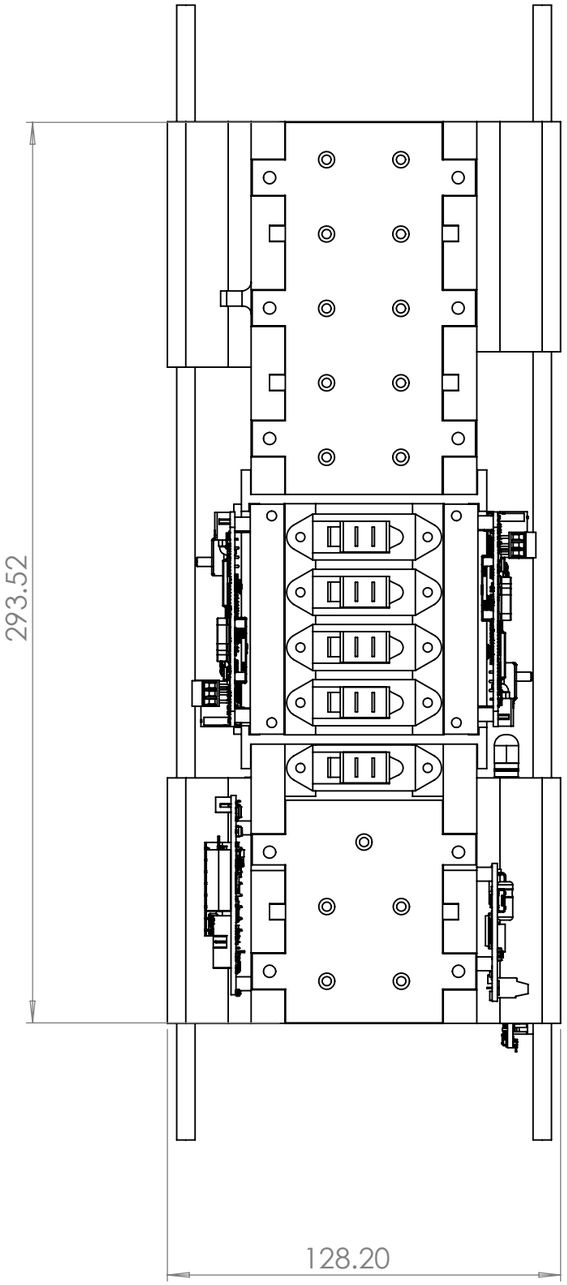
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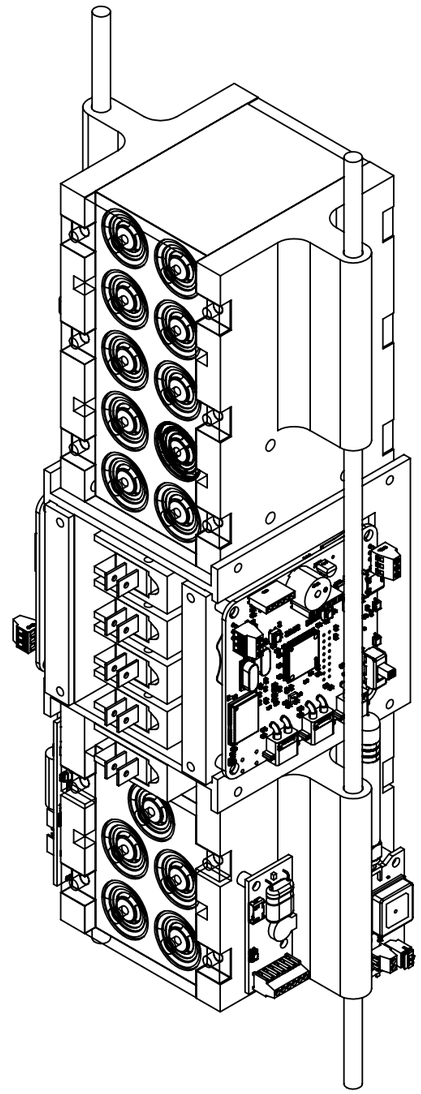
TITLE: Battery Enclosure, Servo Shaft, Shaft Bearing

MATERIAL: 3D Printed ABS

DWG NO. | A3

WEIGHT: | SCALE:2:1 | SHEET 6 OF 6





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TITLE:
AVIONICS BAY INTEGRATED

MATERIAL:
ABS

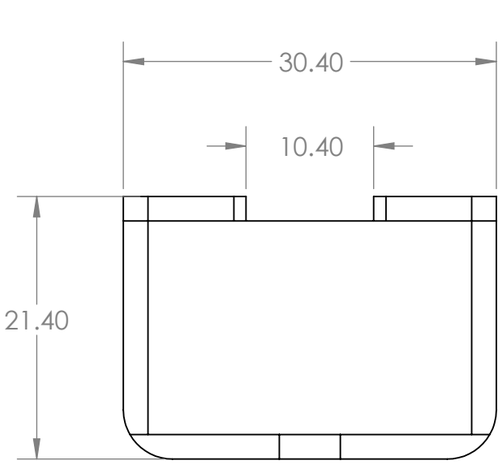
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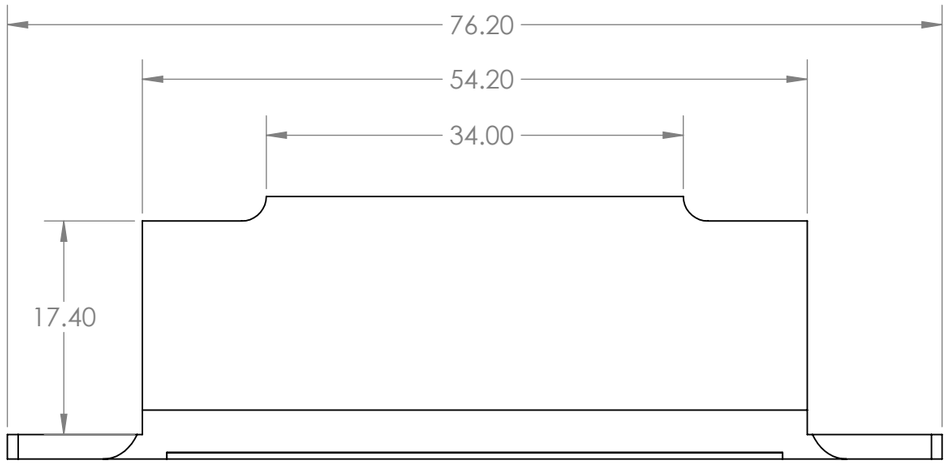
A3

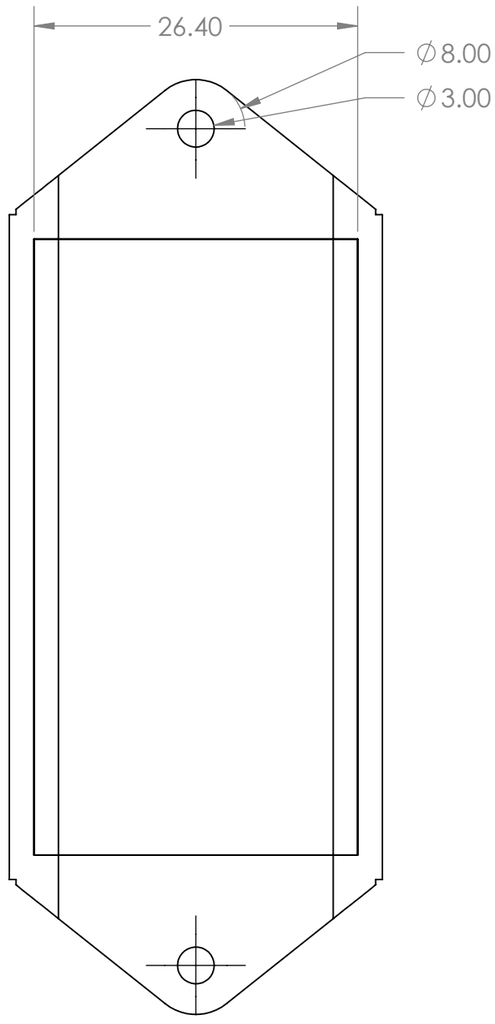
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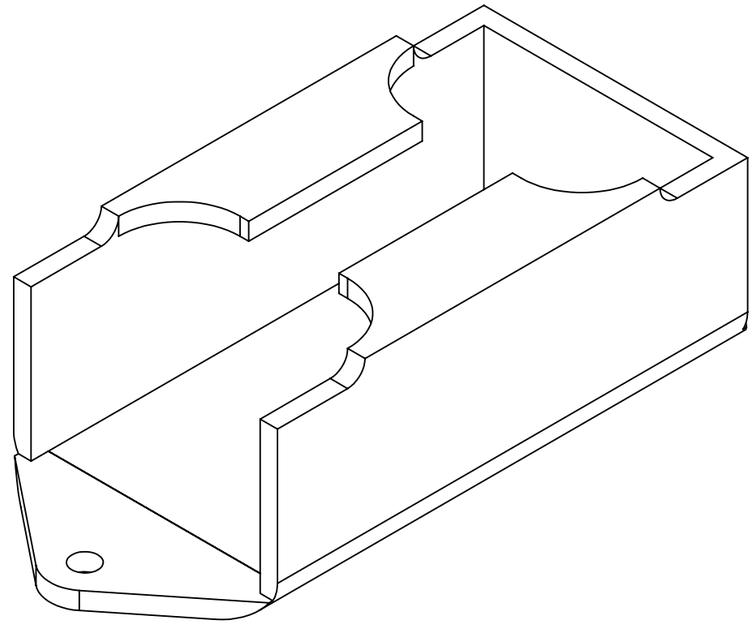
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SHEET 1 OF 2





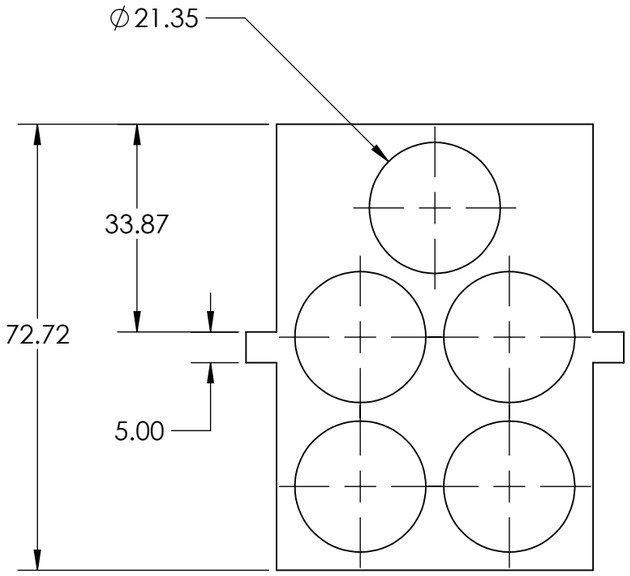


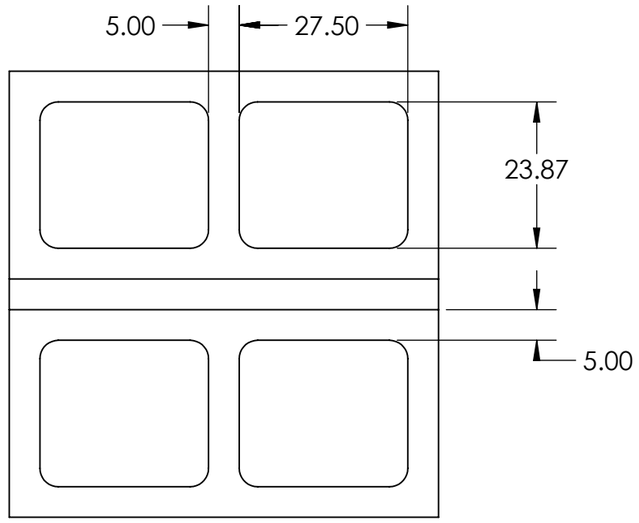


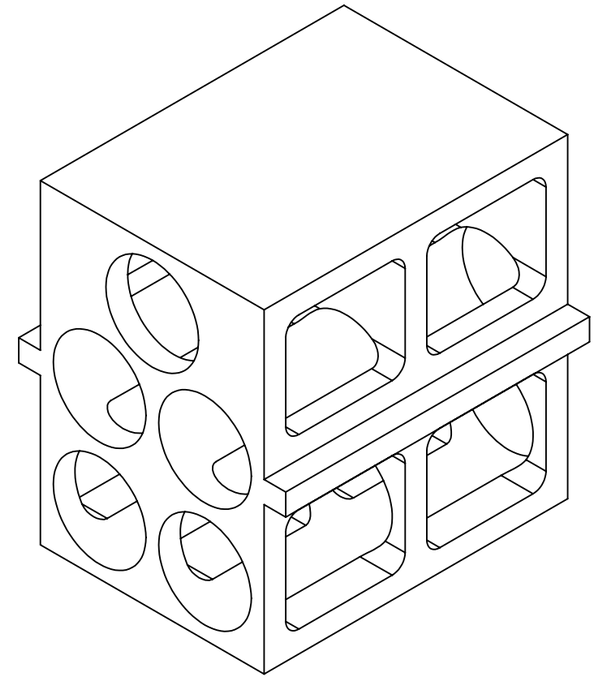
SAMMARD

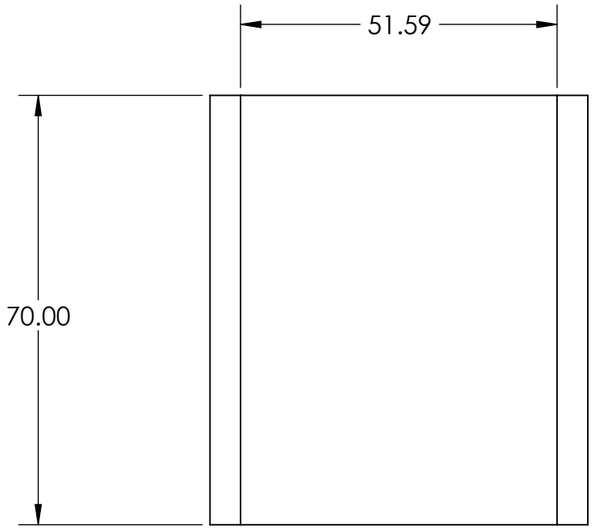
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Proprietary and confidential The information contained in this drawing is the sole property of Team Sammard. Any reproduction in part or as a whole without consent/permission of Team Sammard is prohibited.	TITLE: 9V COTS battery Holder	
MATERIAL: ABS	DWG NO.	A3
WEIGHT:	SCALE:2:1	SHEET 1 OF 10

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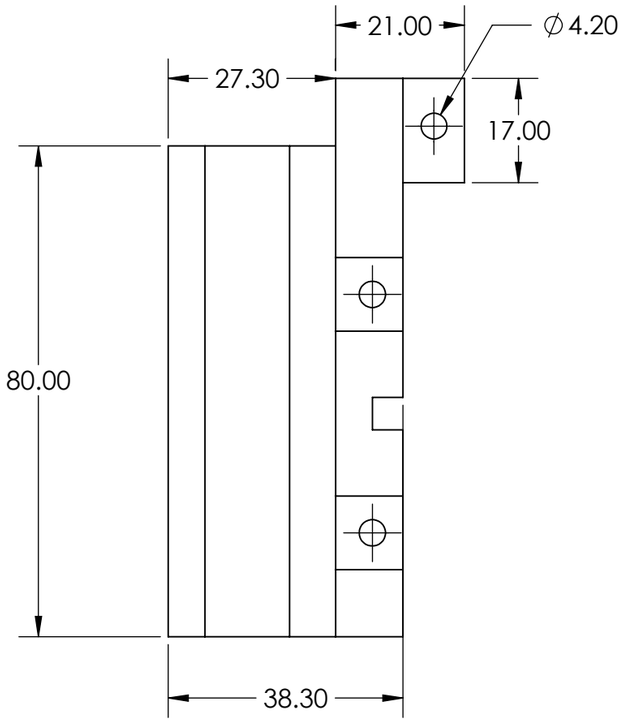


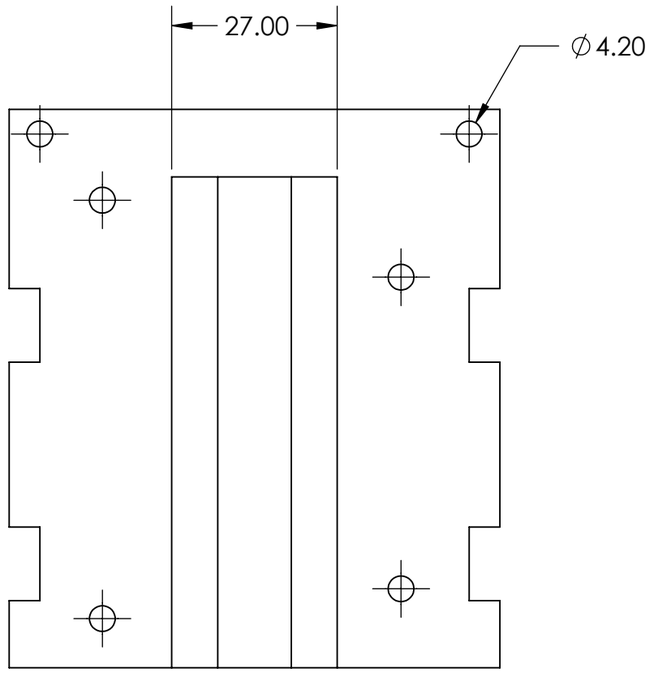


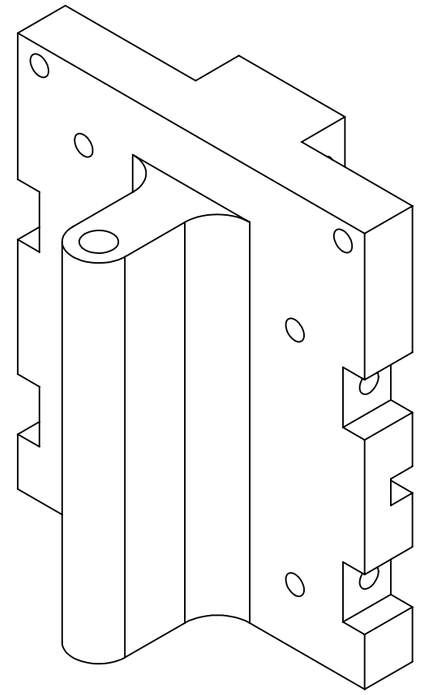


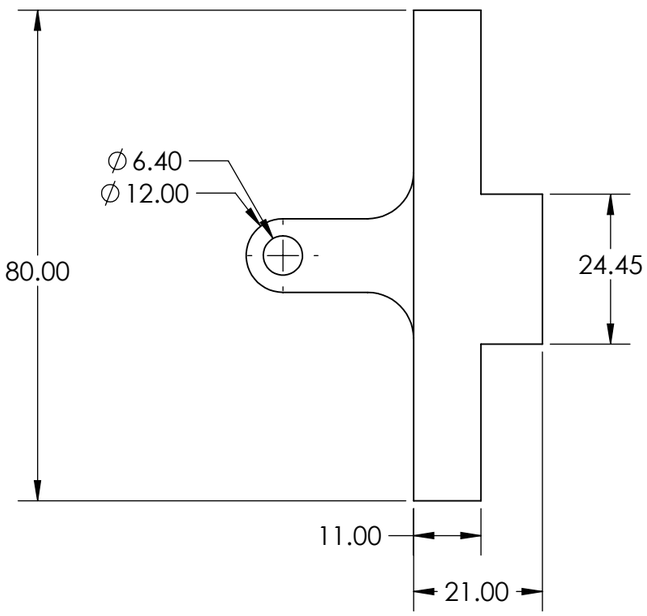
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Proprietary and confidential The information contained in this drawing is the sole property of Team Sammard. Any reproduction in part or as a whole without consent/permission of Team Sammard is prohibited.	TITLE: COTS Battery Pack	
MATERIAL: ABS	DWG NO.	A3
WEIGHT:	SCALE:1:1	SHEET 2 OF 10

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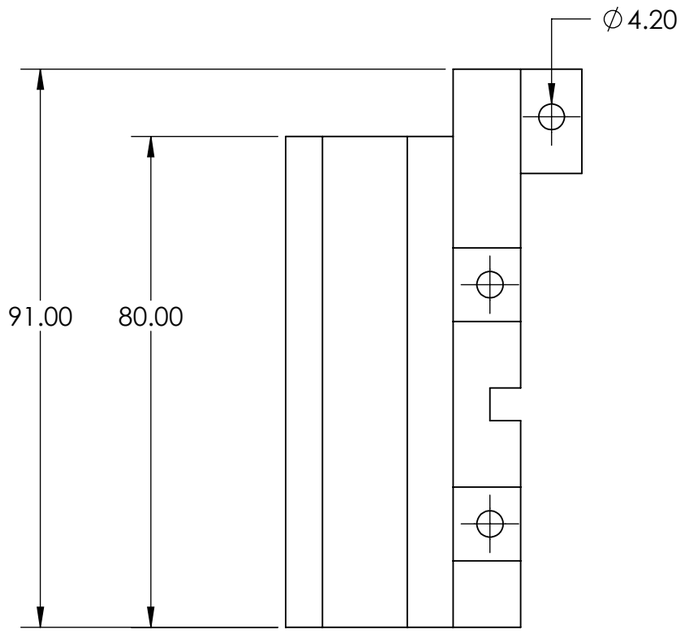
TITLE: Avionics COTS Housing-1

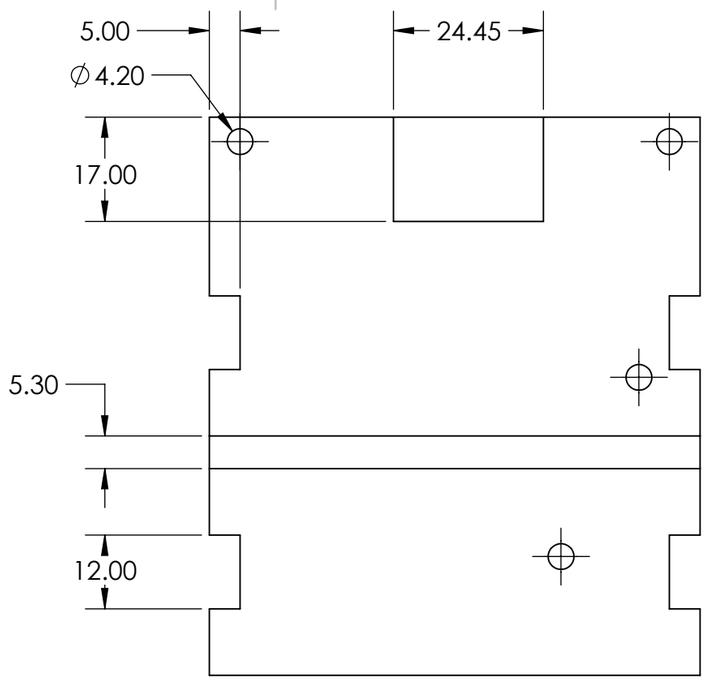
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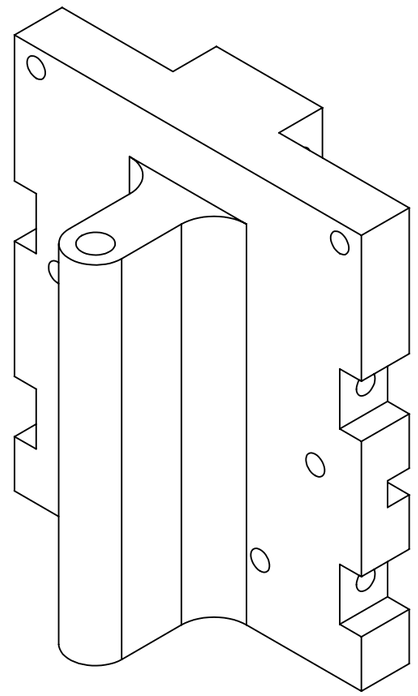
DWG NO. | A3

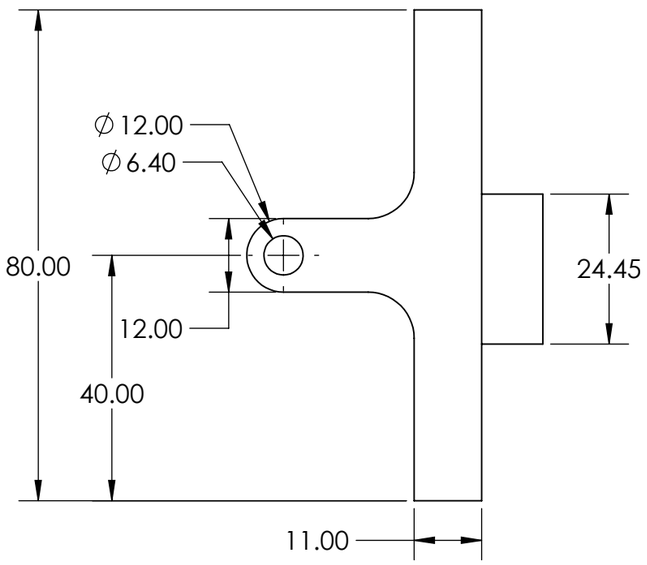
WEIGHT: | SCALE:1:2 | SHEET 3 OF 10

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TITLE: Avionics COTS Housing-2

MATERIAL: ABS

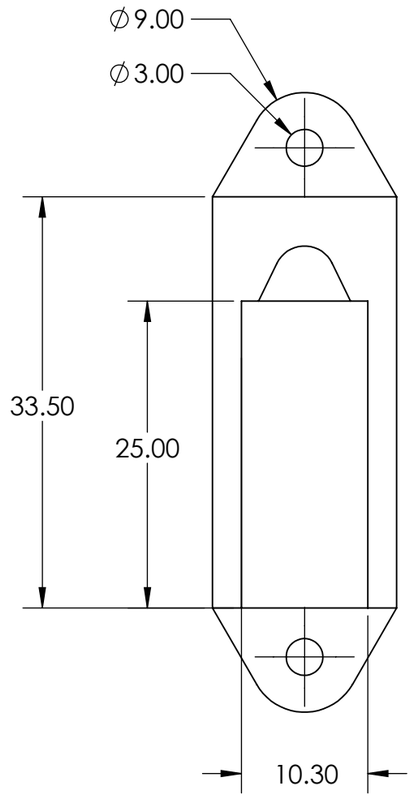
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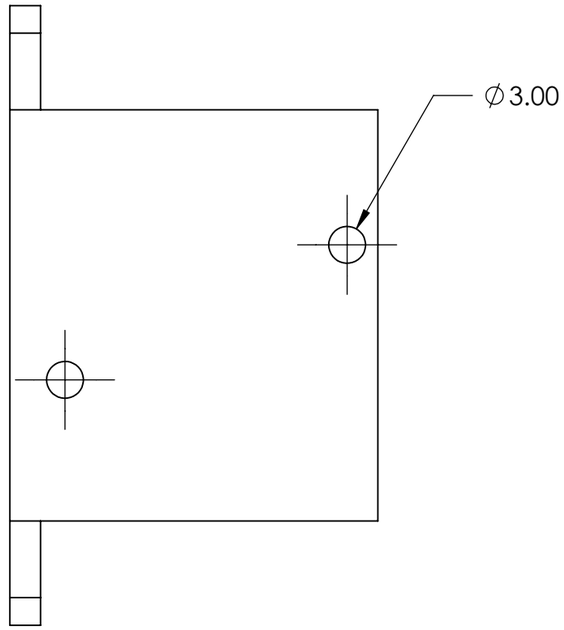
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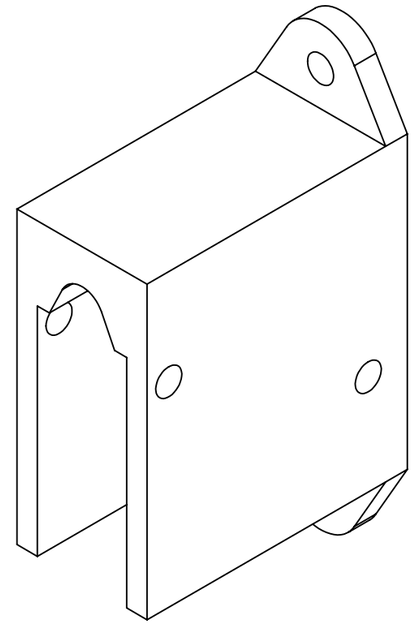
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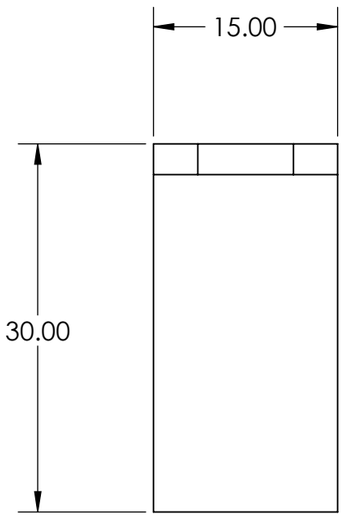
SHEET 4 OF 10

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TITLE: Limit Switch Casing

MATERIAL: ABS

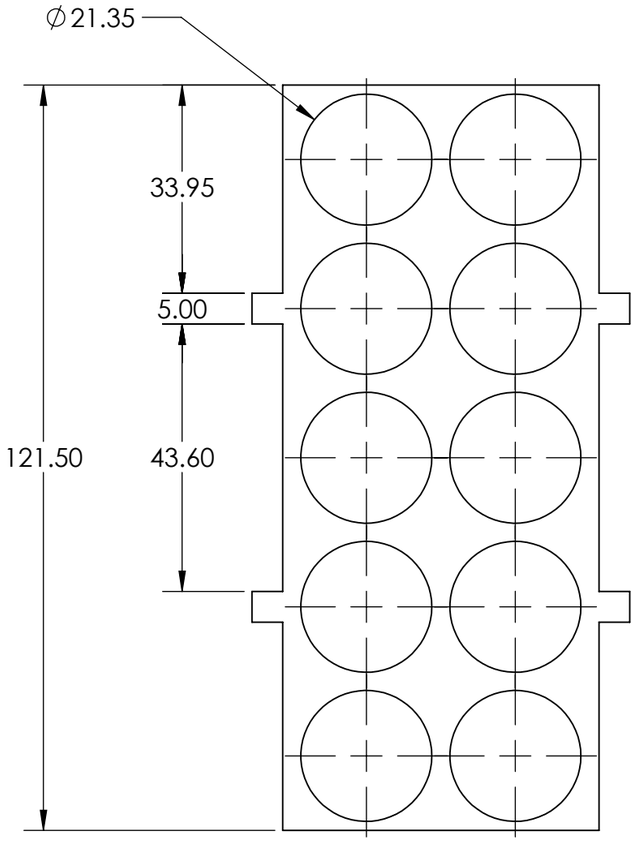
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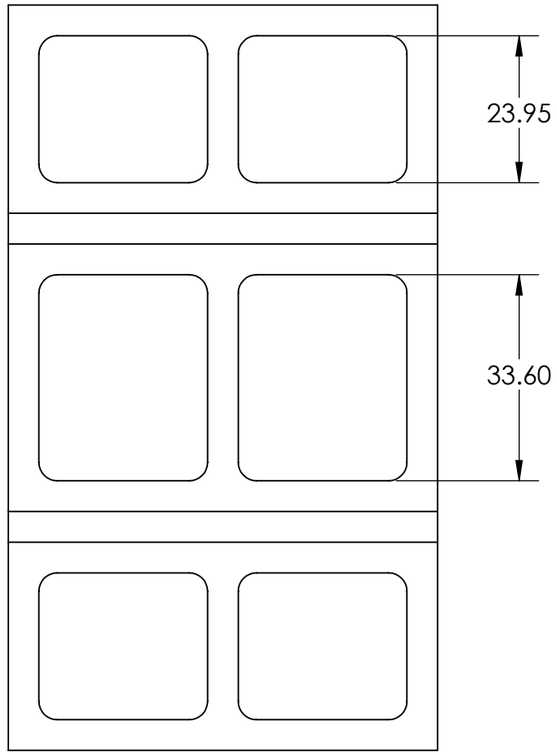
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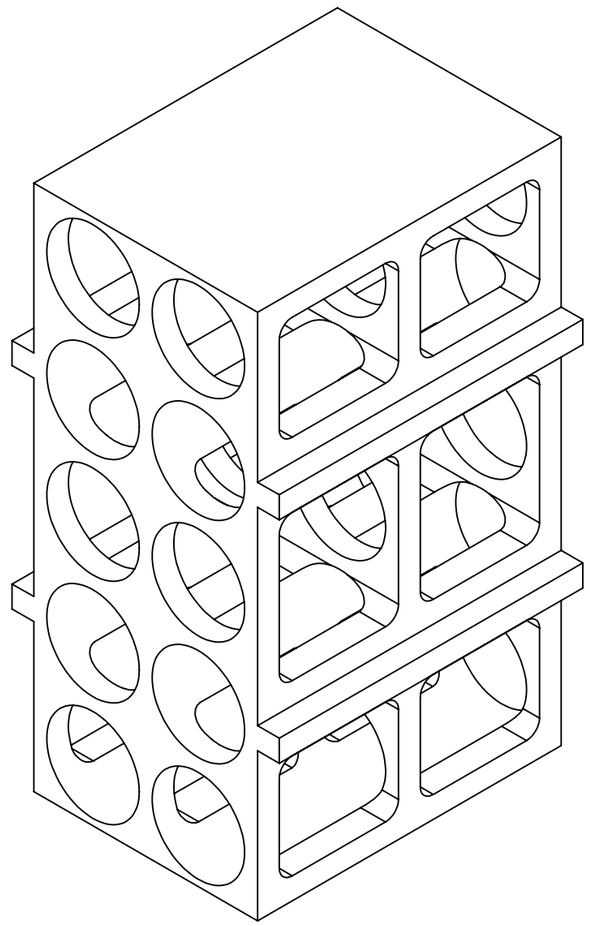
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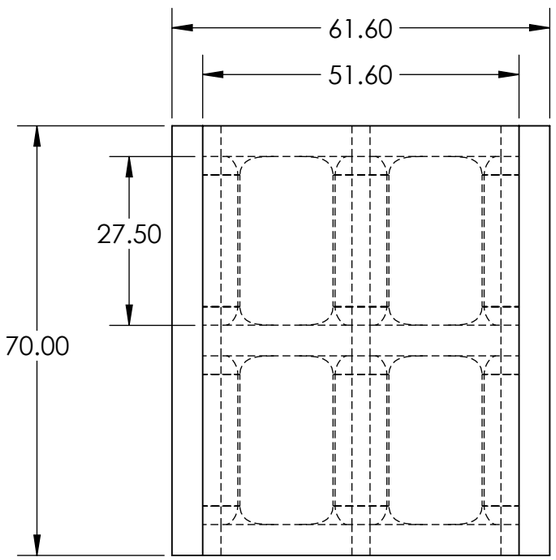
SHEET 5 OF 10

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TITLE: Avionics- Battery Pack

MATERIAL: ABS

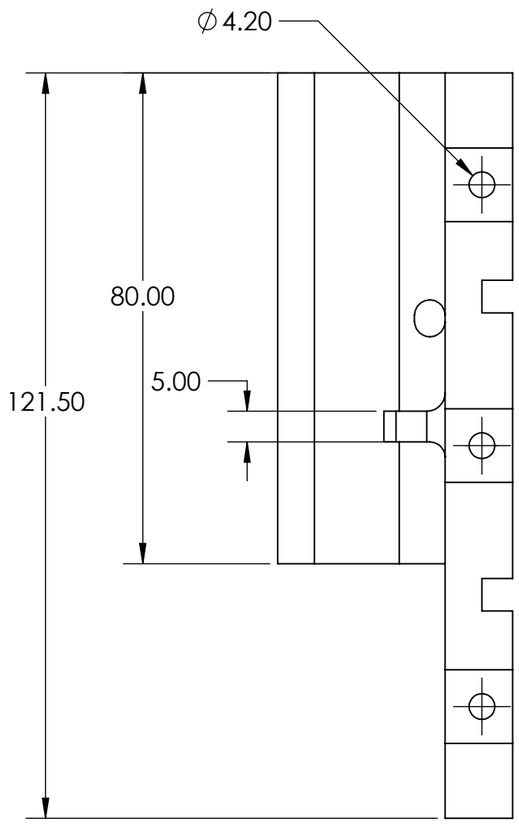
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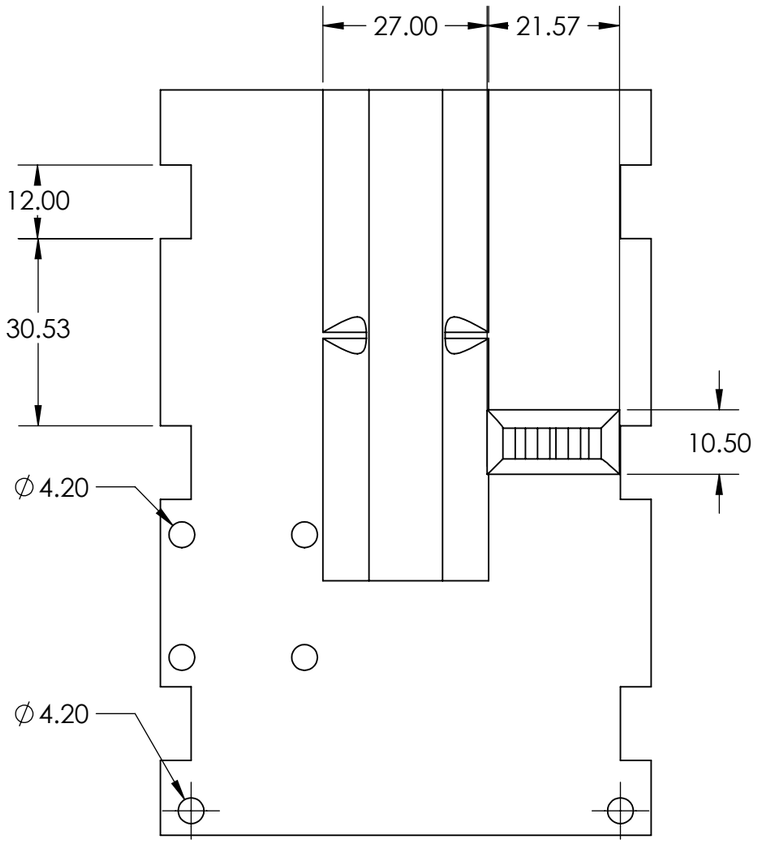
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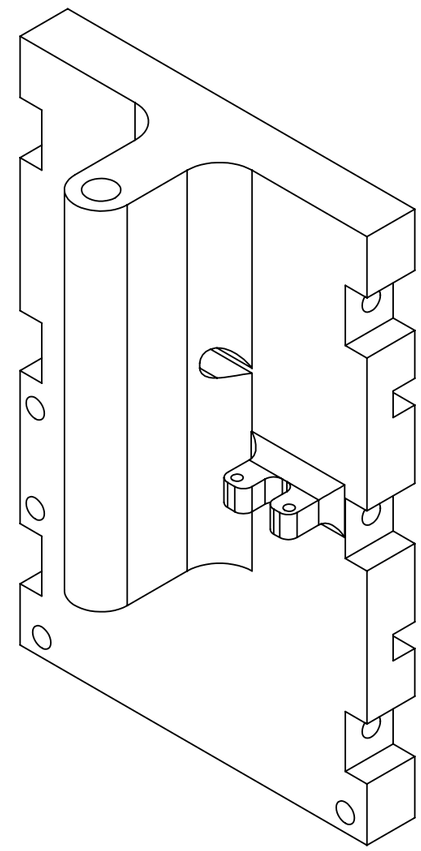
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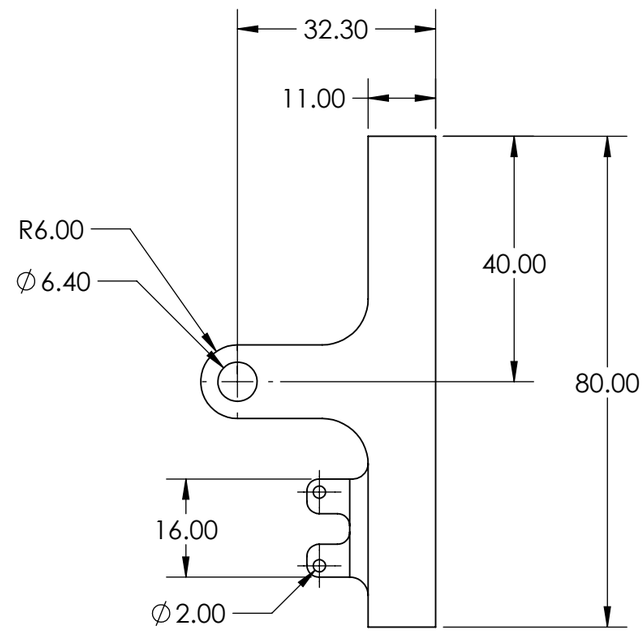
SHEET 6 OF 10

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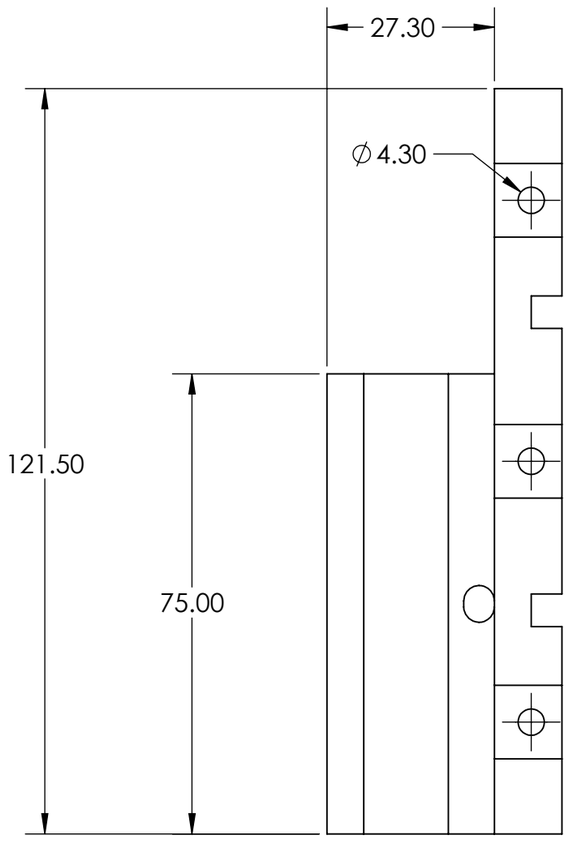


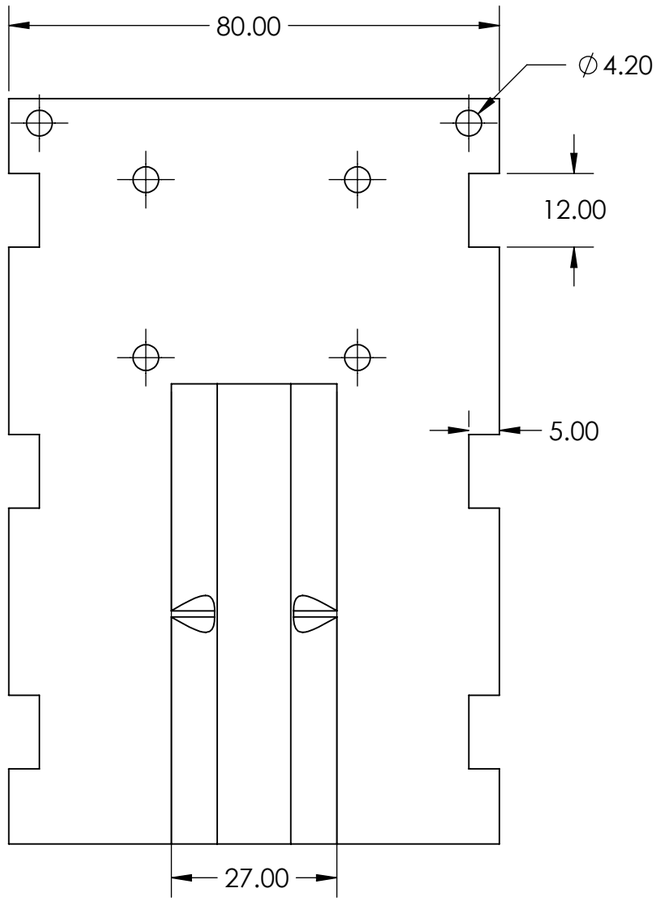


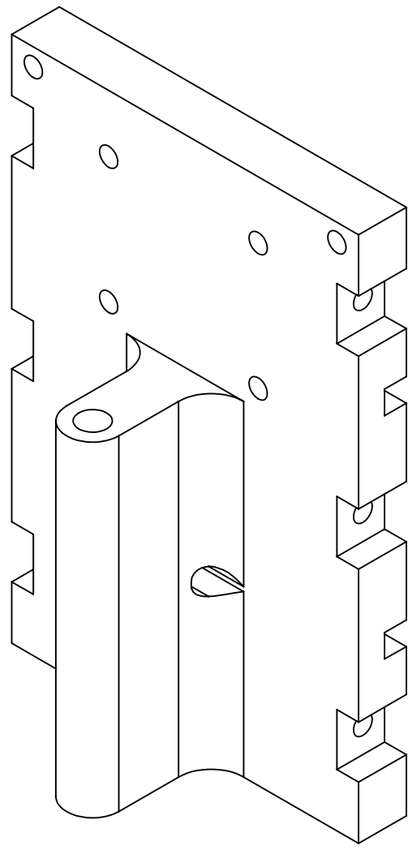


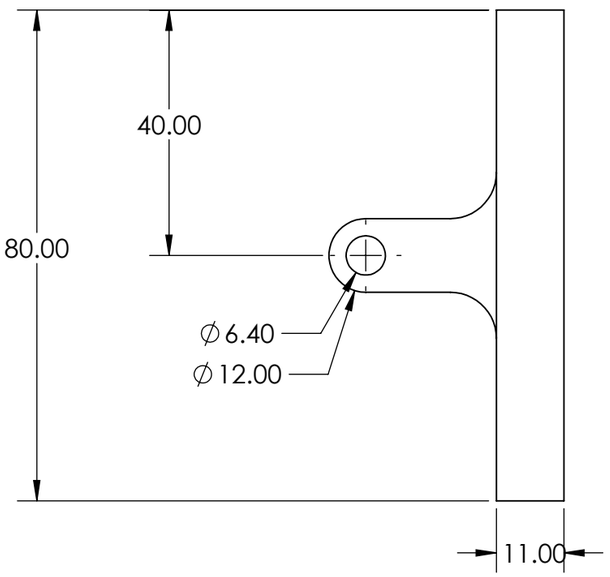
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MATERIAL: ABS	DWG NO.	A3
WEIGHT:	SCALE:1:2	SHEET 7 OF 10

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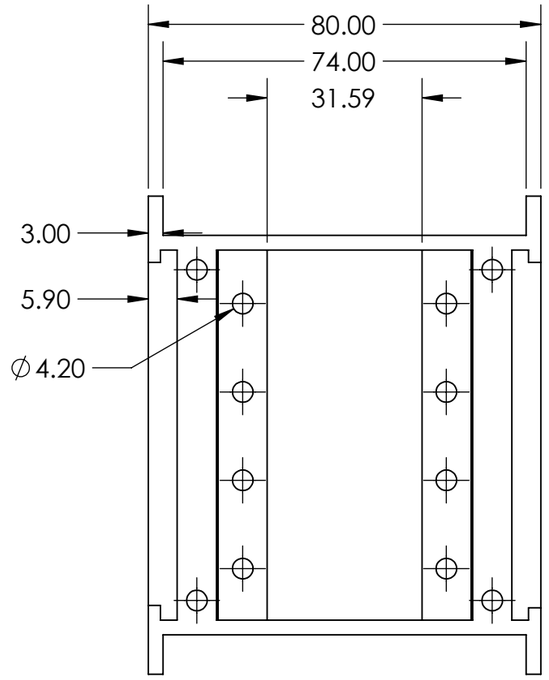
TITLE: Avionics SRAD Section Housing-2

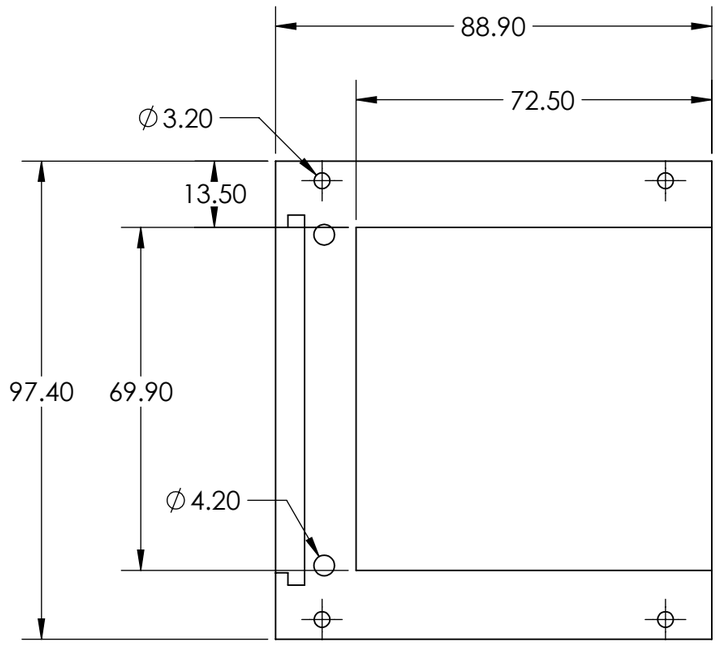
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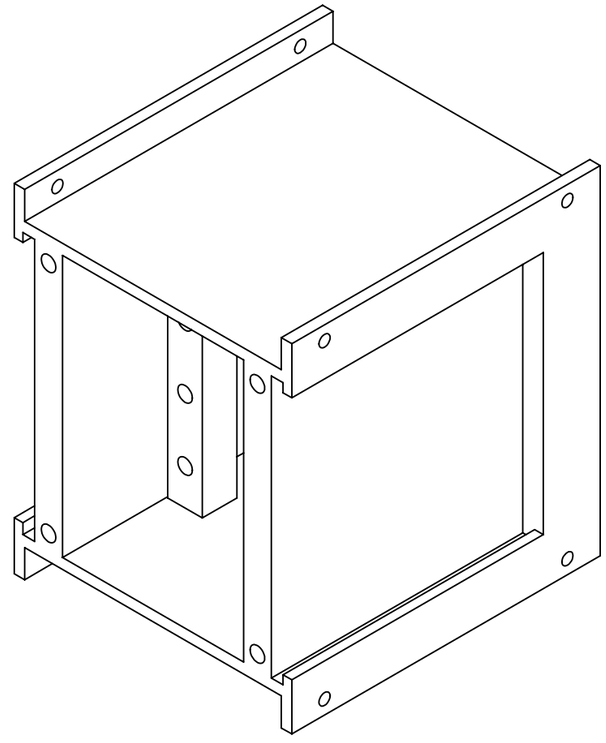
DWG NO. | A3

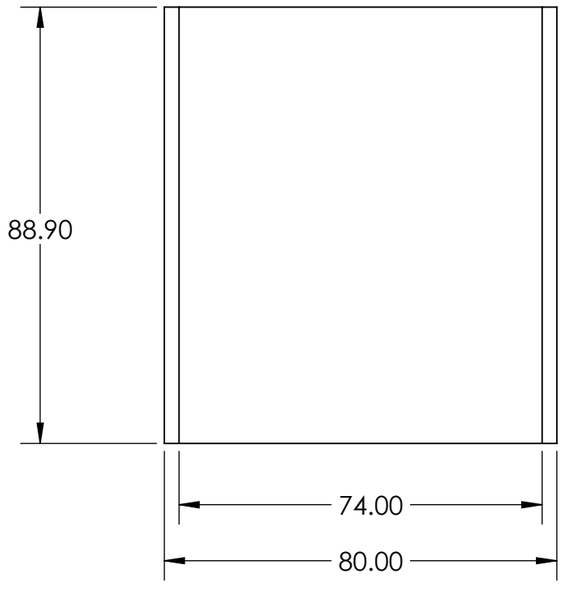
WEIGHT: | SCALE:1:2 | SHEET 8 OF 10

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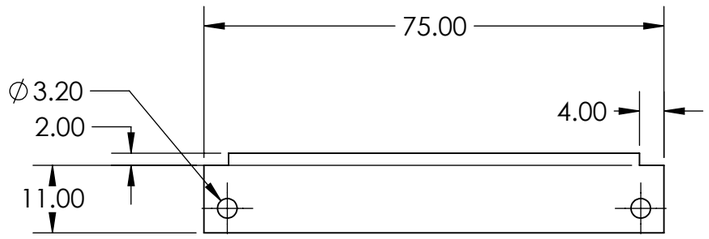


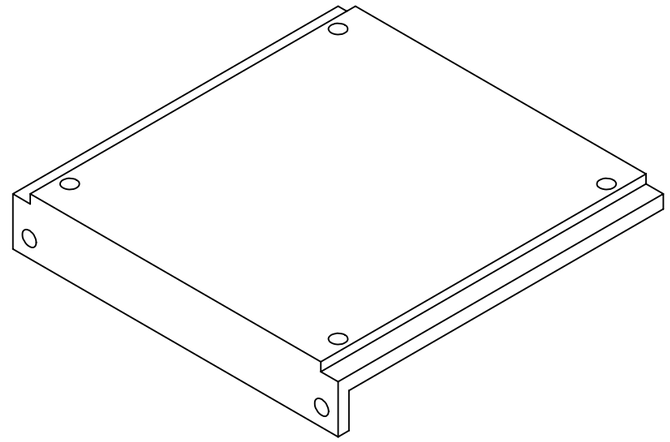


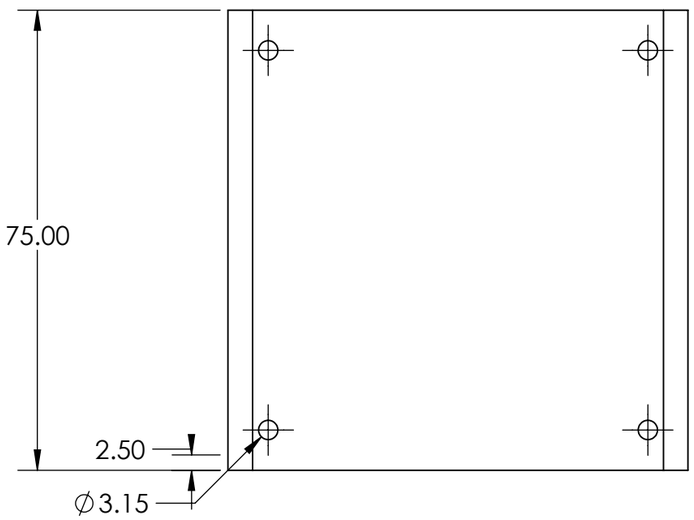


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MATERIAL: ABS	DWG NO.	A3
WEIGHT:	SCALE:1:2	SHEET 9 OF 10

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TITLE: Flight Computer Attachment Plate

MATERIAL: ABS

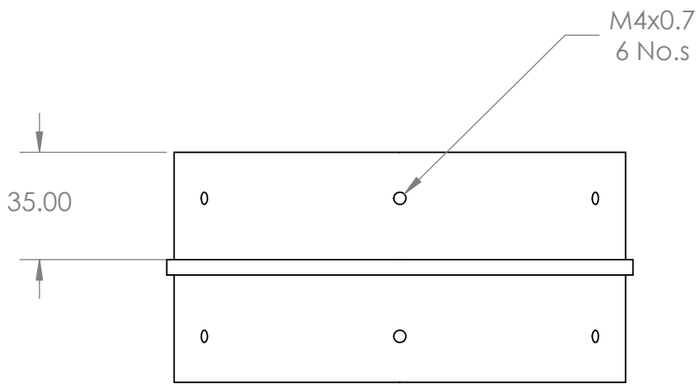
DWG NO. A3

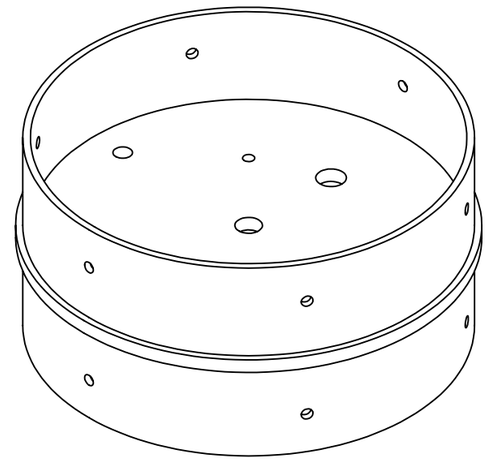
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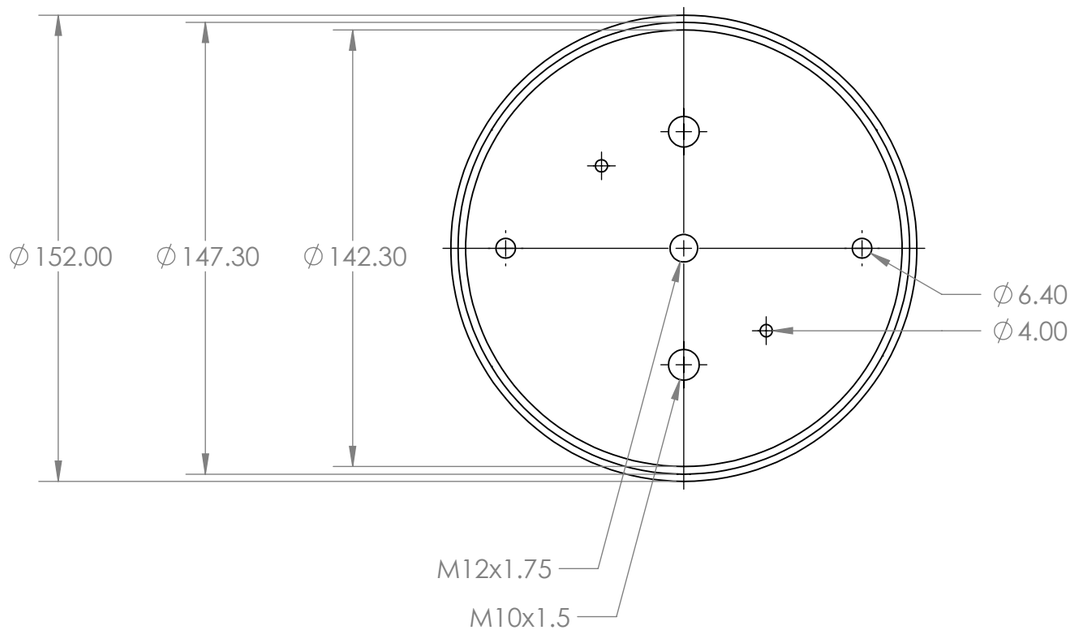
SCALE:1:1

SHEET 10 OF 10

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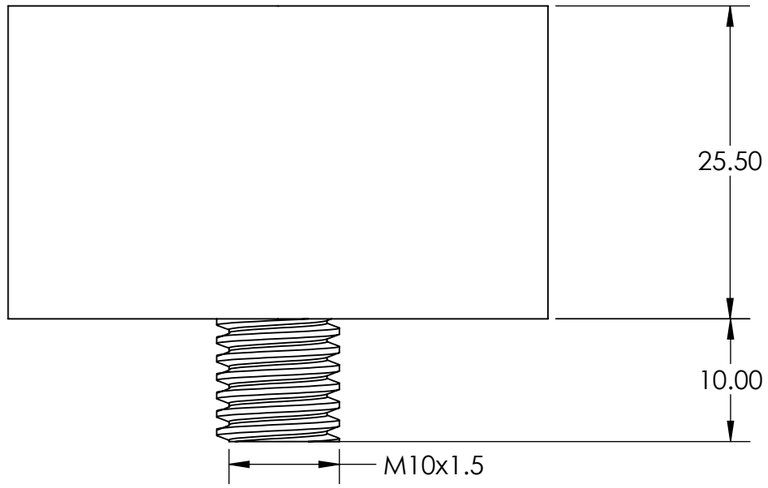


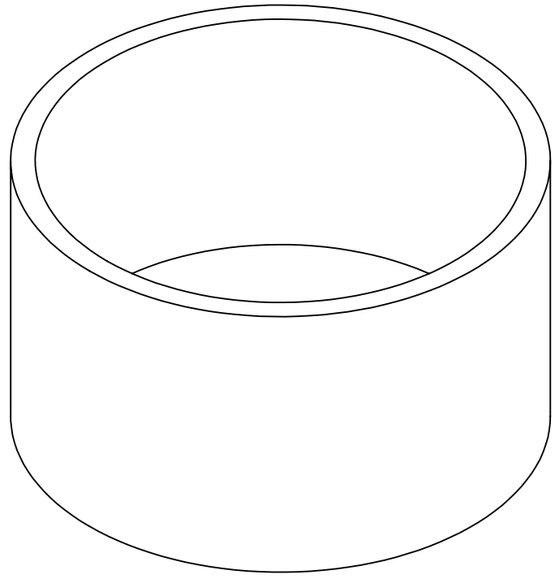


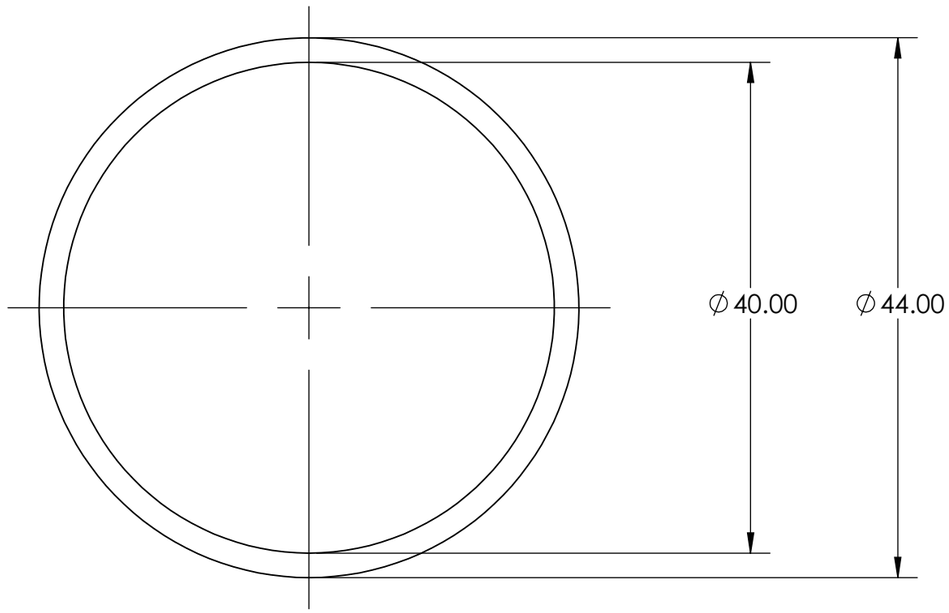
SAMMARD

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Proprietary and confidential The information contained in this drawing is the sole property of Team Sammard. Any reproduction in part or as a whole without consent/permission of Team Sammard is prohibited.	TITLE: Recovery Coupler	
MATERIAL: Aluminium	DWG NO.	A3
WEIGHT:	SCALE:1:2	SHEET 1 OF 4

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TITLE: Charge well Recovery Coupler

MATERIAL: Aluminium

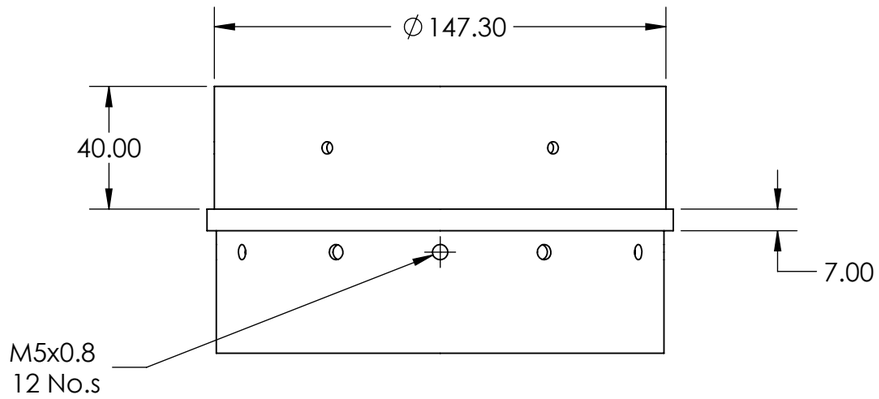
DWG NO. A3

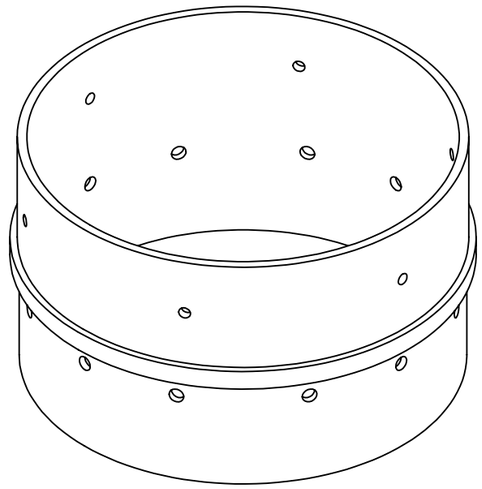
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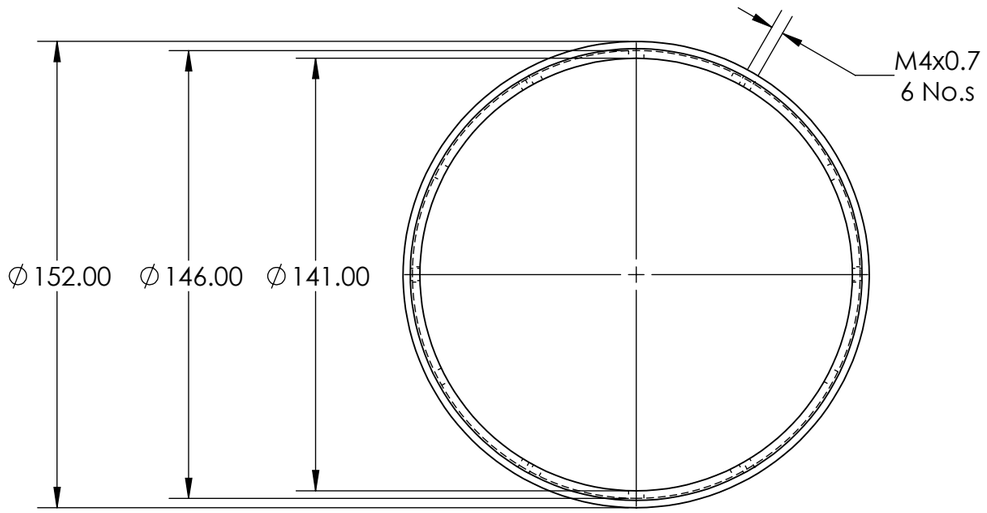
SCALE:2:1

SHEET 2 OF 4

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TITLE: Fin Can Coupler

MATERIAL: Aluminium

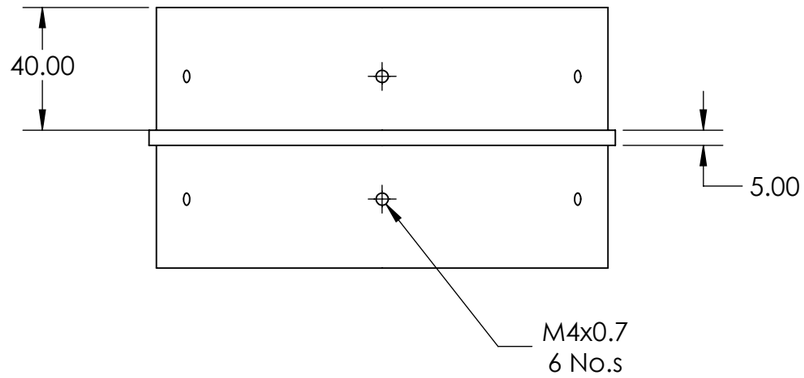
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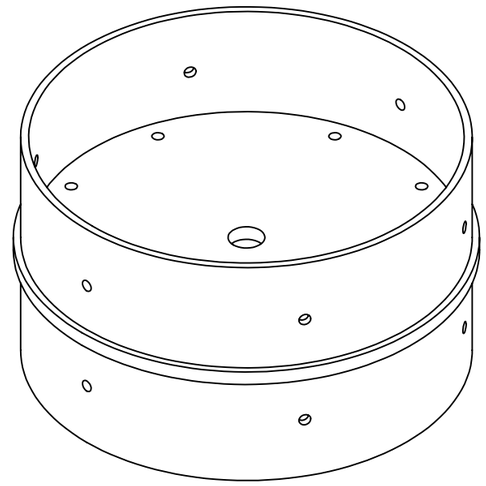
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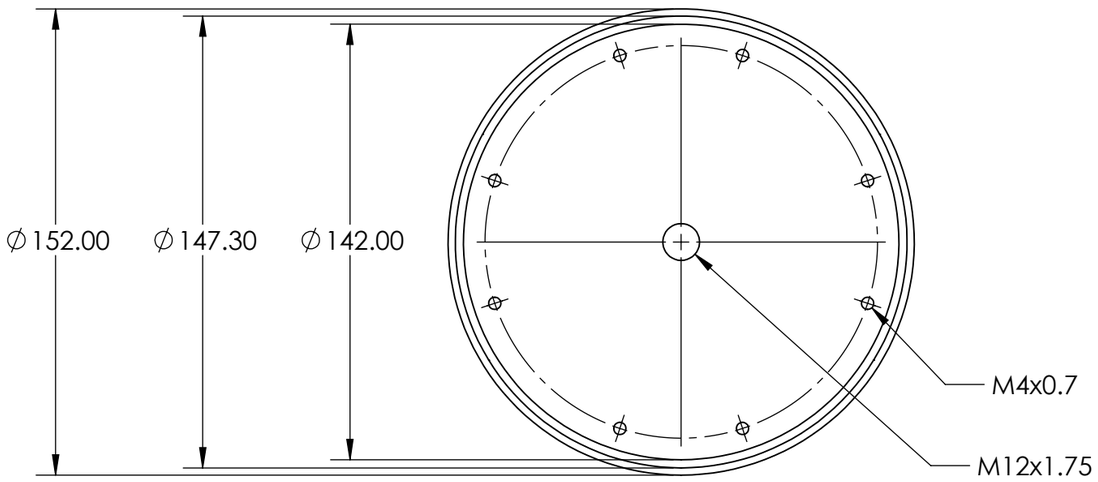
SCALE:1:2

SHEET 3 OF 4

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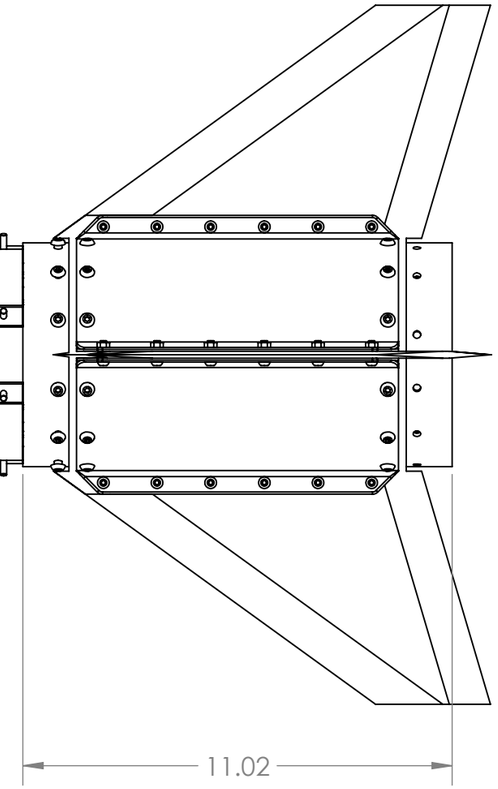
TITLE: Nose cone Coupler

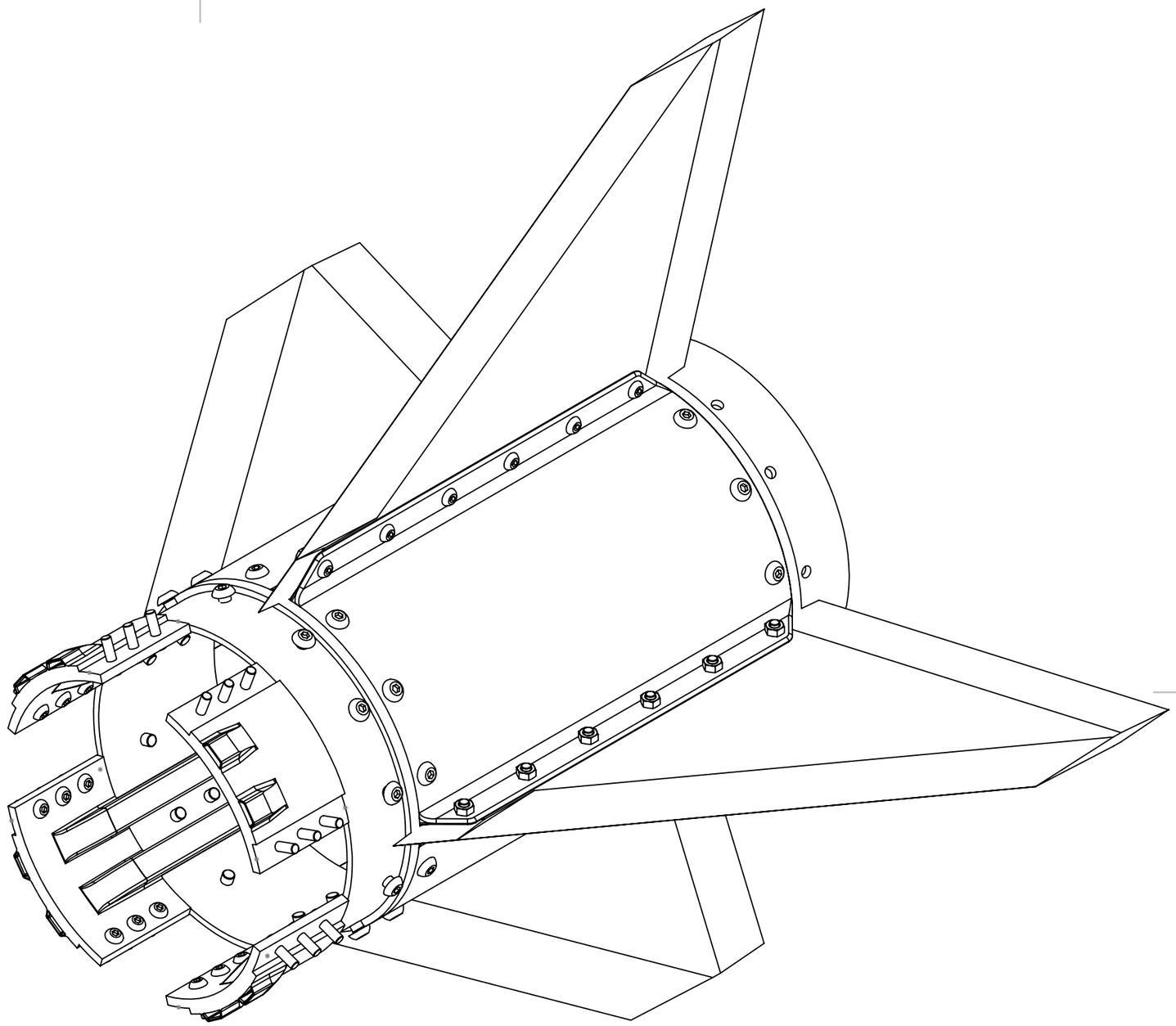
MATERIAL: Aluminium

DWG NO. A3

WEIGHT: | SCALE:1:2 | SHEET 4 OF 4

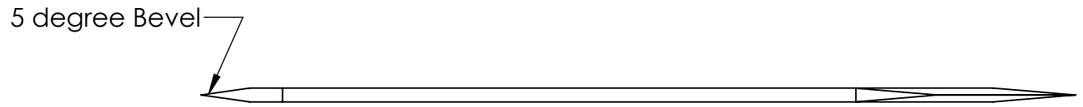
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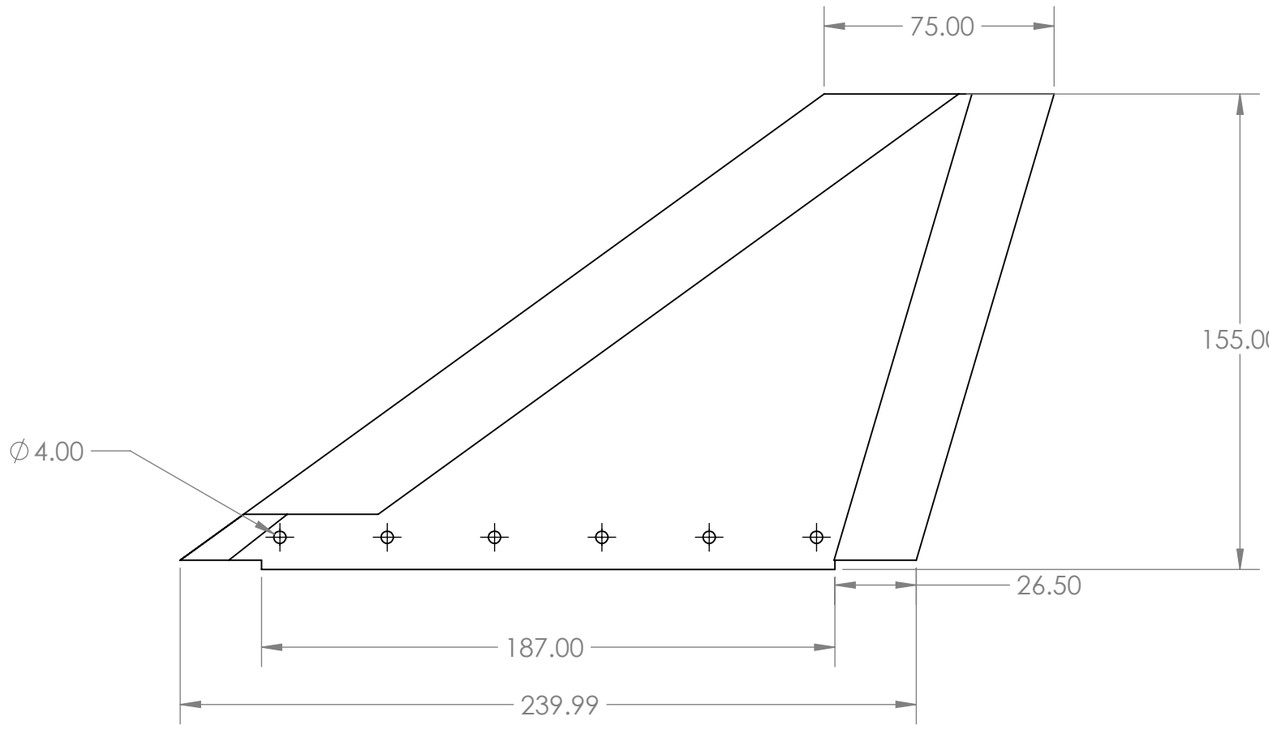




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MATERIAL:	DWG NO.	A3
WEIGHT:	SCALE:1:8	SHEET 1 OF 1

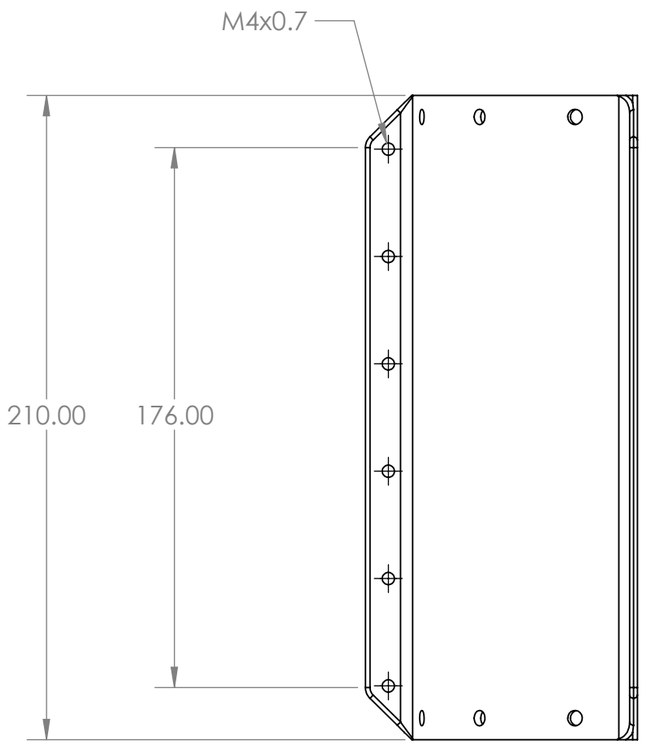


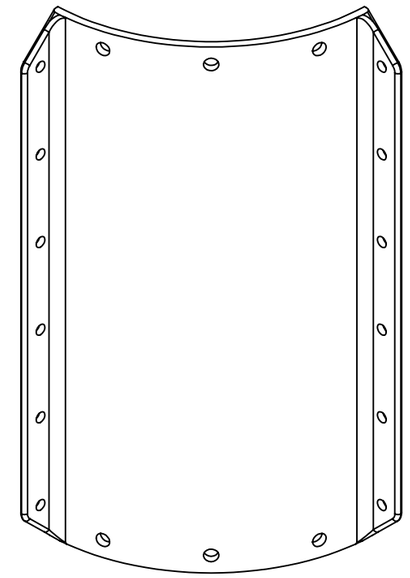


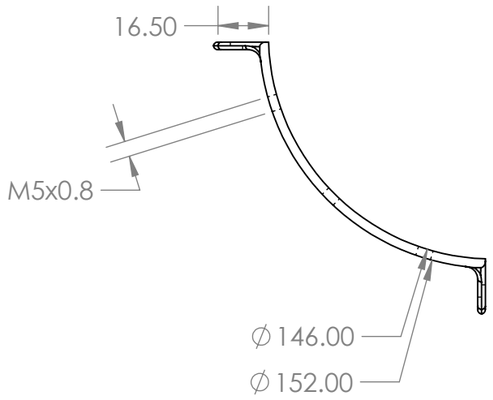
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MATERIAL: Carbon Fibre	DWG NO.	A3
WEIGHT:	SCALE:1:2	SHEET 1 OF 1

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TITLE: Fin Can Plate

MATERIAL: Aluminium 6061-T6

DWG NO. A3

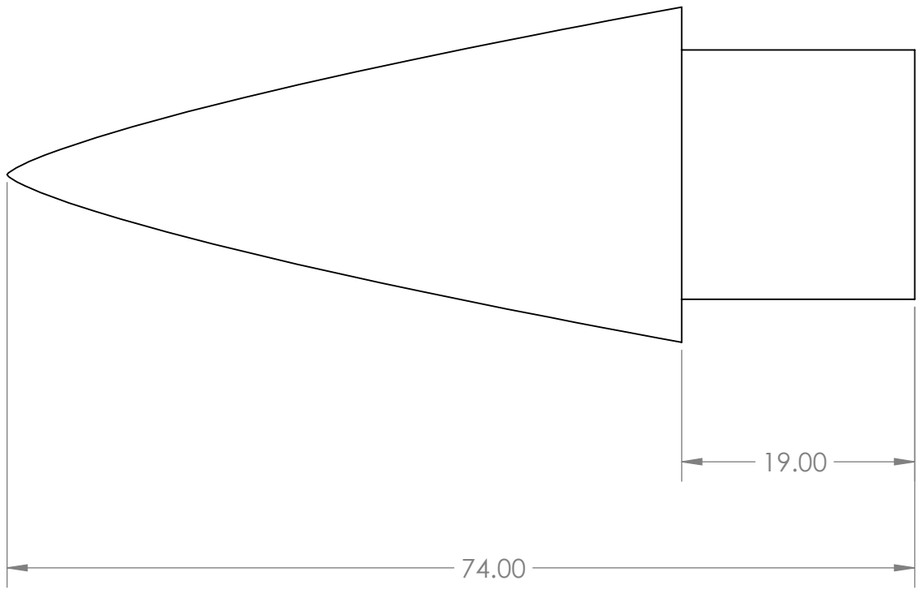
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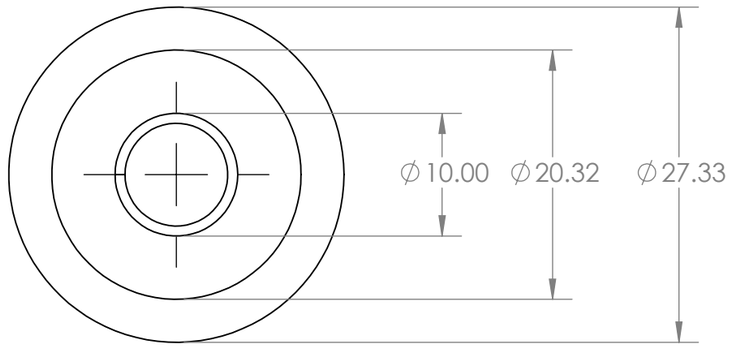
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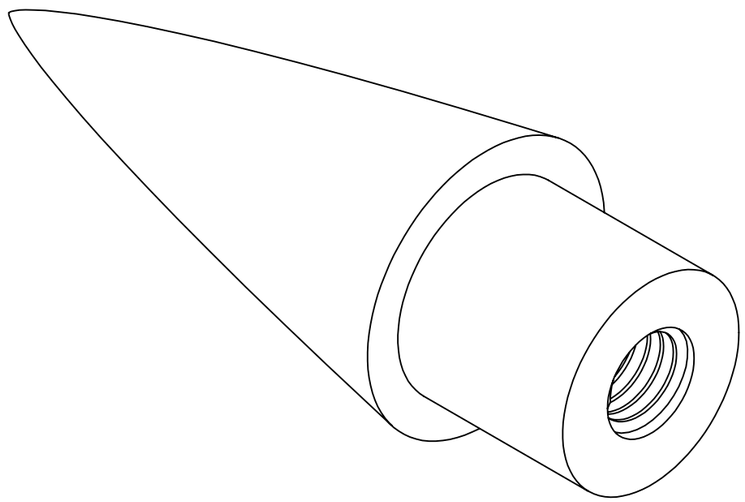
SHEET 1 OF 1

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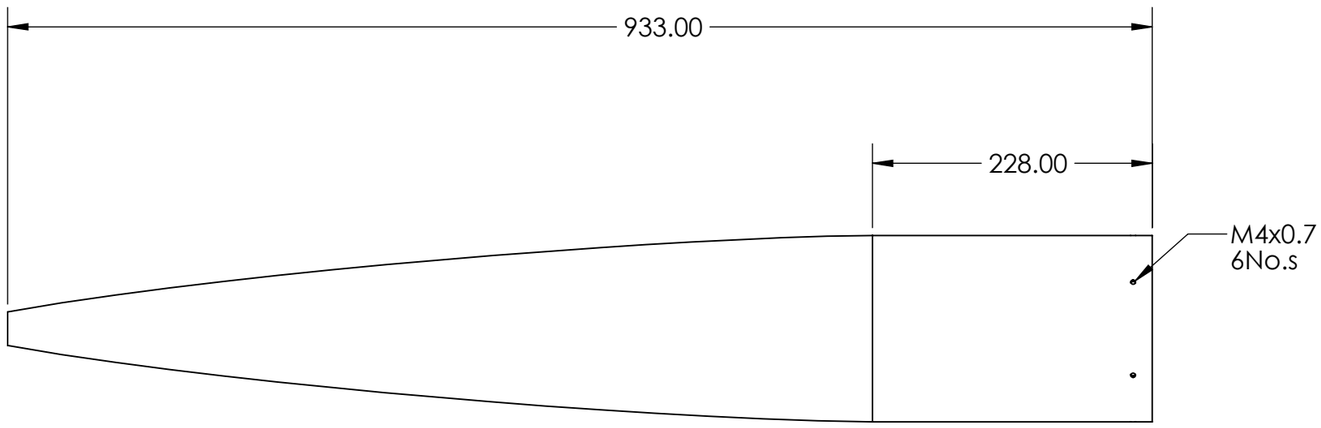


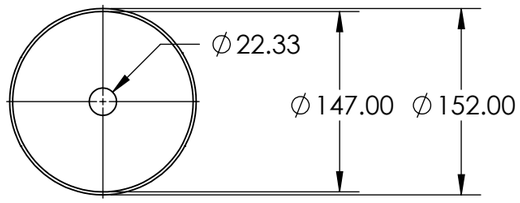


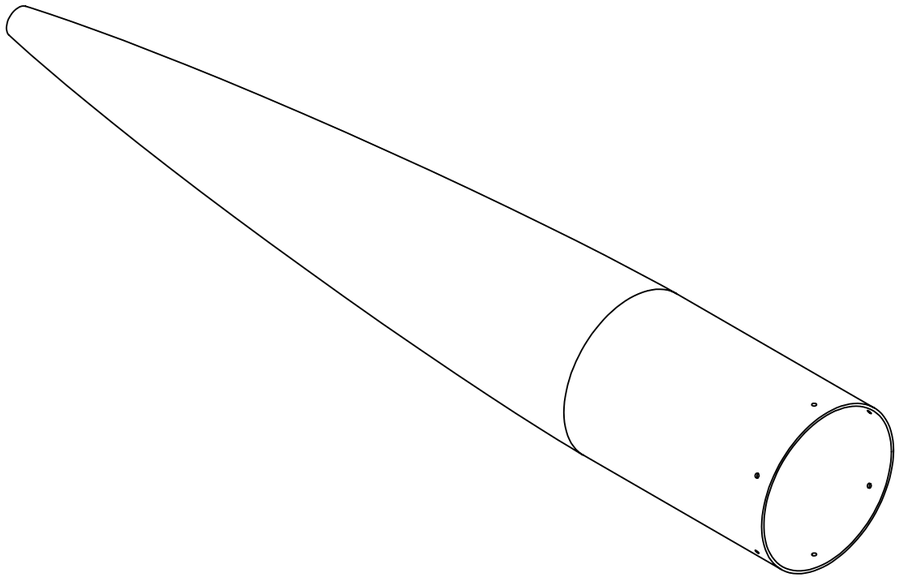
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Proprietary and confidential The information contained in this drawing is the sole property of Team Sammard. Any reproduction in part or as a whole without consent/permission of Team Sammard is prohibited.	TITLE: Nose Cone Tip	
MATERIAL: Aluminium	DWG NO.	A3
WEIGHT:	SCALE:2:1	SHEET 1 OF 3

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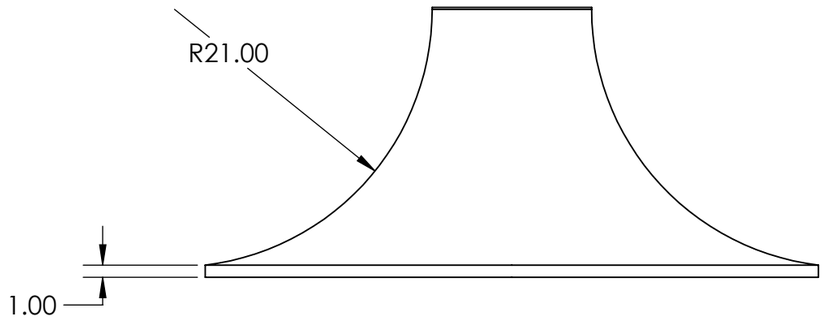


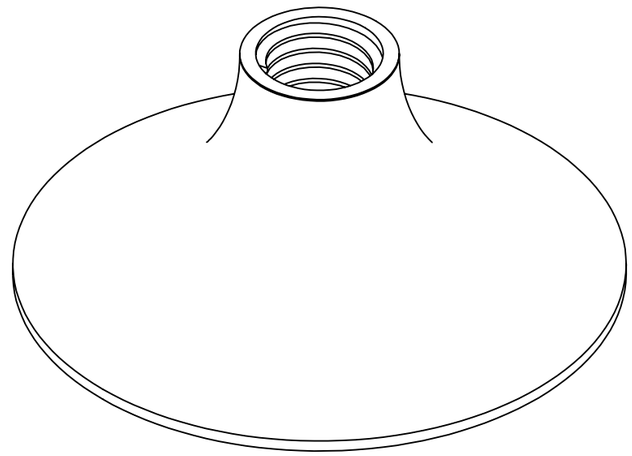


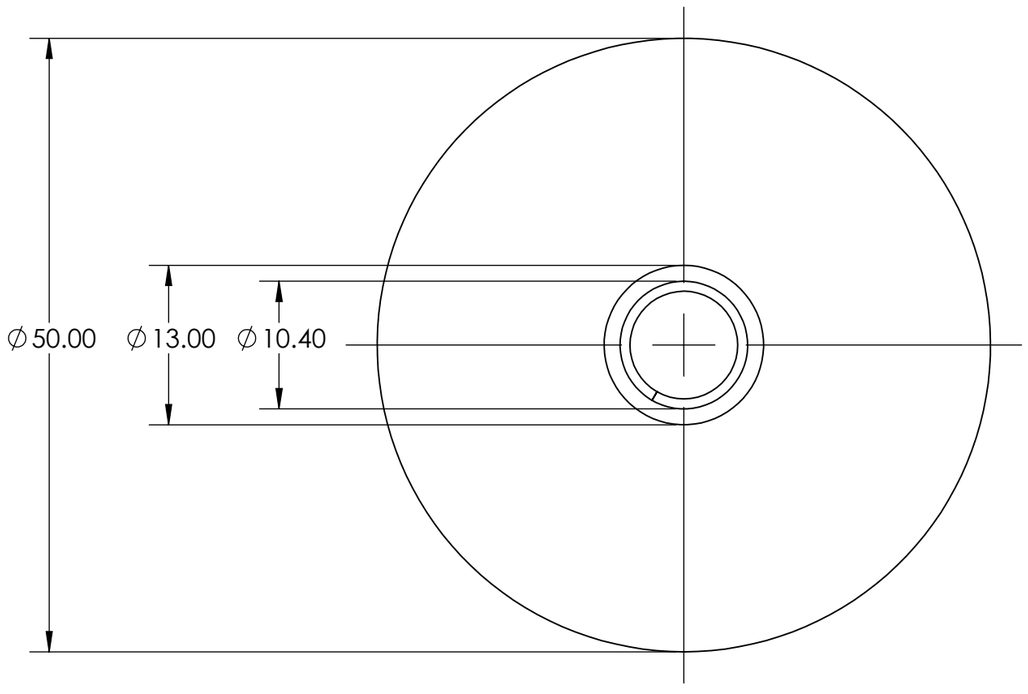


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MATERIAL: Fibreglass	DWG NO.	A3
WEIGHT:	SCALE:1:5	SHEET 2 OF 3

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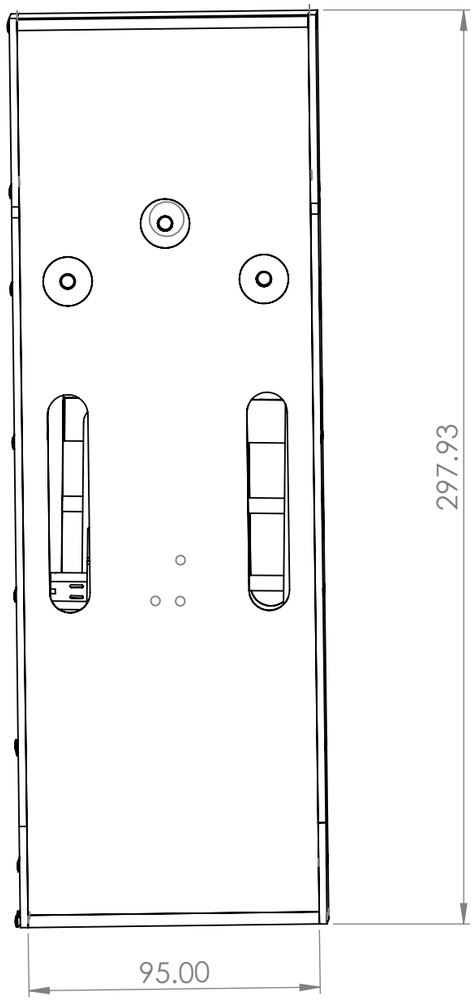


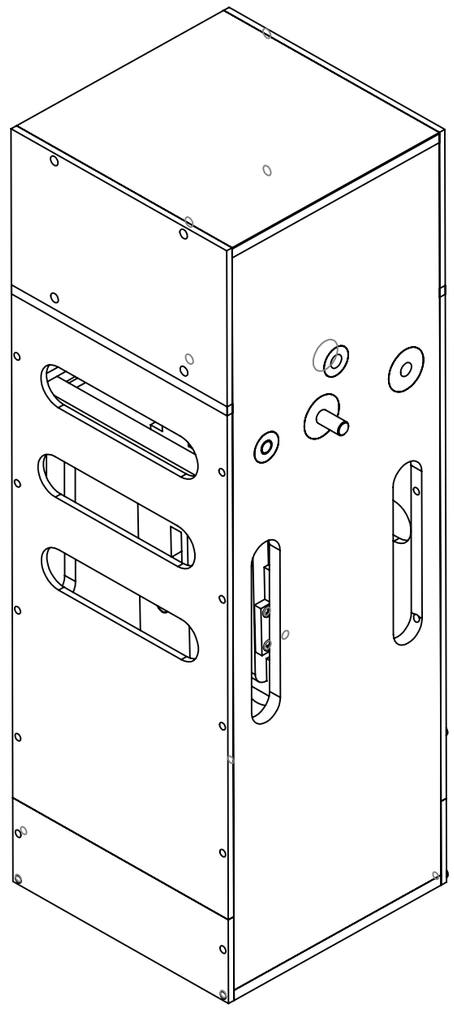




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MATERIAL: Aluminium	DWG NO.	A3
WEIGHT:	SCALE:2:1	SHEET 3 OF 3

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TITLE: PAYLOAD ASSEMBLY

MATERIAL:

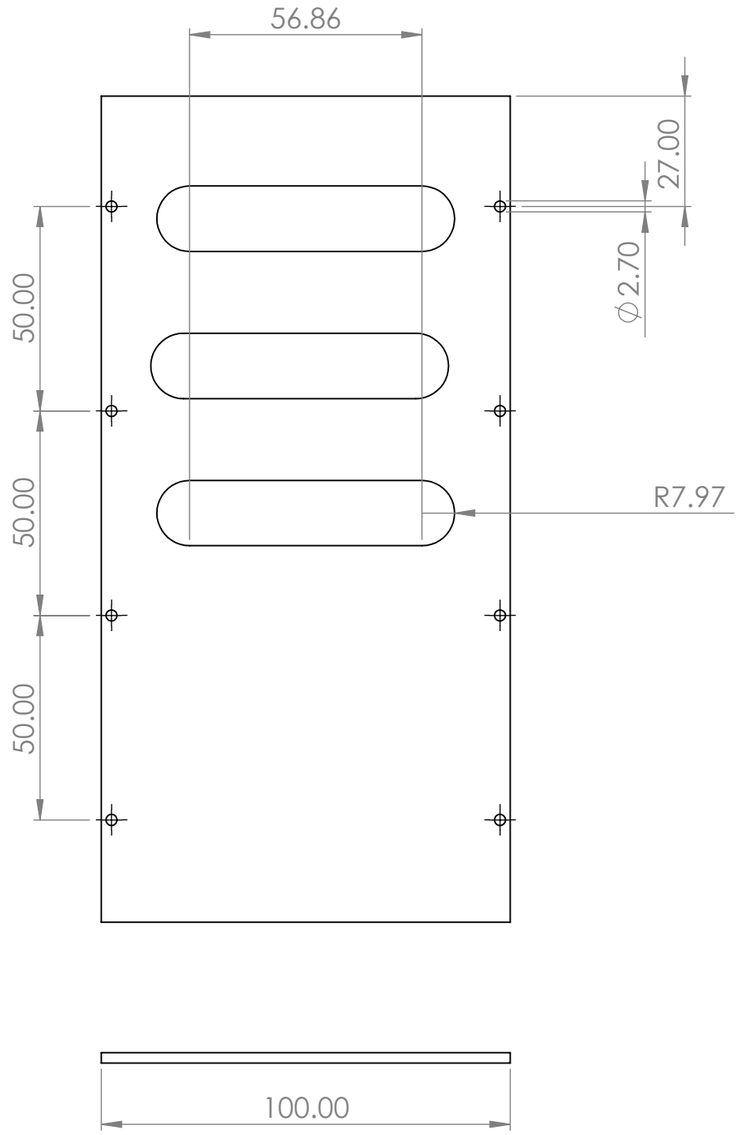
DWG NO.

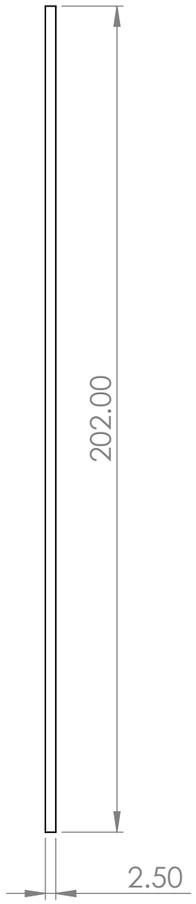
A3

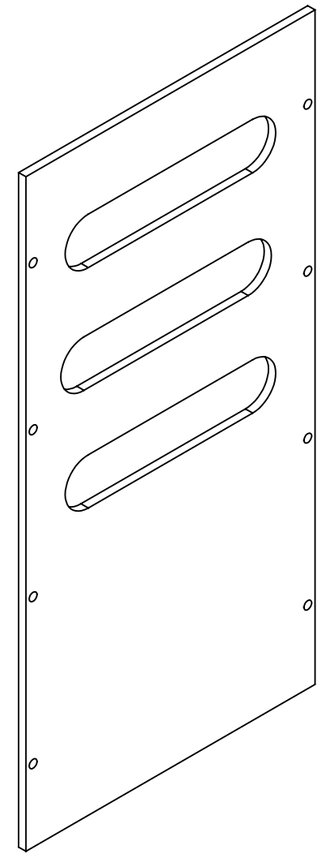
WEIGHT:

SCALE:1:5

SHEET 1 OF 1







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TITLE:
PAYLOAD SIDE SUPPORT

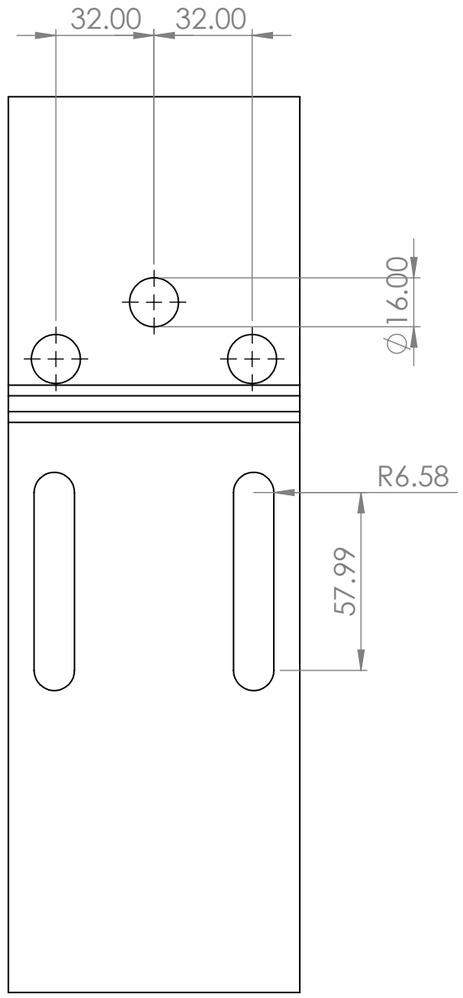
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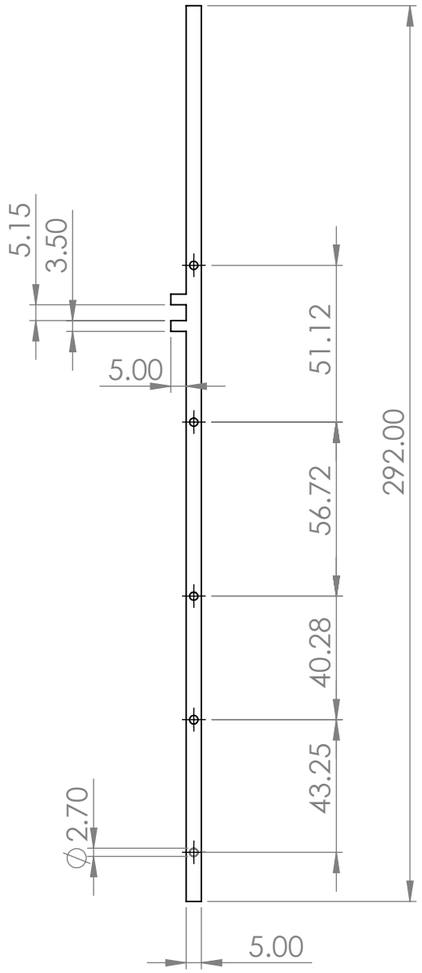
DWG NO. A3

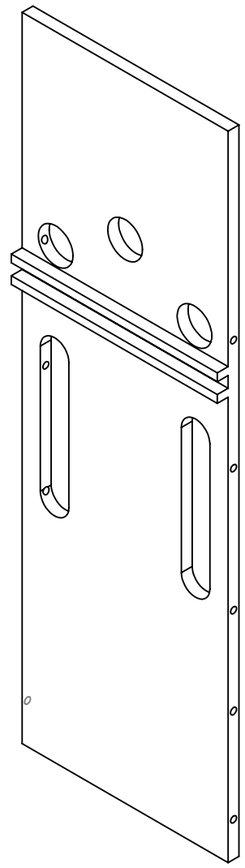
WEIGHT:

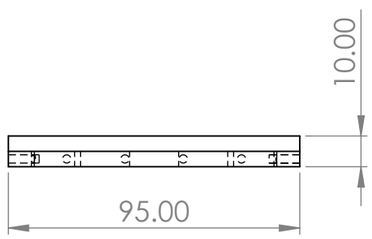
SCALE:1:2

SHEET 4 OF 22









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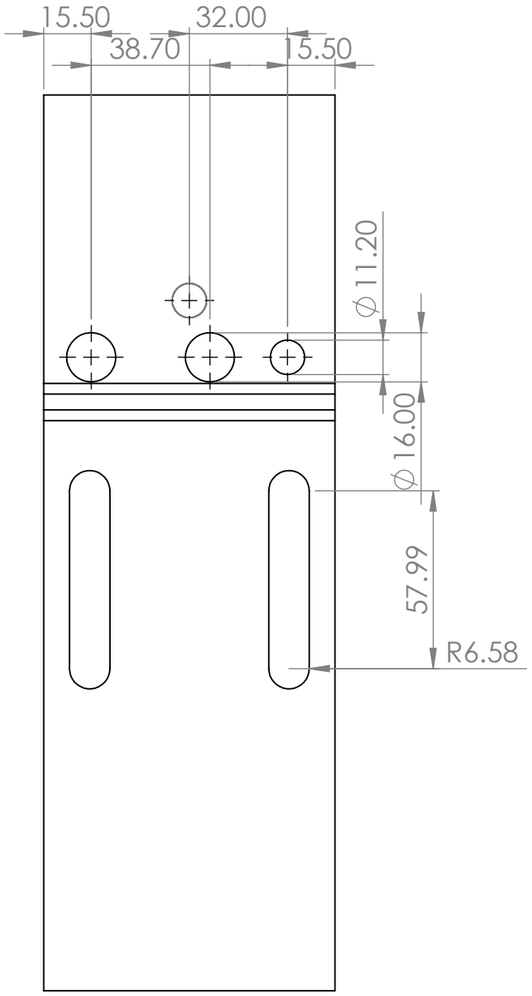
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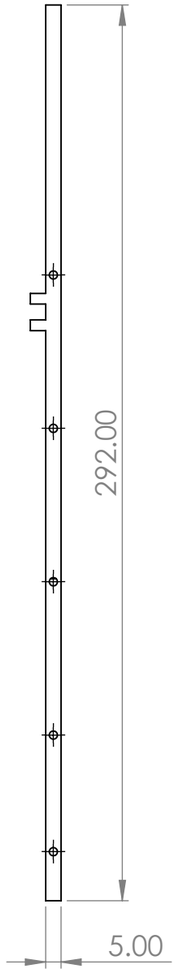
TITLE:
PAYLOAD SIDE SUPPORT

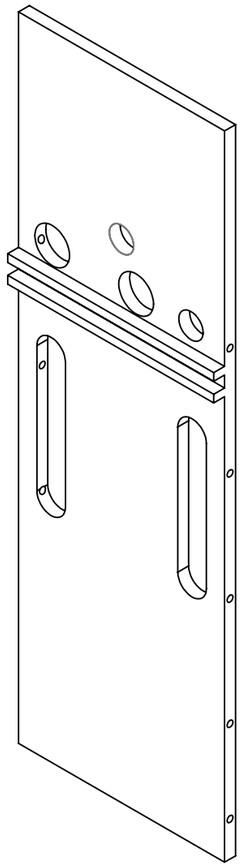
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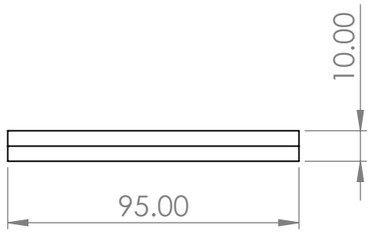
DWG NO. | A3

WEIGHT: | SCALE:1:2 | SHEET 5 OF 22











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TITLE:
PAYLOAD SIDE SUPPORT

MATERIAL:
ABS

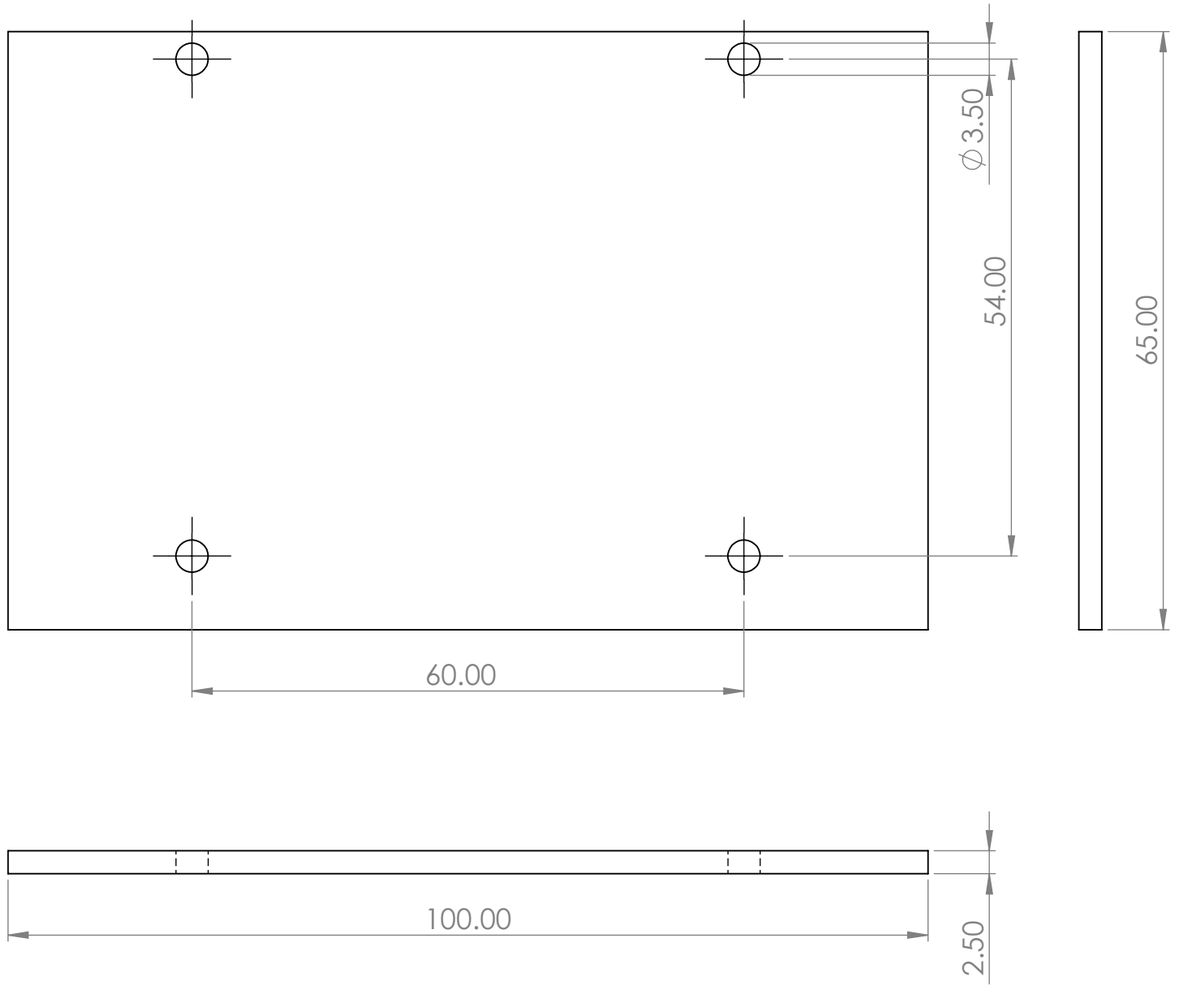
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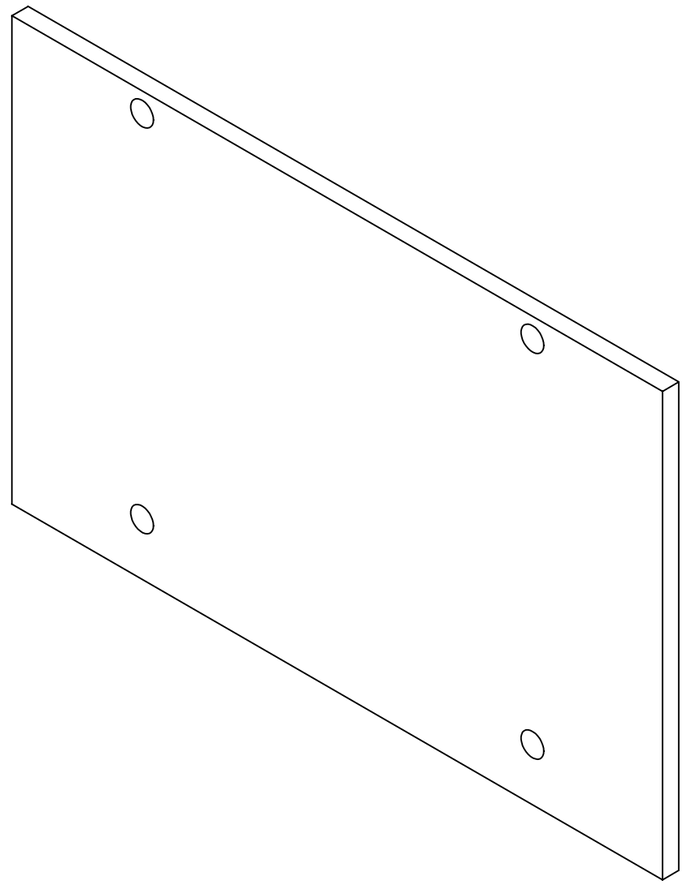
A3

WEIGHT:

SCALE:1:2

SHEET 6 OF 22





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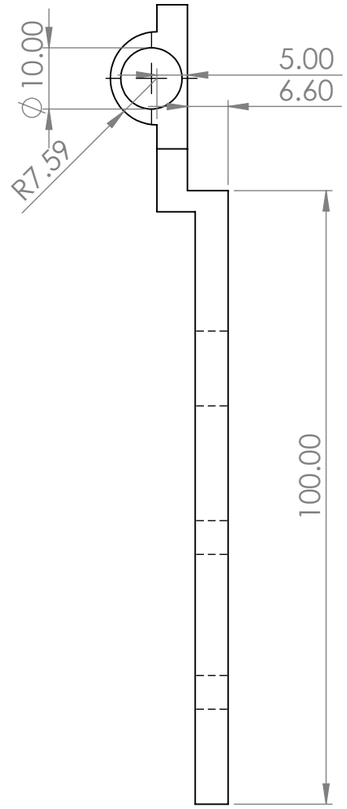
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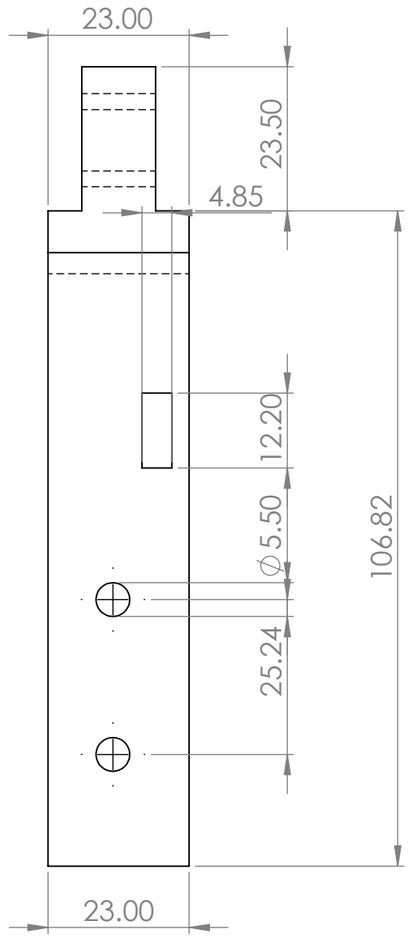
TITLE: PAYLOAD SIDE SUPPORT

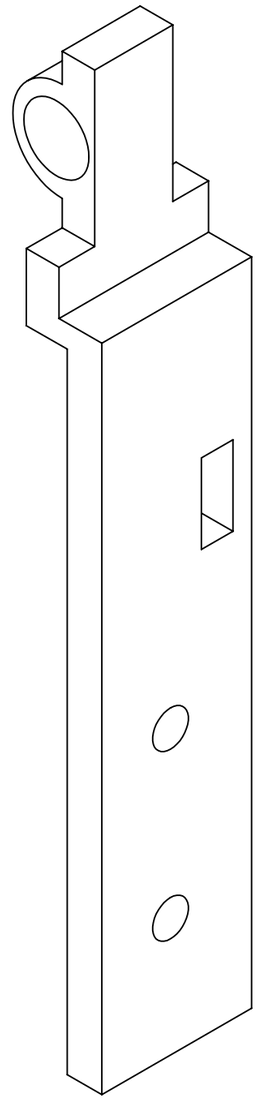
MATERIAL: ABS

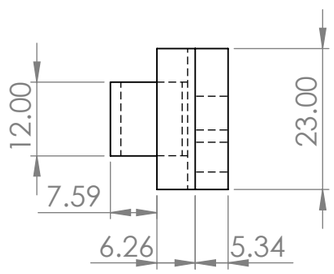
DWG NO. | A3

WEIGHT: | SCALE:2:1 | SHEET 7 OF 22









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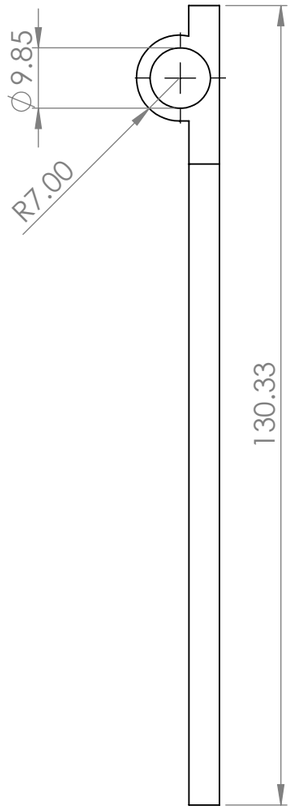
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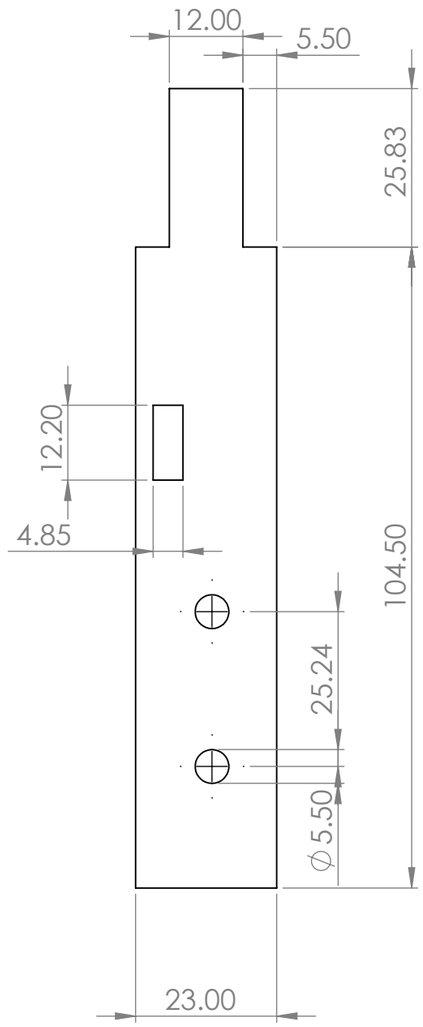
TITLE: PAYLOAD ARM_1

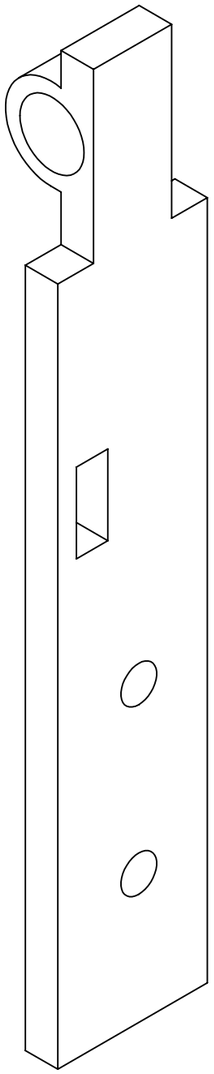
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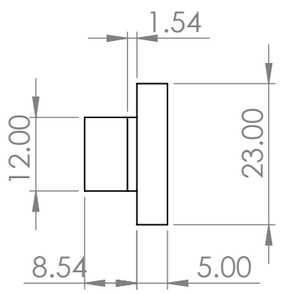
DWG NO. | A3

WEIGHT: | SCALE:1:1 | SHEET 8 OF 22









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TITLE: PAYLOAD ARM_2

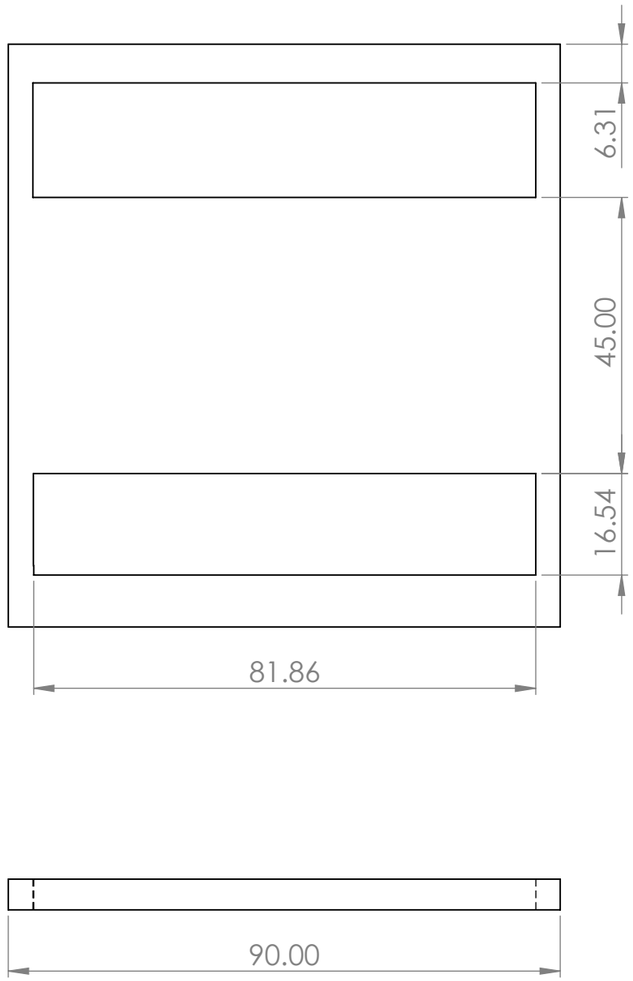
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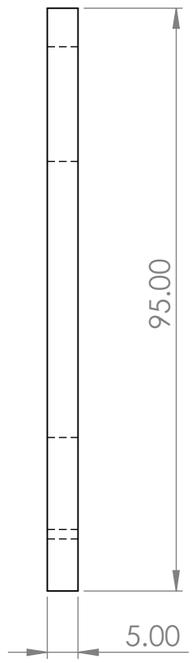
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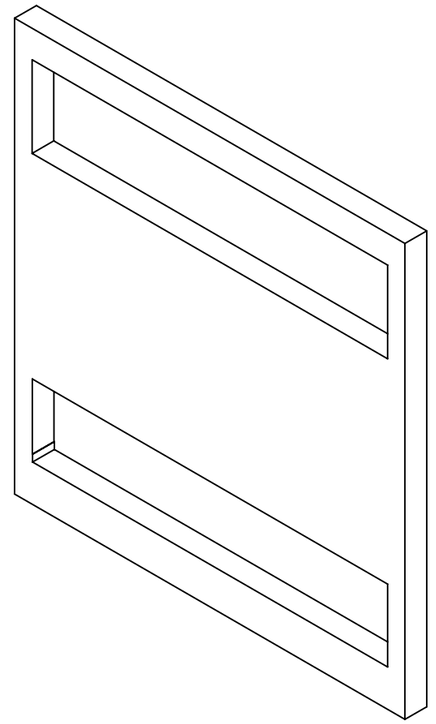
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SCALE:1:1

SHEET 9 OF 22







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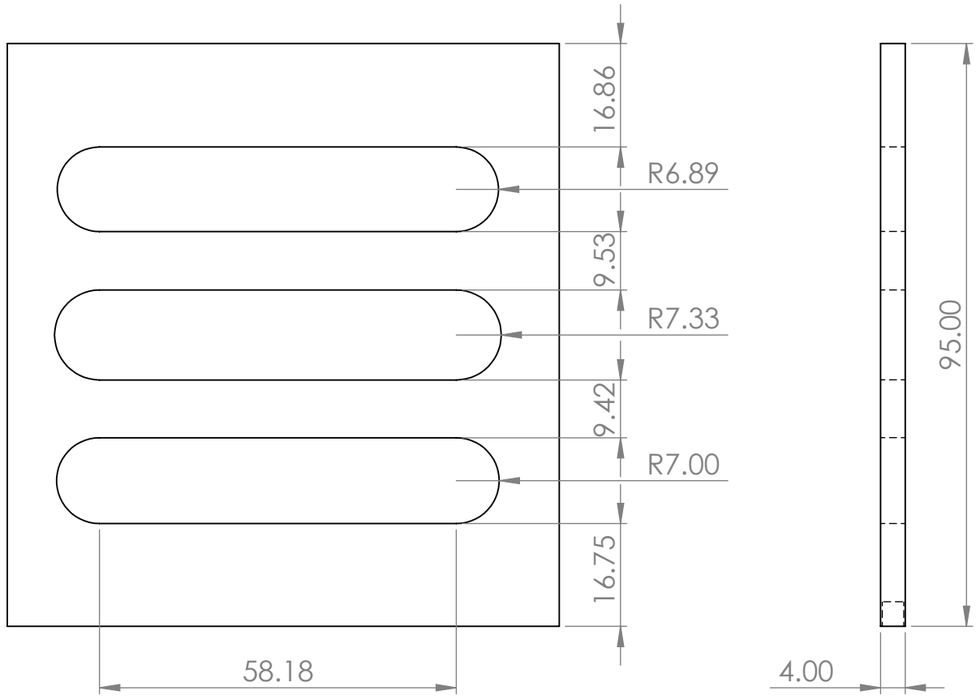
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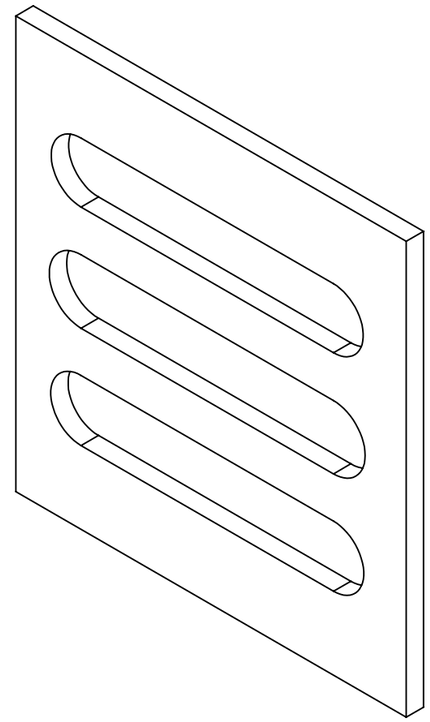
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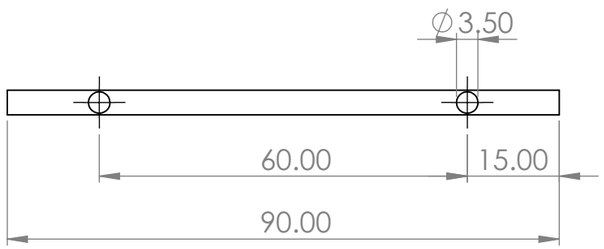
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DWG NO. | A3

WEIGHT: | SCALE:1:1 | SHEET 10 OF 22







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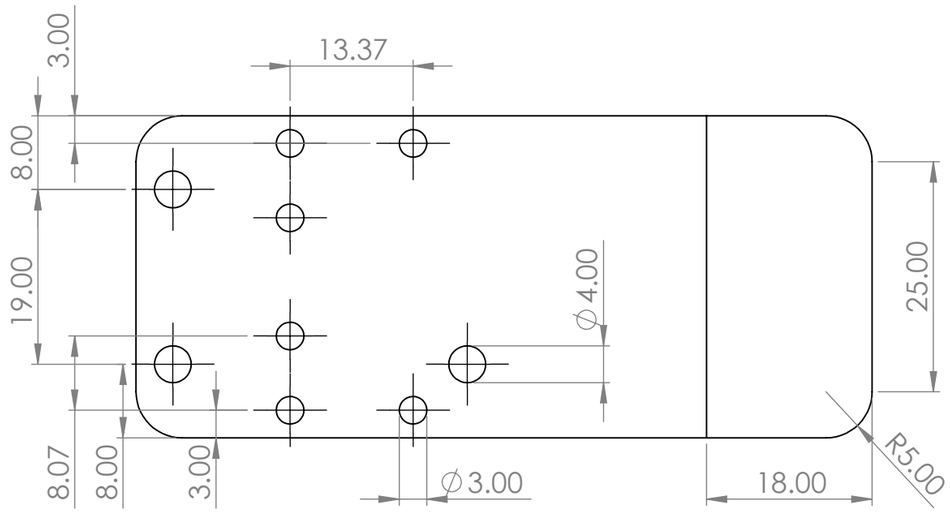
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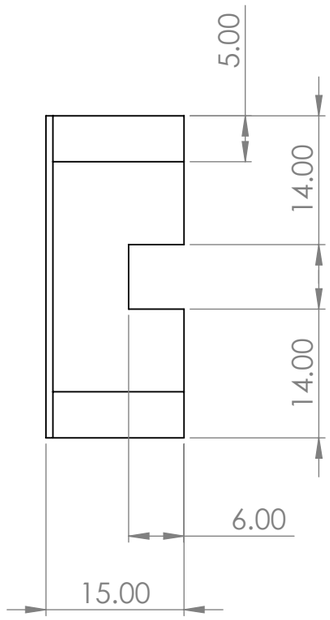
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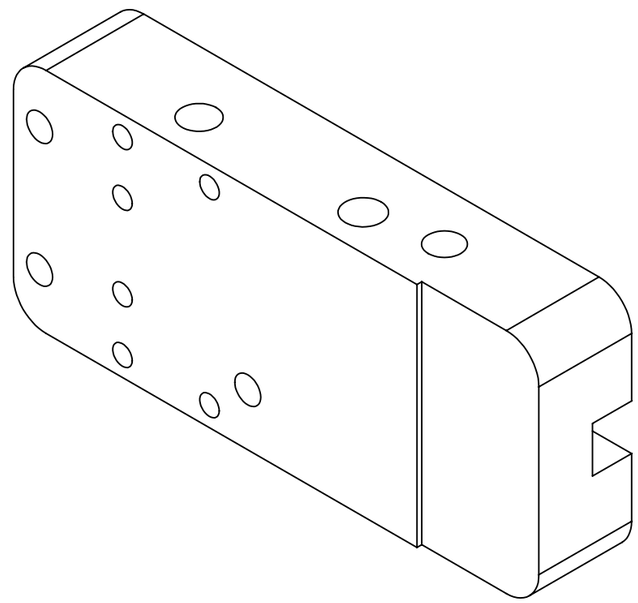
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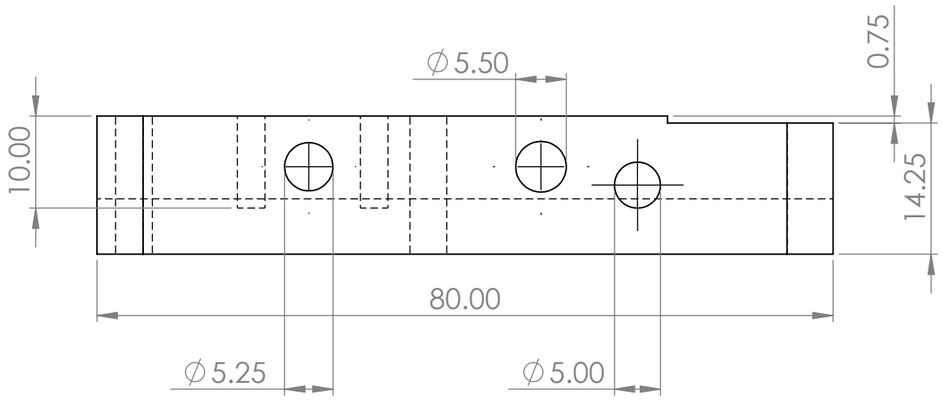
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WEIGHT: | SCALE:1:1 | SHEET 11 OF 22









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PAYLOAD EXTRUDER BASE

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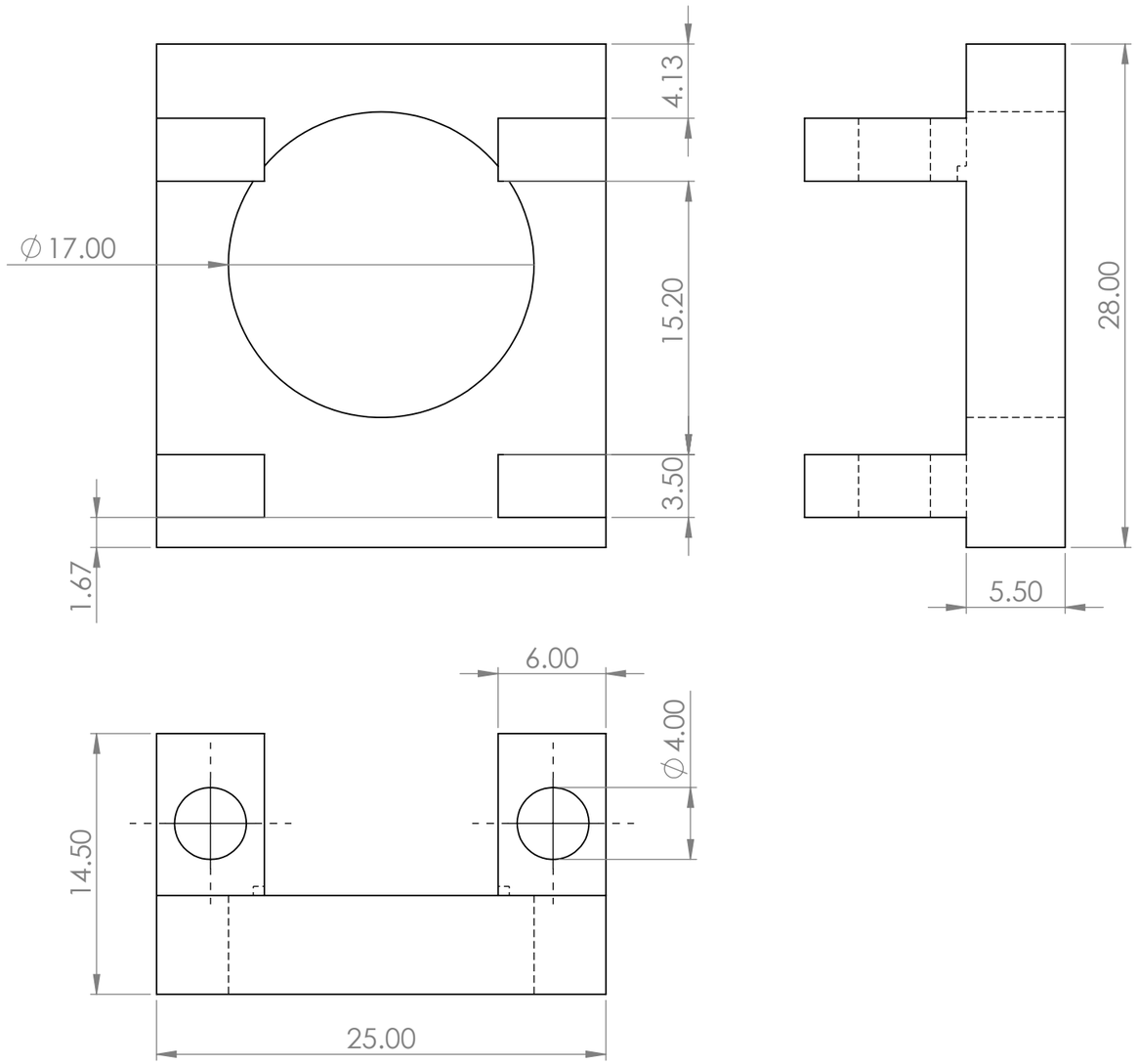
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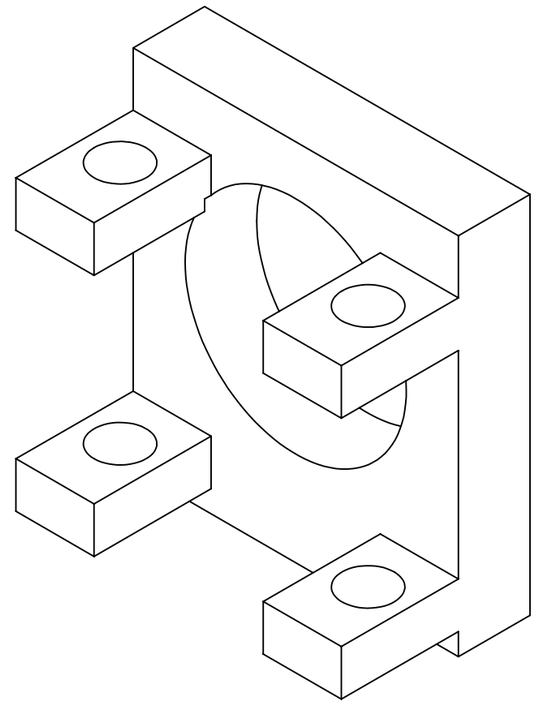
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WEIGHT:

SCALE:2:1

SHEET 12 OF 22





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TITLE:
PAYLOAD EXTRUDER MOUNTING BRACKET

MATERIAL:
ABS

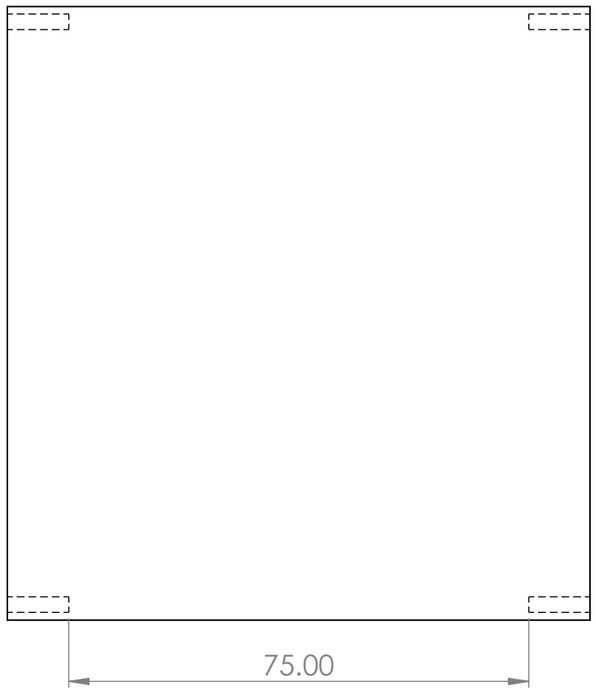
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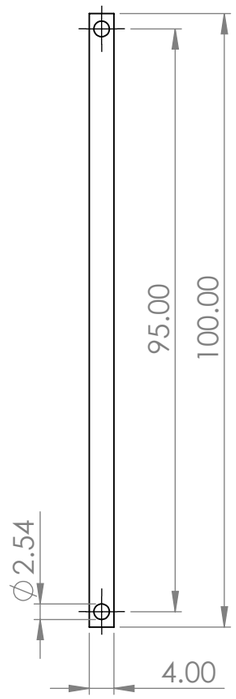
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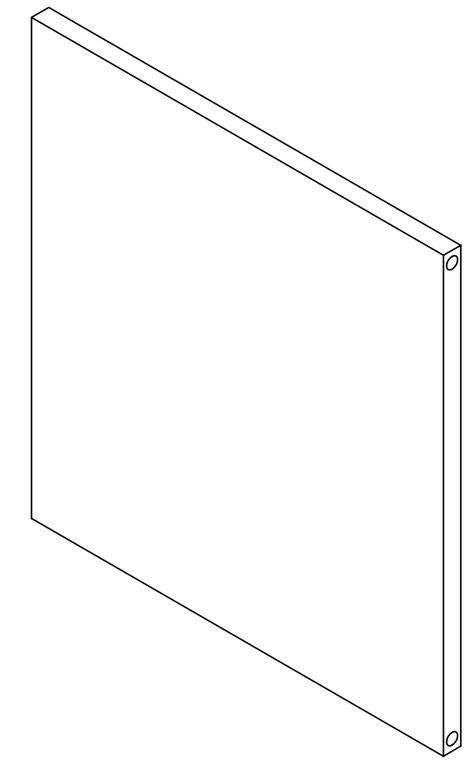
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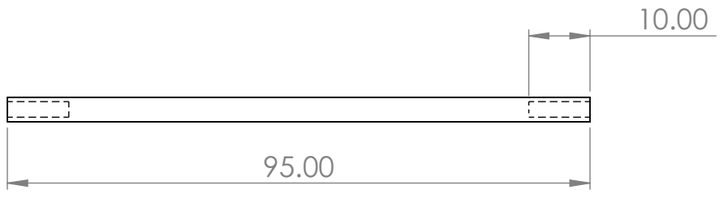
SCALE:2:1

SHEET 13 OF 22









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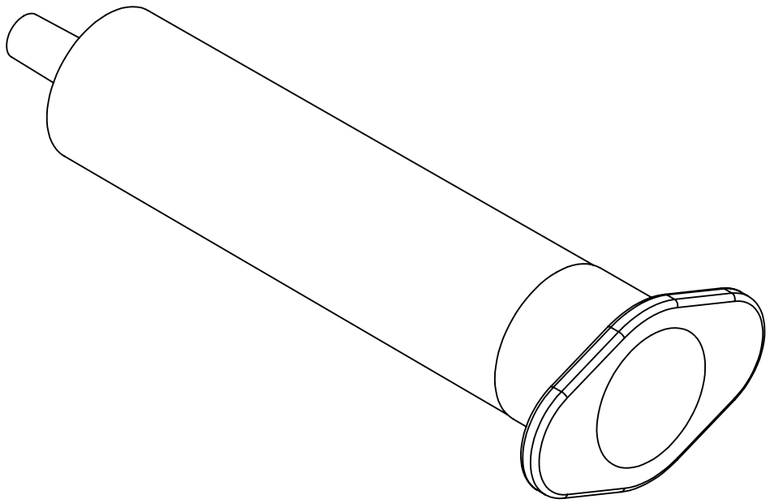
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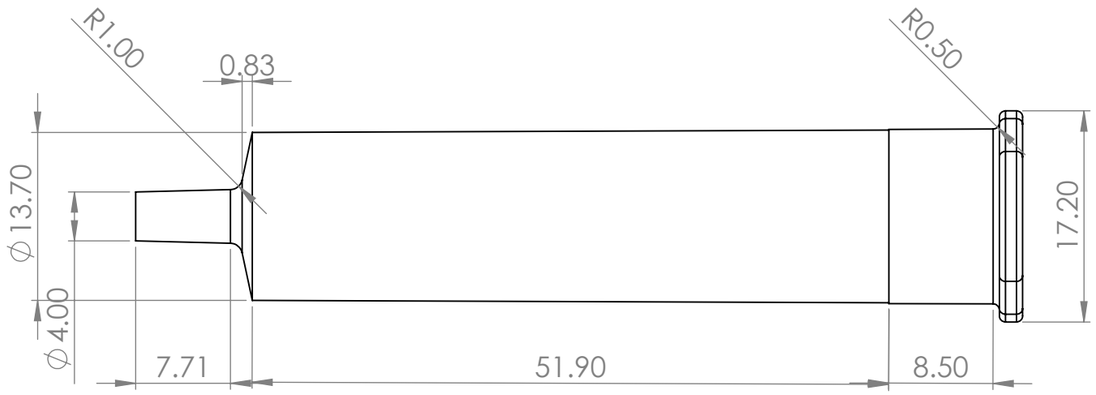
TITLE: PAYLOAD BASE PLATE

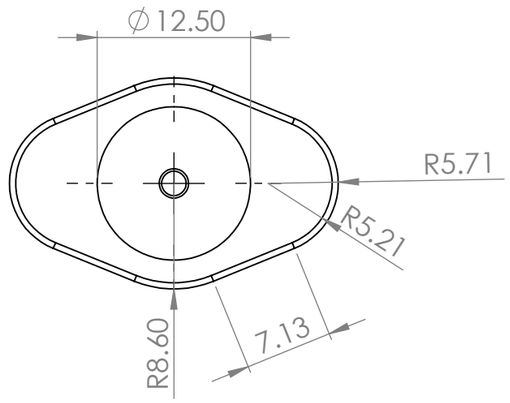
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DWG NO. | A3

WEIGHT: | SCALE:1:1 | SHEET 14 OF 22







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TITLE: PAYLOAD
5ML SYRINGE

MATERIAL: PP

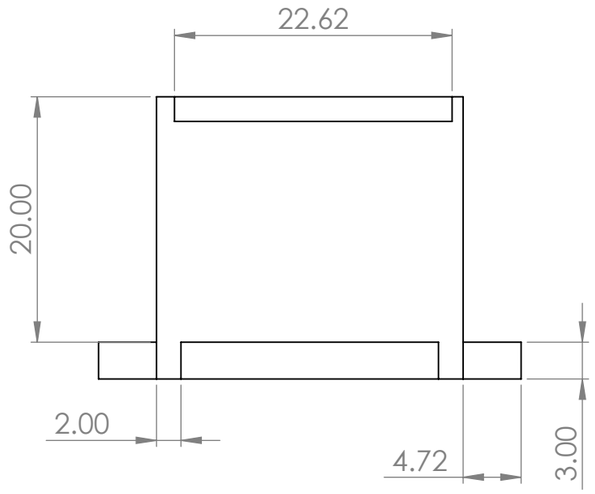
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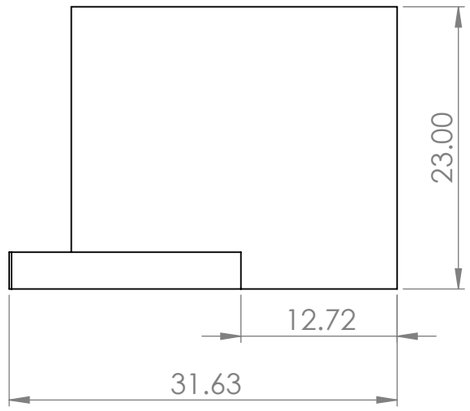
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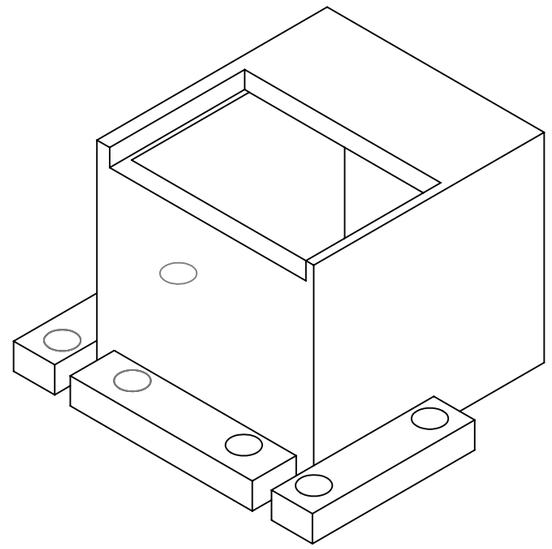
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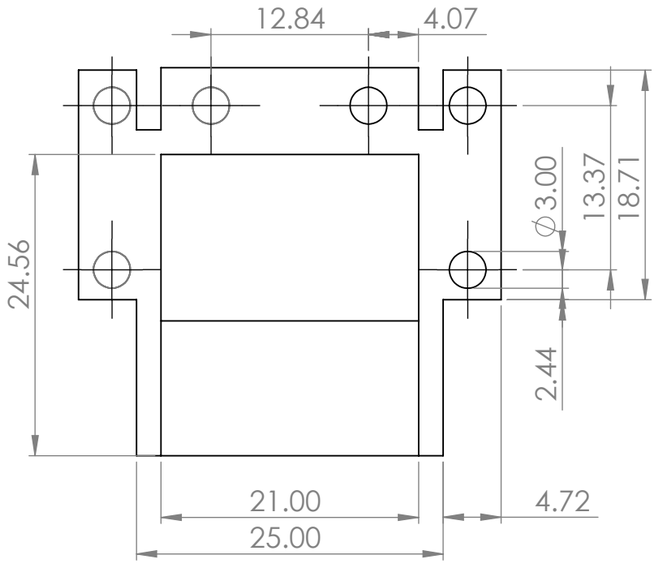
SCALE:2:1

SHEET 15 OF 22









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TITLE: PAYLOAD MOTOR MOUNT

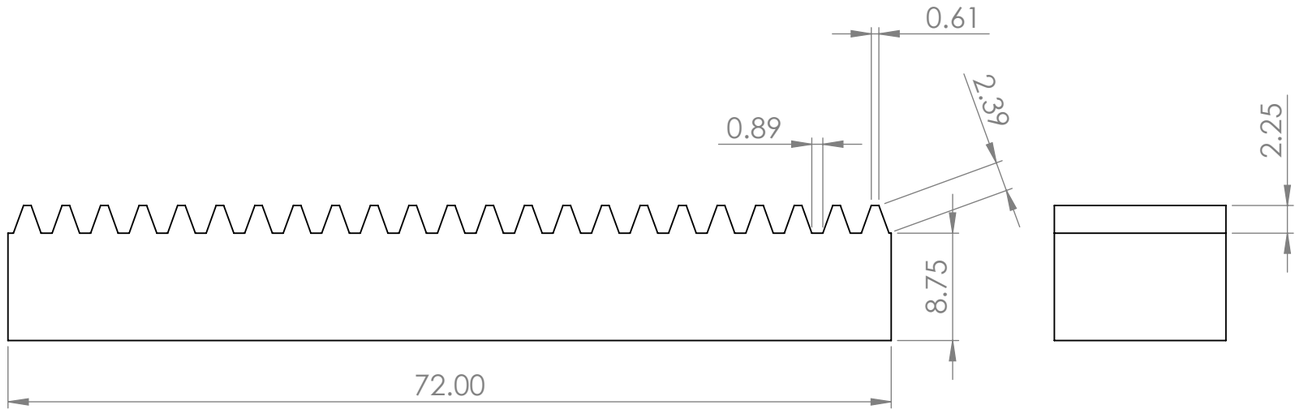
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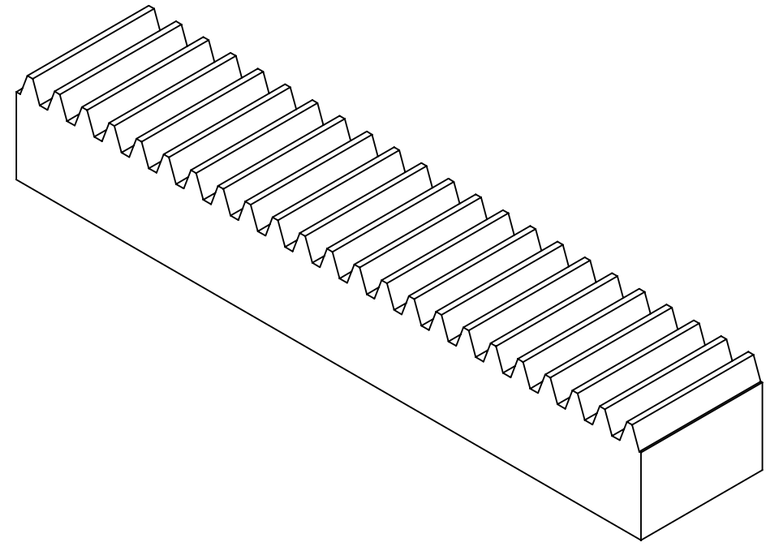
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WEIGHT:

SCALE:2:1

SHEET 16 OF 22





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TITLE: PAYLOAD RACK

MATERIAL: ABS

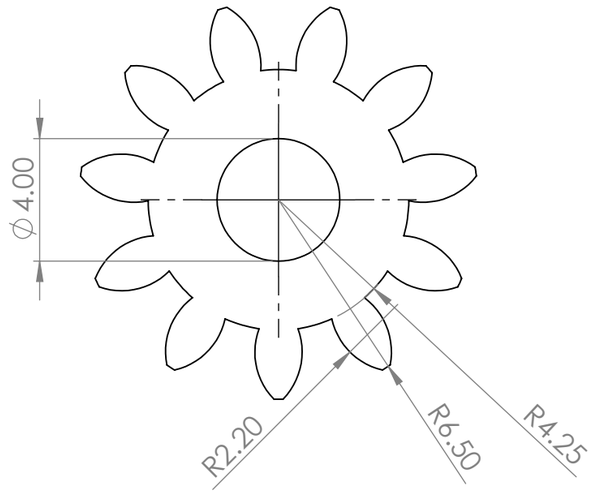
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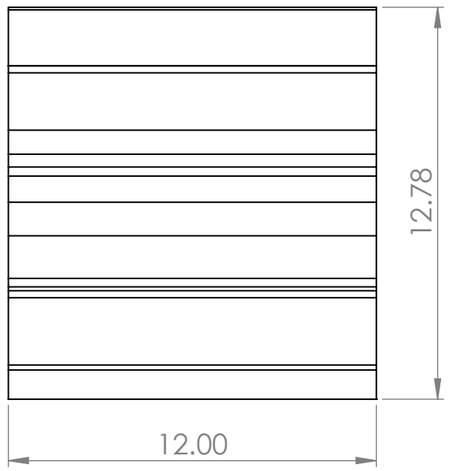
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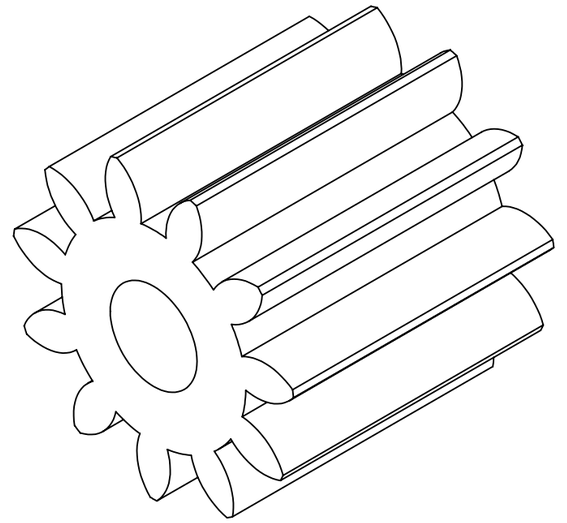
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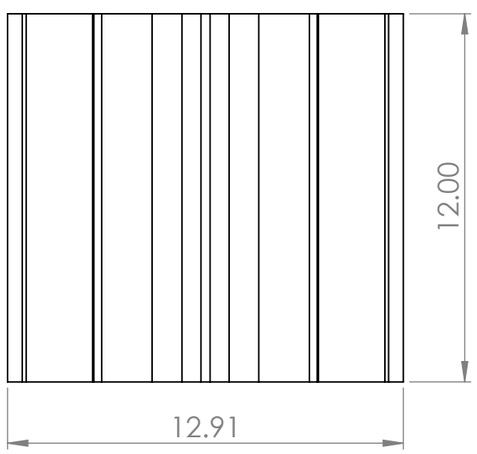
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SHEET 17 OF 22









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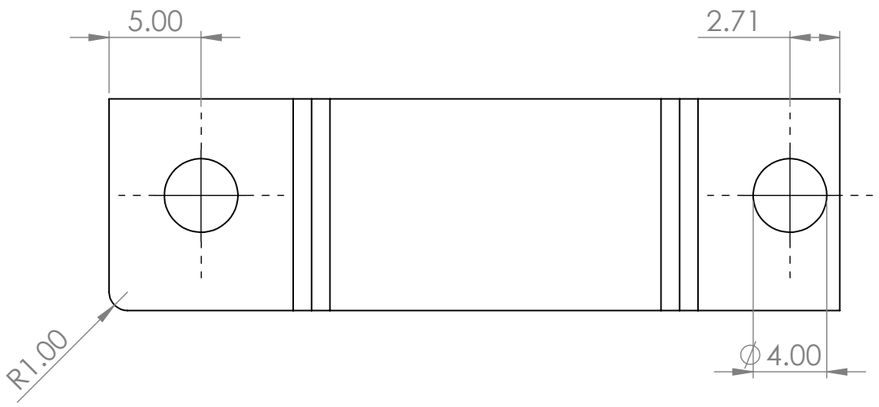
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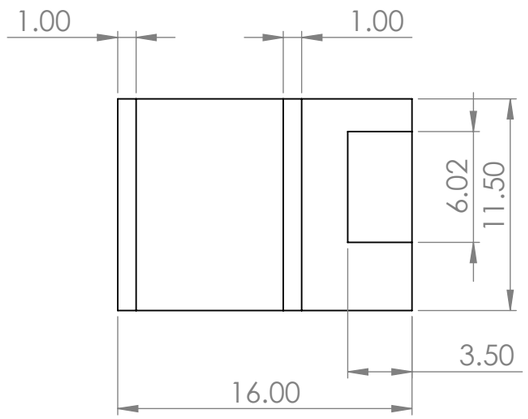
TITLE: PAYLOAD PINION

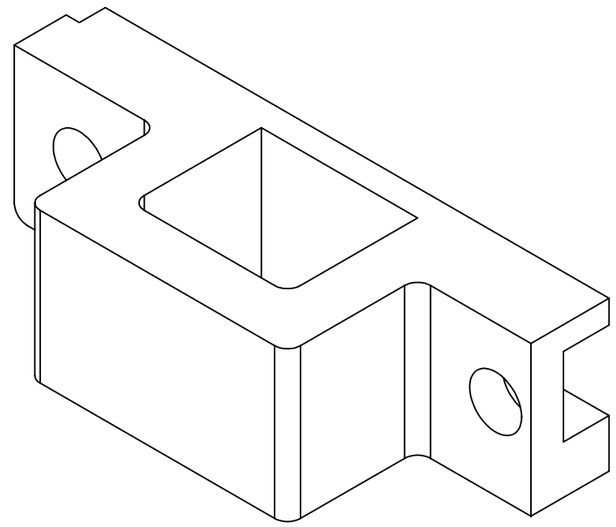
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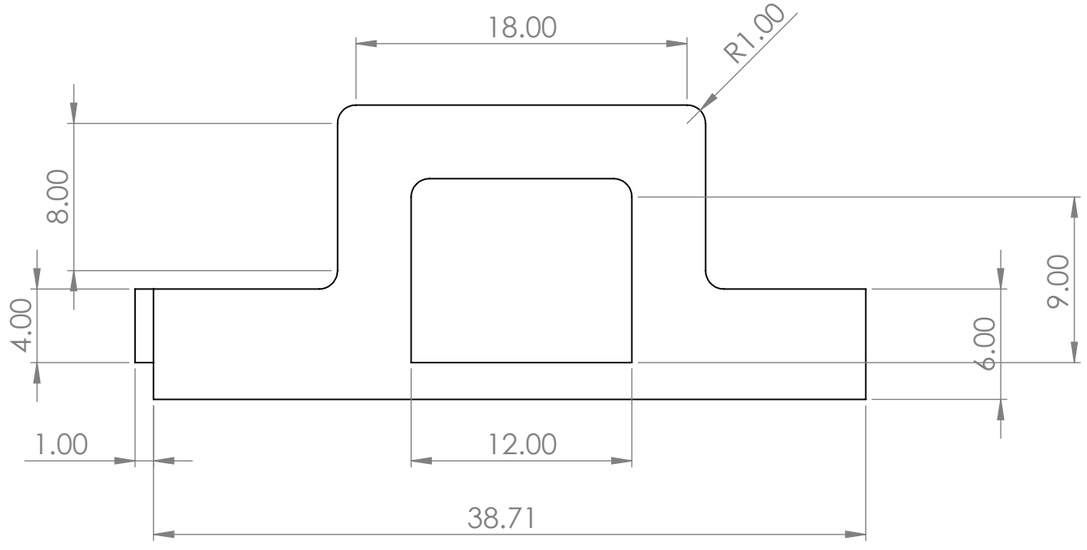
DWG NO. | A3

WEIGHT: | SCALE:5:1 | SHEET 18 OF 22









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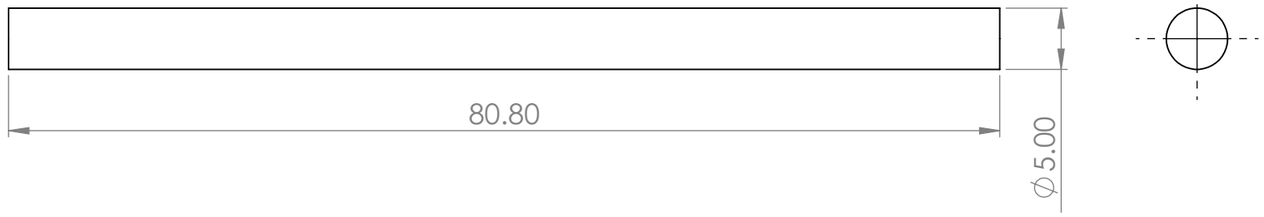
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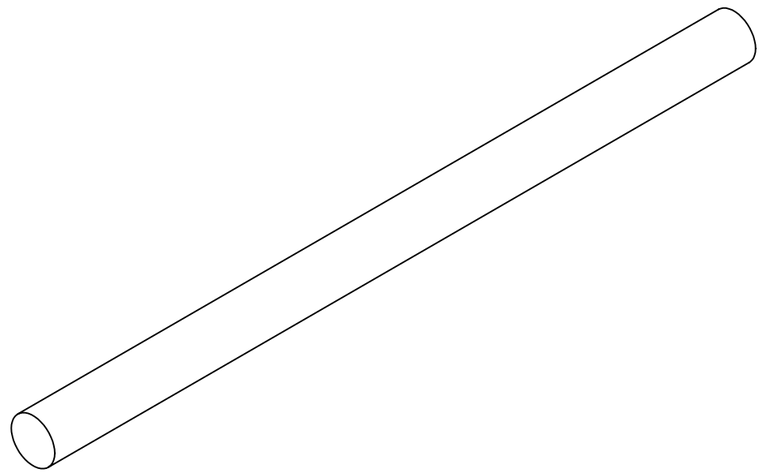
TITLE: PAYLOAD MOTOR MOUNT

MATERIAL: ABS

DWG NO. | A3

WEIGHT: | SCALE:2:1 | SHEET 19 OF 22





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MATERIAL:
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WEIGHT:

DO NOT SCALE DRAWING

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TITLE:
PAYLOAD LINEAR SHAFT

DWG NO.

A3

SCALE:1:1

SHEET 20 OF 22

VIII. REFERENCES

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